MAGNETOM Vida

Online Help System

XA10A

Document Version

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Service Level: 1

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Adminportal arrangement

Access Area

The navigation is loaded by clicking on the according service task tab within the access area. The currently selected process tab is highlighted.

Navigation

Tasks are started by clicking on the corresponding entry in the navigation area.

Caption Bar

The caption bar is displayed above the task area and displays the actual navigation as title above the task area.

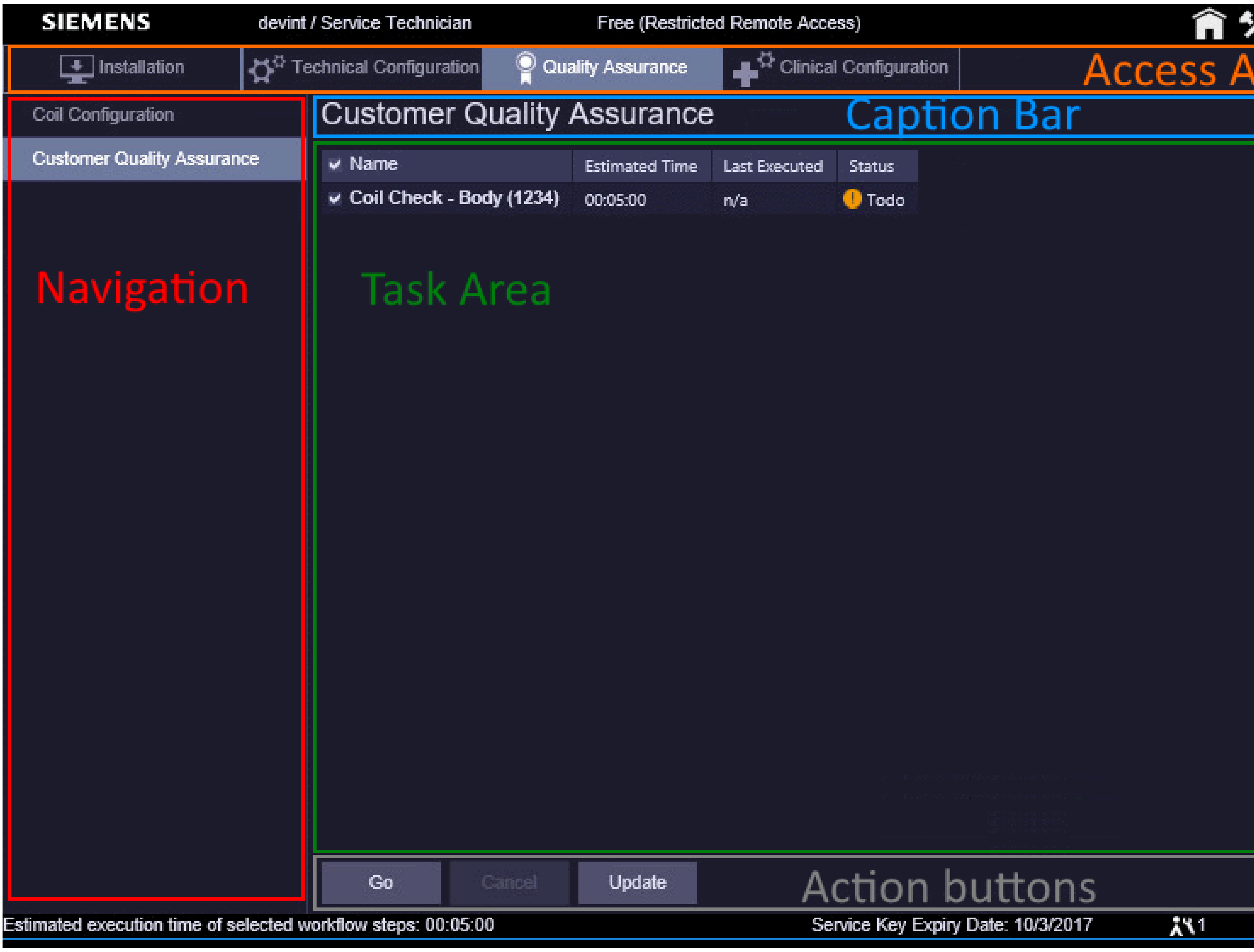
Task Area

The task area is the space occupied by the service task and its controls.

Action buttons

Action buttons are placed into the action bar located below the task area.

Fig. 1: Adminportal arrangement



2.1
TestTools

2.1.1
General

Help ID: F1903933-9843-4A6B-A14F-D0AC3BEDA174
Service Level: 5

Introduction

Purpose

TestTools offers access to programs, which are used by the customer service engineer (CSE) to find the reasons for the following problems:

- poor image quality
- hardware failure
- linearity and stability problems
- Tuneup/QA measurements that are out of specification

The general purpose of TestTools is to identify defective service parts (spare parts, field replaceable units FRUs). The tests do not need to identify which subcomponent of the part causes the malfunction.

TestTools combines all hardware tests into fully automated procedures as much as possible. This means that a full system function test can be performed, including stability and linearity tests. However, you have the option of selecting individual components for the test session.

Concept, Test Categories

In comparison to former software versions, tests are no longer executed by selecting individual components or FRUs, but by choosing functional units. For example "TX" includes all components which create, transmit, and amplify RF signals. As far as possible, redundant measurements are omitted by applying all necessary evaluations on a single data set. For the TX-Box for example, the maximum power, the linearity as well as the reflection of RF power are checked simultaneously.

In addition, tests are grouped on the TestTools platform according to their runtime. For this, the following three “automatic” categories are available:

- **Fast**

  Tests to obtain a quick overview of all parts of the system in a few minutes. It should be possible to find components which are completely out of order.

- **Normal**

  Tests that perform a more detailed check, including tests of subcomponents. By performing these tests it should be possible to find the source of reproducible problems occurring during normal patient measurements.

- **Stability**

  These tests should only be necessary for finding long term stability problems that are important mainly for fMRI measurements. Thus, total measurement times in the range of hours need to be accepted. Interactive tests should be avoided as far as possible.

Additional to the runtime categories, the following test categories are available:

- **Interactive**

  Here the user has to perform manual actions e.g. disconnecting cables. Before the test is started, work instructions, questions, or information are usually displayed in a separate window. This window has to be confirmed before the test actually starts. Cancelling is also possible, which aborts the test immediately.

  For interactive tests the name of the test currently being performed is displayed in the header of the pop-up window.

- **Multi Nuclei**

  Multi Nuclei tests are only available when the system is equipped with the Multi Nucleus option (spectroscopy, X- nucleus).

  Here only the TX chain is tested with frequencies that can be specified in the user interface. The nucleus used for this test group is selected using a drop down list.

> i To run these tests a valid RF characteristic for the MNO transmit path has to be available. Otherwise the test will fail.

Test Selection

By selecting "parent" groups (test categories), e.g. "TX" or "RX", all components/units assigned to this group are selected automatically.

Each individual test can still be selected or deselected to save troubleshooting time. If one button is deselected, the parent is deselected as well because the whole group is no longer selected. This also works vice versa: when you select all tests, the parent is selected automatically as well.

Test Overview

The following table shows the currently available test, its short description, and the necessary prerequisite(s), such as a phantom or the Service Plug.

The list is subject to change during development and makes no claim to be complete or correct.

Tab. 1 Tests available within TestTools

Category	Functional Unit	Test Name	Description	Prerequisites
Fast	Status	Status	Status of all available components followed after a ping to the MaRS.	none
	Control	PCIe	Check of the components that are connected to the optical PCIe links.	none
		PCIe FOC Light Intensity	Check of the FOC transmitter and receiber to isolate connection problem between the components	none
		GPU Accelerator	Check of the GPU-Accelerator board in the MaRS. This is an optional component available at systems with 64 or more receive channels.	none
	TX	Power	Check of RF power and power reflection at BC.	large spherical phantom
	RX	Receiver	Check of receive signal amplitude of all RX channels.	none
	Gradient	Current Actual Value	Check of the **C**urrent **A**ctual **V**alues of the current for all axes.	none

Category	Functional Unit	Test Name	Description	Prerequisites
Normal	Control	Performance	Check of the data rate and performance of the MaRS.	none
	TX	BC Tune/Detune	Check of the BC PIN-diodes.	large spherical phantom
		BC Tuning	Reflection and transmission of the two BC parts as a function of the frequency (analog to QA).	large spherical phantom
		TX Hybrid / Phase Check	Check the TX Hybrid (CP/EP switch) in the TX-Box (systems with one Tx channel). Check the modulators and measurement receivers (systems with two Tx channels).	large spherical phantom
		TX Stab/Lin	Check of the linearity and short time stability of the TX-Box.	large spherical phantom
		Power intrapulse	Check of the stability of the transmit pulse itself.	large spherical phantom
	RX	RX Stab/Lin	Check of the linearity and short time stability of the receive paths.	none
		RCCS Matrix	Check of all possible signal paths (nodes) in the RCCS.	none
	Gradient	Current Actual Value Stab/Lin	Check of the linearity and short time stability of the **C**urrent **A**ctual **V**alues of the current for all axes.	none
		Grad Risetime	Check of the gradient rise times and current sensors of the GPA.	large spherical phantom
		Spike	The QA procedure Spike Check is performed.	none
Stability	TX	TX Longtime Stab	Check of the longtime stability of the transmitted power.	large spherical phantom
	RX	Receiver Longtime Stab	Check of the longtime stability of the receiver paths.	none
		RCCS Matrix Stab	Check of the signal stability of all RCCS nodes.	none

Category	Functional Unit	Test Name	Description	Prerequisites
Interactive	ISO Ring	ISO Ring	The complete ISO ring is checked for correct functioning.	none
	TX	Power interactive	Check of max. RF power and power reflection at the BC with the B1-field regulation switched on. The use of the loader enables the use of more power than the corresponding test in the "Fast" category. This is an optional test as the Body Loader is not delivered with the systems anymore. Just in certain cases HSC MR might request the results of this test. It is not required to fully test the system.	MR Phantom 5300 ml together with Body Loader [1]
	TX	Power intrapulse interactive	Check of the stability of the transmit pulse itself, this time with full power.	MR Phantom 5300 ml together with Body Loader [2]
	TX LC	Power LC interactive	Check of max. RF power and power reflection of the LCTX-path with the B1-field regulation switched on.	Service Plug
	LC Plug	for socket 1..8: LC Plug IF, LC Plug Pin, LC Plug LO, LC Plug 10V, LC Plug 3V, Coil Interrupt	Check of the LC Cabling, the LC_DYN (LC PIN-diode signals), the LO signals, the IRQ signal path, and the power supply wires from the IF_RCCS through the coil sockets at the patient table.	Service Plug
	PMU	PMU Transmit	Check of the ECG test signal transmitted by the PERU and the signal of the PPU.	PERU and PPU[3]
	PTAB	PTAB functionality	Check of switches and sensors of the patient table.	none

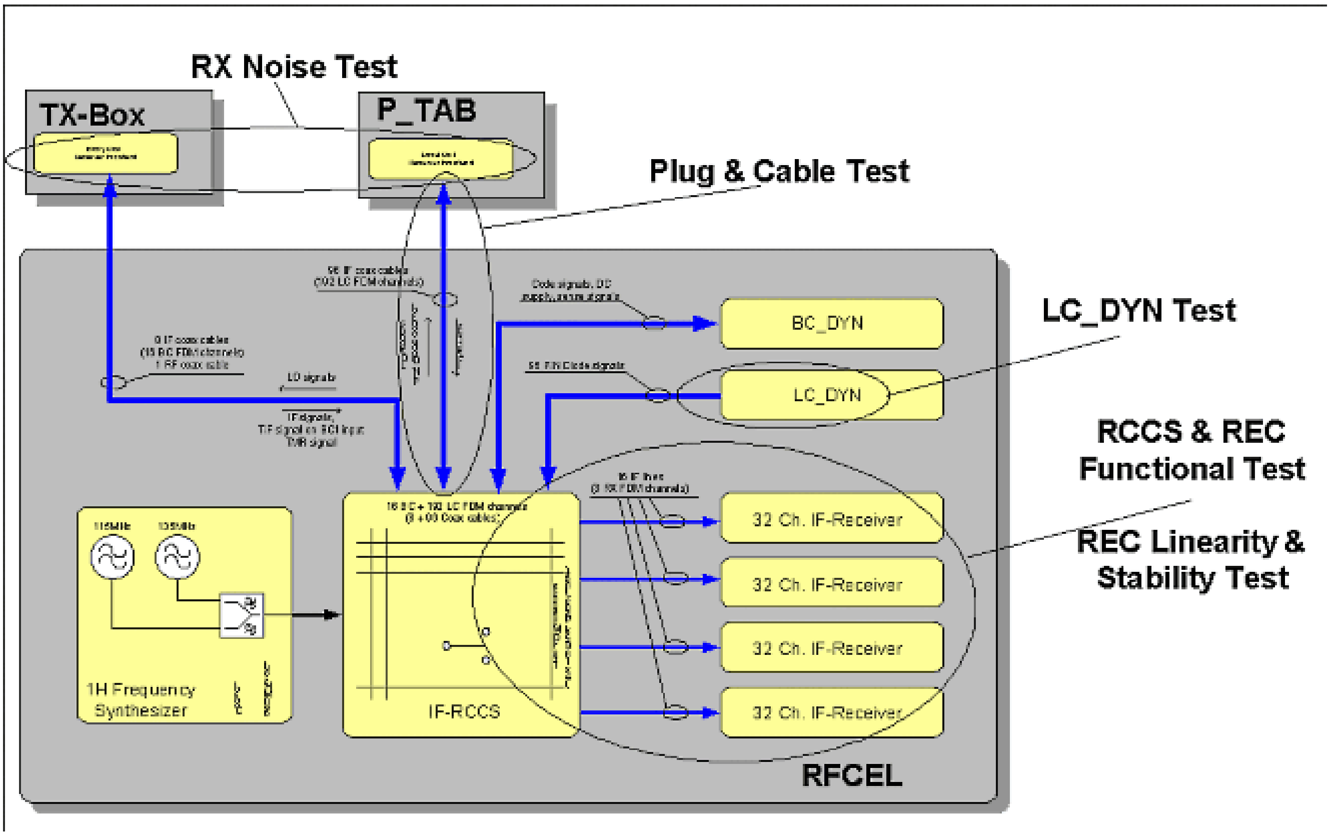
Category	Functional Unit	Test Name	Description	Prerequisites
Multi Nuclei	TX Stability	MNO TX Stab/Lin	Check of linearity and stability of the transmitted power for the selected nucleus. The nucleus is selected in a pull-down menu.	Service Plug
	MNO TX	MNO Power Dummy	Check of max rf power for all supported nuclei. Power is transmitted to the MNO dummy.	Service Plug
	MNO TX LC	MNO Power LC Service Plug	Check of the RF power transmit path for a local MNO TX coil.	Service Plug
	MNO Plug 2	LC Plug LOX Plug 2	Check of the LC RX Cabling and the LOX signals. The LOX test is currently configured only for plug 2.	Service Plug

1. Not delivered with the system, test is for special purposes only.
2. Not delivered with the system, test is for special purposes only.
3. Clinical or Service Patient required

The “MR Phantom 5300 ml” is not delivered with the system and thus not available at every site.

The following picture shows a graphical overview of the RF tests:

Fig. 2: Overview TestTools Tests



Notes

Please read the following notes carefully!

1. It is not possible to run a test or even a normal sequence when the licenses for the MaRS, GPA, and RFPA (TX-Box) are invalid or missing.

   If no tests can be started, please check for the above.

2. Please ensure that the system is switched on for at least 20 minutes before running any functional or interactive RF-tests to allow the temperature to stabilize in the control cabinet.

   Additionally, the doors of the cabinets have to be closed for linearity and stability measurements.

3. If the test is aborted with "No results available," it may be that the source of the problem is not in the component(s) currently tested but in any other compoent of the measurement system. Thus please check the event log for more informations.

4. While running tests it is possible that the gradient system pulses with low amplitudes. This is noticeable through the "knocking" of the gradient coil and is completely normal.

User Interface

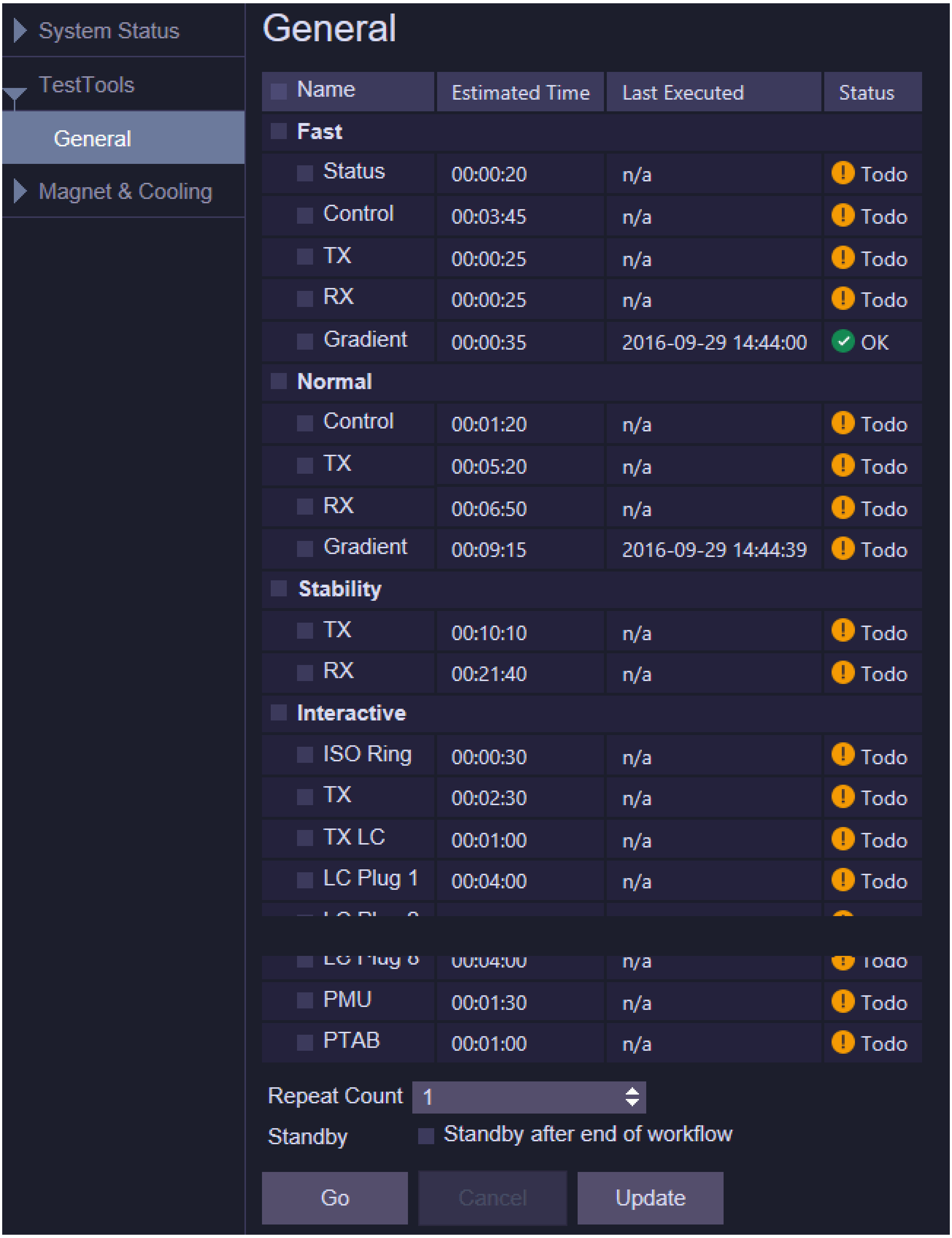
General

The TestTools General user interface (UI) can be accessed from the service software main window, which is opened from the syngo user interface .

The TestTools platform can be divided into

- Navigation Frame (left side of the Admin portal window)
- Task Area (center/right side of the Admin portal window)
- Footer Bar (bottom action bar on bottom of the Admin portal window)

Fig. 3: TestTools General



Navigation Frame

The navigation frame provides navigation within TestTools and its modules.

Only one item in the navigation frame can be selected at a time.

The contents of the function frame (right side) depends on the selection in the navigation frame.

Task Area

The function frame is divided into the following controls and sections.

- **Test Category**

  Basic differentiation between different test duration and interactive tests.

  Available are **Fast**, **Normal**, **Stability**, and **Interactive**. The additionally possible **Multi Nuclei** is only visible when the system is equipped with the Multi Nucleus option (MNO).

- **Functional Unit**

  Function blocks such as measurement control and MaRS ("Control"), receive ("RX"), and transmit ("TX") path without further differentiation of the components/FRUs within these functions.

  By selecting the test category, e.g. **Normal**, all functional units allocated to this category are selected automatically. Each individual functional unit can still be selected or deselected separately. If one checkmark is deselected (e.g. to save time), the parent is deselected as well because the whole group is no longer selected.

  After selecting the required tests, the execution is started by pressing the **Go** button.

- **Repeat Count**

  This field is used to set the number of test cycles. The selected tests are repeated as many times as specified by the Repeat Count.

  The repetition number is valid for one test run only and has to be set again before the next **Go**. Also in case a test session was aborted it is necessary to set the value again. The default value for the repetition counter is "1" (one). To take over a new value either click somewhere in the TestTools window (not a button or link of course), or press **<return>**.

> Interactive tests are performed only once even if the repeat counter is larger than one.

- **Nucleus Selection**

  The nucleus selection in displayed below the test selection. Here the nucleus to be tested can be selected. This selection has to be performed before the tests are started.

  The selection in the popup is free, means any combination of selected or unselected nuclei is allowed.

  Every nucleus corresponds to a certain frequency which will affect, e.g., the automatic selection of the correct RFPA / TX-Box.

  The nuclei 3He, 19F, 7Li, 13C, 23Na, 31P, 123Xe, and 17O can be used for customer measurements and thus also for tests.

> The usage of 1H is for the MNO RFPA not released, the results may be out of specification!

> It is necessary to select the nucleus/frequency prior to the start of the test. If the desired nucleus is set, it will stay until it is changed again, or "**Default**" is pressed.
>
> This should be remembered during troubleshooting.

> The display of **Estimated Time** is not corrected if more than one nucleus is selected.
>
> Instead, only the run time for a single nucleus is shown.

- **Standby**

  The check box "Standby after end of workflow" works in the same way as within Tune-up and QA.

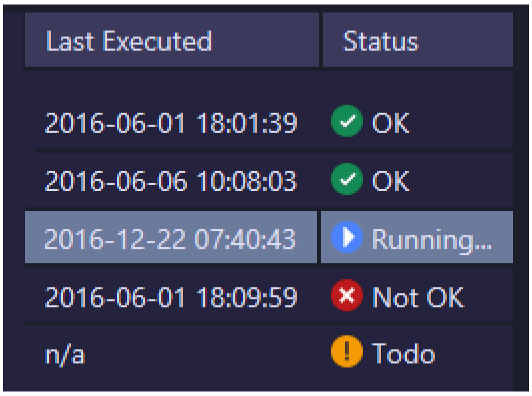
  By selecting this box the system is set to standby mode after all selected tests are executed.

> Be aware when using the Standby functionality that it is not possible anymore to switch on the system remotely.

- **Test Status**

  The information below “Last Executed” show the date and time the test was peformed the last time. This “memory” is not cleared e.g. when the service software is terminated. As the TestTools results are part of the “dynamical data” this might be back until the installation of the system. Thus even that date might appear here.

Fig. 4: Possible Status of Tests



For the status of the tests four different entries are possible:

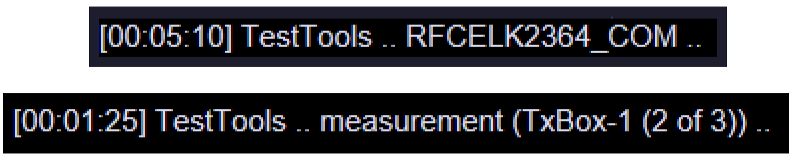
- **Running**
  Shows which test is currently running.
- **OK**
  The last time the test was performed it was successfully terminated.
- **Not OK**
  The last run of the test failed.
- **Todo**
  The test was never performed before, independent from the service activities at the system. As the TestTools results are part of the “dynamical data” this might be until back to the installation of the system.

Footer Bar

The footer bar contains the following fields and buttons:

- **Status** (left)
  In this part of the footer bar, the status of the current test is displayed. It shows information messages from the test programs. The test programs use this to inform the service engineer of what is currently occurring. Its main purpose is to inform the user about the progress of the running test program. In this way the user always knows the status of the test session and the single tests.

Fig. 5: TestTools Status Workflow Examples



- **Estimated Time** (center)

  The **Estimated Time** part of the footer bar shows the estimated time the system will need to perform the selected tests. The Estimated Time is dynamically updated if the tests selection is changed. If a repeat factor is set, this is also taken into account.

  The timer is synchronized with the total estimated test duration at the start of each test and thus counted down. This countdown is not constantly running but for the time the test requires to be performed. E.g. if a test takes two minutes after the end of this test the Estimated Time counter is reduced by two minutes. In the time between nothing happens, i.e. Estimated Time is not constantly counted down by seconds and minutes.

  When a measurement is running next to this field, the approximate remaining measurement time is displayed in brackets and acts like a timer. The program begins with a countdown once the measurement time exceeds approx. 10 seconds. Otherwise, the timer is not started at all. The application software for the customer proceeds in the same way.

> The displayed time is not precise, especially not with interactive tests.The time the user needs for interactions cannot be calculated or determined precisely. The given time is a minimum time and may be longer when the test is actually performed.

> The display of **Estimated Time** is not corrected if more than one nucleus is selected.
>
> Instead, only the run time for a single nucleus is shown.

- **Go** (left)

  Click the **Go** button to start the selected tests.

  The **Go** button is not available during the test execution.

  Contrary to previous software versions the output window is not opened automatically at the beginning but at the end of the test session.

- **Cancel** (center)

  is used to abort the tests or measurements in progress.

  This button is dimmed when no test session is running.

> It may take a while until the tests are actually stopped (approx. 50 seconds, up to two minutes) - please be patient.

- **Update** (right)

  Updates the contents of the task area.

TestTools Workflow and Usage

General

The automatic Service Patient is registered before the first measurement generating raw data is performed. It is deregistered when the last test of the TestTools workflow is finished, no matter whether it is a local or a remote session. This way it is guaranteed that the Service Patient is automatically active for the TestTools measurements only.

> With NUMARIS/X the Service Patient is handeled completely different compared to previous software versions. The automatic Service Patient is registered for the Tuneup-, QA-, and TestTools workflows at their start and always de-registered at the end of the workflow.

Every time a TestTools procedure is started it creates an entry within the Message Viwer. This entry contains the name of the TestTools procedure and its return status. The entries can be used e. g. for when an AutoReport is necessary.

After the end of the tests it is not necessary to reboot the system or measurement control (MaRS) manually. This is done automatically if necessary.

Depending on the test the large spherical phantom and its holder has to be positioned In the magnet center. However, this is only the case for tests of the TX-Box.

For Multi Nuclei tests Service Plug has to be used.

Workflow

The TestTools software architecture is based on the workflow concept. The actual management of the user interface and all actions are controlled by the workflow engine.

The selected tests are started with the **Go** button in the bottom.

Usage

Basically the test session uses the following workflow:

1. At the beginning of the test session in the navigation, one of the items is selected if necessary.
2. Then the selection of the test category or functional unit is made.
3. In case Multi Nuclei tests have to be performed, the required nucleus must be selected before the test is started. For this use the available drop down list.
4. After selecting the required tests and the required nucleus (if applicable), the tests are started by pressing the **Go** button in the footer bar.
5. After starting the test session using the **Go** button, the system checks the necessary system conditions/prerequisites, the so-called PreConditions (refer also to next chapter). This means that for interactive and RF tests the necessary pop-up(s) with work

instructions and/or information for the user appear. They have to be confirmed or cancelled. If they are cancelled the test is not started; the workflow is aborted and returns to the previous window.

6. Error messages and further instructions/information may also appear in subsequent pop-ups and have to be confirmed (or cancelled).

> The selection of tests is reset after the test workflow.

Prerequisites / PreConditions

Before the start of a test the prerequisites for the tests are checked and all necessary actions are displayed in a pop-up. For each test program these prerequisites are specified as a set of PreConditions that have to be fulfilled before the test can start.

A prerequisite is a certain state of the MR system. For transitions between the states, pop-up messages can be defined. The advantage of using prerequisites is that the pop-up messages can be automatically rearranged. E. g. a test requires a re-cabling of the system. If the following test requires the same cabling, no additional pop-up is needed in between. However, if only the second test is run, the pop-up is required and thus displayed.

Mainly only the interactive tests generate pop-ups with work instructions and/or information for the user. These have to be confirmed or cancelled. If they are cancelled the test is not started; the workflow is aborted and returns to the previous window. For other test categories than interactive the PreCondition above is displayed, but only in case the RFPA/TX-Box is involved. This PreCondition can be performed by the customer, too.

With TestTools no automatic phantom positioning is necessary; this function is only used with TuneUp and QA. There the required phantom setup is automatically handled by the programs. That means that the table is automatically moved if the wrong phantom is in the magnet center. This includes the "no phantom" setup, where the patient table is moved out of the magnet.

RF - System Overview

> Although the graphics below show the MAGNETOM Aera/Skyra RF system they are basically valid for MAGNETOM Vida, too.

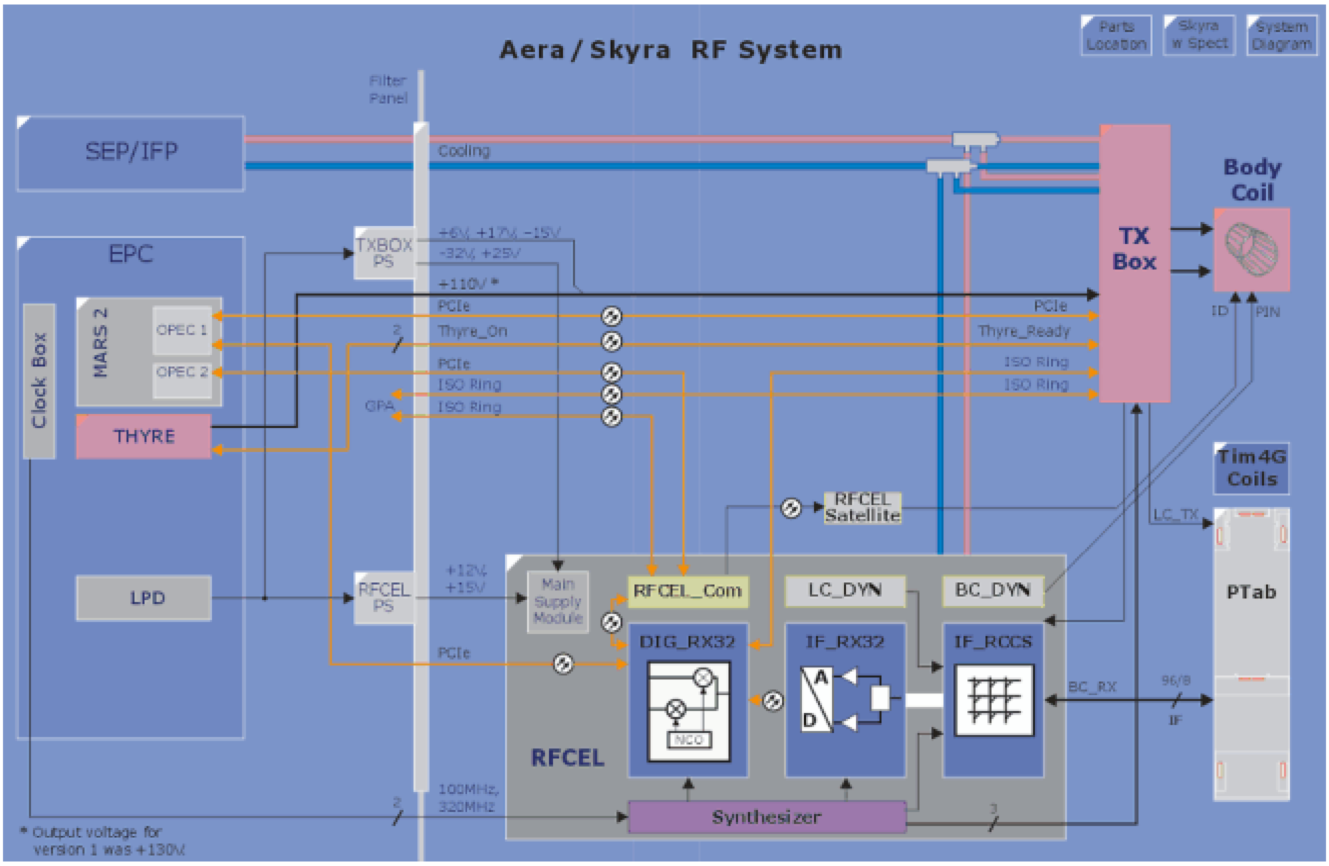
The RF System hardware has been designed with several built-in loops to allow automated testing of RF components.

The 32 channel systems can be described in the following way:

- only one OPEC card (OPEC-Tx) is installed;

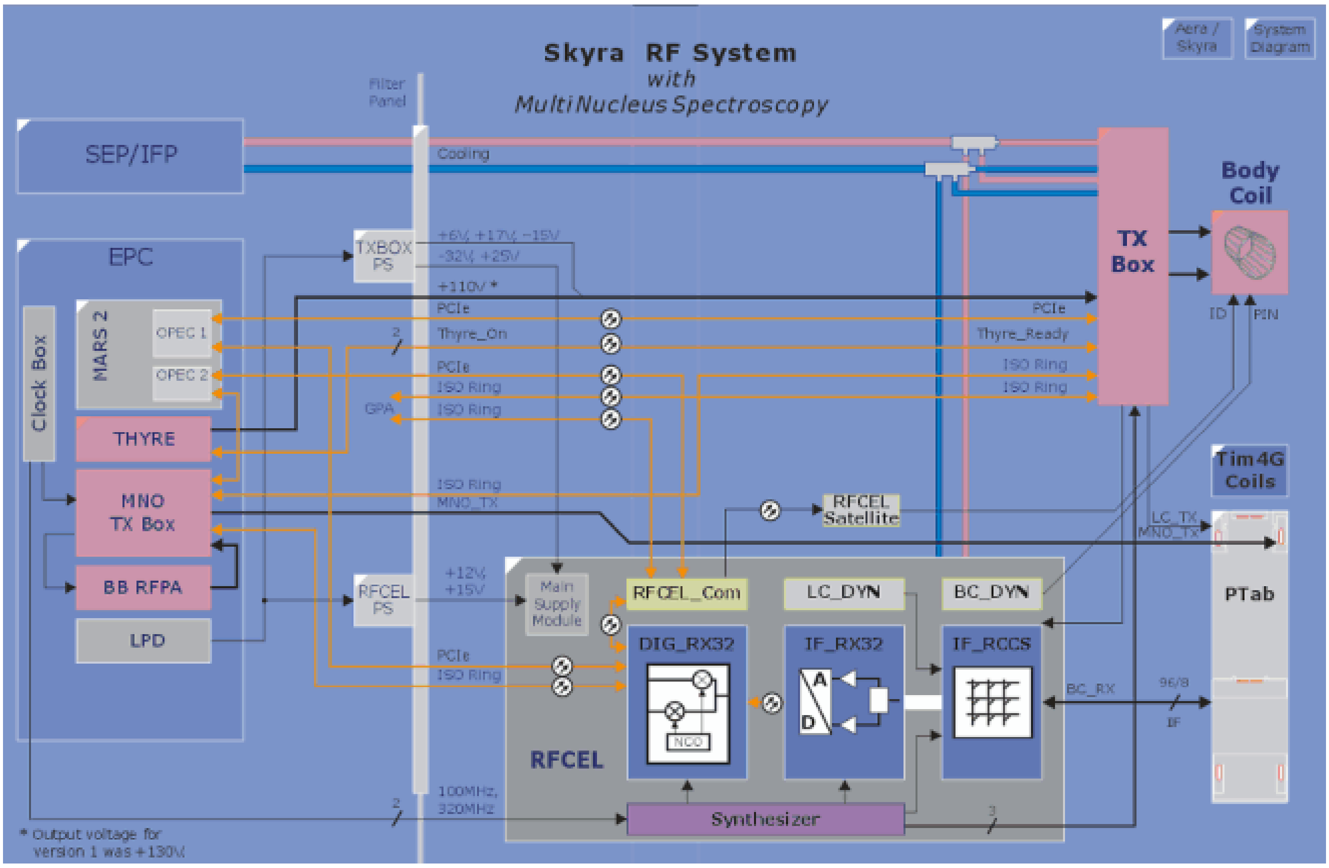
- only three fiber optic lines between DigRX und IF_RX, the fourth is not used and also not tested;
- all four lanes of the OPEC-Tx card are used, no free lane is available: DigRX, GraCon, RFCelCom, and TX-Box. Thus no MNO is possible.

Fig. 6: Overview RF-System MAGNETOM Aera/Aera24, Skyra/Skyra24, Prisma, Prisma^{fit}, Avanto^{fit}, and Skyra^{fit}



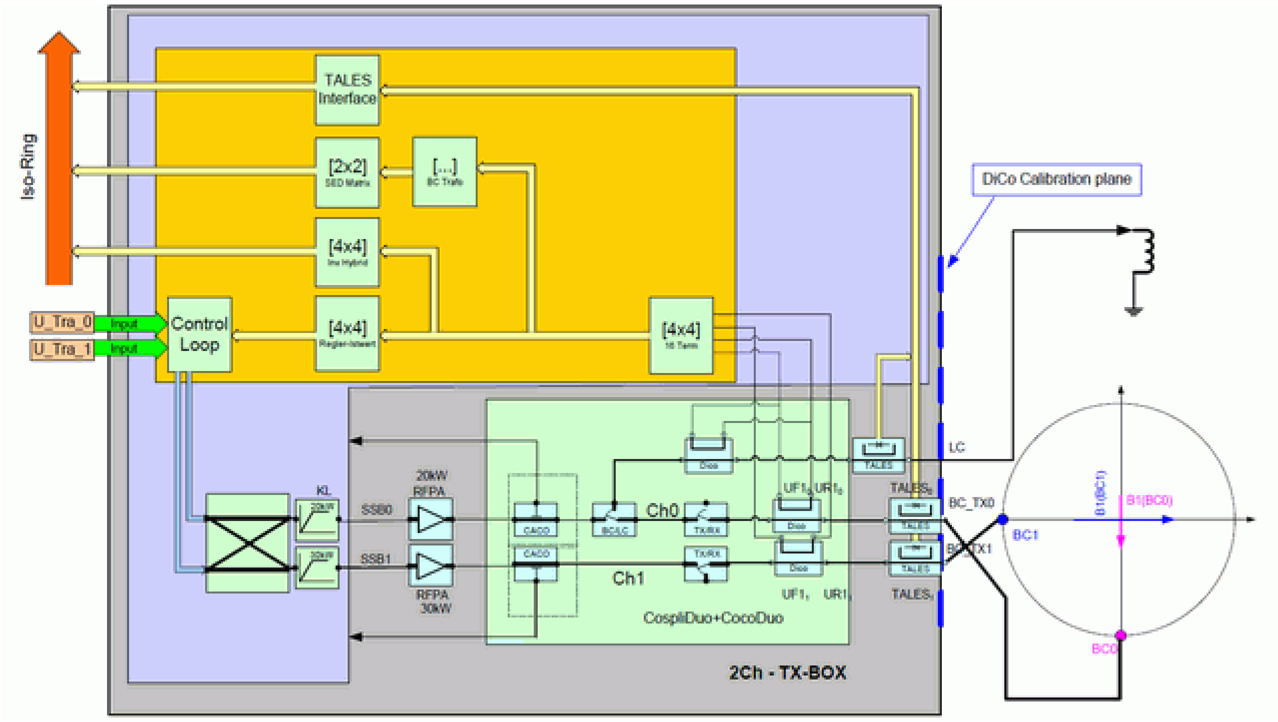
For the 32 channel systems no Multi Nucleus option is available.

Fig. 7: Overview RF-System Skrya/Skyra24, Prisma, Prisma^{fit}, Avanto^{fit}, and Skyra^{fit} with Multi Nucleus Option



For systems with the 2-channel parallel transmit option (pTX) a different TX-Box is used:

Fig. 8: 2-Channel TX-Box (parallel transmit option, pTX)

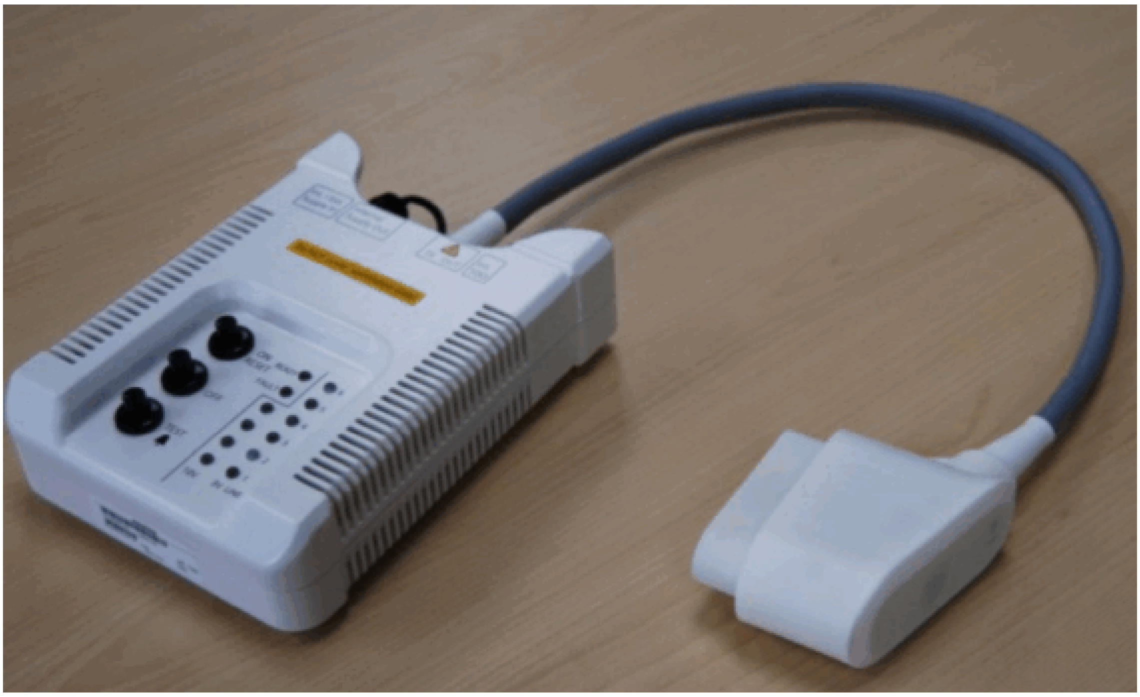


Service Plug

General

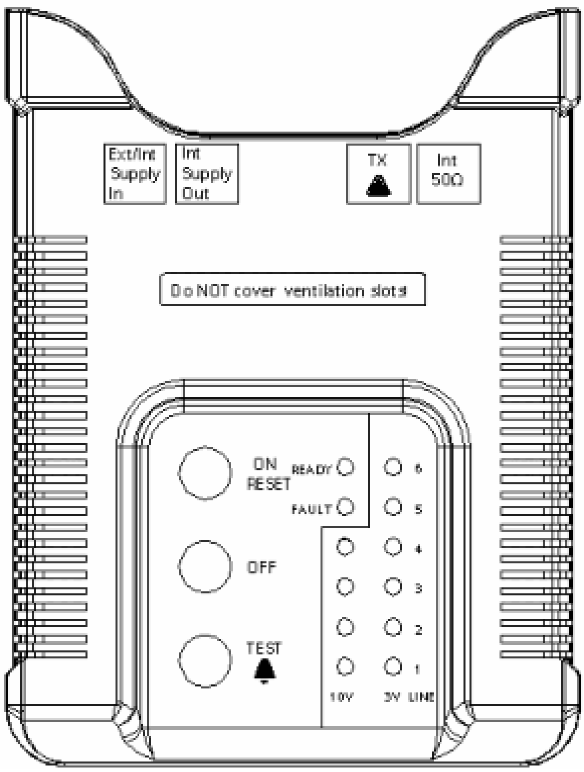
The Service Plug is necessary to test the coil sockets at the patient table and the different interconnections of the energy chain. Included in the Service Plug kit are the adapters needed for troubleshooting these interconnections.

Fig. 9: Service Plug MAGNETOM Aera, Skyra, Prisma, Avanto fit, Sykra fit, and Vida



The front cover of the Service Plug gives an overview over the offered functions, it shows several control buttons, LEDs, and connections:

Fig. 10: Service Plug Front Cover



Function Overview

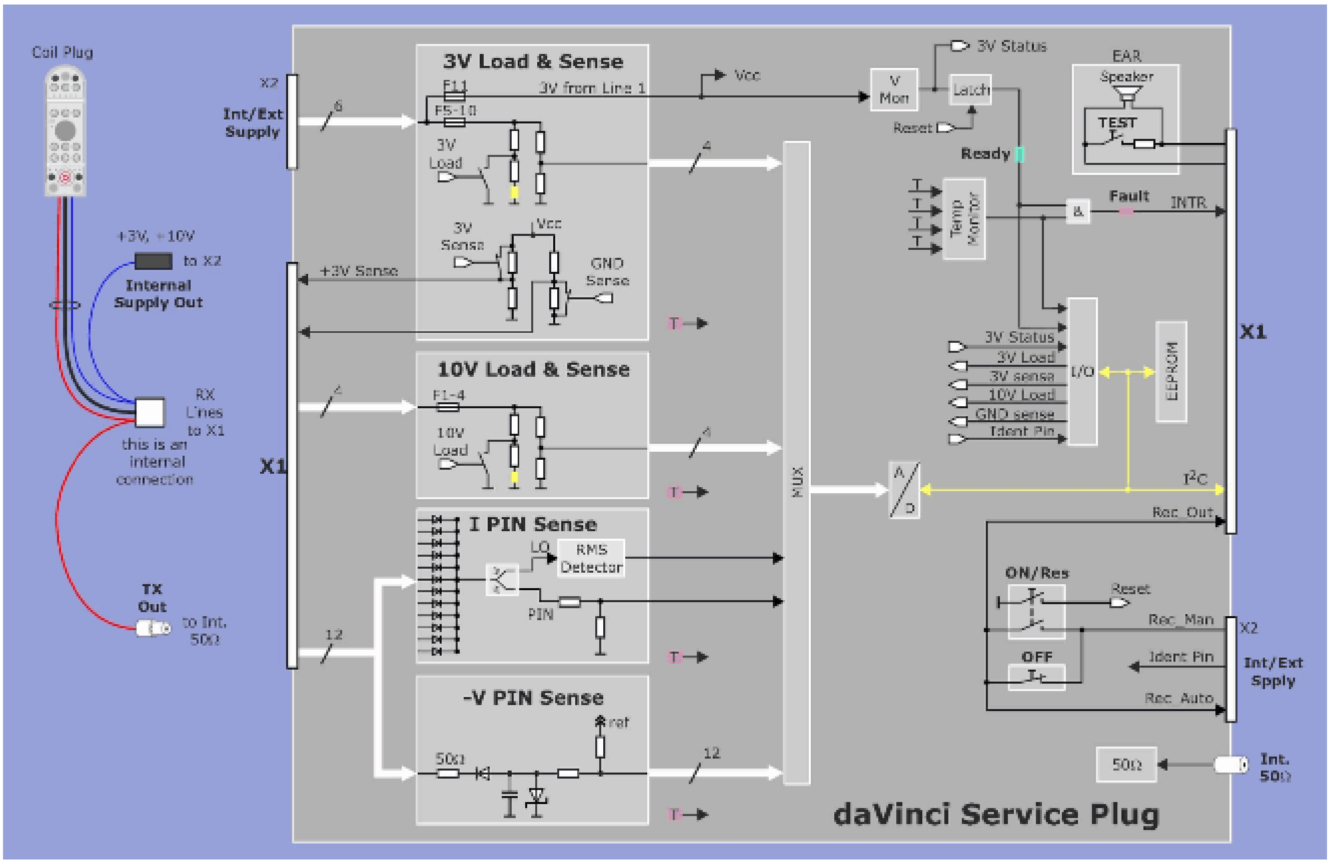
- The Service Plug and the adapters are used together with the TestTools test "LC Plug" to troubleshoot problems of the so-called "energy chain". If the MNO option is installed, the Service Plug is necessary for the Multi Nuclei tests and enables the system transmitting frequencies other than ^{1}H.

  The test of the coil sockets is done automatically. No user interaction like checking/observing illuminated LEDs and son on is necessary. Just plugging the Service Plug to the patient table is enough. TestTools starts the necessary tests and reads out all information digitally from the Service Plug. The evaluation is automatically done.

  The communication to the system is realized trough the I^{2}C interface (like for the local coils).
- LEDs provide basic information about the Service Plug status to the operator. These LEDs are independent from any Software or Service Software and can be evaluated immediately after connecting the Service Plug to the MR system.
- Additionally, it is possible to check patient alarm and audio signal to acoustic transducer (for patient head set).

> Although the graphics below show the MAGNETOM Aera/Skyra RF system they are valid for MAGNETOM Vida, too.

Fig. 11: Service Plug MAGNETOM Aera, Skyra, Prisma, Avanto^{fit}, and Sykra^{fit}



Controls, LEDs, and Connections

Controls

The available controls of the Service Plug are:

- **ON/RESET**

  has two functions:

  a) Switch on the Service Plug when an interconnection of the energy chain is tested.

  NOTICE

  The 3 V power supplies are not switched automatically when testing at interconnections. Potential damage of fragile interconnection pins due to arcing!

  » **Manually switch on the power supply with the ON/RESET button AFTER plugging the Service Plug.**

  » **Manually switch off the power supply with OFF button BEFORE disconnecting the Service Plug.**

  When testing at Manual and Direct Connect sockets of the patient table top and at the Docking Station, pressing this button is NOT necessary. These plugs have the recessed pin which turns on the 3 V power supply automatically when plugging.

  b) The second function of the button is to reset an error condition of the Service Plug, e.g. an internal error of the electronics or a too high temperature inside the Service Plug. If the Service Plug is connected to the coil sockets **ON/RESET** is the only button which can be used.

- **OFF**

  is used to switch off the Service Plug when testing an ERNI-interconnection of the energy chain. These interconnections do not offer the function of the recessed pin like the coil sockets do. So, the power supplies has to be manually turned off with this button prior of disconnecting the Service Plug. Otherwise, the fragile ERNI pins might get damaged by arcing.

LEDs

The LEDs provide basic information about the Service Plug's status to the operator without any software evaluation involved. The LEDs are operational directly after plugging the Service Plug. The following LEDs are available:

- **READY** (green)

  The Service Plug is in normal operation: the supply voltage of 3 V is above the minimum value necessary for operation (approximately 2.7 V). It also means that the internal temperature is below the warning level.

  If the LED is off this indicates that the supply voltage is not sufficient for the operation of the Service Plug electronics.

- **FAULT** (red)

  If the LED is on an internal error occurred, the 3 V supply is not sufficient, or the internal temperature has reached or exceeded the alarm level. In that case all running service procedures are stopped.

  The fault condition can be cleared by use of the **ON/RESET** button as long as the error does not persist.

- **10 V 1...4** (4 yellow LEDs)

  They indicate that the input voltage from the four 10 V supply lines is present. The specification itself is tested with TestTools.

  The LEDs should always light up and stay on as soon as the Service Plug is connected to the MR system and no test is running. These LEDs have no threshold which means they may glow with reduced intensity even if the voltage is lower than specified. So, even a too low voltage is indicated visually. If the green **READY** LED is on, the precise value of the voltage can be tested through TestTools.

  If one or more of the 4 LEDs is/are off it indicates that either the 10 V load is interrupted (i.e. defective), or connected, or the 10 V supply voltage is not present. The 10 V load is only active while the corresponding service test is running.

  If the load appears to be already connected when the Service Plug is plugged into the coil socket, either the Service Plug itself or the system software is faulty.

- **3 V 1...6** (6 yellow LEDs)

  The input voltage from the six 3 V supply lines is present. The specification itself is tested with TestTools.

  The LEDs should always light up and stay on as soon as the Service Plug is connected to the MR system and no test is running. These LEDs have no threshold which means they may glow with reduced intensity even while the green **READY** LED is off. So, even a too low voltage is indicated visually. If the green **READY** LED is on, the precise value of the voltage can be found by performing the corresponding Service Plug tests within TestTools.

  During the voltage test (please refer to the LC Plug tests) the LEDs will go off as soon as the load is switched on.

If the LEDs are off, the following problems might exist:

- If the green **READY** LED is on it is an indication that the main load for the supply voltage is interrupted (i.e. defective) or that the main load is connected. This should only be the case while the corresponding Service Plug test is running.

  If the load appears to be already connected when the Service Plug is plugged into the coil socket, either the Service Plug itself or the system n system software is faulty.
- If the green **READY** LED is also off there is no voltage present at all or only very low voltage is delivered from the 3 V power supply.

Connections

Available connectors of the Service Plug are:

- **Int./Ext. Supply In**

  Here, normally the cable from the "Internal Supply Out" of the Service Plug is plugged in. In case ERNI interconnections of the energy chains are are tested the corresponding plug of adapter ERNI-female (ODU manual-F to ERNI-F) has to be connected to this socket. In this case "**Internal Supply Out**" is not plugged in.
- **Internal Supply Out**

  This socket is the outlet of the power supply lines coming from the IF_RCCS. The plug stays plugged into "**Ext/Int. Supply In**" when the direct and manual coil sockets, or the patient table or docking station connectors are tested.
- **Int. 50 Ohm**

  This is the input to the internal 50 Ohm termination resistor; it is usually connected to "**TX OUT**".
- **TX OUT**

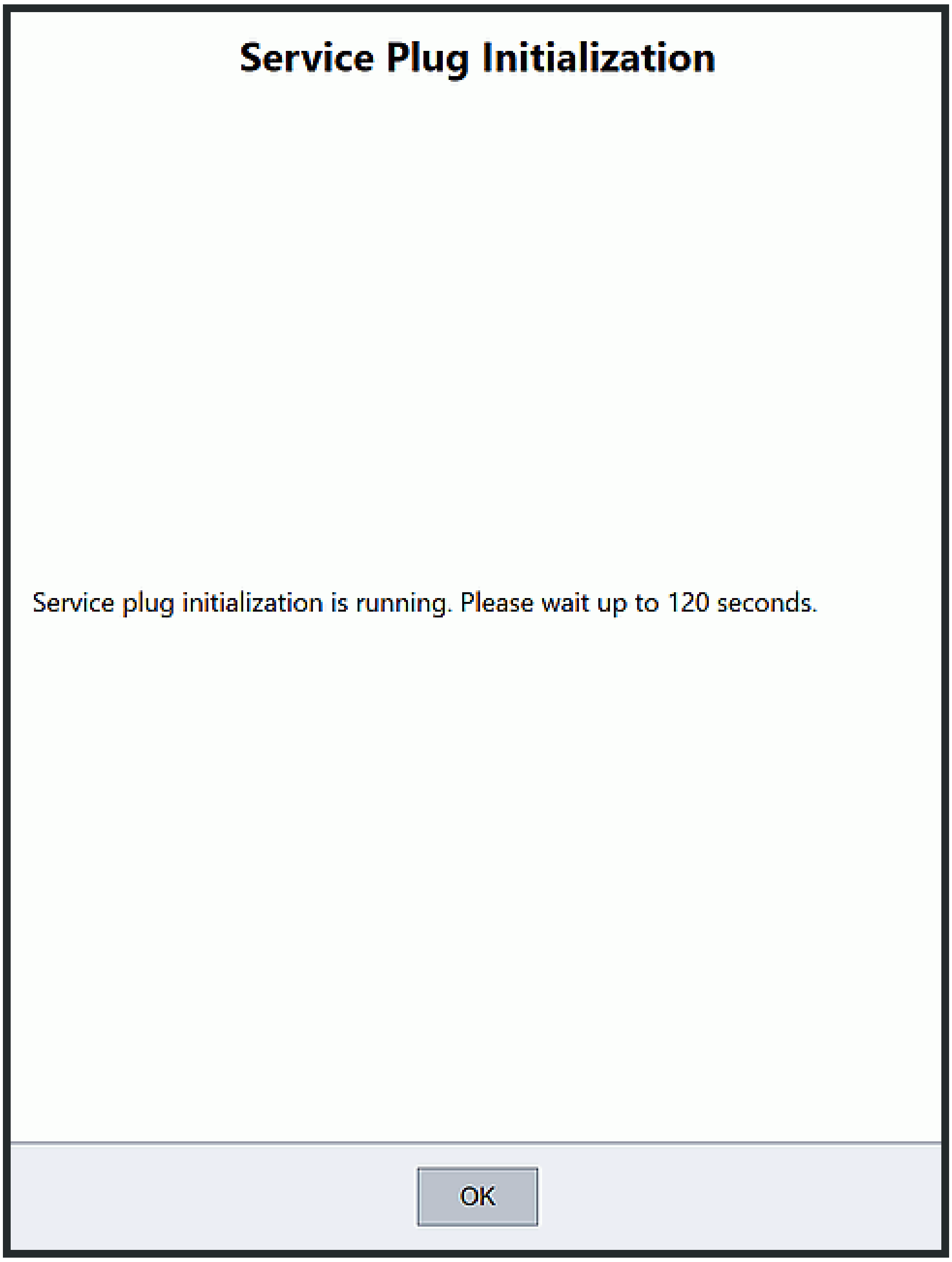
  Output of the LCTX-path from the coil sockets. It is usually connected to the "**Int. 50 Ohm**" inlet of the Service Plug as long as no interconnections are tested.

Download Service Plug Data

After connecting the Service Plug for the very first time at the tested system, the internally stored data is downloaded to the system (e.g. serial and material number, revision level, correction values for the tests). As this is a large amount of data, it can take up to 2 minutes to do this.

> Do not disconnect the Service Plug while the following picture is shown on the Patient Data Display at the magnet.
>
> Otherwise, it results in an incomplete data set on the system and causes the LC Plug tests to fail.

Fig. 12: Download Service Plug Data



The downloaded data is stored at the system for 2 weeks. After that time it is considered outdated and is deleted / marked as invalid.

Therefore when connecting the Service Plug again after that time, the stored data will be downloaded again.

Functions and Tests within TestTools

After connecting the Service Plug and starting the LC Plug test(s) within TestTools the necessary tests are performed and read all information digitally from the Service Plug. The evaluation and display is automatically done in the Service Software.

The Service Plug provides a test possibility for testing all RF channels up to the coil socket concerning

- **3 V and 10 V power supply voltage and sense lines**

  Check of the cabling and the values of the power supply voltage for the preamplifiers of the local coils. Checking is performed while plugging the Service Plug into the coil socket. Refer to the LEDs for this.

  Additionally a check of the cabling (sense-line) and the reaction of the 3 V power supply for changes in the load of the power supply is offered within TestTools.

  The 3 V power supply is routed over 6 wires, the 10 V has 4 wires (ERNI interconnections of the energy chain). The corresponding individual loads and sense networks are integrated in the Service Plug.

- **IF-signal**

  Check of the coupling between the RX channels (crosstalk).

- **LO-signal**

  Check of the amplitude and attenuation of the LO signal.

- **PIN-diode voltage and current**

  Check of the PIN-diode voltages and currents, which includes a check of the correct settings of the LC_DYN.

- **Coil detection and coil interrupt**

  Check of the 2 possible coil detection interrupts: INTR1 (recessed pin) and INTR2 (for TxRx coils).

Adapters and Accessories

The Service Plug kit contains accessories and adapters which are needed for testing the interconnections of the energy chain, both TX and RX.

The adapters are used together with the TestTools test "LC Plug" for problems of the energy chain and the transmit and receive paths.

Notes on ERNI Connections

NOTICE

Carefully connect and disconnect the interconnection plugs and sockets. The connectors are very fragile and break easily.

By mistake, it is easily possible to insert the plug incorrectly, turned by 180 degrees. Risk of damaging the ERNI-interconnection!

» **Pay attention to the correct orientation of the ERNI connectors.**

NOTICE

When testing the interconnections it is necessary to manually switch the Service Plug on and off by using the buttons ON/RESET and OFF.

Otherwise, damage to the interconnections may happen.

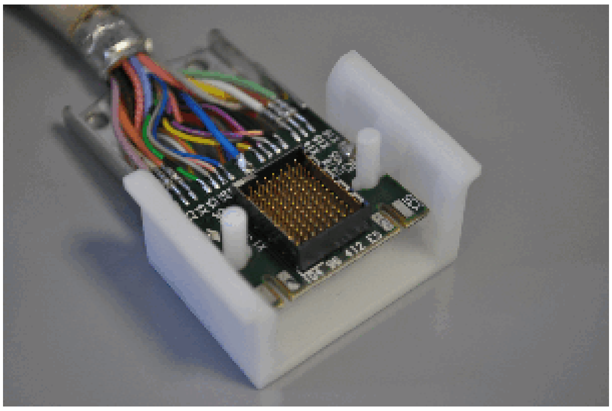
» **Always use the ON/RESET and OFF button to switch the Service Plug on and off.**

Always use the ERNI plugging tools when connecting the ERNI connectors:

- ERNI connector extractor 106 06 518, Tx ODU ejector 106 06 493.
- ERNI plug-in support 106 06 780.
- The ERNI connector extractor tool is also sometimes included in the Service Plug suitcase.

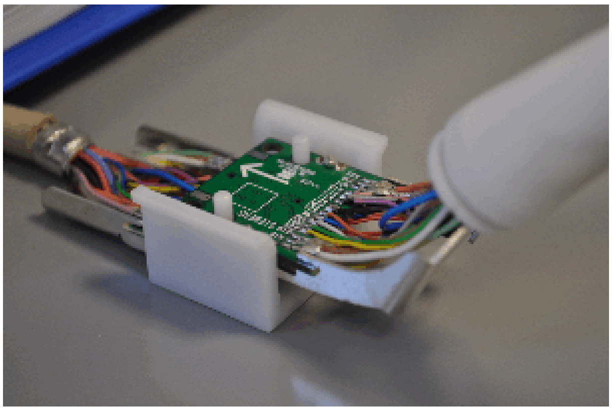
Use of ERNI Plugging Tools

Fig. 13: Use of ERNI Plug-in Support, Step 1



- Position the ERNI plug-in support on a flat place.
- Insert the first part of the ERNI connector.

Fig. 14: Use of ERNI Plug-in Support, Step 2



- Insert the upper part of the ERNI connector and press both ERNI connectors together.

QLA Extension Cable (QLA male to QLA male)

This 5 m cable is used to connect the Service Plug to the

- LC-TX outlet of the TX-Box (used together with the N-male adapter)
- LC-TX path connector of the dockable patient table (used together with the Docking TX adapter)
- LC-TX-path cable inside the coil sockets at the patient table top (used together with the QMA-male adapter)

In all cases connect one end of the QLA extension with the "**Int. 50 Ohm**" input of the Service Plug and the other end to the tested ERNI-interconnection using the appropriate adapter.

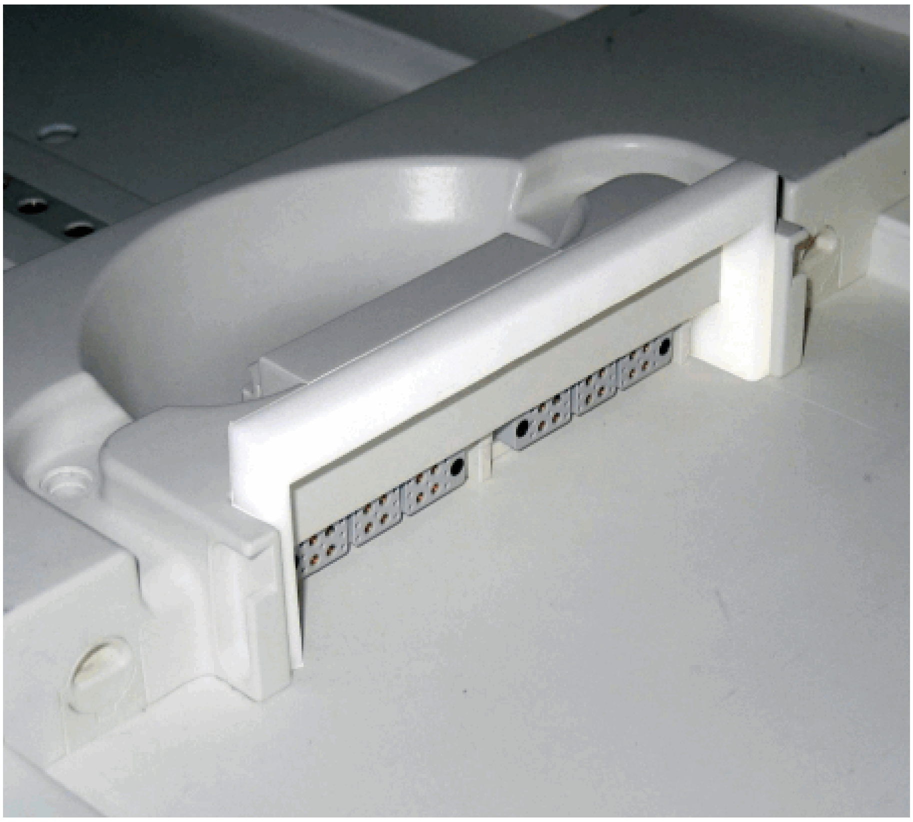
The use of the cable is shown in the chapters below.

Fig. 15: QLA Extension Adapter

Spine Coil Direct Connect Bracket

The bracket is used to lift and fix the flap of the Spine Coil Direct Connect sockets for testing with the Service Plug (depends on the system).

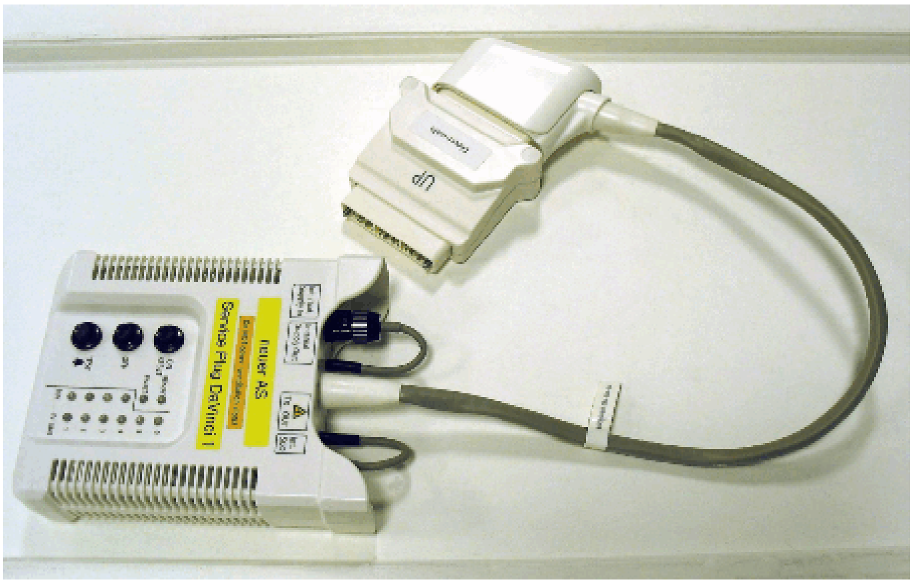
Fig. 16: Use of Holding Bracket for Direct Connect Spine Coil



Direct-male Adapter (ODU Manual Plug female to Direct Plug male)

This adapter connects the Service Plug to the Direct Connect Sockets of the patient table and the docking station (depends on the system).

Fig. 17: Service Plug Configuration with Direct Connect Adapter



The Service Plug switches on and off automatically, the buttons **ON/RESET** and **OFF** are not needed but by use of the “recessed pin” functionality of the coil sockets.

NOTICE

The Direct-male adapter is symmetrical and can be plugged incorrectly (turned by 180 degrees/upside down).

If the Service Plug was connected to a direct socket in the opposite way (adapter upside down), the measurement system is not operational anymore. This is indicated by the Service Plug LEDs, they are not constantly on but blinking.

» **Always turn the flat part of the Direct-male (ODU manual-F to Direct-M) adapter to bottom-side.**

» **In case of electronics failure (indication: the LEDs of the Service Plug are flashing), remove the Service Plug and reboot the measurement system to clear the fault.**

i The use of the Direct Connect adapter at the docking station (only at systems with removable patient tables) is described in chapter “Docking TX Adapter“.

ERNI-female Adapter (ODU Manual Plug female to ERNI female)

This adapter is used to connect the Service Plug to the ERNI interconnections of the energy chain. The round plug has to be connected to the inlet **"Int./Ext. Supply In"** of the Service Plug.

Fig. 18: ERNI-female Adapter

The adapter can be used at the following interconnections:

- IF_RCCS
- docking station (if installed at the system): magnet side and table side
- patient table:
  - foot end connections
  - all coil sockets

NOTICE

At the coil sockets, the interconnections, and the IF_RCCS the PIN diode voltages are always present.

» **Be careful while connecting/disconnecting. If you feel more comfortable switch of the LC_DYN (RFCEL) before connecting to avoid short circuits.**

Testing at the ERNI interconnections:

At several interconnections the ERNI connectors are secured with special washers that have to be removed before the connections can be opened.

For assembling the ERNI-interconnection again it is necessary to use a new set of washers as these can be used only once. Additionally they might have to be destroyed to open the connection anyway.

Approximately 30 of these washers are contained in a bag delivered with the system.

When using this adapter the Service Plug supply voltages can not be switched on automatically due to the missing "recessed pins" at the ERNI interconnections.

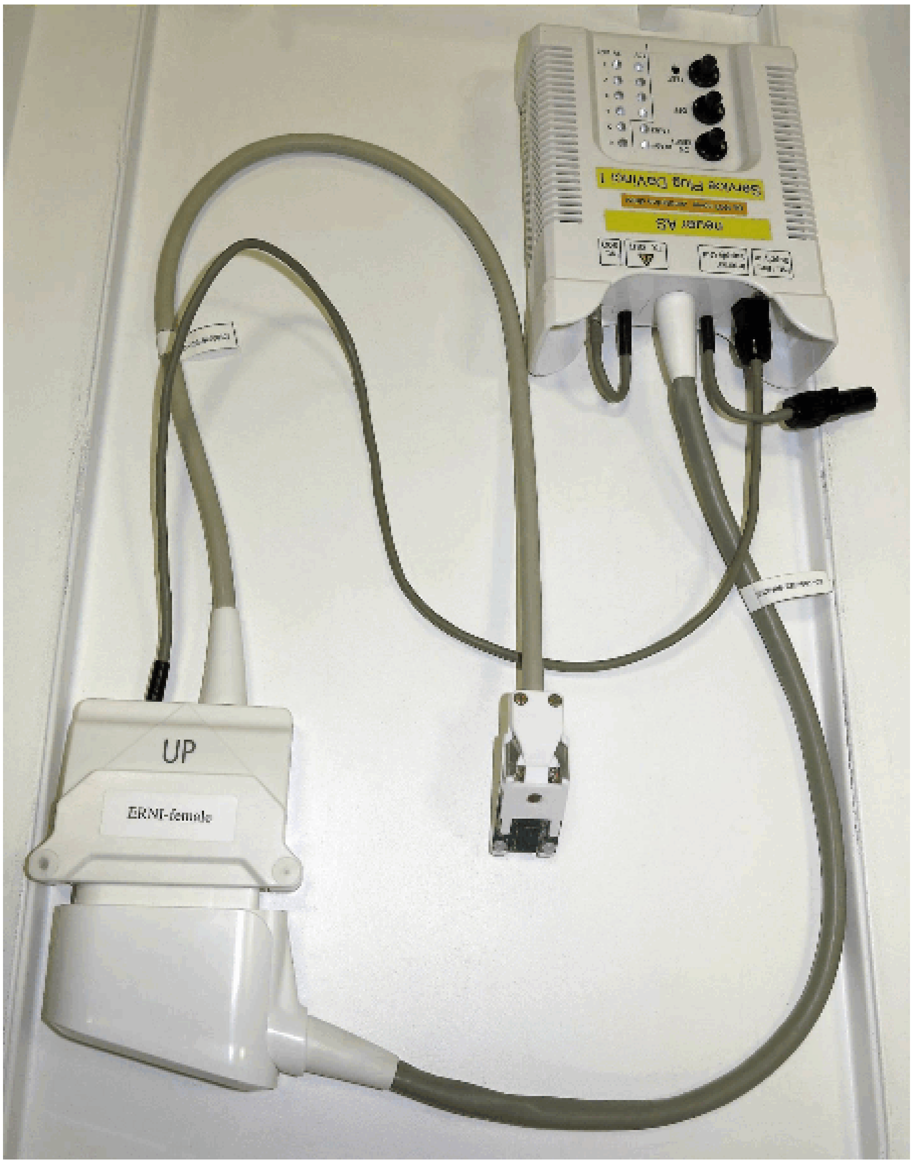
Switch on the Service Plug manually by use of the **ON/RESET**, and switch off by using the **OFF** button.

Pay attention to the input “**Int./Ext. Supply In**”: the cable from the output “**TX Out**” is removed and the supply cable of the adapter plugged in.

Use of the ERNI-female Adapter

Service Plug Configuration:

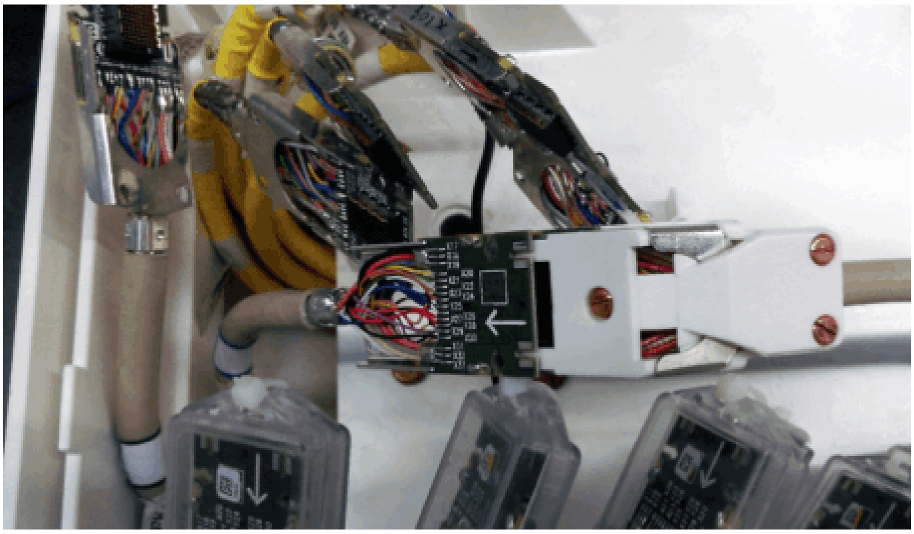
Fig. 19: Service Plug Configuration with ERNI Adapter



Use of the ERNI-female Adapter

Connecting the Service Plug to an ERNI interconnection (example MAGNETOM Aera/ Skyra):

Fig. 20: ERNI Adapter, Detailed View



N-male Adapter (QLA female to N male)

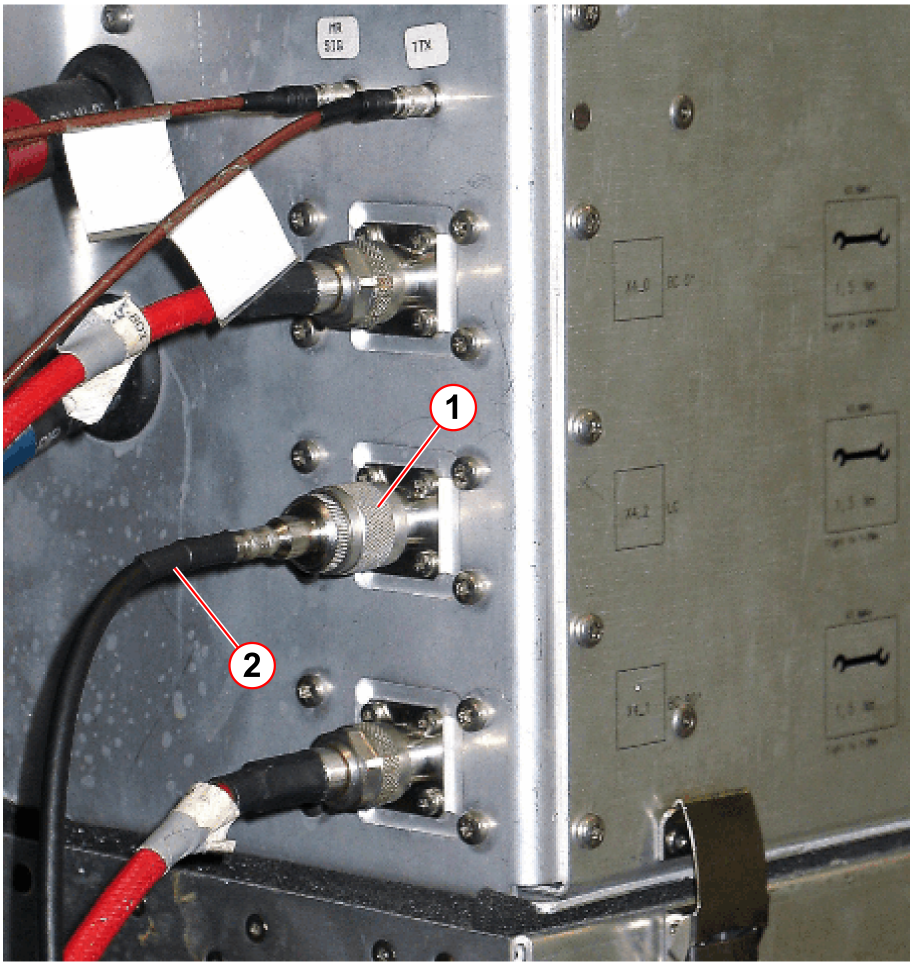
This adapter is used to test the Local Coil transmit path (LC-TX-path) of the TX-Box for transmit/receive coils (TxRx-coils).

Fig. 21: N-male Adapter

For the test set up the following measurement configuration:

- Connect the N-male adapter to the LC-TX path of the TX-Box (X4_2 LC).
- Connect the QLA extension to the N-male adapter.
- Then, plug in the other end of the QLA cable into the Service Plug inlet “**Int. 50 Ohm**”.
- At the end connect the Service Plug to coil socket 1.

Fig. 22: Service Plug LC-TX connected to TX-Box



(1) N-male Adapter
(2) QLA Extension Cable (to Service Plug inlet "Int. 50 Ohm")

Docking TX Adapter (QLA female to ODU TX)

The adapter is used at systems with a removable patient tablet to test the LC-TX path connector of the docking station (depends on the system).

Fig. 23: Docking TX Adapter

For the test connect the Docking TX adapter to the LC-TX local coil transmit path (LC-TX) of the docking station. The orientation of the adapter is not relevant as there is not designated UP or DOWN position.

> **i** At the docking station it is not necessary to switch on the Service Plug manually. It is automatically switched on due to the recessed pin functionality of the used Direct Connect sockets.

QMA-male Adapter (QLA female to QMA male)

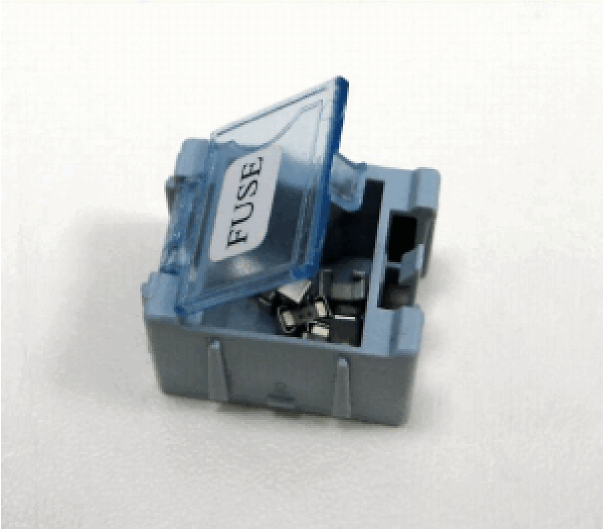
This adapter is used for testing at the LC-TX path cables inside the coil sockets of the patient table and the patient table Interconnection Plate.

Fig. 24: QMA-male Adapter

Spare Fuses

The spare fuses (rating 1 A) can be used in case the fuses of the Service Plug are blown due to a mistake or an internal problem (e.g. a Service Plug control LED does not light up).

Fig. 25: Service Plug Spare Fuses



The fuses are accessible from the bottom of the Service Plug and can be easily exchanged.

Fig. 26: Bottom Side Access to Fuses of the Service Plug

Adapters for MAGNETOM Vida

For MAGNETOM Vida additional adapters are available for use within TestTools:

Fig. 27: Service Plug ERMET Adapter I

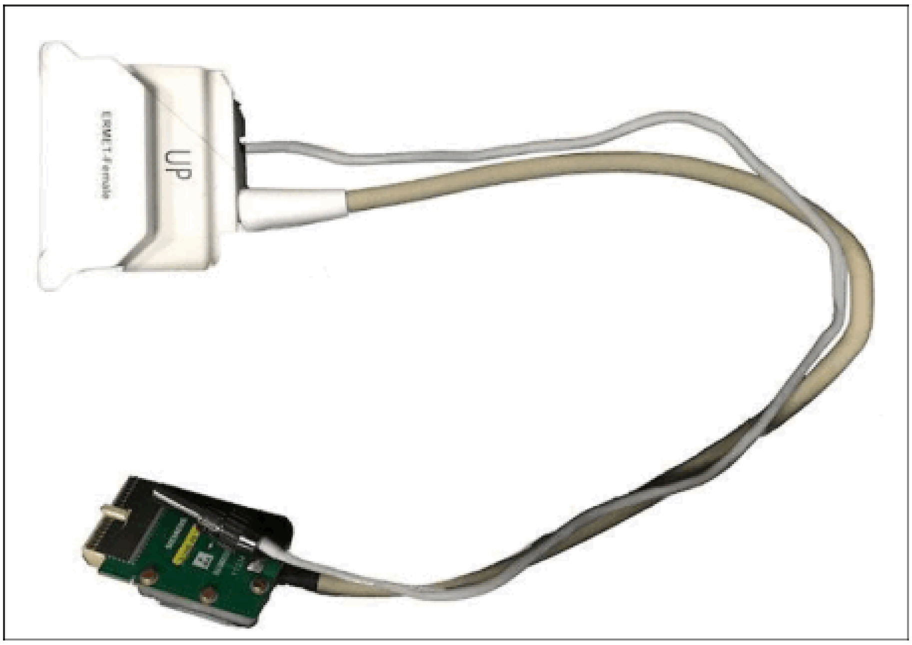
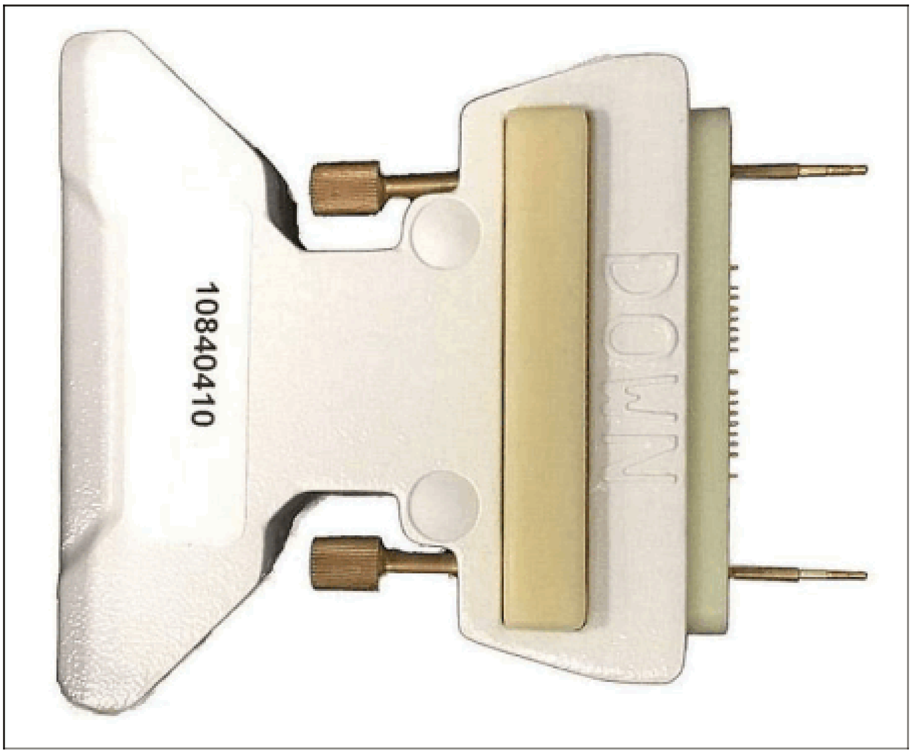


Fig. 28: Service Plug ERMET Adapter II



Adapters for MAGNETOM Spectra

For MAGNETOM Spectra additional adapters are available for use within TestTools.

Notes

NOTICE

With dockable patient tables, the mains supply at the docking station is always present.

» **Be careful when removing the adapter plate of the patient table or connecting the ERNIs in the patient table.**

Before the start of an LC Plug test the status of the Service Plug is evaluated. In case of an error condition like overtemperature or incorrect supply voltage the test is aborted. In that case please reset the Service Plug or unplug/plug it.

In case the 3 V LEDs of the Service Plug are off after a reset of an error condition (e.g. by pressing the **ON/RESET** button) please remove and connect the Service Plug again.

Changes

Initial version.

2.1.2 Expert

2.1.2.1 Fast

Help ID: 08F170A6-AC22-4255-9B5E-9CBE63A82849
Service Level: 7

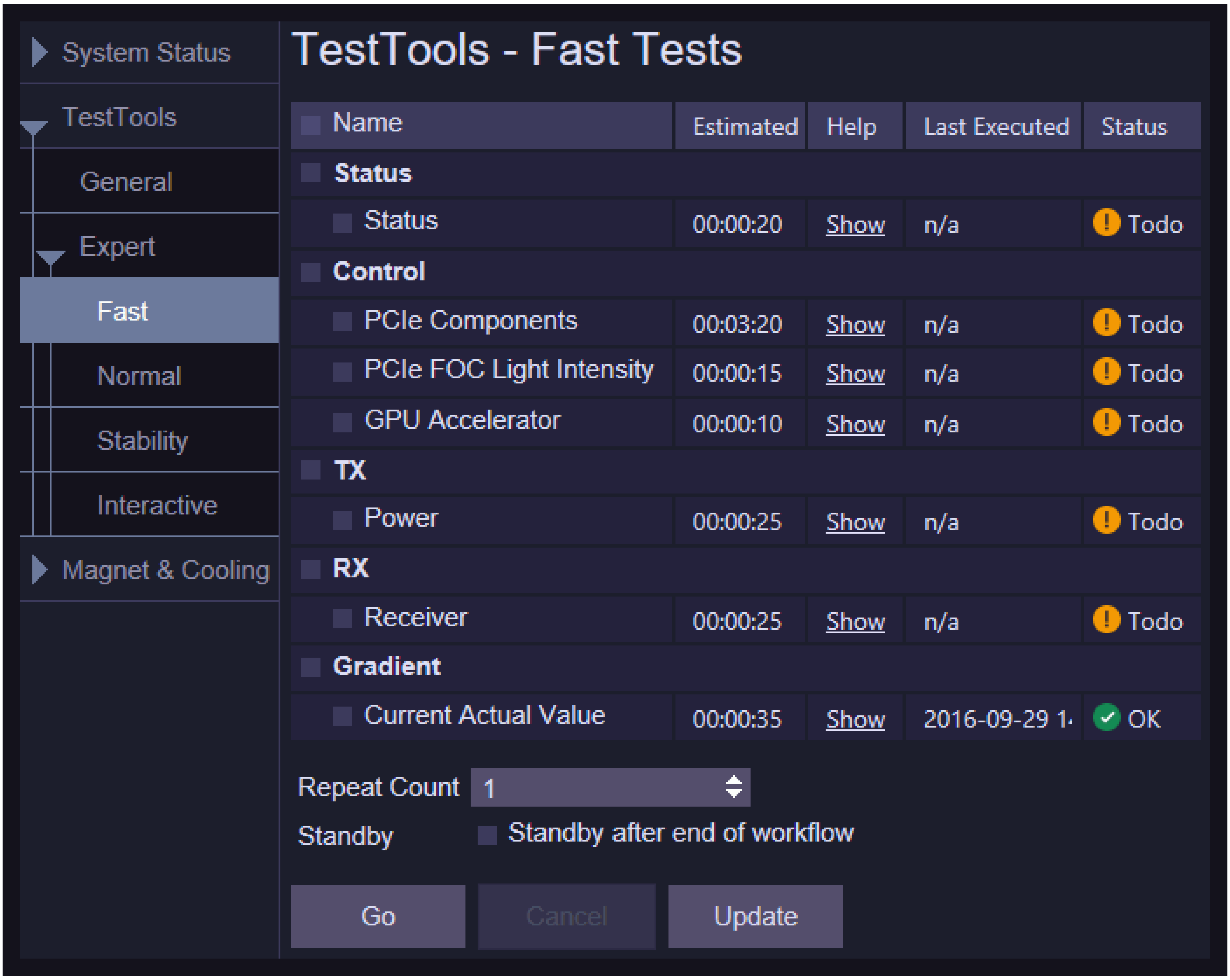
Fast General

In this test category all tests necessary to obtain a quick overview of all parts of the system are collected.

Thus the test session lasts only a few minutes. In this time it should be possible to find components which are totally out of order.

To be able to do this, only very basic tests are done to test the system.

Fig. 29: TestTools Expert Fast Example



2.1.2.1.2 Status

2.1.2.1.2.1 Status

```
Help ID: 76D75CFC-EDE3-4751-A93C-6687BE255CD3
Service Level: 7
```

General

The **Status** test checks the status of all peripheral hardware components supporting the query of their status.

For each processed component a summary of all current errors and warnings is displayed. Should the component have no faults, only an “ok” is printed out.

In case of an error, the error message and all related information are printed out: domain, error ID, message text, service entries “explanation”, “action”, and “information” (if available).

Signal Path

The communication links to the peripheral component are used.

Test

Before querying the status of the peripheral components this test first checks if the MaRS is reachable via network ("ping"). If this fails a message is printed out and the program terminates immediately as in that case the other component will also not be reachable.

Afterwards all peripheral hardware components are queried.

Evaluation

The states of the peripheral components are checked and printed out.

Incorrect states are marked.

The evaluation itself has to be performed by the customer service engineer (CSE). For this the status of the component and the error messages in the **Message Viewer** have to be analyzed.

It depends on the text after that if really an error condition is present. E.g. "resolved", "has not occured", "not active", or "inactive" means that there is no problem.

Please note that the messages that are displayed within TestTools are identical to the messages in the **Message Viewer**.

Conclusions

For this the status of the component and the error messages in the **Message Viewer** have to be analyzed.

In case of multiple errors for a component please check first the configuration of the component in **Tools > Supervision**.

This must be “Real” as “Virtual” will only simulate a communication to that component.

2.1.2.1.3 Control

2.1.2.1.3.1 PCIe Components

Help ID: 1C6AAF96-374C-4154-BB0B-FC60B8CD3D87
Service Level: 7

General

The PCIe test executes the external component tests of all peripheral hardware components supporting self tests. The optical links are used for this. Thus this test checks of the performance and the optical power of the communication links, too.

Before starting the real test, it is first checked whether the MaRS is reachable via network ("ping"). If this fails, a message is printed out and the test workflow aborts immediately as in that case the other component will not be reachable, too.

For each executed test the test results will be shown, including the name of the test, its status and a message showing the test details.

The test execution will be continued even if an error occurs when executing a test.

> Due to internal and independent software configuration it is not defined which physical OptoExpressCard RX (RX-OPEC) is RX-OPEC 1 and which and which one is RX-OPEC 2.
>
> The physical slot number does not necessarily correspond to the board number used within the software.
>
> Thus it is not possible to easily identify the OPEC board which shows an error. E.g. errors indicated for the left OPEC could be the right OPEC, and vice versa.

> Please note that not all listed tests might be available at your system.

> It is necessary to reboot the MaRS everytime any FOC was disconnected and connected again (e.g. for swapping), or when a single component was powered off!

Signal Path

The communication between the MaRS and the peripheral components is done using the optical PCI-Express bus of the MaRS.

Tests

RCCS RFCELK2264_RCCS (IF_RCCS)

- **AllVersion**
  The versions of the FPGAs are read, no analysis is necessary.

RFIS RFCELK2264_RFIS (BC_Dyn & LC_Dyn (PIN Control))

- **FPGAVersion**
  The version of the FPGA is read, no analysis is necessary.

RFCEL_SAT RFCELK2264_SAT (RFCEL satellite)

- **FPGAVersion**
  The version of the FPGA is read, no analysis is necessary.
- **FPGAVersion**
  The version of the FPGA is read, no analysis is necessary.
- **GetStatus**
  The status of the RFCEL_Satellite is evaluated; a bit pattern is read.
  If everything is ok, the status should be 0x00.
  The bits have the following meaning:

Tab. 2 Meaning of the GPA Status Bits

0	Gradient Coil sensors plugged
1	Gradient Coil overtemperature
2	Body Coil overtemperature
3	BORE overtemperature
4	BORE light power fail right
5	BORE light power fail left
6	(reserved)
7	FRONT light power fail red
8	FRONT light power fail green
9	FRONT light power fail blue
10	3.3 V fault
11	5 V fault
12	7.5 V_1 fault

13	7.5 V_2 fault
14	25 V fault
15	16 V fault

Intercom RFCELK2264_INTERCOM (Intercom interface and Intercom board)

- **LWVersion**

  The version of the FPGA is read, no analysis is necessary.

RFCEL_COM RFCELK2264_COM (RFCEL_COM controller)

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **FPGA-Gimli**

  The version of the Gimli FPGA is read, no analysis is necessary.
- **EEPROM Info**

  Reads the serial number and the revision level of the component, no analysis is necessary.
- **ShortLoop**

  A loop test is performed through the Gimli.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteDMALoop** (depends on system type)

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip DMA interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteRingLoop** (depends on system type)

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip ring interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RingStatus**

  The status of the ISO-ring interface is checked. Here the status bits are evaluated for correctness.
- **PLLStatus**

  The status of the PLLs is checked (PLL goes out of lock temporarily).

- **FPGA-Device**

  The version of the FPGA is read, no analysis is necessary.
- **SyncDone**

  This test checks whether the component is in sync with the other components.
- **OpticalRingLoop**

  A loop test through the ISO-ring is performed. This can only be done by the ring-master, which is the RFCEL_Com.

  Test data is sent and received. Afterwards the two data sets are compared
- **MSMTest**

  The status and the temperature of the main module is checked against specification.
- **PTABDockTest**

  This test is related to the dockable table. Depending on the docking status either 0x00 (completely undocked), 0x0f (completely docked), or any value inbetween in case the table is not (yet) fully docked/undocked.

  Check if the values printed out corresponds to the actual status of the table.
- **PTABPositionTest**

  This checks the PTAB position value directly from the CAN-Bus and compares it with the value coming from the RFCEL_COM. The purpose is to check the consistency of the CAN-data.

RFCEL_DSP RFCELK2264_DSP (PALI resp. Stimo functionality, depends on system type)

- **LWVersion**

  The version of the device FPGA is read, no analysis is necessary.
- **EnableDSPPaliTestData**

  PALI test data is switched on. This data is put on the ISO-ring from the TX-Box.
- **DisableDSPPaliTestData**

  PALI test data is switched off.
- **MemoryTest**

  Defined data is written into the memory of the DSPs. Afterwards they are read back and compared with the written data.

  Thus the communication between MARS and DSP, and the function of the memory is tested.
- **PaliDataPath**

  Defined data is read from the IsoRing, afterwards the Pali-data is calculated. Thus the algorithm of the DSP and the communication between the IsoRing and the DSP is tested.

- **StimoDataPath**

  Defined data is read from the IsoRing, afterwards the Stimo-data is calculated. Thus the algorithm of the DSP and the communication between the IsoRing and the DSP is tested.

OPEC_TX OptoExpressCard(20)_Tx (OPEC Tx board)

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **VersionInformation**

  OptoExpressCard(20)_Tx hardware related information are printed out, e.g. Serial-Number, PartNumber, and Revision Level.

RxK2364_64_1

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **FPGA-Gimli** (depends on system /system type)

  The version of the Gimli FPGA is read, no analysis is necessary.
- **EEPROM Info** (depends on system /system type)

  Reads the serial number and the revision level of the component, no analysis is necessary.
- **ShortLoop**

  A loop test is performed through the Gimli.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteDMALoop** (depends on system /system type)

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip DMA interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteRingLoop** (depends on system /system type)

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip ring interface.

  Test data is sent and received. Afterwards the two data sets are compared.

- **RingStatus**

  The status of the ISO-ring interface is checked. Here the status bits are evaluated for correctness.

- **PLLStatus**

  The status of the PLLs is checked (PLL goes out of lock temporarily).

- **FPGA-Device** (depends on system /system type)

  The version of the FPGA is read, no analysis is necessary.

- **SyncDone** (depends on system /system type)

  This test checks whether the component is in sync with the other components.

- **SDRAMTest**

  The 128 MB-DDR memory is tested with different bit patterns. They are first written, then read back and compared.

- **EchoRAMTest**

  The 8 MB-QDR memory (Echo RAM) is tested with different bit patterns. They are first written, then read back and compared.

- **MDHTest**

  Both MOFI-Configuration registers are tested with different bit patterns. They are first written, then read back and compared.

- **VersionInformation**

  Hardware related information for Gilgalad, RxK2304, and IF_RX boards.

- **MofiConfTest**

  Both MOFI-Configuration registers are tested with different bit patterns. They are first written, then read back and compared.

- **CheckClockPulse**

  The clock check is done by reading the RX-status register repeatedly. The corresponding bits "CLK10_Counter" should toggle.

- **MofiStatic** (depends on system /system type)

  Both MOFI-filters generate data by using the test data register value, NCO center and offset frequency with 0 Hz and a special phase offset. The multiplier in the demodulator multiplies the test data with the corresponding sine and cosine output of the NCO. The MOFI filters these products and the results can be read in the different real and imaginary registers of channel 1 through 32 (per RX board). Different test data patterns are used and evaluated.

- **NCOStatic** (depends on system /system type)

  The static NCO-Test is performed for NCO 0 and 1, and also for MOFI1 and MOFI2. The used NCO is set via the NCOSet-Register.

  Both MOFI-filters generate data by using the test data register value, NCO center and offset frequency with 0 Hz and special phase offsets. The multiplier in the demodulator multiplies the test data with the corresponding sine and cosine output of the NCO. The MOFI filters these products and the results can be read in the different real and imaginary registers of channel 1 through 32 (per RX board).

  At the end the read data is compared with the expected data. Different phase offsets are used.

- **RxOnTest**

  Both MOFI-filters generate data (real and imaginary values) by using the test data register value. This generated data is transferred to the control-FPGA, sorted via the Echo-RAM and written into the SDRAM-FIFO. Afterwards the read data is compared with the expected data.

  Some subtests are only done with the first DIG_RX board.
- **SNIF_RX32** (depends on system /system type)

  Reads the serial number and the revision level of the IF_RX board; no analysis is necessary.
- **SNSynthesizer** (depends on system /system type)

  Reads the serial number and the revision level of the Synthesizer; no analysis is necessary.

  This test is only available for the first DIG_RX board.
- **SupplyVoltage (DIG_RX)**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature (DIG_RX)**

  The temperatures of the optical transceivers are checked against specification.
- **SupplyVoltage (IF_RX)**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature (IF_RX)**

  The temperatures of the optical transceivers are checked against specification.
- **ShortLinkTest**

  The pseudo-random-bitpattern-test is started via MOFI and the result is read in from the MOFI-Redimu status register (Test Communication Link to IF_RX32).
- **DMASDRAMCheck** (depends on system /system type)

  Before the test starts some raw data sets are written into the SDRAM-FIFO (DMA-length message, MDH, and data sets for 32 channels). Afterwards a measurement is prepared so that the data is automatically transferred to the GIMLI via the DMA0 channel.

RxK2364_64_2 (depends on system /system type)

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **ShortLoop**

  A loop test is performed through the Gimli.

  Test data is sent and received. Afterwards the two data sets are compared.

- **RingStatus**

  The status of the ISO-ring interface is checked. Here the status bits are evaluated for correctness.

- **PLLStatus**

  The status of the PLLs is checked (PLL goes out of lock temporarily).

- **SDRAMTest**

  The 128 MB-DDR memory is tested with different bit patterns. They are first written, then read back and compared.

- **EchoRAMTest**

  The 8 MB-QDR memory (Echo RAM) is tested with different bit patterns. They are first written, then read back and compared.

- **MDHTest**

  Both MOFI-Configuration registers are tested with different bit patterns. They are first written, then read back and compared.

- **VersionInformation**

  Hardware related information for Gilgalad, RxK2304, and IF_RX boards.

- **MofiConfTest**

  Both MOFI-Configuration registers are tested with different bit patterns. They are first written, then read back and compared.

- **CheckClockPulse**

  The clock check is done by reading the RX-status register repeatedly. The corresponding bits "CLK10_Counter" should toggle.

- **RxOnTest**

  Both MOFI-filters generate data (real and imaginary values) by using the test data register value. This generated data is transferred to the control-FPGA, sorted via the Echo-RAM and written into the SDRAM-FIFO. Afterwards the read data is compared with the expected data.

  Some subtests are only done with the first DIG_RX board.

- **SupplyVoltage (DIG_RX)**

  The supply voltages of the optical transceivers are checked against specification.

- **Temperature (DIG_RX)**

  The temperatures of the optical transceivers are checked against specification.

- **SupplyVoltage (IF_RX)**

  The supply voltages of the optical transceivers are checked against specification.

- **Temperature (IF_RX)**

  The temperatures of the optical transceivers are checked against specification.

- **ShortLinkTest**

  The pseudo-random-bitpattern-test is started via MOFI and the result is read in from the MOFI-Redimu status register (Test Communication Link to IF_RX32).

OPEC_RX OptoExpressCard_Rx (OPEC Rx board)

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **VersionInformation**

  Hardware related information like serial number, material number, and revision level.

TX TxK2364 (TX-Box, also 2-CH and MNO TX-Box)

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **FPGA-Gimli**

  The version of the Gimli FPGA is read, no analysis is necessary.
- **EEPROM Info**

  Reads the serial number and the revision level of the component, no analysis is necessary.
- **ShortLoop**

  A loop test is performed through the Gimli.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteDMALoop**

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip DMA interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteRingLoop**

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip ring interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RingStatus**

  The status of the ISO-ring interface is checked. Here the status bits are evaluated for correctness.
- **PLLStatus**

  The status of the PLLs is checked (PLL goes out of lock temporarily).
- **FPGA-Device**

  The version of the FPGA is read, no analysis is necessary.

- **SyncDone**

  This test checks whether the component is in sync with the other components.

- **SRAMTest**

  The SRAM Test reuses the PCI Remote Ring Loop test after sending a message to switch the loop test data path to include the SRAM like a FIFO (First-In-First-Out).

  The test performs the Remote Loop test for all four queues. The SRAM Status is checked for single bit events.

  In case of an error the following messages might come up as a result of the test:

  - "Warning: single bit error occured (can be corrected)"
  - "Error: multi bit error occured (can NOT be corrected)"
  - "Error: syndrome error occured"

  The used error correction algorithm (HammingCode, Forward Error Correction) is able to detect and correct bit errors.

  - A single bit error can be corrected and is additionally reported, too.
  - If more than a single bit error is detected it cannot be corrected but is reported, too.

- **FlashRomTest**

  This test is done on an unused flash page. It erases the page, then data is written, compared, and then erased again. Then the page is verified to be clean.

- **SPIRom**

  Tests the SPI-ROM interface and ROMs. Data is written, read back, and then compared.

- **ADC Test**

  Tests the ADC of the TX-Box.

GPA GPAK2309 (Gradient Power Amplifier)

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.

- **Temperature**

  The temperatures of the optical transceivers are checked against specification.

- **FPGA-Gimli**

  The version of the Gimli FPGA is read, no analysis is necessary.

- **EEPROM Info**

  Reads the serial number and the revision level of the component, no analysis is necessary.

- **ShortLoop**

  A loop test is performed through the Gimli.

  Test data is sent and received. Afterwards the two data sets are compared.

- **RemoteDMALoop**

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip DMA interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RemoteRingLoop**

  A loop test is performed through the Gimli plus the application FPGA via the chip-to-chip ring interface.

  Test data is sent and received. Afterwards the two data sets are compared.
- **RingStatus**

  The status of the ISO-ring interface is checked. Here the status bits are evaluated for correctness.
- **PLLStatus**

  The status of the PLLs is checked (PLL goes out of lock temporarily).
- **FPGA-Device**

  The version of the FPGA is read, no analysis is necessary.
- **SyncDone**

  This test checks whether the component is in sync with the other components.
- **GCECC-ID**

  Reads the version of the GCECC module, no analysis is necessary.

Evaluation

For the evaluation please refer to the description of the tests.

Conclusions

This depends on the peripheral component.

Please refer to the test report.

> Please perform the interactive ISO Ring test in case the PCIe tests do not give clear results.

2.1.2.1.3.2 PCIe FOC Light Intensity

```
Help ID: 865BE1C1-E0A1-43AF-BA21-10D378AA7AB7
Service Level: 7
```

General

The **PCIe FOC Light Intensity** test checks the LightIntensity of the various transceivers of the peripheral hardware components. Thus this test checks of the performance and the optical power of the communication links.

For each executed test the test results will be shown, including the name of the test, its status, and a message showing the test details.

The test execution will be continued even if an error occurs when executing a test.

> Please note, for the 64-channel receivers the status of the transceivers cannot be retrieved.
>
> Instead the supply voltages and temperatures have to be analysed for troubleshooting.

> It is necessary to reboot the MaRS everytime any FOC was disconnected and connected again (e.g. for swapping), or when a single component was powered off.

Signal Path

The communication between the MaRS and the peripheral components is done using the optical PCI-Express bus of the MaRS.

Tests

The status of the transmitted and received light intensity/power of the optical transceivers is retrieved and evaluated against a specification value/threshold:

- RCCS: RFCELK2364_RCCS
- RFIS: RFCELK2364_RFIS
- RFCEL_SAT: RFCELK2364_SAT
- Intercom: RFCELK2364_INTERCOM
- RFCEL_COM: RFCELK2364_COM
- OPEC_TX: OptoExpressCard20_Tx
- RX: RxK2304_64Ch_1
- RX: RxK2304_64Ch_2
- OPEC_RX: OptoExpressCard20_Rx_1
- TX: TxK2304_2Ch
- TX: TxK2304_MNO
- GPA: GPAK2368

Evaluation

See above.

Conclusions

Please refer to the test report.

2.1.2.1.3.3 GPU Accelerator

Help ID: 0074561F-5F71-4E81-8AAB-75002976C318
Service Level: 7

General

The **GPU Accelerator** test checks the GPU accelerator card of the MaRS computer.

> Currently only systems with 64 and more receiver channels are equipped with this additional board.
>
> The test will be performed with every system. In case the board is not installed the test will terminate with "No accelerator device present and configured". This is ok in case really no GPU is installed. If you get this message with a GPU installed check the setting of the MaRS.

Signal Path

The data communication from the MaRS to the GPU is tested, this is the internal PCI bus.

Test

From the MaRS a part of the memory is transferred via the PCI bus into the memory of the GPU.

There the complete free GPU memory is allocated and the transferred data is multiply copied into the allocated memory.

Afterwards the transferred and the copied data are compared.

At the end the transferred data is copied back into the MARS and finally compared with the originally transferred data.

> In the result output "CUDA" refers to the software driver of the GPU.

Evaluation

See above.

Conclusions

Please refer to the test report.

2.1.2.1.4 TX

2.1.2.1.4.1 Power

Help ID: 8FD3B2E1-93F0-45C2-9324-2278B8B046D8
Service Level: 7

General

Test Description

The **Power** test checks for the maximum RF transmit power and the power reflection at the Body Coil.

As the used sequence can be parameterized, it is possible to have this test in different measurement time duration. This test is therefore used in **Fast, Normal**, **Stability**, and **Multi Nuclei**. The Multi Nucleus capability is optional to the system, therefore the test is only available when the option is installed.

In this chapter the test category **Fast** is described.

Signal Path

The power is transmitted to the BC with the loader placed inside. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. In comparison to Avanto there are two DICOs that are placed after the power splitter. Thus the power for the 0 and the 90 degrees transmit path (BC 0 and BC 1) can be measured separately.

Test

For the test the (new) large spherical phantom has to be in the magnet center.

All data is acquired simultaneously within one measurement using the rf_loop sequence.

The number of RF pulses used for the stability and linearity evaluations depends on the number of measured lines. For the test category **Fast** the measurement time is quite short, just to acquire enough data to be able to evaluate if the TX-Box is in working condition.

Evaluation

The following parameters of the power output are checked at the DICOs:

- maximum power
- deviation of requested and delivered power
- deviation of the power of the two transmit paths (BC 0 and BC 1).

- reflected power for the two parts of the BC (as a function of the transmitted power).

A comparison of TALES and DICO values is not required because the TX-Box performs an online comparison of these values for security reasons.

Conclusions

Please refer to the test report.

2.1.2.1.5 RX

2.1.2.1.5.1 Receiver

```
Help ID: 85049DEF-02B7-474C-82F2-939C7A1D9B22
Service Level: 7
```

General

The **Receiver** test checks the receive signals of all ADCs of the receiver boards by executing MR measurements via the sequence rf_loop (parameterizable sequence). It tries to minimize the number of MR measurements necessary to test the ADCs. Only available/ configured channels are used.

As a result, it returns a list of sub-results each consisting of an MR measurement result and a list of ADC numbers checked by that measurement.

The program supports several test categories (**Fast**, **Normal**, and **Stability**) which do the test in more detail but also require more time:

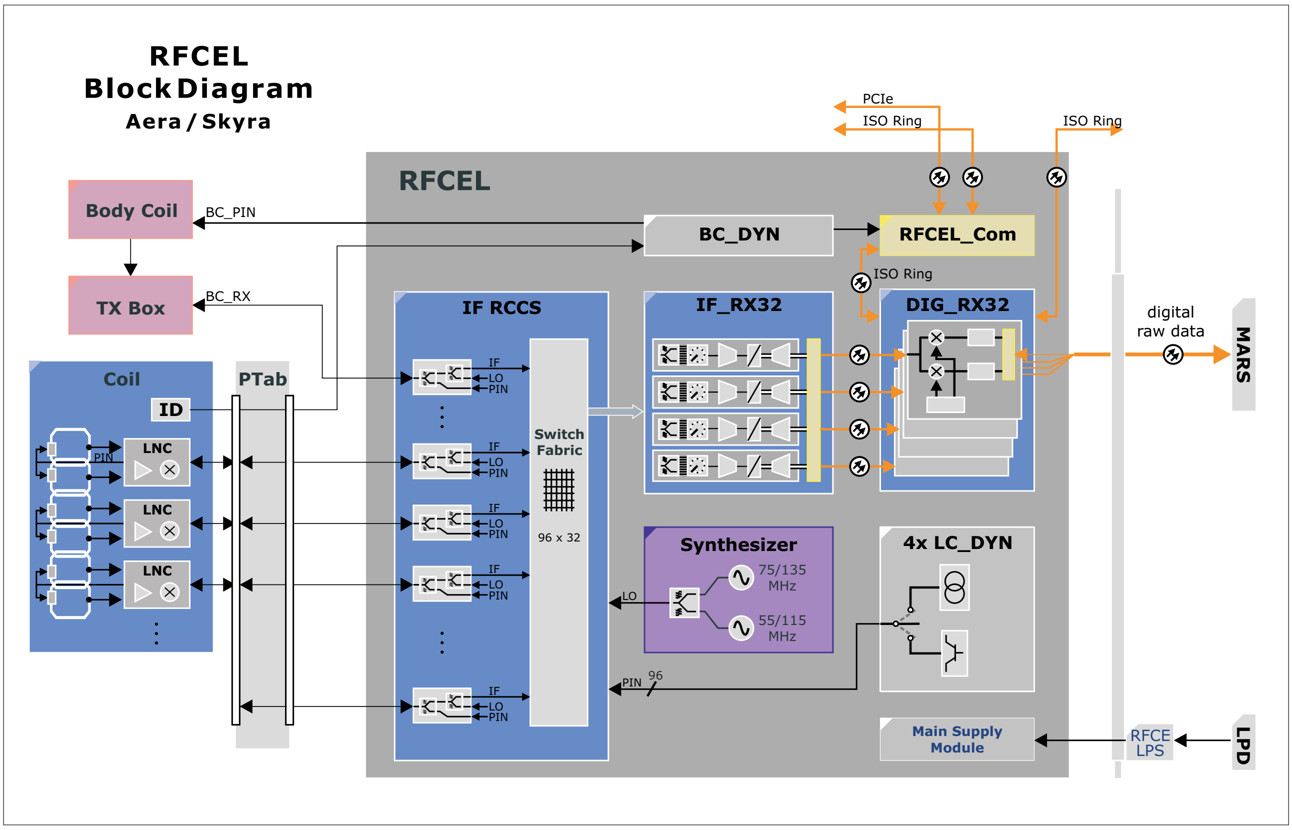
- Receiver (**Fast**)
- RX Stab/Lin (**Normal**)
- Receiver Longtime Stab (**Stability**)

In this chapter the **Fast** mode is described.

> Although the graphics show the MAGNETOM Aera/Skyra RF system they are still valid for MAGNETOM Vida.

An overview of the receive path is shown in the next picture:

Fig. 30: RFCEL Block Diagram



Signal Path

The test concept is based on the possibility of routing a test signal from the modulator of the TX-Box via an arbitrary node of the IF_RCCS to an arbitrary receiver. The signal is measured at the corresponding ADC.

The detailed signal path:

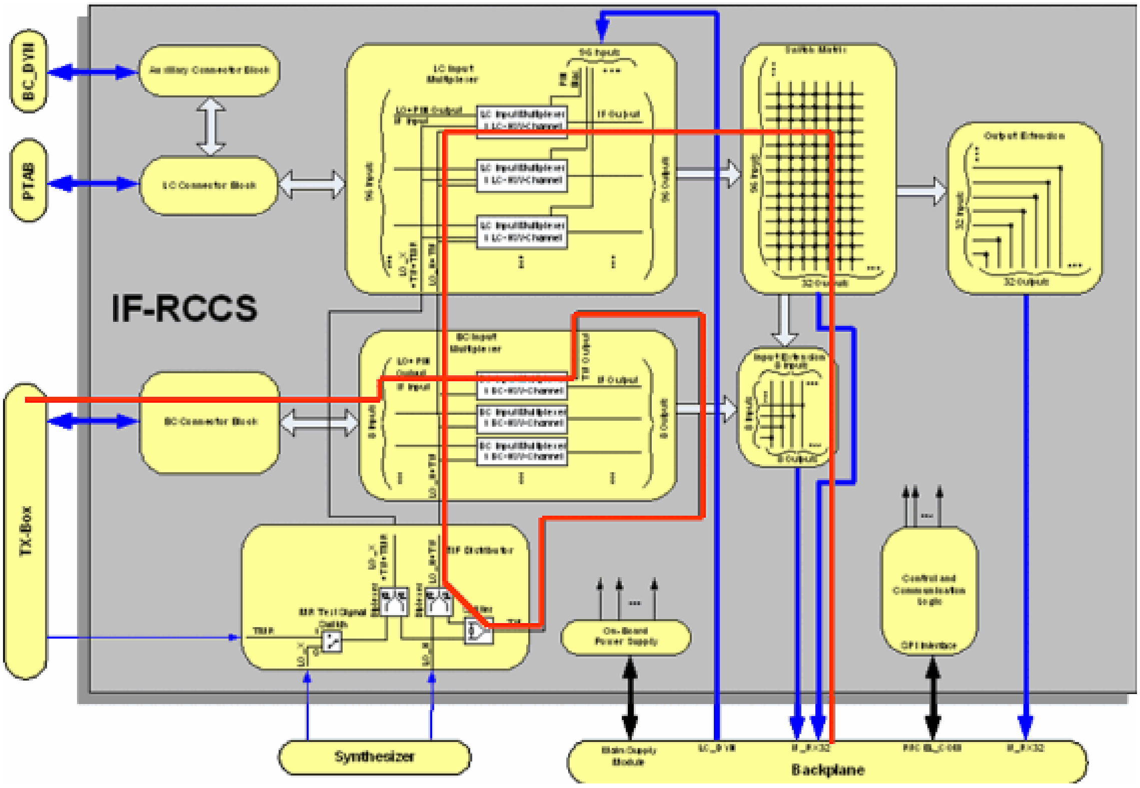
- **TX-Box**

  The TTX-signal of the modulator is routed to the two inputs of the MR signals from BC (without using the amplifier module). The TTX-signal is mixed with the two LO frequencies and thus down converted to the IFs and afterwards combined on one signal line. In the following, this signal is called TIF (Test signal Intermediate Frequency).

- **IF_RCCS**

  From the BC 1 input the signal is routed to all input multiplexers. From each of these multiplexers, the signal can be routed either to a local coil socket of the patient table or to the input of the switch matrix. The outputs of the IF_RCCS are connected to the receivers where the signal is digitized. Before this, the signals are amplified by SGAs (Switchable Gain Amplifiers) using high gain.

Fig. 31: Signal Path RCCS Matrix Test



Test

For the measurement the program sets the following configurable values:

- raw image size (number of lines, columns)
- transmitter amplitude
- repetition time.

Additionally among others the following settings are done:

- all adjustments are disabled
- the RFSWD mode is set to "inactive"
- the frequency is set to "1H"
- the transmitter loop is set to TXLOOP_TTX
- the receiver loop is set to RXLOOP_MRSIG

At the end these settings are reset to the settings that were active before the test was started.

Evaluation

For each ADC the stability (raw data of first sinc-pulse) and linearity (raw data of second sinc-pulse) values are calculated from the corresponding raw data and stored in the evaluation result.

The following magnitude and phase data entities are calculated:

- **Stability**

  min, max, mean values and the peak to peak deviation values
- **Linearity**

  min, max, mean values and the max. deviation values
- **Stability and linearity**

  deviation values of each raw data line (for graphical outputs)

The stability and linearity evaluation results for each ADC are checked if they are in specification.

Afterwards the results are displayed in detail. In the summary table only the failed ADCs are listed to avoid a large table.

The following entities are displayed for each ADC:

- table with stability results containing min, max, mean values as well as peak to peak deviation values for magnitude and phase data
- table with linearity results containing min, max values as well as max. deviation values for magnitude and phase data

Conclusions

Please refer to the test report.

2.1.2.1.6 Gradient

2.1.2.1.6.1 Current Actual Value

```
Help ID: E002408D-5297-432E-ADDB-A2AFB558D156
Service Level: 7
```

General

The **CAV** (Current Actual Value) test checks the actual values used to regulate the gradient amplifier current. For each axis X, Y, and Z a linearity and stability evaluation is performed.

The number of measurement lines is configurable, thus the test can be used in the test categories **Fast** and **Normal**. The run time of the **Normal** test variant is longer as a larger number of lines for the raw data is used. Thus the test shows more precise results for the stability evaluation.

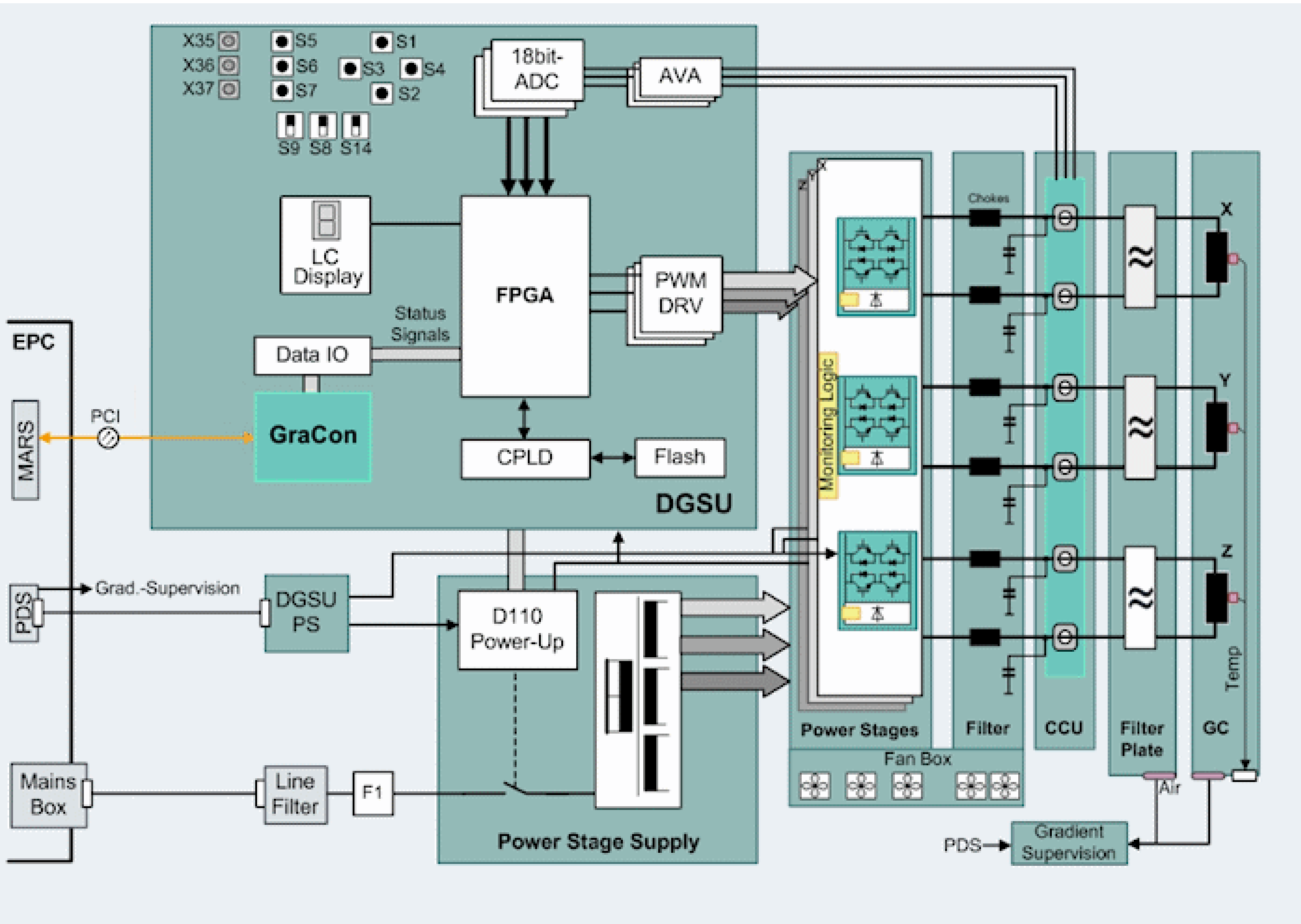
The DAC tests from previous system types are obsolete now as the used digital gradient small signal unit (DGSU, also called Digital Control Component - DCC) has no D/A converter for the nominal values. Apart from the three analogue signals coming from the current sensors, only digital signals are processed in the DGSU.

> This test does not supply a 100 % clear analysis of the stability and linearity of the gradient pulses. This can be done only with a MR-signal (e.g. QA/Stability Check, Calcar versus Calcar_Lemshift, and so on).
>
> Basically the CAV tests checks the "quality" of the regulation.

Signal Path

The actual gradient coil current values are picked up by sensors in the Current Converter Unit (CCU) and fed to the A/D converter in the DGSU separately for every final stage. Together with the nominal current values this information is used as input for the gradient regulator.

Fig. 32: Overview Gradient Amplifier MAGNETOM Vida



Test

The test switches certain signal paths and performs a measurement afterwards. The acquired data is then evaluated, checked against specification values and displayed in graphical form.

Each axis X, Y, and Z is measured separately, that means three measurements are performed; one for each axis.

Evaluation

For each axis the stability and linearity values are calculated from the corresponding raw data and stored in the evaluation result if the corresponding measurement was successful.

The following data is checked:

- **Stability**

  minimum, maximum, and mean values of amplitude and integral data, together with the peak-to-peak deviations from the reference value in percent.
- **Linearity**

  minimum, maximum, and mean values of amplitude and integral data.

For both stability and linearity, a check is performed to check whether the signal is zero respectively above a defined minimum.

Conclusions

Please refer to the test report.

2.1.2.2 Normal

Help ID: 5AE69855-A10F-419F-9732-115460C98066
Service Level: 7

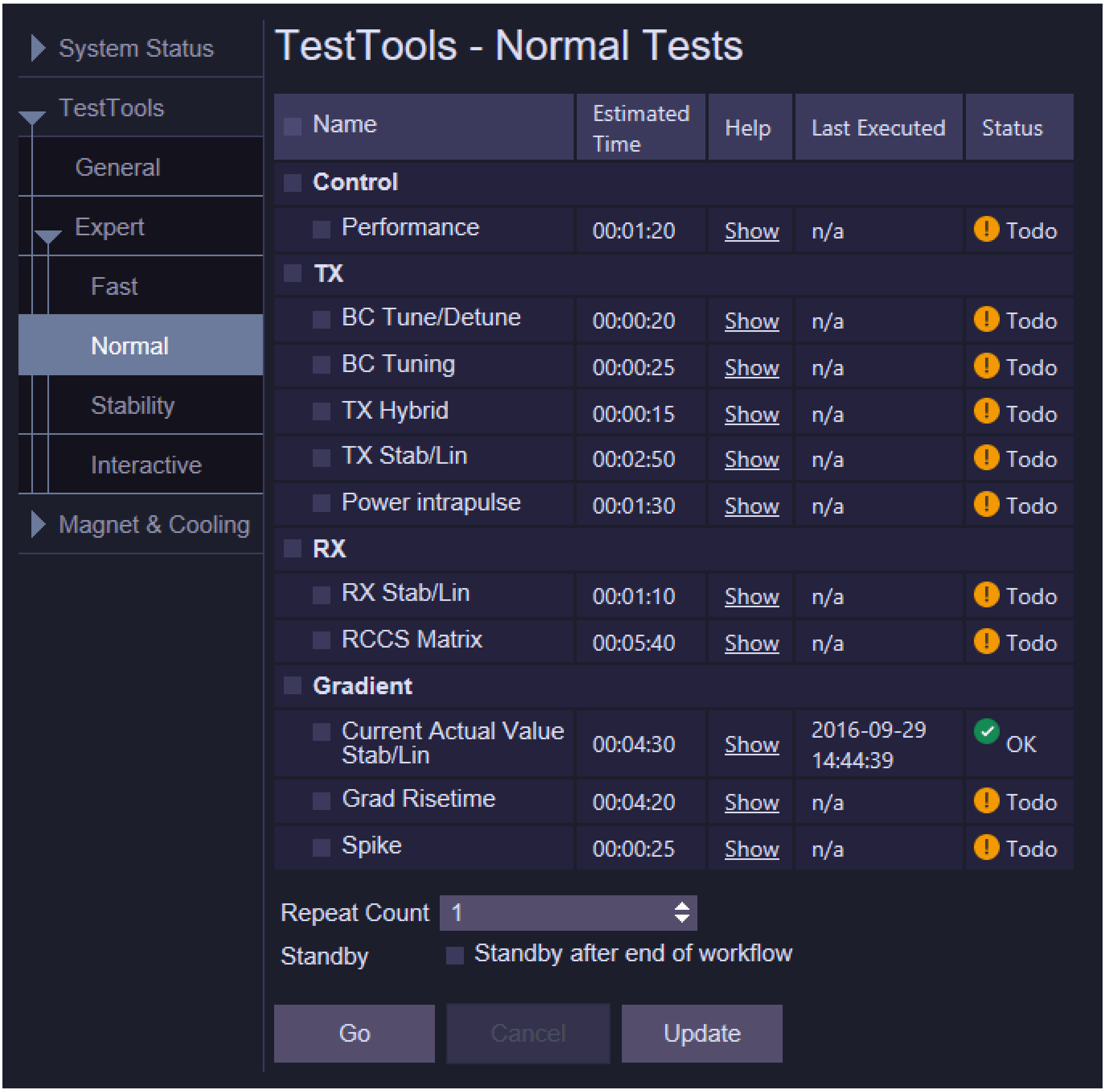
General

In this test category, all tests are collected that are necessary to perform a more detailed check compared to the test category **Fast**. This also includes specific tests of subcomponents within a functional unit.

By performing these tests it should be possible to find the source of reproducible problems occurring during normal patient measurements.

Of course the test session lasts much longer compared to the tests in test category **Fast**.

Fig. 33: TestTools Expert Fast Example



2.1.2.2.2 Control

2.1.2.2.2.1 Performance

Help ID: F786FE60-D6FC-462F-8FC4-3C239A2085E1
Service Level: 7

General

The **Performance** test checks the data rate as well as the overall performance of the MaRS.

For this a special sequence is started. This sequence is designed to produce the required data rate / amount of data. Thus the whole performance of the system including measurement, data transmission, and storage is tested if the specified real-time data rate can be handled by the system.

Test

Before the real "stress" sequence is started, a pre-measurement checks if a measurement can be performed successfully. The used values for the data rate and number of repetitions are usually smaller for this pre-measurement than those for the actual measurement. Both measurements use the special service sequence.

Afterwards the high-performance sequence is started.

If the measurement fails due to a preparation error, the program terminates with an error.

Evaluation

The program is successful if the expected data rate is reached. This means that the measurement was successfully performed and not aborted automatically (too high data rate to process):

The evaluation results of the actual measurement will be output as report table containing the following entities:

- Status of measurement (ok/not ok).
- Reached data rate in MB/sec.

Conclusions

Please refer to the test report.

2.1.2.2.3 TX

2.1.2.2.3.1 BC Tune/Detune

```
Help ID: B272FEF6-927F-48E6-AC85-429A6CD60A3F
Service Level: 7
```

General

The **BC Tune/Detune** test checks the PIN diodes of the Body Coil. Additionally the supply currents and voltages and the S-parameters for the detuned coil are tested.

Signal Path

The RF signal is fed and received to/from both BC systems (0 and 90 degree, referring to BC 0 and BC 1) individually.

Test

For the test the large spherical phantom has to be in the magnet center.

Two measurements are performed to determine the S-parameter matrix of the BC with different BC PIN diode settings; one with the setting “current” and one with the setting “high voltage”. The S-matrix describes the reflections of RF power at the two parts of the BC, as well as the transmission from one part of the BC to the other.

The test consists of two parts:

- the PIN diode signal for the Body Coil is sequentially switched to all available states (low voltage and current).
- measuring the reflections at the Body Coil at two different states: tuned and detuned.

Evaluation

The power reflections at the Body Coil are compared for tuned and detuned.

- tune: BC parameters for tuned state, current from BC_BYN
- detuned: BC parameters for detuned state, voltage from BC_BYN

From the RF signals the S-parameter of the detuned coil is determined.

Conclusions

Please refer to the test report.

2.1.2.2.3.2
BC Tuning

Help ID: 18857FF3-E868-4553-809F-0A2315A3F8CF
Service Level: 7

General

BC Tuning is used to check the central resonance frequency of the two Body Coil systems separately (reflection and transmission).

Additionally the coupling between them is measured (tuning):

- the resonance is given by the frequency for which the reflection of the Body Coil has a minimum
- the decoupling is given by the frequency for which the transmission has its maximum value

The test is identical to the corresponding Tuneup/QA function.

Signal Path

The RF signal is fed and received to/from both BC systems (0 and 90 degree, corresponds to BC 0 and BC 1) individually.

Test

For the test the large spherical phantom has to be in the magnet center.

The measurements are performed over a given frequency range by sending into both systems simultaneously. The signals are received via the normal receive path.

Evaluation

Each Body Coil system's minimal reflection is checked against the specified range, and the transmission factor has to be less than a specified maximum.

Additionally the relative phase difference is evaluated.

As an output a table is shown with the following entries:

- the reflection related values for both BC systems (0 and 90 degree) such as the resonance frequency, the reflection factor, and the difference between both systems
- the transmission related value such as the frequency and the decoupling between both systems

Graphics show the measured values for the transmission and the reflection behaviour as a function of frequency:

- the resulting curves are plotted and the minima as well as the Q-factor are determined
- additionally, the power that is transmitted from one Body Coil part to the other is plotted. The highest transmission value and the corresponding frequency are determined

Conclusions

Please refer to the test report.

2.1.2.2.3.3
TX Hybrid

Help ID: 1A027BAE-C5BC-46CB-A3FB-EC6A241609F9
Service Level: 7

General

The **TX Hybrid** test checks phase difference and the ratio of the amplitudes of the RF signals that are transmitted to the two parts of the Body Coil .

For 3.0 T systems, this measurement is performed for each of the two possible switch settings (CP and EP).

- CP Test: Test for circular excitation mode (hybrid)
- EP Test: Test for the TrueForm mode

Signal Path

The signal from the TX-Box is transmitted to the Body Coil, and the forward wave is measured and the phase between the two signals (double for parallel transmit pTX).

Test

The measurement is performed for each of the two possible switch settings (CP and EP, if applicable).

RF signals are transmitted to the two parts of the Body Coil separately but simultaneously.

Evaluation

From the RF signals the phase difference and the ratio of the amplitudes are determined.

Conclusions

Please refer to the test report.

2.1.2.2.3.4
TX Phasecheck

Help ID: F963FCE0-79D2-4501-814F-663196A1C391
Service Level: 7

General

The **TX Phasecheck** test checks phase difference and the ratio of the amplitudes of the RF signals that are transmitted to the two parts of the Body Coil .

For 3 T systems, this measurement is performed for each of the two possible switch settings (CP and EP):

- CP Test: Test for circular excitation mode (hybrid)
- EP Test: Test for the TrueForm mode

For 2-channel systems (pTX, parallel transmit) without a hybrid this test checks

- the phase relation of the modulator signals from both channels (the whole TX chain gets checked: SW, digital and analog HW, DICOs)
- the phase calibration for the measurement receivers within the TX-Box

There mainly the phase between the two TX-signals is relevant.

Signal Path

The signal from the TX-Box is transmitted to the Body Coil, and the forward wave is measured and the phase between the two signals (double for parallel transmit pTX).

Test

The measurement is performed for each of the two possible switch settings (CP and EP, if applicable).

RF signals are transmitted to the two parts of the Body Coil separately but simultaneously.

Evaluation

From the RF signals the phase difference and the ratio of the amplitudes are determined.

Conclusions

Please refer to the test report.

2.1.2.2.3.5 TX Stab/Lin

Help ID: 6F002BFF-4A55-4901-8494-6FB9601934EB
Service Level: 7

General

The **TX Stab/Lin** test checks the linearity and short time stability of the TX-Box and the power reflection at the Body Coil.

As the used sequence can be parameterized, it is possible to have this test in different measurement time lengths. This test is thus used in **Fast**, **Normal**, **Stability**, and **Multi Nuclei** if applicable (the Multi Nucleus capability is optional to the system).

In this chapter the test category **Normal** is described.

Signal Path

The power is transmitted to the Body Coil. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. In comparison to Avanto there are two DICOs that are placed after the power splitter. Thus the power for the 0 and 90 degree transmit path can be measured separately.

Test

For the test the (new) large spherical phantom has to be in the magnet center.

All data is acquired simultaneously within one measurement using the rf_loop sequence.

The number of RF pulses used for the stability and linearity evaluations depends on the number of measured lines.

Evaluation

The following parameters of the power output are checked:

- deviation of requested and delivered power
- deviation of the power of the two transmit paths (0 and 90 degree)
- reflected power for each of the two parts of the BC
- linearity and phase stability of forward power
- stability of amplitude and phase of forward power

A comparison of TALES and DICO values is not required because the TX-Box performs an online comparison of these values for security reasons.

Conclusions

Please refer to the test report.

2.1.2.2.3.6 Power intrapulse

Help ID: C7D754D2-9D66-448B-94BF-6913499F69E9
Service Level: 7

General

The **Intrapulse** tests check the intra-pulse stability of the RF pulses transmitted to the Body Coil in three different implementations. Besides the tests of the already known pulse stability of the RF power amplitude the new tests can be used to localize defective parts more precise.

The already implemented and available TX-path tests check only the stability and linearity between the center values of the different, subsequent RF-pulses, not within the pulses themselves. This gap was closed with the Intrapulse tests.

The tests themselves are divided in two parts each, one time with the Feedback Loop switched on and one time switched off. Thus the tests are the first choice in case the system has errors regarding Feedback loop errors of the TX-Box during normal imaging, together with error messages coming up at the same time.

For systems with the standard TX-Box the CP mode is used. This results in the splitting of the transmitted power and a reduction of the transmit voltage by about 70 % on each channel (one over square root of 2, which is always with CP, not only with this test).

As for dual channel systems (pTx, parallel transmit) two separate transmitters are used which both have the full requested transmit voltage.

The test is available in different implementations contained in the test categories **Normal** and **Interactive**:

1. Test category **Normal**
   - **Power intrapulse**

     For the test in this category a large spherical phantom in the magnet center is requested (standard Tuneup/QA phantom setup). The disadvantage of this automatic test is that with this setup it is not possible to transmit the maximum power to the Body Coil.For the test in this category a large spherical phantom in the magnet center is requested (standard Tuneup/QA phantom setup).

     The disadvantage of this automatic test is that with this setup it is not possible to transmit the maximum power to the Body Coil.
2. Test category **Interactive**
   - **Power intrapulse interactive**

     is identical to the test within the test category **Normal** but uses the maximum transmit power to the Body Coil. To enable this, the “MR Phantom 5300 ml” with the Body Loader (not delivered with the system) has to be positioned in the magnet center (like for the **Power interactive** test).
   - **Power intrapulse RX-LC interactive**

     is similar but not identical to the test within the test category **Normal**. As also defective local coils can cause a regulation error of the TX-Box, for this test a Rx coil is positioned in the magnet center, instead of the normal BC coil phantom setup. In the first test run this is of course usually the coil which is suspected to be the reason of the errors occurred during patient imaging. The coil is connected and placed in the bore together with a phantom, following the QA-setup described in the user documentation

The second interactive test is only visible and available in the TestTools expert mode and cannot be started through the main user interface (**TestTools / General**). Thus they are only available for Siemens Service.

This is different compared to previous software versions. There the Expert mode was just an other way how the tests are offered to the user in the user interface, no additional test was available.

Signal Path

The RF-power of the TX-Box is transmitted to the BC. The signals are measured through the DICOs in the TX-Box. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. There are two DICOs placed after the power splitter (single channel systems, not with pTX systems). Thus the power for the 0 and the 90 degree transmit path (referring to BC 0 and BC 1) can be measured separately.

Test

The transmit voltages are increased in 8 steps up to the maximum allowed power, which is determined through inline adjustments.

The voltages are calculated in the following way:

- The available transmit voltage range is divided in 8 sections with different sizes to check the attentuators and pre-amplifiers in the TX-Box.
- The used transmit voltages are set to the center of these sections. Basically in this way the transmit voltages are increased by approximately the factor of 2 from step to step.

This results in calculated transmit voltages that are not even numbers like in other TX tests.

In this way the attenuators of the TX-Box are tested, too.

The largest transmit puls(es) might be smaller than expected from this calculation. In that case the measurement system had to clipp the transmit voltage to protect the components of the TX chain. This is printed out in the test report.

All data for one voltage is acquired simultaneously with one measurement using the rf_loop sequence and rectangular pulses. For the complete test 16 measurements are necessary (Feedback Loop on and off).

The following measurement values will be gathered for all measurements:

- Forwarded and reflected power (complex) at DICO
- Forwarded and reflected amplitude at TALES
- Amplitude at the CaCo ("CApacitive COupler", also DriftDetector or PeakDetector called, or "M3" for "third measurement point"), beginning with approximately 80 V transmit voltage

Evaluation

The following values of pulses (transmit voltages) of

- BC 0 with feedback loop switched on
- BC 1 with feedback loop switched on
- BC 0 with feedback loop switched off
- BC 1 with feedback loop switched off

are calculated, checked against limits, and visualized in the report:

- DICO stability: mean, maximum deviation, both forward and reflected
- Reflection factor
- for pTX-systems S-parameters (only for HSC MR relevant): scattering magnitude and phase

The following outputs are not relevant for troubleshooting in the field and intended for internal use only:

- Shift TX register resolution by x bits.
- Got parameter from image header: rows = x
- Got parameter from image header: columns = x
- Got parameter from image header: DwellTime = x
- Got parameter from image header: frequency = x
- Got DICO conversion factor = (x,y)
- Got DICO conversion factor = (x,y)

Conclusions

Please refer to the test report.

2.1.2.2.4 RX

2.1.2.2.4.1 RX Stab/Lin

Help ID: 4D552AAC-4F5A-493E-AC2B-32B942E3AABD
Service Level: 7

General

The **RX Stab/Lin** test checks the linearity and stability of all ADCs of the receiver boards by executing MR measurements via the sequence rf_loop (parameterizable sequence). It tries to minimize the number of MR measurements necessary to test the ADCs. Only available/configured channels are used.

As a result, it returns a list of sub-results each consisting of an MR measurement result and a list of ADC numbers checked by that measurement.

The program supports several test categories (**Fast**, **Normal**, and **Stability**) which do the test in more detail but also require more time:

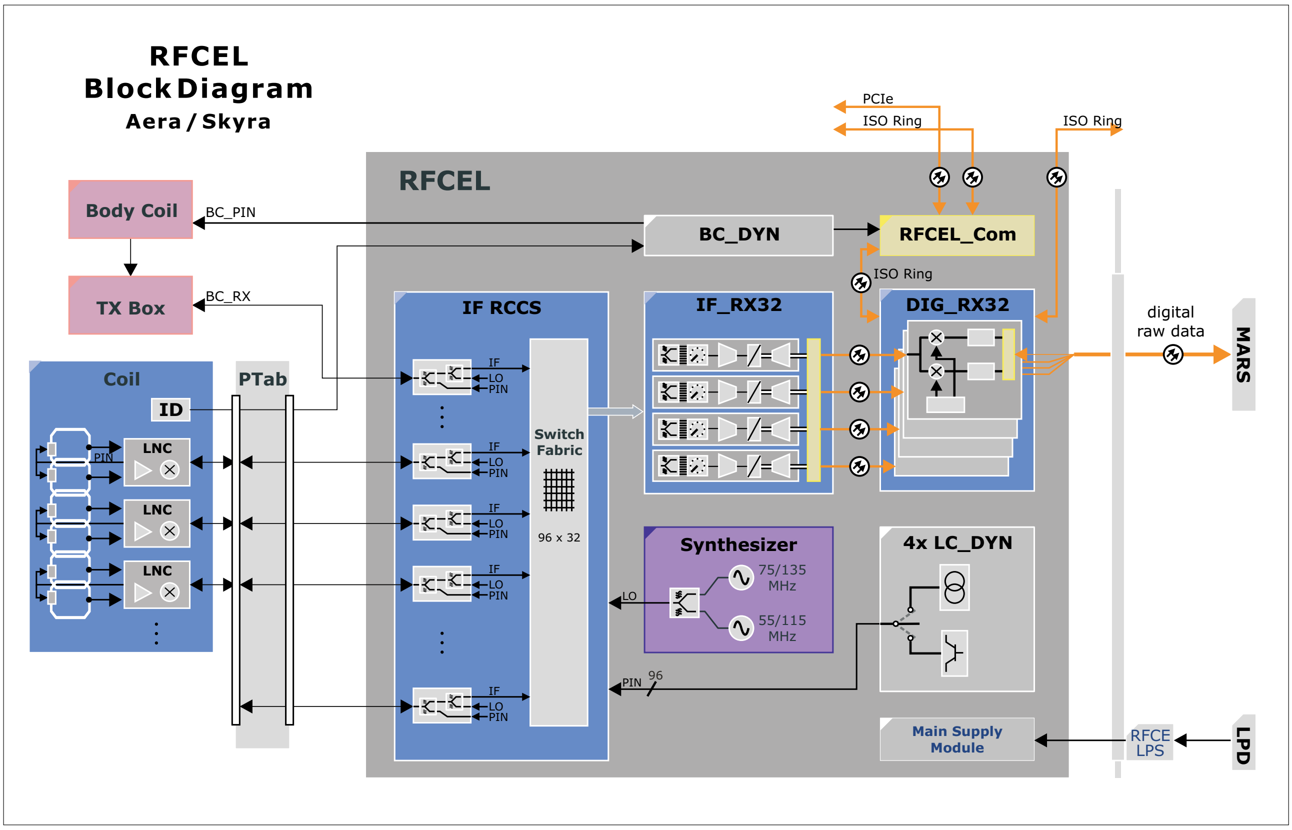
- Receiver (**Fast**)
- RX Stab/Lin (**Normal**)
- Receiver Longtime Stab (**Stability**)

In this chapter the **Normal** mode is described.

> Although the graphics shows the MAGNETOM Aera/Skyra RF system it is still valid for MAGNETOM Vida.

An overview of the receive path can be seen in the next picture:

Fig. 34: RFCEL Block Diagram



Signal Path

The test concept is based on the possibility of routing a test signal from the modulator of the TX-Box via an arbitrary node of the IF_RCCS to an arbitrary receiver. The signal is measured at the corresponding ADC.

The detailed signal path:

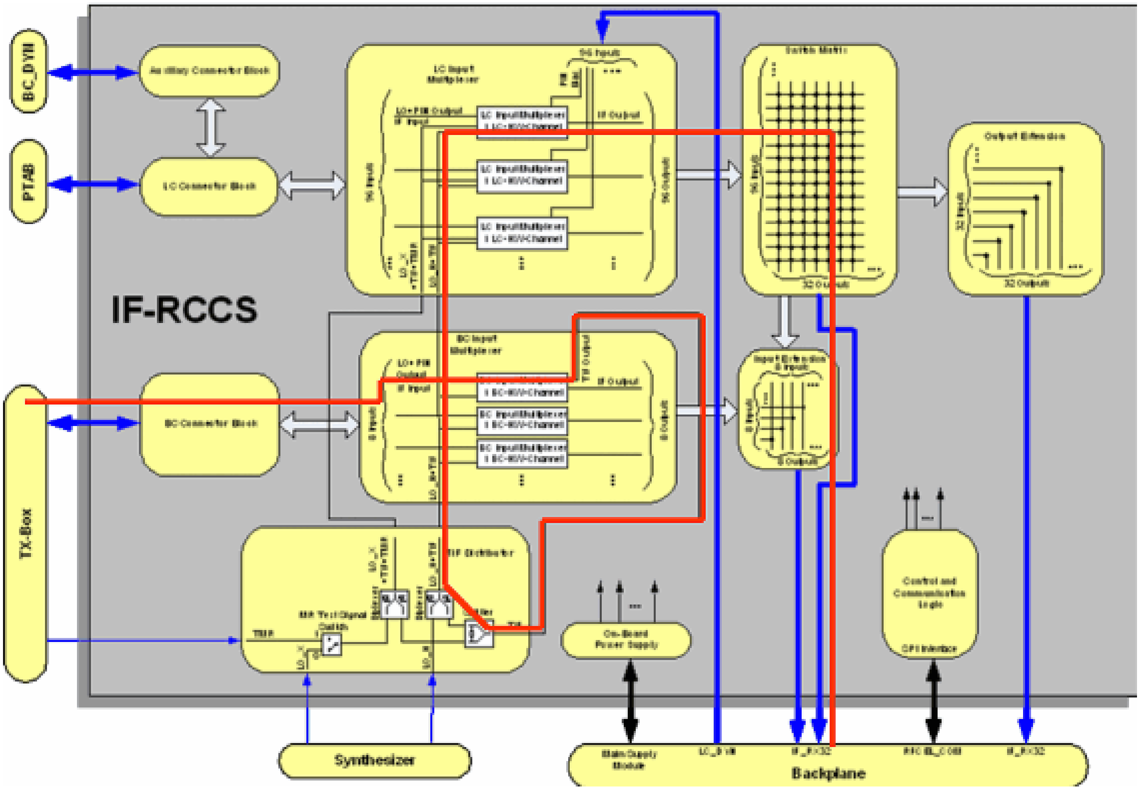
- **TX-Box**

  The TTX signal of the modulator is routed to the two inputs of the MR signals from BC (without using the amplifier module). The TTX signal is mixed with the two LO frequencies and thus down converted to the IFs and afterwards combined on one signal line. In the following, this signal is called TIF (Test signal Intermediate Frequency).

- **IF_RCCS**

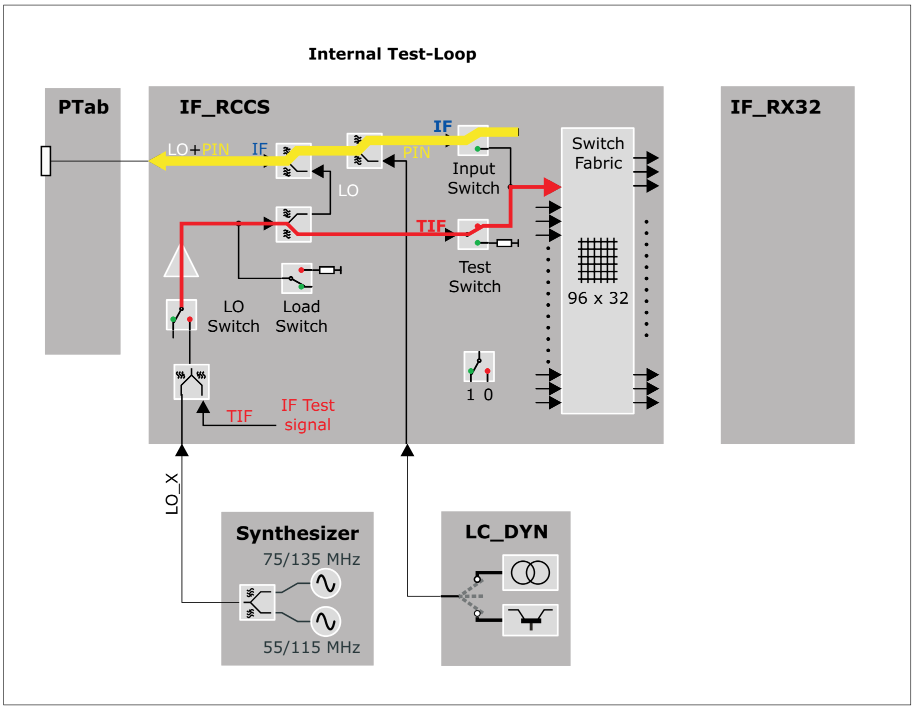
  From the BC 1 input the signal is routed to all input multiplexers. From each of these multiplexers, the signal can be routed either to a local coil socket of the patient table or to the input of the switch matrix. The outputs of the IF_RCCS are connected to the receivers where the signal is digitized. Before this, the signals are amplified by SGAs (Switchable Gain Amplifiers) using high gain.

Fig. 35: Signal Path RCCS Matrix Test



The detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 36: Internal Test-Loop



Test

For the measurement the program sets the following configurable values:

- raw image size (number of lines, columns)
- transmitter amplitude
- repetition time.

Additionally among others the following settings are done:

- all adjustments are disabled
- the RFSWD mode is set to "inactive"
- the frequency is set to "1H"
- the transmitter loop is set to TXLOOP_TTX
- the receiver loop is set to RXLOOP_MRSIG

At the end these settings are reset to the settings that were active before the test was started.

Evaluation

For each ADC the stability (raw data of first sinc-pulse) and linearity (raw data of second sinc-pulse) values are calculated from the corresponding raw data and stored in the evaluation result.

The following magnitude and phase data entities are calculated:

- **Stability**

  min, max, mean values and the peak to peak deviation values
- **Linearity**

  min, max, mean values and the max. deviation values
- **Stability and linearity**

  deviation values of each raw data line (for graphical outputs)

The stability and linearity evaluation results for each ADC are checked if they are in specification.

Afterwards the results are displayed in detail. In the summary table only the failed ADCs are listed to avoid a large table.

The following entities are displayed for each ADC:

- table with stability results containing min, max, mean values as well as peak to peak deviation values for magnitude and phase data
- stability graph with diagrams for magnitude and phase deviation values
- table with linearity results containing min, max values as well as max. deviation values for magnitude and phase data

- linearity graph with diagrams for magnitude and phase deviation values

The diagrams for linearity and stability are not displayed if test category **Fast** is selected, because of too few number of lines.

Conclusions

Please refer to the test report.

2.1.2.2.4.2 RCCS Matrix

Help ID: 1556B389-E0C5-466E-9425-8C6A58348437
Service Level: 7

General

The **RCCS Matrix** test implements a test of the IF_RCCS and checks all nodes of the IF_RCCS switch matrix. The IF_RCCS is part of the RF receive system of the system. It enables the switching of RF receive coil elements to different ADCs. Thus the number of coil elements that are connected to the patient table may exceed the number of ADC channels of the system. During the scan only those elements that are in the field of view are connected to one of the available ADCs.

To reduce the measurement time as many nodes as receiver channels are available are tested simultaneously.

The program has different modes with different run times. Depending on this mode a pre-defined number of RF pulses are routed through each node. In this way it is possible to

- obtain a quick overview of the functioning of all nodes with a single RF pulse per node (**Normal** mode).

  Here only the amplitude is evaluated.
- check the stability or linearity of the signal using a longer measurement time (**Stability** mode).

  Here the stability of amplitude and phase as well as the linearity of the signal are checked.

In this chapter the **Normal** mode is described.

Signal Path

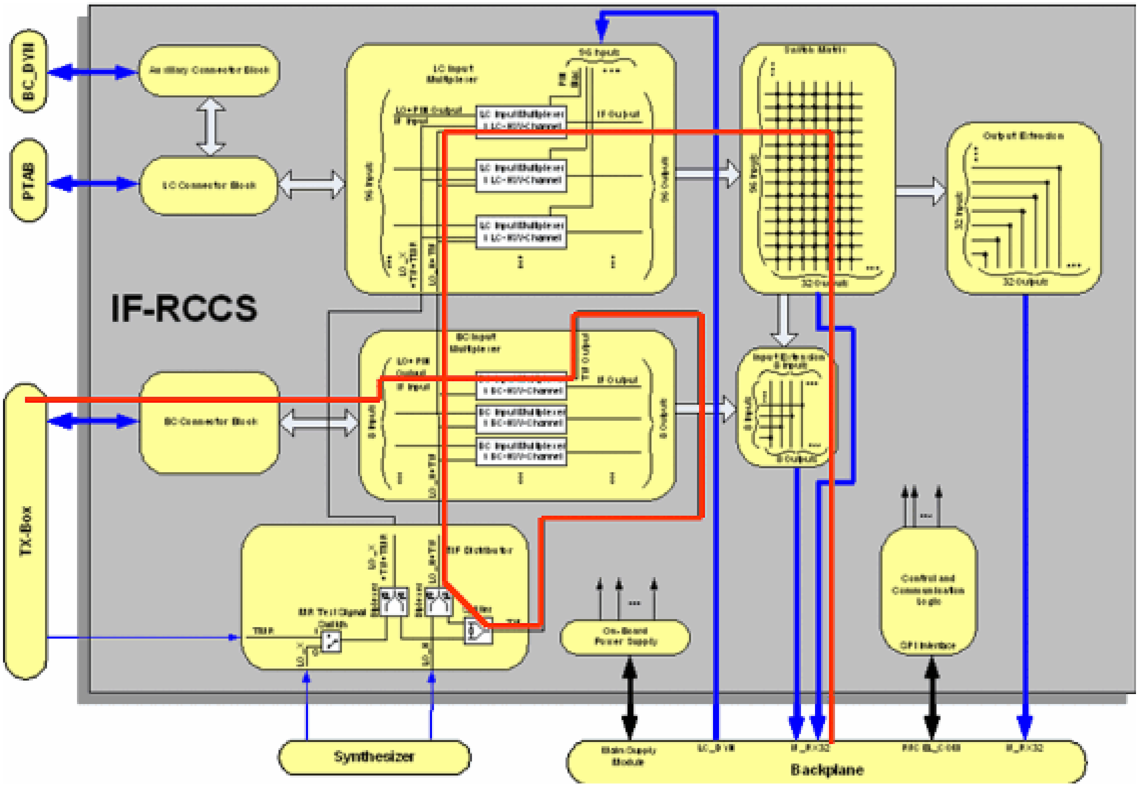
In the following figure the used RF signal loop for the measurement is shown in red.

A test signal TIF (Test signal of Intermediate Frequency) is routed from the TX-Box via the first BC input multiplexer to an arbitrary BC/LC input multiplexer. From there it is fed to the corresponding MUX channel of the IF_RCCS (multiplexer, input channel of the IF_RCCS).

By switching one of the nodes that are connected to that MUX channel, the signal is routed over the selected RX channel to an ADC. Because the system hardware uses two intermediate frequencies, every RX channel is connected to two ADCs. The signal is only evaluated for one ADC, as this is sufficient for checking the function of the node.

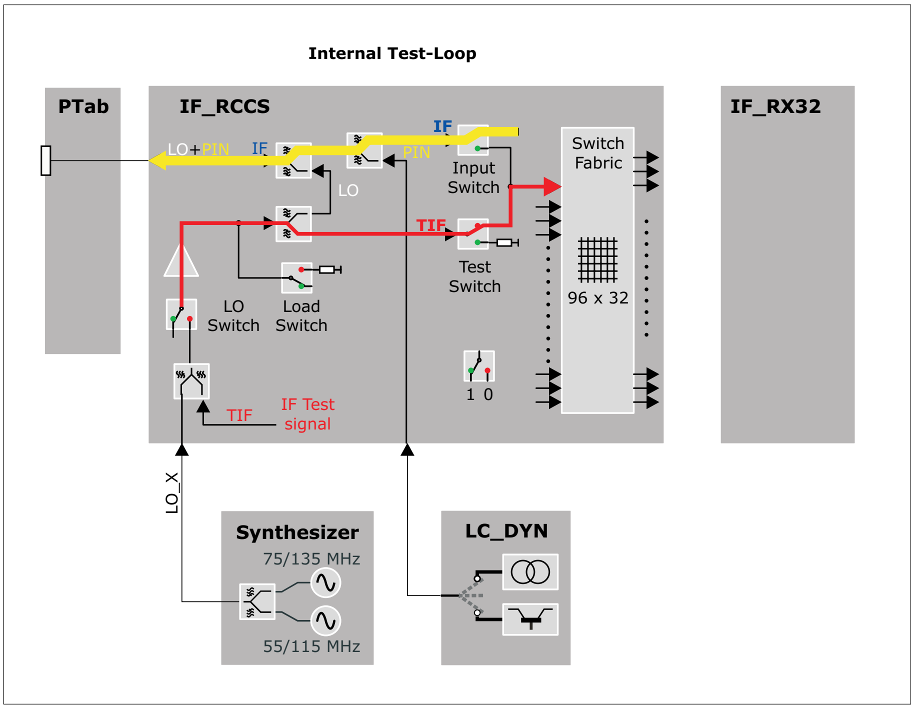
This is done by measuring the attenuation of the TIF signal routed over a certain node.

Fig. 37: Signal Path RCCS Matrix Test



The detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 38: Internal Test-Loop

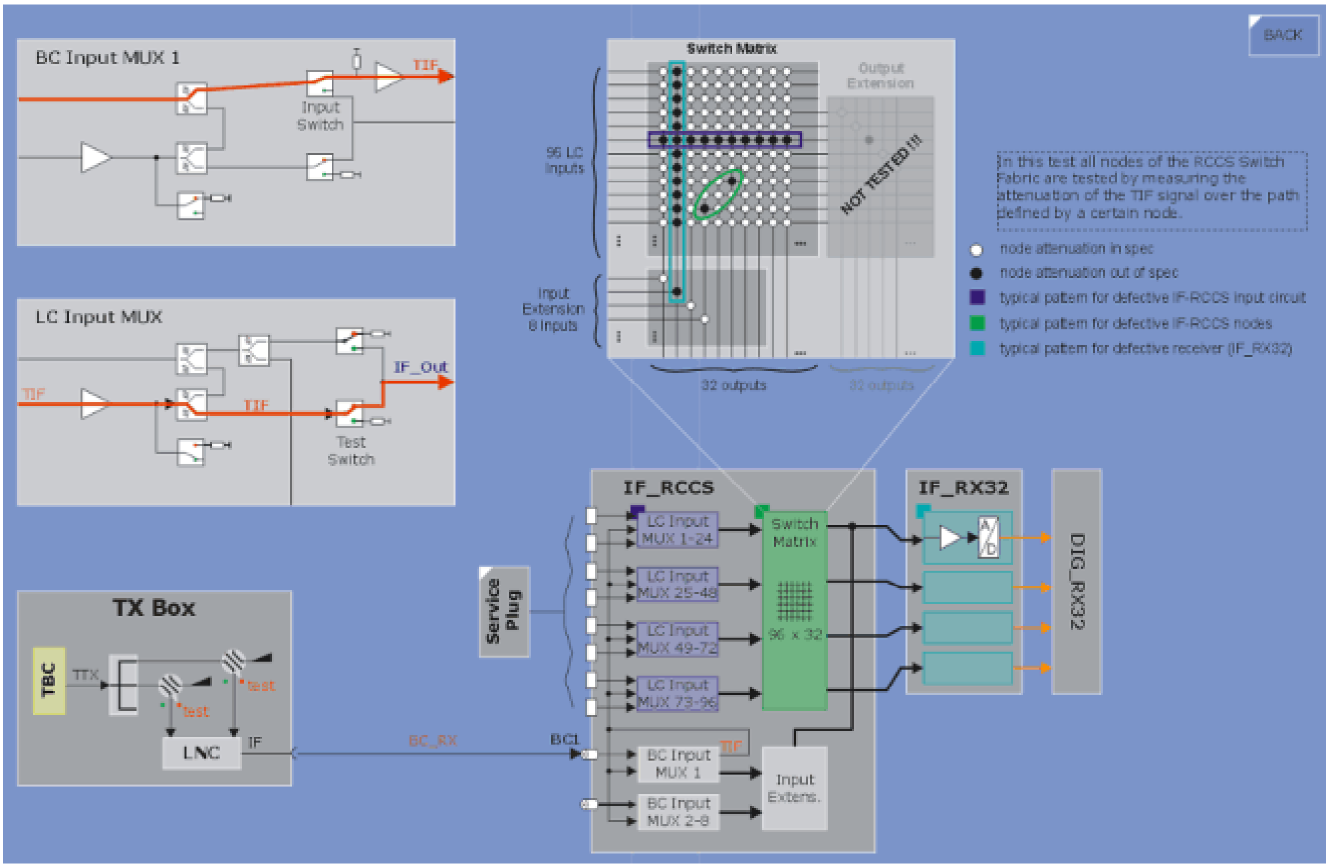


Test

Before a measurement is started, as many nodes as possible are combined to one IF_RCCS setting and then programmed. Up to 64 settings are possible here. The sequence is started with the same number of repetitions as numbers of IF_RCCS settings. For each repetition a different setting is used. Within one setting several nodes are checked as, contrary to previous system types, the test signal can be switched to more than one MUX channel simultaneously. The only restriction is that all nodes within one setting must belong to different MUX and RX channels.

The measurements are performed by the use of the rf_loop sequence. The required TX and RX loop codes as well as the used ADCs are passed to the sequence. The sequence generates two echoes for each line. The first one with constant amplitude is used for stability evaluations, the second one with increasing amplitude for linearity evaluations.

Fig. 39: Overview IF_RCCS Matrix Tests



Evaluation

The evaluation is done by retrieving the measurement data of the series (measurement) from the image data base. Then the corresponding raw data sub set for each node is extracted.

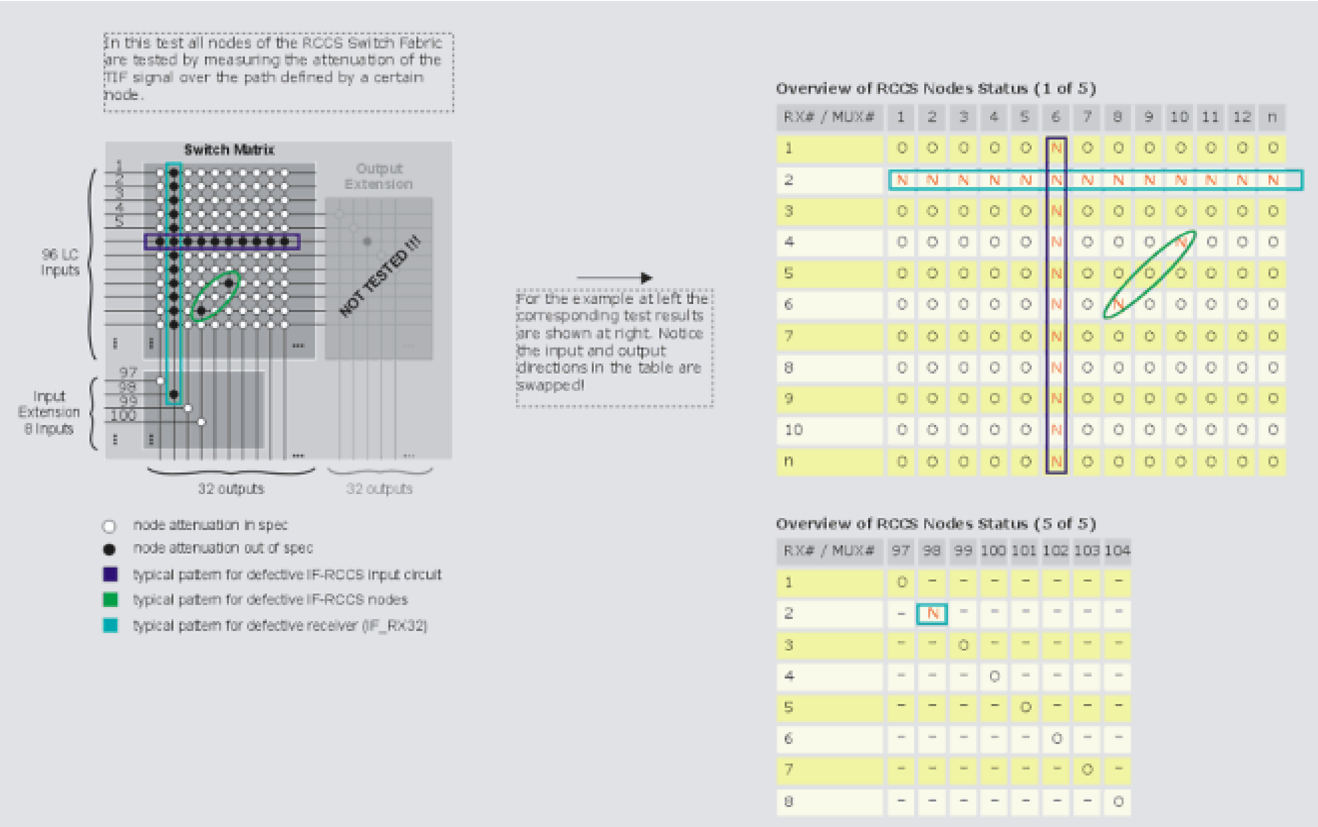
Here the ADC value in the middle of the pulse is calculated, evaluated, and printed out.

The results are displayed as an overview table with a grid.

In every box the status (OK/NOT OK) of one node is shown. For nodes that are NOT OK, a more detailed description is printed at the end of the overview table.

> The graphics show the principle, the actual layout and printouts in the report look different.

Fig. 40: Overview IF_RCCS Matrix Test Evaluation



Conclusions

Please refer to the test report.

2.1.2.2.5 Gradient

2.1.2.2.5.1 Current Actual Value Stab/Lin

Help ID: 074066BE-0508-4EBB-ADD4-FAF9798A35E0
Service Level: 7

General

The **Current Actual Value Stab/Lin** test checks the actual values used to regulate the gradient amplifier current. For each axis X, Y, and Z a linearity and stability evaluation is performed.

The number of measurement lines is configurable, thus the test can be used in the test categories **Fast** and **Normal**. The run time of the latter is longer but the test shows more precise results for the stability evaluation as it uses a larger number of lines for the raw data.

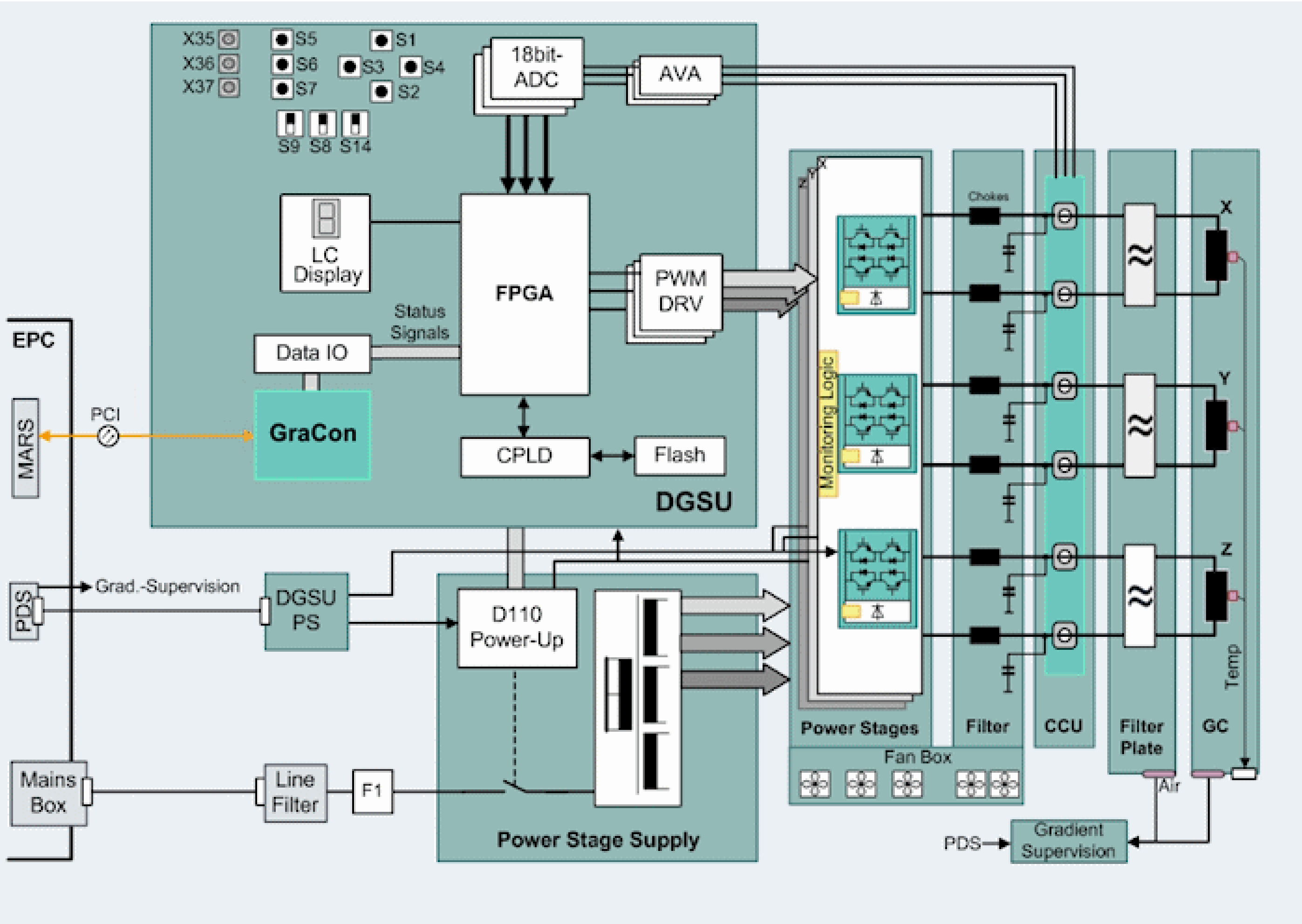
The DAC tests from previous system types are obsolete now as the used digital gradient small signal unit (DGSU, also called Digital Control Component - DCC) has no D/A converter for the nominal values. Apart from the three analogue signals coming from the current sensors, only digital signals are processed in the DGSU.

> This test does not supply a 100 % clear analysis of the stability and linearity of the gradient pulses. This can be done only with a MR-signal (e.g. QA/Stability Check, Calcar versus Calcar_Lemshift, and so on).
>
> Basically the CAV tests check the "quality" of the regulation.

Signal Path

The actual gradient coil current values are picked up by sensors in the Current Converter Unit (CCU) and fed to the A/D converter in the DGSU separately for every final stage. Together with the nominal current values this information is used as input for the gradient regulator.

Fig. 41: Overview Gradient Amplifier MAGNETOM Vida



Test

The test switches certain signal paths and performs a measurement afterwards. The acquired data is then evaluated, checked against specification values and displayed in graphical form.

Each axis X, Y, and Z is measured separately, that means three measurements are performed; one for each axis.

Evaluation

For each axis the stability and linearity values are calculated from the corresponding raw data and stored in the evaluation result if the corresponding measurement was successful.

The following data is checked:

- **Stability**

  minimum, maximum, and mean values of amplitude and integral data, together with the peak-to-peak deviations from the reference value in percent.
- **Linearity**

  minimum, maximum, and mean values of amplitude and integral data.

For both stability and linearity, a check is performed to check whether the signal is zero respectively above a defined minimum.

Conclusions

Please refer to the test report.

2.1.2.2.5.2 Grad Risetime

Help ID: 95CA3436-A38B-4619-9F1E-8EB6A94693BD
Service Level: 7

General

The **Gradient Rise Time** procedure determines the minimal rise time or the so-called maximum slew rate of the gradient system. Additionally, failures of the current sensors such as a baseline artifact can be detected.

Signal Path

The test itself uses an MR experiment. Thus the complete TX and RX chain is used.

Test

For the test the (new) large spherical phantom has to be in the magnet center.

At the beginning of the test, whether the correct phantom is used is checked.

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval five bipolar triangular gradient pulses are switched to generate 5 echoes. The echoes should appear in a nominal position which is defined by the proper pre-dephasing and rephasing of the grad_rise-time sequence.

During the measurement, the pulse amplitudes of all three gradient axes run the range from -Gmax to +Gmax. Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If the echo position deviates from the nominal position, the required slew rate is not reached. The limits can be determined and calculated from the raw data.

During the test, the "regulator warning" will be sent from the GPA. This warning indicates that the GPA is driven at maximum output voltage for a certain time.

The test is identical to the corresponding QA function.

Evaluation

The test is designed in such a way that it also works for an input mains supply with a low voltage that is still within the specifications for the system. Thus the normal results range from 115 to 125 %. With too low a voltage, the result will be less than 100 %.

The check is done for each axis X, Y, and Z and a linearity and stability evaluation will be performed.

The raw data image is analyzed and the gradient rise time calculated.

The data is checked for current sensor failures such as a baseline artifact.

The deviation of the echo position from the nominal position is determined and calculated from the raw data. A mismatch here also points to a defect in the GPA.

The following data is printed out:

- graphics of echo maximum versus gradient slew rate, echo 1 - 5
- statistics of measured values against specifications
- X/Y/Z Gradient Rise Time Results
- X/Y/Z Gradient Rise Time Multiple Echo Output (raw data image)

Conclusions

Please refer to the test report.

2.1.2.2.5.3 Spike

Help ID: A0F8E898-6CF0-4994-8B94-7CE9C3DBCCD4
Service Level: 7

General

The **Spike Check** is a quick test to check the MR imaging system for the occurrence of "spikes" and RF interferences. The function detects and identifies spikes in the raw data. The image data is evaluated for RF interference patterns.

The test is identical to the corresponding QA function.

Signal Path

The complete RX-path is used.

Test

No phantom is required for the test.

A sequence is started that generates mechanical and electrostatic stress by switching the gradient system rapidly in a worst case scenario. At the time period with maximum gradient strength the raw data is acquired (noise measurement).

It also runs through tables covering a wide range of different gradient amplitudes.

This measurement does not apply RF pulses. As a result, it does not generate an MR signal.

Evaluation

The acquired raw data and image data is scanned for spikes.

If spikes are present, the following values are evaluated and displayed:

- total number of spike positions
- magnitude of the highest spike

The same algorithm for searching for spikes in the raw data is applied to the image data to search for RF interferences.

External RF interference can be displayed as a spike in the image data.

Additionally two images are displayed:

- the raw data measured
- the resulting RF interference image.

Conclusions

Please refer to the test report.

2.1.2.3 Stability

Help ID: E13C50B9-A873-4849-B44D-111A4D75A281
Service Level: 7

General

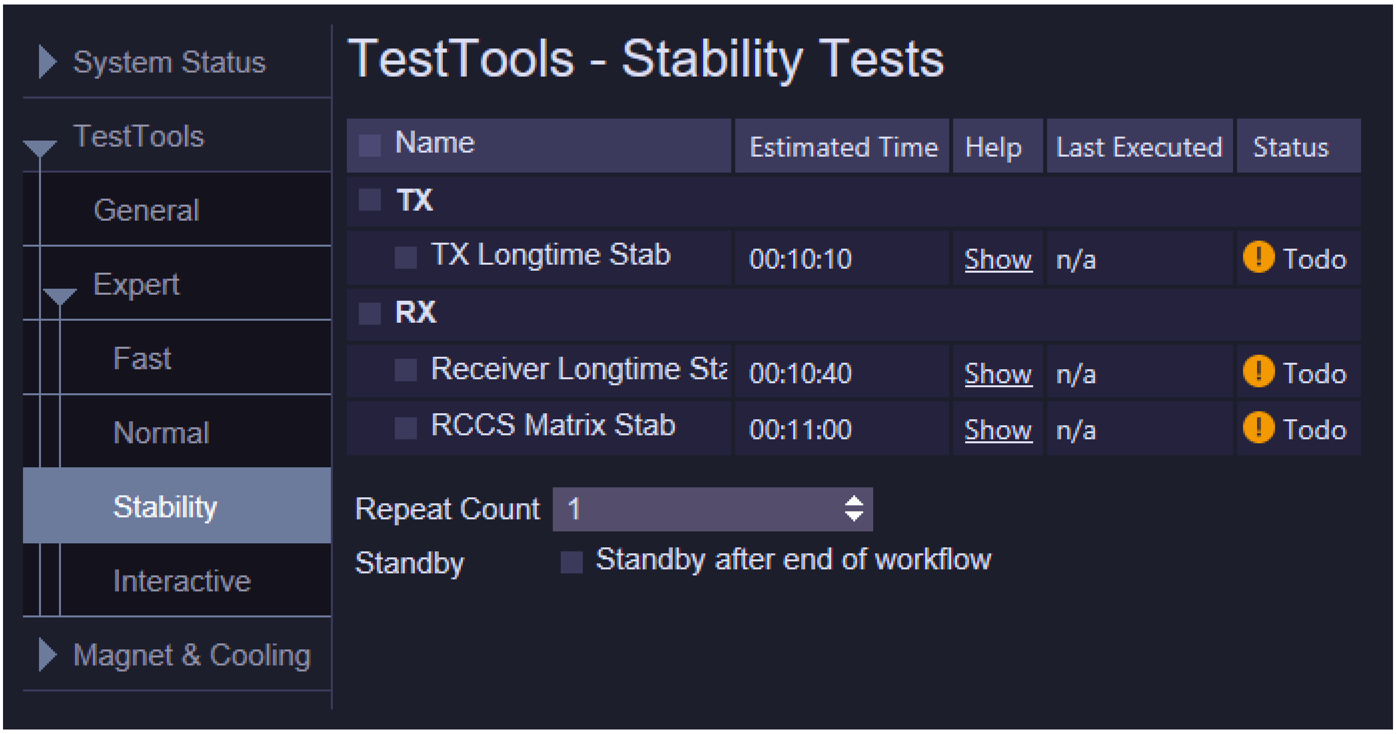
In this test category all tests are collected necessary in order to obtain precise information about the operation of the system when measuring for a long time (e.g. fMRI, BOLD-imaging).

These tests allow you to thoroughly test the linearity and stability. They should only be necessary to find long term stability problems that are important mainly for the above mentioned fMRI measurements. Interactive tests are avoided as far as possible.

It is in the nature of the tests that the measurement time per test is quite long and in the range of 10 to 20 minutes.

No linearity evaluations are executed in the test category **Stability**.

Fig. 42: TestTools Expert Stability Example



Stability (fMRI) Tests

Measurements for Functional MR Imaging (fMRI) take a very long time, i.e. between five and twenty minutes. In order to acquire high quality fMRI images the TX-Box in particular needs to fulfill certain stability requirements.

A special evaluation is applied to the measurement data that is adapted to the special needs of fMRI imaging.

Instead of the normal signal evaluations of the raw data lines, the following are implemented:

- deviation of the integral of the pulse magnitude
- deviation of the average of the pulse phase

> No linearity evaluations are executed in the stability tests.

Pulse Magnitude Integral

The sinc-pulse ($\frac{\sin(x)}{x}$ function) of each measured raw data line is evaluated.

First is the integral of the absolute magnitude of the measured sinc-pulse computed for each line:

$$i_{line} = \sum_{column} \left| rawdata_{column,line} \right|$$

The average of the integrals is computed as a reference value. The integrals are divided by the reference value.

$$ref_i = \frac{1}{lines} \sum_{line} i_{line}$$

$$r_{line} = i_{line} / ref_i$$

The resulting relative integral values are plotted in a diagram. The maximum peak-to-peak value of the relative integral values is shown in a table and compared against a specification value. The test is not ok if the maximum value is greater than the specification value.

Pulse Phase Average

The sinc-pulse of each measured raw data line is evaluated. First is the average of the phase of the measured sinc-pulse computed for each line. The average is computed only for the mid of the pulse, approximately for the part where the pulse magnitude is greater than 25 % of the maximum magnitude value; this excludes the parts of the sinc-pulse where the phase is influenced too much by noise.

$$a_{line} = \frac{1}{endcolumn - startcolumn + 1} \sum_{column=startcolumn}^{endcolumn} Phase(rawdata_{column,line})$$

The average of the phase averages is computed as a reference value. The reference value is subtracted from the phase averages.

$$ref_a = \frac{1}{lines} \sum_{line} a_{line}$$

$$ra_{line} = a_{line} - ref_a$$

The resulting normalized phase averages are plotted in a diagram. The maximum peak-to-peak value of the normalized phase average values is shown in a table and compared against a specification value. The test is not ok if the maximum value is greater than the specification value.

2.1.2.3.3
TX

2.1.2.3.3.1
TX Longtime Stab

Help ID: 156E9408-BFB7-4A09-81E0-19CD00B30C29
Service Level: 7

General

The **TX Longtime Stab** test checks the linearity and short time stability of the transmitted power.

As the used sequence can be parameterized, it is possible to have this test in different measurement time lengths. This test is thus used in **Fast**, **Normal**, **Stability**, and **Multi Nuclei** if applicable (the Multi Nucleus capability is optional to the MR system).

In this chapter the test category **Stability** is described.

Signal Path

The power is transmitted to the Body Coil with the loader placed inside. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. In comparison to Avanto there are two DICOs that are placed after the power splitter. Thus the power for the 0 and 90 degree transmit path can be measured separately.

Test

For the test the (new) large spherical phantom has to be in the magnet center.

All data is acquired simultaneously within one measurement using the rf_loop sequence.

The number of RF pulses used for the stability and linearity evaluations depends on the number of measured lines.

Evaluation

The following parameters of the power output are checked at the DICOs:

- maximum power
- deviation of requested and delivered power
- deviation of the power of the two transmit paths (0 and 90 degree)
- reflected power for the two parts of the BC (as a function of the transmitted power).
- linearity and phase stability of forward power
- stability of amplitude and phase of forward power

Additionally to the evaluations of the shorter program modes, an fMRI-specific evaluation of the data is done in **Stability**:

- **Magnitude deviation**

  maximum deviation for longtime magnitude
- **Stability phase deviation**

  maximum deviation for longtime phase

For further information please refer to the general description of the **Stability** test category.

A comparison of TALES and DICO values is not required because the TX-Box performs an online comparison of these values for security reasons.

Conclusions

Please refer to the test report.

2.1.2.3.4 RX

2.1.2.3.4.1 Receiver Longtime Stab

```
Help ID: A7A69555-E425-4BE8-9EB6-470E0DC684E0
Service Level: 7
```

General

The **Receiver Longtime Stab** test checks the linearity and stability of all ADCs of the receiver boards by executing MR measurements via the sequence rf_loop (parameterizable sequence). It tries to minimize the number of MR measurements necessary to test the ADCs, only available/configured channels are used.

As a result it returns a list of sub-results each consisting of an MR measurement result and a list of ADC numbers checked by that measurement.

The program supports several test categories (**Fast**, **Normal**, and **Stability**) which do the test in more detail but also require more time:

- Receiver (**Fast**)

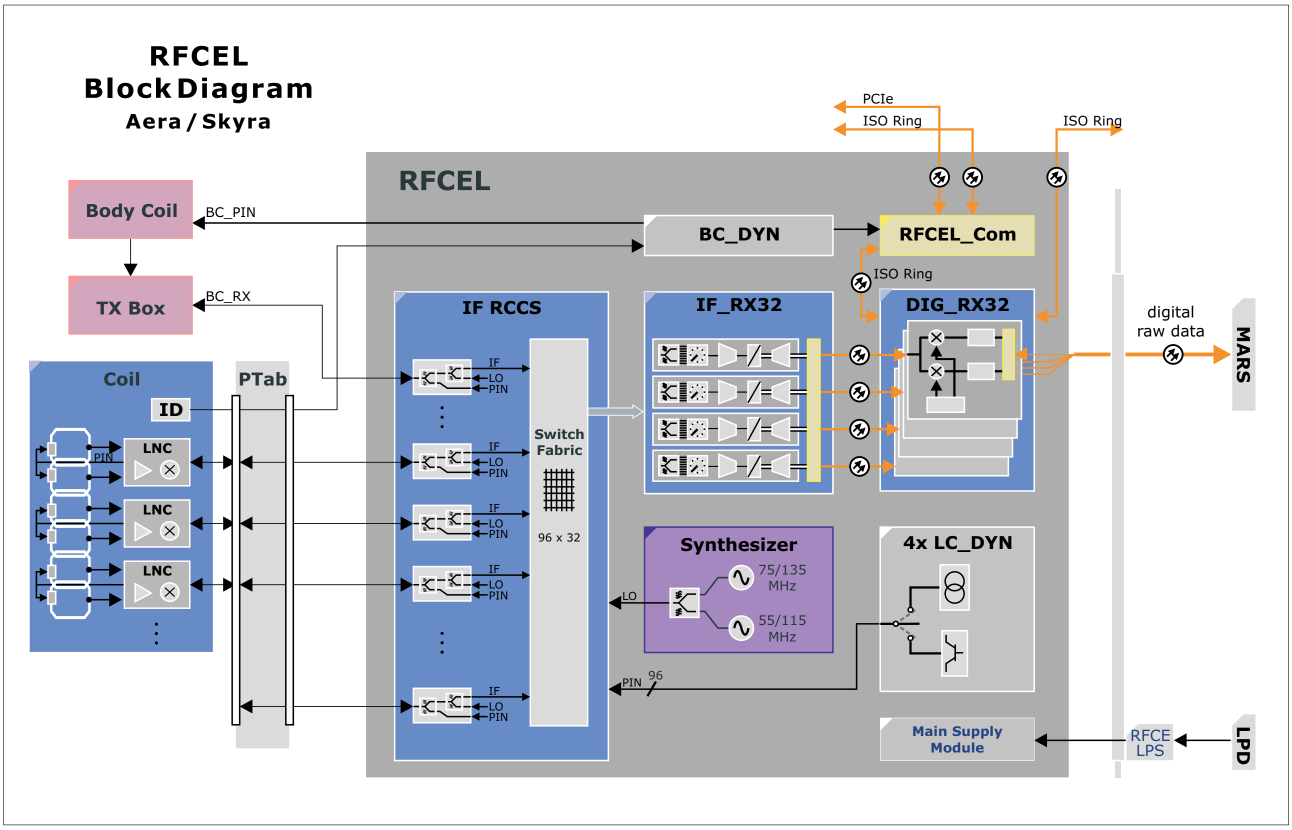
- RX Stab/Lin (**Normal**)
- Receiver Longtime Stab (**Stability**)

In this chapter the **Stability** mode is described.

Although the graphics shows the MAGNETOM Aera/Skyra RF system it is still valid for MAGNETOM Vida.

An overview of the receive path can be seen in the next picture:

Fig. 43: RFCEL Block Diagram



Signal Path

The test concept is based on the possibility of routing a test signal from the modulator of the TX-Box via an arbitrary node of the IF_RCCS to an arbitrary receiver. The signal is measured at the corresponding ADC.

The detailed signal path:

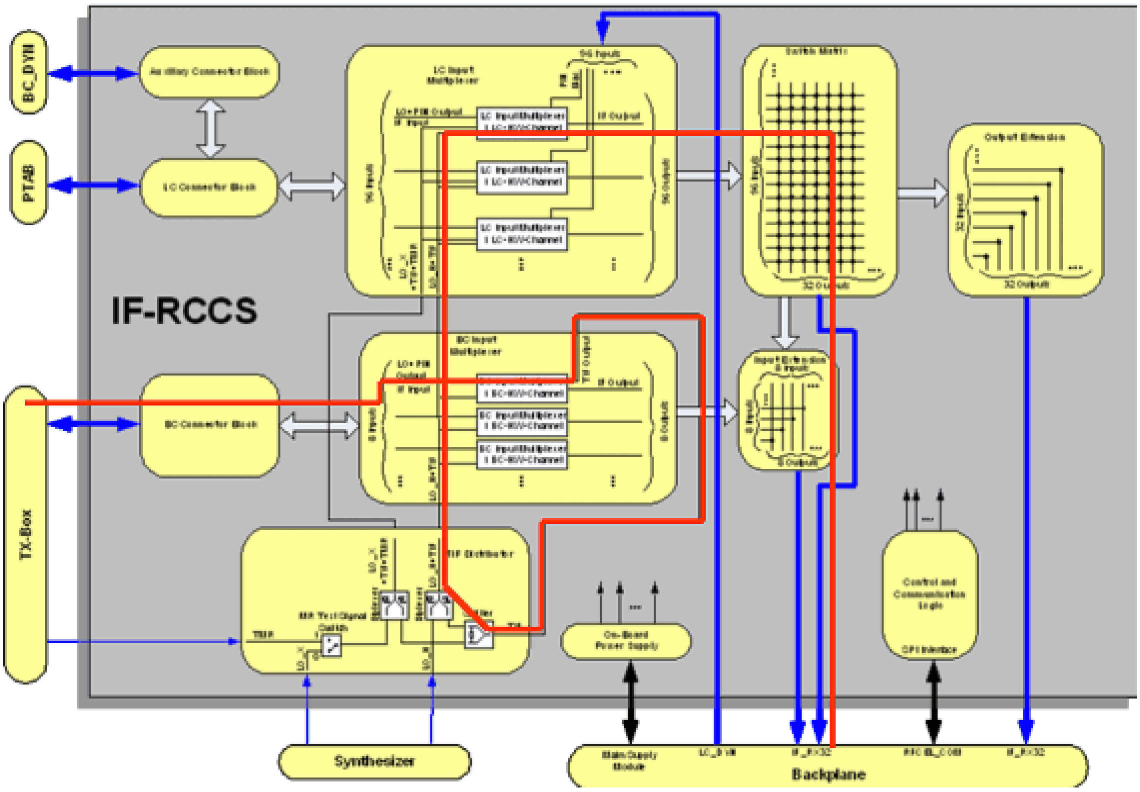
- **TX-Box**

  The TTX-signal of the modulator is routed to the two inputs of the MR signals from BC (without using the amplifier module). The TTX-signal is mixed with the two LO frequencies and thus down converted to the IFs and afterwards combined on one signal line. In the following, this signal is called TIF (Test signal Intermediate Frequency).

- **IF_RCCS**

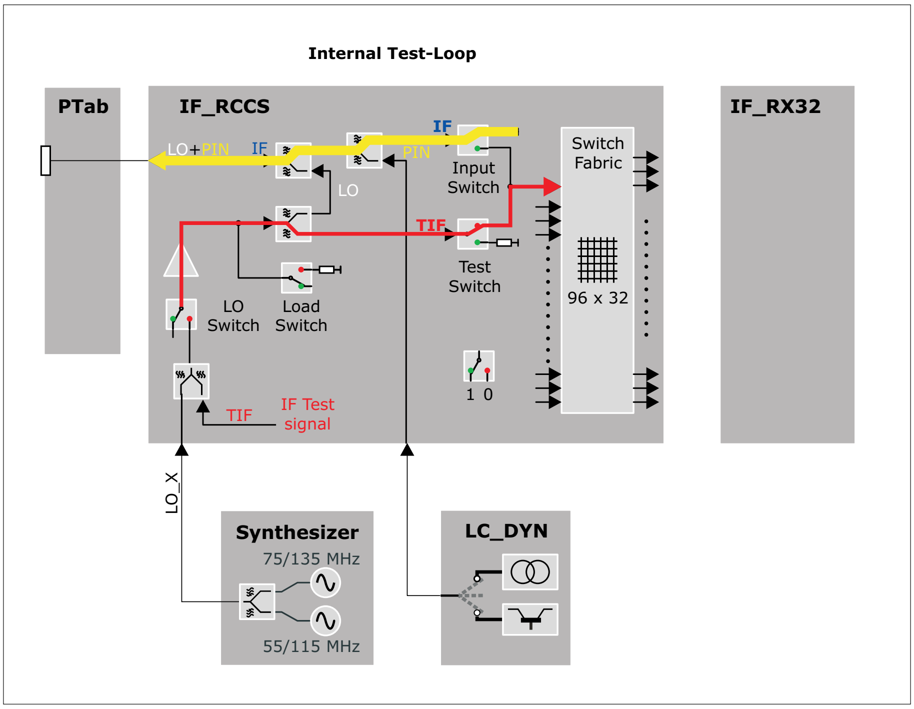
  From the BC 1 input the signal is routed to all input multiplexers. From each of these multiplexers, the signal can be routed either to a local coil socket of the patient table or to the input of the switch matrix. The outputs of the IF_RCCS are connected to the receivers where the signal is digitized. Before this, the signals are amplified by SGAs (Switchable Gain Amplifiers) using high gain.

Fig. 44: Signal Path RCCS Matrix Test



The detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 45: Internal Test-Loop



Test

For the measurement the program sets the following configurable values:

- raw image size (number of lines, columns)
- transmitter amplitude
- repetition time.

Additionally among others the following settings are done:

- all adjustments are disabled
- the RFSWD mode is set to "inactive"
- the frequency is set to "1H"
- the transmitter loop is set to TXLOOP_TTX
- the receiver loop is set to RXLOOP_MRSIG

At the end these settings are reset to the settings that were active before the test was started.

Evaluation

For each ADC the stability (raw data of first sinc-pulse) and linearity (raw data of second sinc-pulse) values are calculated from the corresponding raw data and stored in the evaluation result.

In this test category, an fMRI-specific evaluation of the data is performed. For further information, please refer to the description in the help of the **Stability** test category.

The following magnitude and phase data entities are calculated:

- **Stability**

  min, max, mean values and the peak to peak deviation values
- **Linearity**

  min, max, mean values and the max. deviation values
- **Stability and linearity**

  deviation values of each raw data line (for graphical outputs)

The stability and linearity evaluation results for each ADC are checked if they are in specification.

Afterwards the results are displayed in detail. In the summary table only the failed ADCs are listed to avoid a large table.

The following entities are displayed for each ADC:

- table with stability results containing min, max, mean values as well as peak to peak deviation values for magnitude and phase data
- stability graph with diagrams for magnitude and phase deviation values
- table with linearity results containing min, max values as well as max. deviation values for magnitude and phase data
- linearity graph with diagrams for magnitude and phase deviation values

Conclusions

Please refer to the test report.

2.1.2.3.4.2 RCCS Matrix Stab

```
Help ID: 7F9C51AE-2EBC-4973-8121-F6C6A6987954
Service Level: 7
```

General

The **RCCS Matrix Stab** test implements a test of the IF_RCCS and checks all nodes of the IF_RCCS switch matrix.

The IF_RCCS is part of the RF receive system of the system. It enables the switching of RF receive coil elements to different ADCs. Thus the number of coil elements that are connected to the patient table may exceed the number of ADC channels of the system. During the scan only those elements that are in the field of view are connected to one of the available ADCs.

The measurement procedure of this test is analog to the IF_RCCS Matrix Test with the exception of the number of lines acquired. Instead of one, a configureable number of lines are measured and a stability and linearity evaluation of the signal is performed. Additionally as many nodes as available receiver channels are tested simultaneously.

The program has different modes with different run times. Depending on this mode a pre-defined number of RF pulses are routed through each node. In this way it is possible to

- obtain a fast overview of the functioning of all nodes with a single RF-pulse per node (**Normal** mode).

  Here only the amplitude is evaluated.
- check the stability or linearity of the signal using a longer measurement time (**Stability** mode).

  Here the stability of amplitude and phase as well as the linearity of the signal are checked.

In this chapter the **Stability** mode is described.

Signal Path

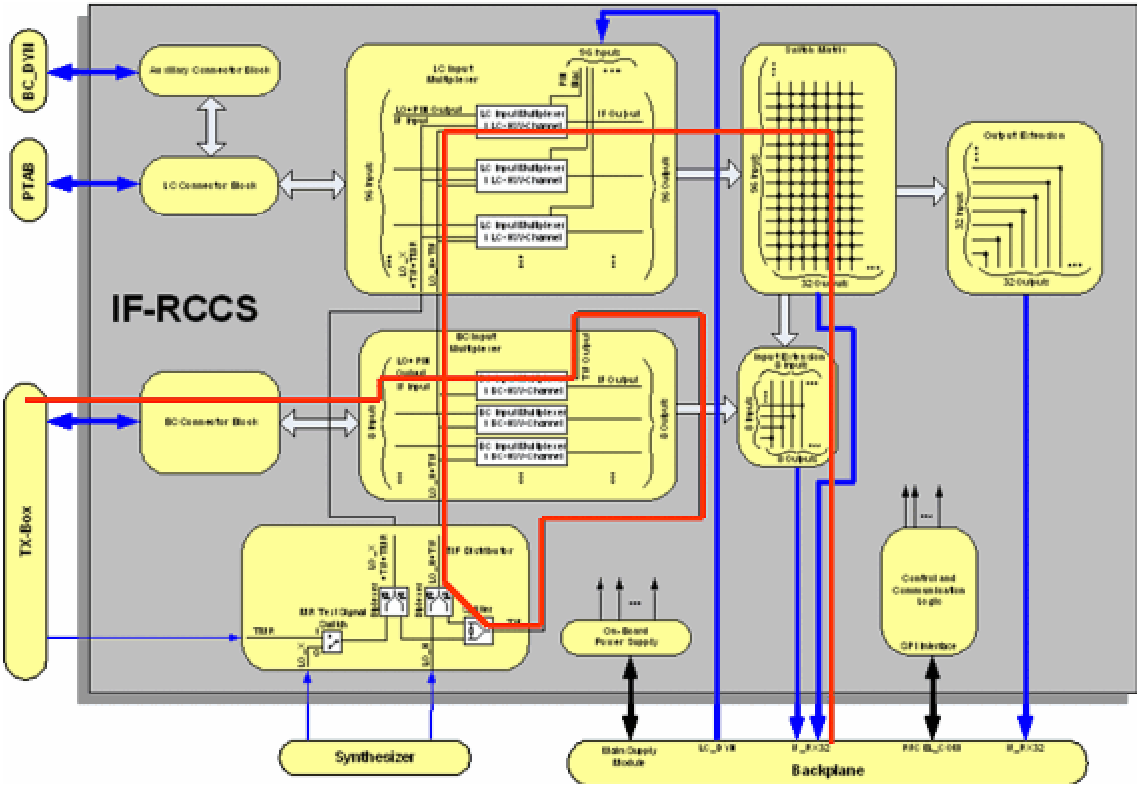
In the following figure the used RF signal loop for the measurement is shown in red.

A test signal TIF (Test signal of Intermediate Frequency) is routed from the TX-Box via the first BC input multiplexer to an arbitrary BC/LC input multiplexer. From there it is fed to the corresponding MUX channel of the IF_RCCS (multiplexer, input channel of the IF_RCCS).

By switching one of the nodes that are connected to that MUX channel, the signal is routed over the selected RX channel to an ADC. Because the system hardware uses two intermediate frequencies, every RX channel is connected to two ADCs. The signal is only evaluated for one ADC, as this is sufficient for checking the function of the node.

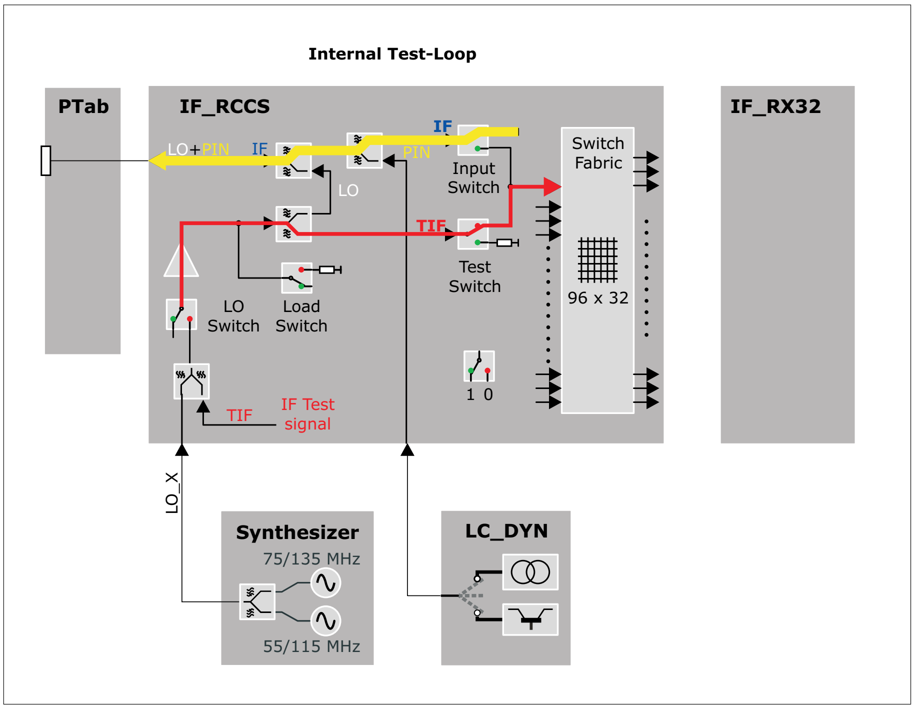
This is done by measuring the attenuation of the TIF signal routed over a certain node.

Fig. 46: Signal Path RCCS Matrix Test



The detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 47: Internal Test-Loop



Test

Before a measurement is started, as many nodes as possible are combined to one IF_RCCS setting and then programmed. Up to 64 settings are possible here. The sequence is started with the same number of repetitions as numbers of IF_RCCS settings. For each repetition a different setting is used. Within one setting several nodes are checked as, contrary to previous system types the test signal can be switched to more than one MUX channel simultaneously. The only restriction is that all nodes within one setting must belong to different MUX and RX channels.

The measurements are performed by the use of the rf_loop sequence. The required TX and RX loop codes as well as the used ADCs are passed to the sequence. The sequence generates two echoes for each line. The first one with constant amplitude is used for stability evaluations, the second one with increasing amplitude for linearity evaluations.

To reduce measurement time, as many nodes as available receiver IF_RCCS output channels are tested simultaneously.

Evaluation

The evaluation is done by retrieving the measurement data of the series (measurement) from the image database. Then the corresponding raw data sub set for each node is extracted.

The following data is evaluated and displayed in a table and partially as diagrams:

- the ADC value in the middle of the pulse
- stability and linearity evaluation of the raw data

The results are displayed as an overview table with a grid.

In every box the status (OK/NOT OK) of one node is shown. For nodes that are NOT OK, a more detailed description is printed at the end of the overview table.

Conclusions

Please refer to the test report.

2.1.2.4 Interactive

```
Help ID: BE0AEA3B-3C2E-4F12-8901-A4156C68BD85
Service Level: 7
```

General

In this test category all tests are collected that require interaction and do not run automatically.

Here the user has to perform manual actions e.g. connecting or swapping cables. These tests are not usable for remote service.

Fig. 48: TestTools Expert Interactive Example

Directly before the test is started and after it is finished, work instructions, questions, or information are usually displayed in the separate window. This window has to be confirmed before the test actually starts. Cancelling is also possible which aborts the test immediately.

> The necessary instructions are displayed directly before the test starts.
>
> This is important to know as different instructions are used for different tests.

Interactive tests have the following workflow:

- At the start of the test, a pop-up with a notification can be displayed. It is also possible that a pop-up with input fields or select boxes comes up.
- The user has the option of starting (**OK** button) or aborting the test (**Cancel** button).
- At the end of the test a further text pop-up with instructions or questions can be displayed. A confirmation is also necessary here.

For interactive tests the name of the test currently being performed shall be displayed in the header of the pop-up window.

2.1.2.4.2
ISO Ring

2.1.2.4.2.1
ISO Ring

```
Help ID: 0C79558E-9FE4-4BCD-9EB8-AF78EFAF7CF1
Service Level: 7
```

General

The **ISO Ring** test checks the ISO Ring for interruptions.

> The test only needs to be executed if the ISO Ring could not be completely initialized and thus the MR system is not ready for measurement.
>
> For these reasons this test is in the "Interactive" test category and is not part of the PCIe tests in the test category Fast.
>
> Additionally it allows interactions like swapping fiber optic cables (FOCs).

Signal Path

The complete ISO Ring is examined.

Test

All field generating and receiver units are connected to the MaRS through optical PCIe connections. Additionally they are connected to each other via the optical ISO Ring. The communication in the ISO Ring breaks down if it is interrupted at any point. To find that point of interruption the ISO ring test initializes the ring step by step by introducing one component after the other. If the ring breaks down, the last introduced component must be the critical one.

Defect FOCs are usually already detected by the PCIe tests. In that case the transmitted or received power could be out of specification.

Also incorrect soldering can be detected by the PCIe test, e.g. extended error correction that exceeds a certain limit after the initialisation.

The ISO Ring test helps to isolate problems that occur already while the initialisation phase of the Ring. For this unplugging/plugging of different cables and so on is possible while this test is running.

> In contrast to the PCIe tests, the ISO Ring test needs a restart of the measurement software on the MaRS afterwards.
>
> At the end of the last test the user can choose between an automatic restart or skip this for further troubleshooting, e.g. by swapping fiber optic cables or swap/exchange/bypass of components. If the test is repeated the same question will be asked again, otherwise the user has to manually reboot the system.
>
> But it is not necessary to reboot the MaRS as long as the ISO Ring test is repeated immediately again, without starting other tests (manually of by use of the repat functionality of TestTools).
>
> Follow the precise instructions of the test pop-up for this.

Evaluation

The test prints out information which component could not be contacted.

Conclusions

Please refer to the test report.

2.1.2.4.3 TX

2.1.2.4.3.1 Power interactive

```
Help ID: 7B4A211B-91EB-45C8-9B4F-E9F591D1C85E
Service Level: 7
```

General

Test Description

The **Power interactive** test checks for the maximum RF transmit power and the power reflection at the Body Coil.

As the used sequence can be parameterized it is possible to have this test in different measurement time duration. This test is therefore used in **Fast**, **Normal**, **Stability**, and **Multi Nuclei**. The Multi Nucleus capability is optional to the system, therefore the test is only available when the option is installed.

This is an optional test as the Body Loader not delivered with the system anymore. Just in certain cases HSC MR might request the results of this test. But it is not required to fully test the system.

Signal Path

The power is transmitted to the BC with the phantom and the loader placed inside.

The signal is measured via the DICOs in the TX-Box. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. In comparison to the MAGNETOM Avanto there are two DICOs that are placed after the power splitter. Thus the power for the 0 and 90 degree transmit path can be measured separately.

Test

For the test, the phantom “MR Phantom 5300 ml” with the Body Loader has to be in the magnet center.

The Body Loader is not delivered with the system and has to be ordered. Thus this test is for special purposes only.

Follow the instructions on the console.

All data is acquired simultaneously within one measurement using the rf_loop sequence.

For the test a loader is requested to enable the transmission of the maximal power to the BC. The maximal power value is requested from the component protection after performing an adjustment. Depending on the BC it may be possible that the maximal power allowed for the test is less than the maximum power that could be delivered by the TX-Box.

The number of RF pulses used for the stability and linearity evaluations depends on the number of measured lines. For the test category **Fast** the measurement time is quite short, just to acquire enough data to be able to evaluate if the TX-Box is in working condition.

Evaluation

The following parameters of the power output are checked at the DICOs:

- maximum power
- deviation of requested and delivered power
- deviation of the power of the two transmit paths (0 and 90 degree, referred to as BC0 and BC1)
- reflected power for the two parts of the BC (as a function of the transmitted power).

A comparison of TALES and DICO values is not required because the TX-Box performs an online comparison of these values for security reasons.

Conclusions

Please refer to the test report.

2.1.2.4.3.2 Power intrapulse interactive

Help ID: 090A3E77-B01C-4F93-B97F-AF36D15DE161
Service Level: 7

General

The **Power intrapulse interactive** tests check the intra pulse stability of the RF pulses transmitted to the Body Coil. The transmit amplitudes of the RF power pulses sent to the Body Coil are increased step by step up to the maximum allowed power. The maximum allowed power is determined through inline adjustments.

This test is implemented in two different implementations, in **Normal** and **Interactive**.

In this chapter the test category **Interactive** is described.

The CP mode is used for single channel systems. For dual channel systems (pTX, parallel transmit) each channel is measured separately.

Signal Path

For the test, the phantom “MR Phantom 5300 ml” with the Body Loader has to be in the magnet center.

The Body Loader is not delivered with the system and has to be ordered. Thus this test is for special purposes only.

Follow the instructions on the console.

All data is acquired simultaneously within one measurement using the rf_loop sequence.

For the test a loader is requested to enable the transmission of the maximal power to the BC. The maximal power value is requested from the component protection after performing an adjustment. Depending on the BC it may be possible that the maximal power allowed for the test is less than the maximum power that could be delivered by the TX-Box.

The number of RF pulses used for the stability and linearity evaluations depends on the number of measured lines. For the test category **Fast** the measurement time is quite short, just to acquire enough data to be able to evaluate if the TX-Box is in working condition.

Test

The power is transmitted to the BC with the Body Loader placed inside.

The signals are measured through the DICOs in the TX-Box. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. There are two DICOs are placed after the power splitter. Thus the power for the 0 and the 90 degree transmit path (referred to as BC 0 and BC 1) can be measured separately.

The following measurement values will be gathered for all measurements:

- Forwarded and reflected power (complex) at DICO

- Forwarded and reflected amplitude at TALES

Evaluation

The following values are calculated, checked against limits, and visualized in the report:

- Pulse characteristics for each transmit voltage
- Reflection factor for each transmit voltage
- S-parameters for each transmit voltage (for pTX-systems)

Conclusions

Please refer to the test report.

2.1.2.4.3.3 Power intrapulse RX-LC interactive

Help ID: 44D9F075-A1BA-4BFF-AC61-C2A5F8B01808
Service Level: 7

General

The **Power intrapulse RX-LC interactive** test checks the intra pulse stability of the RF pulses transmitted to the Body Coil when a local coil is plugged into a coil socket.

The transmit amplitudes of the RF power pulses sent to the Body Coil are increased step by step up to the maximum allowed power. The maximum allowed power is determined through inline adjustments.

The CP mode is used for single channel systems. For dual channel systems (pTX, parallel transmit) each channel is measured separately.

Signal Path

For the test, the tested local coil with its QA-phantom-configuration has to be in the magnet center.

Follow the instructions at the console.

All data is acquired simultaneously within one measurement using the rf_loop sequence.

Test

The power is transmitted to the BC with the local coil placed inside.

The signals are measured through the DICOs in the TX-Box. The DICOs are narrowband detectors which allow the measurement of amplitude and phase of the transmitted power. There are two DICOs are placed after the power splitter. Thus the power for the 0 and the 90 degree transmit path (referred to BC 0 and BC 1) can be measured separately.

The following measurement values will be gathered for all measurements:

- Forwarded and reflected power (complex) at DICO

- Forwarded and reflected amplitude at TALES

Evaluation

The following values are calculated, checked against limits, and visualized in the report:

- Pulse characteristics for each transmit voltage
- Reflection factor for each transmit voltage
- S-parameters for each transmit voltage (for pTX-systems)

Conclusions

Please refer to the test report.

2.1.2.4.4 TX LC

2.1.2.4.4.1 Power LC interactive

Help ID: 77911CF0-6192-400A-8075-676042E0E2EE
Service Level: 7

General

Test Description

The **Power LC interactive** test checks the local coil transmit path of the systen.

This test is similar to the TX Power Test. Instead to the Body Coil the RF power is transmitted to a local transmit and receive coil (TxRx coil). For this test the Service Plug is used as a TX coil.

The test checks the complete 1H LC-TX-path and the socket at the patient table. The component protection is switched on so that the maximum power that is allowed for the coil cannot be exeeded. The test is performed for 1H.

For the test, the Service Plug has to be connected to coil socket 1 at the patient table. Follow the instructions provided during the test run.

This test requires that the transmission cable to the coil socket is terminated with 50 Ohm. This is done by using the 50 Ohm resistor in the Service Plug or a different 50 Ohm load.

> i If the 50 Ohm termination resistor in the Service Plug is defective the test will fail!
> In that case check the resistance with a multimeter first.

If tests shall be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools > Supervision** the supervision has to be set to “Virtual” before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed (“preparation error”).

Before the table is docked again this setting has to be reset to “Real”, then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.

Follow the instructions of the test pop-ups for this.

Signal Path

The power is transmitted from the TX-Box to coil socket 1.

This test requires that the transmission cable to the socket be terminated with 50 Ohm. For this the Service Plug must be connected to coil socket 1 at the patient table. Follow the instructions provided during the test run.

Test

The test consists of two measurements. One RF pulse is transmitted for each measurement:

- in the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor
- the second measurement acquires the signals from the TALES

Evaluation

The following parameters of the power output are checked at the DICOs:

- deviation of requested and delivered power
- reflected power (as a function of the transmitted power)

A comparison of TALES and DICO values is not required because the TX-Box performs an online comparison of these values for security reasons.

The test is passed if the reflection is below 20 %.

Conclusions

Please refer to test report.

2.1.2.4.5 LC Plug 1

2.1.2.4.5.1 LC Plug IF

Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

> If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to "Virtual" before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed ("preparation error").
>
> Before the table is docked again this setting has to be re-set to "Real", then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.
>
> Follow the instructions of the test pop-ups for this.
>
> For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

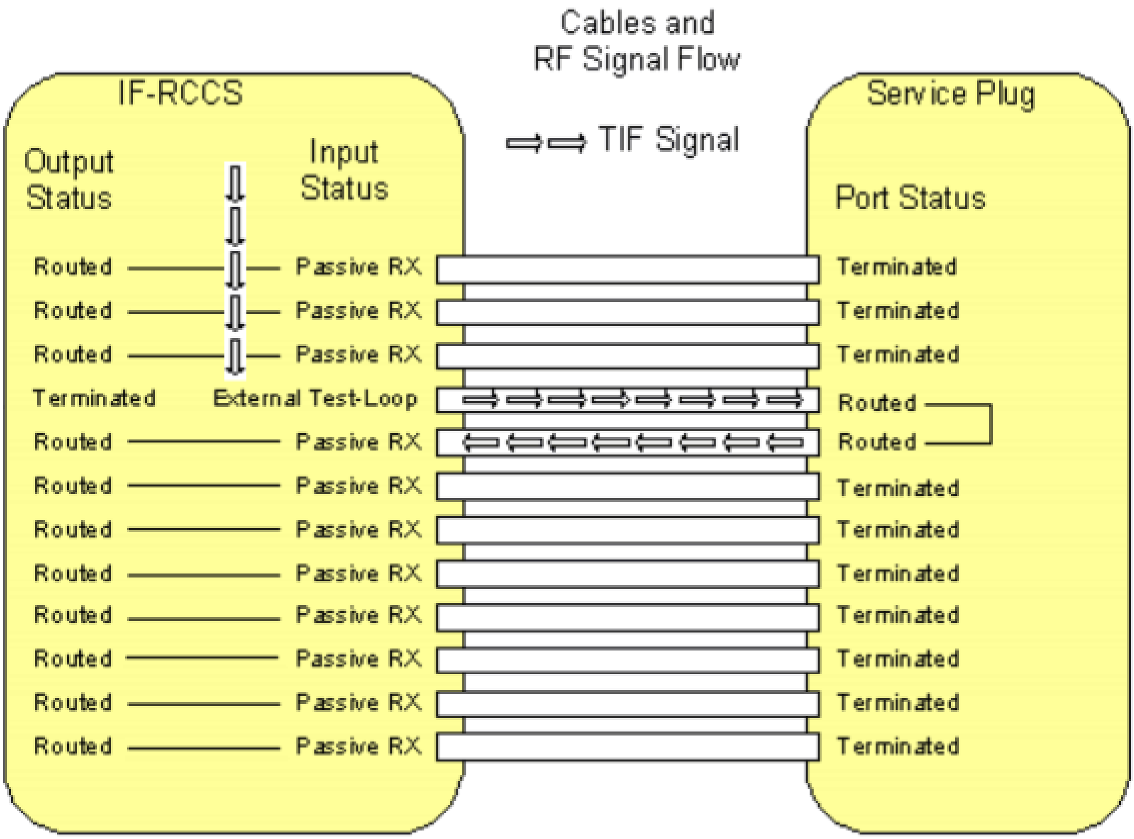
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

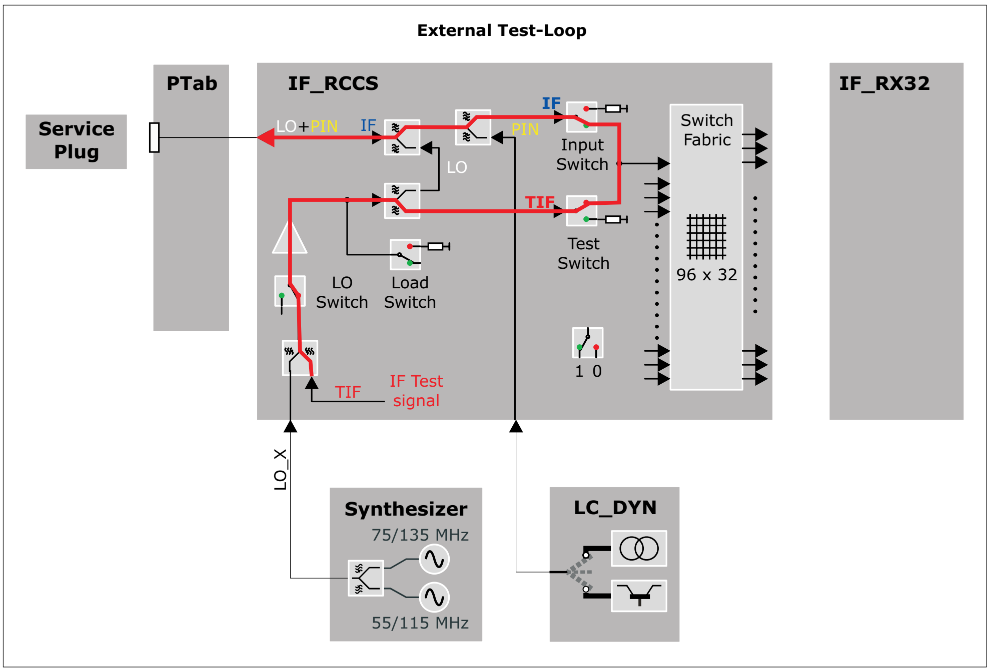
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 49: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 50: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step “n” a test signal is transmitted through the channel “n” and received through the other channels, i.e. all channels except channel “n” are routed back to the receiver. For both channels “n” and “n + 1” used in step “n”, the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not "TX" but indicated by a red "LO"
- the swapped pin is not "LO" but indicated by a red "HI"

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 51: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 52: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 53: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 54: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.5.2
LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.
- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.
- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.
- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

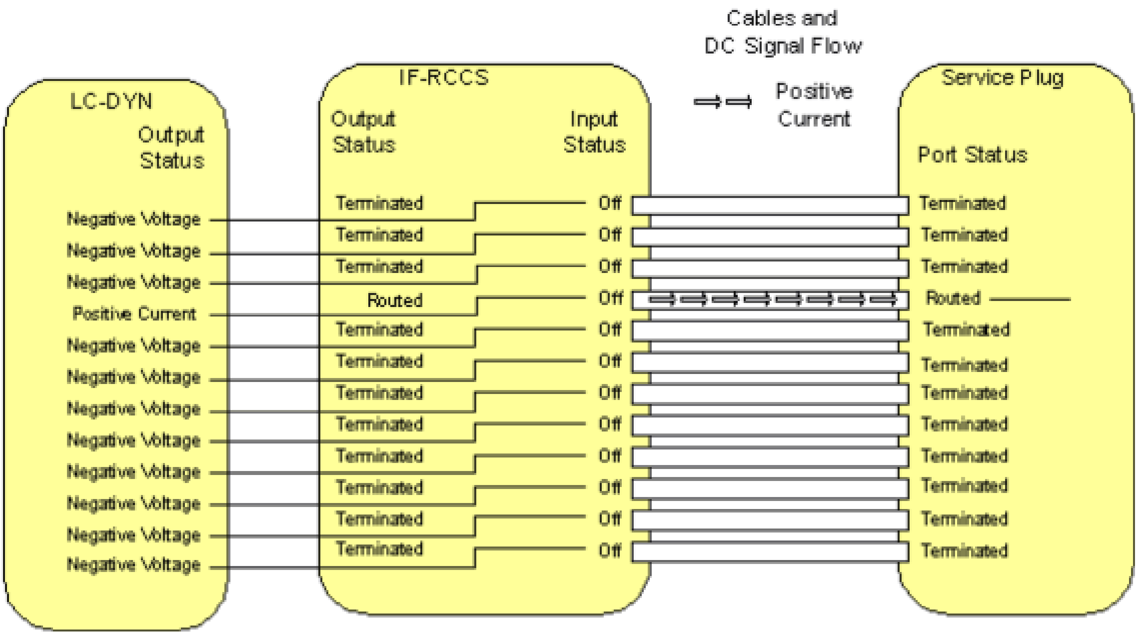
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 55: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

i The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

> The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.
>
> They show the principle, the actual layout in the test reports looks different.

Fig. 56: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 57: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 58: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 59: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 60: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.5.3
LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

Please note that the dBm output is not realized yet. Thus only ADC values are displayed.

Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

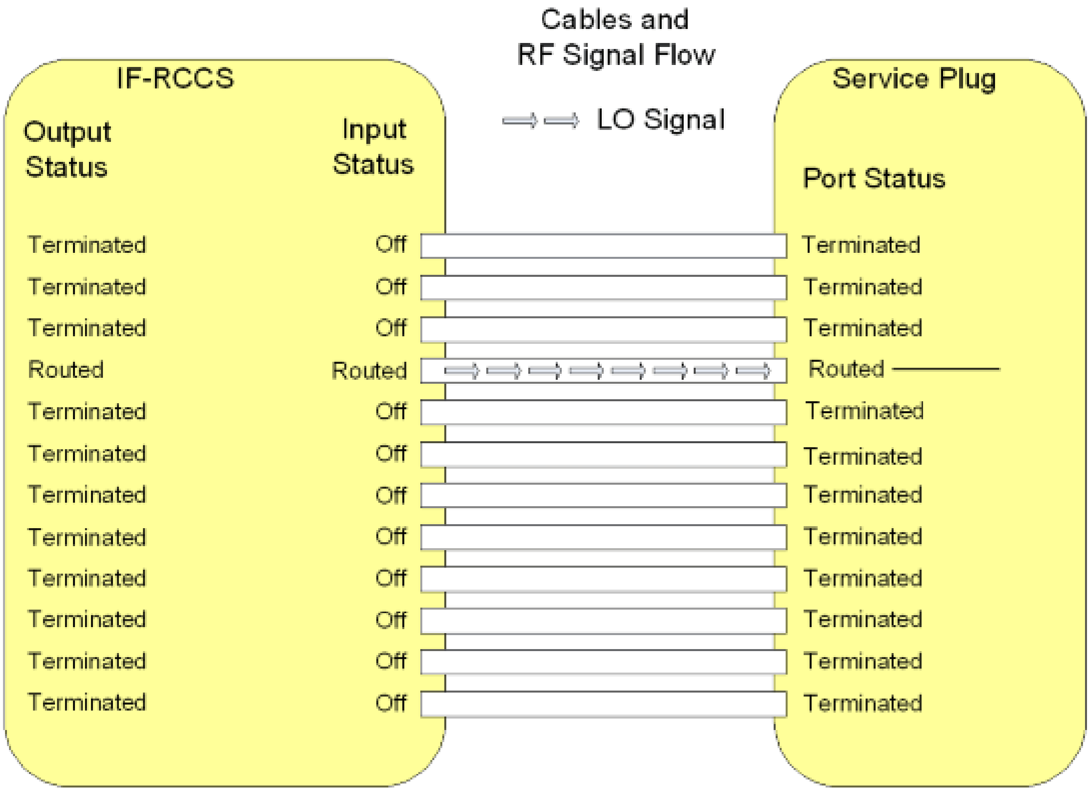
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

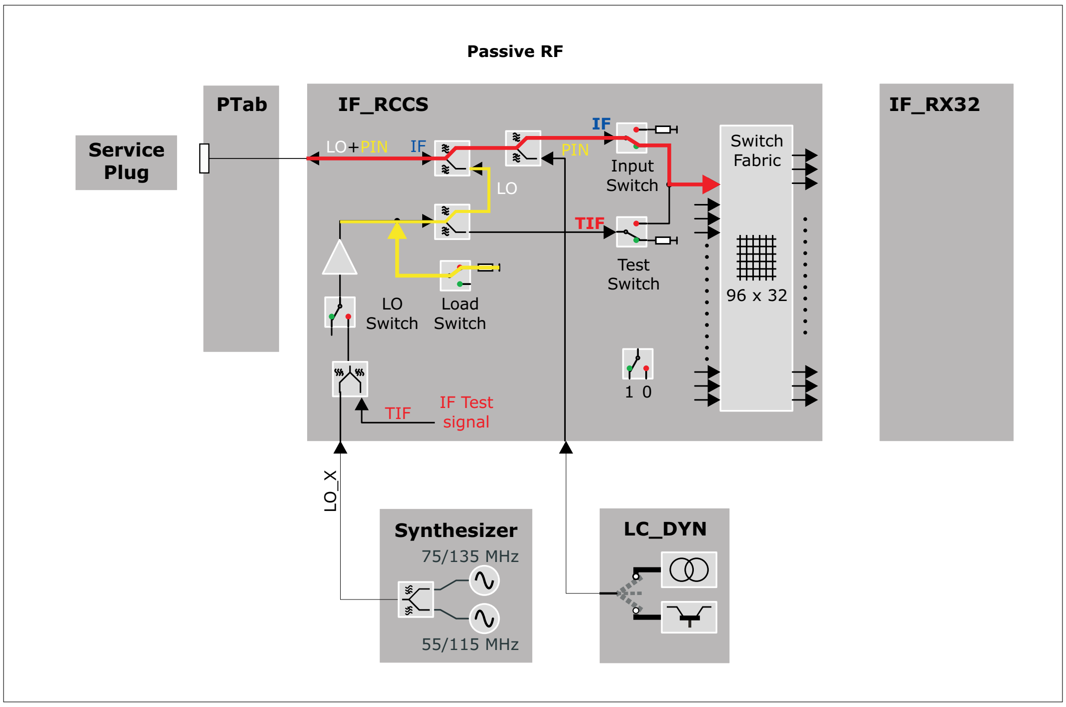
Corrected values are not yet read and checked.

Fig. 61: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 62: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

> Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.5.4
LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.5.5 LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.5.6
LC Plug Coil Interrupt

Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.

2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hardwired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.6
LC Plug 2

2.1.2.4.6.1
LC Plug IF

Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to “Virtual” before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed (“preparation error”).

Before the table is docked again this setting has to be re-set to “Real”, then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.

Follow the instructions of the test pop-ups for this.

For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

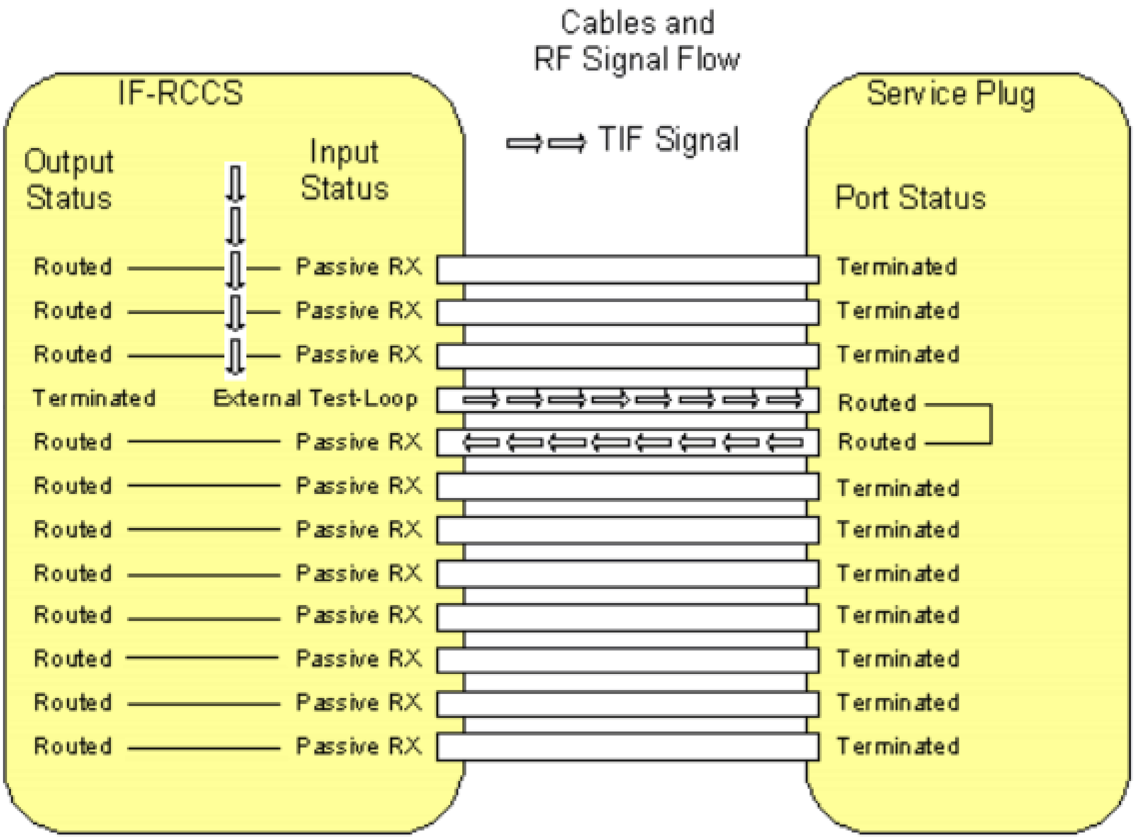
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

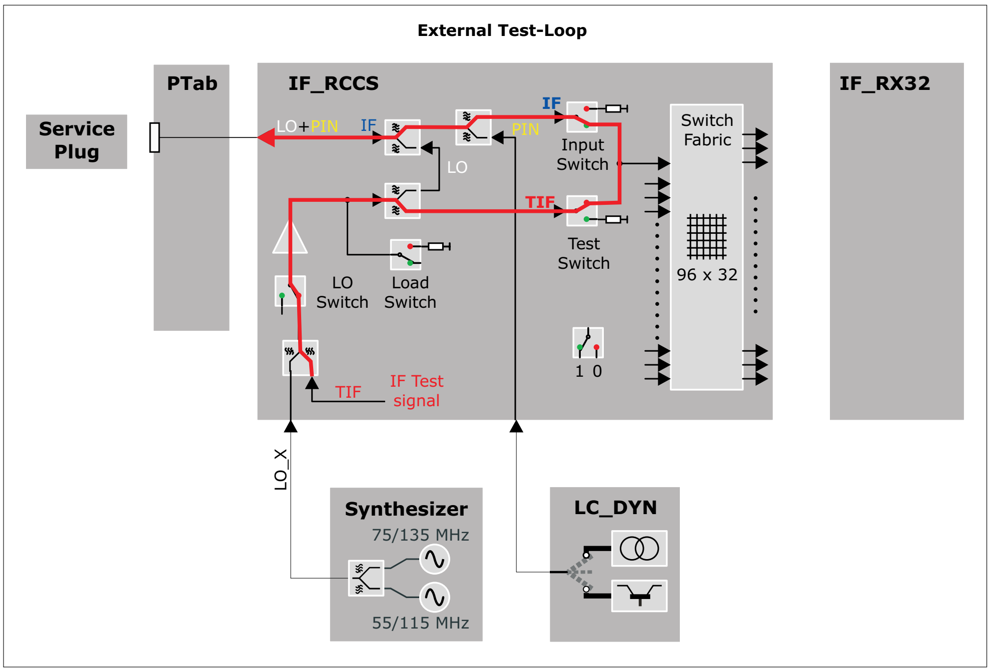
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 63: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 64: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step “n” a test signal is transmitted through the channel “n” and received through the other channels, i.e. all channels except channel “n” are routed back to the receiver. For both channels “n” and “n + 1” used in step “n”, the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by
- the transmit pin is not “TX” but indicated by a red “LO”
- the swapped pin is not “LO” but indicated by a red “HI”

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 65: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 66: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 67: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 68: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.6.2
LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.

- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.

- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.

- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

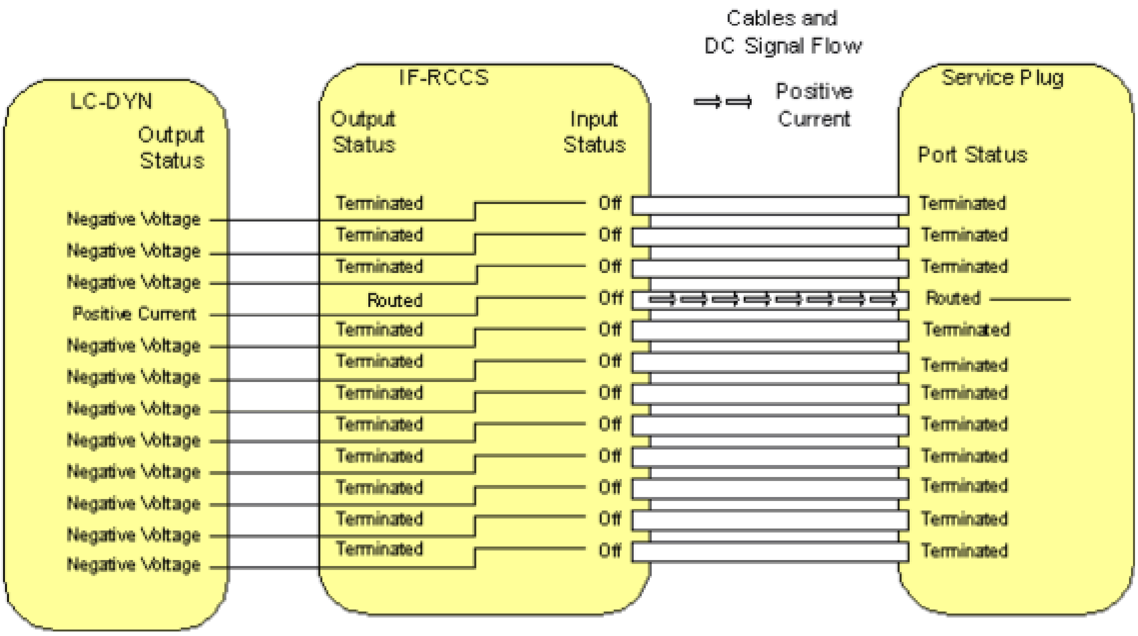
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 69: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

> i The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 70: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 71: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 72: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 73: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 74: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.6.3
LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

Please note that the dBm output is not realized yet. Thus only ADC values are displayed.

Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

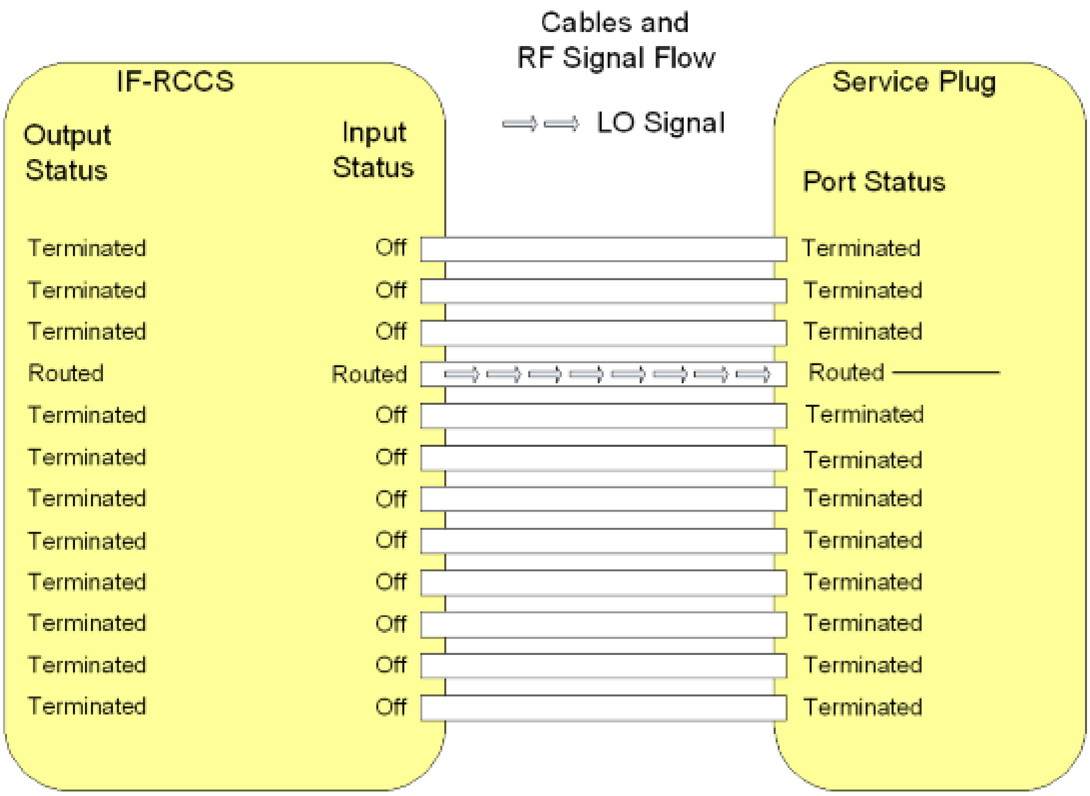
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

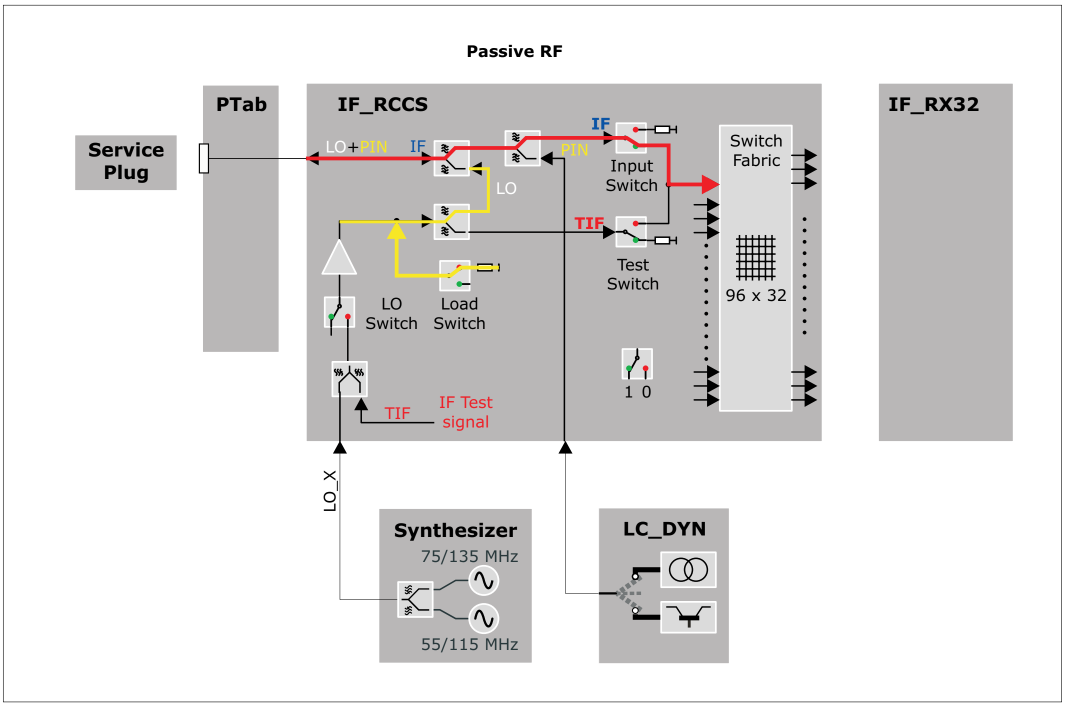
Corrected values are not yet read and checked.

Fig. 75: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 76: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.

Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.6.4
LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.6.5
LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.6.6 LC Plug Coil Interrupt

Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.
2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hard-wired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.7
LC Plug 3

2.1.2.4.7.1
LC Plug IF

Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

i

If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to "Virtual" before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed ("preparation error").

Before the table is docked again this setting has to be re-set to "Real", then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.

Follow the instructions of the test pop-ups for this.

For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

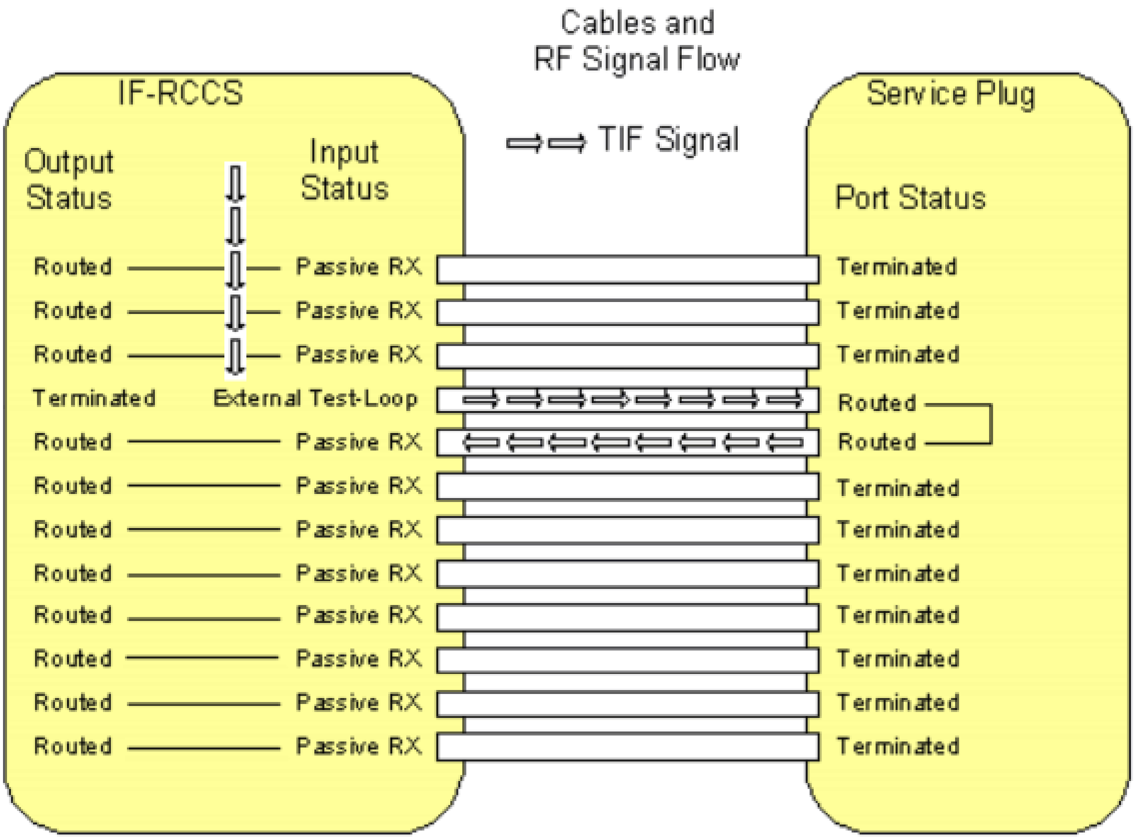
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

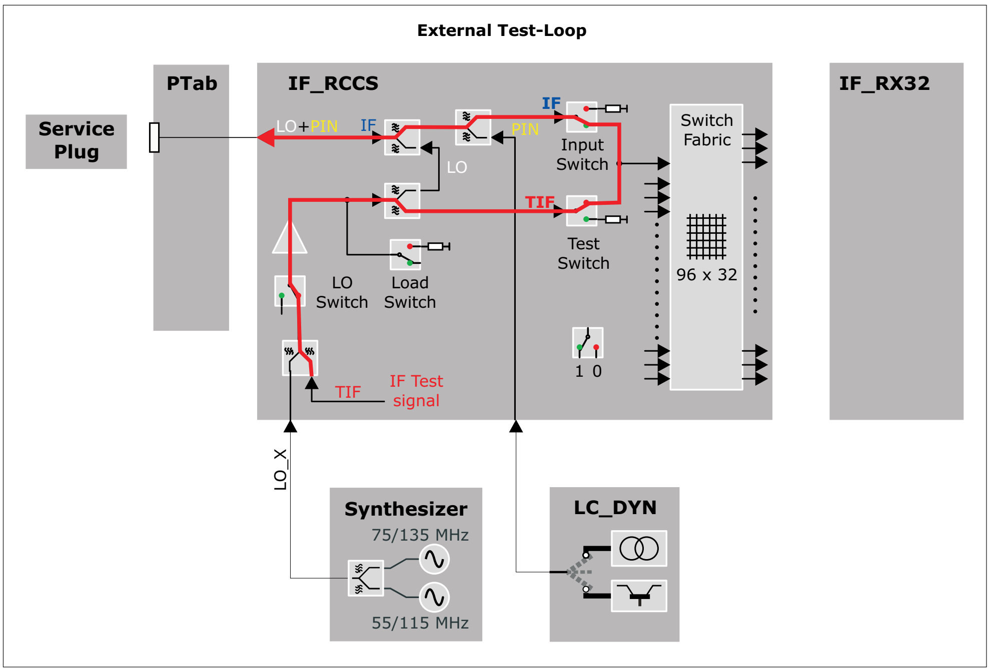
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 77: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 78: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step "n" a test signal is transmitted through the channel "n" and received through the other channels, i.e. all channels except channel "n" are routed back to the receiver. For both channels "n" and "n + 1" used in step "n", the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not "TX" but indicated by a red "LO"
- the swapped pin is not "LO" but indicated by a red "HI"

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 79: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 80: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 81: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 82: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

> Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.7.2
LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.
- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.
- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.
- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

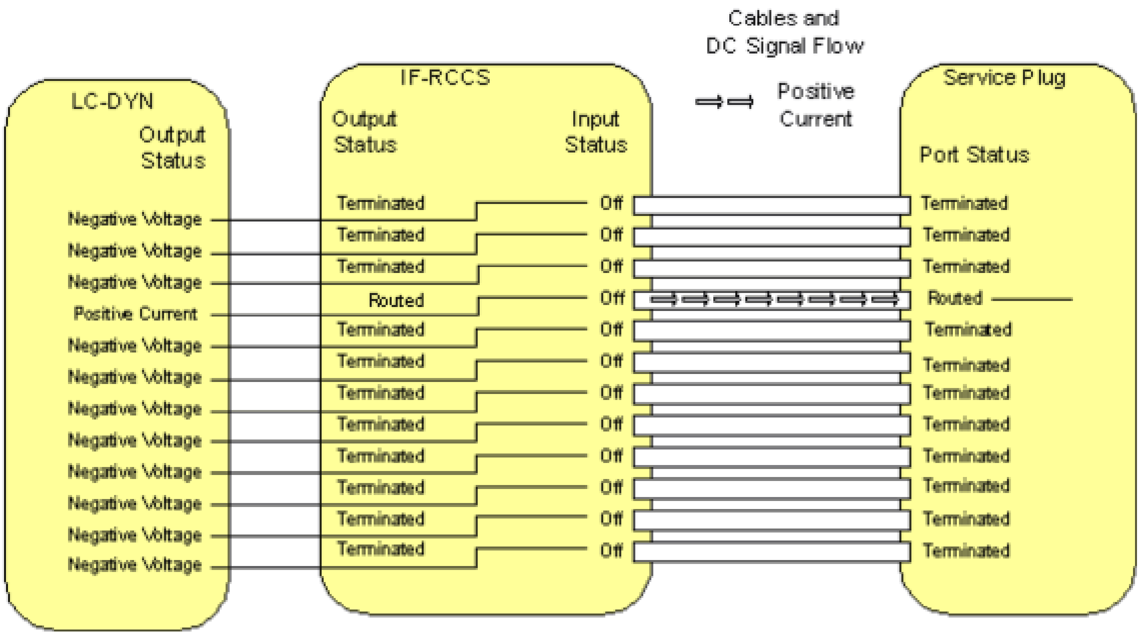
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 83: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

> The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 84: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 85: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 86: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 87: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 88: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.7.3 LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

> Please note that the dBm output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

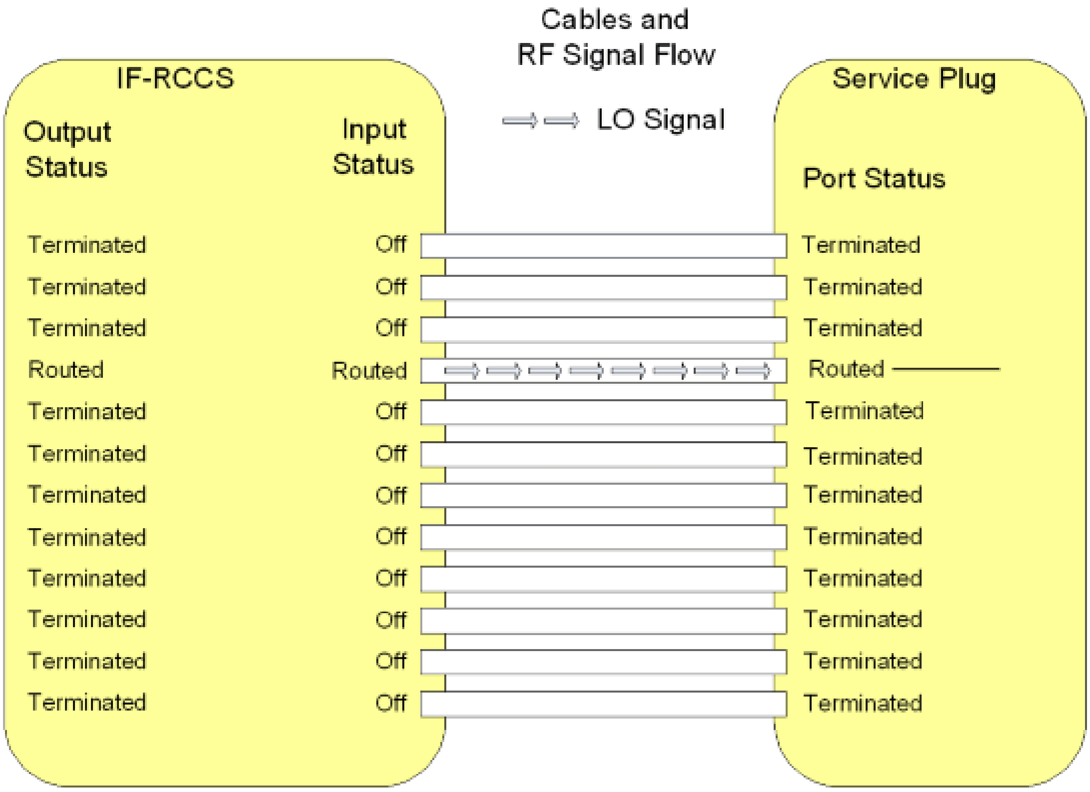
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

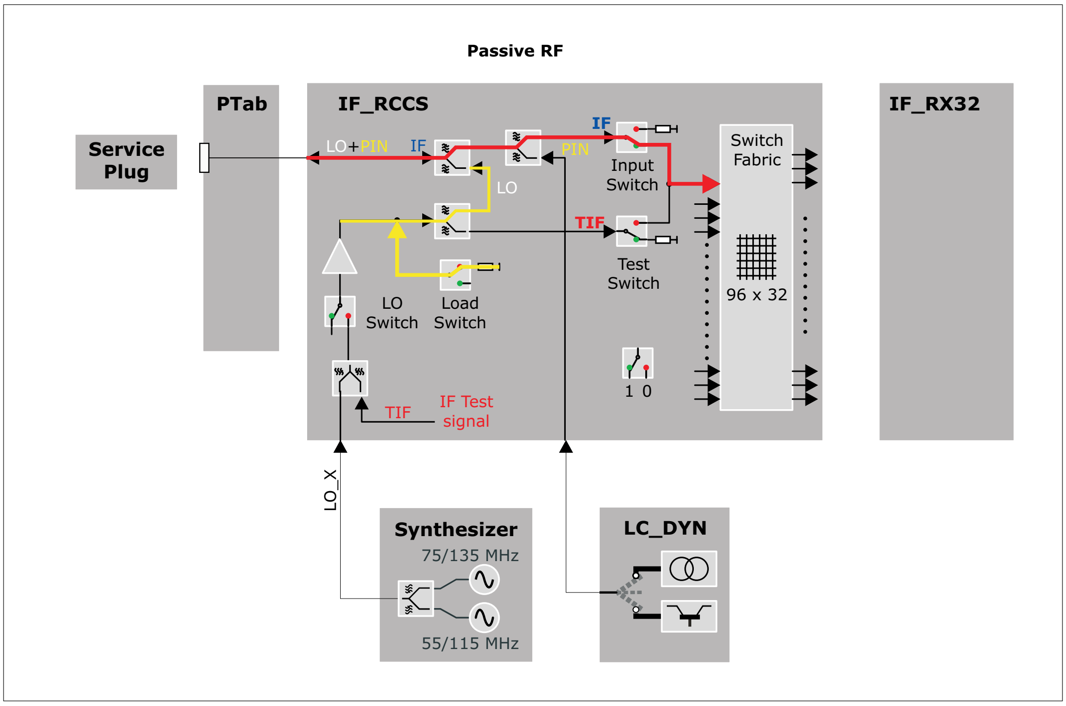
Corrected values are not yet read and checked.

Fig. 89: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 90: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

> Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.7.4
LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.7.5
LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.7.6 LC Plug Coil Interrupt

Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.

2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hard-wired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.8
LC Plug 4

2.1.2.4.8.1
LC Plug IF

Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to “Virtual” before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed (“preparation error”).

Before the table is docked again this setting has to be re-set to “Real”, then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.

Follow the instructions of the test pop-ups for this.

For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

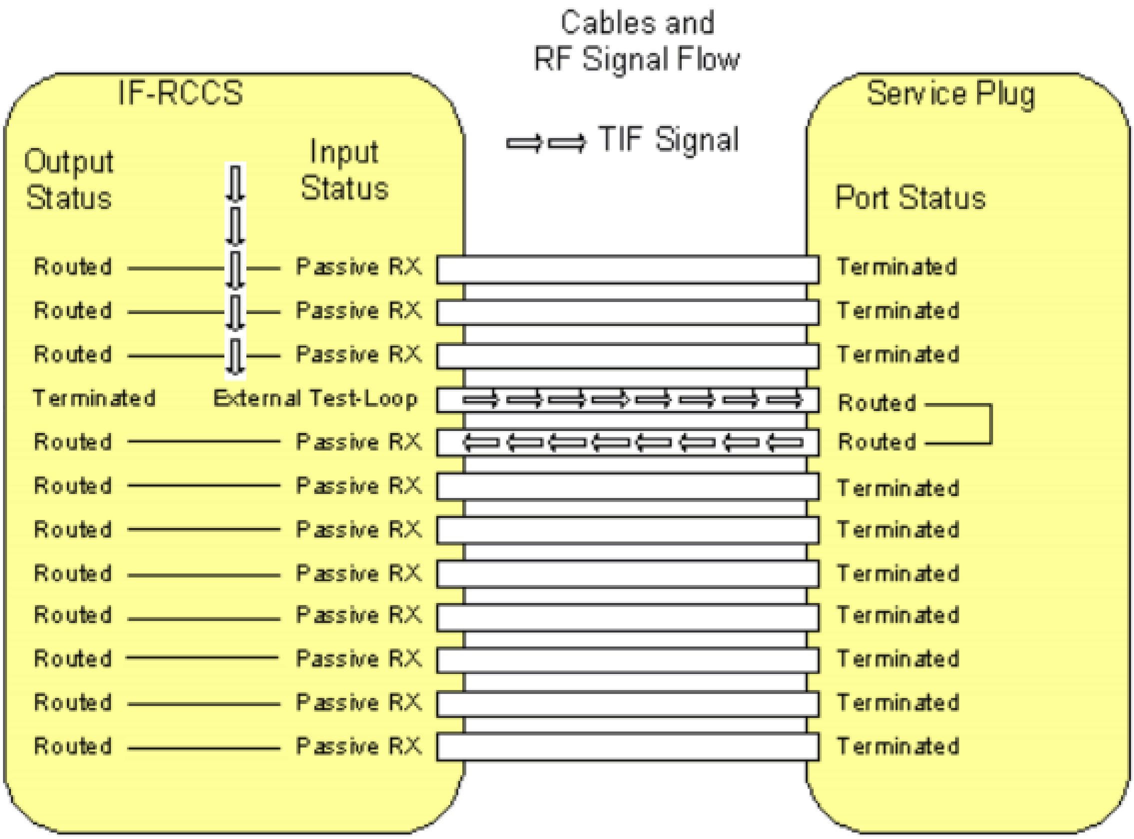
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

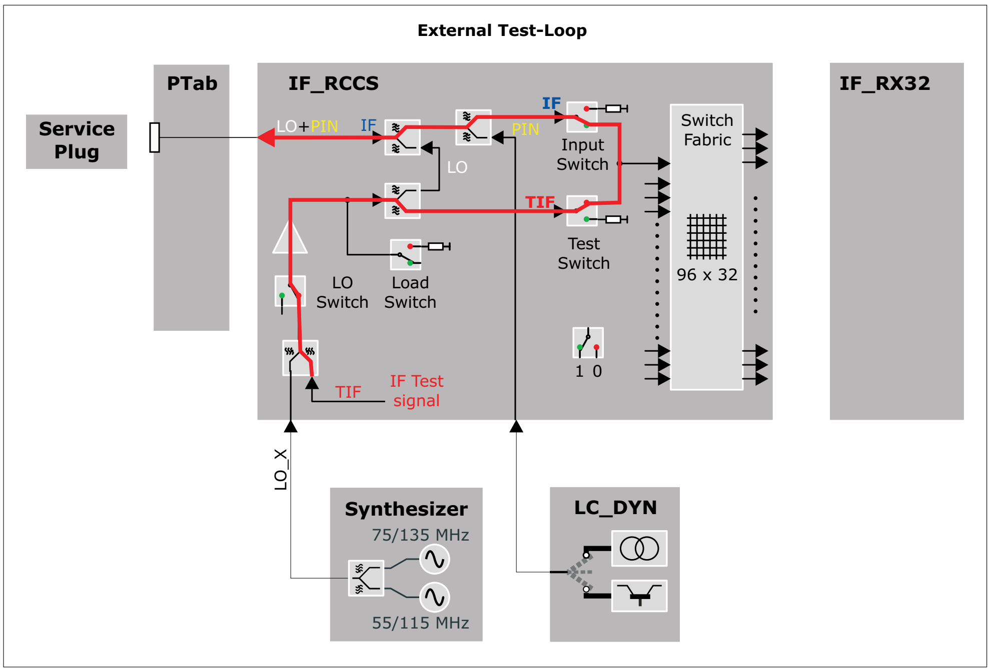
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 91: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 92: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step "n" a test signal is transmitted through the channel "n" and received through the other channels, i.e. all channels except channel "n" are routed back to the receiver. For both channels "n" and "n + 1" used in step "n", the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not “TX” but indicated by a red “LO”
- the swapped pin is not “LO” but indicated by a red “HI”

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 93: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 94: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 95: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 96: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.8.2 LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.
- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.
- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.
- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

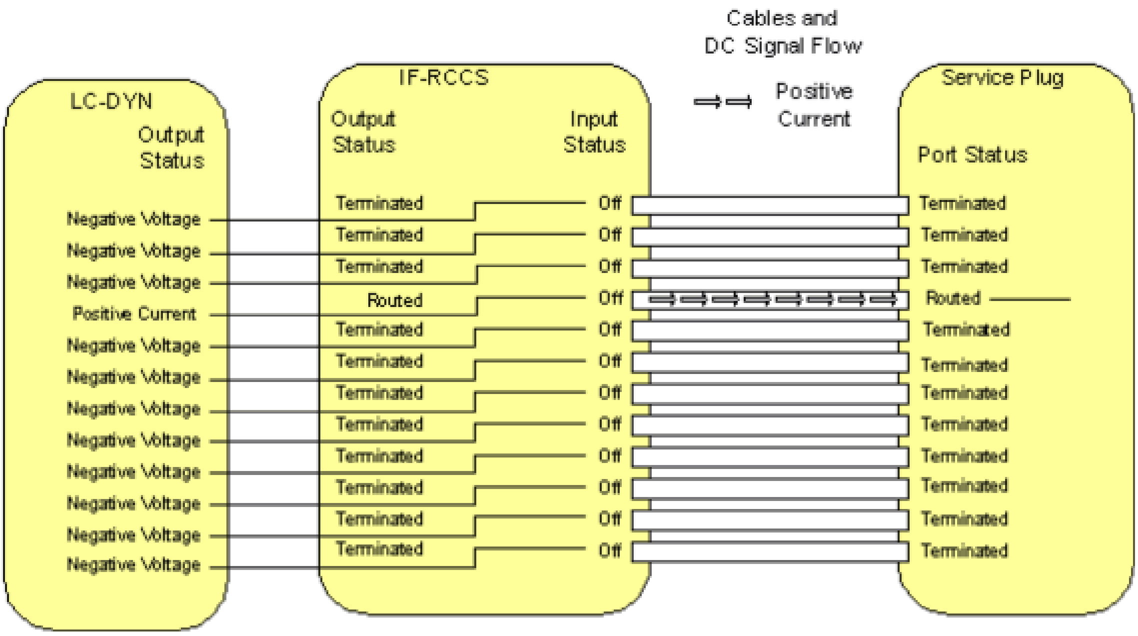
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 97: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

> i The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 98: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 99: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 100: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 101: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 102: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.8.3
LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

> Please note that the dBm output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

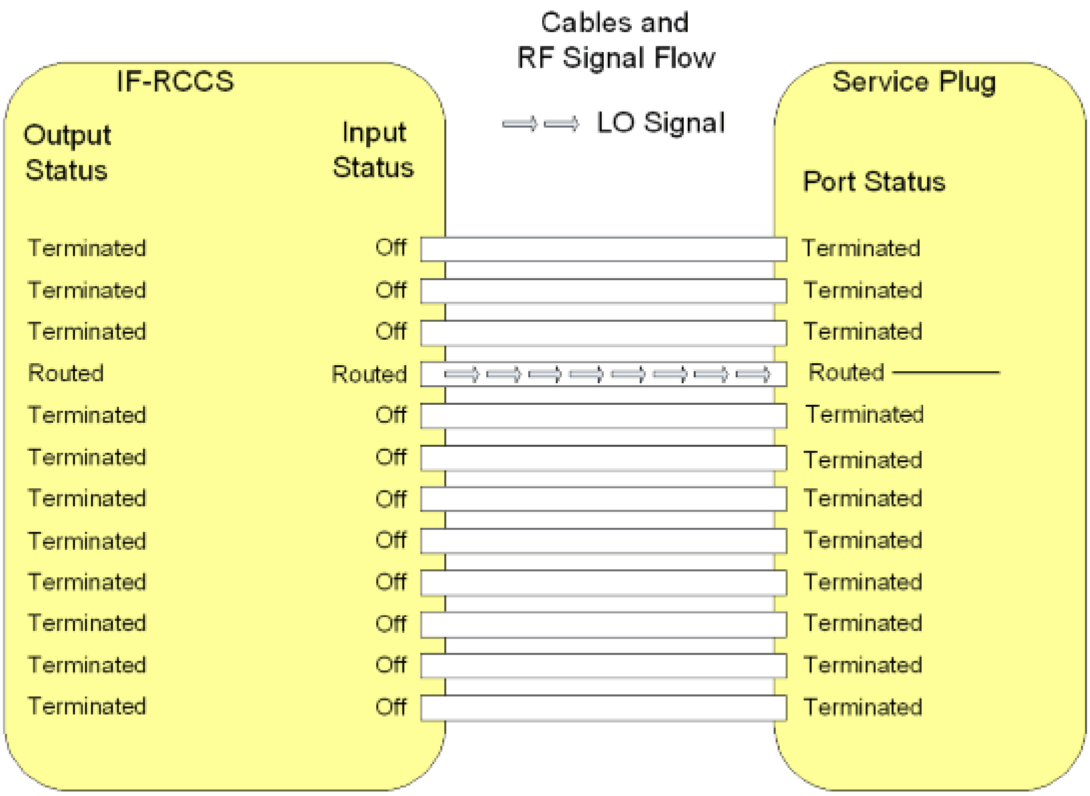
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

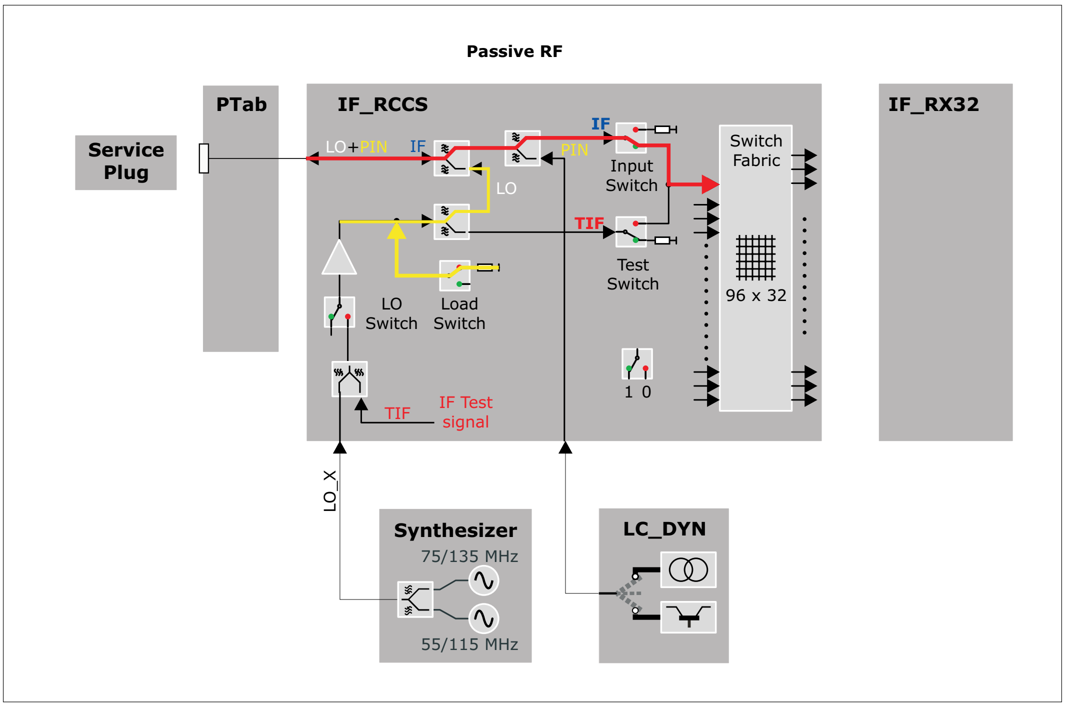
Corrected values are not yet read and checked.

Fig. 103: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 104: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.

Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.8.4 LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.8.5
LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.8.6 LC Plug Coil Interrupt

```
Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7
```

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.
2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hard-wired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.9
LC Plug 5

2.1.2.4.9.1
LC Plug IF

```
Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7
```

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

> If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to "Virtual" before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed ("preparation error").
>
> Before the table is docked again this setting has to be re-set to "Real", then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.
>
> Follow the instructions of the test pop-ups for this.
>
> For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.

- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).

- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

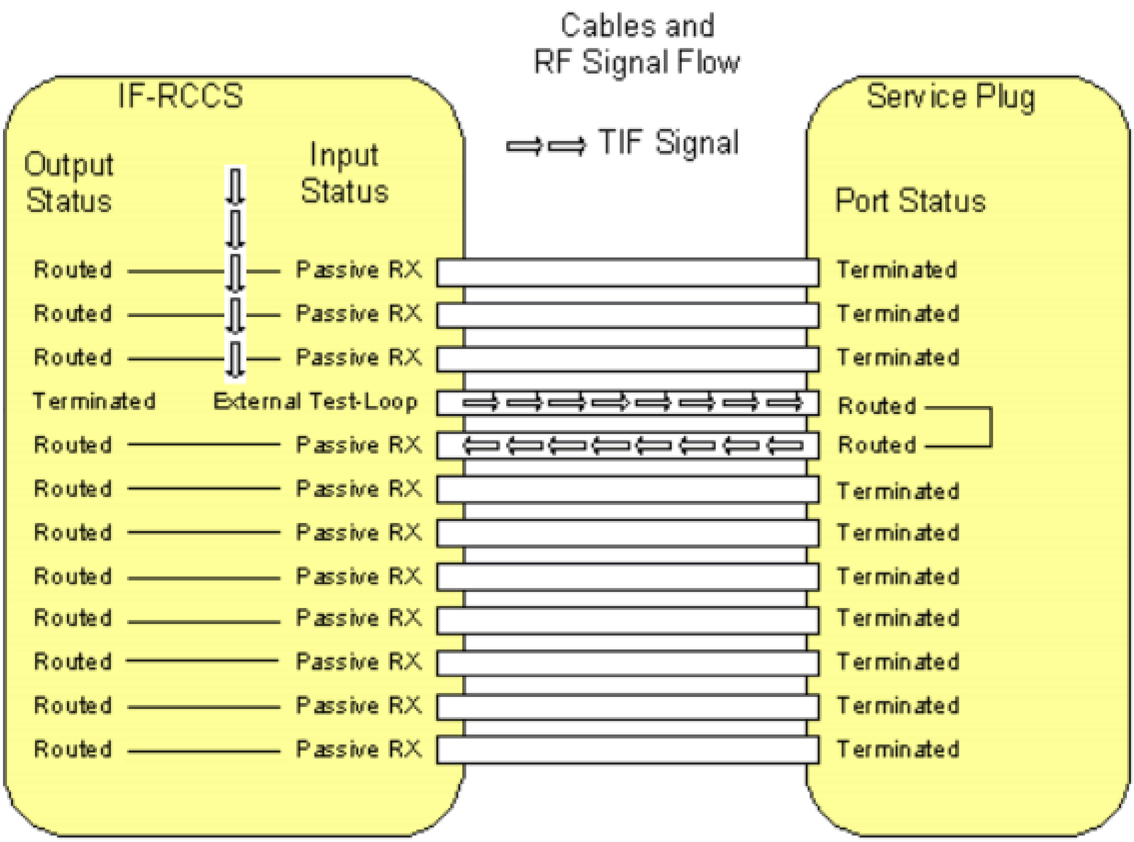
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

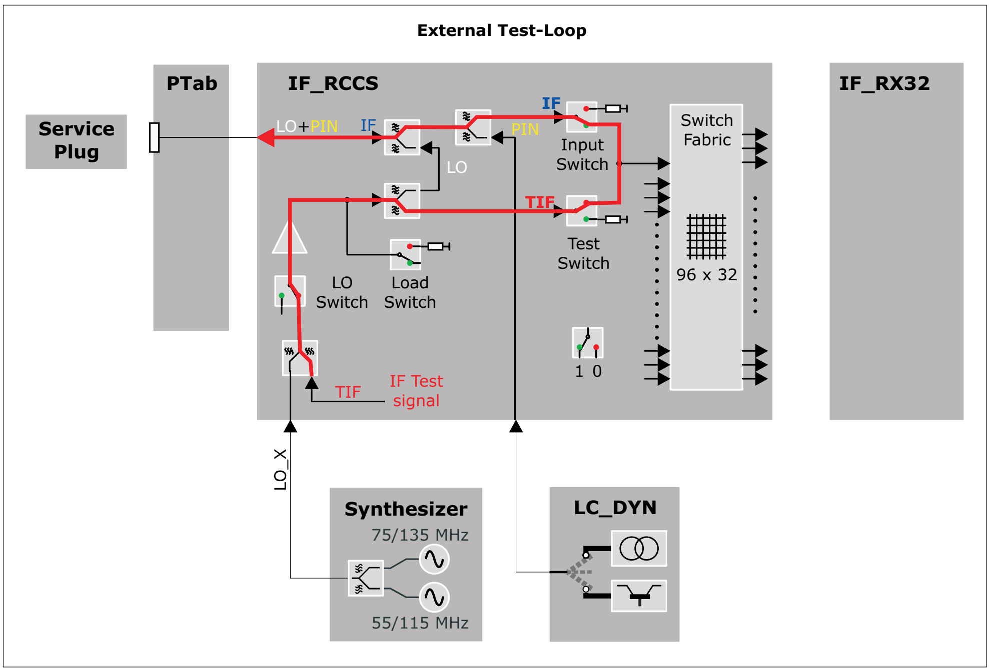
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 105: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 106: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step “n” a test signal is transmitted through the channel “n” and received through the other channels, i.e. all channels except channel “n” are routed back to the receiver. For both channels “n” and “n + 1” used in step “n”, the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not "TX" but indicated by a red "LO"
- the swapped pin is not "LO" but indicated by a red "HI"

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 107: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 108: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 109: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 110: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

> Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.9.2
LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.

- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.

- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.

- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

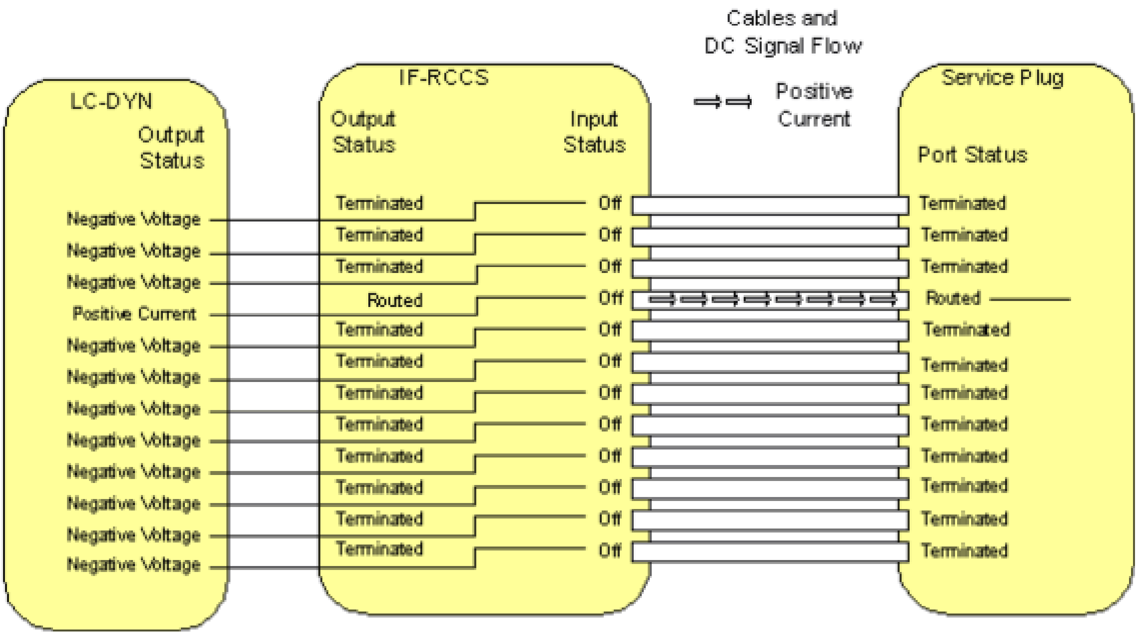
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 111: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

> The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

> The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.
>
> They show the principle, the actual layout in the test reports looks different.

Fig. 112: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAL	OK	OK	OK	OK	ROK

Fig. 113: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 114: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 115: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 116: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.9.3
LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

> Please note that the dBm output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

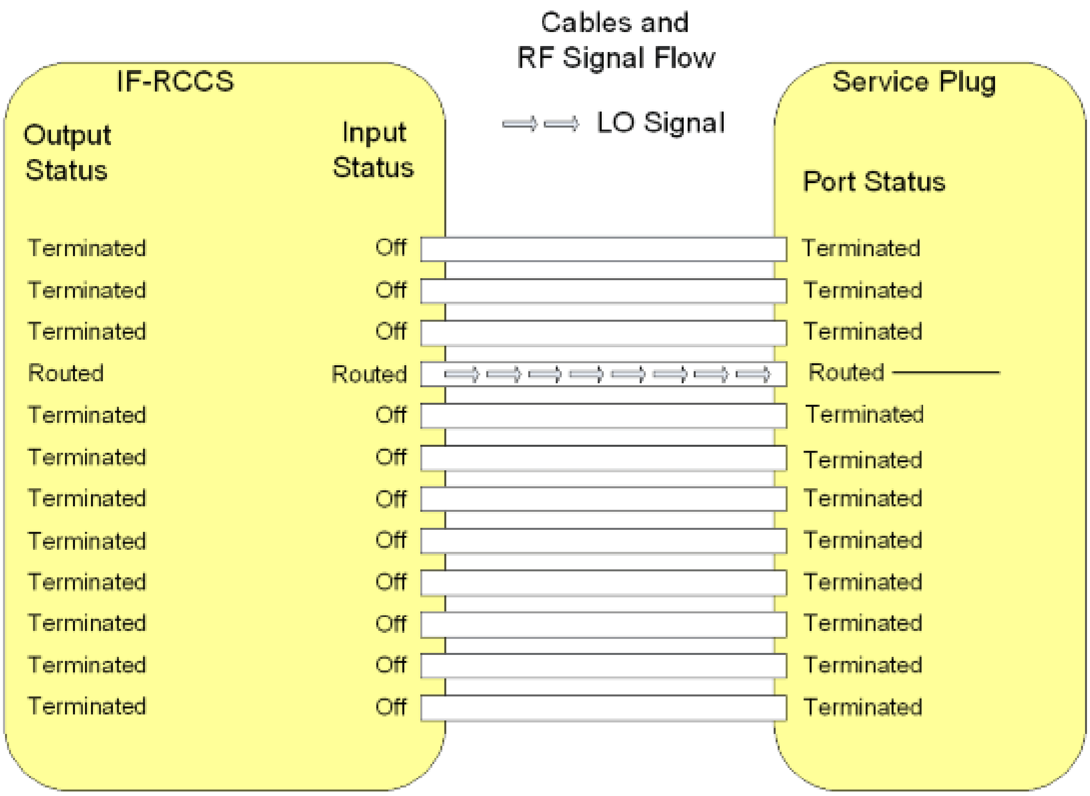
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

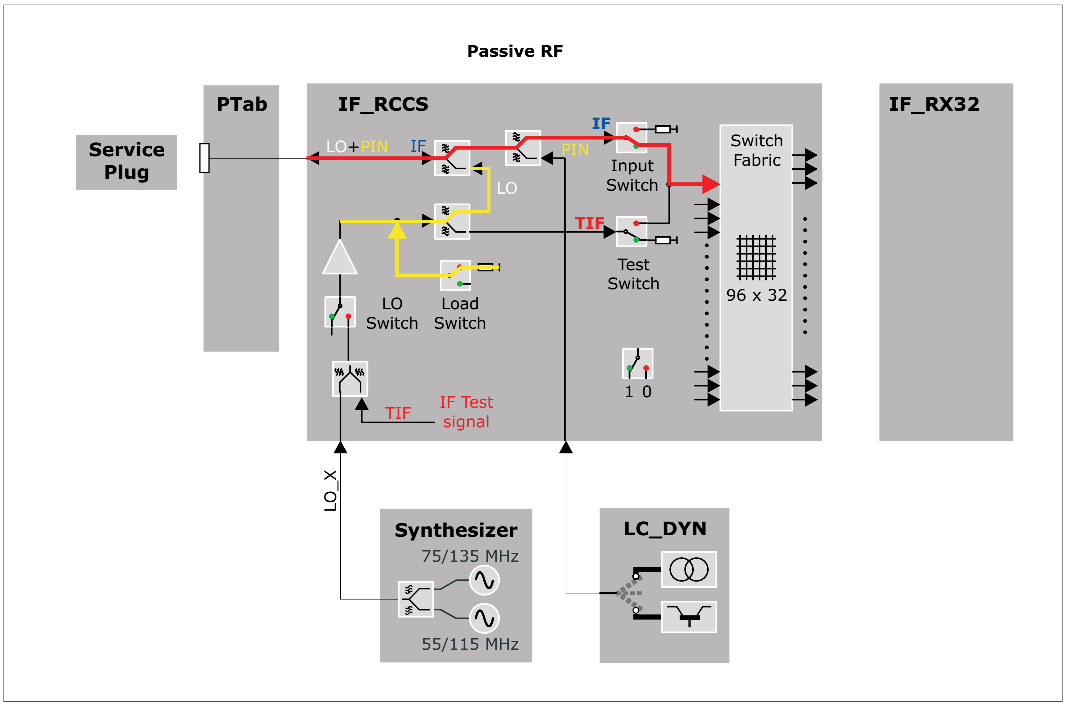
Corrected values are not yet read and checked.

Fig. 117: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 118: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

> Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.9.4
LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.9.5
LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> i Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> i The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.9.6 LC Plug Coil Interrupt

Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.
2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hard-wired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.10
LC Plug 6

2.1.2.4.10.1
LC Plug IF

```
Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7
```

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

> If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to "Virtual" before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed ("preparation error").
>
> Before the table is docked again this setting has to be re-set to "Real", then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.
>
> Follow the instructions of the test pop-ups for this.
>
> For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

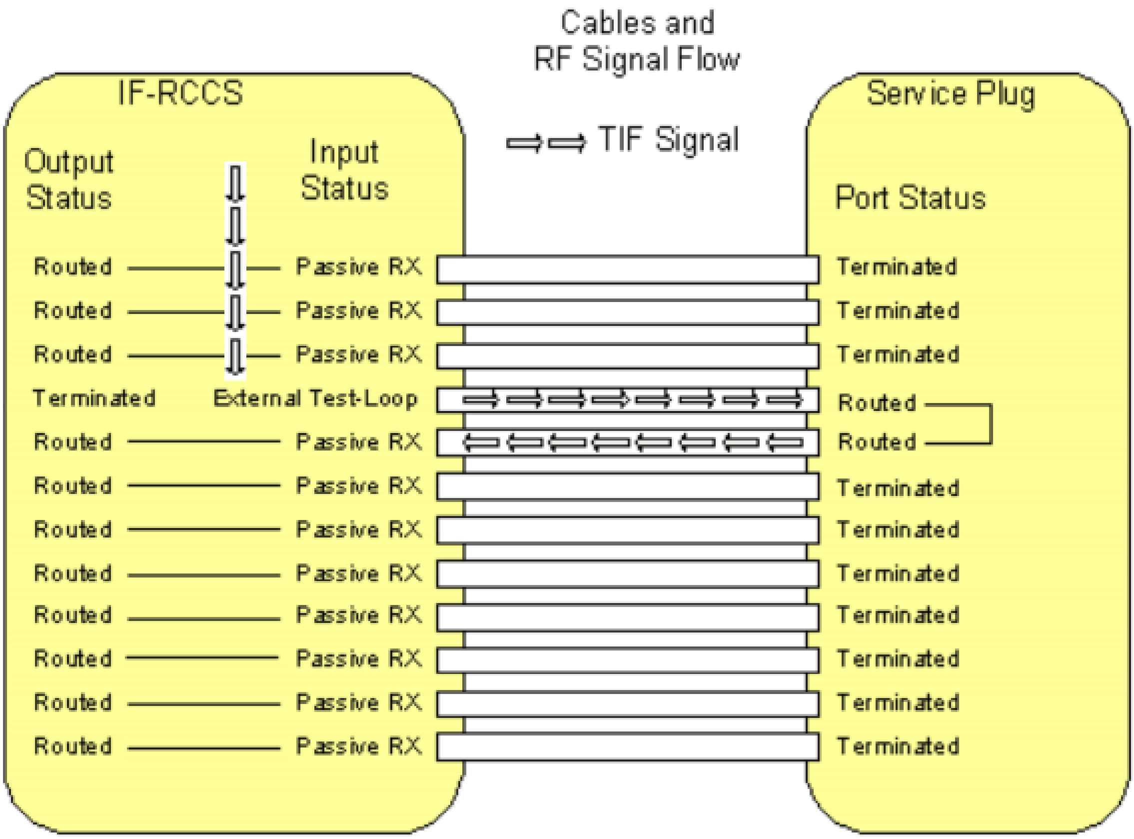
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

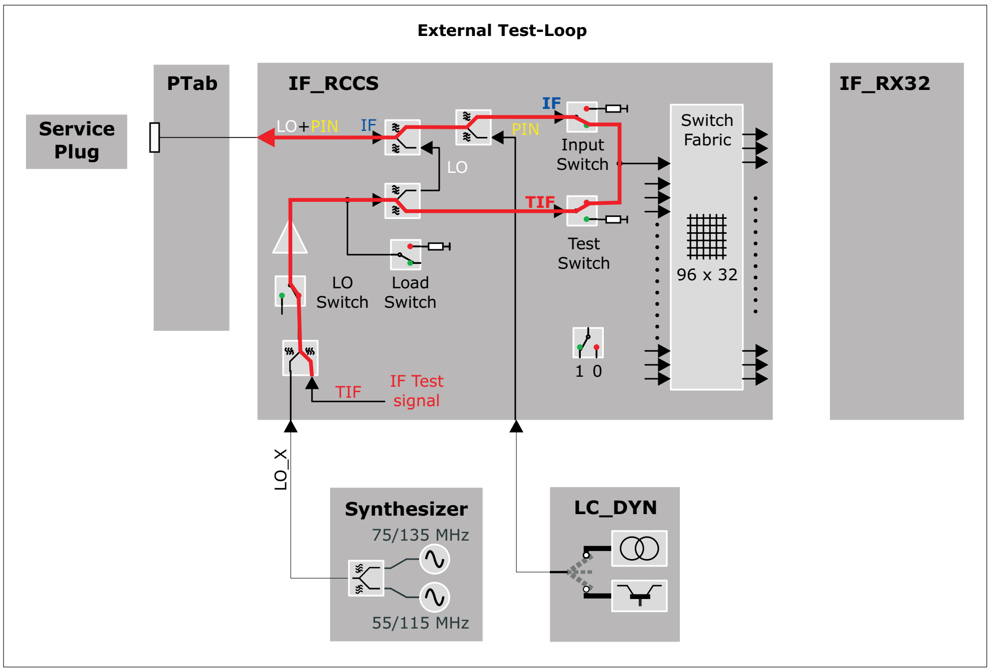
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 119: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 120: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step "n" a test signal is transmitted through the channel "n" and received through the other channels, i.e. all channels except channel "n" are routed back to the receiver. For both channels "n" and "n + 1" used in step "n", the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not “TX” but indicated by a red “LO”
- the swapped pin is not “LO” but indicated by a red “HI”

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 121: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 122: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 123: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 124: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.10.2 LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.
- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.
- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.
- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

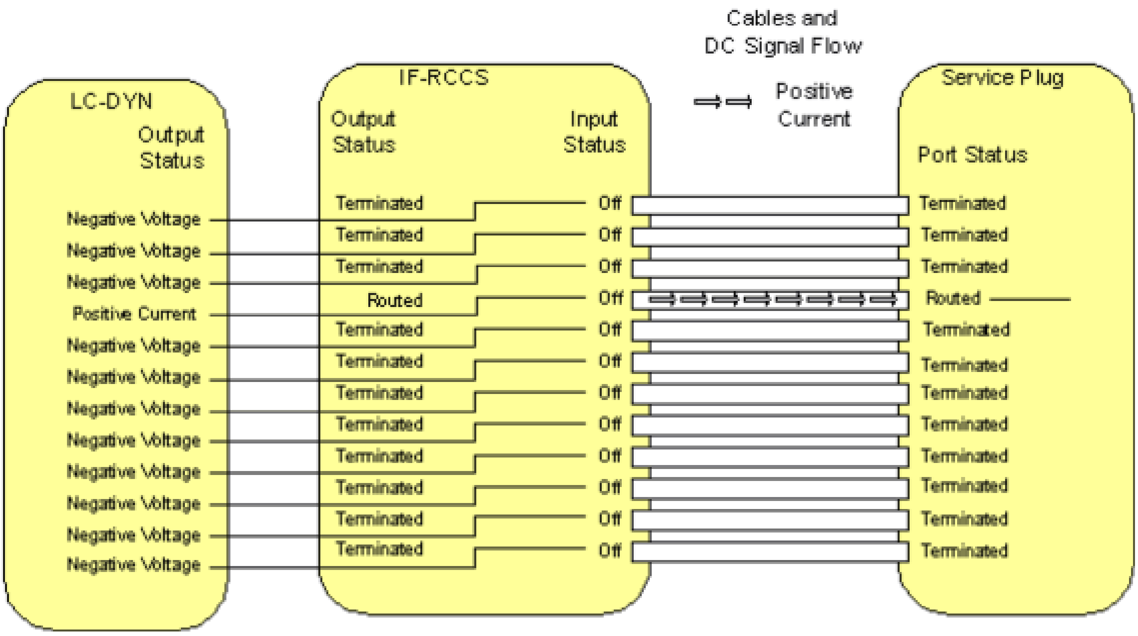
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 125: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

i

The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 126: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 127: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 128: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 129: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 130: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.10.3 LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

Please note that the dBm output is not realized yet. Thus only ADC values are displayed.

Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

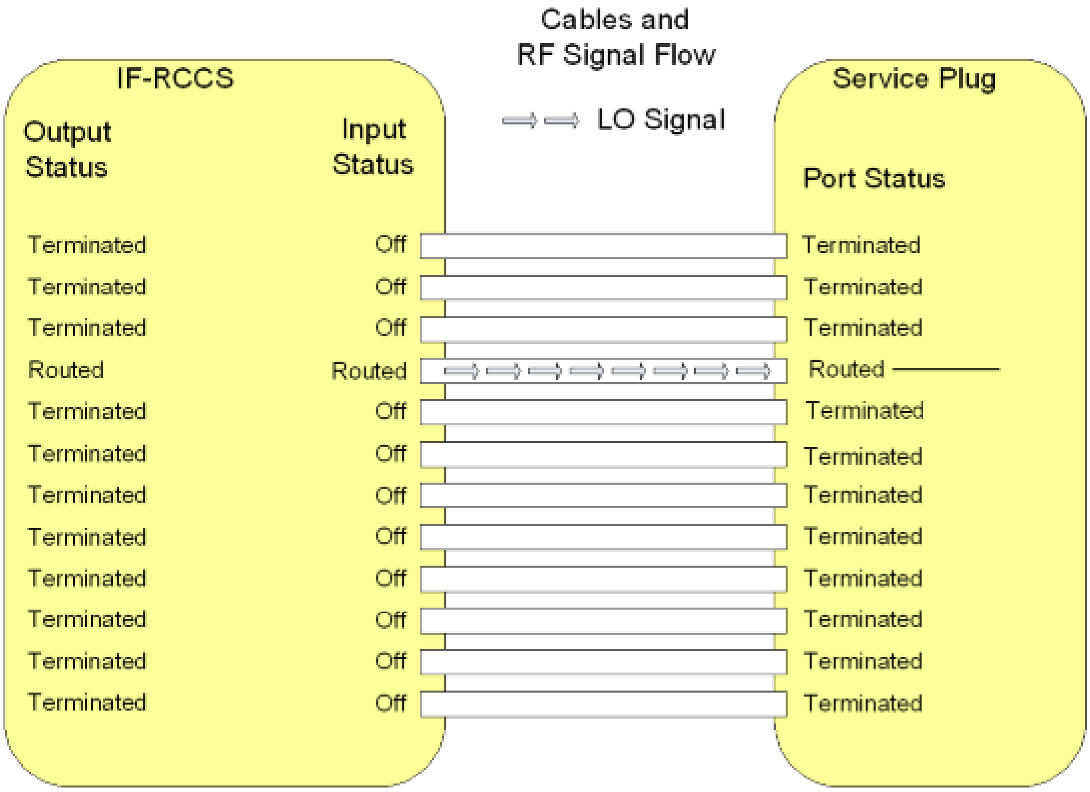
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

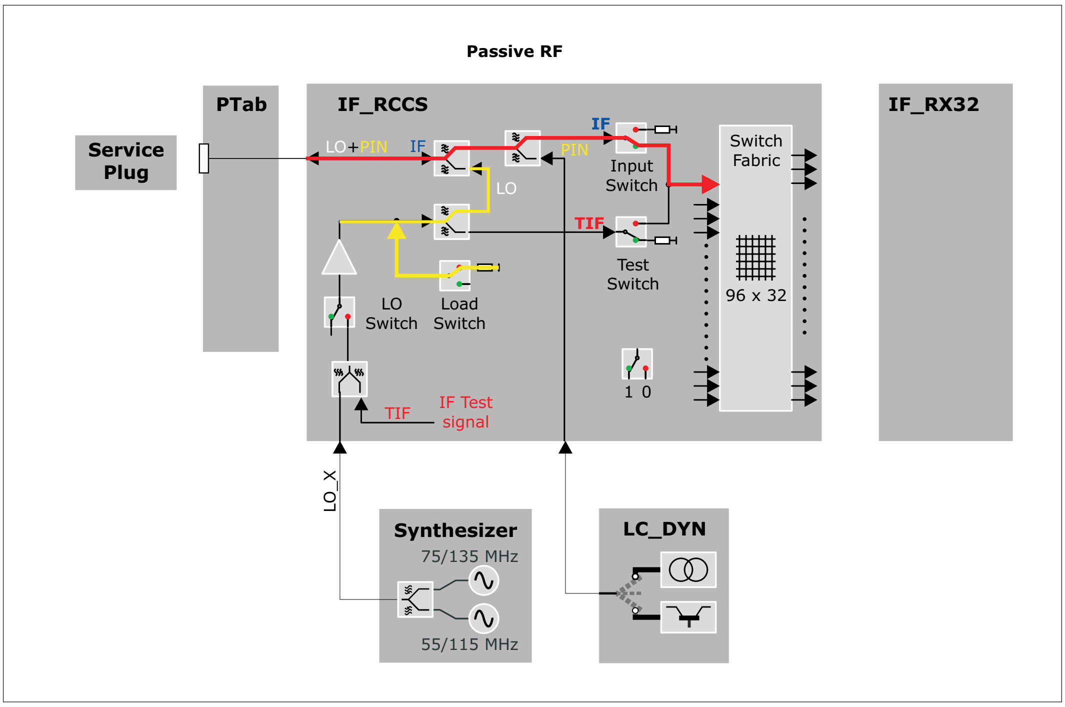
Corrected values are not yet read and checked.

Fig. 131: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 132: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step "n" an LO test signal is transmitted through the channel "n" and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel "n" used in step "n" the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.

Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.10.4 LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.10.5 LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.10.6 LC Plug Coil Interrupt

Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.

2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hard-wired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.11
LC Plug 7

2.1.2.4.11.1
LC Plug IF

Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to “Virtual” before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed (“preparation error”).

Before the table is docked again this setting has to be re-set to “Real”, then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.

Follow the instructions of the test pop-ups for this.

For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

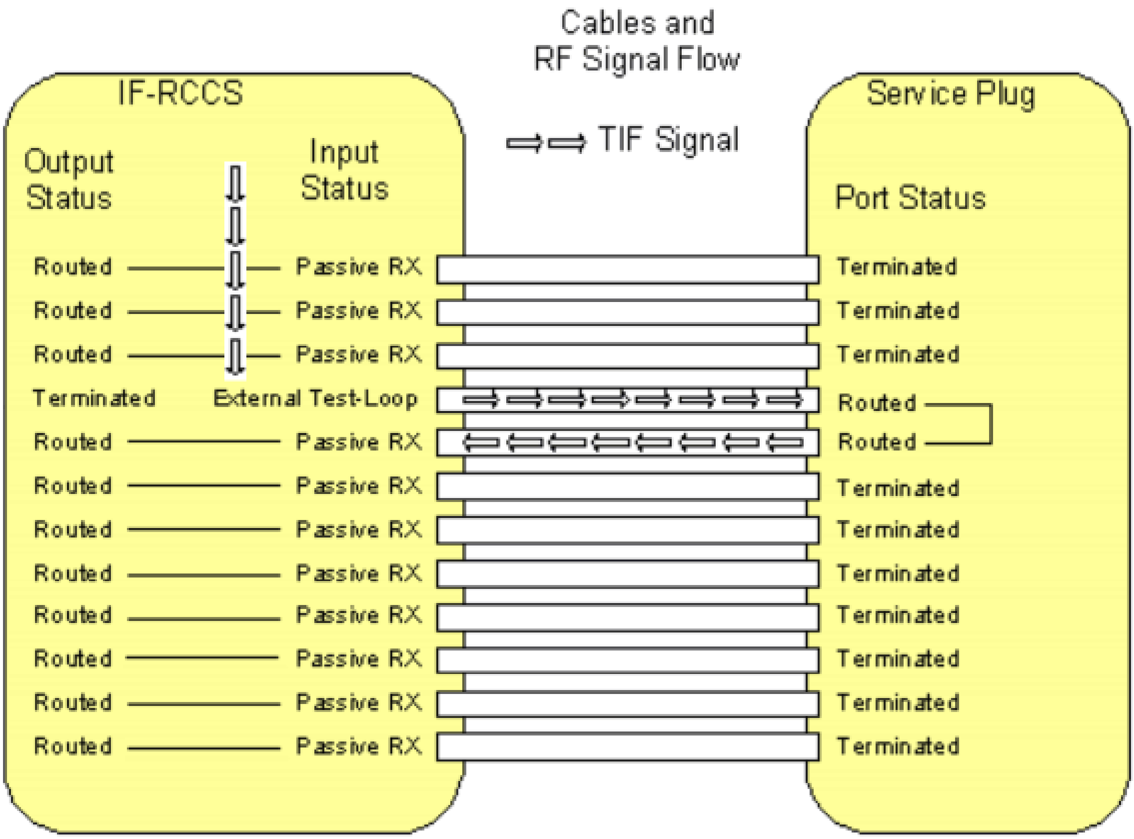
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

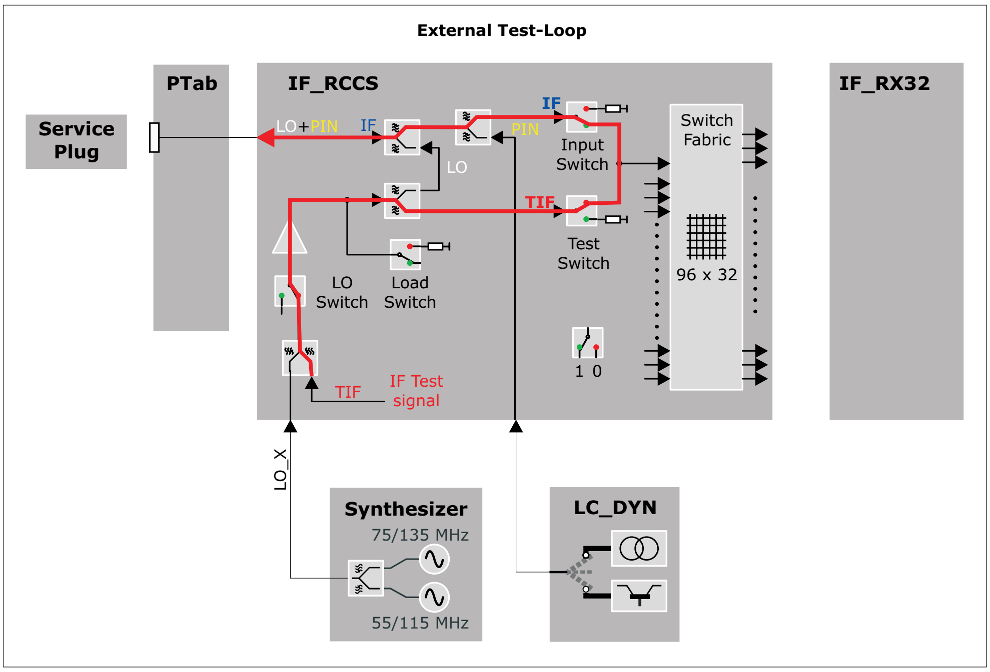
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 133: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 134: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step “n” a test signal is transmitted through the channel “n” and received through the other channels, i.e. all channels except channel “n” are routed back to the receiver. For both channels “n” and “n + 1” used in step “n”, the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not "TX" but indicated by a red "LO"
- the swapped pin is not "LO" but indicated by a red "HI"

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 135: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 136: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 137: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 138: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

> Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.11.2
LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.

- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.

- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.

- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

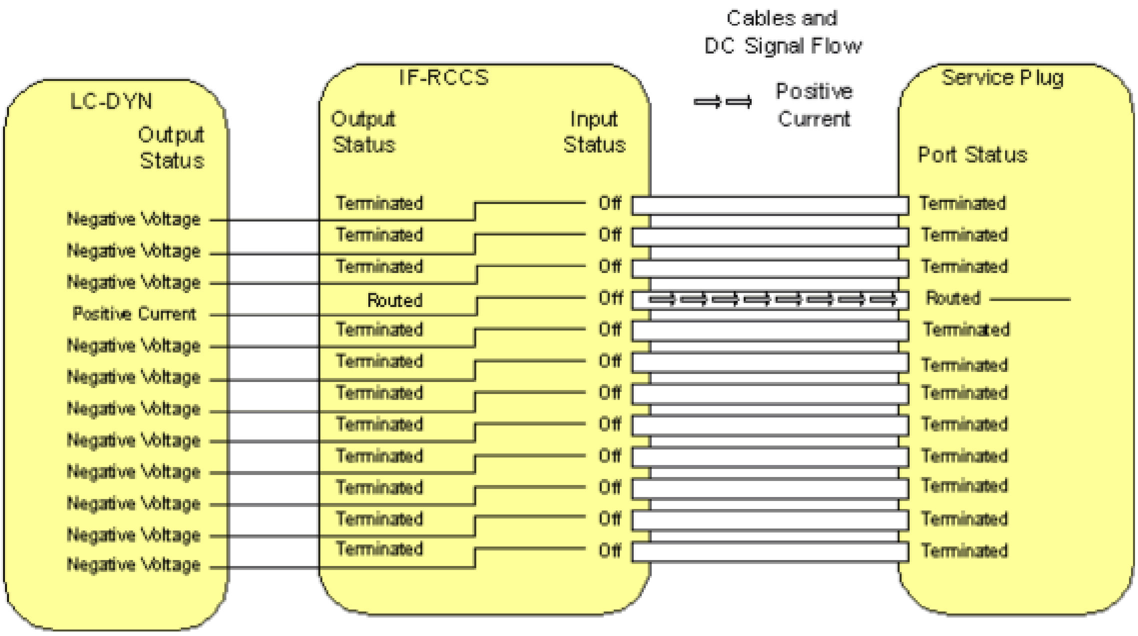
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 139: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

> The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

> The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.
>
> They show the principle, the actual layout in the test reports looks different.

Fig. 140: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 141: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 142: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 143: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 144: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.11.3 LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

Please note that the dBm output is not realized yet. Thus only ADC values are displayed.

Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

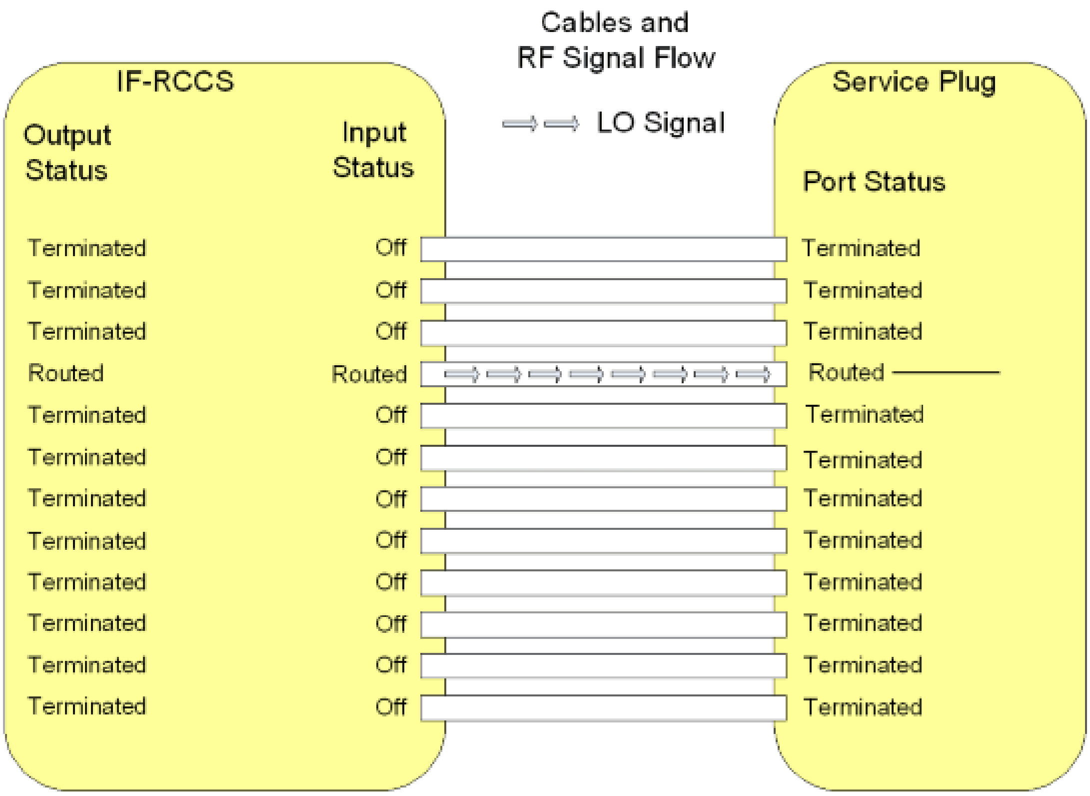
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

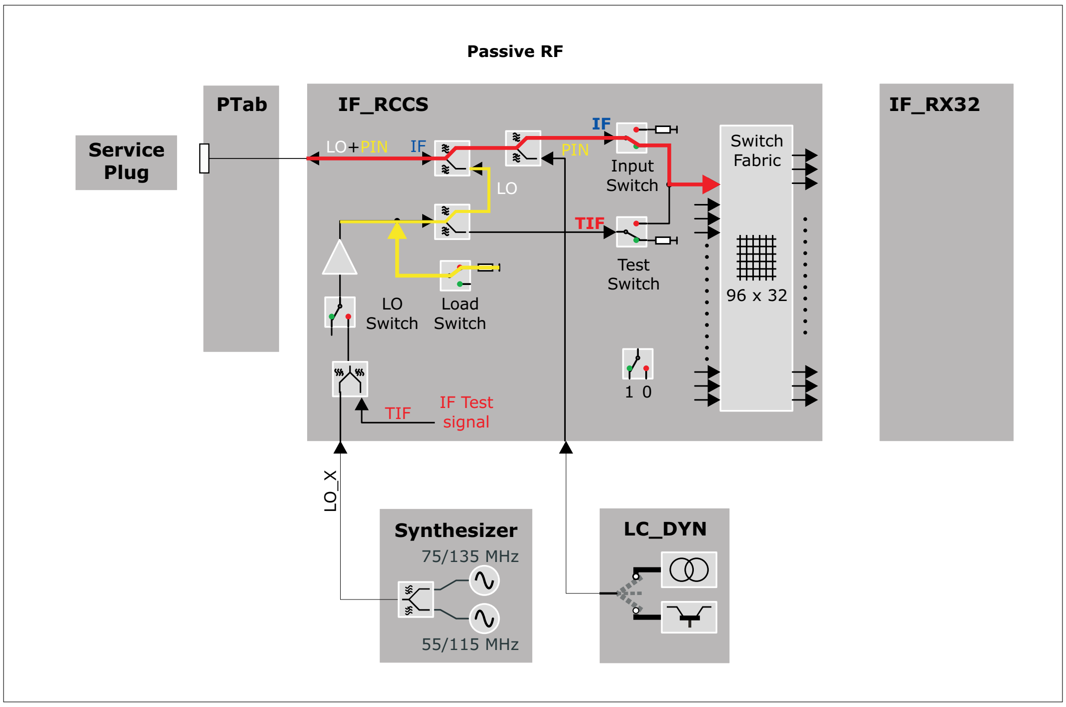
Corrected values are not yet read and checked.

Fig. 145: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 146: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

> i
>
> Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.11.4
LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> ℹ Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> ℹ The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.11.5
LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.11.6 LC Plug Coil Interrupt

Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.

2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hard-wired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.12
LC Plug 8

2.1.2.4.12.1
LC Plug IF

```
Help ID: 7D248033-55B2-41E0-8FF1-83CC91C7A4F6
Service Level: 7
```

General

The **IF Receive Signal** test checks the coupling (cross-talk) between the RX channels. It is also possible to detect broken and swapped cables between the IF_RCCS and the LC sockets.

> If tests will be performed with an undocked dockable table, the supervision of the Patient Table has to be changed before the table is undocked. In **Tools /Supervision** the supervision has to be set to “Virtual” before the table is undocked. Otherwise the test will not be started and an appropriate error meassge will be displayed (“preparation error”).
>
> Before the table is docked again this setting has to be re-set to “Real”, then the table can be docked. Afterwards the MR Scanner has to be rebooted manually.
>
> Follow the instructions of the test pop-ups for this.
>
> For testing the sockets at the patient table or the interconnections it is not necessary to set the supervision to virtual as long as the table is docked.

Signal Path

The following signal paths and signals are used:

- The TIF signal from the TX-Box is routed via the IF_RCCS to one pin of the Service Plug.

  This is done by switching the corresponding LC input multiplexer of the IF_RCCS to the mode **External Test Loop**.
- Inside the Service Plug the signal is routed to the next pin of the plug, e.g. from pin 2 to pin 3.

  For this, the PIN diode signals of the two connected inputs (channel and its neighbor channel) are switched to current mode and thus short cutted.

  At the same time all the other PIN diode signals are switched to voltage mode and thus disconnected.

  That means the TIF signal enters the plug by one channel and has to leave it by the neighbor channel (wrap around for the last pin, 12 -> 1).
- All signals coming from the pins of the Service Plug, except the one that is used for sending the TIF signal, are routed to the corresponding receiver by switching the corresponding LC input multiplexers of the IF_RCCS to the mode **Passive RX**.

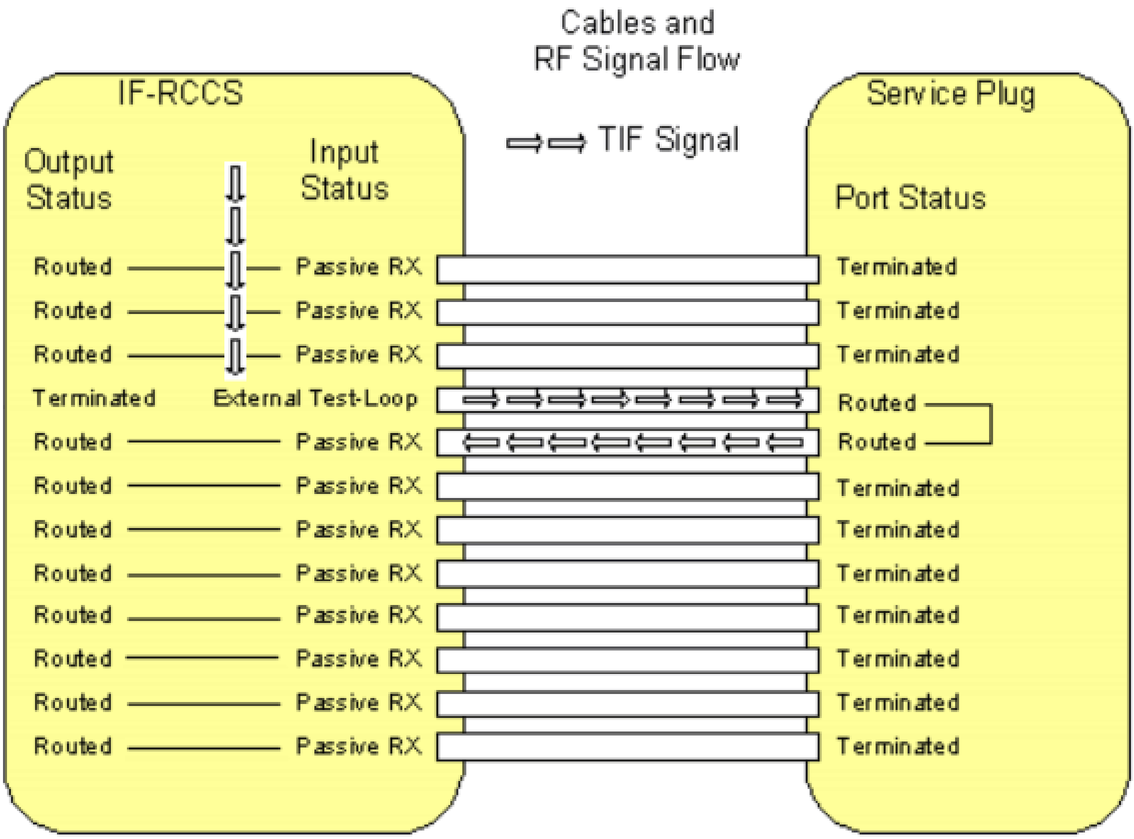
  This is done by activating appropriate nodes inside the IF_RCCS.

- The received signals of all Service Plug pins are evaluated. Only the one to which the TIF signal is routed must show a pulse, the remaining ones only noise.

  If cables from the IF_RCCS to the LC sockets are broken, no pulse is detected. Thus the predefined signal pattern is broken.

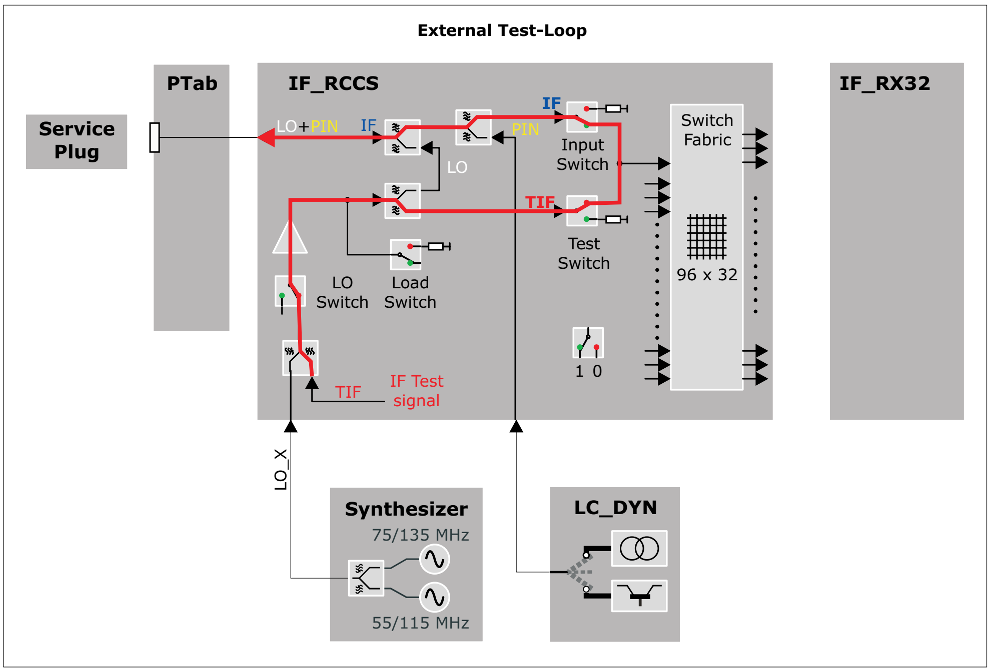
At the end of the test, the settings of the PIN diode signals are reset.

Fig. 147: Signal Path IF Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 148: Test Loop for IF-test (External Test Loop)



Test

The test is able to find the following errors:

- defective cables from the LC_Dyn to the LC connectors
- swapped cables (together with the LC Plug LO test)

The IF signal test for one coil socket is performed in 12 basic steps. In step “n” a test signal is transmitted through the channel “n” and received through the other channels, i.e. all channels except channel “n” are routed back to the receiver. For both channels “n” and “n + 1” used in step “n”, the corresponding PIN diodes in the Service Plug have to be switched to current mode by the LC_DYN.

Each pin of the LC plug is tested by performing an rf_loop measurement.

For each tested pin the ADC values received for each pin on the LC plug are calculated and stored in the evaluation result if the corresponding measurement was successful.

The ADC value is calculated from the complex value contained in the first line and the middle column of the stability raw data matrix.

The test results are reported in an output table as matrix where each line shows the results of a tested pin by displaying the received signal on each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining measurements even if one or more measurements could not be executed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

In the next figures the test result patterns for a socket with a variety of failures is shown. For a failure such as unwanted coupling usually two or four values arranged in a characteristic pattern will be out of specification. For a failure such as increased attenuation usually two values arranged in another characteristic pattern will be out of specification.

By analyzing the actual numeric values the operator has the option of drawing more conclusions about the nature of the failure. This is especially helpful in the cases where no characteristic failure patterns can be recognized by highlighting only the values which are not in specification. Therefore the test procedure also gives the operator the possibility of reading the actual values.

Swapped cables are indicated by

- the transmit pin is not “TX” but indicated by a red “LO”
- the swapped pin is not “LO” but indicated by a red “HI”

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 149: LC Plug IF - Unwanted Coupling between Channel 2 and 3

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	ERR	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 150: LC Plug IF - Unwanted Coupling between Channel 5 and 8

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	ERR	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	ERR	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	ERR	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	ERR	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Fig. 151: LC Plug IF - Unwanted Coupling between Channel 1 and 12 (!)

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	ERR
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO
Tx Pin 11	ERR	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

It is also possible to detect a higher resistance of a cable (e.g. higher resistance at a connection). In that case the attenuation is higher:

Fig. 152: LC Plug IF - Unwanted Attenuation in Channel 10

	Rx Pin 1	Rx Pin 2	Rx Pin 3	Rx Pin 4	Rx Pin 5	Rx Pin 6	Rx Pin 7	Rx Pin 8	Rx Pin 9	Rx Pin 10	Rx Pin 11	Rx Pin 12
Tx Pin 1	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 2	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 3	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO	LO
Tx Pin 4	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO	LO
Tx Pin 5	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO	LO
Tx Pin 6	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO	LO
Tx Pin 7	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO	LO
Tx Pin 8	LO	LO	LO	LO	LO	LO	LO	TX	HI	LO	LO	LO
Tx Pin 9	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO	LO
Tx Pin 10	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	ERR	LO
Tx Pin 11	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX	HI
Tx Pin 12	HI	LO	LO	LO	LO	LO	LO	LO	LO	LO	LO	TX

Due to the internal structure of the test here also the neighbour channel is detected as erroneous.

2.1.2.4.12.2
LC Plug PIN

Help ID: F26B86C1-A96D-45F9-859E-68C5C69B5BC6
Service Level: 7

General

The Service Plug provides test hardware for testing PIN-diode-voltage and current mode for all PIN-diodes of a coil socket via the **PIN diode current/voltage test**. In this way the check of the correct settings of the LC_DYN is included.

The following modes can be tested

- low current (about 10 mA)

  The PIN diode signal of the tested pin is set to high current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against the configurable specification values for high current.
- high current (about 100 mA)

  The PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The current signal received at the ADC of the service plug is checked against configurable specification values for low current.
- negative voltage (about -31 V)

  While the PIN diode signal of the tested pin is set to low current, all other pins of the Service Plug are set to voltage. The voltage signals of these pins are checked against configurable specification values for the negative voltage. The voltage signal of the pin that is switched to current contains an internal reference voltage and is checked against a configurable specification value for this internal reference voltage.

Signal Path

For each tested pin this involves three checks:

- **High current check**

  PIN selection of tested pin is set to high current, all other pins of the LC-plug are set to voltage.
- **Low current check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage.

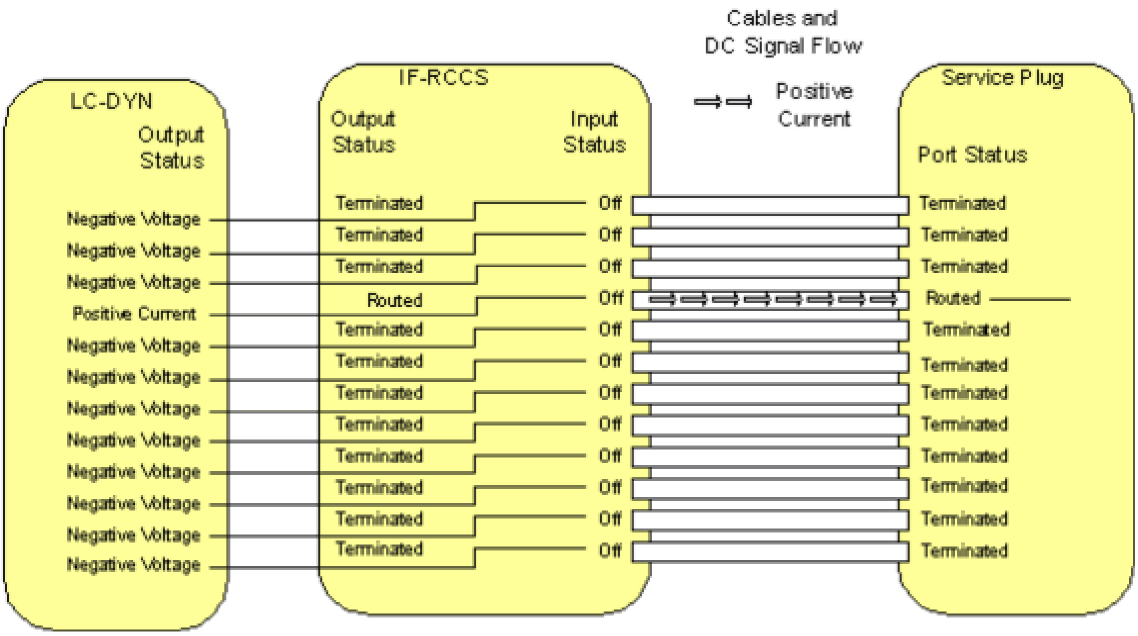
- **Negative voltage check**

  PIN selection of tested pin is set to low current, all other pins of the LC-plug are set to voltage. The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the PIN that is switched to low current, contains an internal reference voltage of the service plug and is checked against a specification value.

  This reference voltage is used to check the correct functioning of the ADC.

Fig. 153: Signal Path PIN Diode Test



Test

The test checks the correct voltages and currents of the signals for the PIN-diodes.

For each pin of the LC plug the PIN signals (low and high current, voltage) are tested by reading the signal received at the ADC of the service plug.

The PIN current/voltage test for one coil socket is performed in 12 basic test steps. In test step “n”, the current of channel “n” is tested. For channel “n” the LC-DYN has to be switched to current mode and for all other channels the LC-DYN has to be switched to voltage mode.

Then for each of these basic test steps 13 subtests are performed:

- The first subtest consists of testing the current value for the channel selected by current mode in the corresponding LC-DYN channel against specified limits. The channel is tested in two operating modes: high current and low current.
- The 12 following subtests consist of the testing of the negative voltage values for all 12 channels against specified limits. At the channel selected by current mode there is

obviously no negative voltage so an internal reference voltage is tested against specified limits.

For each tested pin this involves three checks:

- High/low current

  The current signal received at the ADC of the service plug is checked against the configurable specification values for high/low current.
- Negative voltage

  The voltages of the signals of all PINs are determined by the ADC of the service plug and checked against configurable specification values.

  The voltage value of the pin that is switched to low current, contains an internal reference voltage of the service plug and is checked against a configurable specification value.

  This reference voltage is used to check the correct functioning of the ADC.

The test results are reported in output tables as matrix where each line shows the results of a tested pin by displaying the received high and low current signals of the tested pin and the voltage signals of each pin of the LC socket.

The test tries to collect as much information as possible, e.g. it continues with the remaining checks even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

> The voltage ADC signal read for the tested pin contains the internal reference voltage of the service plug. For the internal reference voltage the corrected signal in [V] is currently not usable, only the raw ADC signal can be used. The deviation of the ADC signal from the internal reference voltage will be read from the service plug.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report

The test results for a socket are displayed as a table. For simplicity the first table does not show any numeric value. By analyzing the actual numeric values the operator has the possibility to draw more conclusions about the nature of the failure. Therefore the test procedure also gives the operator the possibility to read the actual deviation values for the tested currents and voltages.

In the next figures the test result pattern for a socket with a variety of failures is shown. The specified limits for the tested currents and voltages are defined for the hardware requirements of the MR system. Since the PIN networks in the Service Plug have a very simple structure they will be operational for a larger range of current and voltage values as requested for the MR system. Therefore even when the test procedure detects a failure the current or voltages may be sufficient for the operation of the PIN diodes of the Service Plug. If this is the case additional IF and LO signal tests need to be performed.

The examples below are taken from the NUMARIS/4 printouts for a better visibility of the error examples.

They show the principle, the actual layout in the test reports looks different.

Fig. 154: LC Plug PIN - Failure in the negative Voltage of Channel 7, most likely the LC_DYN is defective as some Current is clearly present

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	FAIL	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	FAIL	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	ROK

Fig. 155: LC Plug PIN - Failure in high Current Mode on Channel 10, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	FAIL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 156: LC Plug PIN - Failure in high and low Current Mode on Channel 8, most likely the LC_DYN is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	FAIL	FAIL	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

Fig. 157: LC Plug PIN - Channel 2 and 3 are swapped (cabling)

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

The following pattern is likely to mean no defect of the system but of the Service Plug itself. In this case the same pattern should appear with every socket.

Fig. 158: LC Plug PIN - Failure in Reference Voltage on Channel 5, most likely the Service Plug is defective

	High Current	Low Current	Volt. Pin 1	Volt. Pin 2	Volt. Pin 3	Volt. Pin 4	Volt. Pin 5	Volt. Pin 6	Volt. Pin 7	Volt. Pin 8	Volt. Pin 9	Volt. Pin 10	Volt. Pin 11	Volt. Pin 12
Tx Pin 1	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 2	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 3	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 4	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK	OK	OK
Tx Pin 5	OK	OK	OK	OK	OK	OK	FAIL	OK	OK	OK	OK	OK	OK	OK
Tx Pin 6	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK	OK
Tx Pin 7	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK	OK
Tx Pin 8	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK	OK
Tx Pin 9	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK	OK
Tx Pin 10	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK	OK
Tx Pin 11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK	OK
Tx Pin 12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	ROK

2.1.2.4.12.3
LC Plug LO

Help ID: E2038F82-B229-476C-83CE-4C1FCB2CB0BE
Service Level: 7

General

The **LO Signal Test** checks of the signal level of the LO signal.

For each pin of the Service Plug the following three checks are performed:

- The PIN diode signal is switched to current and the amplitude of the LO signal is determined by the ADC of the plug and checked against a specification value.
- The LO signal is switched to single ton mode to eliminate the amplitude fluctuations of the beat frequency of the two tone signal.
- For multi nuclei measurements the LO signal can be switched to different frequencies, but only the frequency of 1H is tested in order to check the cables.

After this a measurement is performed for every pin of the plug. All LO values are compared and the maximum deviation is checked against a specification value.

Please note that the dBm output is not realized yet. Thus only ADC values are displayed.

Subsequently, the signal levels of the channels will be calculated in dBm from the ADC values in conjunction with a reference signal.

Signal Path

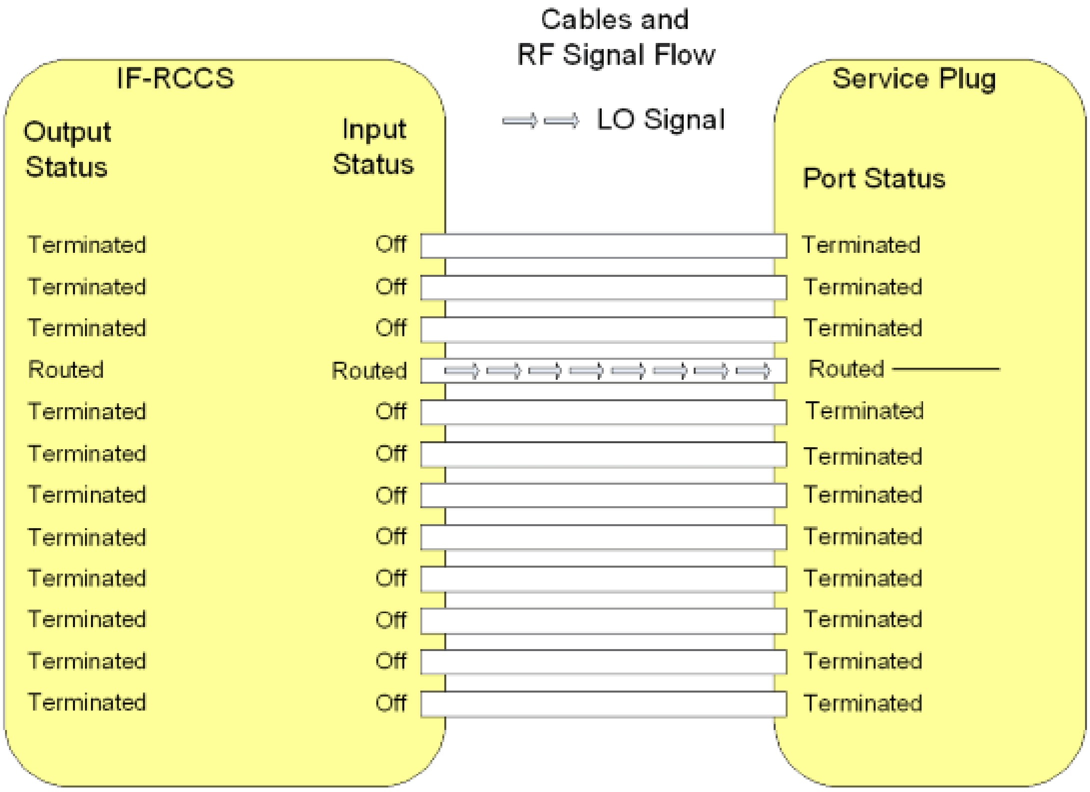
For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

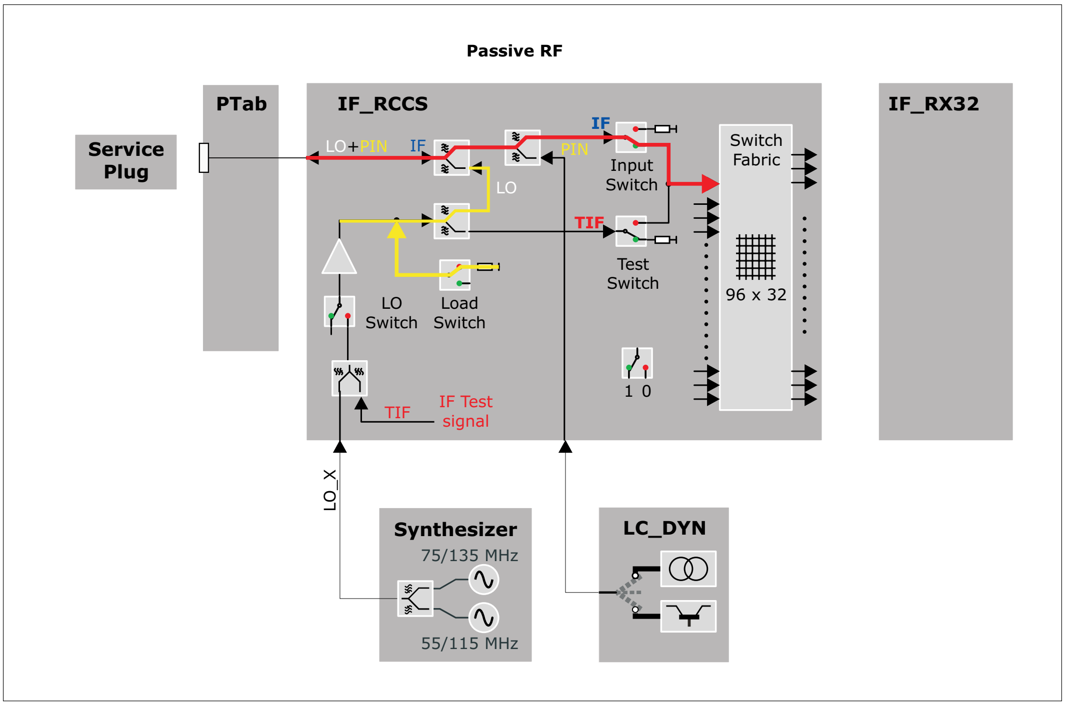
Corrected values are not yet read and checked.

Fig. 159: Signal Path: LO Signal Test



The following detailed view into the IF_RCCS shows the setting of the internal switches.

Fig. 160: Test Loop for LO-Tests (Passive RF)



Test

The test checks the maximum deviation of LO signals between two pins.

The test for one coil socket is performed in 12 basic steps. In step “n” an LO test signal is transmitted through the channel “n” and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel “n” used in step “n” the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

> Please note that the planned attenuation output is not realized yet. Thus only ADC values are displayed.
>
> Subsequently, the attenuation of the channels will be calculated from the ADC values plus a reference signal.

2.1.2.4.12.4
LC Plug 3 V Power Supply

Help ID: F23BDF4F-2773-4AD8-92CF-1600EAAB1FF1
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.

2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.12.5 LC Plug 10 V Power Supply

Help ID: 37591009-6FFC-4A74-A732-283FF00307FE
Service Level: 7

General

The **3 V and 10 V Supply Voltage Tests** checks the 3 V and 10 V supply voltages for the local coils.

The supply voltages are measured for each supply line separately. The maximum number of pins over which the 3 V is routed is six, with 10 V four (ERNI interconnections of the energy chain). Therefore the corresponding individual loads and sense networks are provided.

The Service Plug determines the voltage of each line under load or no load condition and checks these values against specifications. No load condition means an open line whereas load condition means that the line is switched to ground over a resistor simulating the power consumption of a local coil.

Besides other signals each coil socket is equipped with two contacts for the 3 V power supply of the local coil and sense lines for 3 V and ground. This enables the LC_DYN to raise the supplied voltage according to the power consumption of the local coil that is connected. This compensates the voltage drop in the cables that is dependent on the current. To test the integrity of the sense circuits a small voltage drop can be added into each sense line by software controlled switches in the Service Plug. The Service Software checks, if the 3 V power supply reacts with a corresponding small voltage shift.

> Contrary to the 3 V power supply the 10 V are not combined or divided in the ERNI interconnections. The 4 wires run from the IF_RCCS through inside the coil.

Signal Path

The power supply outputs of the IF_RCCS through the coil sockets are tested.

> The connection between the BC_DYN and the IF_RCCS including the backplane of the RFCEL cannot be tested!

Test

The following described test sequence is first performed for the 10 V and then for the 3 V supply voltage:

1. Before the test starts, the status of the Service Plug is evaluated. In case an error like overtemperature is already present, the test is aborted.
2. As next the test switches the internal power supply loads of the Service Plug off and off. This is visible at the 3 and 10 V LEDs on the front plate: if the loads are not activated the LEDs are on. As soon as the loads are activated during the test, the LEDs go off.

   The default and after-reset status for the 3 and 10 V load is "Load Off". This is always the status before the test starts. When the measurement is finished, the load is switched to the default status again to prevent the Service Plug from overheating.
3. With the third part of the voltage test the 3 V sense-lines and the reaction of the power supply are tested. Thus this sub-test is not available at the 10 V supply voltage test. An additional load is applied that forces the power supply to first increase the supply voltage in order to compensate the additional voltage drops from the higher current. This additional voltage (approx. 0.1 V) is detected and evaluated. After that the decrease of the voltage is checked with a different load to force the power supply to decrease the supply voltage by approx. 0.1 V. For the sense test the voltage of pin 1 is evaluated.

   For the 10 V, no sense lines are available that can be checked. Thus it is only checked if the BC_DYN is able to supply the socket with high power (approx. 60 % of the maximum power) and whether the voltage drop is below the specification or not.
4. As a last step for the 10 V supply voltage the “Load On” condition is tested. Additional loads are applied to the supply lines to test the maximum supply current and the reaction and stability of the supply voltage itself.

Evaluation

The result data of the test are read out from the Service Plug via the Service Software and finally displayed in the test report.

The test reads the voltages of all 3 V power supply lines through the Service Plug when the internal resistor in the service plug is

- not connected (load-off)
- connected (load-on)

Additionally all positive and negative 3 V sense voltages are evaluated. Every value is checked against the expected voltage, the deviation is calculated in % and show in the results-table. "Value" in the table stands for this deviation, the allowed range is +/- 6.0 %.

The test is in specification when all deviations are in the allowed range.

The debug mode offers additional tables with the following parameters:

- expected voltage
- measured voltage
- aquired ADV values

Conclusions

Please refer to the test report.

2.1.2.4.12.6
LC Plug Coil Interrupt

```
Help ID: 691695F1-EEB4-4B4C-AE4A-57F388373260
Service Level: 7
```

General

The **Coil Interrupt Test** checks the interrupt cabling for each coil socket.

In general the sockets at the patient table offer two types of interrupts:

1. **interrupt 1** (INTR1) is used for the coil recognition.

   This interrupt is triggered by the BC_Dyn when a local coil is plugged into a coil socket at the patient table.

   The two recessed pins close an open loop circuit, to power up the 3 V supply for the respective coil socket. This is done because hot plugging the coil to the system must be prevented.

   This interrupt is used for coil recognition purposes.
2. **interrupt 2** (INTR2) exists that can be triggered by the local coil itself while it is plugged to a socket.

   This interrupt signals the measurement control information about the coil, like

   - coil overtemperature
   - change of coil orientation within the magnetic field (head/foot orientation)
   - coil change event at an already connected Flex Coil Interface

As soon as an interrupt signal was received by the system, it tries to read data from all EEPROMs on all coil sockets. If a new coil is detected or an existing one was removed, the system recognizes this by the additional data coming from the EEPROM of the new coil or from missing data for a coil which was removed.

An interrupt signal is also sent while the scan is running and a coil unplugged. This causes the scanner to abort the measurement immedialely.

The Service Plug triggers both interrupts when it is connected to a coil socket:

1. The first interrupt is implicitly tested when the system recognizes the Service Plug correctly.
2. The second interrupt is submitted by the Service Plug when connecting, too (hardwired in the plug).

A special behaviour is used with TxRx coils. When they are plugged into the coil socket, the standard INTR1 interrupt is generated. But contrary to other coils also INTR2 is generated for safety reasons, too. In this way the correct function of the coil is checked.

As the Service Plug is considered as a TxRx coil from the measurement system, also the Service Plug has to send this signal. This is done automatically when plugging into a coil socket.

Signal Path

This test retrieves the interrupt status information of the RFCEL_Com component to check the complete signal path from:

PTAB socket - cabling PTAB IF_RCCS - IF_RCCS - cabling IF_RCCS - BC_Dyn - RFCEL backplane - RFCEL_Com.

With mobile patient tables the intermediate plugs at the docking port and are additionally checked.

Test

Similar to the coil recognition this test is implicitely performed when connecting the Service Plug a coil socket. No active test is started in TestTools.

Evaluation

For the evaluation of the test it is checked whether the expected second interrupt was "received" for the Service Plug and noticed in the measurement system.

Conclusions

Please refer to the test report.

2.1.2.4.13 PMU

2.1.2.4.13.1 PMU Transmit

Help ID: 66675C03-ADC6-4D77-AFDB-072452455CA1
Service Level: 7

General

The interactive **PMU Transmit** test checks the functions of the Physiological Measurement Unit of the system:

- the wireless transmission of the physiological signals from the PERU (Physiological ECG and Respiratory Unit) and the PPU (Peripheral Pulse Unit) to the PDAU (Physiological Data Acquisition Unit) in the RFCEL
- the data transfer from the PDAU in the RFCEL to the MaRS
- the data transfer from the MaRS to the host and thus to the Patient Data Display (PDD) at the magnet

The test activates the sending of defined pulses (test signals) from the PMU, the signal being displayed optically in the Physiological Display (syngo MR window on host), and the Patient Data Display.

The test result is manually/visually evaluated by the service technician.

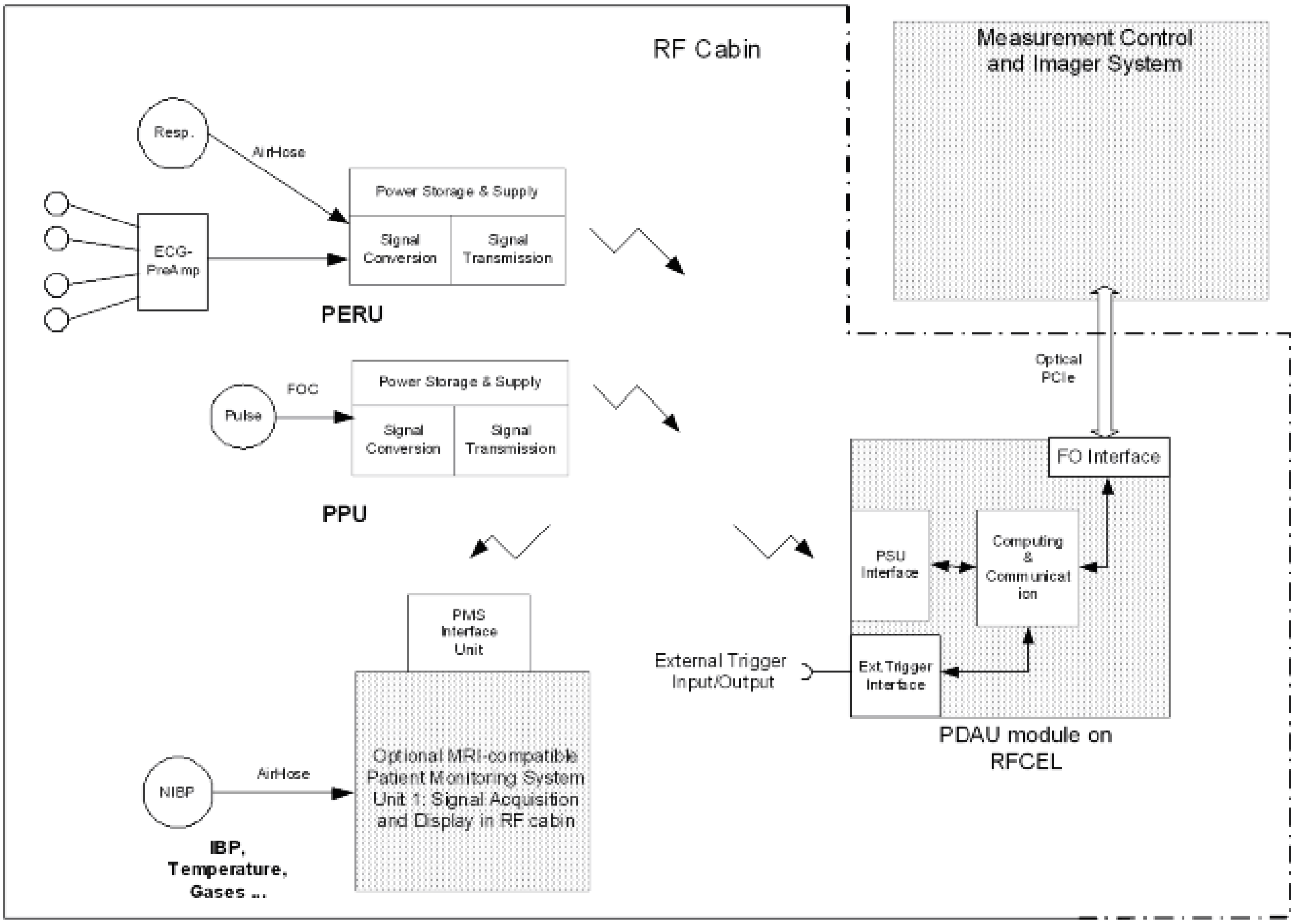
> Communication and status tests of the PMU are contained in the test category **Fast**.

Signal Path

At the PERU and PPU, test signals are generated and sent to MaRS via the PDAU module on the RFCEL_COM board.

The signal follow the paths shown in the next picture:

Fig. 161: PMU - PDAU Signal Flow



Test

Follow the instructions of the test pop-up.

The test starts with an instruction to place the PERU and PPU onto the patient table close to its head end. The test will not work in case the PMU sensors are not placed on the patient table. Without that the software trigger algorithm is not able to synchronize with the test signal and the test will fail. This table position is used from the software trigger algorithm for the learning phase of the ECG trigger signal (amplification receive signal, amplitude of ECG signal).

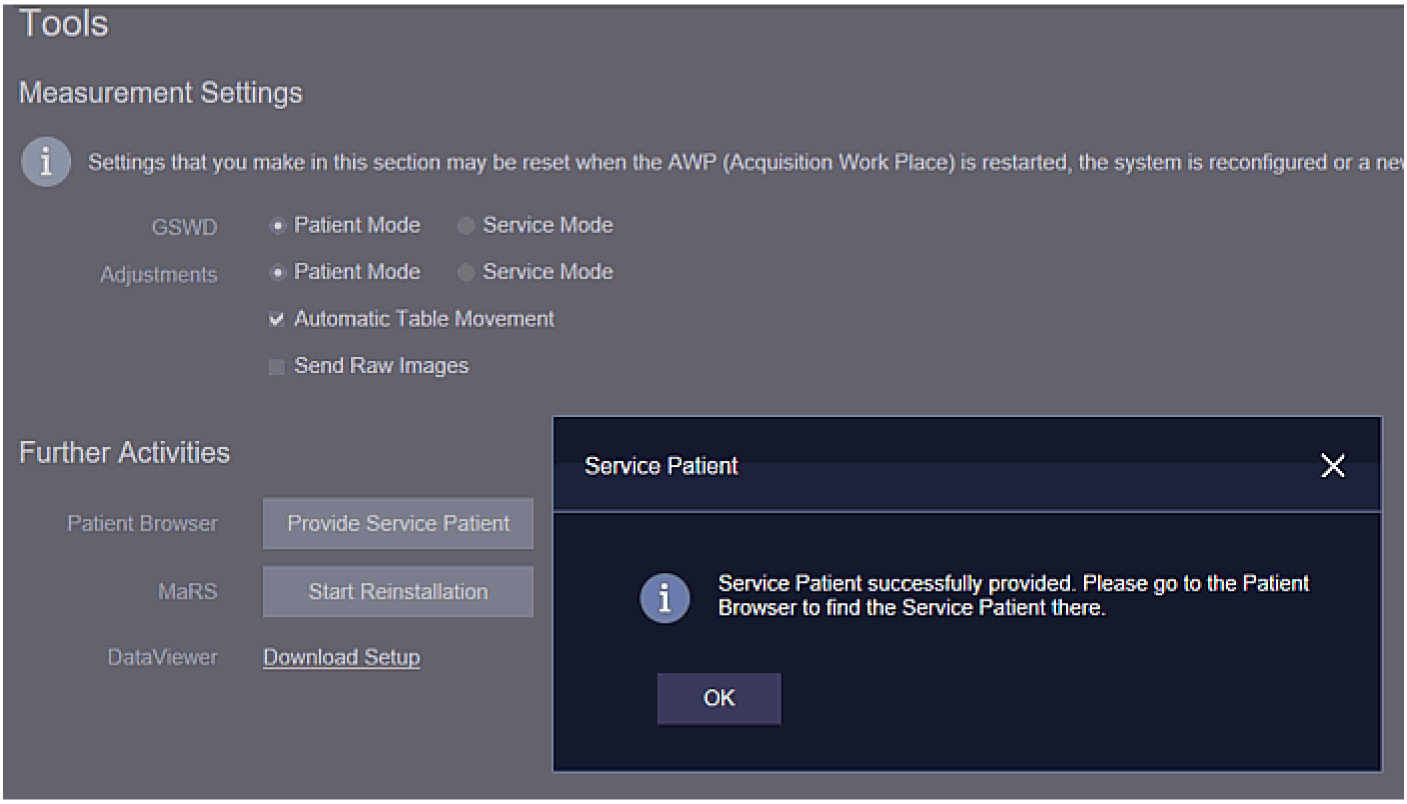
Depending on the tests started before the PMU test, this behavior may not be recognized by the user. For instance, if the interactive patient table test runs before, the table top could be already completely out of the magnet bore and the PMU test will work as long as no defect is present.

Before the test is started, a clinical patient or the service patient has to be registered. Otherwise the physiological display is not accessible.

For the activation navigate to **Tools button / Tools / Further Activities / Provide Service Patient**.

Afterwards register the Service Patient in the patient browser and fill in or confirm all undefined settings until the **Exam** button is released.

Fig. 162: Activating the Service Patient



The setup of the Physiological Display is as follows, the popups are not displayed simultaneously, but displayed here for easier instructions. The popup windows appear when you approach the white triangles, respectively click on them.

Fig. 163: Software Setup PMU Transmit Test

After the confirmation of this dialog, the test activates test signals with defined pulses in the PERU and PPU. The signal is transmitted from the sensors to the PDAU.

A physiological submenu is shown on the display at the magnet and on the Physiological Display at the console. After an initial set-up time, the test signals will show up in the Physiological Display on the host and on the Patient Data Display at the magnet.

The user is then queried through a dialog whether the test was successful or not. After the confirmation of this popup, the test signal is switched off.

The results of the PMU check are printed in a report table containing the test result (ok, failed) from the user's interactive input.

For sporadic mis-functions of the ECG triggering start the test at different positions of the Patient Table, or e.g. by moving the table towards and through the magnet after the test has started.

After the trigger signal is displayed at the host and the Patient Data Display at the magnet, it is possible to move the patient table. Check the correct triggering by centering the PERU and PPU.

If the PMU sensors are moved too much towards the service end of the magnet, the connection might get lost as the antenna is at the patient end of the magnet.

Evaluation

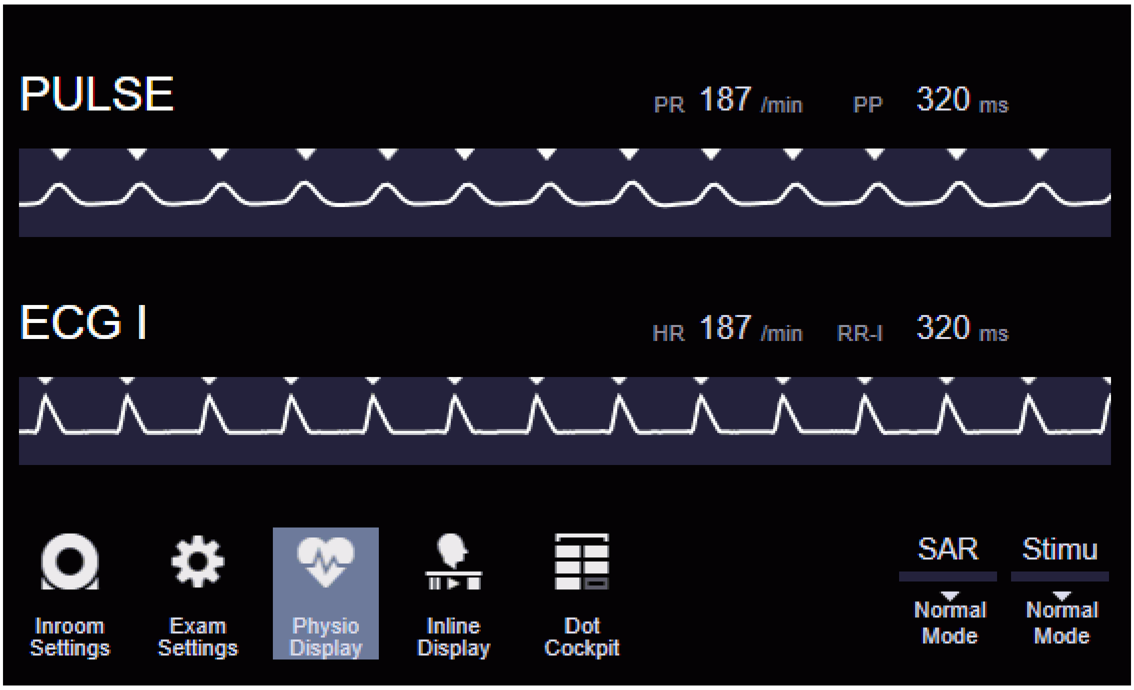
The evaluation is done interactively from the user as the system is not able to make an automatic evaluation of the PMU signal.

The test is successful if the following expected results are fulfilled:

- at the host the PMU signals are visible in the Physiological Display
- on the Patient Data Display at the magnet the signals are displayed

The Physiological Display appears as shown in the following picture:

Fig. 164: Correct Trigger Pulses of PMU Transmit Test



In the event of an electrode fault (not all electrodes are applied properly), the PERU generates an internal triangular test signal that is added to the analog path of the ECG channels. These are used for verification and service purposes. The test signals are only visible to the user if they are requested.

In the event of an application fault (e.g. finger adapter is not attached to a patient's finger), the PPU generates a test signal as well.

Conclusions

Please refer to the test report.

2.1.2.4.14
PTAB

2.1.2.4.14.1
PTAB functionality

Help ID: 8F533EE0-DFB1-42E4-A367-CF7CCEE61C25
Service Level: 7

General

The **Ptab Functionality** test checks the switch functionality of the patient table, if required at different position of the table top.

Signal Path

The patient table sends the current status of the switches to the host. There the output at the Patient Data Display is generated.

Test

The interactive PTAB test checks switches and sensors of the table.

First a dialog box is shown to provide explanations for the test:

- Pressing **Cancel** will cancel the test before it actually starts
- **OK** will start the test

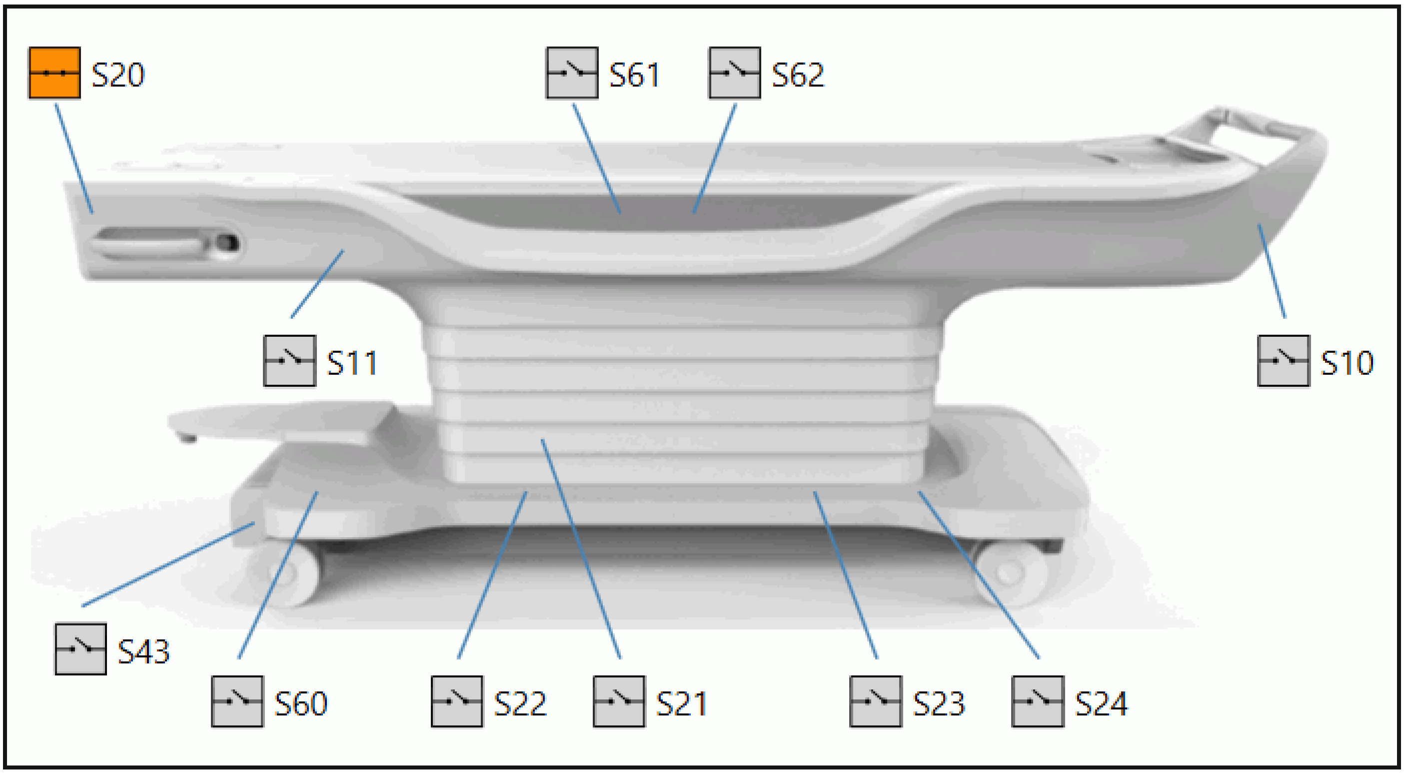
After this a new dialog box is shown.

After finishing the test, the user has to select **Yes** here if the tests were successful. Otherwise the user has to select **No**.

After the test is started, the user can perform actions on the patient table such as swinging up and down the side railings, pressing the emergency button, or pressing the table stop.

The Patient Data Display shows the switches of the patient table in a graphical display. Moving the table and parts of it (holder, table top, emergency button, and so on) will activate or release these switches. At the magnet display, the changing states of the corresponding table switches are displayed. If the switch is closed (activated) the display of that switch will change to an orange color for easier identification.

Fig. 165: TestTools Test of Patient Table Switches



> The displayed picture is independent from the actually installed table. At systems with a fixed table only some of the switches can be tested, of course.
>
> Distinguishing between the normal mobile table and a motorized one is not possible within TestTools at the Patient Data Display.

In this way the functionality is tested and the user has to check whether the correct status is shown.

Evaluation

The evaluation of the test has to be done by the tester.

The program exit status depends on the pushed button of the second pop-up window:

- "Success" if the **Yes** button was selected
- "Out of Spec" if the **No** button was selected

Conclusions

Please refer to the test report.

2.1.2.5
Multi Nuclei

Help ID: 3B9FD02E-AF67-4BE4-A315-518F1FC4E4C9
Service Level: 7

General

Purpose

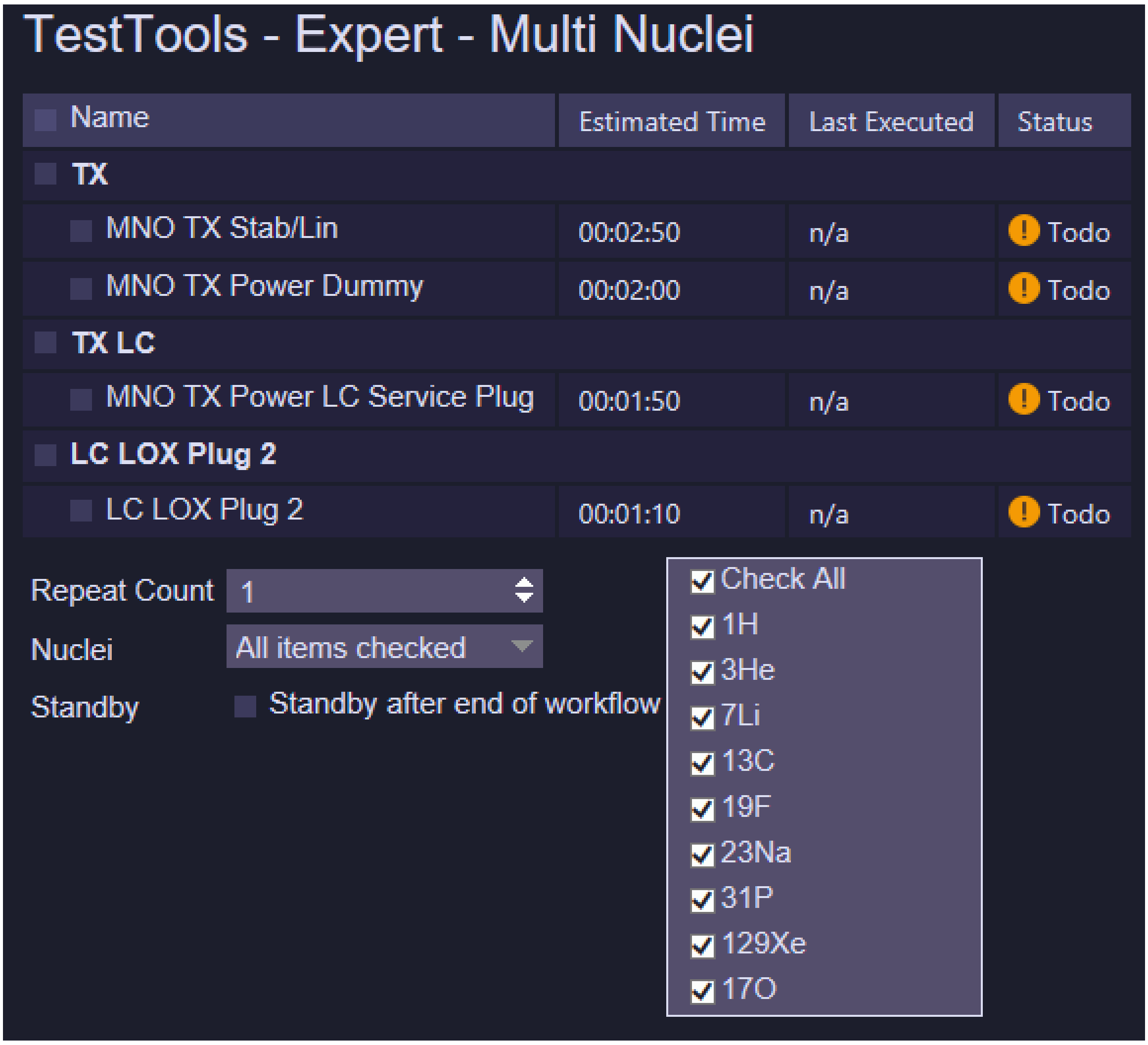
In this test category the tests are collected necessary to troubleshooting problems at system that are equipped with the Multi Nucleus Option (abbreviated MNO).

Basically only the transmit chain (RFPA/TX-Box, cables, coil sockets) is tested here with frequencies that can be specified in the user interface.

> Multi Nuclei tests are only available when the system is equipped with the Multi Nucleus option (MNO/MKO, spectroscopy, X-nucleus).

> Please note that for systems with the 2-channel parallel transmit option (a different TX-Box) the numbering of the RFPAs changes.
>
> There RFPA1 and 2 are internally used for the TX-Box whereas the broad band amplifier (“Dressler”) is named “RFPA3”.

Fig. 166: Example of TestTools "Multi Nuclei" Expert Mode



Nucleus Selection

The nucleus selection in displayed below the test selection. Here the nucleus to be tested can be selected. This selection has to be performed before the tests are started.

The selection in the popup is free, means any combination of selected or unselected nuclei is allowed.

Every nucleus corresponds to a certain frequency which will affect, e.g., the automatic selection of the correct RFPA / TX-Box.

> It is necessary to select the nucleus/frequency prior to the start of the test. If the desired nucleus is set, it will stay until it is changed again, or "**Default**" is pressed.
>
> This should be remembered during troubleshooting.

The display of **Estimated Time** is not corrected if more than one nucleus is selected. Instead, only the run time for a single nucleus is shown.

To speed up testing it is also possible to selecet "Check All". In that case all available nuclei are tested one after another.

The following tests use all nuclei in any case, independent which nucleus is selected:

- MNO Power Dummy
- MNO Power LC Service Plug

Nuclei

The nuclei 3He, 19F, 7Li, 13C, 23Na, 31P, 123Xe, 17O, and of course 1H can be used for customer measurements and thus also for tests.

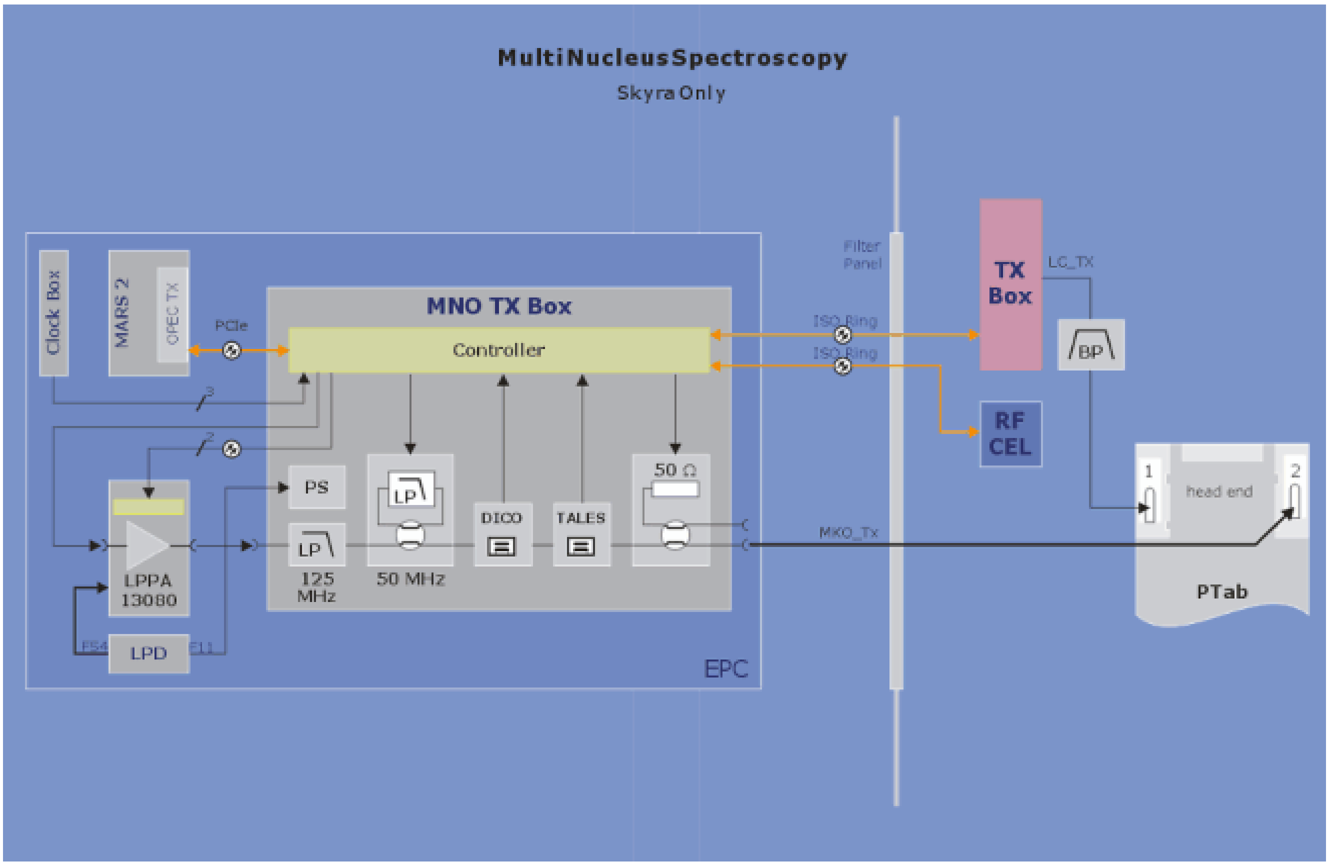
The usage of 1H is for the MNO RFPA not released, the results may be out of specification!

Overview Multi Nucleus RFPA

Although the following graphic shows "Sykra only" it is still valid for MAGNETOM Vida.

The following picture shows an overview of the used Multi Nulceus RFPA (Dressler):

Fig. 167: *Overview Multi Nucleus RFPA*



Hardware Differences

Th hardware uses the early conversion of the received MR signal to an intermediate frequence of about 10 MHz.

This conversion is performed in the LNCs (Low Noise Converters) that are located in the local coils (LC) and in the TX-Box for the Body Coil. Every LNC processes the signals of two LC elements. The two signals are each mixed with a different LO frequency (e.g. for 3T: 115 MHz and 135 MHz) which results in two IF signals with frequencies of about 8 and 12 MHz. Both IF signals are transmitted over one cable from the LNC through the IF-RCCS to the IF-Receiver. Inside the Receiver both IF-signals are separated and digitized.

The multi nuclei RX concept is a little bit different. It needs two LO signals:

- LO1 that is distributed over the first eleven pins of each coil socket
- LOX that is only available at pin 12

The LO1 signal is a one tone signal with a nucleus dependant frequency. It is used to mix up the frequency of an X nucleus to a frequency of about 160 MHz.

The LOX signal is a two tone signal of 75 and 85 MHz. A MNO coil doubles both frequencies to 150 and 170 MHz. The signals of two coil elements each with 160 MHz are now mixed with these two frequencies and distributed over one cable. The resulting frequencies are again about 8 and 12 MHz like for 1H.

2.1.2.5.3 TX

2.1.2.5.3.1 MNO TX Stab/Lin

Help ID: 21EAC207-ABA9-4130-8A90-5B1B85E116F8
Service Level: 7

General

The **MNO TX Stab/Lin** test checks similar parameters as the **Power Test** but this time the MNO RFPA is checked, except the comparison of the two transmit paths (0/90 degrees) as there is only one Tx path available. The test checks the linearity and short time stability of the transmitted multi nuclei power.

All data is acquired simultaneously within one measurement.

As the used sequence can be parameterized, it is possible to have this test in different measurement time lengths. This test is thus used in **Fast, Normal, Stability**, and **Multi Nuclei** if applicable (the Multi Nucleus capability is optional to the system).

Since the body coil is mono-resonant and supports only ^{1}H, the local coil TX path must be used for this test. Besides that, the test is comparable with the TX-Box power test.

> The display of **Estimated Time** is not corrected when more than one nucleus is selected. Instead, only the run time for a single nucleus is shown.

> In order to run this test a valid RF-characteristics for the MNO transmit path is required.
> Without that the test will fail in any case.

Signal Path

The output of the RFPA is switched to the dummy load (manually or automatically, depending on system type).

Besides of the the complete LC-TX-path is used.

Evaluation

The following values are determined for the selected nucleus

- maximal power
- linearity and phase stability of forward power
- stability of amplitude and phase of forward power
- reflected power for the two parts of the Body Coil (as a function of the transmitted power).

With 3 T systems a comparison of TALES and DICO values is not required because the TX-Box performs an online comparison of these values for security reasons.

Conclusions

Please refer to the test report.

2.1.2.5.3.2 MNO TX Power Dummy

Help ID: C4276CB5-D33B-4AFD-81A0-230568D7E4C7
Service Level: 7

General

This test checks the MNO RFPA for the maximum reachable power. The maximum possible amplitude is taken from the RF Chara (Tim systems).

As the component protection is switched off in able to request the maximum power from the MNO RFPA, the measurement has to be performed with a dummy load. The power is not sent to the local coil/Service Plug.

The maximum achievable power is determined by the RF characteristic that is measured within the TuneUp. For the power detection the DICO is used. The test can be carried out for all available nuclei separately or automatically all together but in separate measurements.

Three measurements are done with increasing amplitudes to approach the maximum transmit voltage.

> To speed up testing all available nuclei are tests at once. This is independent from the selected nucleus.
>
> All available nuclei are tested one after another.

> In order to run this test a valid RF-characteristics for the MNO transmit path is required.
>
> Without that the test will fail in any case.

Signal Path

The output of the RFPA is switched to the dummy load (manually or automatically, depending on system type).

Besides of the the complete LC-TX-path is used.

Evaluation

For the power test, the following values are determined. The test is carried out for all available nuclei.

- maximal power
- deviation of requested and delivered power
- reflected power

Conclusions

Please refer to the test report.

2.1.2.5.4 TX LC

2.1.2.5.4.1 MNO TX Power LC Service Plug

Help ID: 46FD554D-4FD9-4C6B-8C70-9DCC1B2ED0DD
Service Level: 7

General

The **MNO TX Power LC Service Plug** test is similar to the normal Power LC Test.

The component protection is switched on to limit the power in order to protect the coil/ Service Plug.

The test is carried out for all nuclei at once that are supported by the coil, the selection of the nucleus is ignored. This means for the Service Plug that all nine nuclei are tested.

Three measurements are done with increasing amplitudes to approach the maxiumum transmit voltage.

The maximum achievable power is determined by the RF characteristic that is measured within the TuneUp. For the power detection the DICO is used.

> To speed up testing all available nuclei are tests at once. This is independent from the selected nucleus.
>
> All available nuclei are tested one after another.

In order to run this test a valid RF-characteristics for the MNO transmit path is required. Without that the test will fail in any case..

Here the Service Plug has to be connected to the coil socket 2 at the patient table.

The transmit path is switched to the dummy.

Signal Path

The output of the RFPA is routed until the local Tx coil/Service Plug. In the Service Plug the 50 Ohm termination resistor is used.

If a coil is used a phantom has to be inside the coil and the coil centered in the magnet.

Evaluation

The following values are determined:

- deviation of requested and delivered power measured at DICO
- deviation of requested and delivered power measured at TALES
- ratio of reflected to forward power at DICO and TALES

Conclusions

Please refer to the test report.

2.1.2.5.5 LC LOX Plug 2

2.1.2.5.5.1 LC LOX Plug 2

Help ID: C8F1AEE5-538B-411C-BCBE-C1CF1C14AF75
Service Level: 7

General

The **LX LOX Plug 2** test checks of the signal level of the MNO LO signal, here maximum deviation of LO signals between two pins.

The test is performed in 12 basic steps. In step "n" an LO test signal is transmitted through the channel "n" and evaluated in the Service Plug. The value of the LO level is transmitted to the MR system via the I^2C bus. For channel "n" used in step "n" the corresponding PIN diode in the Service Plug has to be switched to current mode by the LC-DYN. For all other channels the LC-DYN has to be switched to voltage mode.

For each socket the result of this test consists of 12 values for the LO signal which all have to be within specified limits.

Additionally the maximum deviation of LO signals between 2 pins is calculated and checked.

The test results are reported in two output tables. The first table contains the read LO signal values, where each line shows the results of a tested pin. A second table reports the maximum deviation of the LO signals between two pins.

The test tries to collect as much information as possible, e.g. it continues with the remaining tests even if one or more checks could not be executed.

At the end of the test the PIN diodes are reset.

This test is available for each LO frequency. Note that the spectroscopy LO frequencies are available only in channel 12 of a coil socket. So in that case only 1 test step is needed.

Before the start of a LC Plug test the status of the Service Plug is evaluated. In case of an error condition like overtemperature or incorrect supply voltage the test is aborted. In that case please reset the Service Plug or unplug/plug it.

In case the 3 V LEDs of the Service Plug are off after a reset of an error condition (e.g. by pressing the **On/Reset** button) please unplug and plug in the Service Plug again.

For the description of the Service Plug itself please refer to the corresponding chapter of the online help.

Signal Path

For each pin on the LC plug the test is performed by setting the PIN selection of the tested pin to low current, all other pins of the LC plug are set to voltage.

The corresponding IF_RCCS MUX channel modes are set to imaging for the tested pin and to off for all other pins.

Then the LO signal is read from the service plug.

Corrected values are not yet read and checked.

Evaluation

The data is analyzed and displayed.

Further information can be found in the test and conclusion chapters.

Conclusions

Please refer to the test report.

2.2 Magnet & Cooling

2.2.1 Magnet Status

Help ID: C3A0AF34-464B-4F6A-AC8D-E3DEFE0E92BA
Service Level: 3

Magnet values and status information are displayed in the Magnet & Cooling platform.

Fig. 168: Magnet Status

Magnet Status Information

- Display of the helium level in liter.

- Helium measurement
  » **Start** for performing a helium measurement.

ERDU Information

- Display of magnet specific real time values
- Display of the last ERDU test.

MSUP Information

- Turns the EIS heater on/off
  » Drop down button for turning the EIS heater on/off
- Sets the pressure heater mode
  » Drop down button for turning the pressure heater on/off, or setting it into automatic mode (requires Full Access for RDIAG).
- Downloads all MSUP logfiles to the host
  » **Download** for downloading the MSUP logfiles to the host.
- Resets the LVQD flag
  » **Reset** for resetting the LVQD flag to “Waiting Trigger” (requires Full Access for RDIAG). Enables the MSUP to generate a new LVQD log file.

i Reset the LVQD before ramping the magnet to field after a quench. Otherwise there is no possibility to generate a new LVQD file in case of a quench.

Tab. 3 Abbreviations

LVQD	**L**ow **V**oltage **Q**uench **D**etection
ERDU	**E**mergency **R**un**D**own **U**nit
FCL	**F**ixed **C**urrent **L**ead
EIS	**E**xternal **I**nterference **S**creen

Heater / Carbon Resistor Values

Further Activities

- **Start**
  » Starting the helium measurement.
- **Download**
  » Downloading the MSUP logfiles to the HOST.

- **Reset**
  - » Resetting the LVQD to “Waiting Trigger” (requires Full Access for RDIAG). Enables the MSUP to generate a new LVQD log file.

Footer Bar

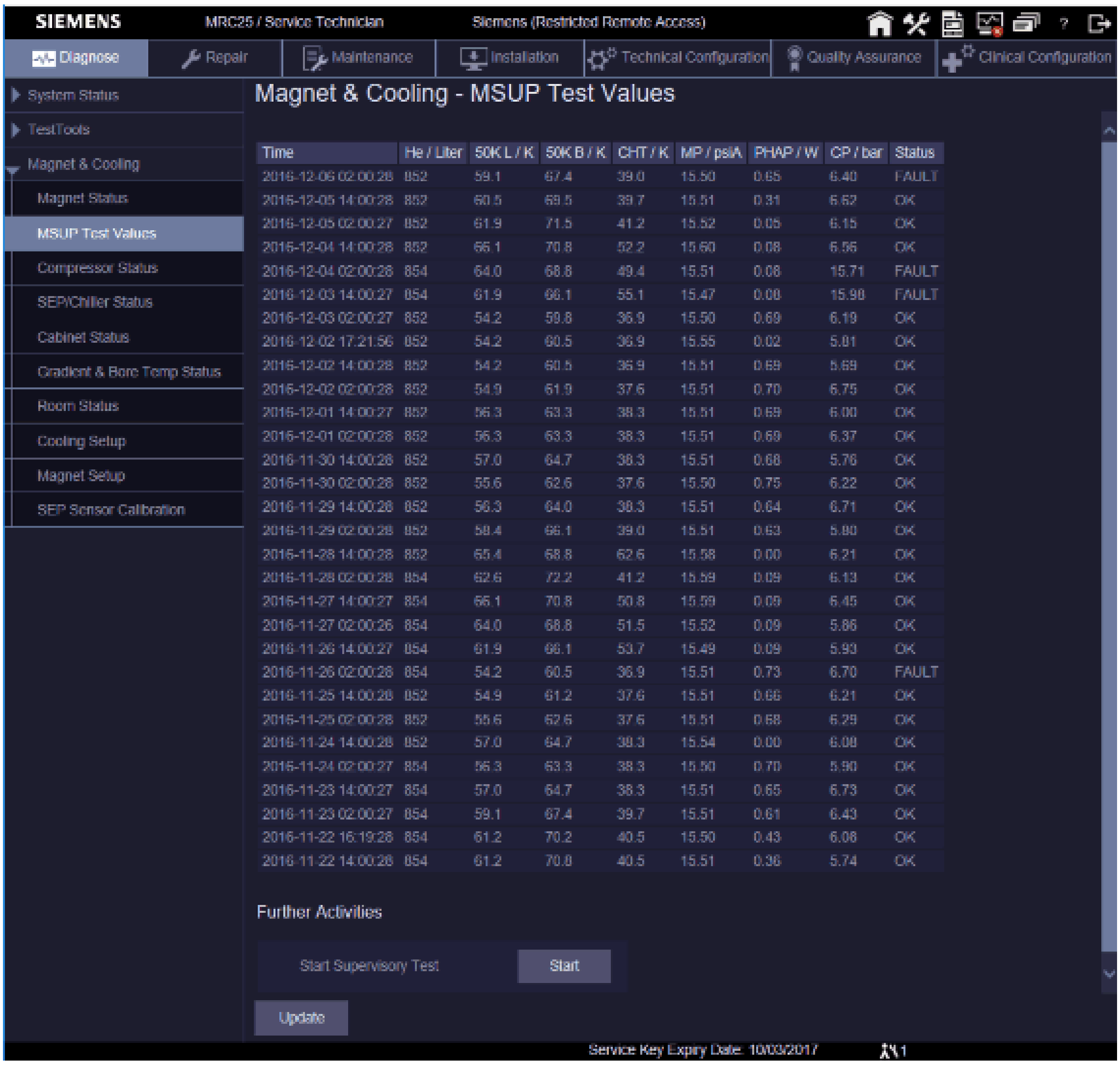
The footer bar provides the following buttons:

- **Save to HW**
  - » Saves the current values (requires Full Access from RDIAG).
- **Update from HW**
  - » Updates the page content.
- **Save as Default**
  - » Save the values as default values (requires Full Access from RDIAG).
- **Load Default Values**
  - » Load the default values (requires Full Access from RDIAG).

2.2.2 MSUP Test Values

Help ID: 2013C55D-57B0-4783-9834-4EFDA93D3DB2
Service Level: 3

Fig. 169: MSUP Test Values



- **Start**
  - » From MR console:Starting the supervisory test[1] (requires Full Access from RDIAG)

Following magnet values values are displayed:

	Description
Time	Date and time
He / Liter	Helium level (Liter)

1. Helium level measurement, EIS reset and battery test

	Description
50K L/K	Shield temperature **50K L**ink (K)
50K B/K	Shield temperature **50K B**ore (K)
CHT / K	**C**old **H**ead **T**emperature (K)
MP (psiA)	**M**agnet **P**ressure (psiA)
PHAP / W	**P**ressure **H**eater **A**verage **P**ower (W)
CP / bar	**C**ompressor **P**ressure (bar)
Status	Status OK / FAULT

2.2.3 Compressor Status

Help ID: 8EBE6E66-1D19-4B78-B409-C70A9962BD31
Service Level: 3

Status Information

The platform shows information about:helium gas pressure and temperature of the compressor

- Status information
- Pressure Information
- Temperature information

Fig. 170: Compressor Status

Further Activities

- Compressor
  - » Drop down list for turning off the compressor. The compressor will turn back within 10 min to 60 mn automatically.
- Compressor Static Pressure Test
  - » **Start**

    Compressor is switched off for stabilisation for about 60 min. The compressor static pressure is logged with time stamp. Compressor is restarted automatically.
- Download
  - » **Download**

    The compressor logfiles are downloaded to the HOST.

2.2.4 SEP/Chiller Status

Help ID: 9D442FC7-0BFA-4D99-A8E1-A13FFC61D155
Service Level: 3

The MR system can have a SEP cabinet or a Chiller with an IFP.

System with SEP

Fig. 171: SEP / Chiller Status on

Primary Water

Information about the primary water circuit.

Secondary Water

Information about the secondary water circuit

Sensor information

Information about the sensor

General

Information about ECO mode: on / off

ECO Mode

- System is on
- Triggers when the system is for 5 minutes idle
- Triggers when the patient table is in home position
- Reduction of pump speed
- Water temperatur set-point 20°C

Night Mode

- System is off
- Triggered by CAN heartbets
- Reduction of pump speed
- Water temperature set-point 24°C

System with Chiller

Fig. 172: SEP / Chiller Status 2

Chiller information

Information about the Chiller

General

Information about ECO mode: on / off

ECO Mode

- System is on
- Triggers when the system is for 5 minutes idle
- Triggers when the patient table is in home position
- Reduction of pump speed
- Water temperatur set-point 20°C

Night Mode

- System is off
- Triggered by CAN heartbets
- Reduction of pump speed
- Water temperature set-point to switch control mode.

IFP information

Information about the IFP

Free Cooling Unit (FCU)

The free cooling unit is optional.

Booster Unit

The Booster Unit is optional.

2.2.5 Cabinet Status

Help ID: B488FE68-800D-4AB5-A93A-59AC249696B7
Service Level: 3

Fig. 173: Cabinet Status

EPC

- Air temperature inside the cabinet
- Status information about the fans in the GPA and the EPC.
  » Information about the rotation of the fans are in the "Export...xml- file

RFCEL

- Air temperature information.
- Status information of the fan.

TX_Box

- Information about the water supply temperature..
- Information about the air temperature.

2.2.6
Gradient & Bore Temp Status

Help ID: 8B90CB62-ACA1-46DE-ABAA-74725BC15B5A
Service Level: 3

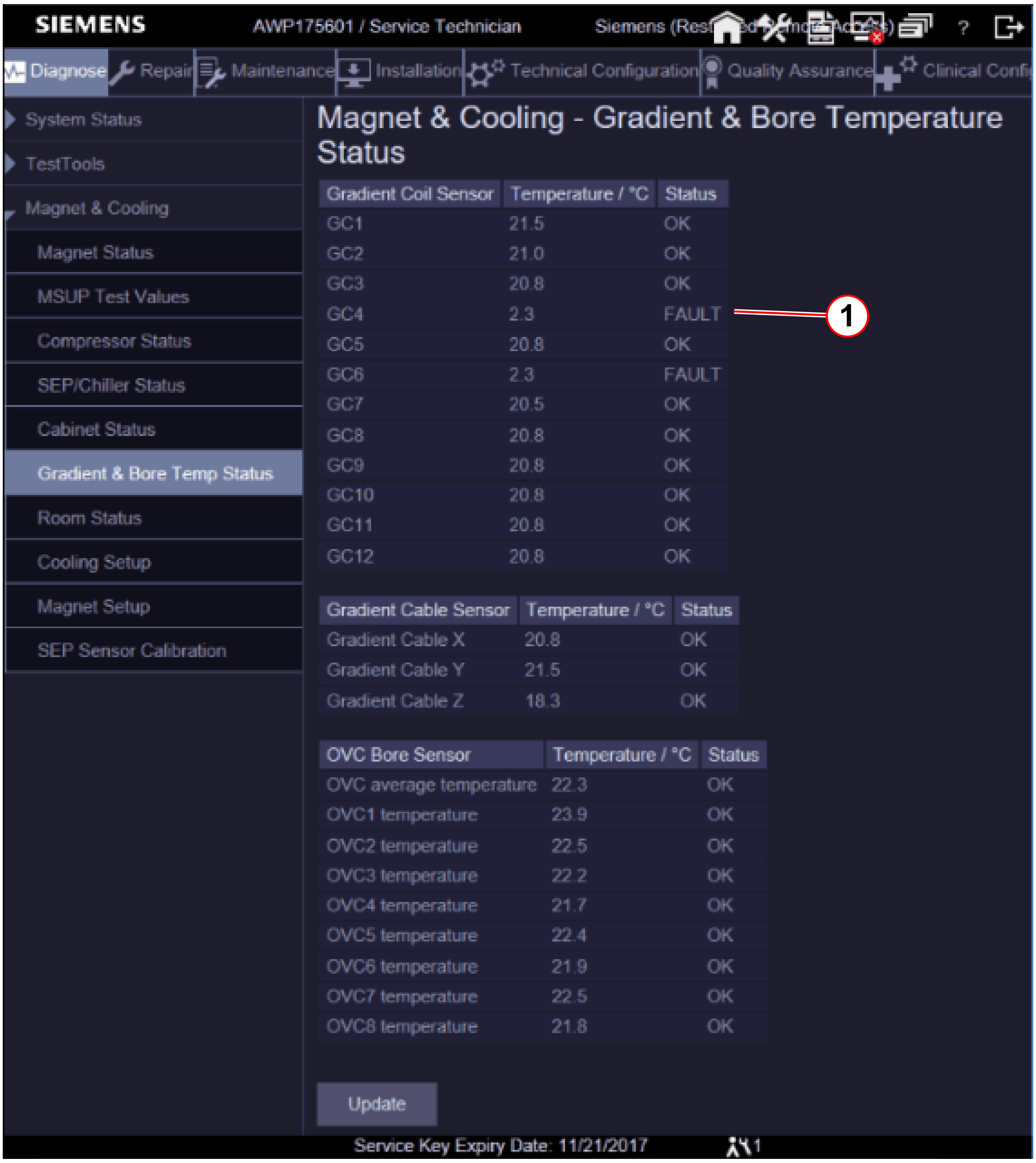
Fig. 174: GC / Bore Temperature

Gradient Coil Temperature Sensor

Temperature information about the gradient coil:

- Sensor (Sensor name)
- Temperature (˚C)
- Status (OK / FAULT)

Fig. 175: GC Temperatur Sensor



(1) Sensor Fault

Defective Coil Temperature Sensor

A defective sensor is indicated as FAULT in the status column.(→ 1/Fig. 175 Page 292)

Defective GC Temperature Sensors:

When booting the MR system, a test program for the RF_SAT runs on the host.This program reads the current temperature at all GC sensors and checks whether there are any errors.

The following temperature thresholds have been defined as fixed values in the program:

- T < 10˚C: NTC defective (Broken or missing cable).

- T > 140 °C: NTC defective (or short circuit).
  » Evaluation logic determines whether the failed sensor will permit continued operation of the scanner.

The following conditions have to be met:

- The 3 sensors at the gradient cable coaxial connectors must not be defective (GC1, GC2 and GC3).
- At least sensor GC4 or GC5 must work.
- At least sensor GC6 or GC7 must work.
- At least sensor GC8 or GC9 must work.
- At least sensor GC10 or GC11 or GC12 must work.
- Nine sensors must work in sum.
- A maximum of 3 sensors may be defective.
  » Redundant sensors are included in the SW code. Defective sensors generate a warning in the event log, and are identified in Magnet&Cooling / Cooling Status with FAULT. Permitted sensor failures do not result in system downtime.

Gradient Cable Sensor

Temperature information about the gradient cable.

The sensors are located near the connection plate on the Magnet.

Temperature warning threshold is set to 65°C.

OVC Bore Sensor

8 flat NTC sensors are in the OVC bore, behind the gradient coil.

RFCEL_SAT averages all 8 OVC sensors and 8 GC sensors every 649 ms.

These average values are used to determine a new value of B0 shift.

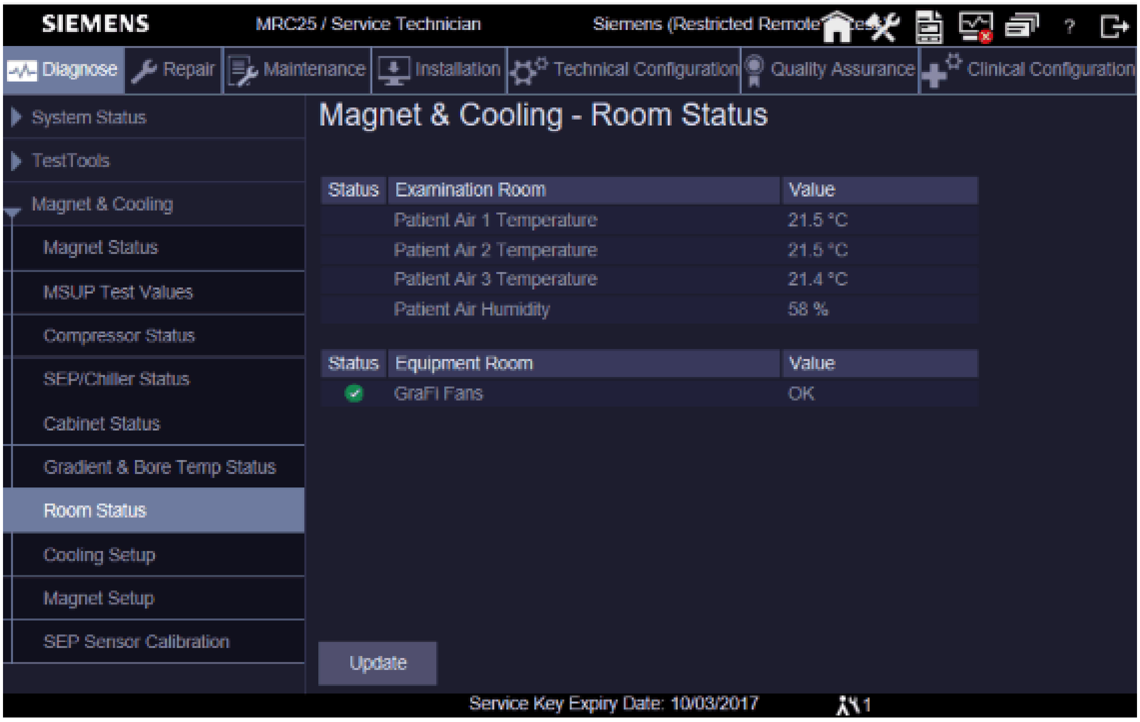
If more than 4 OVC temperature sensors are defective the B0 compensation is disabled.

A warning message is generated.

2.2.7 Room Status

Help ID: 50F6A29A-BCA8-446C-8E09-E301BB003188
Service Level: 3

Fig. 176: Room Status



Examination Room

- Information about the patient air temperature and patient air humidity.
  - Sensor of Patient Air 1, 2, 3 and Patient Air Humidity are on the board of the patient air fan.
- Optional: Information about the room temperature and humidity (only if Humidity Option is installed)

Equipment Room

- Status information about the Grafi fans
- Optional: Information about the room temperature and humidity (only if Humidity Option is installed)

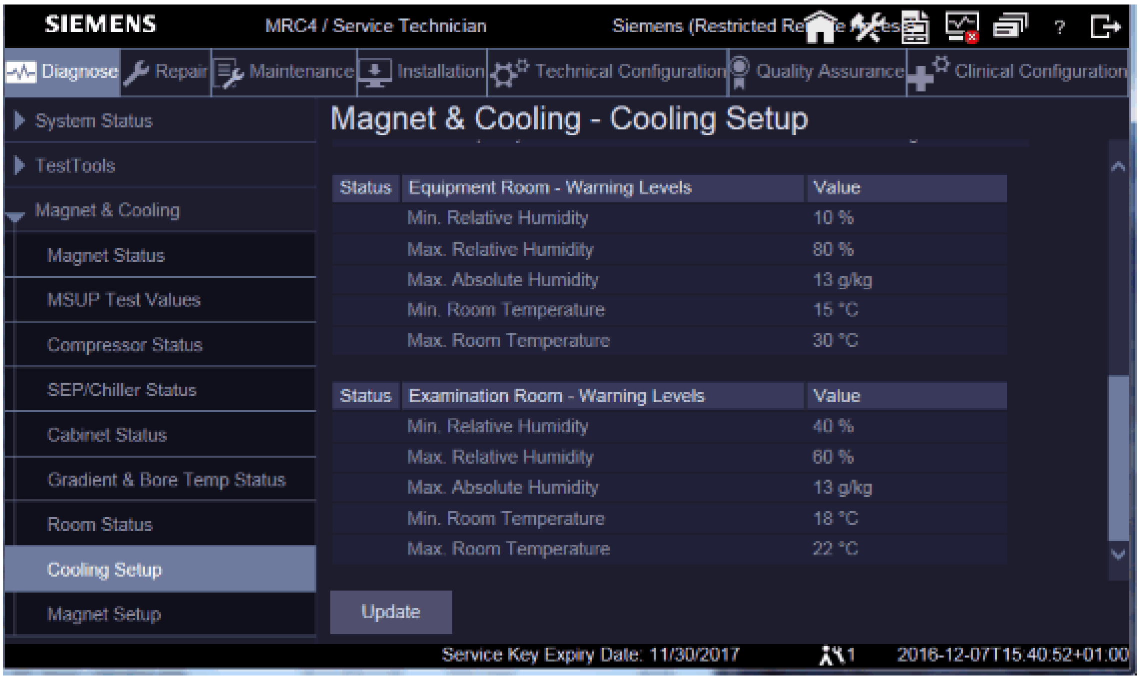
2.2.8
Cooling Setup

Help ID: 5FFA639C-7ACA-4605-B1B9-22E2B88E19E4
Service Level: 3

System with SEP

Fig. 177: Cooling Setup SEP

Fig. 178: Option - Cooling Setup



Platform for setup of:

- Primary and secondary water circuit of the SEP.
- Temperature warning levels of the EPC air temperature.
- Cooling Fluid Properties of the primary water circuit
  - » Enter 0 % or a value between 35 % and 40 %
- Warning levels of the Humidity option in the examination room and the equipment room.(Optional)

System with Chiller and IFP

Fig. 179: Cooling Setup Chiller

Platform for setting of:

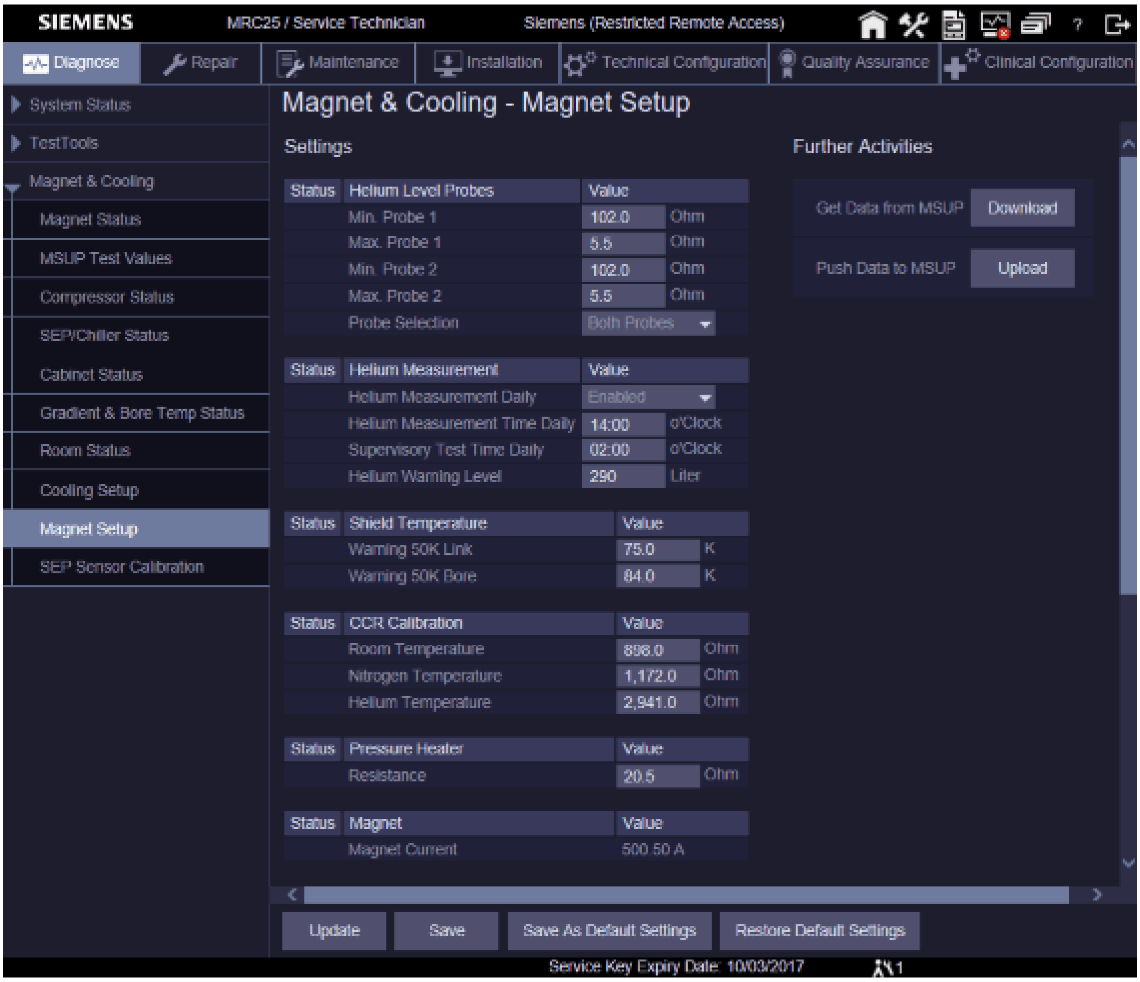
- Water temperature set point and water pressure.
- Temperature warning levels of the EPC air temperature.
- Cooling Fluid Properties
  - » Enter 0 % or a value between 35 % and 40 %
- Warning levels of the Humidity option in the examination room and the equipment room. (Optional)

2.2.9 Magnet Setup

Help ID: 12ECCC55-95CD-4F34-88FC-D033C45CA4B6
Service Level: 3

Magnet Settings

Fig. 180: Magnet Setup Settings



Helium Level Probes

- Probe resistance for 0% and 100%
- Probe selection

Helium Measurements

> System should measure the helium level twice per day.
> - Helium Measurement Daily
>   - » enabled
> - Helium Measuremnet Time Daily
>   - » e.g. 14:00 o’clock
> - Supervisory Test Time Daily
>   - » e.g. 2:00 o’clock
>
> The system should transfer the helium measurement twice per day via Auto Report.

- The **Helium Measurements Daily**
- Set Helium Measurement Daily to “enabled”
- Setting of the daily helium measurements
  - » 1st measurement at 2:00 am (Supervisory test)
  - » 2nd measurement at 2:00 pm (Helium measurement)
- Setting the Helium warning level threshold in liter.

Shield Temperature

- Setting of temperature warning levels of the 50k shield.

CCR Calibration

- Resistance for CCR sensors

Pressure Heater

- Resitance for the pressure heater.

Magnet Current

- Display of the magnet current.

Configuration

Fig. 181: Magnet Setup Configuration

Magnet Information

Information about magnet type, serial number.

MSUP Information

Information about MSUP serial number.

Alarmbox Information

Information about Alarm box serial number.

Further Activities

- **Download**
  - » Downloads the magnet setup/calibration data to the HOST.
- **Upload**
  - » Uploads the magnet setup/calibration data from the HOST to the MSUP.

2.2.10 SEP Sensor Calibration

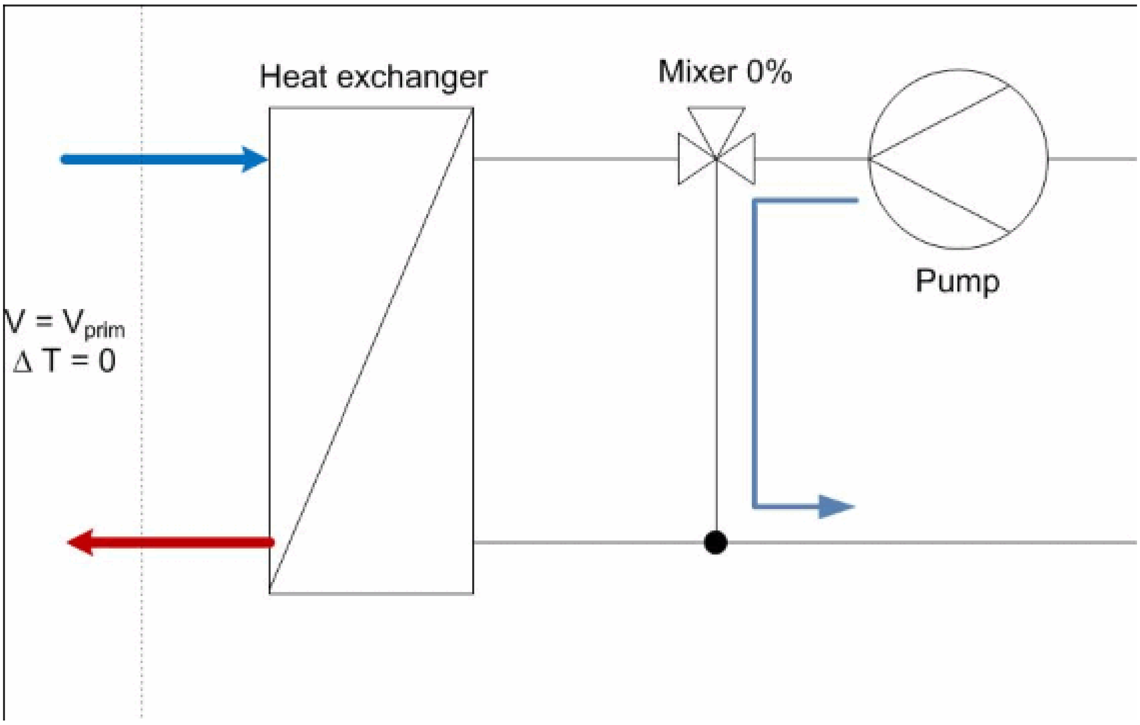
Help ID: C633113A-D189-4E11-8954-F608547D2765
Serviced Level: 3

The primary outlet temperature sensor has to calibrated during start-up, SEP controller exchange and sensor exchange.

Prerequistes

- The system is switched on.
- The primary water temperature difference is 0 °Celsius.
- The mixer will bypass the secondary water at the heat exchanger.

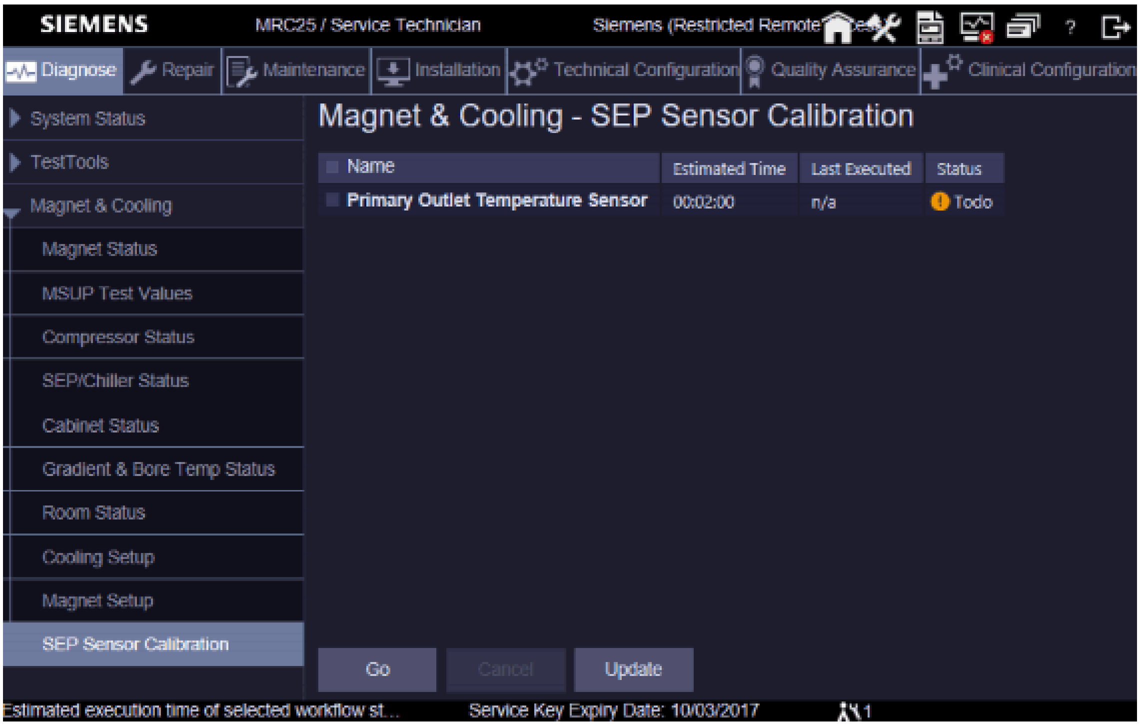
Fig. 182: Sensor Calibration



Workflow of the Sensor calibration

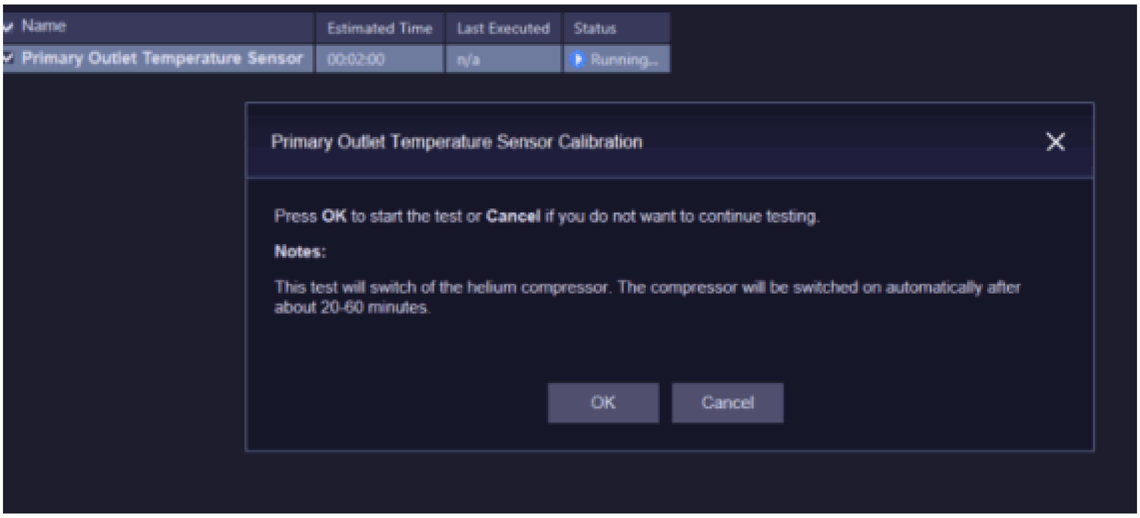
- Open the SEP Sensor Calibration platform.

Fig. 183: SEP Sensor calibration



- Start the workflow with **GO**.

Fig. 184: SEP sensor calibration



The compressor is switched off during sensor calibration and will be switched on automatically again after about 20 to 60 minutes.

- At the end the output of the calibration workflow will be displayed:.

Fig. 185: SEP sensor calibration

3.1 Coil Configuration

Help ID: AC53C5B9-B1D9-4B27-8555-01B6A2321912
Service Level: 1

Within the “Coil Configuration” you can manage your MR systems coils.

You have the possibility to add, edit, or delete coils in the “Coil Configuration”.

To register new coils, please plug the coils onto the patient table and press **Add Plugged Coils**.

To delete coils from you configuration use **Delete**

Only configured coils will be listed in the **Quality Assurance > Coil** workflow.

Fig. 186: Coil Configuration


MR Configuration - Coil Configuration

Name	Serial Number	Part Number	Revision	Registration Date
FlexLarge 4	1739	10835335	03	2016-09-01 10:06
FlexSmall 4	1731	10835327	03	2016-09-01 10:47
ShoulderLarge 16	1866	10500087	05	2016-09-01 10:56
ShoulderSmall 16	1860	10500088	05	2016-09-01 11:01
Spine 32	2173	10496545	09	2016-09-01 11:07
Body 18	1231	10835325	02	2016-09-01 11:27
Body 18	1617	10835325	02	2016-09-01 11:45
HeadNeck 20	2148	10496500	09	2016-09-01 12:15
HeadNeck 20 TCS	202	10835555	00	2016-09-01 12:33
Body	1006	10676584	00	2016-09-01 14:48
Breast 18 F	1148	10606916	02	2016-09-02 14:59
TxRx CP Head	1034	10606826	02	2016-09-02 16:42
Spine 72 RS	202	10835688	00	2016-09-08 08:10
Spine 72 RS	203	10835688	00	2016-09-08 08:36
Spine 72 RS	201	10835688	00	2016-09-08 09:10

To register new coils, please plug the coils onto the patient table and press [Add Plug

Add Plugged Coils

3.2 Shim

3.2.1 Shim - General

Help ID: 46A7A82A-C6D7-45B8-9C8C-BED652C9CBFF
Service Level: 3

General

Shimming

Superconductive magnets for MR imaging systems must have a homogeneous magnetic field at the required strength. To this end, a shimming technique is used to reduce field inhomogeneity to a minimum.

Underlying Physics

The primary objective of MR systems is to provide diagnostic images for medical purposes. Images of the body can be derived from the information obtained by exciting protons in a magnetic field using a radiofrequency signal. As the protons relax, they release their own radio signal that is detectable and provides information about the chemical contents of the body. The image quality depends on the homogeneity of the magnetic field: the better the homogeneity, the better the image quality. Shimming is an integral part of the magnet installation. It compensates for unit fabrication spread and environmental effects on the system homogeneity.

User selection

The Terminal Program is used for controlling the magnet power supply.

The following shows the correct setup of the Terminal Program.

Terminal Program Setup

Terminal Program must be configured and started **before the 3600 MPSU is switched on.**

3.2.1.2 Magnet Shim

Help ID: 86BF2F55-5481-4532-B078-DFCEBE88F075
Service Level: 3

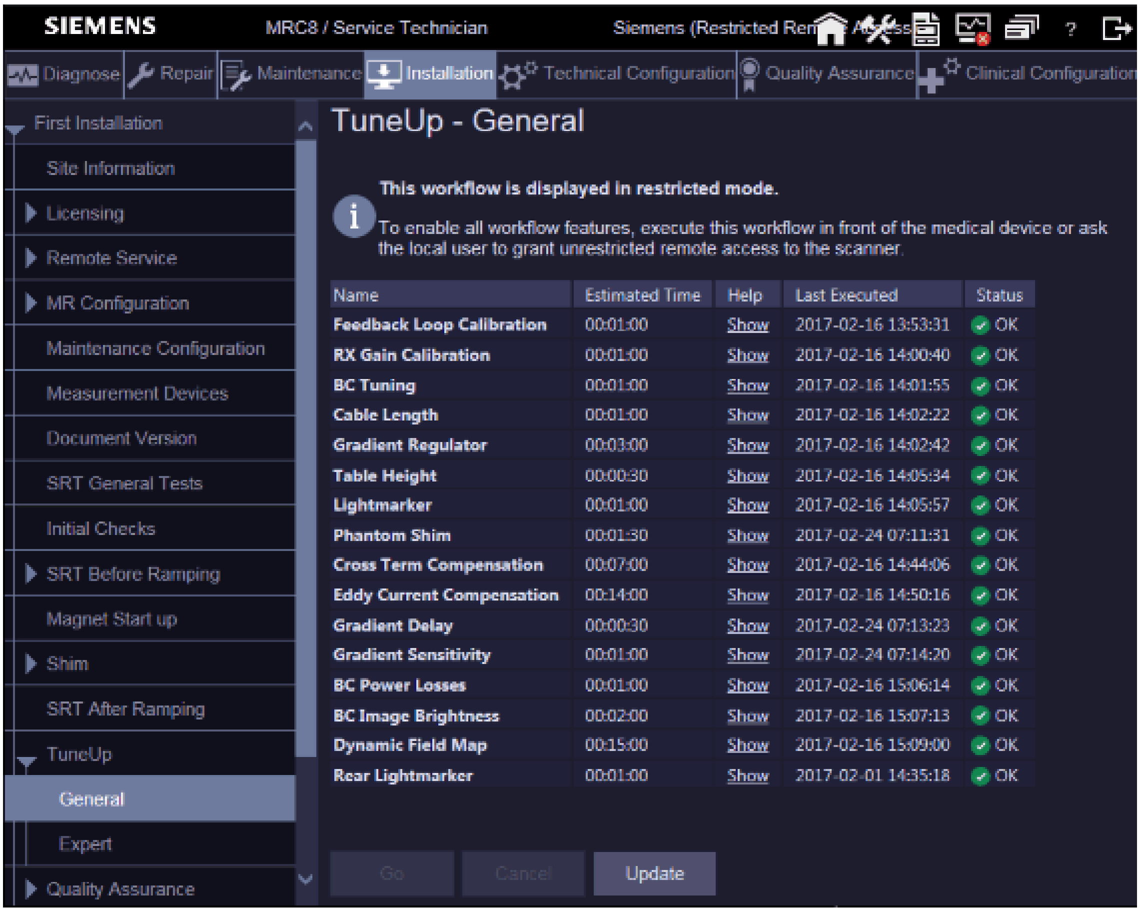
Prerequisites for the Shim Workflow

> **i** The Feedback Loop Calibration has to be done before starting the shim workflow.

- Select First Installation

- Select TuneUp / General
- Start **Feedback Loop Calibration**

Fig. 187: Feedback Loop Calibration

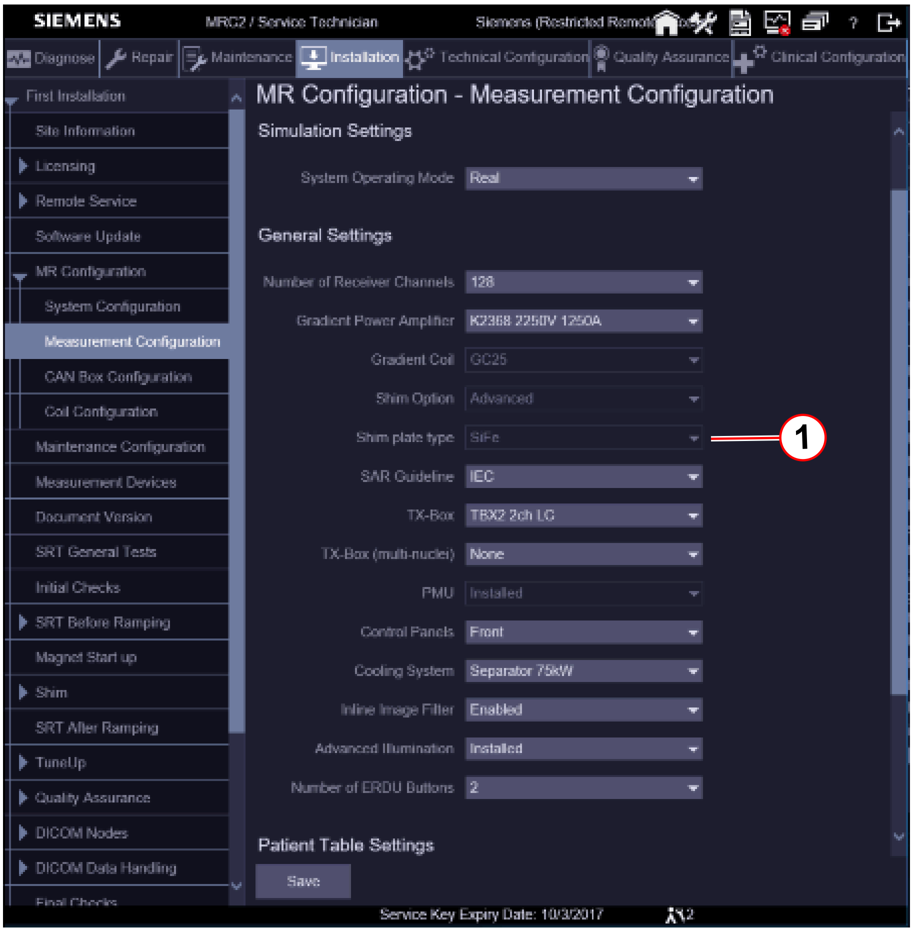


Name	Estimated Time	Help	Last Executed	Status
Feedback Loop Calibration	00:01:00	Show	2017-02-16 13:53:31	OK
RX Gain Calibration	00:01:00	Show	2017-02-16 14:00:40	OK
BC Tuning	00:01:00	Show	2017-02-16 14:01:55	OK
Cable Length	00:01:00	Show	2017-02-16 14:02:22	OK
Gradient Regulator	00:03:00	Show	2017-02-16 14:02:42	OK
Table Height	00:00:30	Show	2017-02-16 14:05:34	OK
Lightmarker	00:01:00	Show	2017-02-16 14:05:57	OK
Phantom Shim	00:01:30	Show	2017-02-24 07:11:31	OK
Cross Term Compensation	00:07:00	Show	2017-02-16 14:44:06	OK
Eddy Current Compensation	00:14:00	Show	2017-02-16 14:50:16	OK
Gradient Delay	00:00:30	Show	2017-02-24 07:13:23	OK
Gradient Sensitivity	00:01:00	Show	2017-02-24 07:14:20	OK
BC Power Losses	00:01:00	Show	2017-02-16 15:06:14	OK
BC Image Brightness	00:02:00	Show	2017-02-16 15:07:13	OK
Dynamic Field Map	00:15:00	Show	2017-02-16 15:09:00	OK
Rear Lightmarker	00:01:00	Show	2017-02-01 14:35:18	OK

Passive Shim Workflow

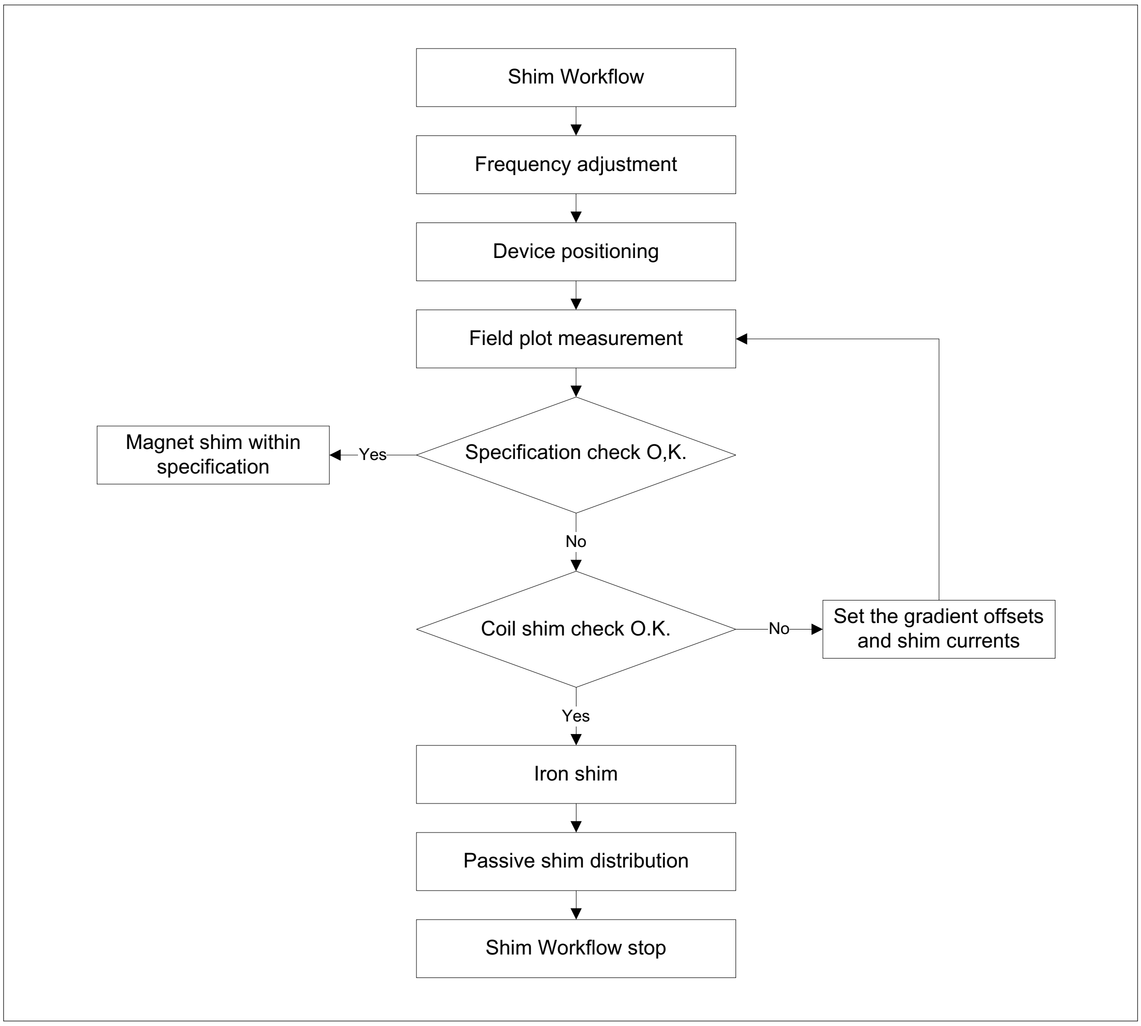
> Magnetom Vida systems are shimmed with Silicon iron (SiFe).
>
> Check the correct shim plate settings. Refer to: MR Configuration / Measurement Configuration / Shim plate type / SiFe.

Fig. 188: Shim plate type



(1) SiFe

Fig. 189: Shim Workflow



The following individual steps are performed in the shim workflow within each workflow mode.

1. Frequency adjustment

- Check to make sure the ASD is connected and no other coils.
- Perform standard frequency adjustment.
- Perform extended frequency search if standard frequency adjustment fails.

  » If the extended frequency search converges, it is shown in the report how to change the magnet current to arrive at the nominal resonance frequency of the system. Once the current is changed, the standard adjustment should converge with the next execution of the program.

2. Device positioning

- Check to make sure the ASD is connected and no other coils.
- Measure frequency of FID signal in all probes for two gradients of opposite strength.
- Check the polarity.
- Calculate position and check whether it is in specification.

- If the position is not in specification, the user is asked to reposition the ASD and then measure again.

3. Field plot measurement

- Check to make sure the ASD is connected and no other coils.
- Ensure that the gradient offsets are applied, and, if applicable, the shim currents as well.
- Check the table position.
- Perform EIS reset.
- Perform adjustment.
- Perform field plot measurement.
- Check data.

4. Specification check

- Create IQShim input file.
- Check to make sure the shim is in specification.
- Set “magnet homogeneity” in the installation protocol according to result.
- Decide whether to continue with coil shim (5.1) or iron shim (5.2) in case shim is not in specification.

5.1 Coil shim

- Create new gradient offsets and shim current.
- Save new offsets/currents if confirmed.

5.2 Iron Shim

- Create an IQShim input file.
- Analyse IQShim output file.
- Report the passive shim distribution result.

Array Shim Device (ASD) kit

Contents of the array shim device box:

- Array Shim Device
  - » Can be used for 1.5T and for 3T
- ASD adapter
  - » Can be used for 1.5T and for 3T
- Array shim cable
  - » Can be used for 1.5T and for 3T
- Shim iron
  - » **Silicon iron for 1.5T and 3T**

Plate type	Part number
Plate 1	10105269

Plate type	Part number
Plate 2	10105268
Plate 3	10105267
Plate 4	10105266
Isolation shim	10117093
Packing shim	8395563

» **Cobalt iron for 3T**

Plate type	Part number
Plate 1	8395522
Plate 2	8395530
Plate 3	8395548
Plate 4	8395555
Isolation shim	10117093
Packing shim	8395563

Do not mix up the cobalt iron and the silicon iron!

Fig. 190: Aera/Skyra Array Shim Device

Fig. 191: Aera/Skyra ASD

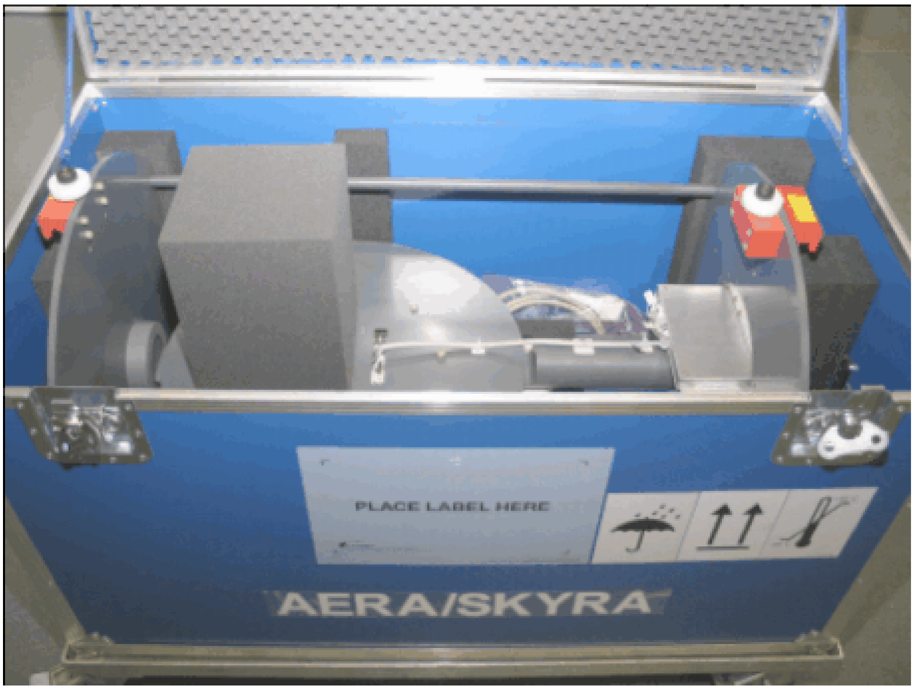


Fig. 192: Aera/Skyra Array Shim Device

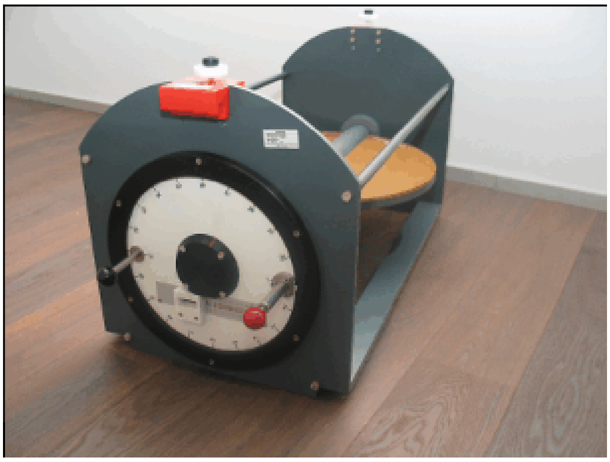
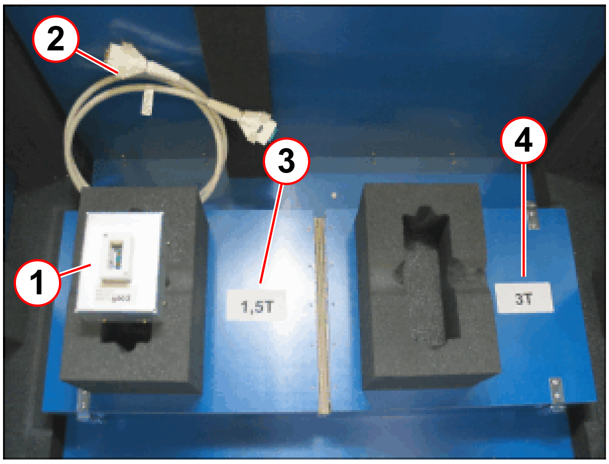


Fig. 193: Array Shim device kit



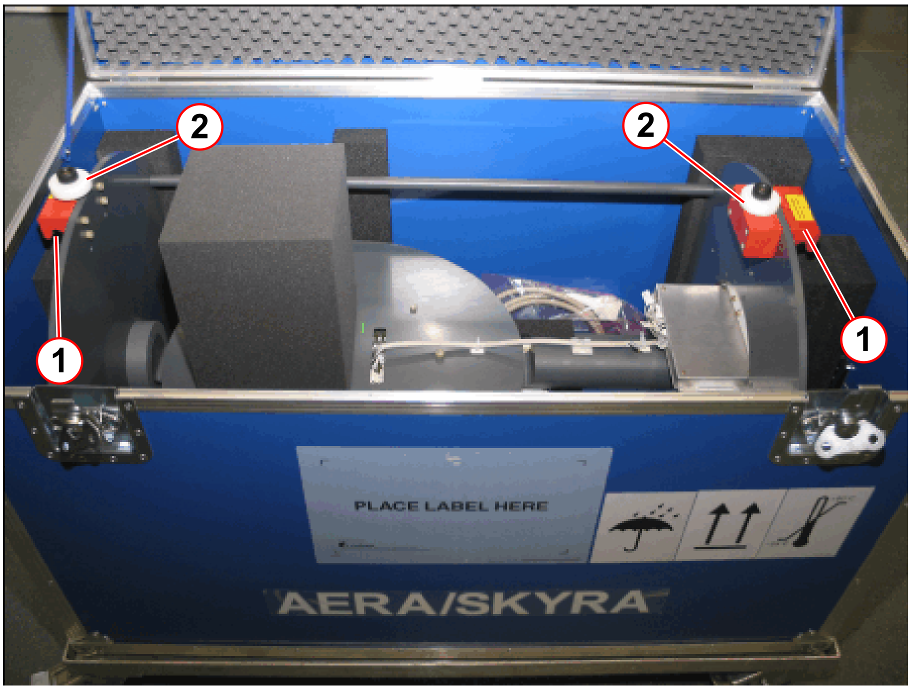
(1) ASD adapter 1.5T, 3.0T
(2) Cable
(3) Shim iron box 1.5T
(4) Shim iron box 3.0T

Installation of Device

- Pick up the array shim device by the two red handles, lift it up carefully, and remove it from the crate.

> To prevent damage to the array shim device, use only the two red handles for transport.
> Two persons are necessary for lifting the array shim device.

Fig. 194: Array shim device



(1) Handle
(2) Bolt

- Move the patient table into the lowest position.
- Slide the array shim device into the magnet.

  Slide it up to the correct position (behind the guide roller 2).

Fig. 195: ASD Position

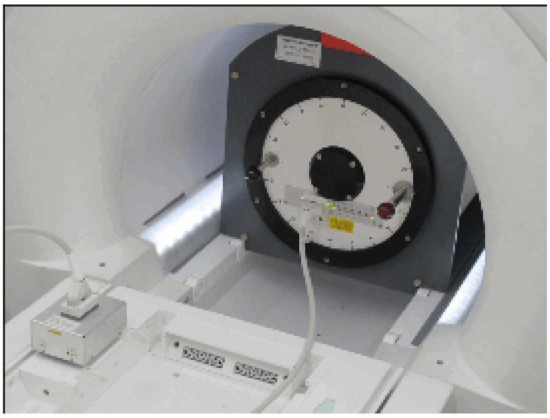
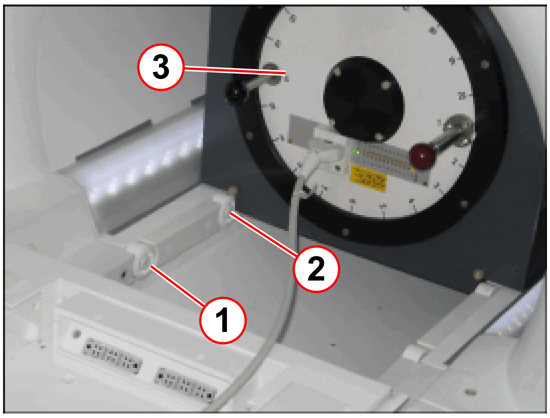
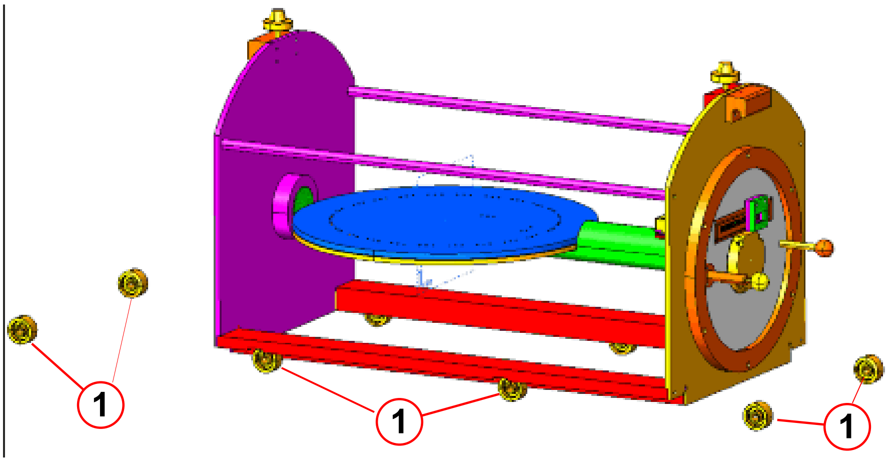


Fig. 196: ASD Position



(1) Guider Roller 1
(2) Guide Roller 2
(3) ASD

Fig. 197: Magnetom Skyra ASD position

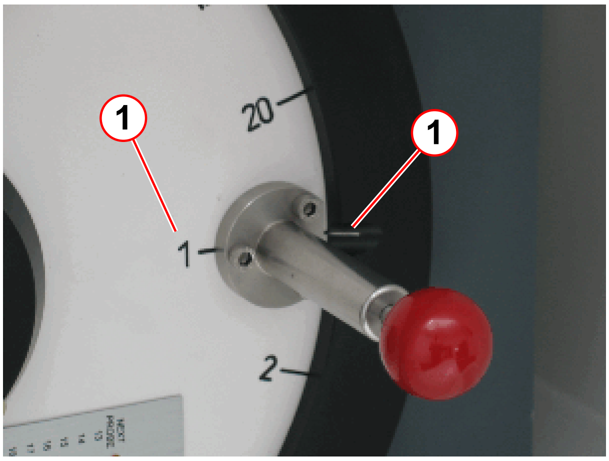


(1) Guide rollers

The array shim device is positioned roughly now..

- Set the array shim device to **position 1**.

Fig. 198: ASD



(1) Position 1

- Pull out the red front knob of the positioning pin, and turn the shim device into position 1.

NOTICE

Do not press the large rotary push button of the control panel if the ASD is installed.

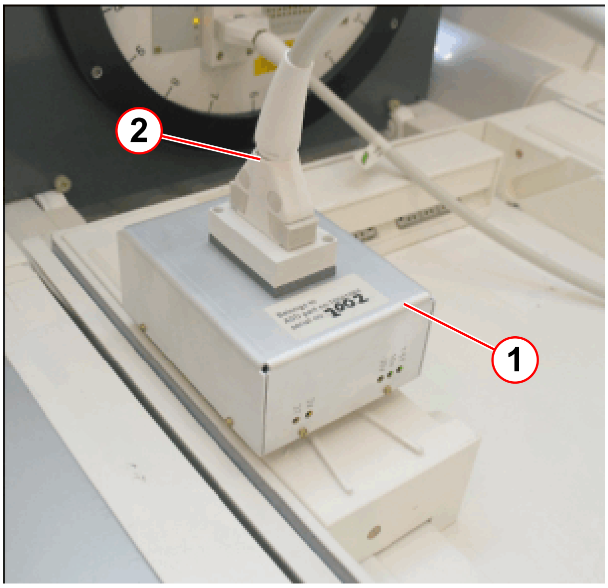
Pressing the large rotary push button will move the patient table to the isocenter automatically.

» **Do not press the large rotary push button if the ASD is installed.**

» **Use the dual stage short range buttons for UP/IN and for OUT/DOWN movement.**

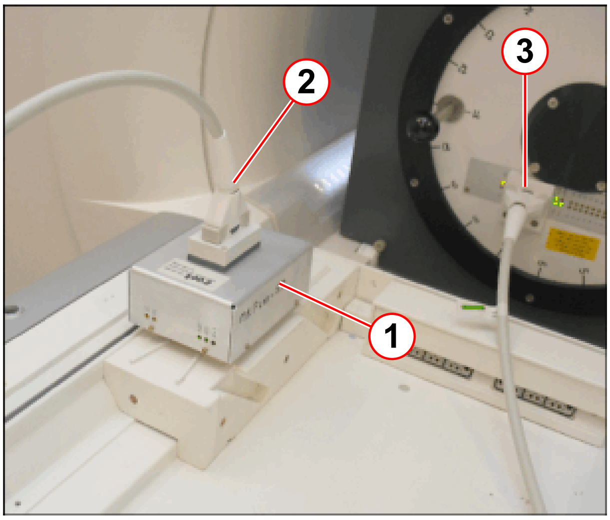
- **Move the patient table into the home position**. The table is slightly magnetic and has to be shimmed as well.
- Connect the ASD adapter to connecter 1 or connector 2 of the patient table.
- Connect the array shim cable to the ASD adapter.
- Connect one end of the array shim cable to the multiplexer of the ASD.

Fig. 199: ASD adapter



(1) ASD adapter
(2) Cable

Fig. 200: ASD adapter

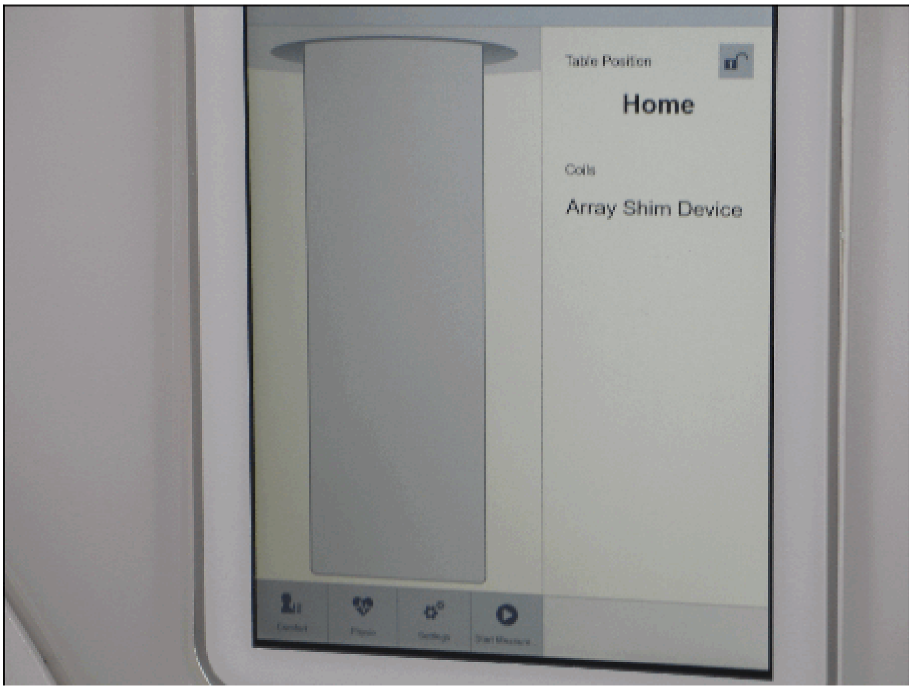


(1) ASD adapter
(2) Cable
(3) Multiplexer

To maintain the correct position of the array shim device, it should not be removed from the magnet during shimming.

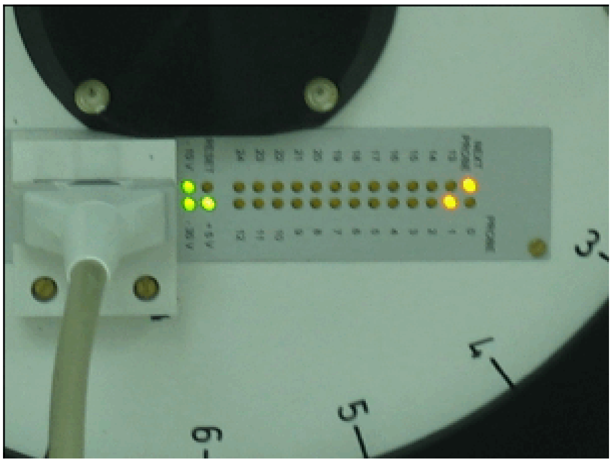
- Indication of the Array Shim Device on the PDD.

Fig. 201: Patient Data Display



- Check the LEDs on the ASD

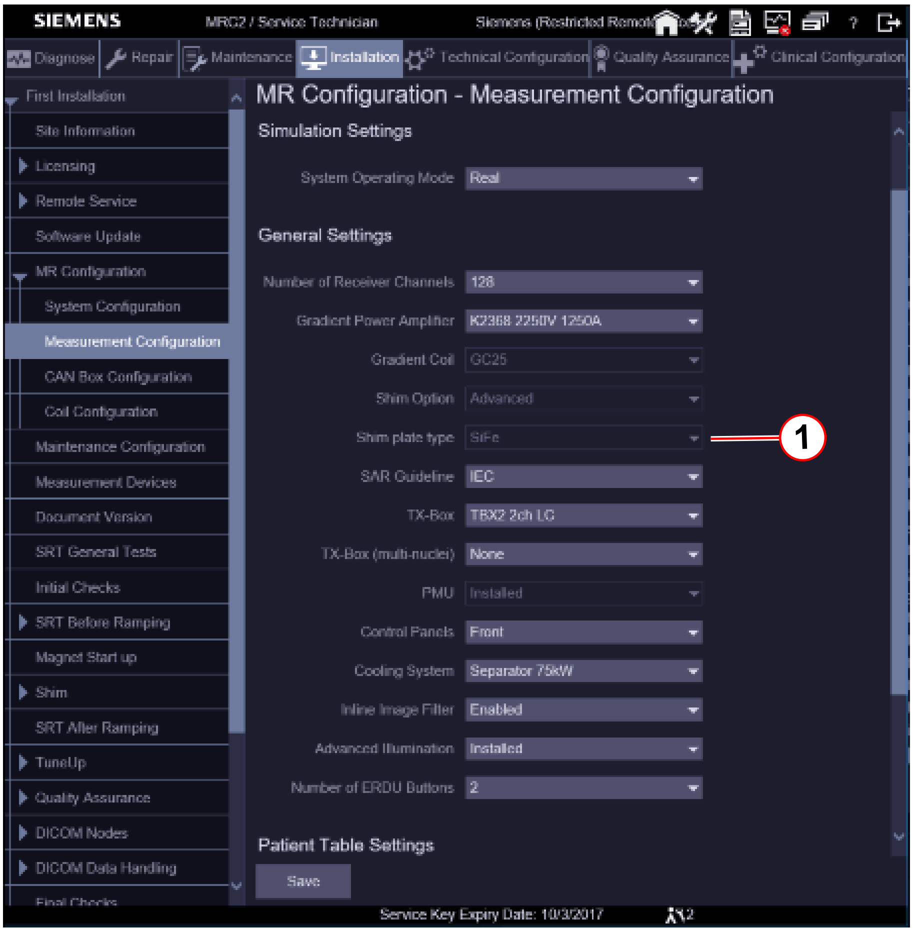
Fig. 202: ASD



Array Shim Measurement

> Magnetom Vida systems are shimmed with Silicon iron (SiFe).
>
> Check the correct shim plate settings. Refer to: MR Configuration / Measurement Configuration / Shim plate type / SiFe.

Fig. 203: Shim plate type



(1) SiFe

The array shim measurement consists of two parts:

- The frequency search that uses the center probe of the array shim device (Frequency range = Center frequency + / - 100kHz).
- The measurement of the 50 cm sphere in 20 steps.

Each measurement consists of 24 measurement points on a 50 cm circle.

The probe signals measured are converted into frequency data and saved to a field data record at the end of the measurement.

This field data record is used for calculating the shim data.

Prerequisites

- MR system is switched on.
- Magnet is at field.
- After the magnet is ramped up, wait for the following amount of time before taking the measurement:
  - At least 30 minutes for the bare and first iteration
  - At least 1 hour for subsequent iterations

- Patient table is in the top position.

Starting the shim workflow

- Start the Service Software UI at the MR console.
- Select **Installation / First Installation / Shim / General**.

Fig. 204: Shim Workflow

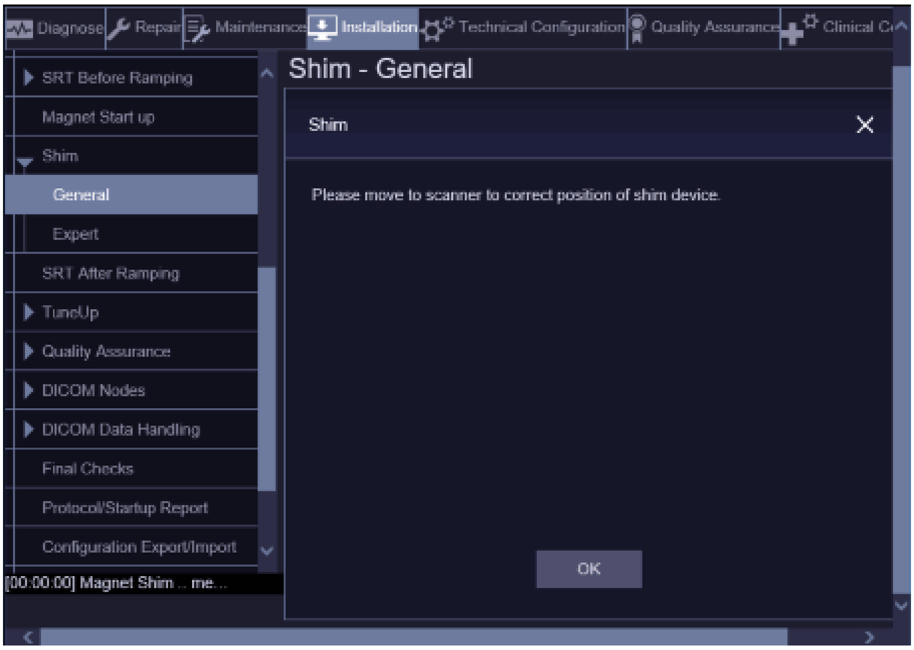
- Start the shim workflow with **Go**.
  » Workflow starts

The shim workflow starts with:

1. Frequency adjustment.
2. Array shim device positioning.
   - The system applies a +1mT/m gradient and measures the frequency at planes 12 and 13.
   - The system applies a -1mT/m gradient and measures the frequency at planes 12 and 13 again.
   - The shim program checks whether the Z gradient is correctly wired.
   - The shim program calculates the offset from the gradient center.

- The shift is displayed on the PDD.

Fig. 205: Shim Workflow

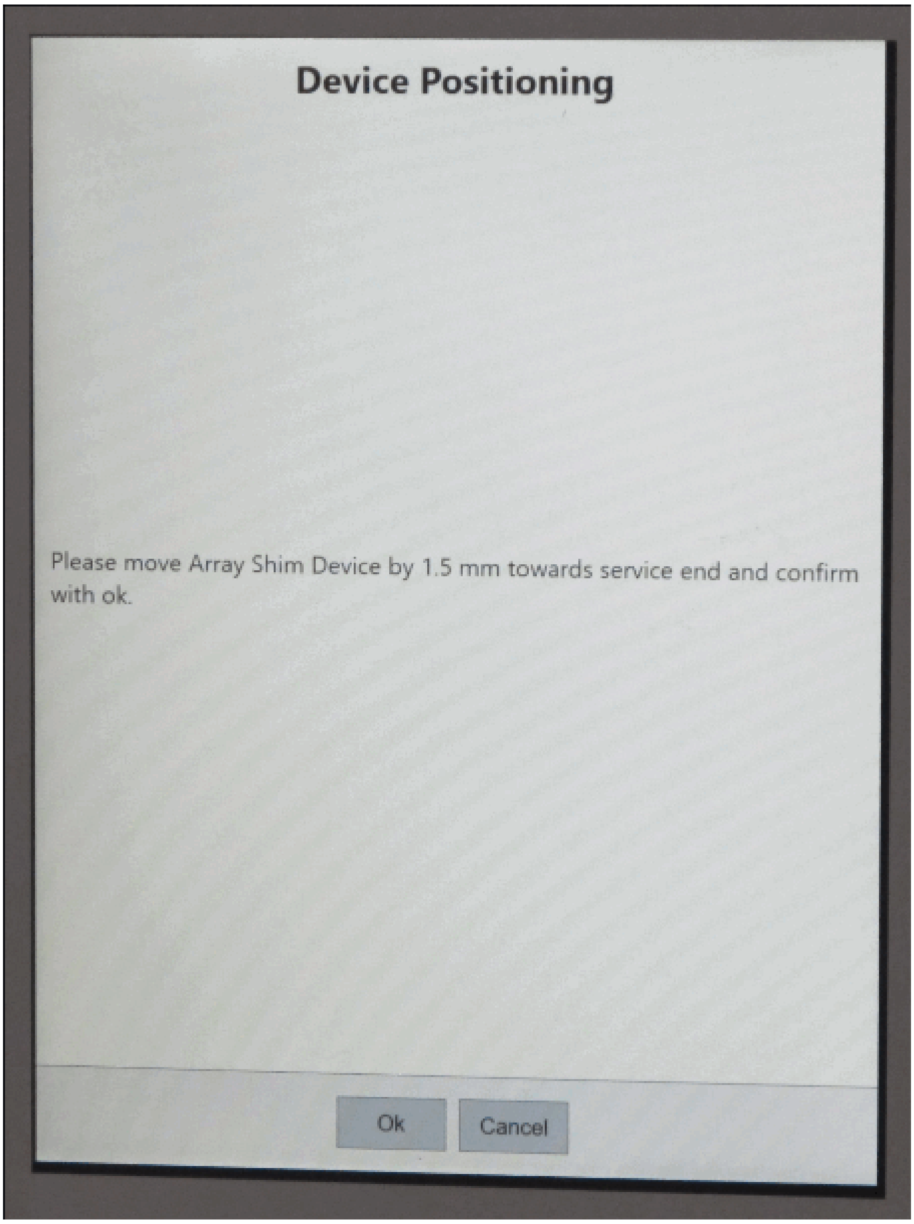


- Press OK and go to the Patient Data Display (PDD).

  » Move the ASD to the correct position (towards service end or towards patient end)

  Press **Ok**

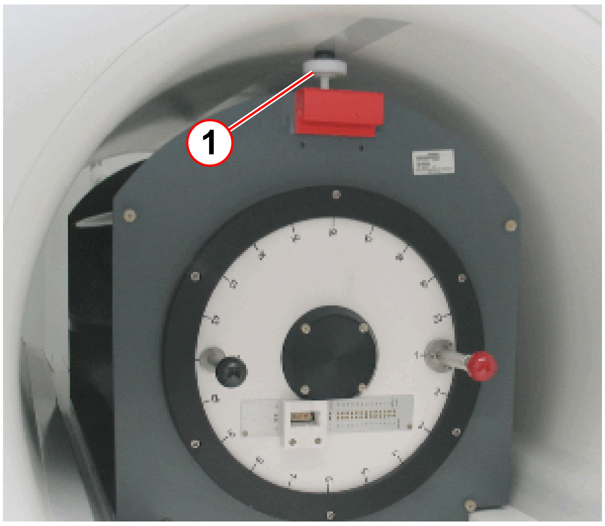
Fig. 206: Device Positioning



» Repeat until the deviation is within +/- 1 mm.

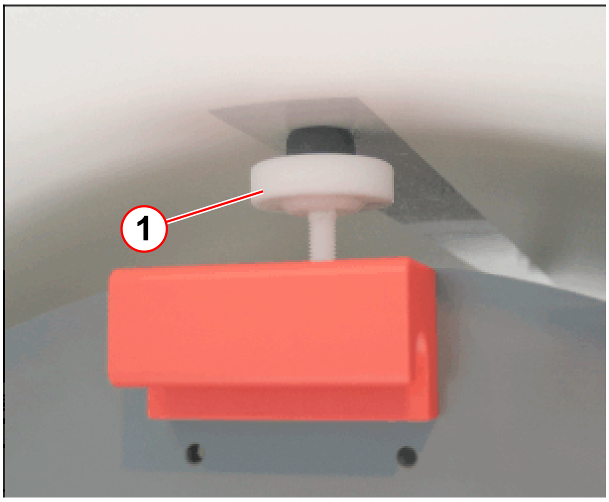
- Secure the ASD by tightening bolt 1 in the resonator at the patient end.
- Secure the ASD by tightening bolt 2 in the resonator at the service end.

Fig. 207: ASD patient end



(1) Tightening bolt 1

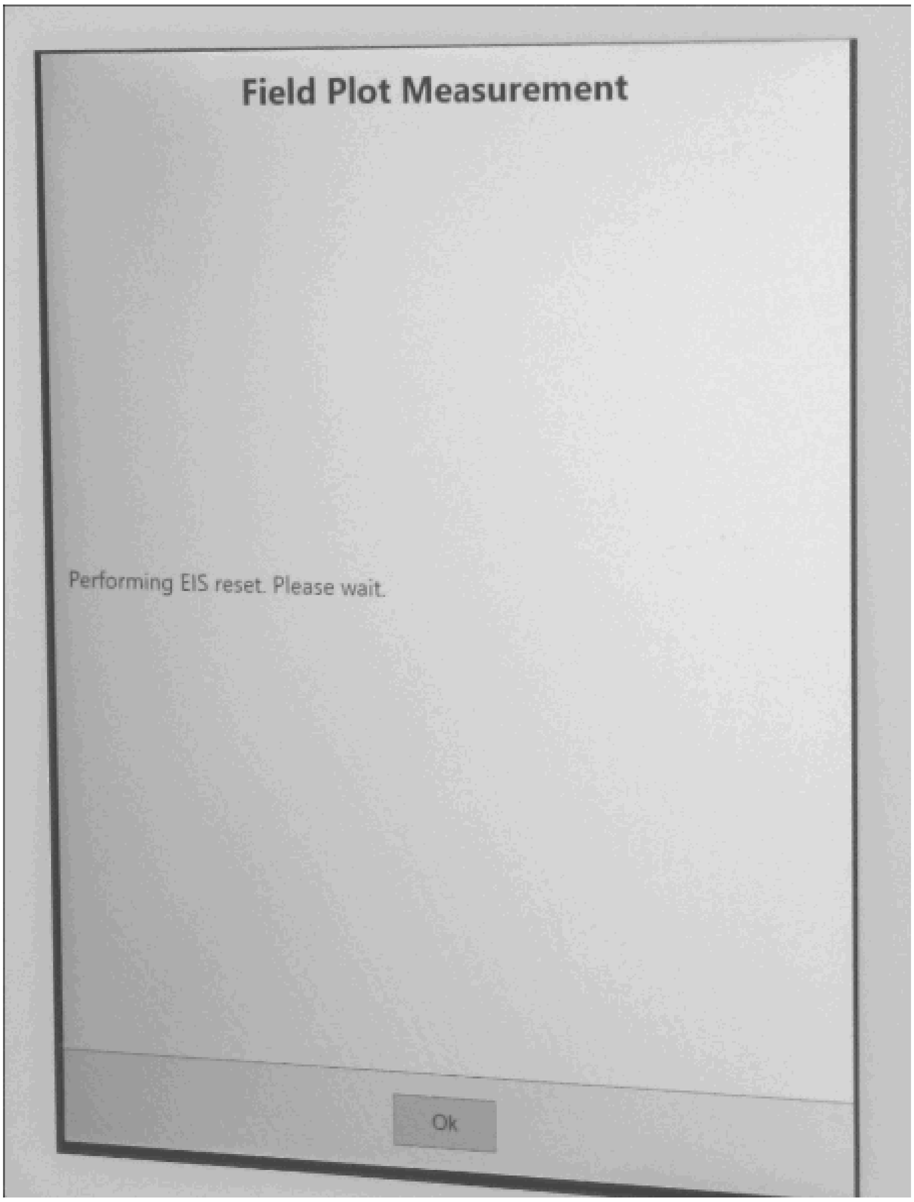
Fig. 208: ASD service end



(1) Tightening bolt 2

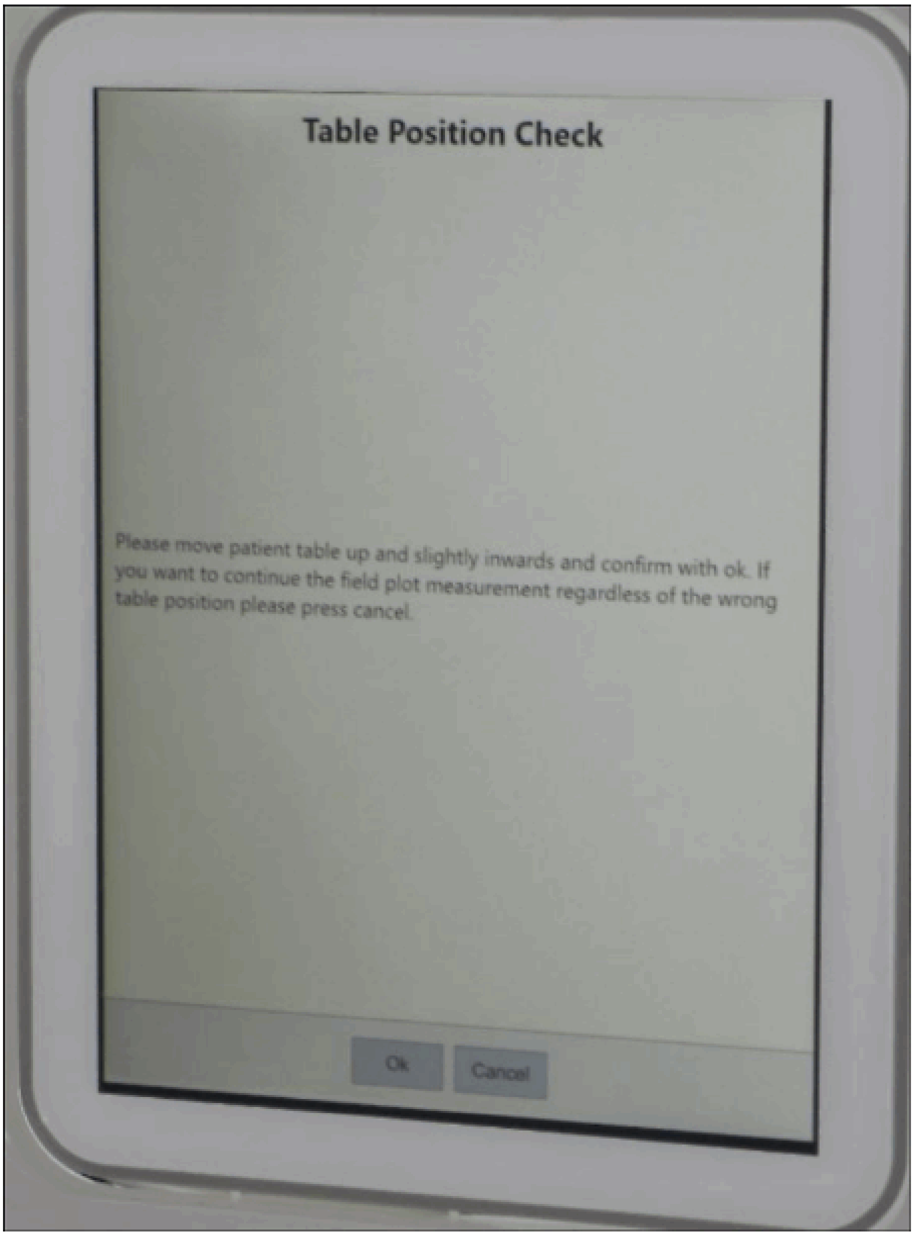
3. EIS reset is performed by the Shim workflow.

Fig. 209: Field Plot Measurement



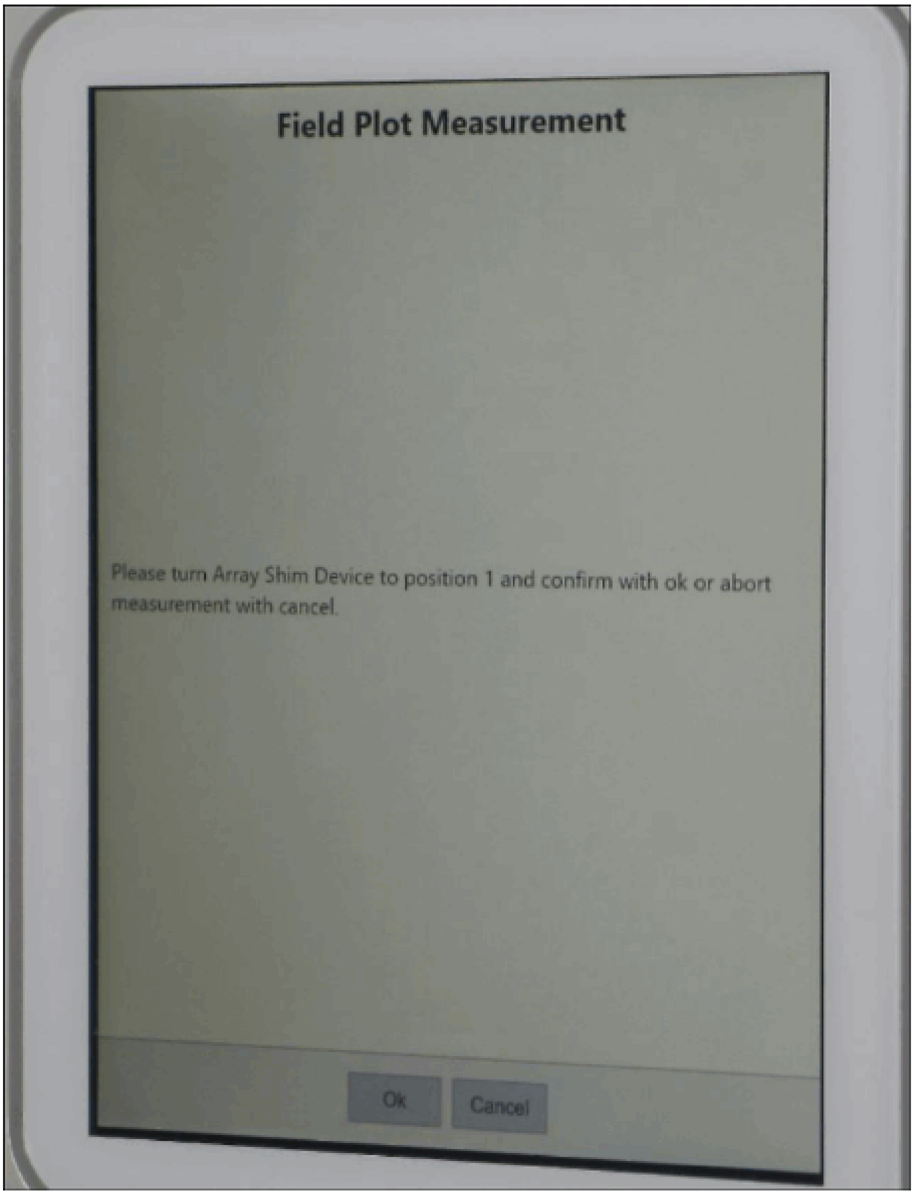
4. Position Check of the patient table
   - If the patient table is not in the correct position following information appears.

Fig. 210: Table Position Check



5. Perform array shim measurement from the patient table operating panel.
   - Start the measurement from position 1.

Fig. 211: Field Plot Measurement

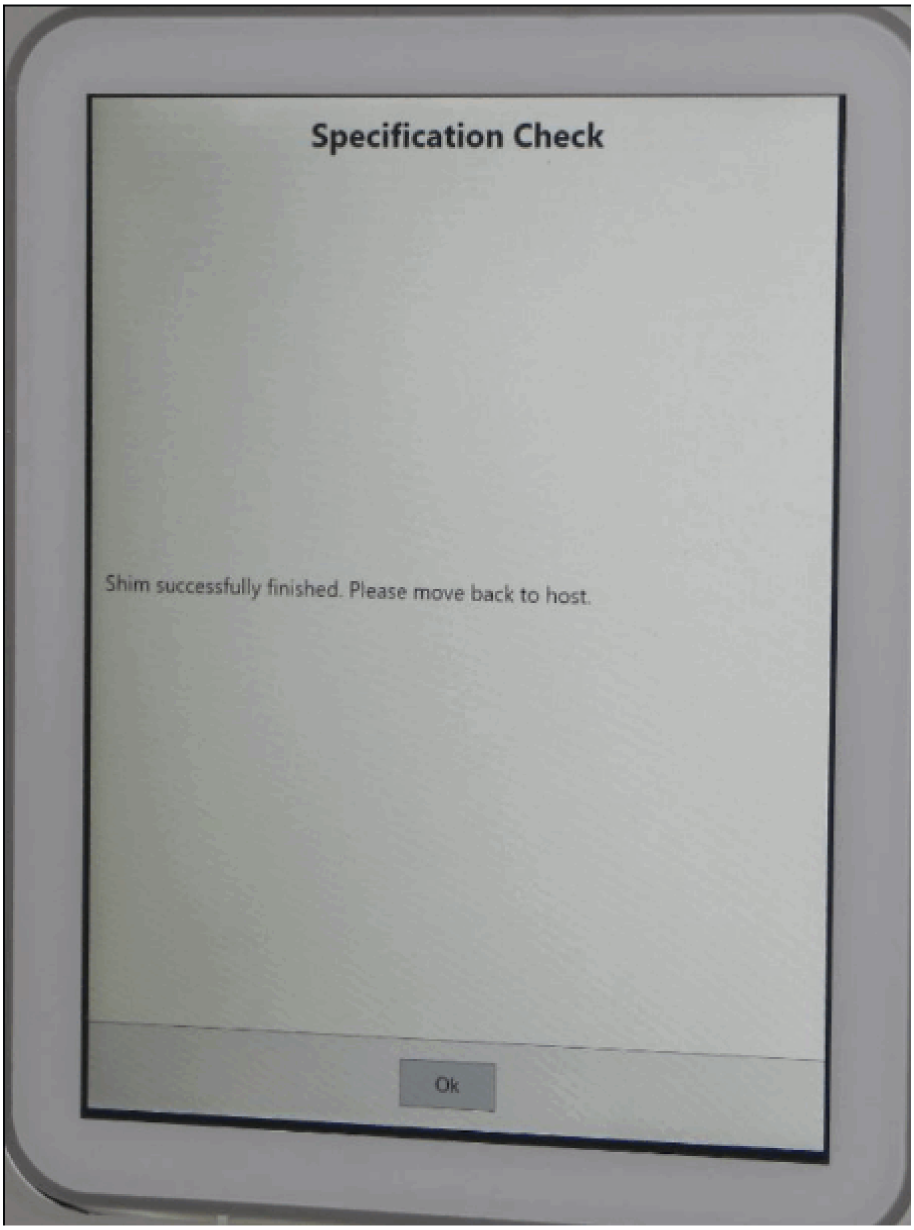


   » Press **Ok**.
   - Turn the ASD device to position 2.
   » Confirm with OK.
   - Repeat this procedure up to position 20.

The measurement has been completed. You can now return to the MR operating console.

The shim program checks the specification values and calculates the passive shim distribution.

Fig. 212: Specification Check



6. Specification check
   - Creates IQShim input file.
   - Checks whether shim is in specification.
   - Decides whether to continue with iron shim in case the shim is not in specification.
7. Coil shim
   - The shim program calculates the gradient offsets and shim currents (E-Shim option)

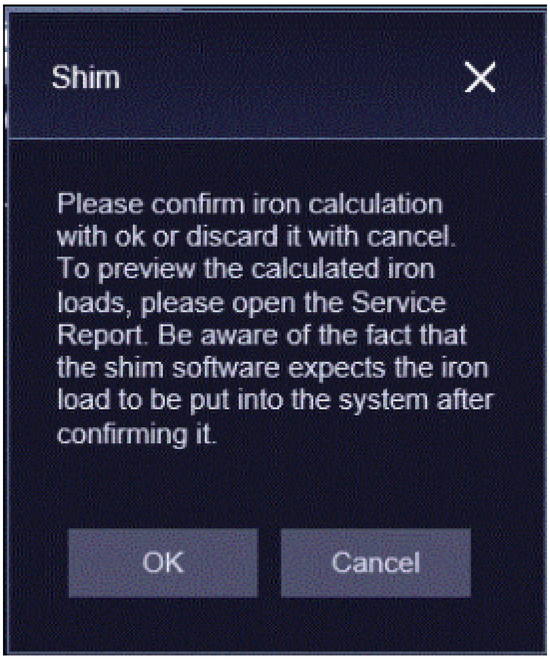
8. Iron shim
   - The shim program calculates the passive shim distribution.
     » Confirm the IQShim output.

When you press OK button, the shim software expects the iron load to be put into the magnet.

OK: The calculated iron will be stored in the database if the OK button is pressed.

Cancel: The calculated iron will not be stored in the database if the Cancel button is pressed.

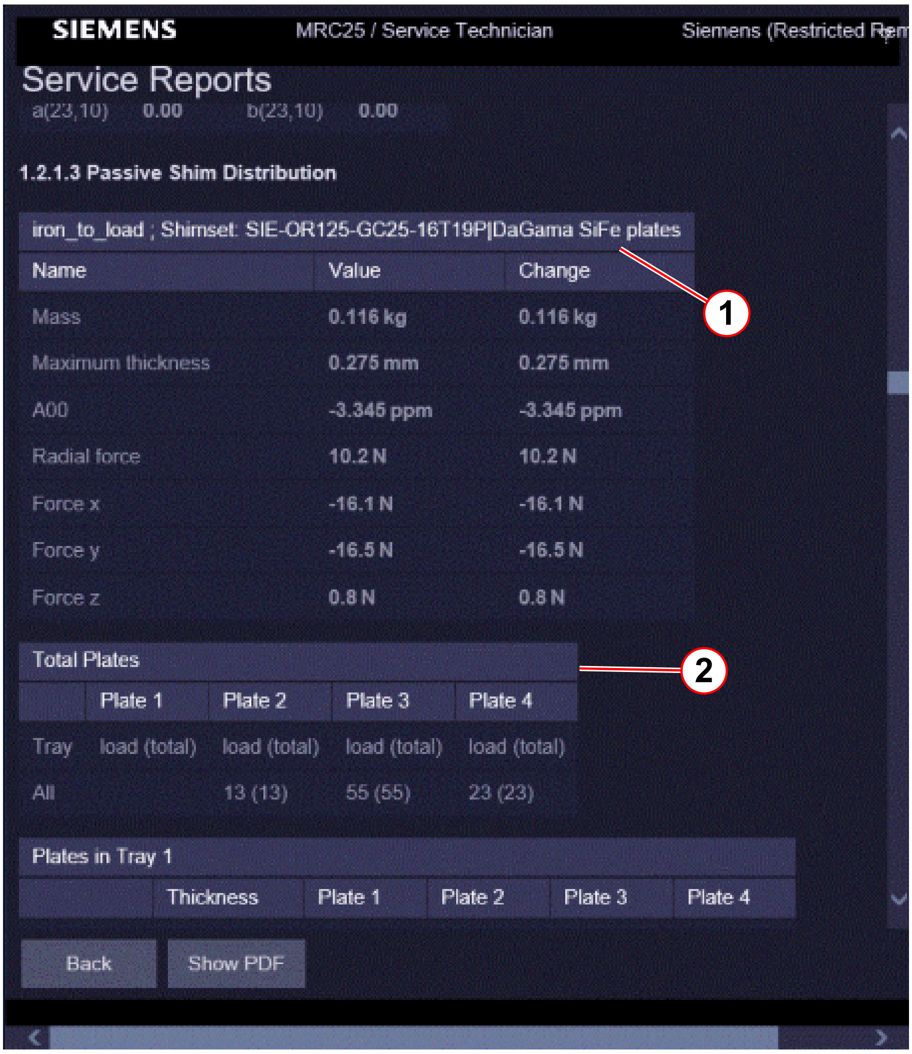
Fig. 213: Confirmation Iron Calculation



The result is saved in the "Service Report" or in the "Shim " directory.

9. Checking the correct shim plate type. (→ 2/Fig. 214 Page 329)
   Checking the calculated passive shim distribution.

Fig. 214: Iron Calculation Report



(1) Shim Plate Type
(2) Total Shim Plates

> **i** Values out of specification are highlighted.

Preparation for Shim Plate Loading

- Print out the distribution list if possible

- Count and sort the number of shim plate 1, shim plate 2, shim plate 3 and shim plate 4.(→ 2/Fig. 214 Page 329).

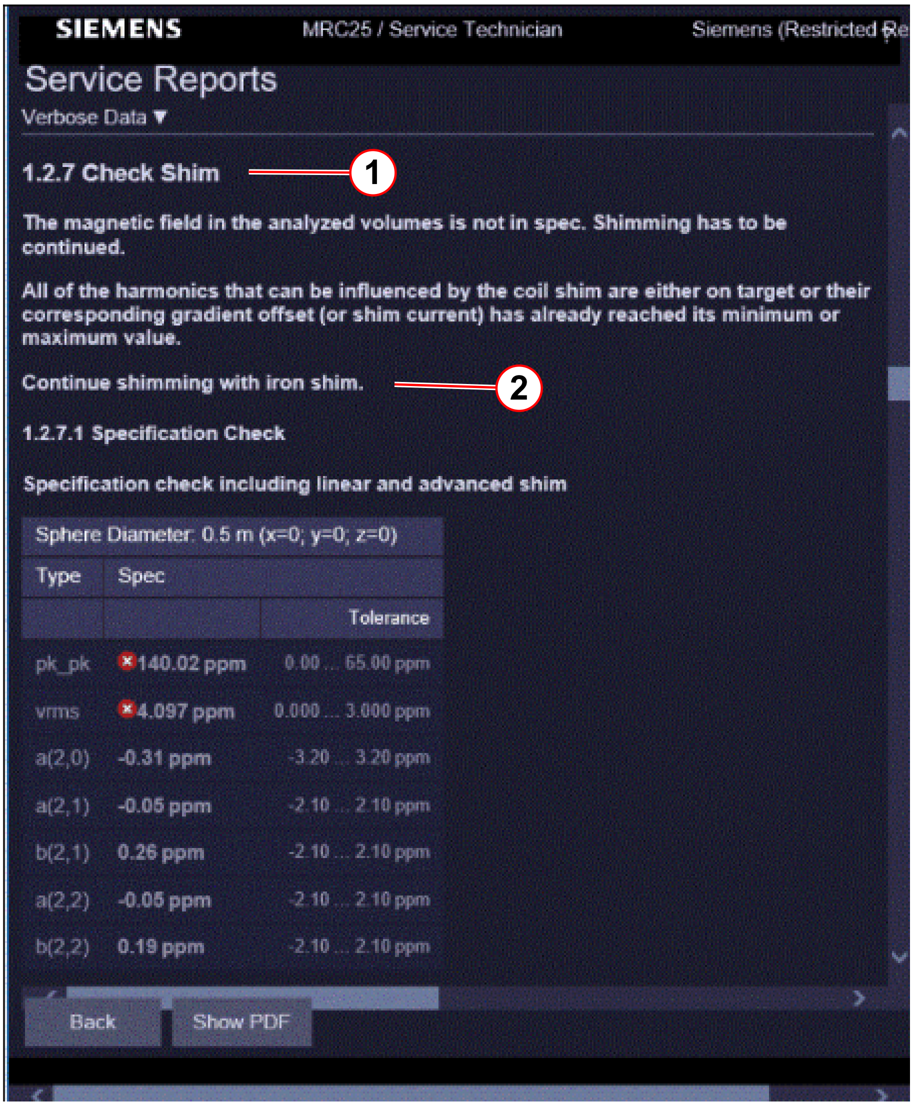
Fig. 215: Iron total

Total Plates				
	Plate 1	Plate 2	Plate 3	Plate 4
Tray	load (total)	load (total)	load (total)	load (total)
All	16 (431)	45 (98)	98 (209)	41 (77)

(1) Plates to load
(2) Plates in total

- Run to zero the magnet for removing and installing the shim trays.
- Repeat the shim workflow until the homogeneity is in specification. (→ 2/Fig. 216 Page 331)

Fig. 216: Shim Check

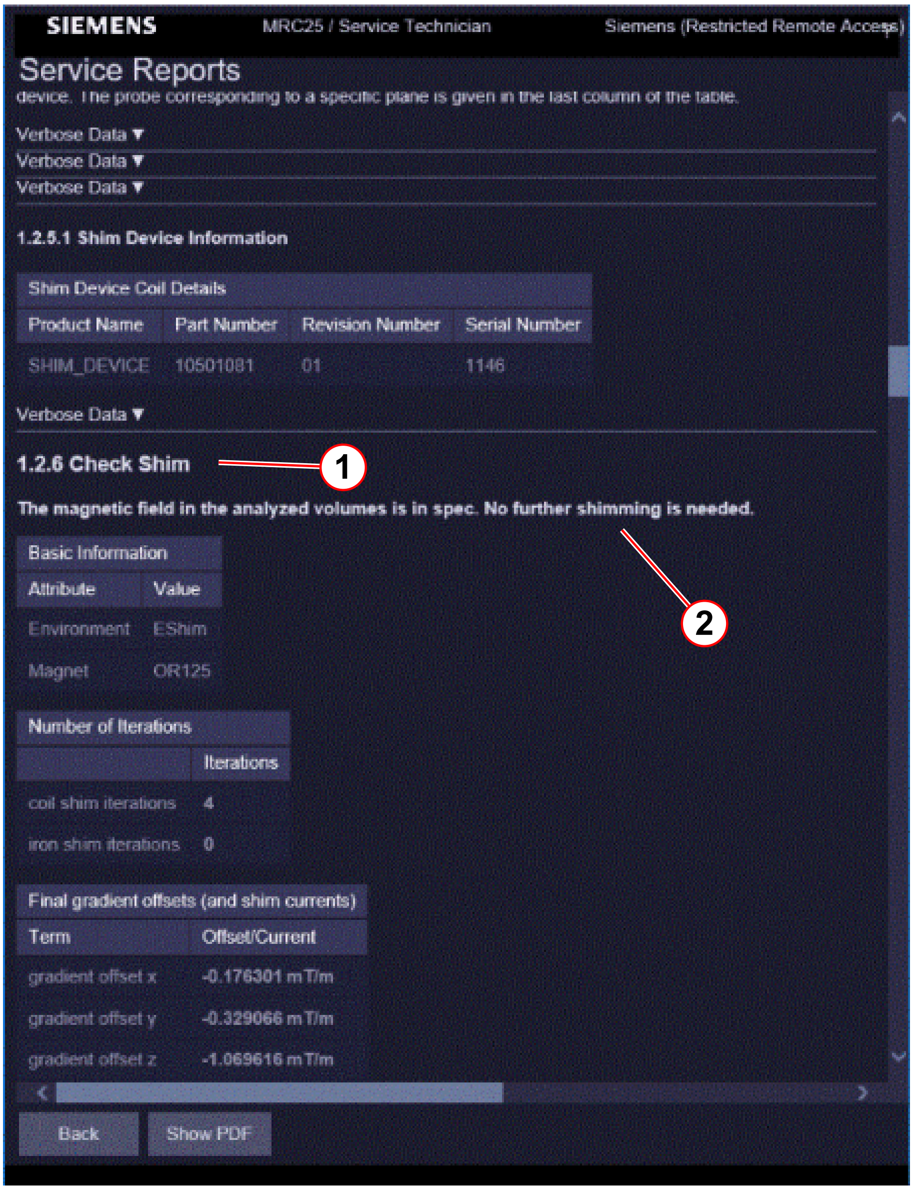


(1) Check Shim
(2) Not in specification

Homogeneity is within specifications.

- When the magnet is shimmed within specifications, the shim is completed.
  - » Output of the IQShim if the magnet shim was successful
  - » No further shimming is needed. (→ 2/Fig. 217 Page 332)

Fig. 217: Shim Check



(1) Check Shim
(2) Shimming is not needed

Removing the shim trays

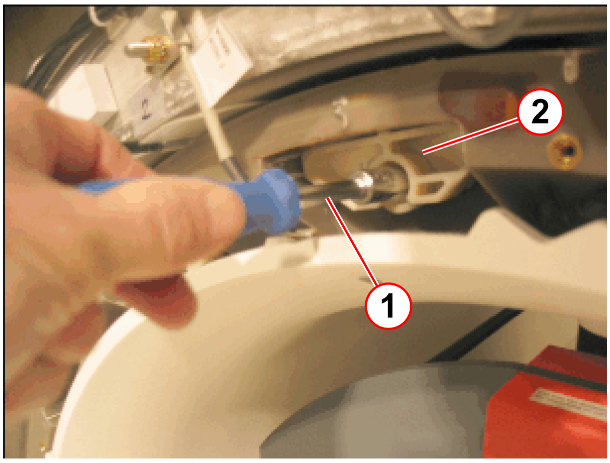
> **⚠ WARNING**
>
> **The force created by the fringe field of the magnet will pull out the shim plates and move them to a different location in the shim tray.**
>
> **Observe the warning. Otherwise death or serious injury can occur.**
>
> **» Ramp down the magnet before removing and installing the shim trays.**

Ramp down the magnet before removing the shim trays.

The shim trays are accessible from the patient end.

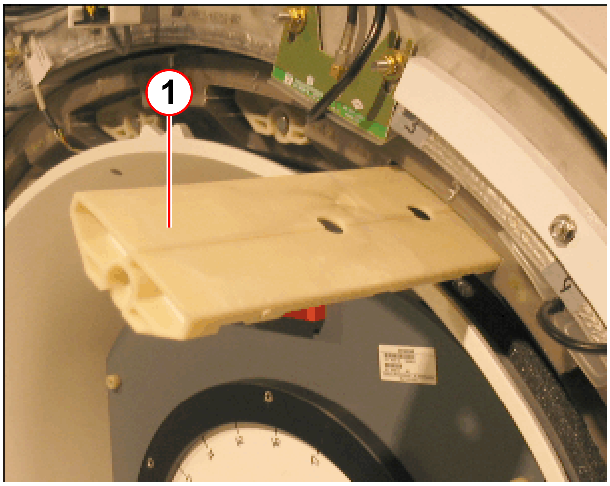
- Remove the funnel.
- Loosen the shim trays with a 10mm socket wrench.
- Remove the mounting screws of the shim trays.

Fig. 218: Shim tray



(1) Socket wrench
(2) Shim tray

Fig. 219: Shim tray



(1) Shim tray

- Carefully pull the shim trays from the gradient coil.

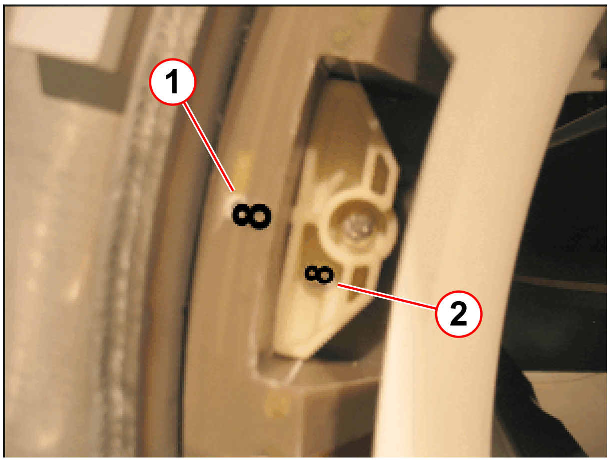
  Each shim tray is identified by a number in front, ranging from **01** to **16**.

When you install the shim trays, ensure that the magnet is ramped down.

When you install the tray, ensure that the number matches the number embossed on the gradient coil.

Tighten the trays with the screws.

Fig. 220: Shim tray



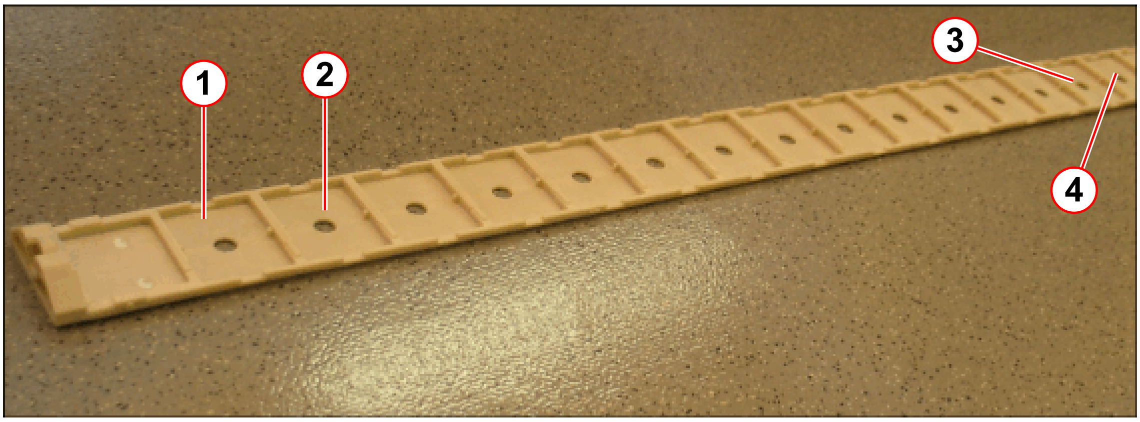
(1) GC label
(2) Shim tray label

Installing the shim iron

Shim kit

The figure shows the shim tray with the position of the trays.

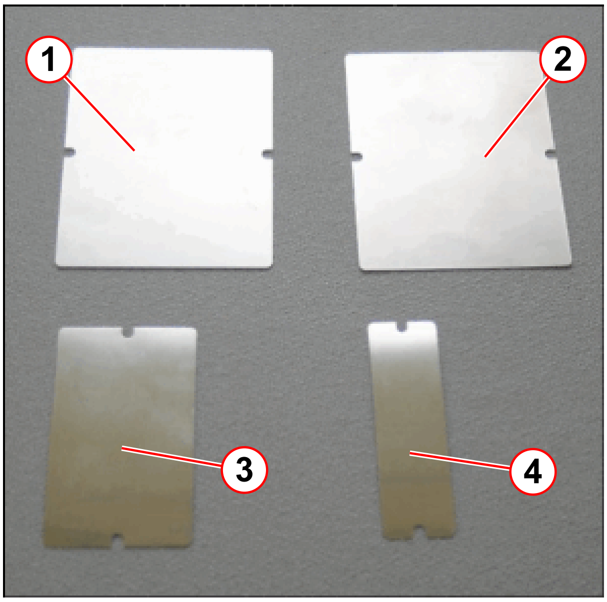
Fig. 221: Shim tray



(1) Position A
(2) Position B
(3) Position P
(4) Position R

The figure shows the shim plates and the packing shims.

Fig. 222: Shim plates



(1) Plate 1 (64.8 x 80 x 0.3 mm)
(2) Plate 2 (64.8 x 80 x 0.1 mm)
(3) Plate 3 (64.8 x 40 x 0.05 mm)
(4) Plate 4 (64.8 x 20 x 0.05 mm)

Fig. 223: Packing shims (75 x 55 x 1 mm, polypropylene molding)

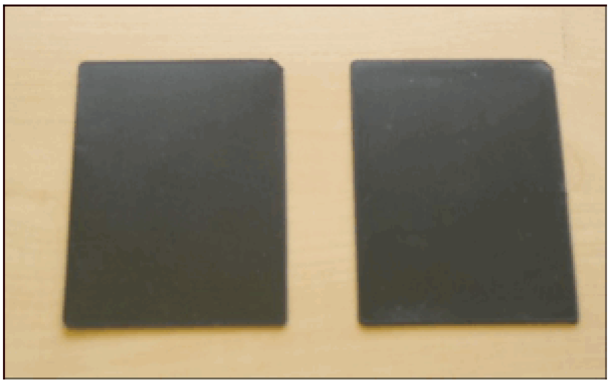
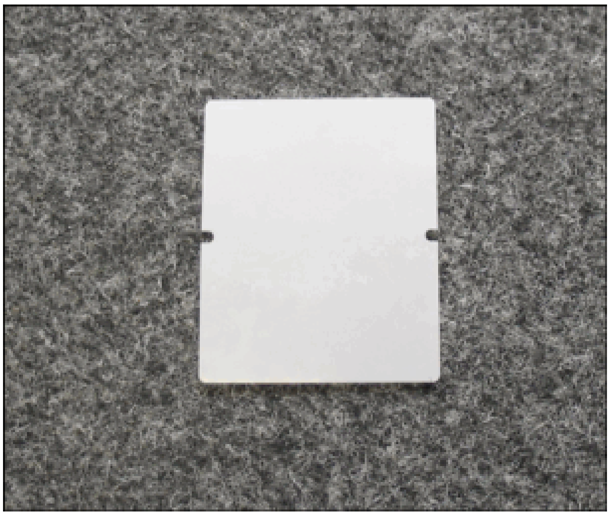


Fig. 224: Insulation shim (80 x 64.8 x 0.19 mm, polyester packer)



Shim plate distribution

Fig. 225: Plates in tray

Plates in Tray 1					
	Thickness	Plate 1	Plate 2	Plate 3	Plate 4
Position	change (total)	load (total)	load (total)	load (total)	load (total)
A	0.000 (0.000)				
B	0.287 (2.987)	0 (9)	2 (2)	3 (3)	1 (1)
C	0.000 (0.000)				
D	0.000 (0.000)				
E	0.000 (0.000)				
F	0.000 (0.000)				
G	0.000 (0.000)				
H	0.000 (0.075)			0 (3)	
J	0.000 (0.237)		0 (2)	0 (1)	0 (1)
K	0.025 (0.025)			1 (1)	
L	0.000 (0.000)				
M	0.000 (0.212)		0 (2)		0 (1)
N	0.000 (0.013)				0 (1)
P	0.000 (0.000)				
R	0.000 (0.000)				
S	0.000 (0.000)				
T	0.000 (0.000)				
U	0.287 (2.987)	0 (9)	2 (2)	3 (3)	1 (1)
V	0.850 (0.850)	2 (2)	2 (2)	2 (2)	
Total Plates		2 (20)	6 (10)	9 (13)	2 (5)

The table shows the number and position of the shim plates:

The number without brackets is the number of plates to be loaded for the actual iteration.

The number in brackets shows the total number of plates.

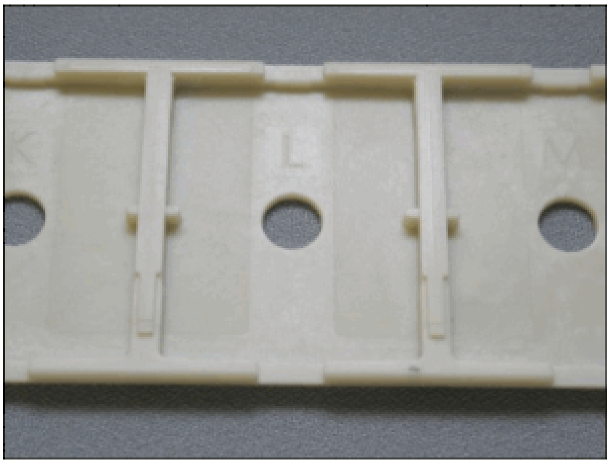
The first position of the shim trays (mounting location at the shim tray) is identified with:

» Magnetom Vida: **A** followed by B, C ,D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V

> Empty pockets should not be covered. The covers of empty pockets may make noise.
> Store remaining covers on-site for re-shimming.

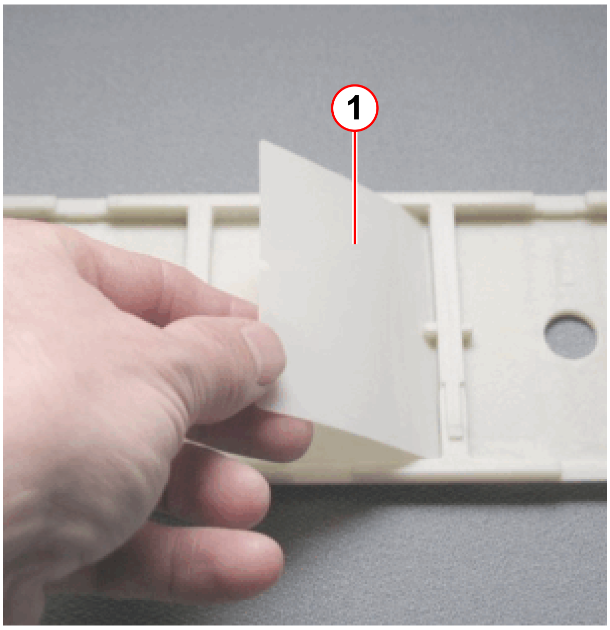
- Install the shim plates as shown in the following figures.

Fig. 226:



- Select the correct shim tray number (1 - 16)and the correct pocket position (A - V).

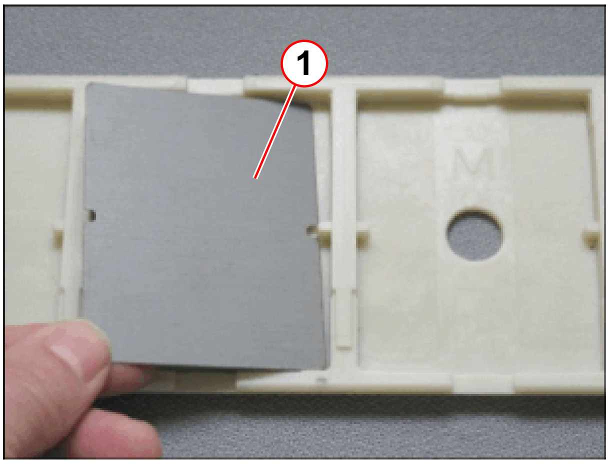
Fig. 227: Insulation shim



(1) Insulation shim

- Install an insulation shim first.

Fig. 228: Shim tray

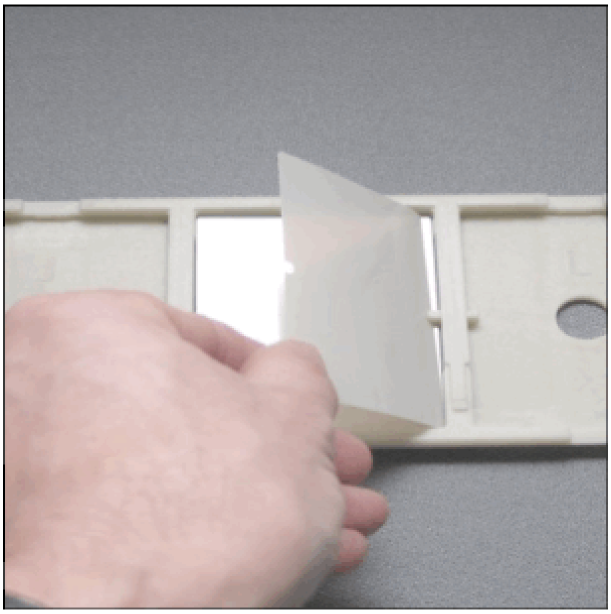


(1) Shim plate

- Install the required plate size
  » Use one layer of insulation shim between each shim plate.

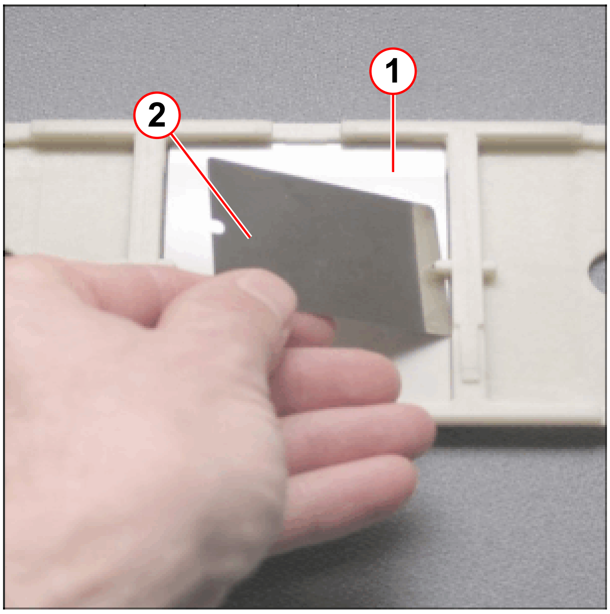
> Use one insulation shim between each shim plate.

Fig. 229:



- Install one insulation shim between each shim plate.

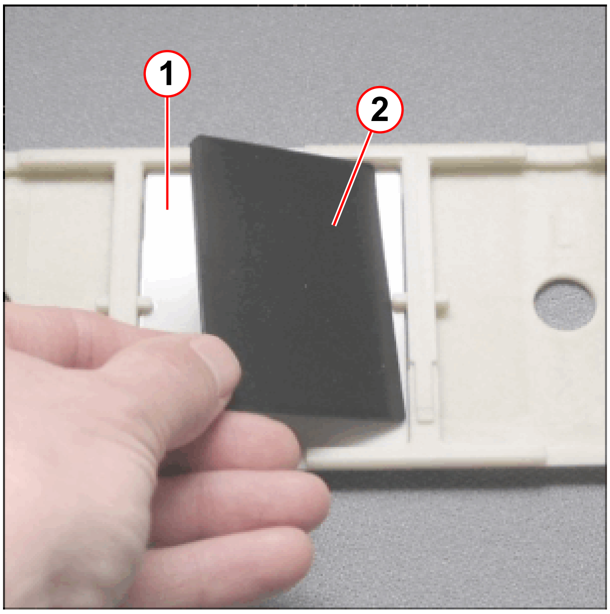
Fig. 230: Shim tray



(1) Insulation shim
(2) Shim plate

- Install the required shim plate.

Fig. 231: Shim tray

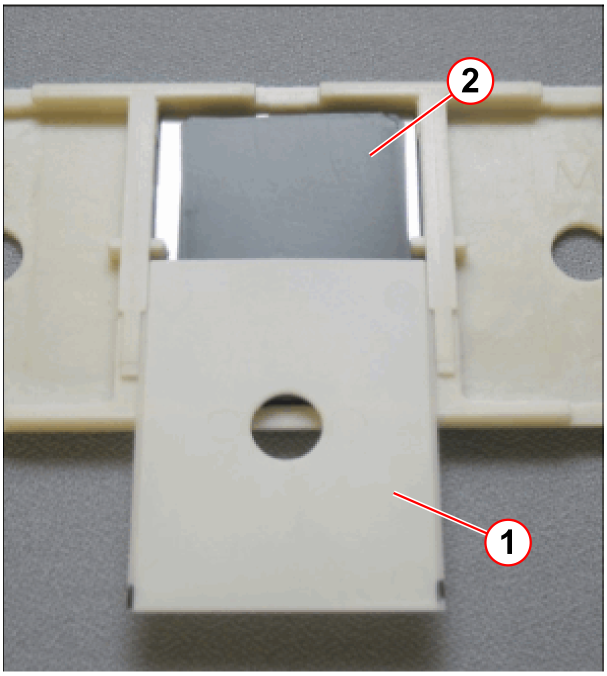


(1) Insulation shim
(2) Packing shim

- Cover the last shim plate with an insulation shim.
- Fill the space between the shim plates and the cover with packing shims.

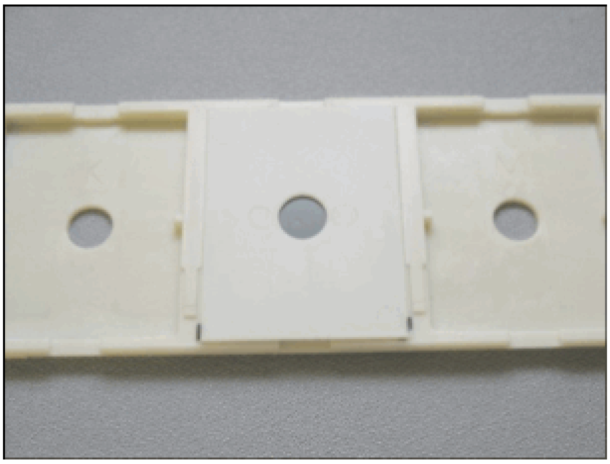
Depending on the space between the shim plates and the cover, use **1 mm packing shims** to keep the shim plates from moving in the shim trays.

Fig. 232: Shim tray



(1) Cover
(2) Packing plate

Fig. 233:

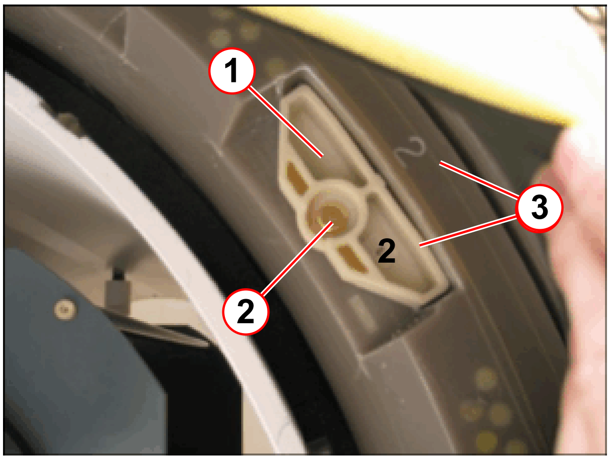


- Close the pocket with the cover.

> Ensure that the shim plates are securely seated in the shim trays. Loose shim plates cause spikes.

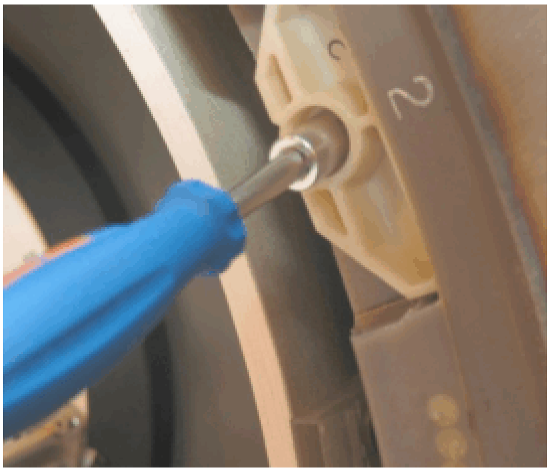
- Install the shim trays in the gradient coil.
- Tighten the shim tray with the screw.

Fig. 234: Shim Tray



(1) Shim tray
(2) Screw
(3) Number

Fig. 235:



> When you install the shim trays, ensure that the tray number matches the number embossed on the gradient coil.

Setting the magnet current

To compensate for the additional iron and to ensure that the predefined frequency range of (+/- 100 kHz) is maintained for the array shim, a slightly higher magnet current is set when the magnet is ramped for the first time.

This value is the **Operating current/Measured value** listed under **Operating Parameters** in the **Magnet Acceptance Protocol** by **SMT**.

Magnet current setting

1. Iteration

Use the value listed as Operating current / Measured value under Operating Parameters in the Magnet Acceptance Protocol by SMT.

2. Iteration and following iterations

To ensure that the magnet is ramped at 4 year intervals only, the magnetic field should lie a little under the maximum frequency. Use the calculate current listed in the Magnet Shim Report File.

Fig. 236: Magnet Current

1.2 Details

1.2.1 Frequency adjustment

Standard Frequency Adjustment converged. Resonance Frequency: 123.208066 MHz

Target Magnet Current			
	Value	Tolerance	Target
frequency	**123.208066 MHz**	123.125000 ... 123.265000 MHz	**123.260000 MHz**
magnet current	**500.73 A**		**500.94 A**

(1) Actual Magnet Current
(2) Target Magnet Current

Formula to calculate the current for the maximum field strength:

> Maximum frequency for "Tuning Calibration"
> Magnetom Vida: 123.265 000 MHz.

- Use the following mathematical formula to compute the current for the maximum field strength. The formula takes the tolerances of the different magnet power supplies into consideration:

Magnetom Vida: I final = (**123.260 000** MHz x **I** actual) / **Fre** actual

> The lower limit of the frequency is 123.100 000 MHz

Magnetom Vida with Multinuclear Spectroskopy Option (MNO)

> Magnet OR125 range of frequency:
>
> Target frequency: 123,260 000 MHz
>
> Maximum frequency: 123,265 000 MHz +0 kHz
>
> Minimum frequency: 123,125 000 MHz -0 kHz

Use the mathematical formula below to calculate the current to determine the **target frequency.**

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

OR125	**I final = (123,260 000 MHz x I actual) / Freq actual**

Formula for converting the frequency to Tesla

$$2 * P * f = g * B_0$$

P	=	3.14159
f	=	Frequency in Hz
g	=	Gyromagnetic coefficient of proton, 2.67522128 * 10^8 s^{-1} T^{-1}
B_0	=	Magnetic field in Tesla

Vertical position of the patient table

> The vertical position of the patient table can be adjusted with a reflector bracket.
>
> This is necessary if the front cover funnel is dismounted.

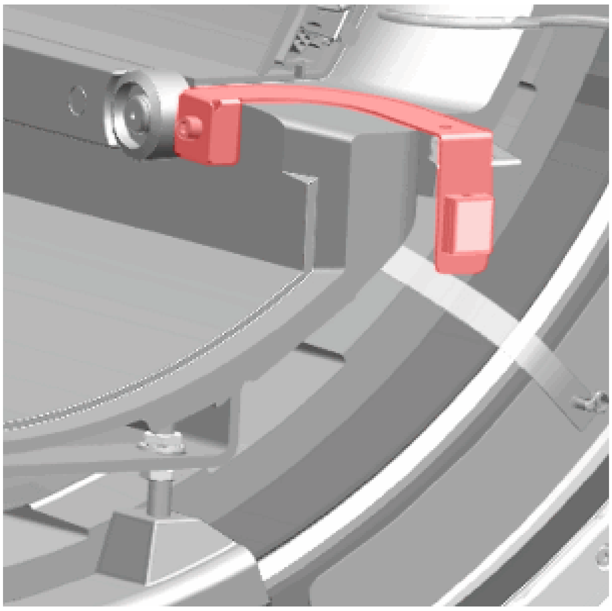
- Take out the reflector bracket from the storgae position behind the front cover.

Fig. 237: Reflector bracket storage position

(1) Reflector bracket

- Fasten the reflector bracket with a screw from the removed funnel.

Fig. 238: Reflector bracket attached to BC



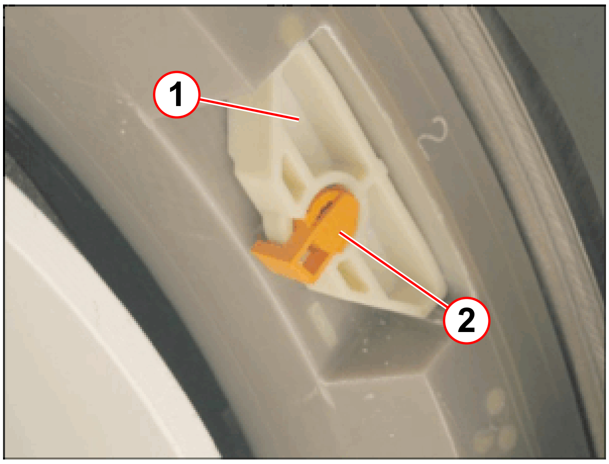
Work after shimming

Locking the shim trays.

> **i** Certain sequences cause extreme transient vibrations of the gradient coil and its surroundings. This could result in components shaking loose and cause severe spiking in the images.

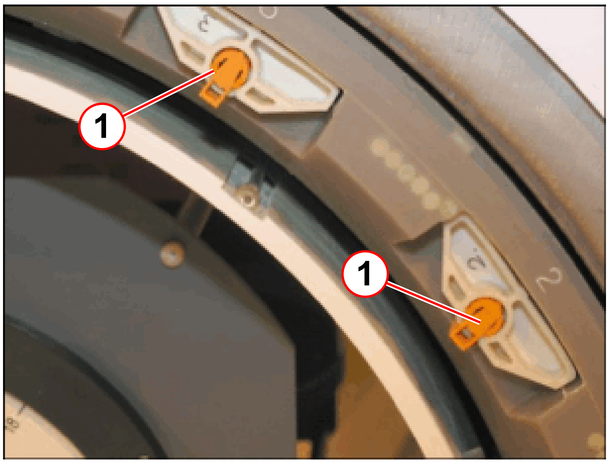
» Therefore, after finishing the passive shim, all fastening screws of the shim trays need to be locked.

Fig. 239: Shim tray



(1) Shim tray
(2) Screw locking device

Fig. 240: Shim tray



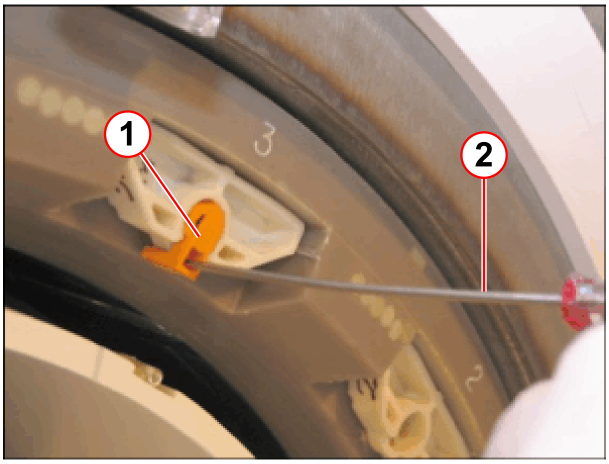
(1) Screw locking device

Unlocking the screw locking device

- Use a flat head screwdriver for unlocking.

Fig. 241: Shim tray

Fig. 242: Shim tray



(1) Screw locking device
(2) Screwdriver

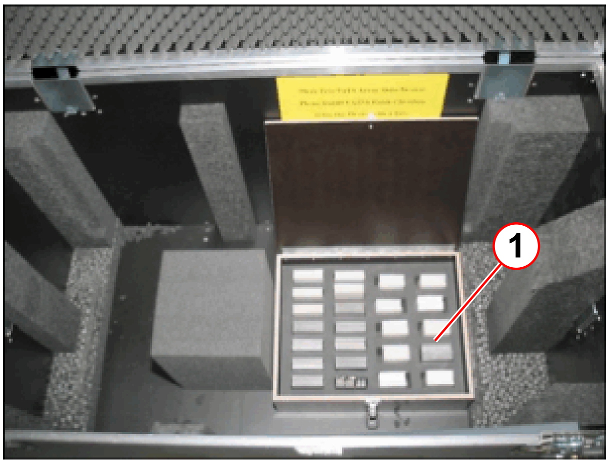
- Install the funnel

Remaining shim plates

The packaged shim plates that are not required should be stored as backup plates in the array shim device kit. These extra shim plates can be used in case the delivered shim plates do not suffice.

Store shim plates only in their original packaging in the array shim device kit.

Fig. 243: Array shim device kit



(1) Shim plates

Fig. 244: Packaged shim plates

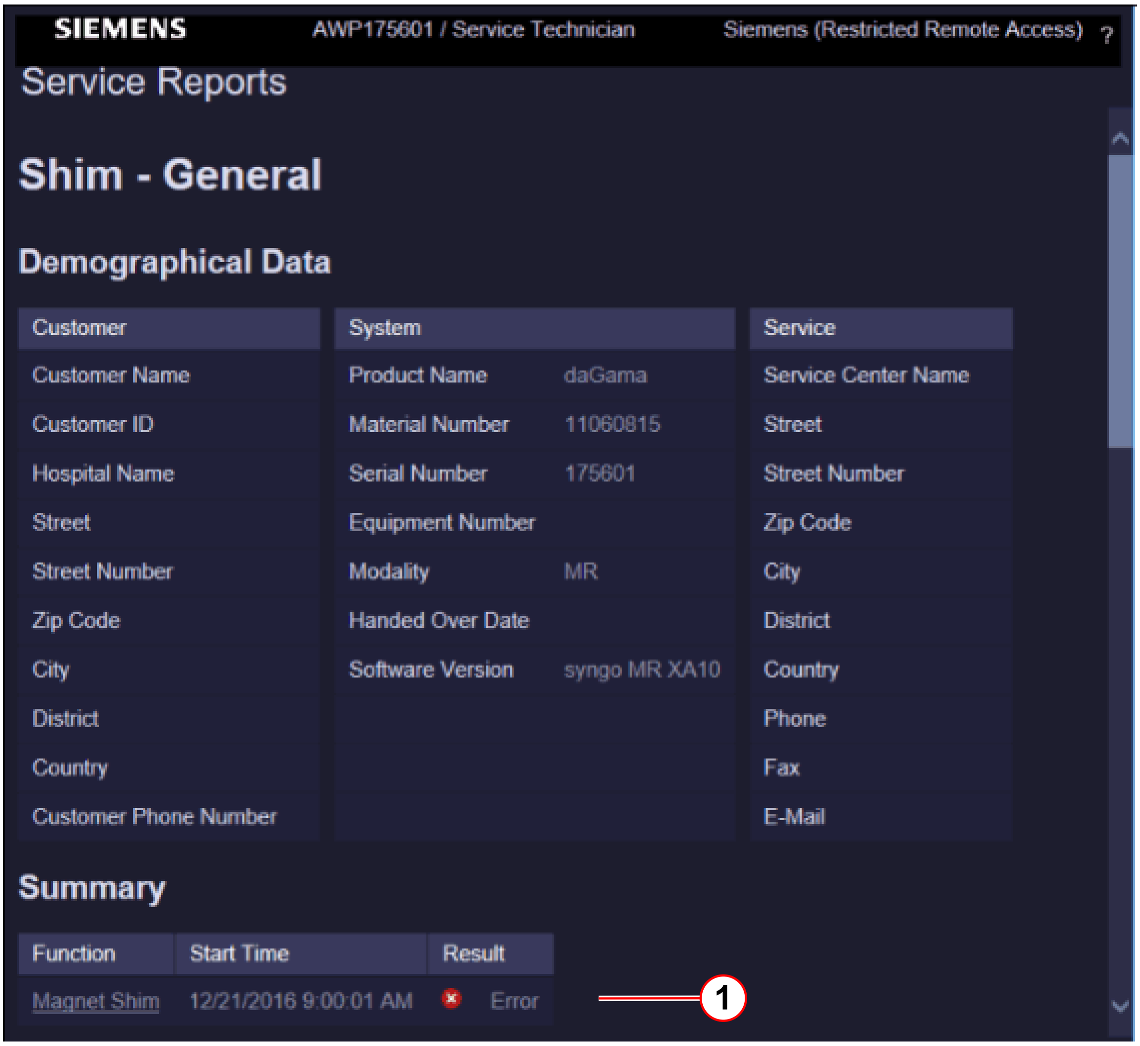
Error Handling

What to do if shim workflow fails

The following test should be performed.

1. Check the shim output for error messages.
   » “Service Reports / Shim - General”

Fig. 245: Shim - General

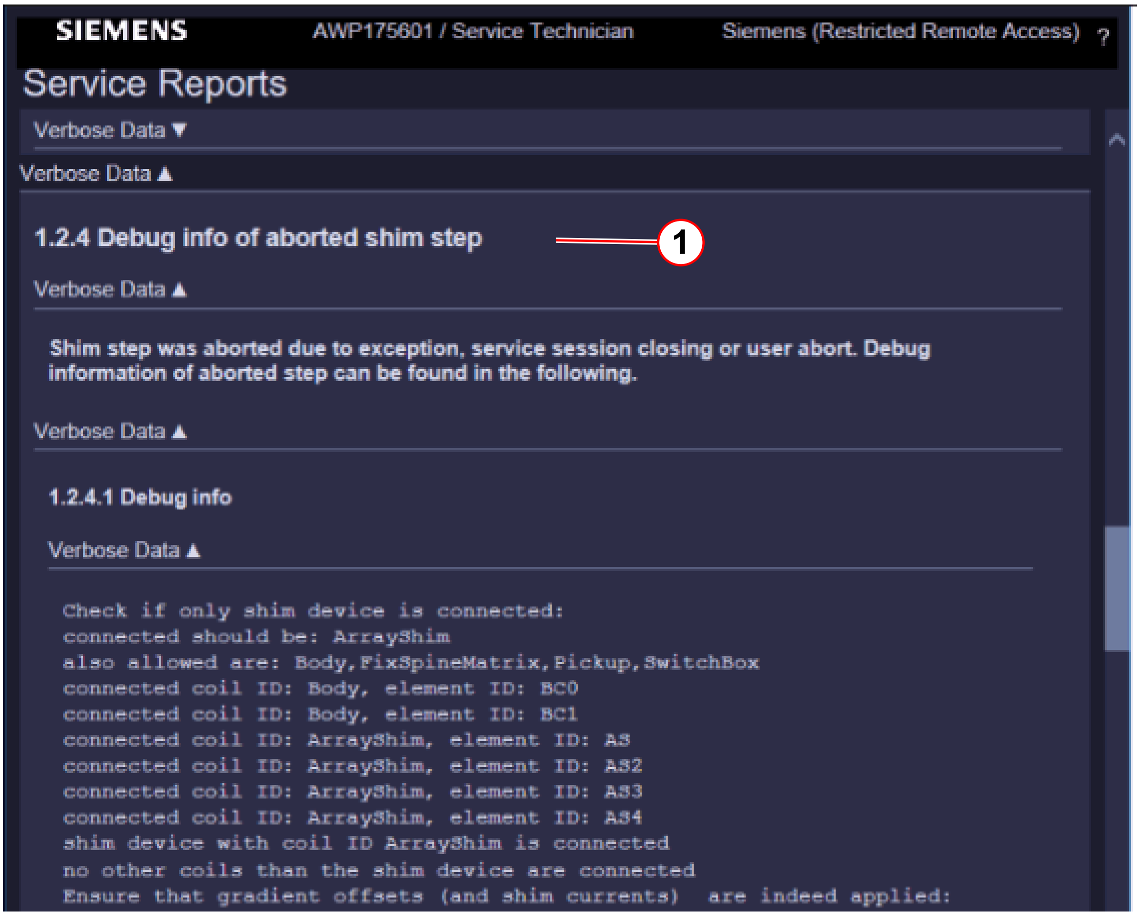


(1) Error

Debug mode

The debug mode shows detailed information.

Fig. 246: Service Report



(1) Debug Information

Shim support

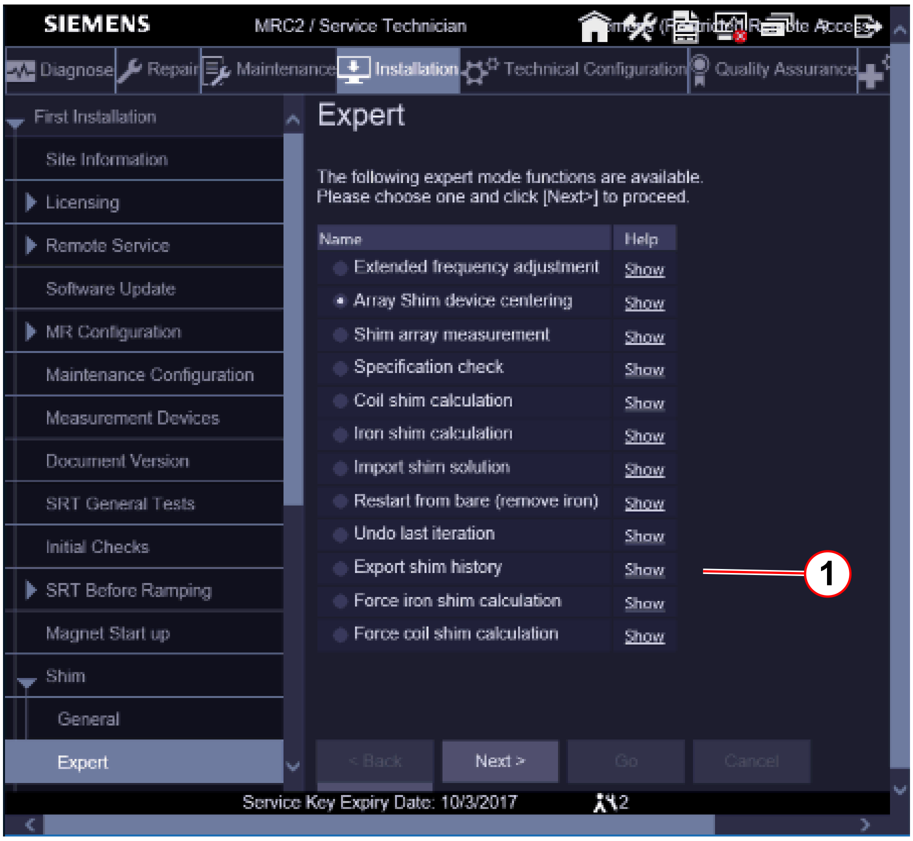
Send the shim folder as a ZIP file to the following address to get shim support:

» magnet.hsc.med@siemens.com

Open the Shim Expert mode:

Select: **Export shim history, Fig.58, Pos1**

Fig. 247: Shim Expert



(1) Export Shim History

This mode saves the current shim history, i.e. shim settings file and all IQShim input and output files, to a “MagnetShim-zip” file.

The zip file is saved in the directory: **K:\Seso\ShimExport\MagnetShim-yyyy-mm-dd-hhmmss.zip**

3.2.2 Shim - Expert

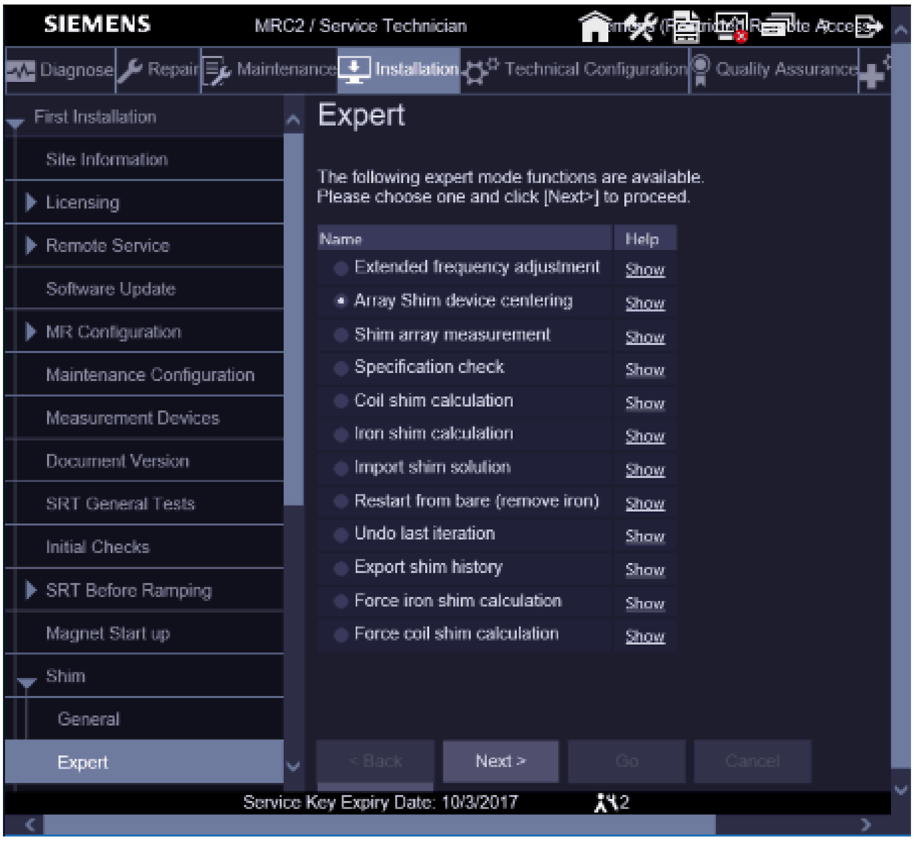
Help ID: EB5078AF-7B40-4D9D-89AF-AA2E282632E1
Service Level: 7

The Expert shim platform allows the individual workflow steps to be started.

Special functions are helpful for troubleshooting.

- Choose the required workflow step.
- Click **Next**

Fig. 248: Shim Expert



3.2.2.1 Extended frequency adjustment

Help ID: 6D7C9E5B-3B13-4ED4-9640-0724C6B5019F
Service Level: 7

An extended frequency adjustment is performed.

3.2.2.2 Array Shim device centering

Help ID: 58A216C8-A96F-4B33-A7D2-079519950845
Service Level: 7

- A gradient centering check is performed.
- Output of required Array Shim Device shift.
- No field measurement afterwards.

3.2.2.3 Shim array measurement

Help ID: CDA931B8-9056-4D95-9FDD-07BDCD61AE1F
Service Level: 7

- Field plot measurement without gradient centering.
- PTAB position check.

- EIS reset.

3.2.2.4 Specification check

Help ID: DFE046D0-340D-4EDE-942A-8BF0F2AFE2CC
Service Level: 7

- Using the existing field plot.
- Run IQShim program.
- Output of harmonics and specification values.

3.2.2.5 Coil shim calculation

Help ID: B3C2E0FD-A78E-4E8D-A64B-604F890EF92A
Service Level: 7

- Using the existing field plot.
- Coil offset calculation is performed.
- Check and eccept or cancel the result.

3.2.2.6 Iron shim calculation

Help ID: F978548E-41D2-412D-930D-83FFACE9C05C
Service Level: 7

- Using the existing field plot.
- Iron shim calculation is performed.
- Check and accept or cancel the result.

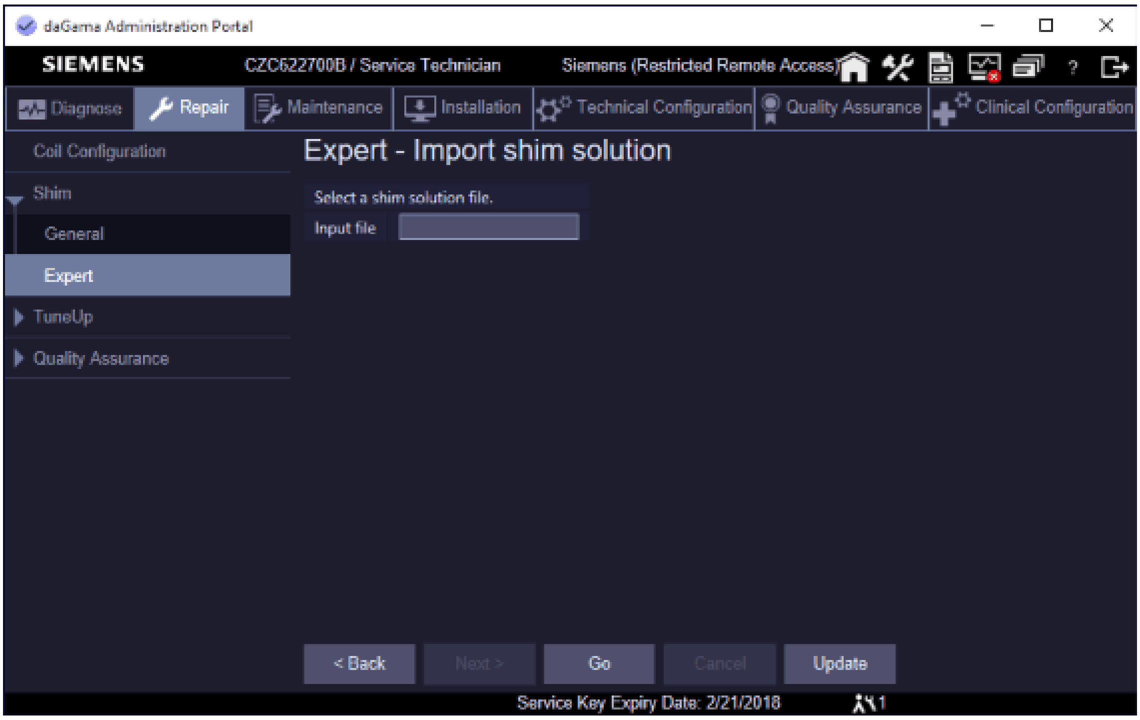
3.2.2.7 Import shim solution

Help ID: EE46BD6D-75B7-4463-B8D4-CFAE47AB3601
Service Level: 7

It may happen that a standard calculation of IQShim is not appropriate for shimming a magnet under certain circumstances. In this case, the current shim history can be sent to a shim expert. The expert performs an IQShim calculation with special parameters and sends the result back, i.e. IQShim output xml file.

The solution can be imported with Expert - Import Shim Solution.

Fig. 249: Import Shim Solution



3.2.2.8 Restart from bare (remove iron)

Help ID: 23BEB07C-6700-418F-AFCE-FDE36AC8785E
Service Level: 7

Remove all iron from the magnet

- Remove all shim iron from the magnet.
- The loaded iron is set to zero in the data base.
- The offset current is set to zero in the data base.

  The old data still remain in the data base for reference.

  Re-set action is logged.
- Iron iteration count is reset to zero
- The shim status is set to “To Do”.

3.2.2.9 Undo last iteration

Help ID: 2C2E208B-0B1A-4159-9B05-521DCFDE12F2
Service Level: 7

- Remove the last iron iteration.
  » The Iron count is updated
- Old data remained in the database for reference.
- Shim status is set to “To Do”

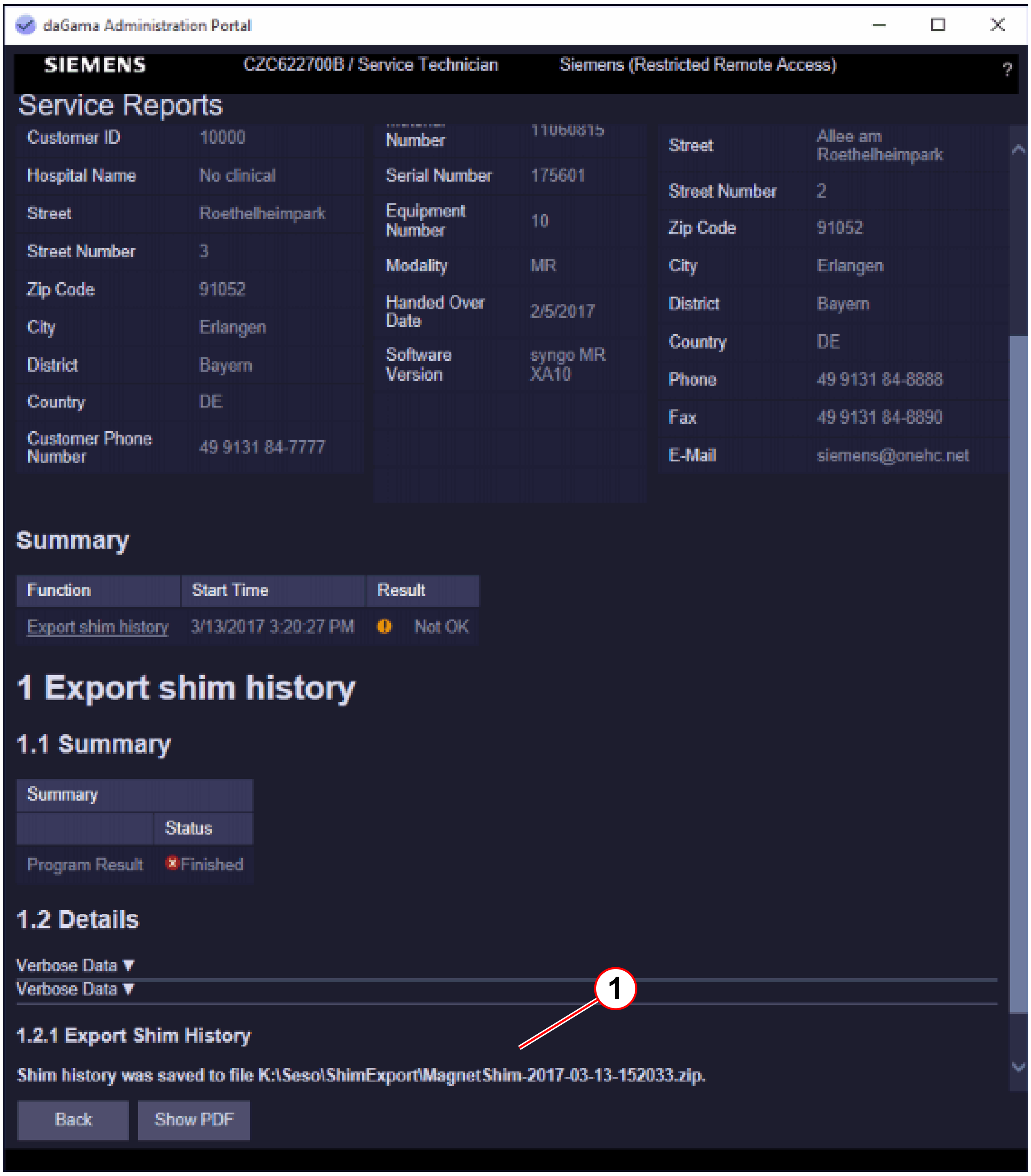
3.2.2.10
Export shim history

Help ID: A584E976-AA36-4210-BFF5-50D6E3190E7A
Service Level: 7

- The content of the Magnet Shim working directory is zipped automatically
- The zip file is saved in the directory: **K:\Seso\ShimExport\MagnetShim-yyyy-mm-dd-hhmmss.zip**

  Refer to Service Reports / Export shim history, Fig 1, Pos1.

Fig. 250: Export Shim History



(1) Log Directory

3.2.2.11 Force iron shim calculation

```
Help ID: 1F72B798-45D8-424E-9C70-1D35B3BE1634
Service Level: 7
```

- The force calculation flag is set in the iqshim_input.xml file.
- Plot checks are ignored.
- Iron calculation is performed.
  » Confirm or cancel the result.
- Shim workflow is set to “To Do”.
- Perform a new shim iteration

3.2.2.12 Force coil shim calculation

```
Help ID: D99B8108-6521-4082-A8BB-96A08505879F
Service Level: 7
```

- The force offset flag is set in the iqshim_input.xml file.
- Plot checks are ignored.
- Offset calculation is performed.
  » Confirm or cancel the result.
- Shim workflow is set to “To Do”.
- Perform a new shim iteration

3.3 TuneUp

3.3.1 TuneUp - General

Help ID: 9A54DF7F-CE66-4190-863F-B65D390E1BA5
Service Level: 3

Introduction

General TuneUp contains a fixed set of procedures (workflow) for calibrating system parameters

Each individual procedure automatically saves data in the corresponding system parameter files, log files and database files.

TuneUp Workflow

- General TuneUP is separated into modules, each routine runs as a stand-alone process.
- After start, each procedure reads the specification and configuration parameters automatically.
- TuneUp procedures determine if their measured results are in or out of specifications.
- TuneUp procedures write their individual results to the Service Report.
- The status is returned to the main TuneUp platform and the corresponding status field in the Service UI is updated.

After pressing the **[Go]** button in the TuneUp UI, the **procedures start and perform typical tasks**:

- Each procedure registers the Service Patient (if not already registered).
- If a clinical patient is registered, a pop-up informs the operator to de-register the clinical patient before proceeding.
- The correct phantom setup is displayed by a pop-up and confirmed by the operator (if applicable). If QA is performed remotely, automatic table move is disabled and the operator receives a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Table movement:
  - If **TuneUp is performed locally** at the system, the **table moves automatically** to center the required phantom in the measurement field (change between large/small spherical phantom).
  - If **TuneUp is performed remotely**, **automatic table move is disabled**. The operator gets a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Configuration and specification values are read.

- Measurements are performed.

TuneUp Function Status

Tab. 4 Possible TuneUp Function Status

Status	Description
To Do	The TuneUp procedure has not yet been performed yet and is pending. OR: The TuneUp procedure has to be performed again because another TuneUp procedure has requested its execution.
Ok	The TuneUp procedure has been carried out successfully.
Not Ok	The related parameters/results were found to be out of specifications during the last operation.
Error	An error occurred during the last operation.
Cancelled	Procedure canceled by operator.

Evaluation/Results

After end of the TuneUp measurement, evaluations are performed automatically:

- The procedures determine if the measured results are in or out of specifications.
- Determined parameters are saved to system files if the procedure results are in specifications.
- The results are written to the Service Report.
- The results status is returned to the main TuneUp platform and the corresponding status field is updated.

TuneUp Verification

All **TuneUp** workflow procedures have their corresponding verification routines in the **Quality Assurance** workflow menu.

TuneUp/TuneUp Dependencies

TuneUp procedures are executed in a listed order because some steps depend on a successful execution of previous steps. For example, the successful execution of “Gradient Regulator” is required for all following TuneUp steps which use an MR-signal.

This means, if the operator runs a Tune Up procedure again, also the status of all depending procedures is reset to **To Do** and they have to be performed again.

In addition, the Software monitors connected MR-system components (for example,TX-Box) for hardware changes. If a change is detected (for example, after parts replacement,) the affected TuneUp adjustment steps are automatically reset to **To Do** status.

TuneUp/QA Dependencies

If a TuneUp procedure is successfully completed and has changed its status, the corresponding verification procedure(s) in the Quality Assurance workflow menu is/are changed to status **To Do**.

TuneUp procedures may also affect multiple QA procedures, for example, after TuneUp Phantom Shim, the QA procedures Phantom Shim Check, Gradient Sensitivity Check, BC Image Brightness and Long Term Stability Check have to be performed.

3.3.1.2 Feedback Loop Calibration

Help ID: C2FE7D73-8E4B-4758-8858-5A5CD39E46B0
Service Level: 3

General

The TX-Box has a built-in feedback loop to compensate for instabilities of the TX B1-field.

The procedure **Feedback Loop Calibration** is used to calibrate internal feedback loop parameters of the TX-Box.

From experience it is known, that the feedback signal phases and the feedback delay depend on the TX-Box loadware version. Thus, an individual phase- and delay-adjustment must be performed in TuneUP although the feedback phase has been initially adjusted during TX-Box fabrication. Improper feedback phase and delay results in suboptimal dynamic behavior and can, in the worst case, even cause oscillation.

For non-pTx systems, a second measurement through the LC path is needed in case the phase difference between LC and BC channels needs calibration.

For systems with multiple transmit channels (pTX), the relative phases of the various channels are free parameters anyhow which are continuously controlled by the measurement system.

After a loadware change at the MR system, it might be necessary to repeat the phase- and delay-adjustment by running Feedback Loop Calibration in TuneUP. The procedure reads out the current delay value of the TX-Box. Then, the phase of second TX-channel is determined at minimum, middle and maximum operation frequency. From the measured phase values, the new delay value and a new file "nb_hybrid_compen.dat" is calculated and loaded into the flash memory of the TX-Box.

Measurement

BC Path

The service sequence “tuncal” is invoked, which runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box. Then, the Software analyzes the frequency dependence of the signal phases.

A linear curve is fitted to the frequency dependency of the signal phases. From the slope of this line, the feedback loop delay is obtained. In order to enhance the precision, the results obtained from both Body Coil elements are combined.

LC Path

A hardware flag determines, if a second measurement through the LC path is needed or not. That measurement calibrates the phase difference between LC and BC channels.

For the measurement, the Service Plug is connected and the service sequence “tuncal” is run through the LC TX-path.

The obtained data set is analyzed in the same way as for the BC TX-path. The difference of the BC and LC phases at the center frequency contains the desired value.

Restart MaRS

After the successful determination of the feedback control loop parameters, a restart of the MaRS Software is required (only needed in TuneUp mode).

The boot process starts the activation of the feedback loop parameters in the flash memory of the TX-Box. Afterward, the feedback control loop is ready for normal operation.

Evaluation/Results

Results in Report:

Tables:

- Feedback loop delay:

  - Delay evaluation

  - Dispersion of delays from single BC elements

- Restart MaRS Software Results

Plots (debug mode):

- BC phase plot

Modified parameters:

- Feedback loop delay

- Feedback loop phase deviation

are saved

3.3.1.3 RX Gain Calibration

Help ID: 66C1477B-627A-4488-A469-B5EC0946F2E6
Service Level: 3

General

The individual receive channels in the RX-electronics have slightly different electrical/ frequency characteristics which must be corrected. Otherwise, these variations would cause amplitudes deviations of the individual receive signals leading to image brightness variations in the final image.

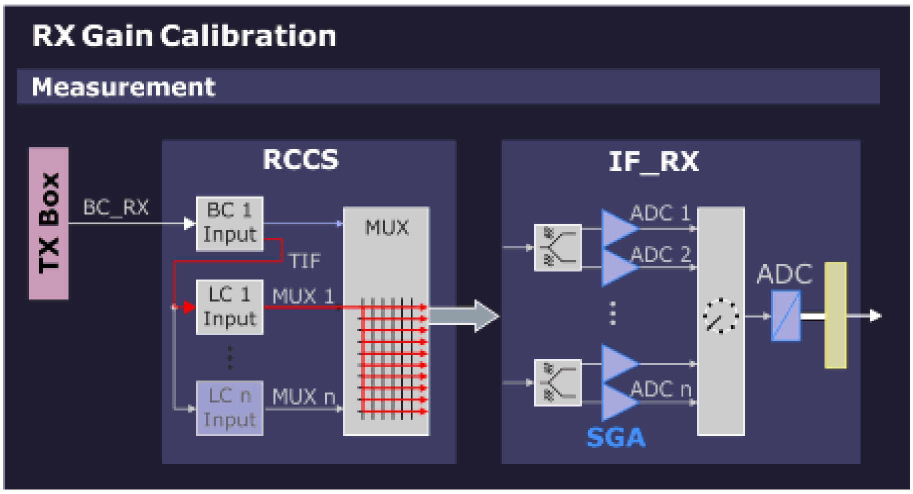
Therefore, it is necessary to normalize the signal intensity of each receive channel so that the image intensity of the final image is the same regardless which receiver channel(s) were used for imaging.

The receiver-related deviations are determined with the RX Gain Calibration with 2 measurements:

- **Channel correction** - determine deviations caused by the switch matrix (in RCCS).
- **High/Low gain correction** - determine amplitude and phase deviations between high- and low-gain settings of the switchable-gain amplifiers (SGA) of the IF_RX board(s).

Measurement

Fig. 251: RX Gain Calibration - Measurement



A fixed calibration signal (BC_RX) is generated in the TX Box and transferred to the RCCS over coax cable. This signal is then fed through the RCCS-internal local coil input multiplexer to the switch matrix where it is sequentially routed through to each of the SGAs and ADCs on the IF_RX boards.

There are 2 measurements made per channel at high- and low-gain settings.

Evaluation/Results

The correction factor of the measured amplitudes for the RX-channels and the quotient of the high/low gain modes are evaluated.

Results in Report:

Tables:

New calibration factors

Plots:

Measured signals

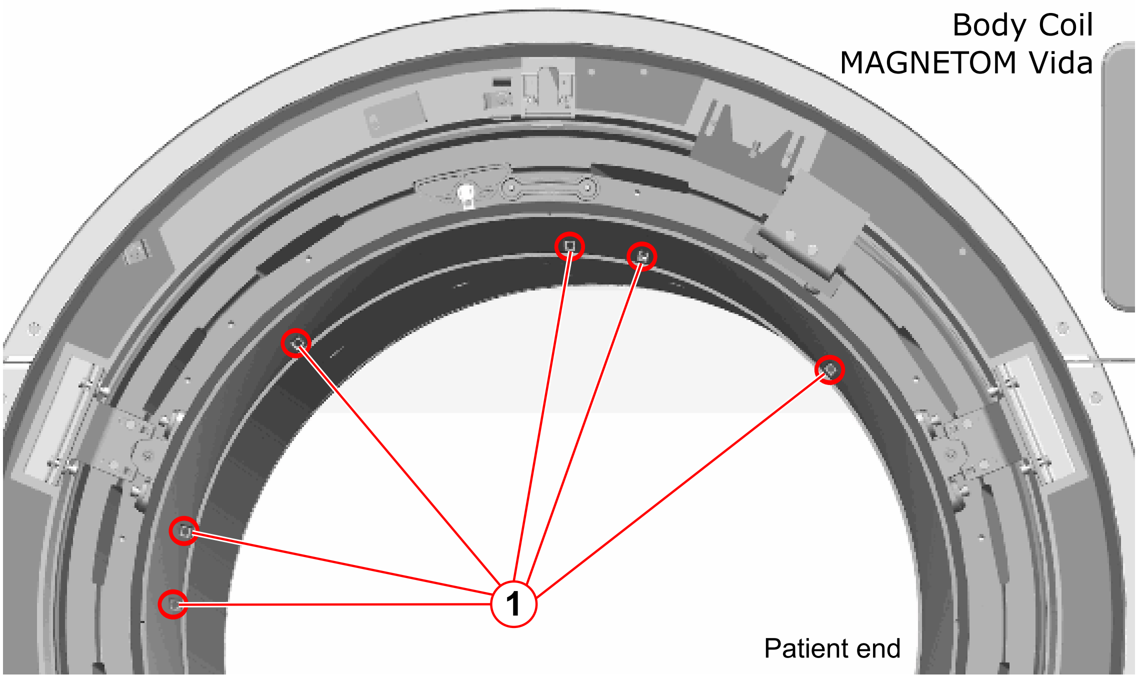
Modified parameters:

The RX calibration factors for all RX channels in high/low gain mode are saved.

3.3.1.4 BC Tuning

Help ID: 03DC2B49-7E45-4948-BAC2-00780F56D2DB
Service Level: 3

Fig. 252: Body Coil - BC Tuning



(1) Location of Trimmer Capacitors

Safety

WARNING

In case a BC Mechanical Position Adjustment at magnet rear-side (service end) is required, you will work close to Gradient Coil connections. These cables carry high voltage!

Risk of electrical shock!

» **Switch-off the GPA with switch S7/ DGSU and circuit breaker F1 in the GPA cabinet.**

» **Subsequently, wait at least 20 minutes to discharge GPA capacitors before starting any BC Mechanical Adjustment.**

» **However, for only adjusting the BC Tuning trimmer capacitors (patient end), the GPA cabinet can be left switched-on.**

NOTICE

The trimmer capacitors have 32 full turns each.

Risk of damage!

» **Do not use force when reaching the capacitor end stops to prevent damage!**

General

The **BC Tuning** procedure is used to measure and, if necessary, adjust the central resonance frequencies of the Body Coil resonator systems and minimize the coupling between them.

A high efficiency of RF power transmission to the Body Coil requires sufficiently low reflection factors at the TX-Box output.

This condition is usually met, if the reflection factors of the Body Coil resonator systems BC0 and BC1 are high (without load) and equal. However, with load, these reflection factors shall be low.

The other requirement is, coupling between both resonators shall be minimum (that is, high decoupling values).

Measurement

BC Reflection

First, a **BC Reflection** measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals, covering the full transmit operating frequency range. The result of each measurement is the reflection or r-factor = Ur/Uf which is plotted for both Body Coil systems BC0 and BC1 along the frequency axis.

The evaluation determines:

- that the tuned frequency of each coil is within a specified range, that is, not too far-off from the operating range center.
- that the center frequencies of both coil systems BC0 and BC1 are tuned closely to each other, that is, the delta frequency is low.
- that the reflection factors are within expected limits. These values provide useful information about the quality (Q-factor) of the BC resonant circuits. They also depend on the coil load and cannot be adjusted.

  The Software detects the resonant frequency always at the point where the reflection of the BC has its minimum.

Transmission

In a second step, a **Transmission** measurement is performed with transmitting a signal into the BC0-system and measuring the coupled signal in the BC1-system and in the opposite direction. The transmission factor describes the amount of RF-coupling between the two Body Coil systems, mainly due to coil construction but also affected by other factors, for example, mechanic position adjustment of the Body Coil within the Gradient Coil Faraday Shield.

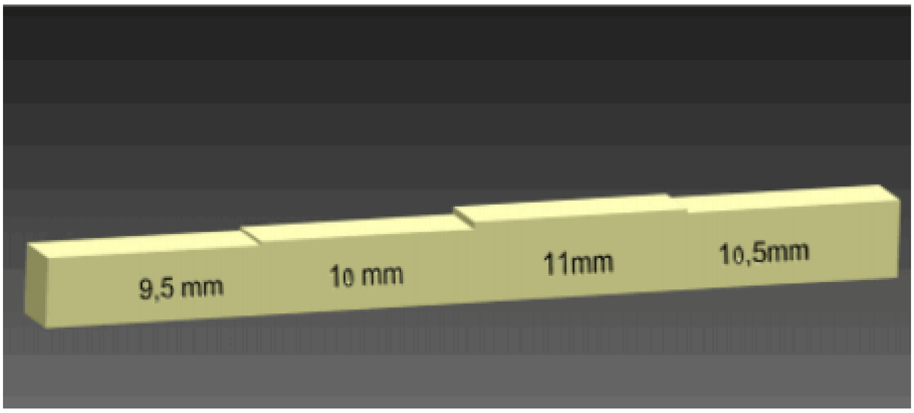
The evaluation checks the transmission value, expressed in decibels (dB), to be within expected limits. **A high negative dB-value indicates minimum coupling**, that is, a good decoupling between BC0 and BC1 system.

Adjustment

Prerequisites

- For BC mechanical position adjustment, the BC feeler gauge (material number 106 76 562) is required.

Fig. 253: BC Feeler gauge (10676562)



- To access the trimmer capacitors at the **Body Coil front-side** remove the service access cover and the front funnel insert.

- Take out the reflector bracket from its storage position behind the magnet front cover.

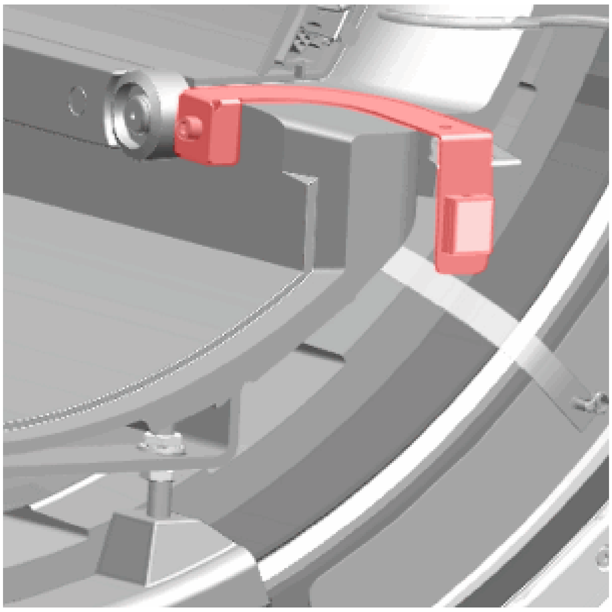
Fig. 254: Reflector bracket storage position

(1) Reflector bracket

> The reflector position defines the drive-in height for the patient table and is adjustable. At first use, check for the correct adjustment to avoid patient table collision.

- Fasten the reflector bracket with a screw from the removed front funnel.

Fig. 255: Reflector bracket attached to BC



Generally, BC Tuning is performed in 3 Steps:

- Step 1: Reset (BC Mechanical Position + Trimmer Capacitors)
- Step 2: Adjust BC Mechanical Position
- Step 3: Adjust Trimmer Capacitors

Exception for new installations or if BC Tuning is already close to specification:

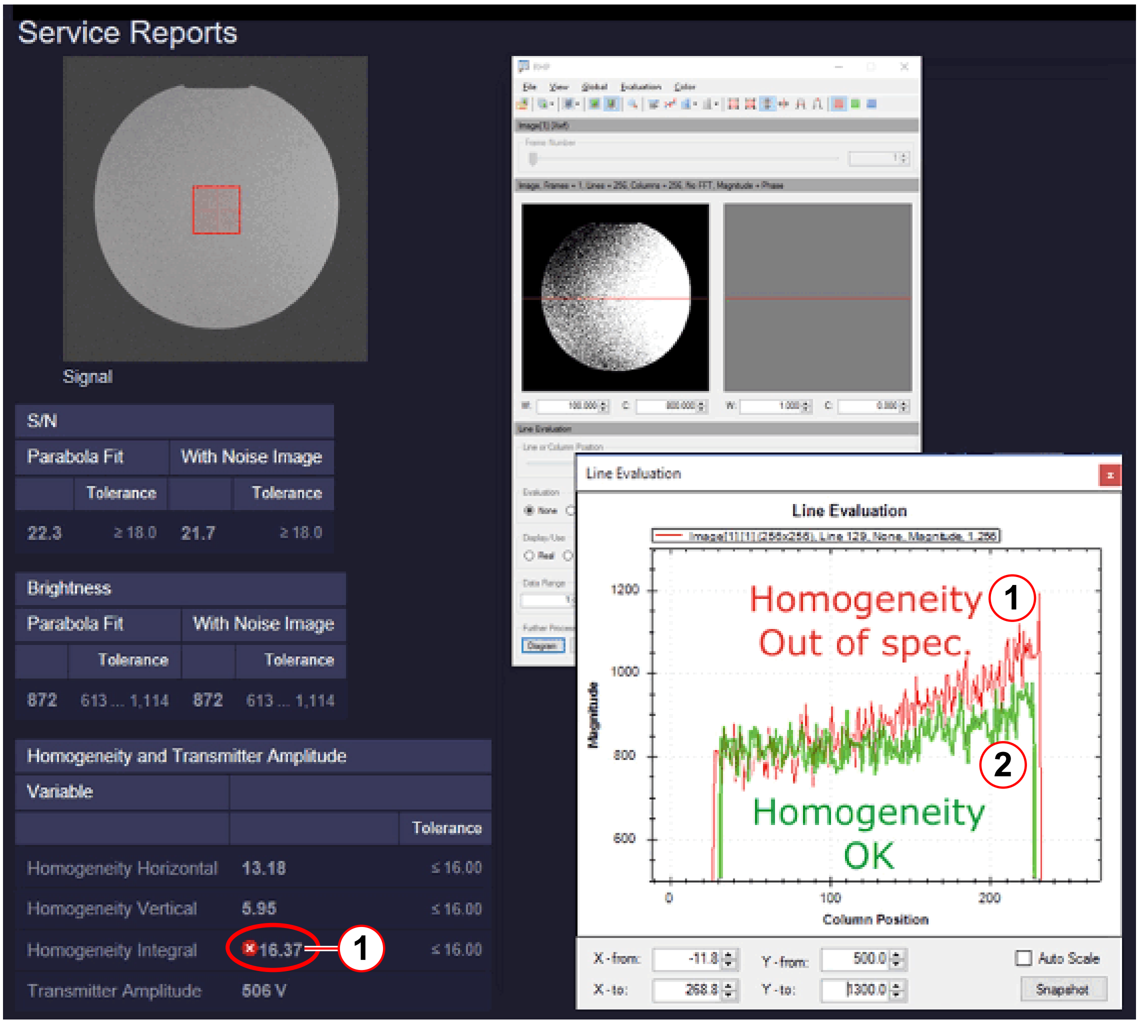
- Skip Steps 1 and 2.
- Perform Step 3 “Adjust Trimmer Capacitors” directly.

Always perform the complete BC Tuning workflow (Step 1-3) after:

- BC was replaced.
- BC Tuning: Transmission value is less than -15dB.
- BC Tuning: Reflection factor difference for BC0 and BC1 is greater than 0.1.
- QA Image Brightness Check fails with homogeneity results out of spec.

- Strong asymmetry in RF-field (B1-field) distribution (for example, strong inhomogeneity in QA Image Brightness Check, transversal orientation).

Fig. 256: QA/ BC Image Brightness Check - Results



(1) Homogeneity NOT OK
(2) Homogeneity OK

Step 1 - Reset

Reset any changes and establish the Body Coil default set up.

Switch off GPA

- Switch off the GPA with switch S7/ DGSU and circuit breaker F1 in the GPA cabinet.
- Wait at least 20 minutes to discharge GPA capacitors before starting BC Mechanical Adjustment!

Set Trimmer Capacitors to Center Position

- Turn all trimmer capacitors carefully in clockwise (“CW”) direction until mechanical stop.

- Now, turn 16 full rotations back in counter-clockwise (“CCW”) direction.

Set BC Mechanical Position to “Center”

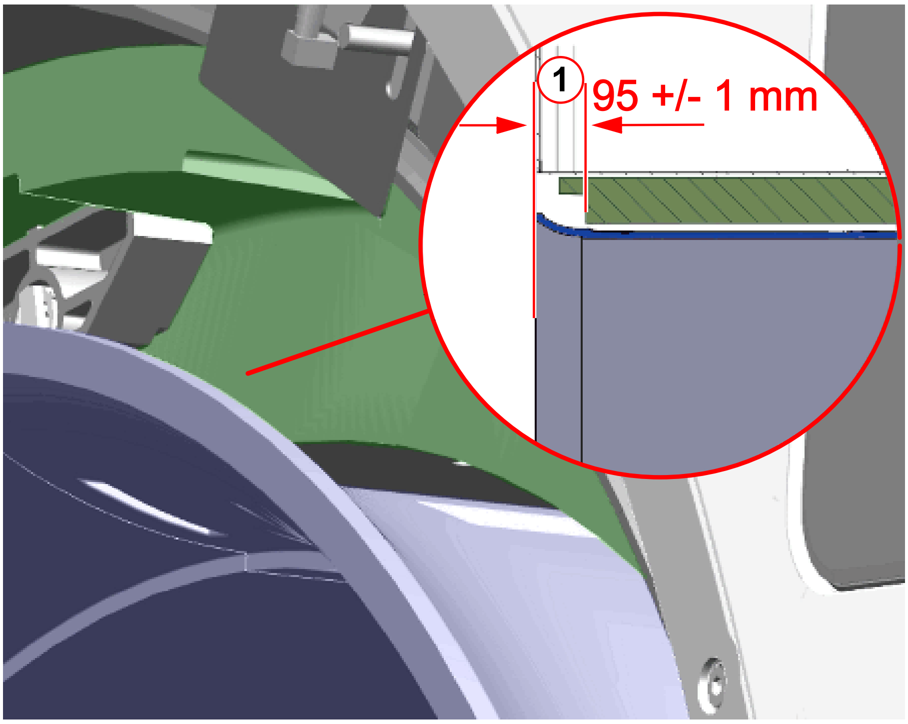
> The Body Coil is aligned with respect to the inner bore of the Gradient Coil (the RF-shield).

> For BC mechanical position adjustment, the BC feeler gauge (material number 106 76 562) is required.

At **Patient end** adjust fixed BC distances to the Gradient Coil:

- In **Z-direction**, adjust **distance of 95 +/-1 mm** between BC tube front surface and recess in GC (see graphics below).

Fig. 257: Body Coil - Front View

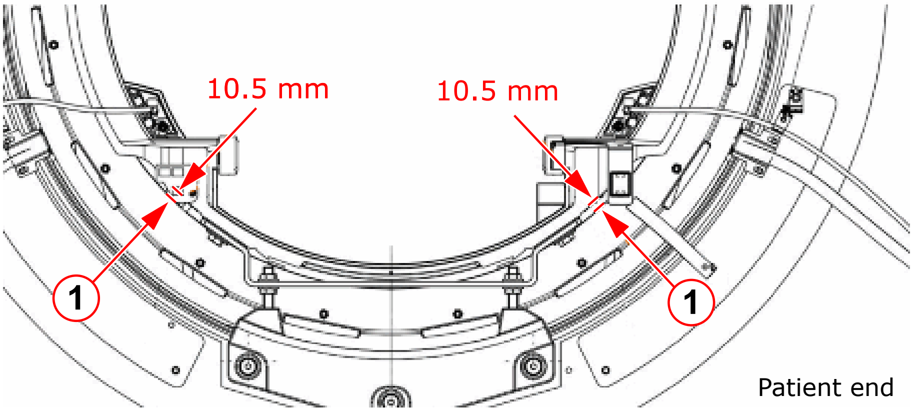


(1) Z-distance 95 +/-1 mm

- In radial direction, **adjust the 2 left/right distances equally to 10.5 mm**.

  Measure at the reference points next to the BC support (from the RF-shield to the 2 flat reference surfaces on the BC bore tube).

Fig. 258: Body Coil - Front View

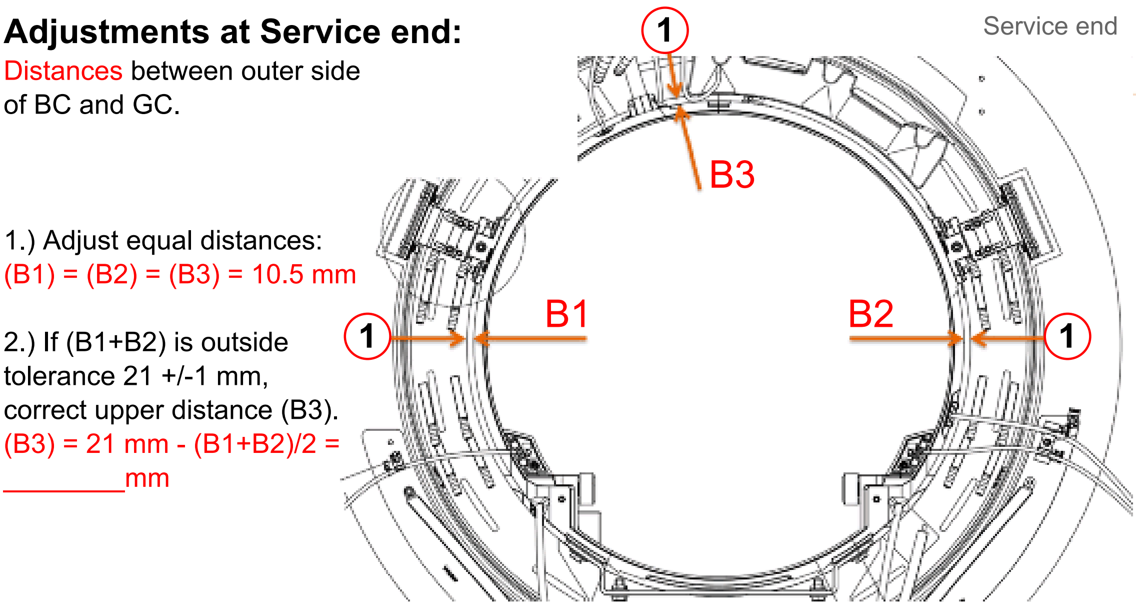


(1) Left/right distance 10.5 mm

At **Service end** center the BC radially inside the GC:

- Adjust **equal left/right distances (B1 = B2) =** 10.5 ± 0.5 mmbetween BC and GC.
- Check the **upper distance (B3)** between BC and GC; **(B3) =** 10.5 ± 0.5 mm.
- Check all distances again **(B1 = B2 = B3) =** 10.5 ± 0.5 mm
- **If (B1 + B2) is outside tolerance** 21 ± 1 mm, correct upper distance: **(B3) = 21 mm – (B1 + B2)/2**

Fig. 259: Body Coil - Service end



(1) Measurement points for distance BC-GC

At service end, the lower BC position (at 6 o'clock) cannot be directly measured because the BC support blocks access.

Due to mechanical tolerances, the actual distance values may slightly differ from the nominal value of 10.5 ± 0.5 mm.

Some calculation is required then, see description above.

Step 2 - Adjust Mechanical Position

The Body Coil Tuning is sensitive to its exact mechanical position inside the Gradient Coil.

A further optimization of the BC mechanical position might be necessary to achieve a successful BC Tuning adjustment with trimmer capacitors.

Switch off GPA

- Switch off the GPA with switch S7/ DGSU and circuit breaker F1 in the GPA cabinet.
- Wait at least 20 minutes to discharge GPA capacitors before starting BC Mechanical Adjustment!

Criteria for a good BC Mechanical Adjustment

1. BC Tuning results (no trimmer adjustments performed yet):
   - **Transmission value better than -25dB**
   - The **reflection factors** for BC0 and BC1 are **maximum**
2. Homogeneous RF-field (B1-field):
   - **QA / BC Image Brightness Check is in spec.**

The BC mechanical adjustment influences the reflection and transmission factors (BC Tuning results) and the B1-homogeneity in the unloaded coil.

Therefore, a balance might be required to satisfy both requirements.

In most cases, a good adjustment of reflection and transmission values also leads to a good B1-field distribution.

Check of BC Tuning Results

Run BC Tuning with the defined QA setup and check:

- Are the “factors (at minimum)” in specification?
  - » Under normal conditions, these **reflection factors without load should be greater than 0.55**.
  - » Otherwise, if the reflection factors without load are much lower then 0.55, the BC mechanical position might be incorrect or additional losses are present (for example, defective capacitors at the Faraday Shield of the GC).
- Is the “factor (at minimum)” **difference for BC0 and BC1 less than 0.1**?

- Is the **transmission value better than -25dB** (that is, larger negative value)?
  » If all conditions are OK, continue directly with Step 3.
  » Otherwise, continue with further optimization below.

Further Optimization of Mechanical Position

At Service end, try:

- Move BC up 1 mm

  or
- Move BC down 1 mm

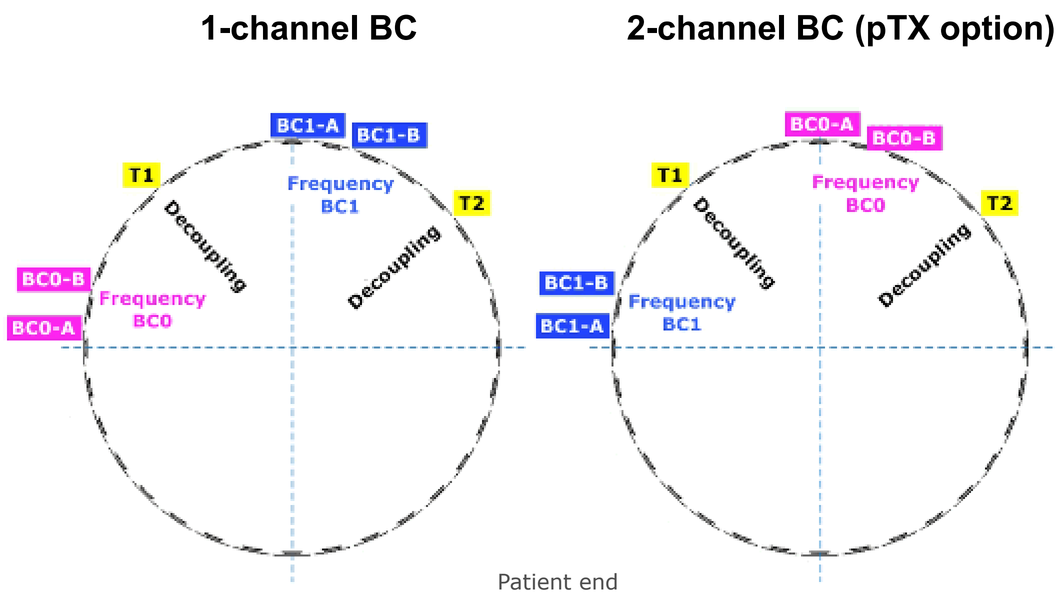
  or
- Move BC left 1 mm

  or
- Move BC right 1 mm
  » Check after each change if the three BC Tuning Results (see above) have improved.

Step 3 - Adjust Frequency and Decoupling

Adjustment of Trimmer Capacitors

Fig. 260: Trimmer Capacitor assignment



> The Trimmer Capacitor assignment is different for 1-ch BC or 2-ch BC (pTX Option)!

- The BC has **6 trimmer capacitors** which are accessible from patient end.

- The **trimmer capacitors** which belong to the same function (for example, decoupling T1, T2) **are arranged in pairs**.

  Turn a capacitor pair synchronously in the same direction and with same number of turns.
- **Turning a capacitor in clockwise direction** (“CW”) increases its capacitance value which **decreases frequency**.
- More detailed adjustment information is shown at the magnet display during the BC Tuning workflow.

Evaluation/Results

Results in Report:

- The reflection-related values for both (BC0 and BC1) systems (for example, resonance frequency, reflection factor and the difference between both systems).
- The transmission-related values (frequency and decoupling between both systems).
- The new and old phase difference.
- Diagrams showing the values for transmission and reflection as a function of frequency.

3.3.1.5 Cable Length

Help ID: C129565B-356D-4AE3-8F27-3D3ADE42E986
Service Level: 3

General

The **Cable Length** measurement determines the individual length of the BC transmit cables. This information is needed for the extended B1-field regulation loop of the TX-Box.

For an optimal operation of the TX B1-field regulation loop, precise knowledge of the phase situation at the Body Coil is necessary. These signal phases depend mainly on the known RF-properties of the Body Coil itself. However, they also depend on the length of the BC transmit cables which induce an additional phase shift into the TX-path. The actual BC transmit cable lengths may vary slightly (production tolerances) and must be determined at the system by a precise measurement of the induced phase shifts.

Afterward, the exact phase situation at both Body Coil systems is known. This information is used to better analyze and compensate B1-field fluctuations which can easily occur during a measurement (e.g. a BC impedance change caused by thermal drift effects).

Measurement

The Cable Length measurement uses the service sequence rf_loop. An RF pulse is fed to both Body Coil channels with the Body Coil switched to ‘detune’ mode. The forward and reflected signals are taken from the internal directional couplers (DICO) of the TX-Box and analyzed by the monitoring Software.

From these results, the so-called S-matrix elements which describe the characteristics of the BC transmit path are determined. These values allow, together with the known Body Coil RF-properties, to calculate the cable lengths for both Cody Coil channels in terms of the induced signal phase shifts.

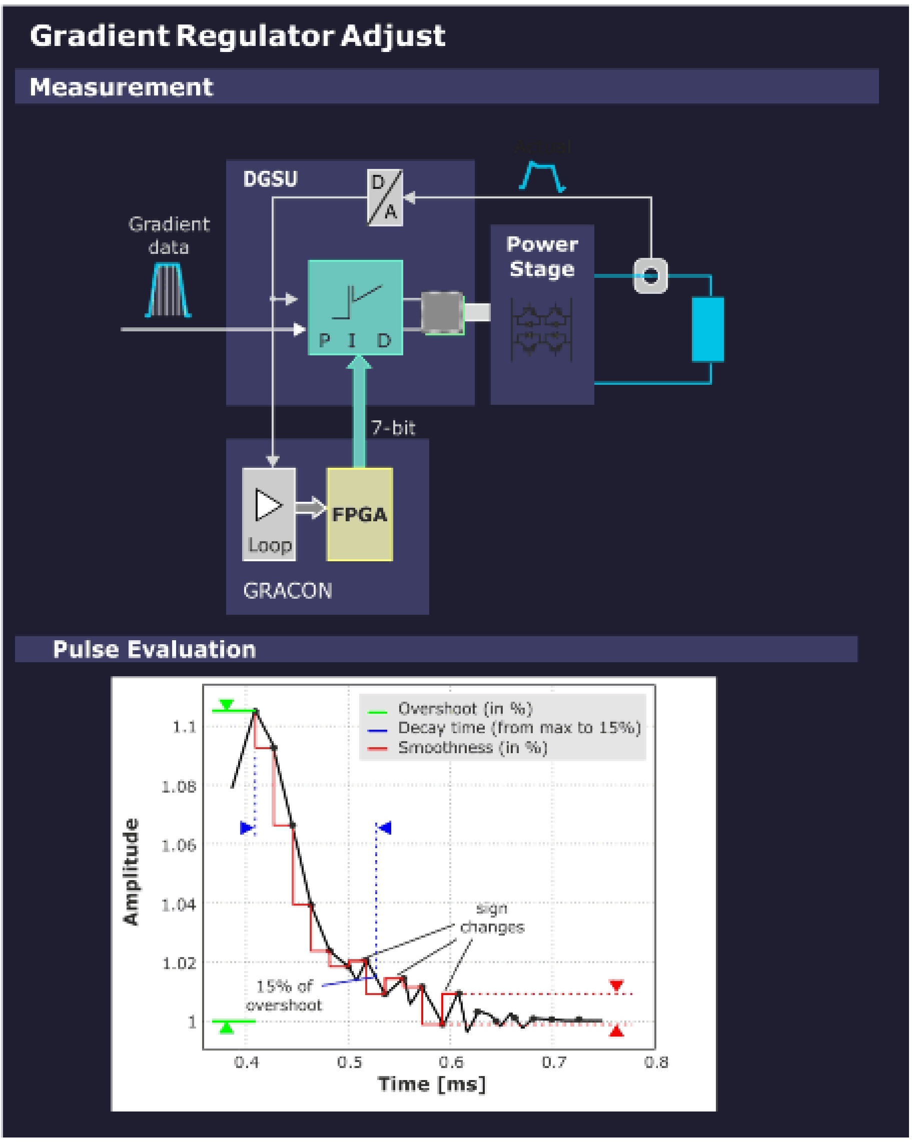
Evaluation/Results

In the report, the measured phase shifts for both Body Coil channels together with their previous values are displayed. The latter ones are read from the BC coil files. In case the phase shifts lie within a configurable specification range, they are saved permanently to the BC coil files.

3.3.1.6 Gradient Regulator

Help ID: 69D468F4-E746-4AD2-A81A-AF7BC2D67D70
Service Level: 3

Fig. 261: Gradient Regulator - Measurement



General

The **Gradient Regulator** adjustment determines the set of PID (P = proportional; I = integral; D = differential) regulator values that reproduce the applied gradient waveforms best. This is achieved iteratively for each axis by repeating the following steps until the results are in specification:

- The system response to an applied waveform is measured.
- New PID values are calculated and applied.

Gradient Regulator is an iterative procedure to improve the PID settings step by step to find the optimum solution. An evaluation is performed after each measurement. The results are plotted before the next iteration starts.

The regulator components are corrected one after another in the order P, I and D. The parts of the PID that have not been subject to the optimization already are deactivated. The initial value for each part (P, I and D) are read from the configuration.

After all regulators have been tuned, it is checked if the measured gradient waveform is in specification. If so, the regulator values are set permanently. If an error occurs or the program is aborted, the old regulator values (saved before the optimization) are restored to the GPA.

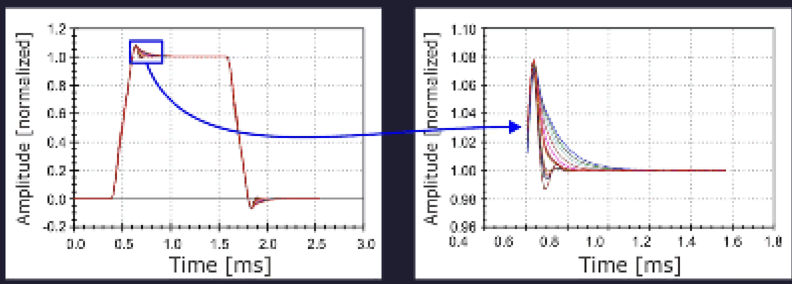
Measurement

- The sequence uses a gradient table to generate the gradient pulses so that the selected gradient (depending on the selected orientation) is switched from the maximum negative to the maximum positive gradient amplitude pulse in linear steps. Each gradient in the table is switched on using a ramp up and ramp down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.
- Based on the maximum gradient amplitude pulse, the program calculates the characteristic overshoot that is detected just after the ramp up time. The amplitude of this overshoot is determined as a percentage relative to the nominal gradient amplitude. Also, the time it takes for this exponential-like overshoot to return to a specified value is calculated. In addition, a smoothness calculation is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.
- Finally, each line in the gradient table is integrated and the resulting mean value for each line is fitted to a linear function. The difference between the data and the linear fit and the maximum deviation are returned as quality parameters.

Evaluation/Results

Fig. 262: Gradient Regulator - Results

Results



Gradient Regulator Results			
Axis	P	I	D
X-Axis	91	90	39
Y-Axis	92	87	38
Z-Axis	92	97	40

P Controller

P Values X-Axis						
Iteration	P	I	D	Overshoot	Decay Time	Smoothness
1	70	127	0	6.291%	0.355ms	0.396%
2	78	127	0	2.513%	0.355ms	0.393%
3	91	127	0	0.789%	0.355ms	0.387%

- PID gradient regulator parameters
- optimization parameters overshoot, decay time and smoothness
- diagrams showing
  - The complete gradient current pulse (pulse form)
  - The enlarged overshoot portion and current ripple (top portion)

Modified/saved parameters:

- PID regulator values of the GPA

3.3.1.7 Table Height

Help ID: 52AB3696-7EB8-453A-8B1D-70BD022B4C0E
Service Level: 3

General

The **Table Height** TuneUp procedure is used to determine the reflector height which is used to find the correct drive-in height for horizontal patient table movement.

The reflector is mounted at the magnet cover front-side and is recognized by an optical sensor at the patient table. The actual reflector position can differ due to mechanical adjustment tolerances but also due to different magnet installation heights.

Knowing the Table Height (i.e. actual reflector position), the Software can optimize the search strategy looking for the reflector (move table up or down?). This saves time for the operator.

Measurement

- The patient table is moved into the Home Position to “wake up” the vertical drive.
- The host computer reads out the actual vertical height value from the patient table controller. A system-specific number of increments is subtracted to determine the “middle of reflector height” position.
- The “middle of reflector height” value is saved at the host computer.
- Prior to each following vertical travel request, the Software can compare the actual vertical table height with the known, determined Table Height (i.e. “middle of reflector height”). Depending on the result, the search strategy to find the reflector is optimized (i.e. table first moves up- or downward).

Evaluation/Results

Results in Report:

- The status of the automatic procedure is displayed.

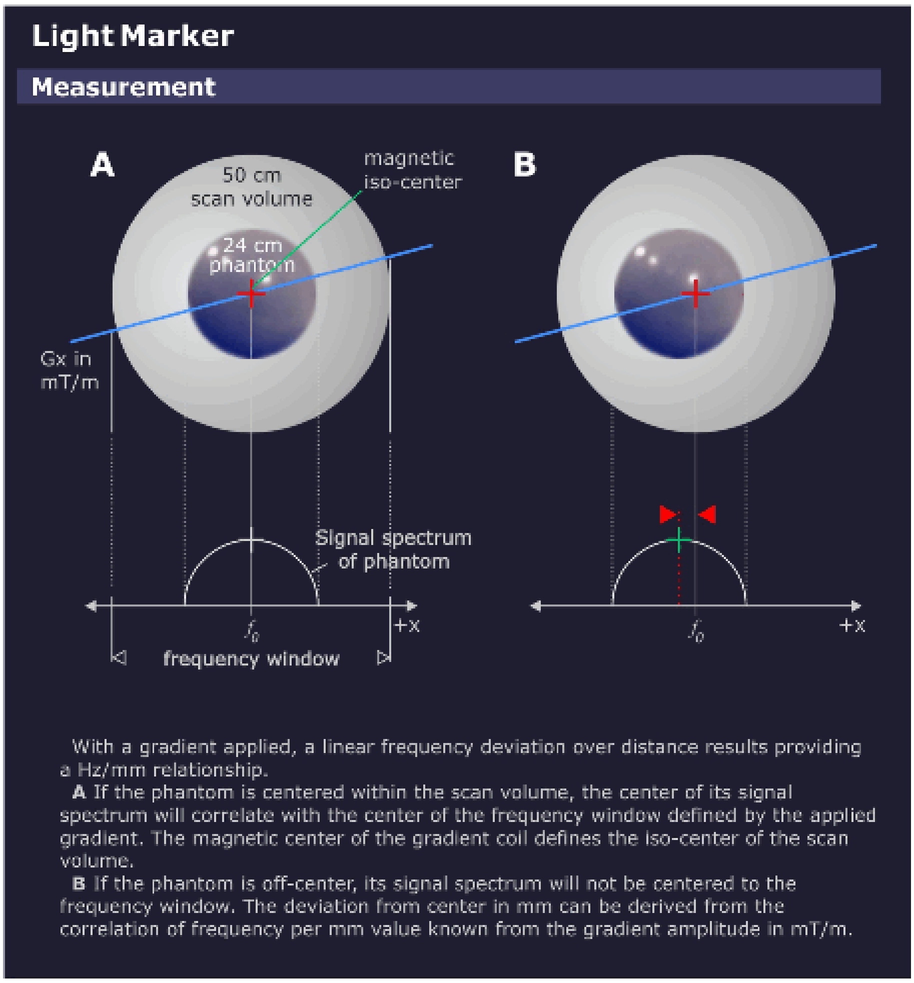
Modified/saved parameters:

The last known “middle of reflector height” is stored in the system.

3.3.1.8
Lightmarker

Help ID: 48907CEE-0E67-4BD6-A412-B297B56D1476
Service Level: 3

Fig. 263: Lightmarker - Measurement



General

The **Light Marker** procedure determines the distance between light marker and gradient center.

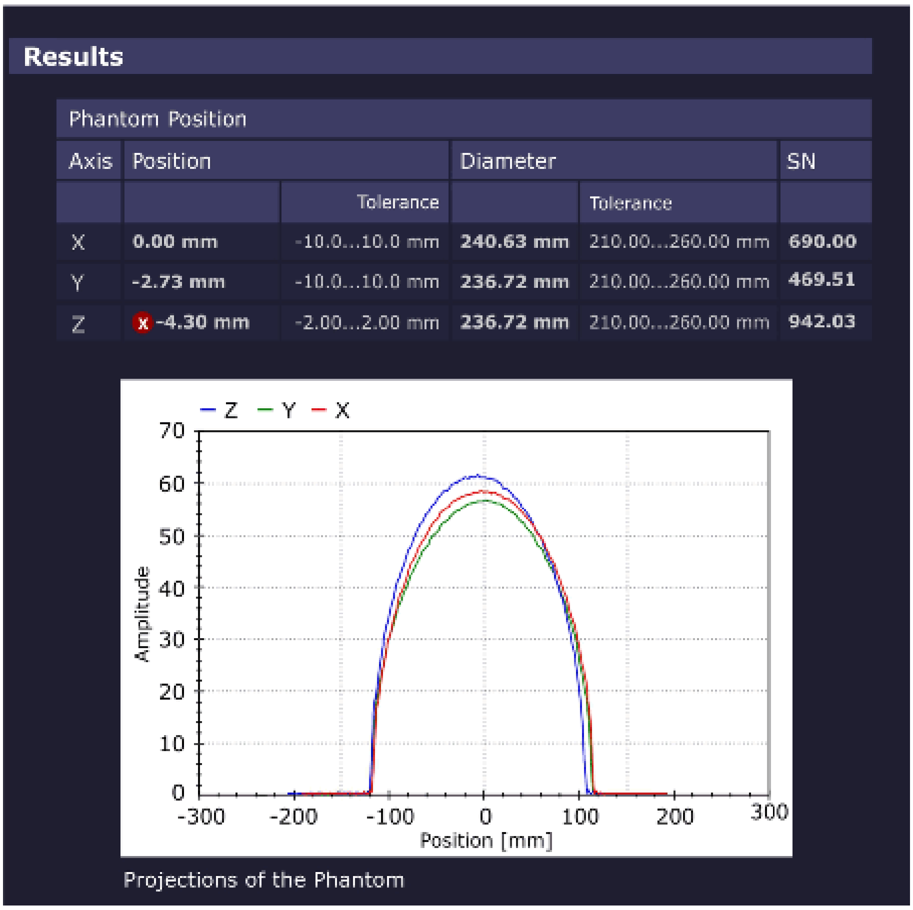
The distance of phantom center to gradient center is evaluated and used to correct the light marker distance.

Measurement

The measurement uses a double echo sequence. The first echo is acquired without applying a readout gradient resulting in an FID. The length of the FID is a rough check of the fundamental imaging volume homogeneity. During the second echo, a readout gradient is applied resulting in a projection of the image onto the readout gradient axis.

Evaluation/Results

Fig. 264: Lightmarker - Results



- The projected image of the phantom is determined, giving the position for the respective axis. This yields a new translation vector to the real gradient isocenter.
- The result of the z-axis measurement is used to determine the table position which is corrected afterward. The table is automatically moved to the correct gradient center. This procedure is repeated iteratively until the phantom position is in specifications.

Results in Report:

- phantom position coordinates.

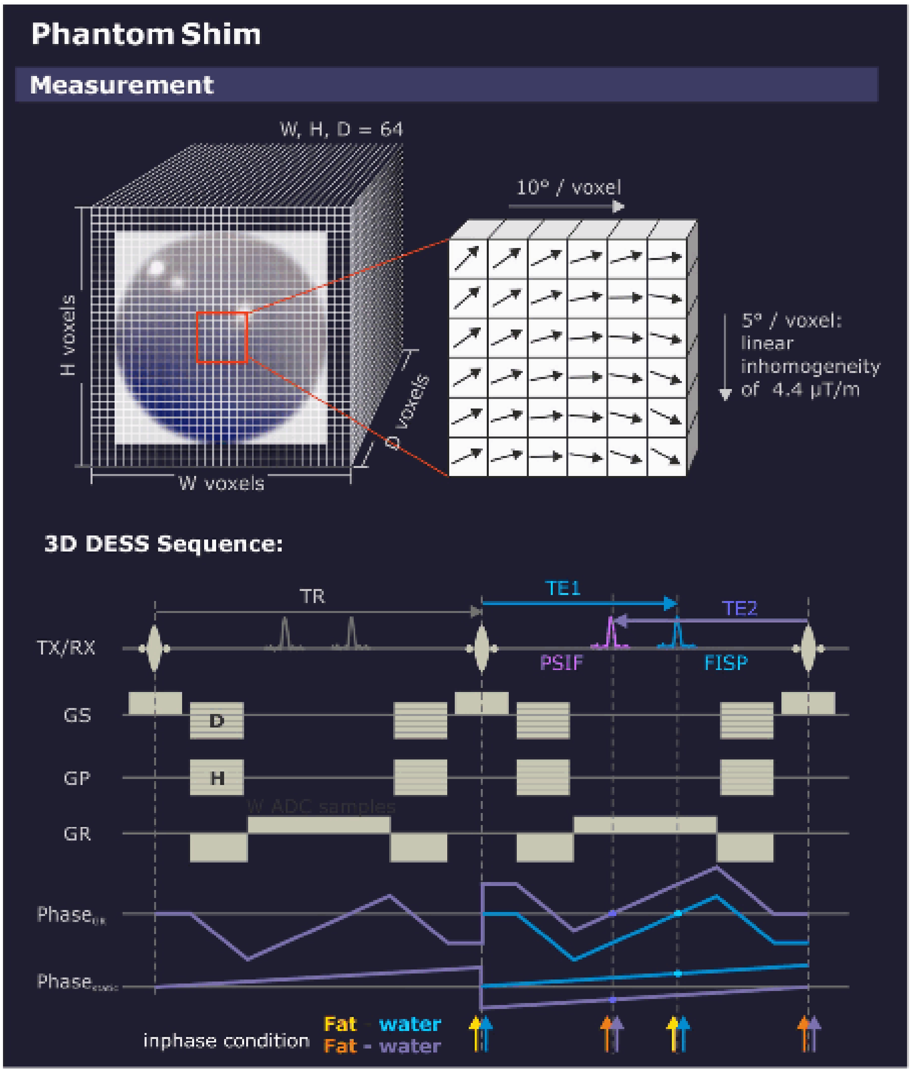
Modified/saved parameters:

- distance from light marker to the gradient isocenter.

3.3.1.9
Phantom Shim

Help ID: 2B324732-3689-4E3A-8FD7-E32FECD8F6CA
Service Level: 3

Fig. 265: Phantom Shim - Measurement



General

The **Phantom Shim** procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be exactly determined based on the frequency distribution in the volume elements making up the phantom image.

Afterward, new gradient offsets and shim coil currents (option) are calculated to improve the field homogeneity. The shim currents are improved iteratively.

The correction currents determined here also serve as default values for the 3D-Shim procedure used for patient imaging.

Measurement

Phantom Shim takes a 3D-measurement, a so-called "Field Map". The resulting voxels are cubes, i.e. the resolution is isotropic. The echo times are chosen for fat and water in phase.

In order to reduce hardware effects (e.g., gradient delays or multiple coil element phase errors), two echoes are measured in one 3D-Dess-sequence (Double Echo Steady State containing an FISP and a PSIF echo) and the phase difference of both echoes is analyzed. Then, the measurement is repeated a second time with opposed gradient polarities to eliminate eddy current effects.

The shim currents are improved iteratively.

Evaluation/Results

Fig. 266: Phantom Shim - Results

Results

Field Terms

Field Term	Value	Tolerance		
Bpp	**1.360**			
Brms	**0.093**			
A10	**-0.020**	-0.70	0.70	ppm
A11	**0.018**	-0.70	0.70	ppm
B11	**0.003**	-0.70	0.70	ppm
A20	**0.046**	-0.70	0.70	ppm
A21	**0.005**	-0.70	0.70	ppm
B21	**0.003**	-0.70	0.70	ppm
A22	**-0.011**	-0.70	0.70	ppm
B22	**0.010**	-0.70	0.70	ppm

Gradient Offset

Axis	Gradient Offset (new)	Gradient Offset (old)	Gradient Offset (change)
GradX	**-0.156** mT/m	-0.155 mT/m	-0.000 mT/m
GradY	**0.311** mT/m	0.311 mT/m	-0.000 mT/m
GradZ	**1.161** mT/m	1.161 mT/m	0.000 mT/m

Shim Currents

Shim Channel	Shim Current (new)	Shim Current (old)	Shim Current (change)
A20	**-2375** mA	-2372 mA	-3 mA
A21	**-161** mA	-160 mA	-1 mA
B21	**72** mA	72 mA	-0 mA
A22	**31** mA	29 mA	2 mA
B22	**41** mA	43 mA	-2 mA

The phase is calculated for each volume element. Based on the phase, the magnetic field can be expanded in terms of spherical harmonics. To reduce calculation times, voxels with little or no signal are discarded.

The individual terms are - up to a sensitivity factor - the parameters to compensate the field terms, including the system frequency, the gradient offsets, and the shim currents.

The impact of the changed currents is checked implicitly with the next iteration to ensure that the homogeneity has improved; a negative impact would suggest wrong cabling that could be checked in more detail with Expert Mode.

Results in Report:

- Field terms
- MR frequency
- Gradient offsets
- Shim currents (optional)

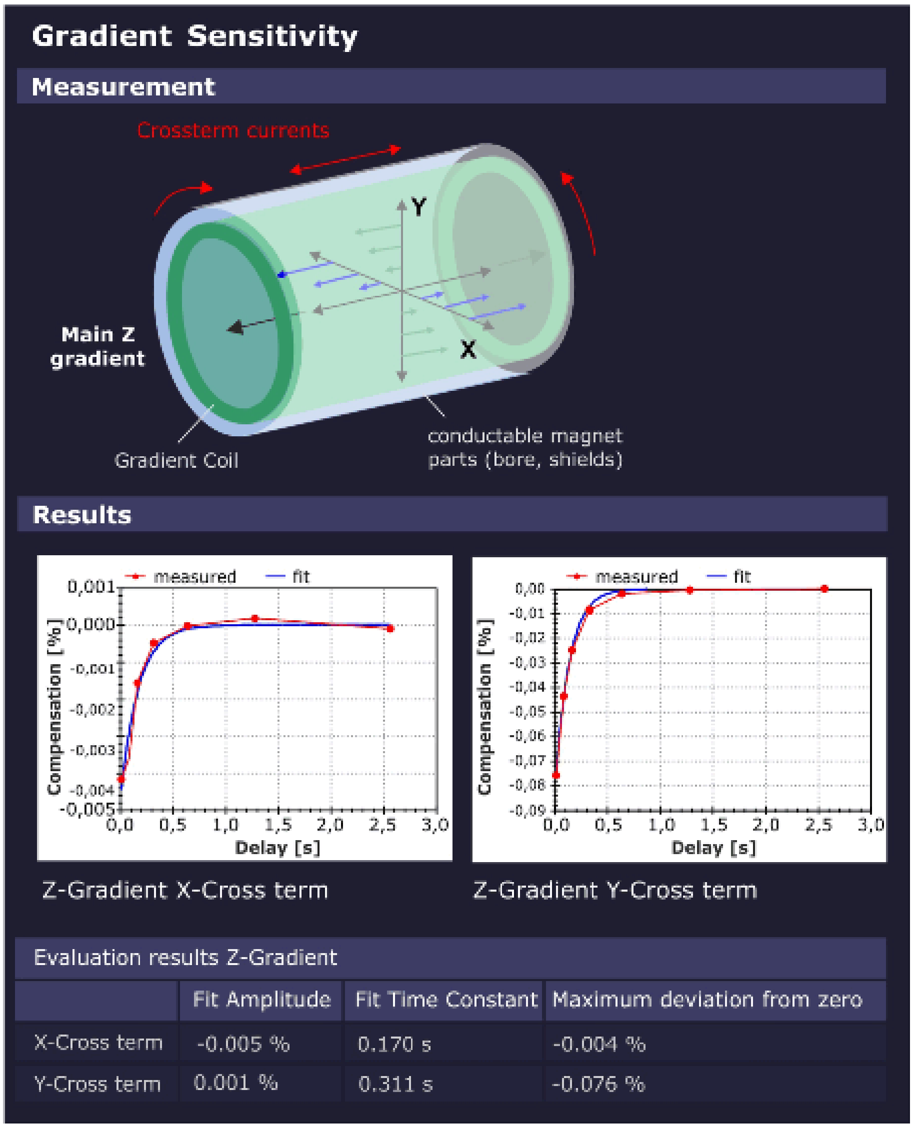
Modified/saved parameters:

- Gradient Offsets and Shim Currents

3.3.1.10
Cross Term Compensation

Help ID: 6E349297-0783-4926-8406-B493D5B788D7
Service Level: 3

Fig. 267: Cross Term Compensation



General

The operation of a single gradient, e.g., Z gradient, produces eddy currents mainly in the Z axis, but also small amounts of eddy currents which are coupled over the conductive magnet components onto the other 2 gradient axes (e.g. X and Y).

These small orthogonal “Cross Term” eddy currents decay exponentially in time, in general with only 1 time constant.

Measurement

The **Cross Term Compensation** procedure measures the decay of these Cross Term eddy-currents and calculates their amplitude and time constant. These data are used later during imaging sequences to apply appropriate, inverse correction gradients in the orthogonal gradient axes.

The result is a much cleaner main gradient with very small residual Cross Terms, improving the final image quality.

Evaluation/Results

Results in Report:

- Cross Term Amplitudes and Time Constants.
- Diagrams visualizing the corresponding Cross Term amplitudes.

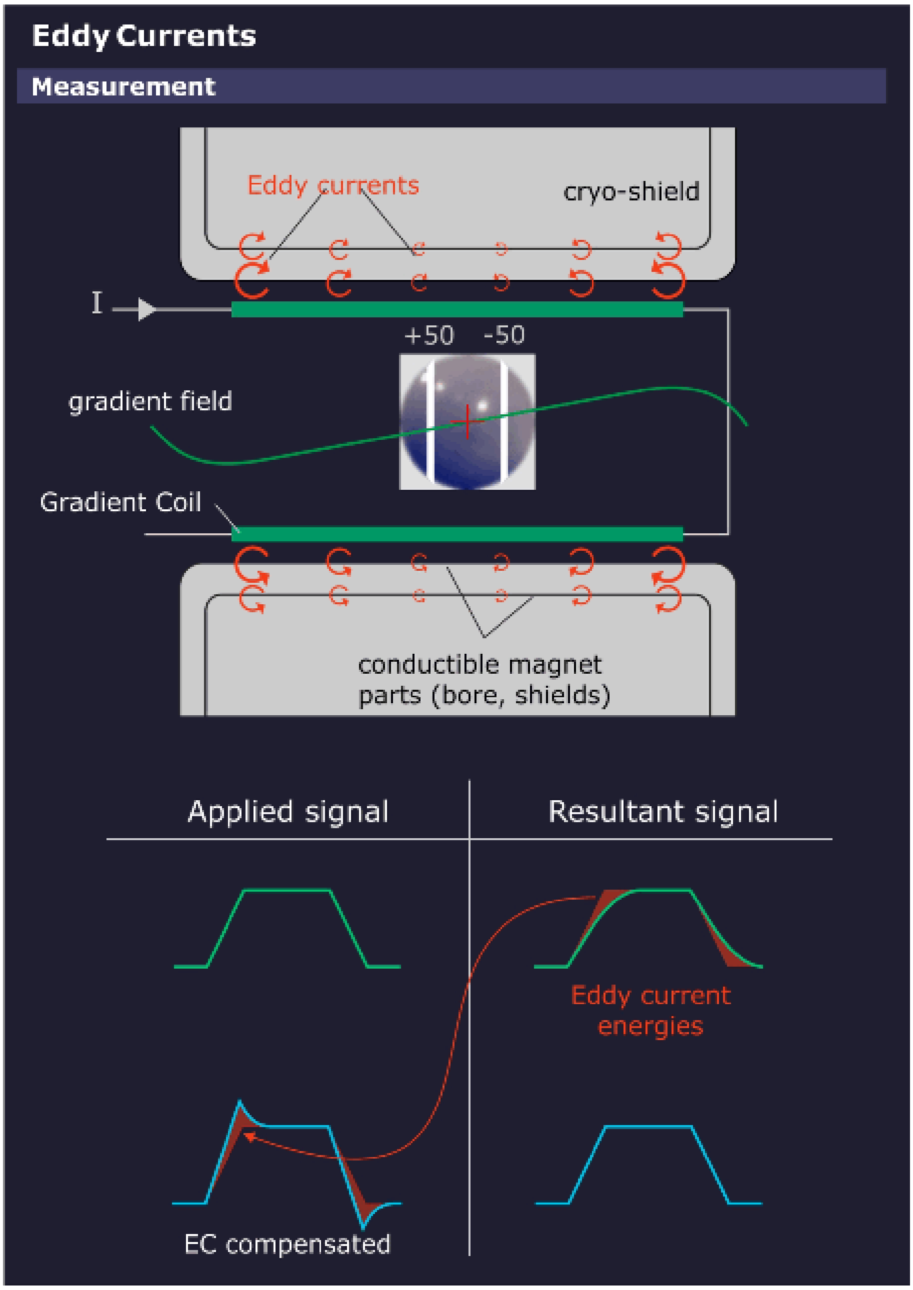
Modified/saved parameters:

- Cross Term compensation parameters

3.3.1.11 Eddy Current Compensation

Help ID: D1587435-76CB-4D71-A21B-091AB06092A9
Service Level: 3

Fig. 268: Eddy Current Compensation - Principle



General

Eddy Current Compensation (ECC) measures the eddy currents induced by all 3 gradients. The measurement requires the **small spherical phantom** rather than the normal body phantom.

The dynamic gradient fields produce eddy currents in all surrounding conductive structures (mainly the Body Coil, magnet bore and the cryogenic shields), resulting in magnetic fields which oppose and distort the applied gradient fields. Eddy currents in the warm components (Body Coil and magnet bore) have relatively short decay times, whereby eddy currents developed in the cold cryo-shields can have relatively long time constants. The eddy currents are analyzed for their time constants and amplitudes to achieve a good compensation. The compensation is made by adding the reciprocal of the measured eddy currents to the gradient pulse, thus neutralizing their effects. For accurate compensation, 5 time constants are used.

Spatial Dependency of Eddy Currents

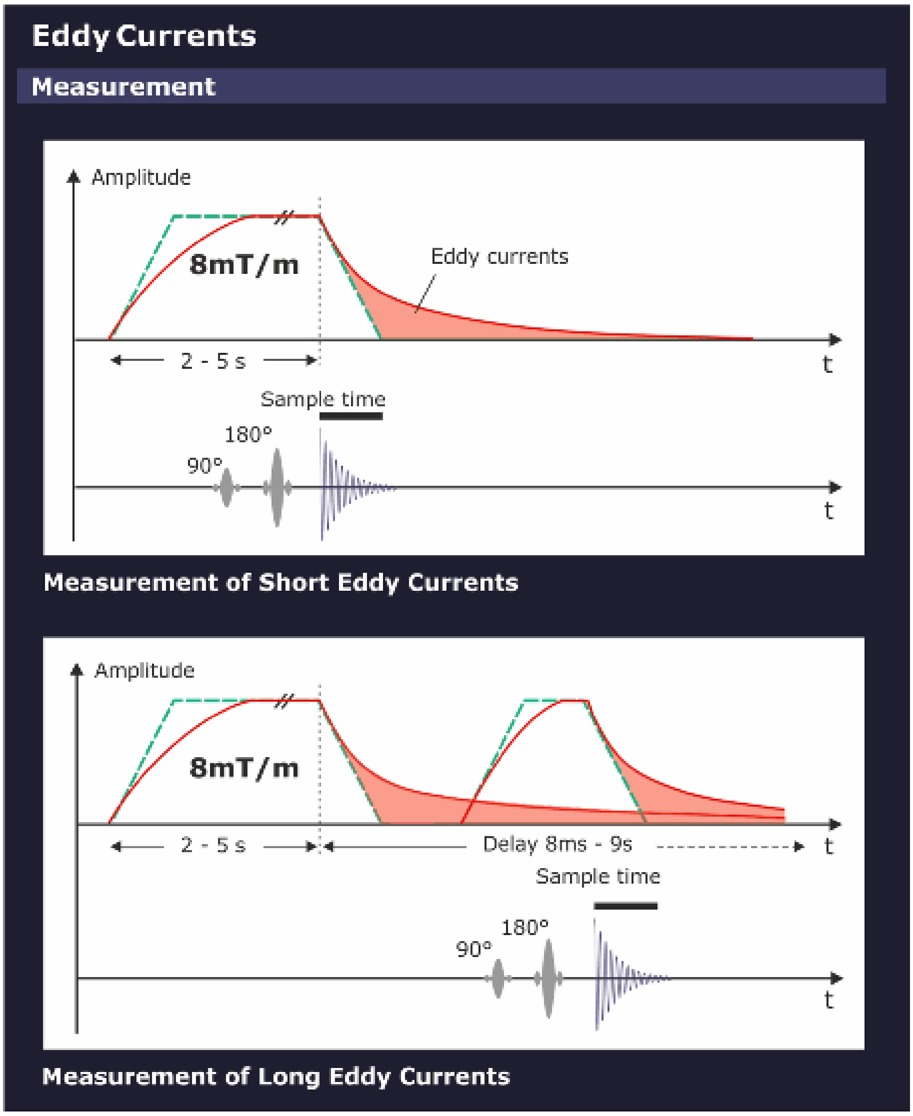
Experience has shown, that for smaller slice shifts it is sufficient to express the eddy fields into a 0^{th} and a 1^{st} order term.

- The **0^{th} order term** can be compensated by changing the synthesizer frequency during measurements.
- The **1^{st} order term** can be compensated by predistorting the gradient waveforms for the GPA.

Term	Description
0^{th}	This term arises from an **asymmetry** of the gradient coil with respect to the magnet bore and cryo shields. This term is space independent and present in the complete imaging volume and adds to the nominal B_0-Field. The amplitude (x) is given in the unit μTm/mT. The time constant of the most significant 0th term component is about 500ms, although there are shorter time constant components as well.
1^{st}	This is the most important term as it has the same symmetry as the gradient field itself. The amplitude is given in % of the applied gradient pulse.
2^{nd} or above	There are also high ordered eddy currents present. But they are usually small and negligible as long as the slice shifts and off-center zooms are not too large. No compensation for the high order eddy currents can be made.

Measurement

Fig. 269: Eddy Current Compensation - Measurement



A **Preparation Measurement** checks for correct positioning of the phantom in the magnet center and for sufficient field homogeneity. This is performed via a double echo sequence.

The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity.

The second echo uses a readout gradient and calculates the phantom position in the readout direction.

The sequence for the measurements of the eddy currents generates several spin echoes with varying delay times.

The measurement is made at the moderate slice shifts of 0 ± 50 mmto prevent contributions from higher order terms.

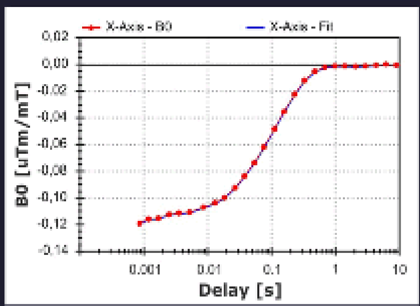
Evaluation/Results

Fig. 270: Eddy Current Compensation - Results

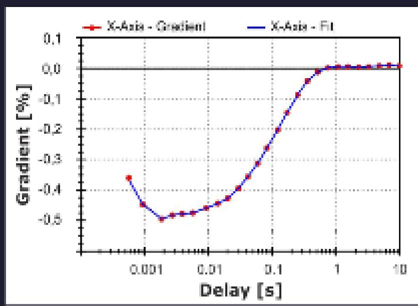
Results

ECC X-Axis Fit Results, Step 1

Range	B0		Gradient	
		Tolerance		Tolerance
less than 10ms	x **0.132 uTm/mT**	0.00...0.10 uTm/mT	x **0.541%**	0.00...0.10%
more than 10ms	x **0.116 uTm/mT**	0.00...0.10 uTm/mT	x **0.486%**	0.00...0.10%
2-100ms	x **0.059 uTm/mT**	0.00...0.03 uTm/mT	x **0.309%**	0.00...0.040%



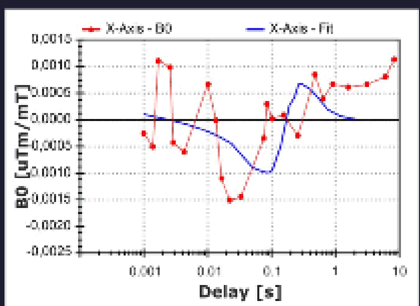
X-Axis B0 Step 1



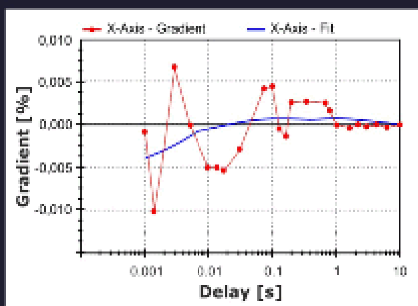
X-Axis Gradient Step 1

ECC X-Axis Fit Results, Step 2

Range	B0		Gradient	
		Tolerance		Tolerance
less than 10ms	**0.001 uTm/mT**	0.00...0.10 uTm/mT	**0.010%**	0.00...0.10%
more than 10ms	**0.002 uTm/mT**	0.00...0.10 uTm/mT	**0.005%**	0.00...0.10%
2-100ms	**0.003 uTm/mT**	0.00...0.03 uTm/mT	**0.011%**	0.00...0.040%



X-Axis B0 Step 2



X-Axis Gradient Step 2

The 0th and 1st order compensations are calculated.

The decision how many iterations are necessary, is based on the expected improvement for the actual fit.

Results in Report:

- The tables shows for 3 typical time domains (<10ms, >10ms, 2 - 100ms):

  - Relative gradient strength to compensate the first order eddy currents

  - Relative change in B_0 for the 0th order eddy currents.

- Diagrams show the measured data against the fit for both:
  - B_0 eddy currents
  - 1st order eddy currents

Modified/saved parameters:

- Eddy current compensation parameters

3.3.1.12 Gradient Delay

Help ID: 902BC568-F068-432E-9F4E-47CB1EAE3317
Service Level: 3

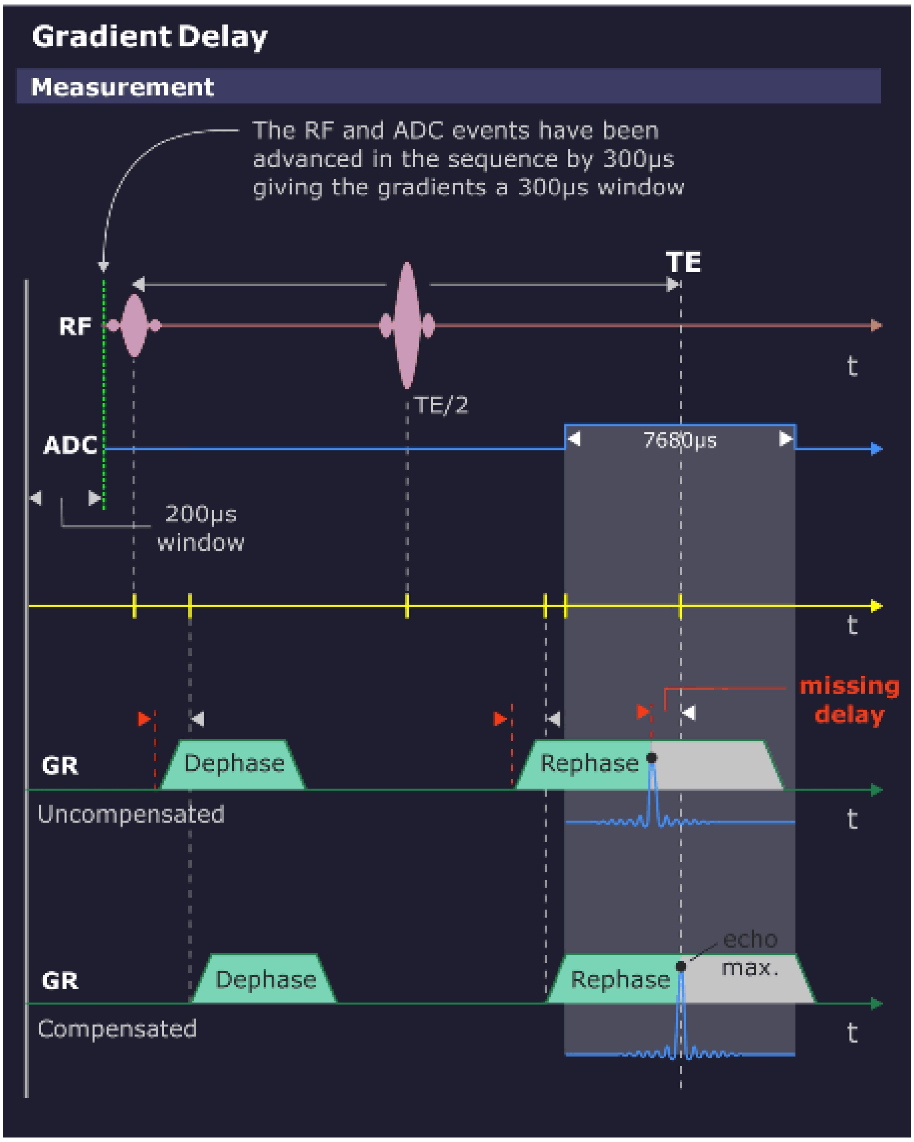
General

The gradient system has a small time delay between the time the gradient is switched on digitally by the measurement control and the time the actual gradient field is present. This delay is in the microseconds range, e.g. 30µs and mainly caused by the coil inductance.

The **Gradient Delay** procedure determines the individual delay time which can slightly differ for each gradient axis. A precise timing of gradient pulses together with the RF pulses is very important for optimum image quality.

Measurement

Fig. 271: Gradient Delay - Measurement



The measurement uses a Spin Echo sequence without phase encoding, hence only 1 line is measured. The measured gradient axis acts as the readout gradient and is switched on once between the 90°- and the 180°-pulse and then again during the readout time of the Spin Echo.

The first gradient pulse causes a dephase of the magnetization vector in the direction of the gradient axis. The amount of dephase is proportional to the time-amplitude integral of the gradient pulse. Subsequently, a 180° refocusing pulse is applied normally resulting in a Spin Echo at TE/2 time later. However, due to the dephasing of the vector by the first gradient pulse, the echo only appears in the center of the ADC window (which is also centered exactly around the TE time point) when an equal rephasing gradient has been applied at the proper time so that the rephasing is finished at the center of the ADC cycle.

For a non-corrected Gradient Delay, the 2 gradient pulses are time-delayed. This fact does not influence the dephasing effect of the first gradient pulse, however, the delayed second gradient pulse causes a rephasing delay and thus a time-delayed echo signal. From the echo signal delay the required correction time can be determined.

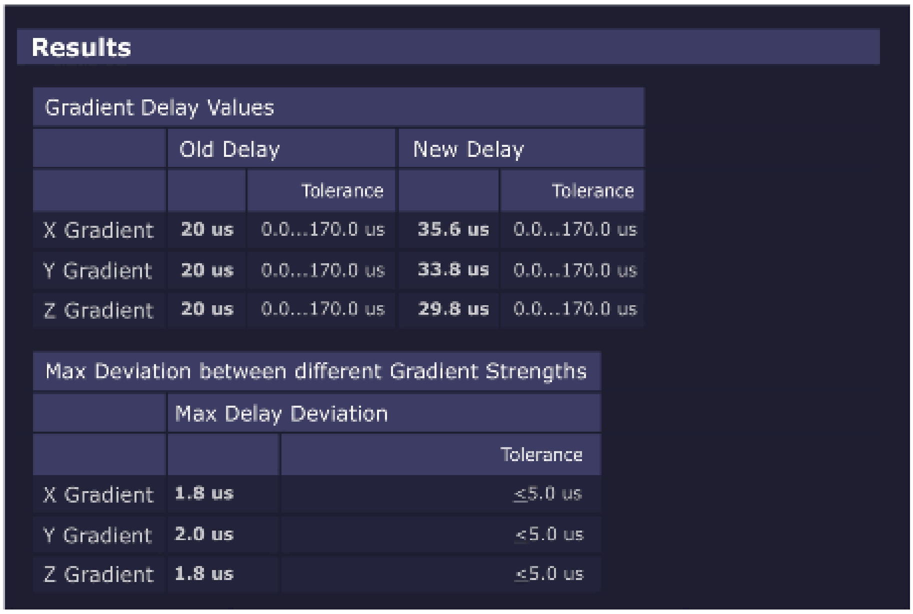
Difference Deviation Check

Each gradient is measured at 3 different gradient strengths - 1.5, 4 and 8 mT/m - giving 3 Gradient Delays for each gradient. For an ideal gradient system, all delays are the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, i.e. high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, there is some kind of nonlinearity present in the gradient system.

Evaluation/Results

Fig. 272: Gradient Delay - Results



Results

Gradient Delay Values

	Old Delay		New Delay	
		Tolerance		Tolerance
X Gradient	**20 us**	0.0...170.0 us	**35.6 us**	0.0...170.0 us
Y Gradient	**20 us**	0.0...170.0 us	**33.8 us**	0.0...170.0 us
Z Gradient	**20 us**	0.0...170.0 us	**29.8 us**	0.0...170.0 us

Max Deviation between different Gradient Strengths

	Max Delay Deviation	
		Tolerance
X Gradient	**1.8 us**	≤5.0 us
Y Gradient	**2.0 us**	≤5.0 us
Z Gradient	**1.8 us**	≤5.0 us

Results in Report:

- current and calculated Gradient Delays
- maximum deviation values.

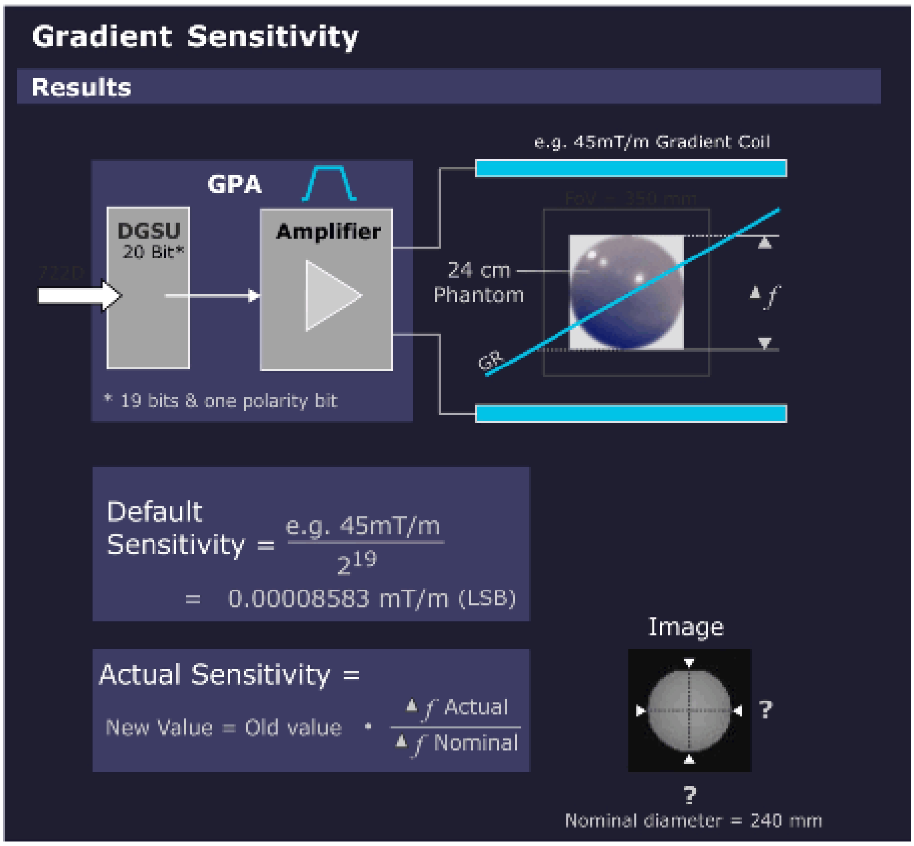
Modified/saved parameters:

- Gradient Delay values

3.3.1.13 Gradient Sensitivity

Help ID: 2E17EBC4-A846-41F7-B775-BA7D410952D4
Service Level: 3

Fig. 273: Gradient Sensitivity - Measurement



General

The **Gradient Sensitivity** procedure calibrates the gradient sensitivities so that the system uses the correct gradient strengths and measure the correct image size in all orientations..

The patient table distance (distance light marker to gradient coil center) is also calculated.

If the phantom is not centered in the magnet within a specified tolerance, the gradient sensitivities cannot be calculated. Repeat the measurement after correcting the table position.

Measurement

- A standard imaging sequence (FLASH with distortion correction) is repeated 3 times, 1 measurement for each orientation.
- The 3 images are analyzed to determine the exact location of the center of the phantom and the phantom diameters. By swapping the standard readout and phase encoding directions in1 orientation, the center and diameter of the phantom for each gradient direction can be determined, once with the gradient as a readout encoder and

again with the gradient as a phase encoder. The average of these 2 results determines the final diameter values.

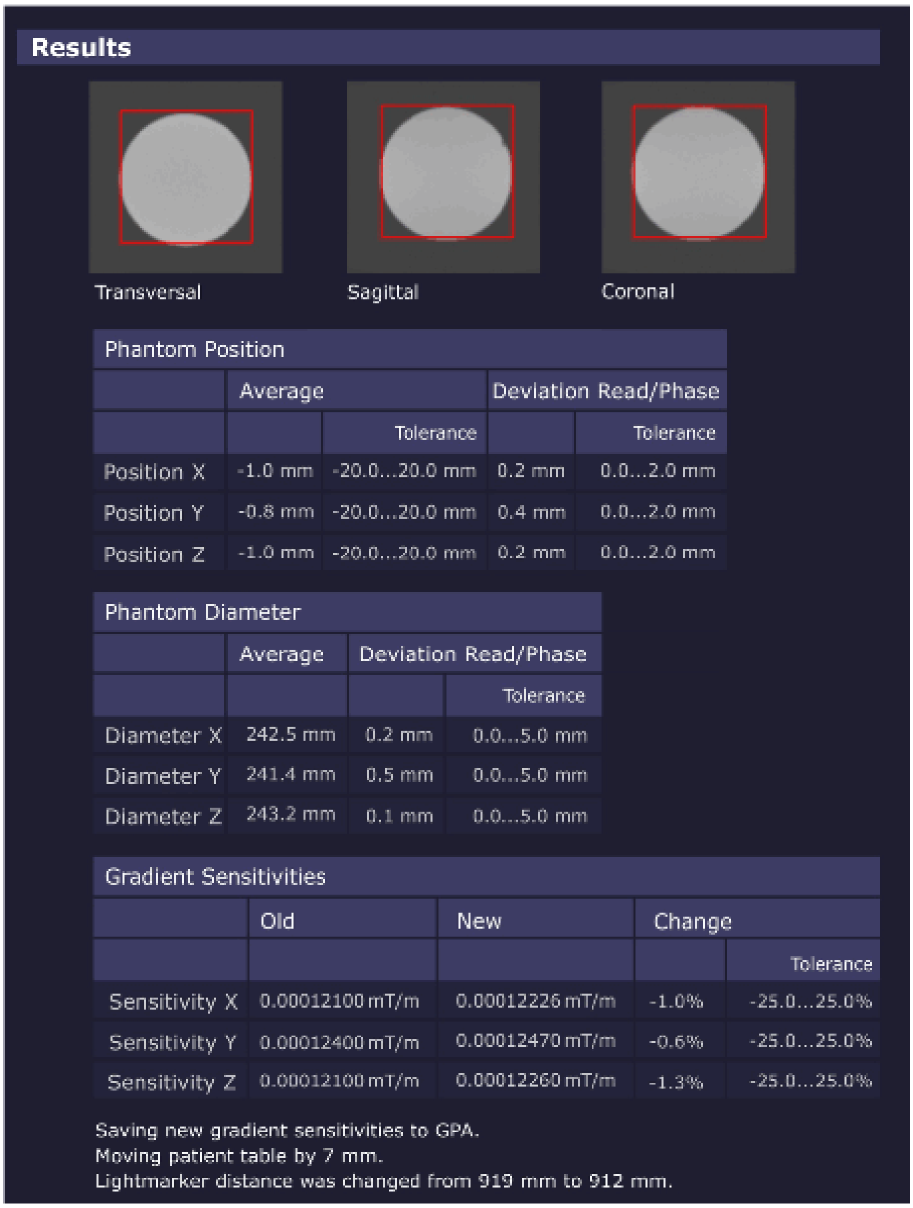
- The measured values are evaluated and compared with the well-known diameters of the phantom for calculating the gradient sensitivities.

The gradient sensitivity is defined as the gradient strength per ADC unit (e.g. 0.00009 mT/m per 1 ADC unit).

> The patient table position is corrected automatically. In case of problems (e.g. if the table moves in the wrong direction), check the Z gradient polarity.

Evaluation/Results

Fig. 274: Gradient Sensitivity - Results



Phantom Position				
	Average		Deviation Read/Phase	
		Tolerance		Tolerance
Position X	-1.0 mm	-20.0...20.0 mm	0.2 mm	0.0...2.0 mm
Position Y	-0.8 mm	-20.0...20.0 mm	0.4 mm	0.0...2.0 mm
Position Z	-1.0 mm	-20.0...20.0 mm	0.2 mm	0.0...2.0 mm

Phantom Diameter			
	Average	Deviation Read/Phase	
			Tolerance
Diameter X	242.5 mm	0.2 mm	0.0...5.0 mm
Diameter Y	241.4 mm	0.5 mm	0.0...5.0 mm
Diameter Z	243.2 mm	0.1 mm	0.0...5.0 mm

Gradient Sensitivities				
	Old	New	Change	
				Tolerance
Sensitivity X	0.00012100 mT/m	0.00012226 mT/m	-1.0%	-25.0...25.0%
Sensitivity Y	0.00012400 mT/m	0.00012470 mT/m	-0.6%	-25.0...25.0%
Sensitivity Z	0.00012100 mT/m	0.00012260 mT/m	-1.3%	-25.0...25.0%

Saving new gradient sensitivities to GPA.
Moving patient table by 7 mm.
Lightmarker distance was changed from 919 mm to 912 mm.

The images are analyzed to determine the exact location of the center of the phantom and the phantom diameters. The found diameters are checked against the known phantom diameters of 240 mm. The quotient between both gives the correction factor to the actual gradient sensitivity.

Results in Report:

- measured phantom diameter in all 3 dimensions
- measured phantom centers in all 3 dimensions
- gradient sensitivities
- phantom images in all 3 slice orientations

ResulModified/saved parameters:

- Gradient Sensitivity

3.3.1.14 BC Power Losses

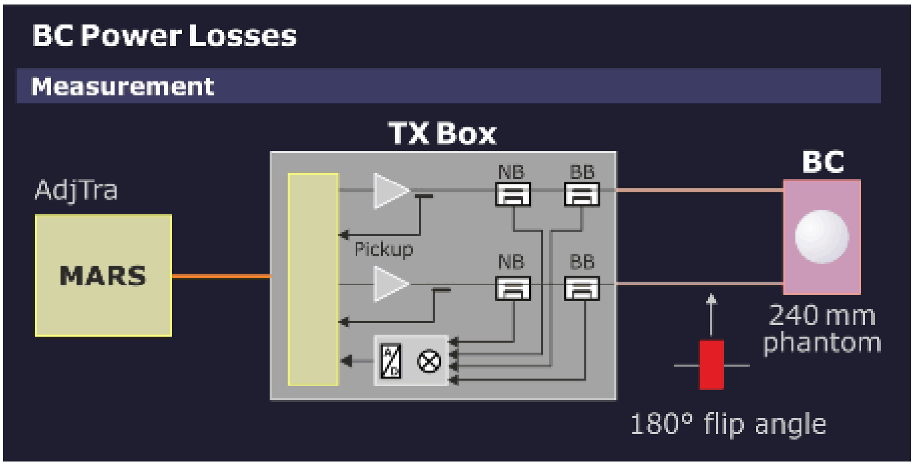
Help ID: 0FBDAC47-1539-4FBF-AD8E-9EE97D158876
Service Level: 3

General

The procedure **BC Power Losses** measures the amount of power loss in the Body Coil and connecting cables. This value is used by the RF Safety Watchdog (RFSWD) monitor to obtain a more accurate SAR measurement.

Measurement

Fig. 275: BC Power Losses - Measurement



- The Coil Power Loss measurement switches the RF Safety Watchdog (RFSWD) into the patient mode.
- At first, a complete inline adjustment with a non-volume selective frequency adjustment is performed to adjust transmitter and DICO. During the transmitter adjustment, the TX-Box internal TALES power reference value is measured and the values of the directional couplers (DICOs) for both transmitting systems.
- The subsequent measurement verifies the position and size of the phantom to ensure that the correct 240 mm spherical phantom is used.

- From the difference of the given theoretic Adj. Tra value for the used phantom and the actual (higher) measured power values, the Software can calculate the Coil Power Losses in the Body Coil and connecting cables.
- For pTX systems (i.e. systems with 2-ch BC; aka TX True Shape option), the Y-Matrix, which describes the characteristics of BC transmit path, is determined by an additional measurement. This provides an alternative means to determine the Coil Power Losses for arbitrary BC excitations.

Evaluation/Results

Fig. 276: BC Power Losses - Results

Results

Coil Power Loss for Polarisation Mode CP

Phantom position and diameter				
	X		Z	
		Tolerance		Tolerance
Center	**-0.82 mm**	-20.00...20.00 mm	**-0.24 mm**	-20.00...20.00 mm
Diameter	**240.79 mm**	220.00...260.00 mm	**240.46 mm**	220.00...260.00 mm

Rf Safety Common	
Name	Value
Sigma Phantom	**1.00**
Reference Amplitude (AdjTra)	**391.73 V**

Rf Safety (Type Sum)		
Name	Value	
		Tolerance
Coil Power Loss	**1,623.30 W**	1200,00...4,000.00 W
Previous Coil Power Loss	**1,000.00 W**	

Rf Safety (Type YMatrix)		
Name	Value	
		Tolerance
Coil Power Loss	**1,625.95 W**	
Coil Power Loss Consistency	**0.16 %**	-10.00...10.00 %

Coil Power Loss for Polarisation Mode EP

Phantom position and diameter				
	X		Z	
		Tolerance		Tolerance
Center	**-0.87 mm**	-20.00...20.00 mm	**-0.18 mm**	-20.00...20.00 mm
Diameter	**240.93 mm**	220.00...260.00 mm	**240.27 mm**	220.00...260.00 mm

Rf Safety Common	
Name	Value
Sigma Phantom	**1.00**
Reference Amplitude (AdjTra)	**4356.54 V**

Rf Safety (Type Sum)		
Name	Value	
		Tolerance
Coil Power Loss	**1,985.00 W**	1200,00...4,000.00 W
Previous Coil Power Loss	**1,000.00 W**	

.
.
.

Results in Report:

- tables showing the phantom position.
- tables showing measured and stored Coil Power Loss values
- tables showing the measured reference power from the TALES

Modified Parameters:

- Body Coil Power Losses in CP- and EP-mode

3.3.1.15 BC Image Brightness

Help ID: 51D260CB-C301-47E2-BBD2-F54667156BC5
Service Level: 3

General

This procedure measures the image brightness of the Body Coil using a standard image measurement. A scaling factor (FFT Scale Factor) is determined and set to ensure that the mean image brightness of the pixels within the phantom is close a specified value.

Measurement

A specified protocol is measured.

Evaluation/Results

The correction factor is calculated for the measured signal levels within the phantom to reach a specified brightness value.

Results in Report:

- Serial and material number of BC (from EEPROM on BC).
- New and old Tx Amplitude.
- New and old FFT Scale Factor.

Modified/saved parameters:

- The FFT scale factor for the coil used is changed in the respective coil file.

3.3.1.16 Dynamic Field Map

Help ID: BCF3C1B6-8197-407E-8991-23DE6FD54E73
Service Level: 3

General

Gradient pulses are always accompanied by unwanted time-dynamic field perturbations caused by e.g. eddy current or hysteresis effects. These field perturbations comprise multiple terms with different spatial dependencies.

The 0^{th} and 1^{st} order terms are analyzed and compensated by the ECC and CTC TuneUp procedures.

The purpose of the **Dynamic Field Map** procedure is to measure and analyze the residual higher order pertubations (>2nd order field terms).

Measurement

The dynamic field map service sequence is executed to measure the dynamic field perturbations using the large spherical phantom.

Evaluation/Results

Results in Report:

- peak-peak values in ppm
- Spherical harmonics up to the 5^{th} order

Modified/saved parameters:

Data for image-based correction of dynamic field perturbations in diffusion imaging are saved.

3.3.1.17 Rear Lightmarker

Help ID: 3EA5E53A-1F90-4308-A6D1-B99EBC2C7D9B
Service Level: 3

General

A second light marker in back of the magnet is available as an option. The procedure" Rear Lightmarker" calibrates the distance between the rear light marker and the Gradient Coil magnetic center.

Measurement

The large phantom must be positioned with the help of the rear light marker and then moved to the gradient center. The measurement uses the TuneUP program "Gradient Sensitivity" (service sequence "gradsens") with three orthogonal slice orientations.

Evaluation/Results

An algorithm determines the Z-position of the phantom center in the image. Using this information, the distance from rear lightmarker to Gradient Coil center is updated and saved.

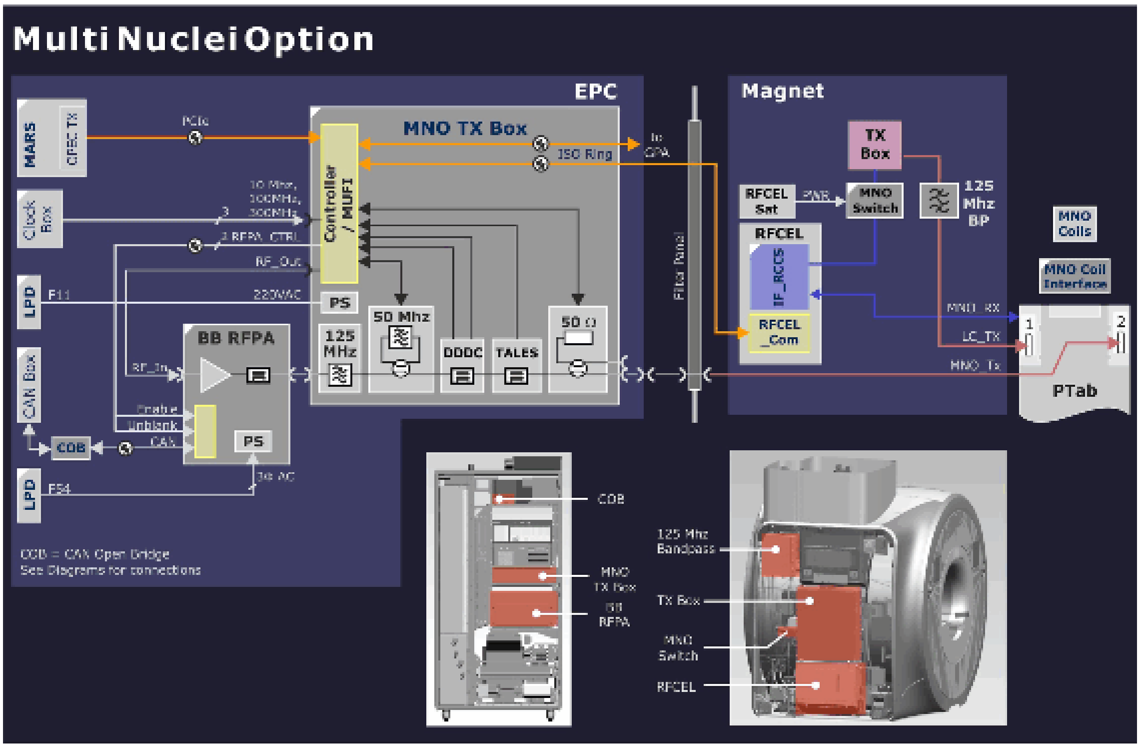
Modified parameters:

- distance from second light marker to gradient center.

3.3.1.18
RF Characteristics MNO

Help ID: 6E1EF408-9472-48A1-87C9-3F22471C6BE0
Service Level: 3

Fig. 277: Multi Nuclei Spectroscopy Option (MNO)



General

The TuneUp step **RF Characteristics MNO** is available only at systems where the Multi Nuclei Option (MNO) for spectroscopy is configured.

These systems have an additional MNO TX-Box and standalone 8 kW broadband RF amplifier (BB RFPA) which can generate hydrogen and several lower x-nucleus frequencies for the MR spectroscopy experiments. Due to the frequency dependency of these MNO components, each individual frequency is calibrated individually with a characteristic curve for each nucleus.

The measurement determines the characteristics (amplitude and phase) at the output of the broadband RFPA. For the measurement, the service plug must be connected to coil plug 2 but no power is transmitted to the plug because the internal 50 Ohm dummy load of the MNO TX-Box is automatically switched to the output of the BB RFPA. Then, the corresponding hydrogen and x-nucleus frequencies are generated at the MNO TX-Box, amplified by the broadband RFPA and fed to the dummy load. Forward and reflected voltages of the BB RFPA are measured at the directional coupler (DICO) and TALES of the MNO TX-Box.

From the measurements, the characteristic curves for each nucleus are calculated and stored in the Software. The data is used to pre-distort the input signal of the BB RFPA with an inverse function, thus a linear output power characteristic can be achieved for all nuclei.

Measurement

The service plug is connected to coil plug 2. The 50 Ohm dummy load of the MNO TX-Box is automatically connected to the BB RFPA output by the Software.

The following measurement steps are performed for every nucleus:

- Calibration voltage search x-nucleus
- RF max. power search x-nucleus
- RF Characteristic measurement x-nucleus

Evaluation/Results

Results in Report:

- Calibration voltage search result
- RF calibration (saved)
- RF maximum power search result
- Measured characteristics (amplitude and phase)
- Calculated characteristic (amplitude and phase)

Modified parameters:

- RF Characteristic data

3.3.2 TuneUp - Expert

Help ID: 244F1380-8367-4997-9BFB-10F63E39A0F3
Service Level: 7

Introduction

Some procedures have an **Expert Mode** platform. These platforms are intended for qualified persons only, because they may provide additional configurations and parameters which require a more detailed knowledge of the underlying physics. In the Expert Mode, an arbitrary coil can be selected for most of the available procedures.

In the Expert Mode, TuneUp is no longer automated. Only one selected procedure at a time is performed.

NOTICE

Expert Mode is used for troubleshooting only!

Any (mis-)adjustments achieved in Expert Mode are also used for standard measurements.

» **For imaging with patients, ensure that all TuneUP and QA steps have been successfully performed in Normal Mode!**

Expert Mode TuneUp procedures may have an impact to the **status** of other TuneUp and QA procedures (dependencies). That means, steps with status “**Done**” may fall back to “**To Do**”.

To reach the “**Done**” status of any TuneUP function, run the procedure successfully in Normal Mode under TuneUp / General.

3.3.2.2 Extended Frequency Adjustment

Help ID: 484CA0A2-E6AC-4DE6-B572-01D58965BD4B
Service Level: 7

Expert Mode

The **Extended Frequency Adjustment** is used to find the system frequency in case the current system frequency is far away from the actual MR resonance frequency.

A larger frequency drift may occur after ramping the magnet (e.g. shim workflow).

Extended Frequency Adjustment can be also used with the Array Shim Device.

Measurement

The procedure performs several non-volume selective frequency adjustments with a moving frequency interval. As a result, the **entire specified frequency range is covered** to find the resonance peak, even if it is far away from the last stored current frequency.

MR system	Lower frequency (Hz)	Upper frequency (Hz)
3 Tesla	123100000	123265000

Evaluation/Results

Results in Report:

- System Frequency

Modified/saved parameters:

- System Frequency

3.3.2.3 Feedback Loop Calibration

Help ID: 6F181F8E-2E47-4CA9-8629-CE1995AA85D7
Service Level: 7

Expert Mode

> This Expert Mode step is only available for single TX channel systems.

General

The TX-Box has a built-in feedback loop to compensate for instabilities of the TX B1-field.

The procedure **Feedback Loop Calibration** is used to calibrate internal feedback loop parameters of the TX-Box.

The **Expert Mode** allows for calibrating/checking either feedback loop parameter independently of the actual system status.

3.3.2.4 BC Tuning Expert

Help ID: 1C2F7900-E019-4610-BF22-EE8E0E613846
Service Level: 7

Expert Mode

The **BC Tuning in Expert Mode** offers the possibility to continue the tuning adjustment even if the previously obtained results already fulfill the specification limits.

This method allows further optimization of the BC reflection and transmission values beyond the defined spec. values.

3.3.2.5 Gradient Regulator Expert

Help ID: 6AAB138D-BB8F-459C-84F4-81BC0DA740CD
Service Level: 7

Expert Mode

In the **Expert Mode**, the operator has more control over the adjustment and setting of the regulator values. The selection of a single gradient axis is possible. By default, the X gradient is selected.

Measurement

Three numerical values (P=proportional portion, I=integral portion, D=differential portion; range 1 to 128) display the regulator settings for the selected gradient. When the gradient is selected for the first time, the existing regulator settings are displayed for this gradient. The values are configurable for the operator.

When selecting **[Go]**, these values are loaded into the gradient system as start parameters for the following iterative measurement. At the end, the new, calculated regulator settings are displayed. These settings are loaded when selecting **[Go]** the next time.

3.3.2.6 Phantom Shim Expert

Help ID: 22052F03-68D8-409A-84AD-FDDCA75B3054
Service Level: 7

Expert Mode

In **Expert Mode,** the operator can perform a single phantom shim iteration.

3.3.2.7 Eddy Current Compensation Expert

Help ID: D9B69F8A-96D7-4019-BAC0-3B48D2D0F5AE
Service Level: 7

Expert Mode

In the **Expert Mode** a coil selection is available.

In addition, the Eddy Current Compensation measurement can be performed for one or more gradient axes (by operator selection).

3.3.2.8 RF Characteristics

Help ID: A3FB5B7B-4700-435B-97C2-91525CAA916C
Service Level: 7

Expert Mode

RF Characteristics In Expert Mode operates similar as in the Normal Mode but with a few additional options.

General

The following additional options are available in Expert Mode:

- Refusal to store RF Characteristics after the measurement.
- Modification of maximum RF power allowed.
- Selection of TX-channel, RF path and nucleus (according to available licenses).
- Option to switch-off the frequency adjustment.
- Option to disable the switching to dummy load.

3.4 Quality Assurance

3.4.1 General Quality Assurance

Help ID: 7DFD7C17-DADA-4F81-B6E1-1AF37D0BBC53
Service Level: 3

Introduction

General Quality Assurance (QA) is an automated procedure for:

- Verification of TuneUP adjustments.
- Checking the overall system quality.
- Image quality checks.

Different from TuneUp, **system parameters are not changed in QA**.

QA Workflow

- General QA is separated into modules, each routine runs as a stand-alone process.
- After start, each procedure reads the specification and configuration parameters automatically.
- QA procedures determine if their measured results are in or out of specifications.
- QA procedures write their individual results to the Service Report.
- The status is returned to the main QA platform and the corresponding status field in the Service UI is updated.

After pressing the **[Go]** button in the QA UI, the **procedures start and perform typical tasks**:

- Each procedure registers the Service Patient (if not already registered).
- If a clinical patient is registered, a pop-up informs the operator to de-register the clinical patient before proceeding.
- The correct phantom setup is displayed by a pop-up and confirmed by the operator (if applicable). If QA is performed remotely, automatic table move is disabled and the operator receives a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Table movement:
  - If **QA is performed locally** at the system, the **table moves automatically** to center the required phantom in the measurement field (change between large/small spherical phantom).

- If **QA is performed remotely, automatic table move is disabled**. The operator gets a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.

- Configuration and specification values are read.
- Measurements are performed.

QA Function Status

Tab. 5 Possible QA Function Status

Status	Description
To Do	The QA procedure has not yet been performed yet and is pending. OR: The QA procedure needs to be performed again because another procedure (from TuneUp) has requested its execution.
Ok	The QA procedure has been carried out successfully.
Not Ok	The related parameters/results were found to be out of specifications during the last operation.
Error	An error occurred during the last operation.
Cancelled	Procedure canceled by operator.

Evaluation/Results

After end of the QA measurement, evaluations are performed automatically:

- The procedures determine if the measured results are in or out of specifications.
- The results are written to the Service Report.
- The results status is returned to the main QA platform and the corresponding status field is updated.

TuneUp/QA Dependencies

If a TuneUp procedure is successfully completed and has changed its status, the corresponding verification procedure(s) in the Quality Assurance workflow menu is/are changed to status **To Do**.

TuneUp procedures may also affect multiple QA procedures, e.g. after TuneUp Phantom Shim, the QA procedures Phantom Shim Check, Gradient Sensitivity Check, BC Image Brightness and Long Term Stability Check have to be performed.

Standby

In QA, a **Standby** function selection is offered. If selected, the system is switched to standby mode after completion of the QA workflow. All units except for the host computer are switched-off.

After using the Standby functionality, it is not possible to switch-on the system again remotely.

3.4.1.2 RF Verify MNO

Help ID: 4210C5CB-BBD0-40ED-9200-0E96102D8567
Service Level: 3

General

The procedure RF Verify MNO performs a verification of the Multinuclei RF Characteristics determined during Tune Up.

Individual measurements for all supported nuclei are performed because of the frequency dependency of the broadband RFPA's power output. Forward and reflected DICO power values are measured and checked against their corresponding error tolerances (amplitude- and phase deviation from last saved RF Characteristics).

Measurement

The RF signal is generated at the MNO-TX Box Controller, amplified by the BB RFPA and routed to the dummy load in the MNO TX-Box.

The DICO factor is calculated and checked. Then, the deviation of the amplitude and phase from the linear characteristic is evaluated.

Evaluation/Results

- DICO calibration voltage
- DICO calibration statistics
- plots show the measured characteristics (amplitude and phase)
- maximum power deviation and peak-to-peak phase deviation

3.4.1.3 Feedback Loop Calibration Check

Help ID: 4172C660-AF1B-46AF-A127-DF43D6E9194F
Service Level: 3

General

The **Feedback Loop Calibration Check** procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the "tuncal" service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box.

Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a “Not Ok”.

Evaluation/Results

Results in Report:

Tables:

- Delay evaluation
- Dispersion of delays from single BC elements

Plots:

- BC phase plot

3.4.1.4 RX Gain Calibration Check

Help ID: 4A3CD412-40FD-43CD-9DD2-BEF0BA7D4C40
Service Level: 3

General

The **RX Gain Calibration Check** verifies the amplification correction factors for all individual receiver channels which were determined during TuneUP / RX Gain Calibration:

- (Residual) RX channel correction factors
- High- / Low-gain factors

Measurement

The procedure uses loop measurements (without MR measurements). A fixed calibration signal is fed through all RX channels which are switched to high- and low-gain. For the measurement, the stored TuneUp parameters for the given RX channel- and high- / low-gain-correction are applied. The generated raw data is analyzed and all gain correction factors and the quotient of the High- / Low-gain modes are determined.

Evaluation/Results

Results in Report:

- Average gain correction factor over all receive channels.
- Gain correction factors for each individual receive channel.
- High- / Low-gain signals for each individual receive channel. Comparison of amplitude- and phase-difference to the stored TuneUp correction values.
- Noise evaluation.

3.4.1.5 BC Tuning Check

Help ID: 12EC839E-96E8-4A6D-8498-C6930BCCCDEF
Service Level: 3

General

The **BC Tuning Check** procedure is used to measure and evaluate the central resonance frequencies of the Body Coil resonator systems as well as the coupling between them.

A high efficiency of RF power transmission to the Body Coil requires sufficiently low reflection factors at the TX-Box output.

This condition is usually met, if the reflection factors of the Body Coil resonator systems BC0 and BC1 are high (without load) and equal. However, with load, these reflection factors shall be low.

The other requirement is, coupling between both resonators shall be minimum (i.e. high decoupling values).

Measurement

BC Reflection

First, a **BC Reflection** measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals, covering the full transmit operating frequency range. The result of each measurement is the reflection or r-factor = Ur/Uf which is plotted for both Body Coil systems BC0 and BC1 along the frequency axis.

The evaluation determines:

- each coil's center (tuned) frequency is within a specified range, i.e. is not too far from the center of the operating range.
- that the center frequencies of both coil systems BC0 and BC1 are tuned within specified limits of each other, i.e. the delta frequency is low.
- that the reflection factors are within expected limits. These values provide useful information about the quality (Q-factor) of the BC resonant circuits. They also depend on the coil load and cannot be adjusted.

  The Software detects the resonant frequency always at the point where the reflection of the BC has its minimum.

Transmission

In a second step, a **Transmission** measurement is performed with transmitting a signal into the BC0-system and measuring the coupled signal in the BC1-system and vice versa. The transmission factor describes the amount of RF-coupling between the two Body Coil systems, mainly due to coil construction but also affected by other factors e.g., mechanic position adjustment of the Body Coil within the Gradient Coil's Faraday Shield.

The evaluation checks the transmission value, expressed in decibels (dB), to be within expected limits. **A high negative dB-value indicates minimum coupling** i.e., a good decoupling between BC0 and BC1 system.

Evaluation/Results

Results in Report:

- The reflection related values for both (BC0 and BC1) systems, like the resonance frequency, the reflection factor and the difference between both systems.
- The transmission related value like the frequency and the decoupling between both systems.
- The new and old phase difference.
- Diagrams showing the measured values for the transmission and the reflection behavior as a function of frequency.

3.4.1.6
Cable Length Check

Help ID: B19E08C7-B349-43A5-99BF-E46BAB8192A4
Service Level: 3

General

The QA step **Cable Length Check** verifies the BC transmit cable phase shifts which have been determined in TuneUp / Cable Length.

The BC transmit cable length may slightly vary due to production tolerances. However, accurate information about the lengths of both BC transmit cables is required for an optimal function of the extended TX B1-field regulation loop in the measurement framework.

Measurement

A measurement with the service sequence rf_loop is performed. An RF pulse is fed to both Body Coil channels with the Body Coil switched to ‘detune’ mode. The forward and reflected signals are taken from the internal directional couplers (DICO) of the TX-Box and analyzed by the monitoring Software.

From these results, the so-called S-matrix elements which describe the characteristics of the BC transmit path are determined. These values allow, together with the known Body Coil RF-properties, to calculate the cable lengths for both Body Coil channels in terms of the induced signal phase shifts.

Evaluation/Results

In the Service report, the actual, measured phase shifts for both Body Coil channels are displayed, together with the old values from TuneUp. The latter ones are read from the BC coil files. Difference values are calculated and checked against the specified tolerance range.

3.4.1.7 Gradient Regulator Check

Help ID: 99738FE4-5E37-4EFF-A222-CE0B23F5D7EE
Service Level: 3

General

The **Gradient Regulator Check** procedure only measures the gradient waveform with the present PID regulator settings and verifies that these settings are correct. The check analyzes several aspects of the actual current output by comparing it to the expected ideal pulse output. In addition, the procedure will measures the linearity of the gradients by stepping through the full range of the gradient strength from negative to positive amplitudes.

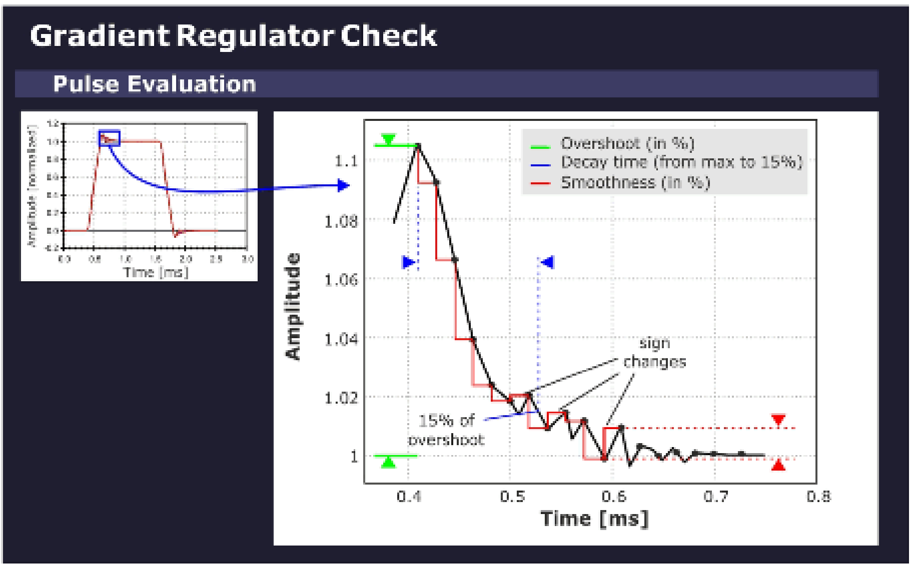
Measurement

The sequence uses a gradient table to generate the gradient pulses. The selected gradient (depending on the selected orientation) steps from the maximum negative to the maximum positive gradient amplitude pulse in linear portions. Each gradient in the table is switched on using a ramp-up and ramp-down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.

Evaluation/Results

Based on the maximum gradient amplitude pulse, the program evaluates the characteristic overshoot that occurs just after the pulse ramp-up. The amplitude of this **overshoot** is determined as a percentage relative to the nominal gradient amplitude. The program also calculates the **decay time** it takes for this exponential-like overshoot to return to a specified value. In addition, a **smoothness calculation** is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.

Fig. 278: Gradient Regulator Overshoot



Results in Report:

- PID gradient regulator parameters.

- Optimization parameters overshoot, decay time and smoothness.
- Gradient current pulse diagram (pulse form).
- Enlarged diagram of overshoot portion and current ripple (top portion).

3.4.1.8 Phantom Shim Check

Help ID: D847AEFF-8EE6-4E1B-83A4-EE3620B311E9
Service Level: 3

General

The **Phantom Shim Check** procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be accurately determined based on the frequency distribution in the volume elements making up the phantom image.

Two measurements are performed with the parameters from the corresponding TuneUp procedure to determine the remaining field terms and the overall homogeneity within the phantom volume. No adjustment of the shim terms is made in this case.

For systems with the advanced shim option, the procedure is performed with shim currents applied.

Measurement

The actual shim status of the system is measured with a 3D-Dess-sequence (Double Echo Steady State with a FISP and PSIF echo). The resulting voxels are cubes which are analyzed

by the Software.

Evaluation/Results

The field homogeneity is evaluated based on the frequency of the voxels. For the check of the B_0-field, it is sufficient to compare the field differences between neighboring voxels. The field difference is obtained by comparing the phases of the magnetization vectors.

The complete data set is evaluated by a differential shim equation. This equation minimizes the difference between the field generated by the3 Gradient Coils (and the optional 5 shim coils) and the measured magnetic field inhomogeneities.

Results in Report:

- Field terms
- MR Frequency
- Gradient Offsets
- Shim Currents (optional)

3.4.1.9 BC Power Losses Check

Help ID: D35D6115-3348-47DF-8FBE-8CBE8F1C7E7C
Service Level: 3

General

A check is performed to verify that the TuneUp of BC Power Losses has been performed successfully. This is done by measuring the Coil Power Loss value again and checking against the Coil Power Loss value in the coil file. Since the default of this value is very low, it would be out of specification if no TuneUp was executed.

In addition, this QA step verifies that the SAR (Specific Absorption Rate) check is activated. The forward and reflected voltages from both TX Box channels are also tested during the measurement.

Measurement

The TuneUp “Coil Power Loss” measurement is performed.

Evaluation/Results

The Coil Power Loss is measured in CP and EP mode.

- Check if the Coil Power Loss value in the coil file is within specifications and comparable to the measured value.
- Read and check the value for the SAR.
- Read and check the forward and reflected voltages from TX-Box and check them for consistency with broadband power meter values.

Results in Report:

- Phantom position and diameter.
- Measured and stored Coil Power Loss values.
- Measured reference amplitude (AdjTra) from the TALES (TX-Box power meter).
- The measured SAR and voltage values.

3.4.1.10 Cross Term Compensation Check

Help ID: 94F6F45A-CD81-4440-9FDE-F6D1AC3824A4
Service Level: 3

General

The **Cross-Term Compensation Check (CTC)** checks the quality of the cross-term compensation previously adjusted in TuneUp.

In QA mode, the parameters of the inverse filter function which compensates the cross-term effect are checked.

Measurement

A CTC measurement is performed.

Evaluation/Results

Results in Report:

- Cross Term Amplitudes and Time Constants.
- Diagrams visualizing the corresponding cross term amplitudes.

3.4.1.11 Eddy Current Compensation Check

Help ID: ACA062B6-B909-438A-A732-F3BF88BA7502
Service Level: 3

General

The **Eddy Current Compensation Check** (ECC) checks the quality of the eddy current compensation adjustment previously performed in TuneUp.

Measurement

The procedure performs a check of the ECC parameters. The existing ECC settings are used for the measurement.

A preparation measurement is performed to check for the correct positioning of the phantom in the magnet center and for sufficient field homogeneity. The used sequence is a double echo. The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity. The second echo uses a readout gradient and calculates the phantom position in the readout direction.

Evaluation/Results

Results in Report:

- B_0 eddy current values for different time ranges.
- 1^{st} order (gradient) eddy current values for different time ranges.

3.4.1.12 Gradient Delay Check

Help ID: 946ABE8C-323C-48C6-B295-5C45C3DD15AC
Service Level: 3

General

The **Gradient Delay Check** procedure determines the delay time between gradient switched on digitally and the time the actual gradient field builds up. Gradient Delay can slightly differ for each gradient axis.

In QA mode, the Gradient Delay adjustment values, previously established in TuneUp, are is verified.

Measurement

Each gradient is measured at 14 different gradient strengths from -19.5 mT/m to +19.5 mT/m. For an ideal gradient system, all delays are the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, that is, high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, there is some kind of nonlinearity present in the gradient system.

Evaluation/Results

Results in Report:

- Current- and calculated- Gradient Delay values in microseconds.
- Maximum Gradient Delay Deviation values.

3.4.1.13 Gradient Sensitivity Check

Help ID: 16C2854F-2DEB-4B6F-BEDA-E6418994A88B
Service Level: 3

General

The **Gradient Sensitivity Check** procedure checks the Gradient Sensitivity values previously adjusted in TuneUp to ensure that the system uses correct the gradient strengths and measures the correct phantom image size in all orientations.

For the measurement, the distortion correction filter is enabled to compensate gradient field non-linearities.

Measurement

- A standard imaging sequence is repeated three times: one measurement for each orientation.
- The three images are then analyzed to determine the exact location of the center of the phantom and the phantom diameters. The standard readout and phase encoding direction for each orientation is swapped to determine the center and diameter of the phantom. The average of these two results determines the final diameter value.
- The measured values are evaluated and compared with the well-known diameters of the phantom for verification of the correctly adjusted Gradient Sensitivities.

Evaluation/Results

Results in Report:

- Measured phantom diameter in all three dimensions.
- Measured phantom centers in all three dimensions.

- Gradient Sensitivities.
- Phantom images in all three slice orientations.

3.4.1.14 BC Image Brightness Check

Help ID: 4B8C4548-E064-495C-B6B2-AC746D517DD4
Service Level: 3

General

This procedure is used to verify the image intensity (brightness) for the Body Coil which was previously adjusted in the TuneUp step BC Image Brightness..

Measurement

The measurement runs the service sequence "Body_Tra_br" and generates images with the Body Coil.

Evaluation/Results

Both Body Coil channels BC0 and BC1 are evaluated independently from each other. Also the sum signal of BC0_BC1 is checked.

From each image the following values are determined:

- Brightness
- FFT Scale Factor
- TX Amplitude
- RX Gain Factor

The brightness value is the signal intensity. The FFT scale factor is a correction factor to adjust the brightness to the middle of the specified value range (min. and max.). The transverse image is used for determining the FFT scale factor.

The transmitter amplitude for the sequence (180 degree flip angle) is also evaluated. This value is coil load specific, so it depends on the phantom.

3.4.1.15 Image Orientation Check

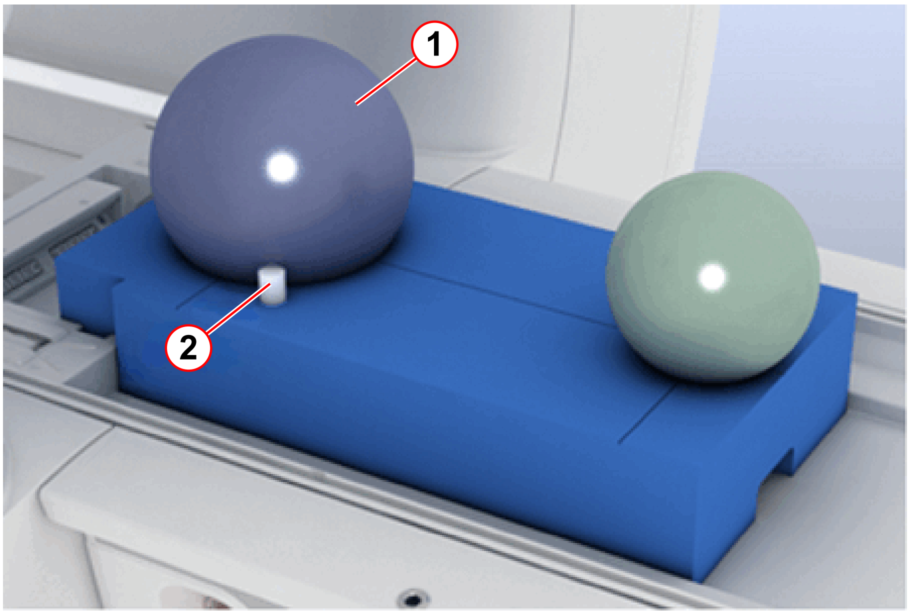
Help ID: F09D8D74-F01E-4A0A-B3FD-A116D3E77C87
Service Level: 3

General

The **Image Orientation Check** verifies the correct position of the phantom setup. It uses an additional, asymmetrically positioned small bottle phantom to determine the correct gradient orientation and polarity.

If any of the gradient power amplifier or -coil connections were swapped or polarities of any gradient axis were wrong, the images would be displayed with incorrect orientation.

Fig. 279: TuneUP_QA Phantom Setup

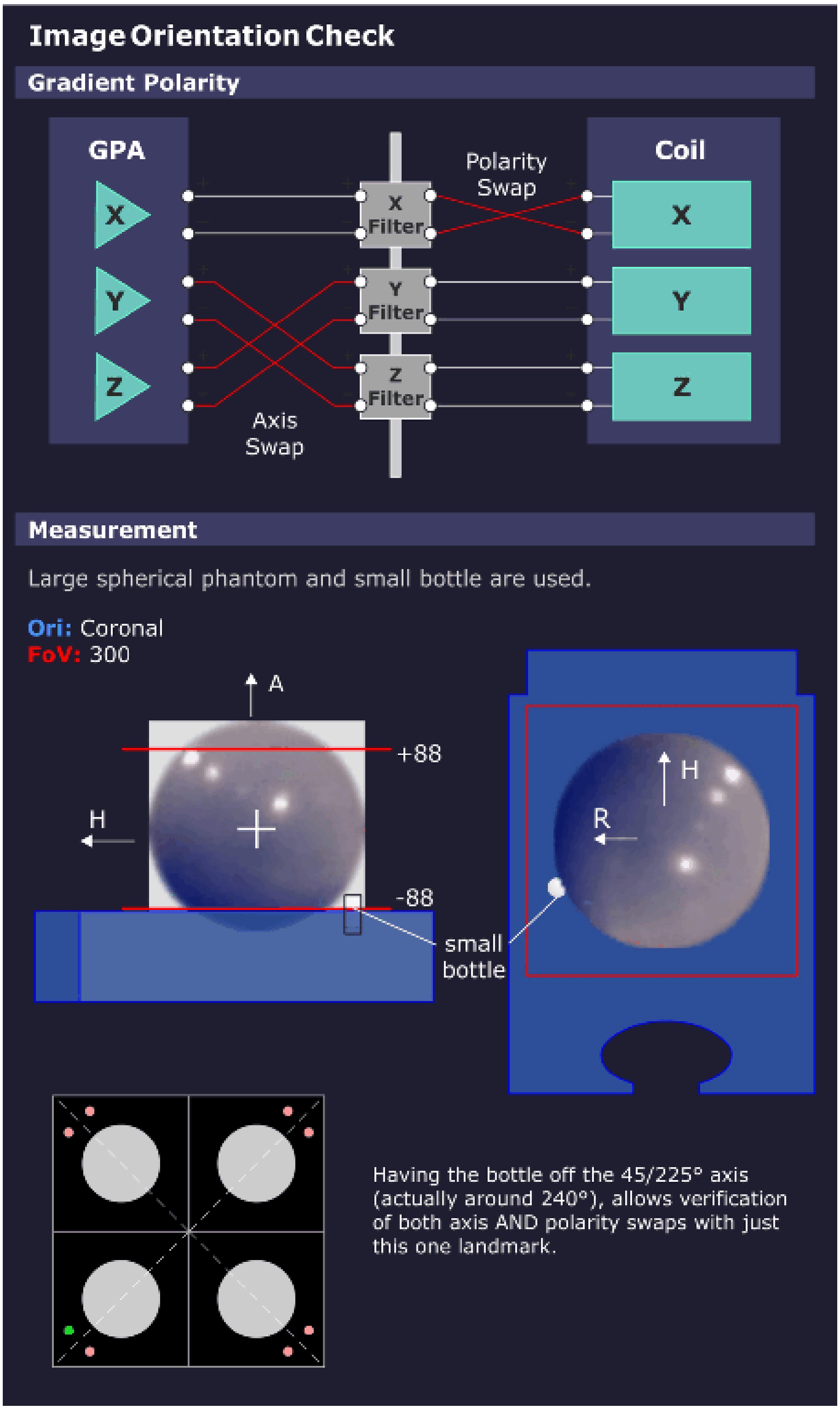


(1) Spherical phantom

(2) Small bottle phantom for Image Orientation Check

Measurement

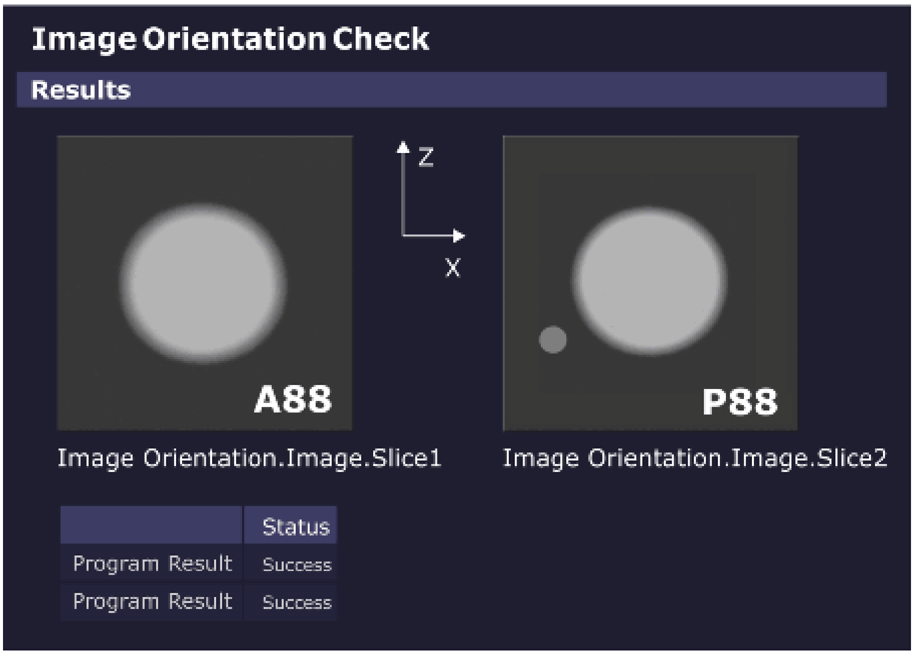
Fig. 280: Image Orientation Check - Measurement



Images are generated from the phantom setup with the additional, asymmetrically positioned small bottle.

Evaluation/Results

Fig. 281: Image Orientation Check - Results



The measured images are automatically evaluated.

If the phantom and bottle are not in the expected position, the procedure returns with the status “Not Ok”.

Results in Report:

- Images showing the phantom geometry (phantom and bottle).

3.4.1.16 Gradient Risetime Check

Help ID: B668AA17-8C5F-4E6C-8908-BB2F522B4F00
Service Level: 3

General

The **Gradient Rise Time Check** measures the minimal gradient Rise Time or the so-called “Slew Rate”.

Measurement

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, five echoes develop because of the bipolar triangle gradient pulses. These echoes should appear, according to the pre-dephasing, exactly at the pulse spikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate could not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Sequence of procedures:

- Verification check of the used phantom.
- Gradient Risetime measurement for all three gradient axes.

Evaluation/Results

Results in Report:

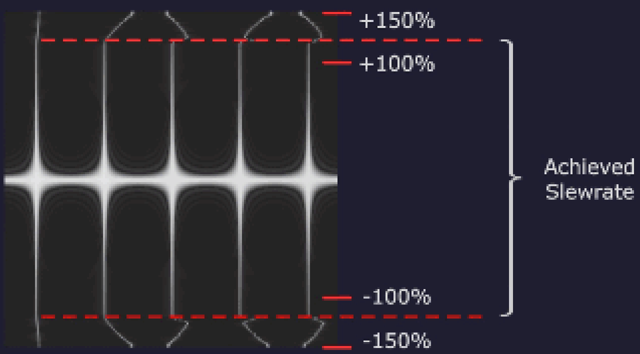
- Graphics of echo maximum versus gradient slew rate, echo 1-5
- Statistics of measured values against specifications
- X/Y/Z-gradient Rise Time results
- X/Y/Z-gradient Rise Time multiple echo output (raw data image, echo 1-5)

Fig. 282: Gradient Risetime Check - Results

Gradient Risetime Check

Results

Gradient Rise Time Results		
Axis	Slew Rate	
		Tolerance
X	112.46 %	≥100 %
Y	111.43 %	≥100 %
Z	112.48 %	≥100 %



X-Gradient Multiple Echo Output

Slewrates and Echopositions						
Echo	Slew Rate Positive Polarity		Slew Rate Neg. Polarity		Diff. of Echo Positions	
		Tolerance		Tolerance		Tolerance
1	**126.68 %**	≥100.00 %	**126.68 %**	≥100.00 %	-0.039	-1.000 ... 1.000
2	**114.67 %**	≥100.00 %	**114.69 %**	≥100.00 %	0.044	-1.000 ... 1.000
3	**113.58 %**	≥100.00 %	**113.58 %**	≥100.00 %	-0.076	-1.000 ... 1.000
4	**112.48 %**	≥100.00 %	**113.58 %**	≥100.00 %	0.041	-1.000 ... 1.000
5	**112.48 %**	≥100.00 %	**112.48 %**	≥100.00 %	-0.039	-1.000 ... 1.000

X-Gradient strength slew rate achieved is 112.483% of specified value
(mean value of the lowest two values of all echoes)

Troubleshooting

In case of problems (result **“Not Ok”**) check the following items:

- Check raw data for spikes. No spikes must be visible.
- Stability/value of mains voltage
- Line resistance
- GPA line transformer and correct line voltage adaptation
- GPA rectifier board
- GPA current sensors (especially if value “Difference of Echo Positions” is out of spec.)
- GPA Power Stages

- Results of QA Regulator Check

3.4.1.17 Spike Check

Help ID: B4E9366D-2E5A-4CE9-969E-3EBDE191949C
Service Level: 3

The **Spike Check** is a quick test to check the MR imaging system for spikes and RF interference.

A "spike stress" measurement is run and its raw data evaluated by a program for spikes. The corresponding image data is evaluated for RF interference.

General

Electrostatic discharges during the measurement may generate some spikes within the raw data. Spikes are individual raw data points or small groups of raw data points with amplitudes above the typical noise level of the MR imaging system. **Spikes** can degrade the MR imaging quality. In this case, the signal-to-noise ratio can be reduced or the image can show a periodic intensity modulation or structure.

One typical source of spikes is the mechanical and electrostatic stress of the MR system caused by the operation of the gradient system.

A simple way to check the MR system for spikes is to run a noise measurement without RF pulses. During the acquisition window massive mechanical and electrostatic stress is generated with rapid switching of gradient pulses with maximum strength.

In case no spikes and no RF interference are present, pure statistical noise shall be received.

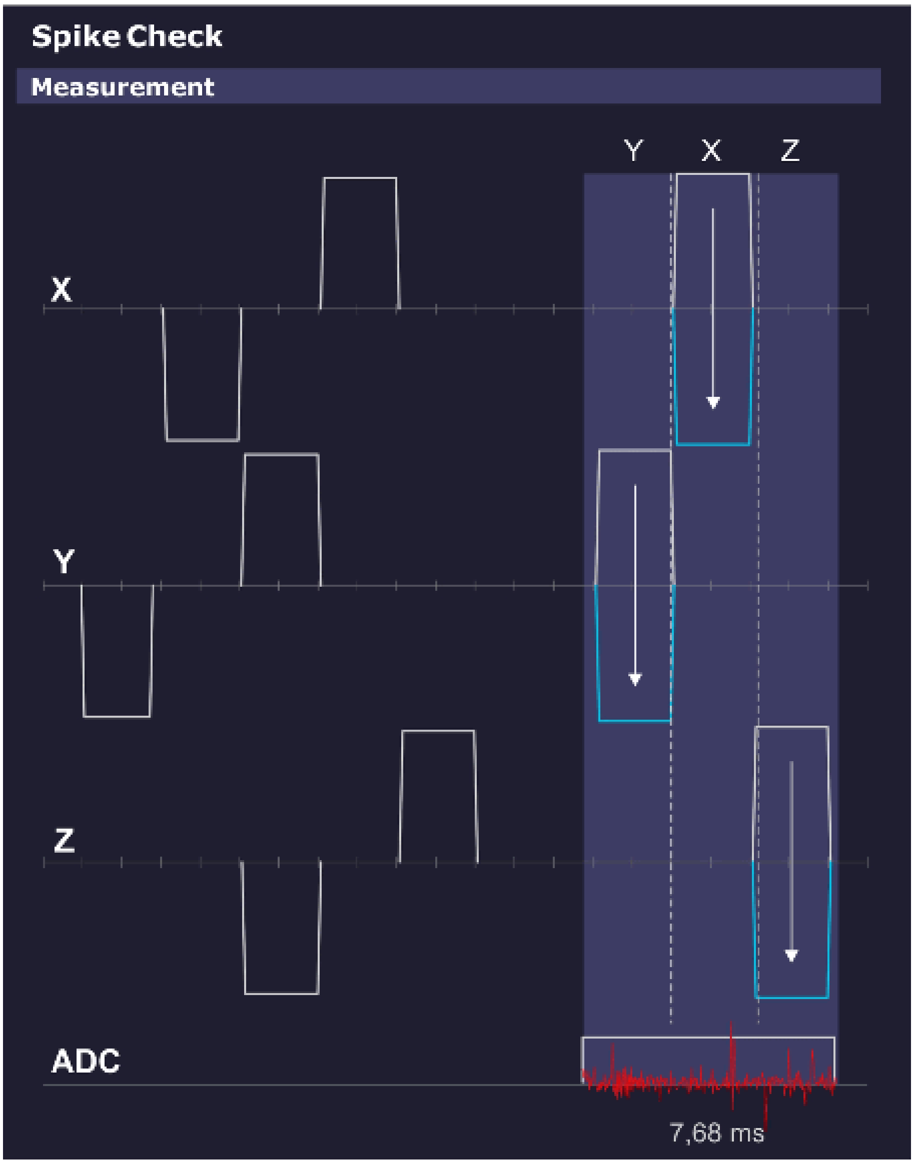
Measurement

This sequence does not apply RF pulses, so no MR signal is generated.

The spike sequence starts with an initial series of strong gradient pulses of both positive and negative polarities on all three gradients intended to cause a buildup of static electrical energy caused by any moving components (cables, covers, etc.).

The ADC is then enabled and again a pattern of gradient pulses with amplitudes ranging from max. positive to max. negative are applied. Any static electrical discharges caused by e.g. moving or loose parts are picked up by the receiving coil (usually the Body Coil), digitized by the ADCs and stored in the raw data.

Fig. 283: Spike Check - Measurement



Evaluation/Results

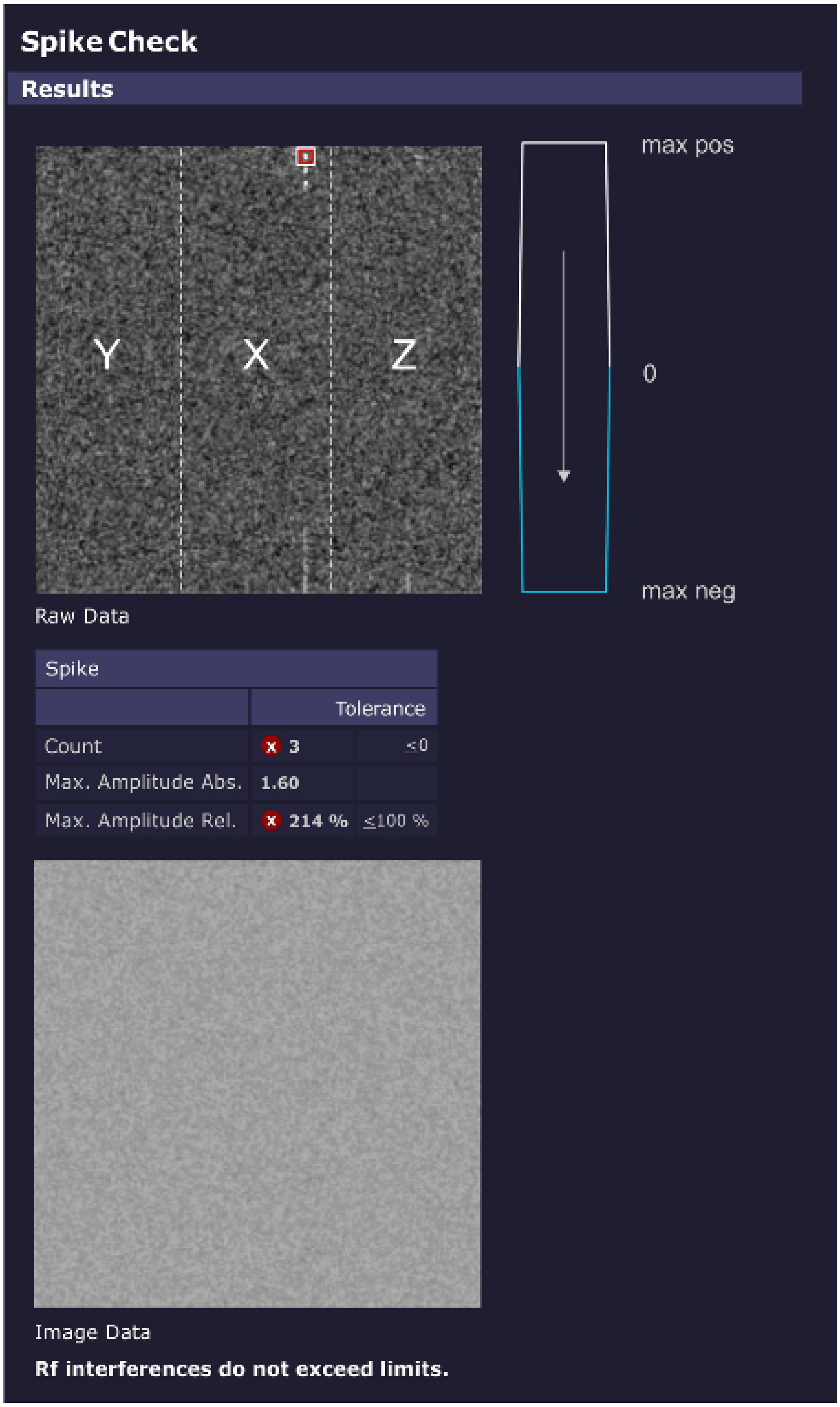
Search for **spikes in raw data** and **RF interference in the image data**.

The evaluation looks for intensities in the raw data set exceeding a threshold (usually noise) and counts both the number and intensities of the spikes and displays them numerically and graphically. The electrical discharges are visible as "spikes" in the raw data.

The position of the spikes in the raw data set relates to the gradient axis that was switched on when the spike occurred. This position may indicate the cause of the spikes.

In case RF interference was present during the acquisition, the disturbances can be seen in the displayed image data.

Fig. 284: Spike Check - Results



Results in Report:

- Raw data image and resulting RF interference image.
- If spikes are found, the following values are displayed:
  - Total number of spikes.

- Magnitude of the highest spike.

3.4.1.18 RF Noise Check

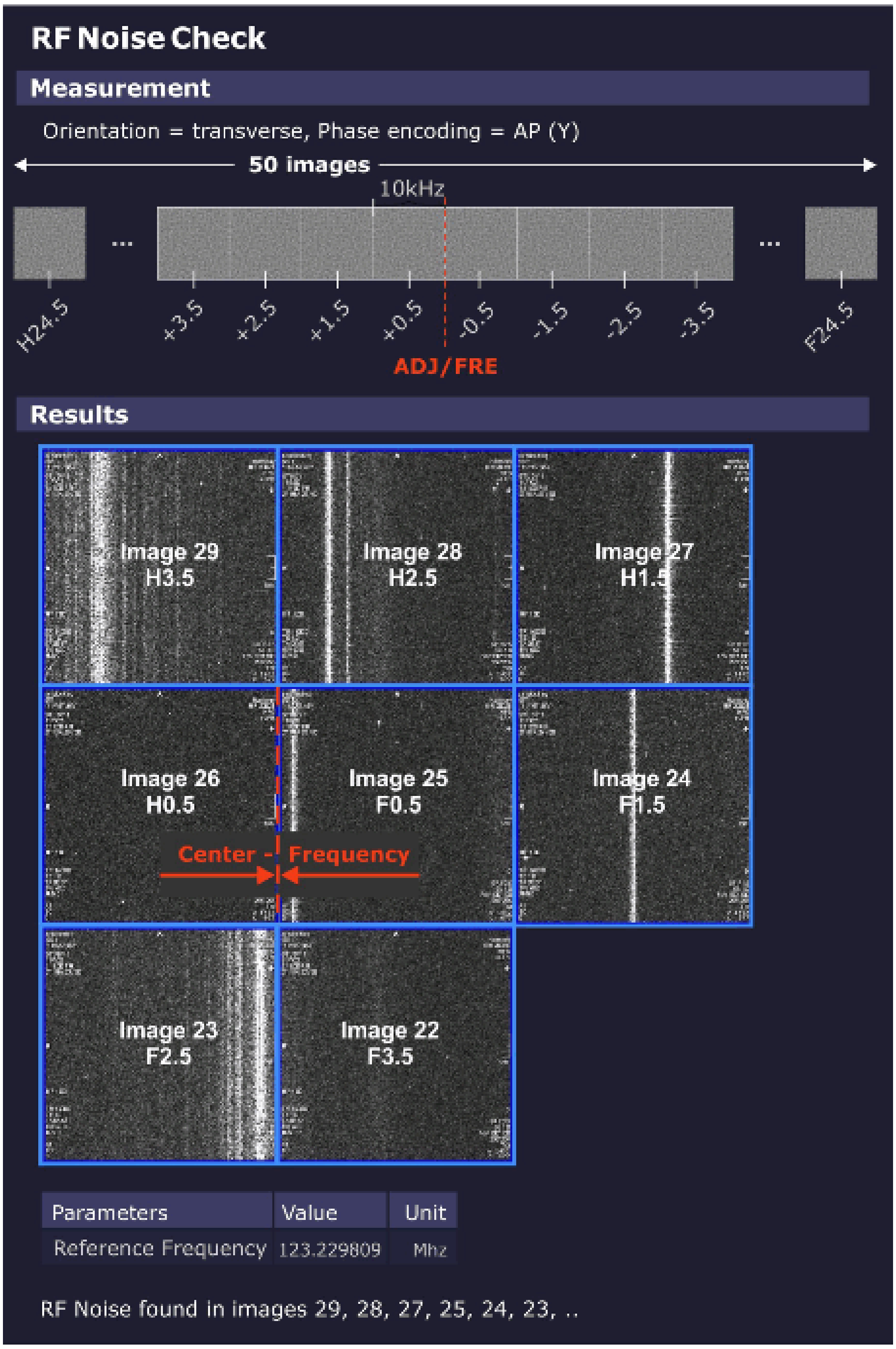
Help ID: AA585552-5E09-43B7-8E85-B9A8394037E2
Service Level: 3

General

The **RF Noise Check** is used to detect RF interferences which can disturb the MR-image.

Measurement

Fig. 285: RF Noise Check



The noise measurement is performed with the service sequence rf_noise. The sequence is without RF or gradient pulses, just an open receiver.

The sequence produces 50 images (i.e., slices). The difference between the images is the variation of the center frequency. Each image covers 10 kHz (256 times 39 Hz) so that the complete receiver bandwidth of +/-250 kHz is covered.

The border between the center slices 25, 26 is the current center MR system frequency.

Evaluation/Results

The evaluation algorithm looks for RF interference in the acquired image data i.e., dots and streak artifacts in phase-encoding direction and noise level.

Results in Report:

- RF interference according to the analyzed frequency bands.

3.4.1.19 PTX Image Check

Help ID: 82D6FBB2-442B-4B71-A95B-FD165371082B
Service Level: 3

General

A parallel transmit (pTX) array consists of 2 or more distinct transmitter channels where amplitude and phases can be modulated independently from each other.

Such a system can generate dedicated local magnetization patterns, e.g. exciting one spatial region of a patient or phantom while leaving other regions untouched.

This mechanism, however, is rather sensible to system instabilities or malfunctions of components in the imaging chain.

In order to detect such deviations, the service procedure runs the dynamic pTX sequence to produce a well-defined local magnetization pattern which is analyzed for image quality.

Measurement

For the measurement, the small spherical 170 mm water-filled phantom is used.

a homogeneous image is produced which allows to verify the correct positioning of the phantom.

The pTX sequence “dyn_pulse” is executed and generates a well-defined target magnetization pattern.

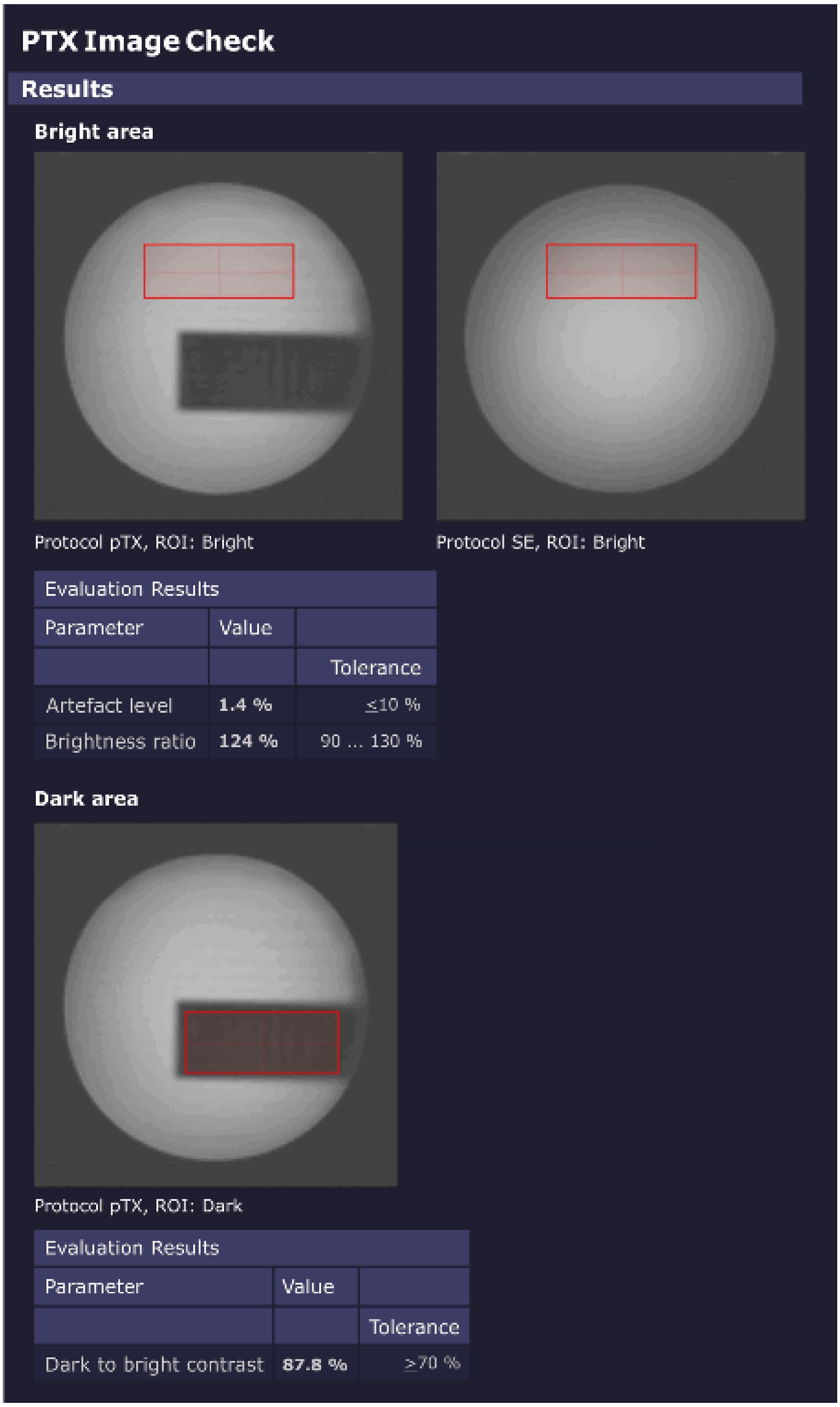
First, a homogeneous image is produced which allows to verify the correct phantom positioning.

Second, a homogeneously excited plane with an embedded dark (non-excited) rectangle is measured.

Subsequently, the Software analyzes the resulting images to detect potential system malfunctions.

Evaluation/ Results

Fig. 286: PTX Image Check - Results



First, a ROI is placed entirely in the **bright region** of the excitation pattern. This ROI is analyzed for relative image brightness (which should be similar to the brightness in the homogeneous image) and artifact level.

A second ROI is placed entirely inside the **dark region** (rectangle). The relative brightness of this second ROI, compared with the brightness of the first ROI, defines the image contrast.

3.4.1.20 Stability Check

Help ID: 796762F3-ECF9-458A-83A1-CAFDEC3BA27D
Service Level: 3

General

The **Stability Check** evaluates the stability of the MR signal under various sequence conditions.

A spin echo and a gradient echo are measured 256 times without phase encoding using fixed sequence parameters.

As a result, all raw data lines look alike (if the system is stable and not taking in account stochastic variations).

The stability of the echo signal across the measured lines is evaluated:

- Variations of the amplitudes
- Variations of phases
- Positions of the echoes
- Frequency perturbations

Measurement

In Standard Mode, a **spin echo** and a **gradient echo** are measured 256 times without phase encoding using fixed sequence parameters:

- TR = 301 ms
- TE = 20 ms

All **three orientations** are measured, so each gradient is performed as readout gradient once:

In addition, both **0 and 50 mm slice shift** positions are measured.

Evaluation/Results

The **spin echo sequence** uses a default setting of TE = 20 ms. Due to chosen TR = 301 ms and the inversion effect of the 180° pulse, it is sensitive to instabilities in the range of a few Hz up to 100 Hz which may be caused by e.g.:

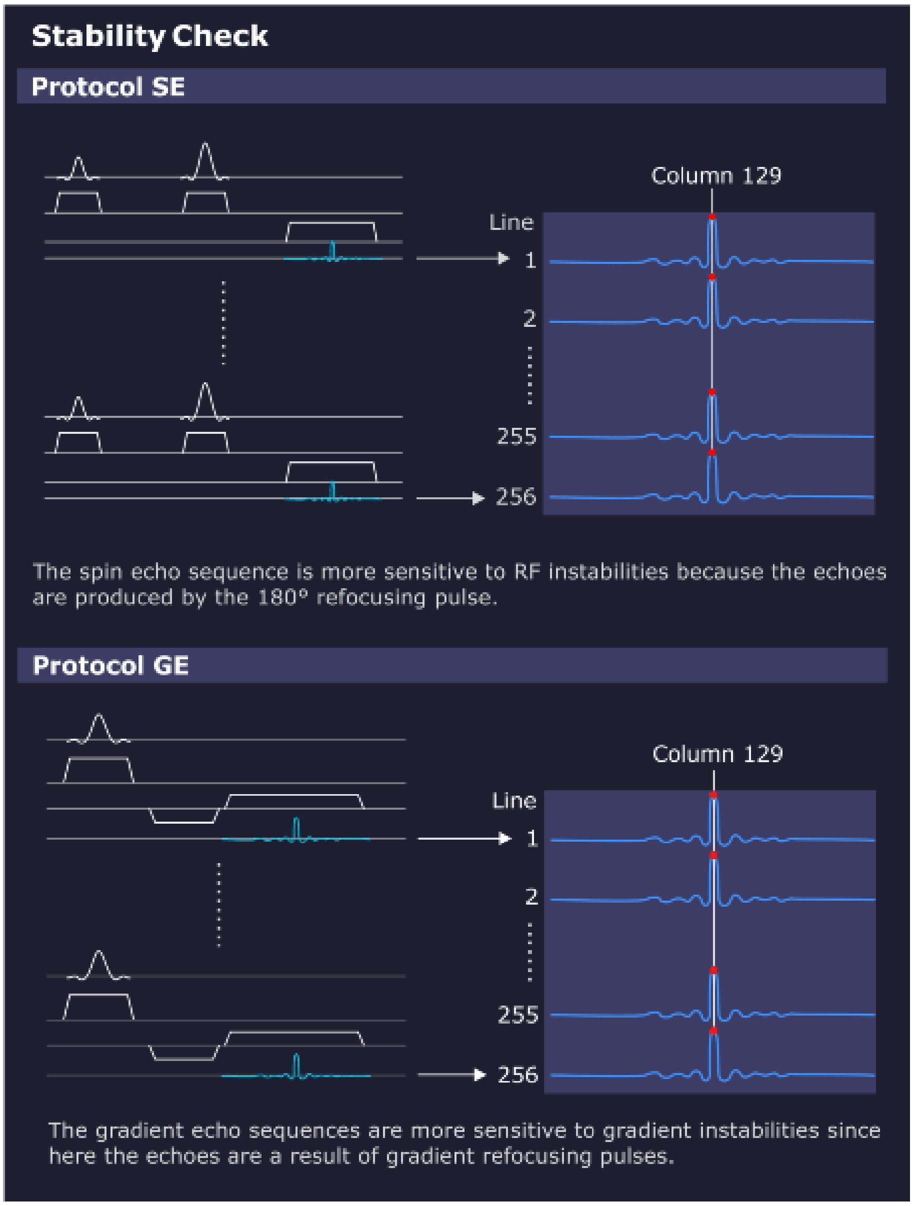
- 50/60 Hz AC power line hum;
- or 16.7 Hz AC trains.

The **gradient echo / flash sequence** also uses a default setting of TE = 20 ms and TR = 301 ms. Due to the chosen parameters and the missing 180° pulse, it is insensitive to 50 Hz line hum. The gradient echo is sensitive from DC up to a few Hz to slow external field interferences caused by e.g.:

- DC-driven local train systems like tram or subway;

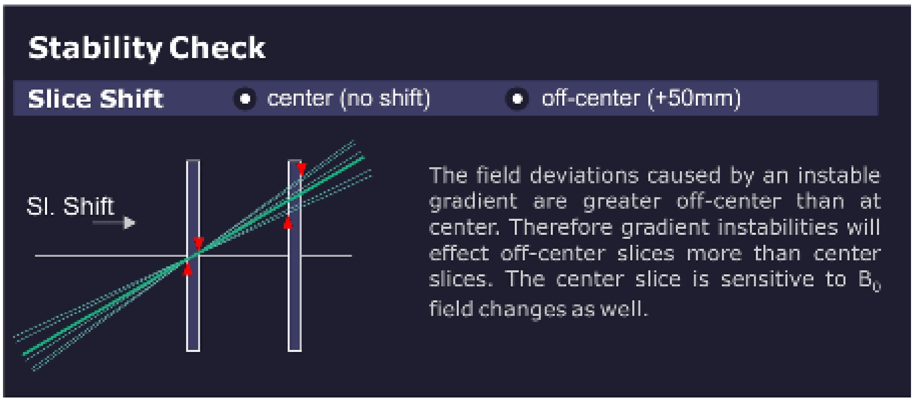
- Magnetic field instabilities;
- or mechanical vibrations.

Fig. 287: Stability Check Evaluation



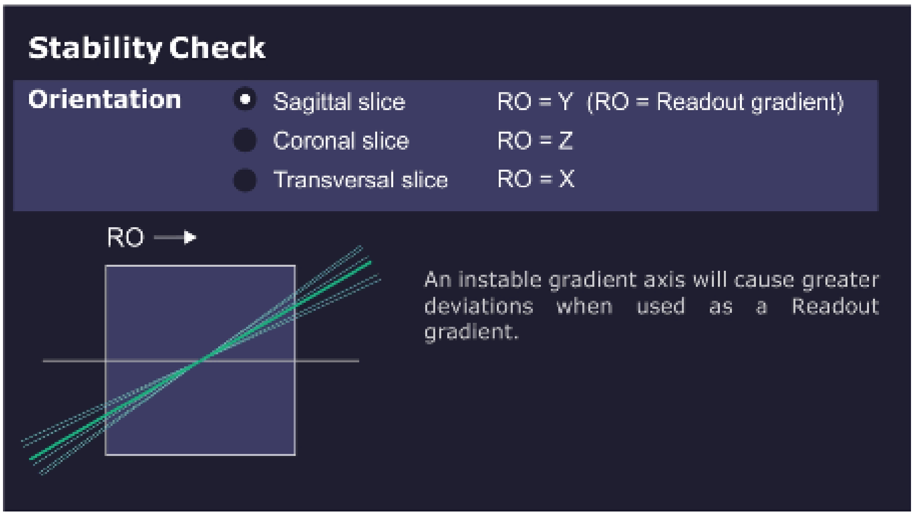
The measurement result of the **center slice** is sensitive to instabilities of the B_0 field. The **50 mm offset slice** is more sensitive to gradient-like instabilities.

Fig. 288: Slice Shift



The **orientation** determines the role of each gradient axis. For example, lets assume there is an instability in the X gradient. The values for the sagittal orientation (X = slice selection) may be in spec. while the spec for the transverse orientation (X = read out) could be out of spec.

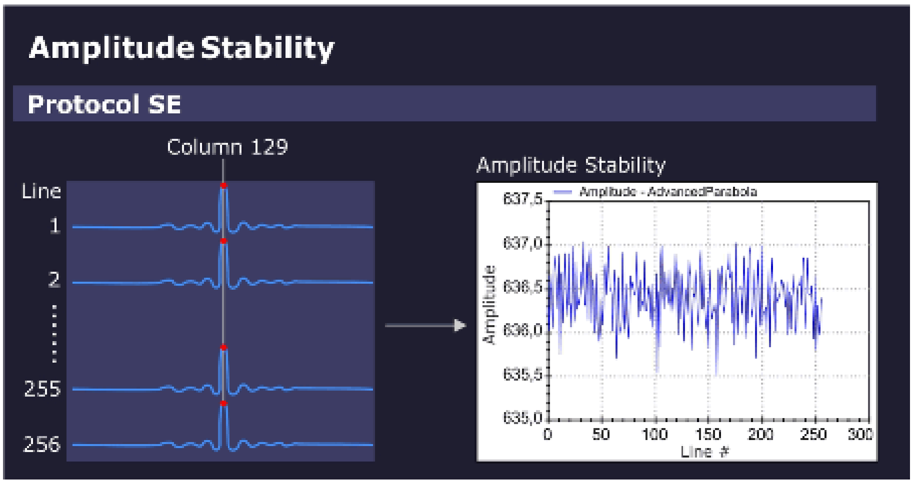
Fig. 289: Orientation



Amplitude stability (spin echo):

The amplitude value in the maximum signal column (e.g. column 129) is displayed in the graphics output.

Fig. 290: Amplitude Stability (Spin Echo)



Phase stability (spin echo):

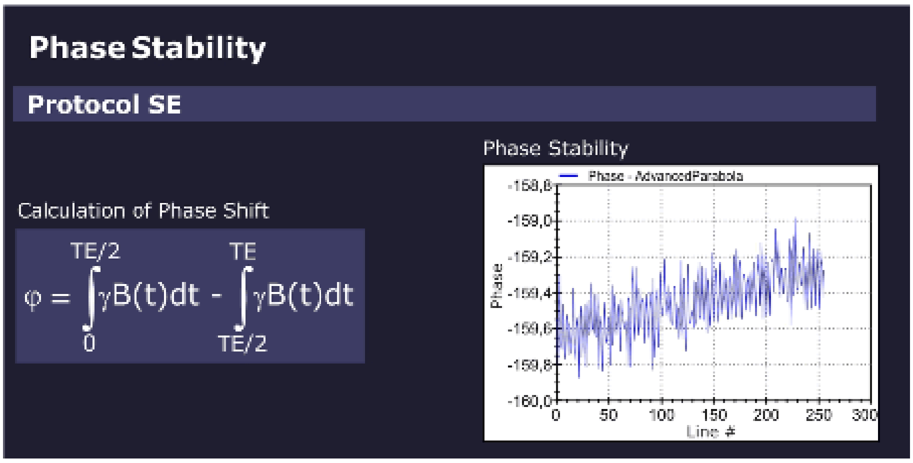
The phase stability of the spin echo is displayed in the graphics output.

The phase shift of a spin echo is determined by the integral across the magnetic field from a 90° to a 180° pulse minus the integral from a 180° pulse to the echo.

For a constant field or a slowly varying field, both integrals cancel out each other.

However, for an oscillating field with a period of TE (20 ms = 1/50 Hz), this expression becomes maximum. In other words, the spin echo measurement with the default setting of TE = 20 ms (and TR = 301 ms) is sensitive to interferences which may be caused by 50/60 Hz AC power line hum or 16.7 Hz trains.

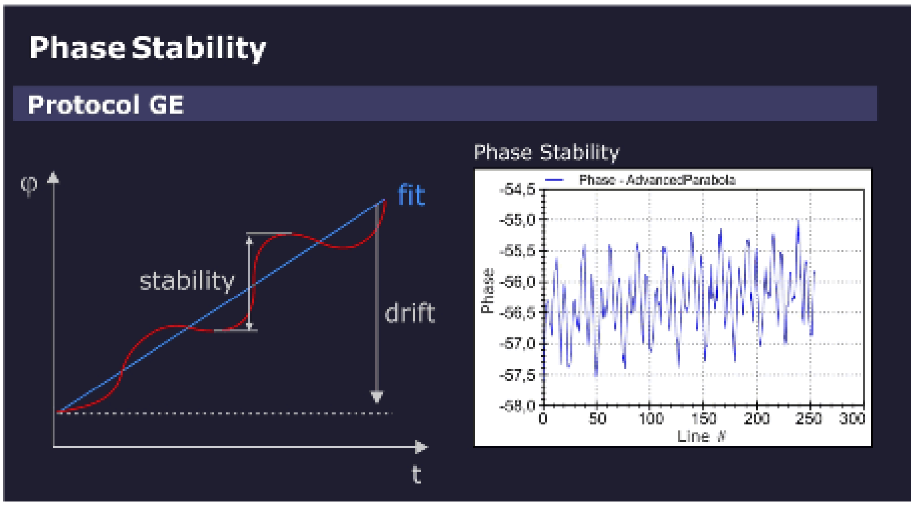
Fig. 291: Phase Stability (Spin Echo)



Phase stability (gradient echo):

For the gradient echo sequence, the phase stability is evaluated by fitting a straight line to the phase data. When you subtract the value of the linear fit from the phase data, the phase stability changes. The slope of the straight line indicates the linear phase drift, caused by slow external field interferences, for example, moving elevators, cars, trucks but also mechanic vibrations.

Fig. 292: Phase Stability (Gradient Echo)



Stability of echo position:

For both measurements, spin echo and gradient echo, the position of the echo peak column is determined for every measurement line and checked against specifications.

The cause of an unstable echo position may be gradient-like interferences from the environment or from the gradient itself.

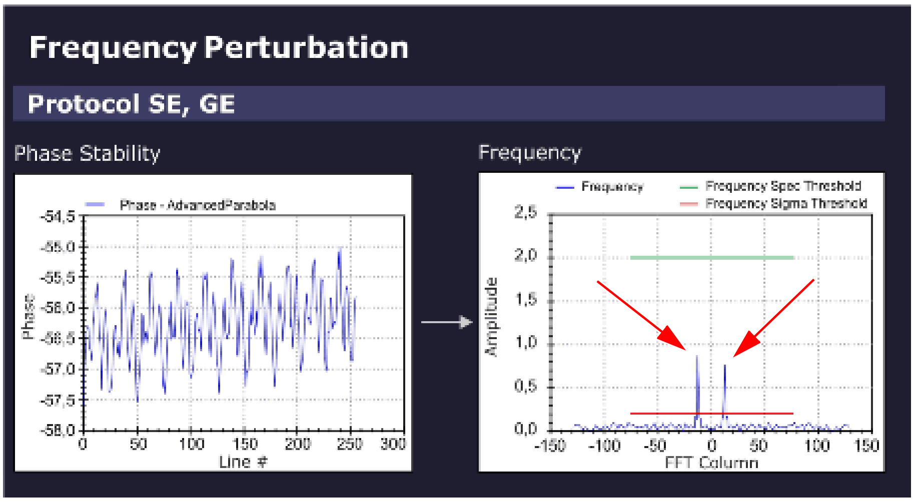
The sensitivity for the direction of the interference is determined by the orientation of the readout gradient.

Frequency perturbation:

The Software differentiates the raw data of the stability measurements and checks for sidebands (ghosts).

In the evaluation result, a frequency perturbation value in percent and the Frequency (peak) position in the FFT Column is displayed.

Fig. 293: Frequency Perturbation (sidebands)



Example: (Frequency Perturbation of 0.8%, Frequency Position -13)

- The sampling rate in the stability measurement is one sampling point every 301 ms (default TR = 301 ms).
- Consequently, the Nyquist-frequency = +- 1/602 ms = +-1.66 Hz. The limiting frequencies at +-128 in the frequency diagram above correspond to +1.66 Hz and – 1.66 Hz, respectively.
- The perturbation frequencies are located at +-13 units, i.e. at 13/128 * 1.66 Hz = +-0.17 Hz.
  » The 300th harmonic of 0.17 Hz yields a 50 Hz interference.

Results in Report:

Measured value	Description
Amplitude Stability	Range of amplitudes (peak to peak)
Phase Stability	Deviation of phase from linear fit
Phase Drift	Drift of phase
Peak Column Stability	Range of peak column
Peak Column Average	Mean peak column
Frequency Perturbation	Relative amplitude of the highest contributing disturber frequency; zero if no frequency is above the relative threshold nor above (noise level * noise factor).
Frequency Position	Position of the highest contributing disturber frequency; zero if no frequency is above the relative (sigma) threshold nor above the (noise level * noise factor).

Troubleshooting

Symptom/Deviation	Possible Cause/Service Action
Amplitude	RF-instability problem, check: ▪ Stability of RF System / TX ▪ Stability of RF System / RX

Symptom/Deviation	Possible Cause/Service Action
Phase	Magnetic field fluctuation, check: ▪ After magnet ramping or after heavy gradient load (e.g. DTI-, EPI- or StabilityLongTermCheck measurement) wait at least 1 hr. before running Stability Check; ▪ Run QA-Expert / Field Stability Check.
Depending on orientation	Instability of a gradient axis, check: ▪ The Gradient System hardware. ▪ Run service sequence "CalcAr" in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.
Independent of orientation	Magnetic field instable or external field interference, check: ▪ The Gradient System hardware. ▪ See hints "Only in gradient echo".

<table>
<tr><th>Symptom/Deviation</th><th>Possible Cause/Service Action</th></tr>
<tr><td>Only slice-shift dependent</td><td>Most likely, a gradient problem, check:
■ The Gradient System hardware
■ especially, the gradient axis used for slice-selection</td></tr>
<tr><td>Only in gradient echo</td><td>1. Static or slowly varying magnetic field problem?
In the vicinity of the site, check for:
- Trains; interference from power lines and rails.
- Elevators.
- Car/truck traffic and parking.
2. Very low amplitude due to static magnetic field (B_o) homogeneity problem?
- Compare amplitude with spin echo result (both amplitudes should be similar).
- Run QA / Phantom Shim Check.
3. Phase drift out of spec., B_o-drift problem?
- System hot? Excessive usage prior to Stability Check measurement (e.g. repeated DTI-, EPI- or StabilityLongTermCheck measurements);
- System too cold? Wait at least 2 hr before measurement.
4. Peak column stability out of spec. i.e., fluctuation of echo position due to gradient field instability?
- Check the Gradient System hardware
- Run service sequence “CalcAr” in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.</td></tr>
</table>

Additional hints:

- Switch off components one by one to isolate the cause of the instability, e.g.:
  - Cold head
  - Lighting in magnet room
  - Shim PSU
  - Disable gradient axes with DIP switch at DGSU board (first each single axis, then complete gradient power amplifier)
- For periodic instability, vary TR and TE (Stability Check / Expert Mode) to confirm or rule out the frequency of interference:
  - Vary TR to avoid being synchronous to the interference.

  - Vary TE to become more sensitive to particular frequencies.
- If an external interference is suspected, repeat measurement at different times e.g. during night.

3.4.1.21 Synthesizer Check

Help ID: 100FDE50-E606-4FB0-B37C-3F6B6E932DAF
Service Level: 3

General

The **Synthesizer Check** procedure is an important test to verify proper operation of the synthesizer.

Validation of the frequency and phase quality of the synthesizer is important for the accuracy of the slice position.

Measurement

- Verification of the frequency and phase operation of the RF reference signal is performed by using the MR signal itself. After demodulation of the received MR signal, the reference signal is compared to the MR signal in terms of frequency and phase.
  - In one test, the reference signal is varied in frequency over the whole range of the MR system.
  - A second test modifies the phase in fixed intervals.

    » With this test, the entire operating range of the oscillator can be tested. The test is performed for one RX-NCO (Numerically Controlled Oscillator) from group 1 and one RX-NCO from group 2. The two RX-NCOs are validated and can be used for testing the other NCOs.
- Loop-measurements between the tested RX-NCOs and the TX-NCOs approve the correct function of the TX-NCOs.
- In additional steps, all other RX-NCOs are tested against the approved TX-NCOs by testing group 1 RX-NCOs and group 2 RX-NCOs separately.

Evaluation/Results

Results in Report:

- Frequency- and phase-deviations for all used NCOs.

3.4.1.22 Long Term Stability Check

Help ID: 8B498548-2A22-4DDB-8B42-C258D604AAF0
Service Level: 3

General

The **Long Term Stability Check** determines the scanner stability which is important for Functional MR Imaging (fMRI).

The fMRI application visualizes activities in the brain. The patient is exposed to a periodic optical, acoustical or other form of stimulation. During the exposure, a series of MR images is acquired. Subsequently, the images are evaluated for signal intensity changes. The stimulated, activated regions of the brain are localized by their typical, very little variations of signal intensity.

However, these little variations are easily superimposed by instabilities of the MR system or the environment. This QA step makes sure, they are below a specified level.

Typical sources of instabilities are:

- Local B_0-field variation caused by temperature variation in the shim unit (shim iron plates) generates frequency instabilities. The result is a virtual movement of the measured object during acquisitions.
- B1-field variations due to instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, also leading to virtual movement of the object.
- Mechanical vibrations, e.g., by the cold head, produce signal intensity variation caused by the movement of the measured object relative to the magnet isocenter.
- Movement of phantom fluid causes phase errors.

Measurement

A series of 512 measurements (EPI_FID sequence) with 11 slices is performed and repeated once again. The first measurement is performed with a low flip angle of 30 degrees for stability evaluation of the B1-field (i.e., caused by instabilities of the TX chain).

The second measurement uses a higher flip angle of 90(70) degrees to minimize the influences of instable RF but focus on B_0-instabilities and gradient-like interferences.

> Wait at least 15 minutes between positioning the phantom and starting the measurement. Even small movements in the phantom liquid have a negative effect to the measurement results.

Evaluation/Results

The Software places an evaluation ROI in the central slice and performs the following graphical and numerical evaluations:

- Mean values of a 15x15 ROI related to the mean value of all measurements as a function of the image number
- Standard deviation of mean values of a 15x15 ROI (linear and logarithmic plot)
- Measurement protocol values

- Statistics of mean values (for ROI size 15x15 and 25x25):
  - Mean Image Intensity
  - Peak to Peak Variation
  - Relative Peak to Peak Variation
  - Standard Deviation
  - Total Relative Linear Drift
  - Standard Deviation from Linear Fit
- Image quality parameters such as SNR and ghosting

3.4.2 Coil Quality Assurance

Help ID: 3020AFB6-8C57-418D-AFDD-9141803D9BF4
Service Level: 3

The QA procedure **"Coil Check"** checks the function and quality of each local coil individually.

> Before Coil QA, configure new coils in **Admin Portal / Technical Configuration / MR Configuration / Coil Configuration.**

Coils can be also added under **Customer QA**.

3.4.3 Quality Assurance - Expert

Help ID: D2F41F74-14FD-41F9-AD9C-601B24C3C3AE
Service Level: 7

Introduction

For some of the QA procedures, an **Expert Mode** is provided.

Depending on the selected QA step, the user interface shows additional parameters on the right side.

> Expert Mode should be only used by qualified operators.
>
> This mode provides additional settings and configurable parameters which require a more detailed knowledge of the underlying physics.

3.4.3.2
RF Verify

Help ID: C7354DE7-20D8-4639-A1DA-911E628FB7AF
Service Level: 7

Expert Mode

In Expert Mode, the measured TuneUp RF Characteristics MNO for hydrogen or the different x-nuclei frequencies can be verified individually.

Available additional options:

- Disable storage of RF Characteristics after the measurement.
- Modification of maximum RF power allowed.
- Selection of TX channel, RF path and nucleus (according to licenses available).
- Option to switch off the frequency adjustment.
- Option to disable the switching to dummy load.

3.4.3.3
Feedback Loop Calibration Check

Help ID: 4433E809-00E7-430B-9826-64F3AD4DFC7B
Service Level: 7

Expert Mode

i This Expert Mode step is only available for single TX channel systems.

Feedback Loop Calibration Check in **Expert Mode** offers the possibility of either checking the feedback loop phase difference, the feedback loop delay, or both values by a single program call up.

General

The Feedback Loop Calibration Check procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the “tuncal” service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box. Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a “Not Ok”.

Evaluation/Results

Results in Report:

- Delay evaluation
- Dispersion of delays from single BC elements
- BC phase plot

3.4.3.4
Phantom Shim Check

```
Help ID: 897AF1C5-8F5A-4AED-B8F7-65DC5BA57946
Service Level: 7
```

Expert Mode

A coil selection is available in the Expert Mode.

3.4.3.5
Gradient Risetime Check

```
Help ID: 16EA805E-B413-4008-B505-8101C4724E20
Service Level: 7
```

Expert Mode

This procedure determines the minimal rise time or maximum slew rate of the gradient system.

The Expert Mode UI allows the user:

- To select the measurement coil.
- To select the gradient axis.
- To enable additional stress pulses for the measurement.

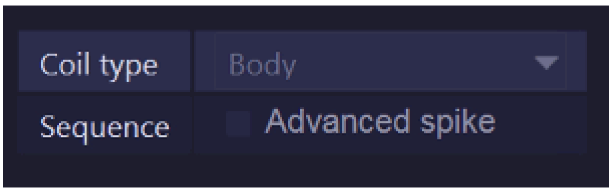
3.4.3.6
Spike Check

```
Help ID: EDA77FDC-B943-4DBD-838C-2D0545E0D309
Service Level: 7
```

Expert Mode

For the **Spike Check** in **Expert Mode**, an additional sequence, **Advanced Spike**, is offered.

Fig. 294: Spike Check - Expert



In some cases, spike artifacts may be visible with customer protocols, but not with the regular spike check. If so, it is advisable to reproduce spikes and troubleshoot with the same setup (coils, protocols, position of patient table, Physiological Measurement Unit, etc.) which is used by the customer.

If the artifacts cannot be reproduced at all and also all typical spike sources (see Services Knowledge Base) are ruled out, contact RSC/ HSC.

NOTICE
The “Advanced Spike” procedure contains a powerful gradient pulse scheme which induces high mechanical stress over a wide frequency range. It shall be therefore used very careful and only when all other means are exhausted. **Risk of hardware damage!** » **DO NOT USE the “Advanced Spike” procedure without guidance from RSC/ HSC!**

3.4.3.7 RF Noise Check

```
Help ID: 3EACA50D-1597-4F09-A708-0AE6878D8DB0
Service Level: 7
```

Expert Mode

The Expert Mode user interface allows selection of a coil.

In addition, the operator can select either select the full frequency range of +/- 250 kHz or focus on a single frequency band to reduce the measurement time for troubleshooting.

3.4.3.8 Synthesizer Check

```
Help ID: 9D7F9FAC-6FC1-475E-8399-8CC21DFA8A4E
Service Level: 7
```

Expert Mode

A coil selection is available in the Expert Mode.

3.4.3.9 Coil Check - Measure

```
Help ID: C7AD132A-5800-41DD-A778-77EB9BA7C47D
Service Level: 7
```

Expert Mode

Coil Check - Measure provides the option to individually select a coil.

In addition, the uniformity evaluation can be enabled or disabled for the acquired QA image.

3.4.3.10 Coil Check - Evaluate series

Help ID: BE987CDD-C2E1-4636-84BD-FED886C9E9D1
Service Level: 7

Expert Mode

Coil Check - Evaluate series provides the option to perform a standard Coil Check evaluation with images that are stored in the database. No measurement is performed in this case, only the appropriate Series ID(s) have to be entered.

In addition, uniformity evaluation for the image(s) can be enabled or disabled.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

3.4.3.11 Field Stability Check - Measure

Help ID: B0C73F74-790C-412B-B992-D1B251EF274D
Service Level: 7

Expert Mode

This procedure checks the stability of the magnetic field and corresponding resonance frequency over time. It is useful to evaluate (for example after magnet ramping) if the field is stable enough for imaging or service procedures.

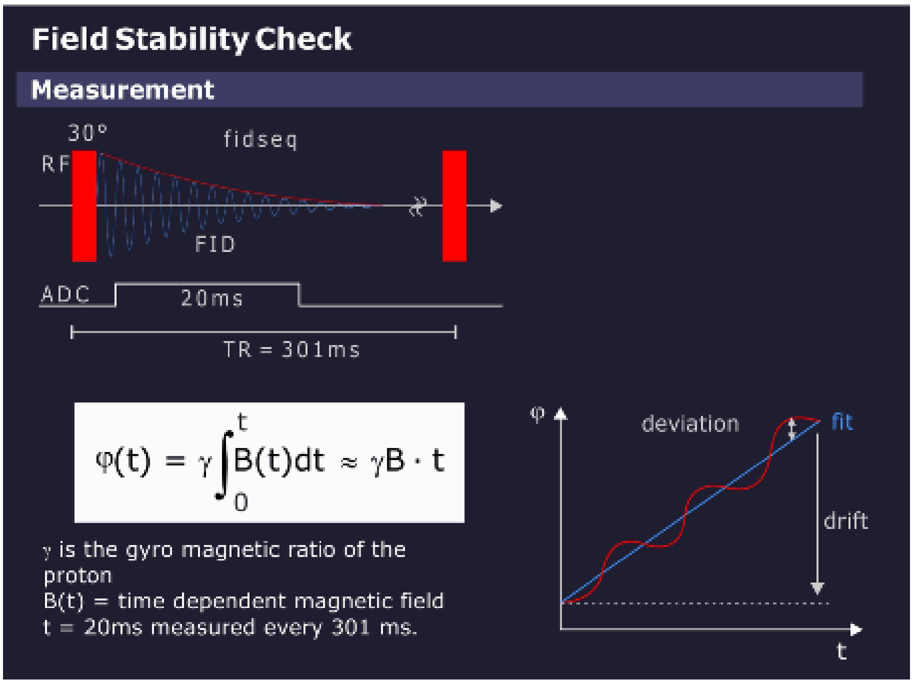
> i The Field Stability Check is only available in Expert Mode.

> i Allow for sufficient magnet stabilization time before performing Field Stability Check.

Measurement

A series of simple FID (Free Induction Decay) measurements is performed, see graphics below.

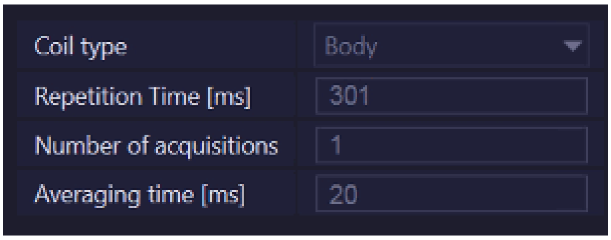
Fig. 295: Field Stability Check - Measurement



A total of 512 FIDs are measured over a 154 second period (repetition time of 301 ms). From each FID, the phase over time is calculated and a linear fit is performed, the slope giving the frequency.

The time range of the fit should be long to get a high accuracy but is usually limited by the length of the FID. A good compromise is 20 ms, which averages over 50 Hz line hum which is usually of no interest in this measurement.

Fig. 296: Field Stability Check - Parameter selection



The following parameter selections are offered:

- Coil type
- Repetition time
- Number of acquisitions
- Averaging time

The measurement time with default parameters is 512 x 301 ms = 154 s.

Evaluation/Results

The result of the field measurement is the field (= frequency) over the measurement time. As a measure of quality, the linear drift over 10 min (= typical measurement time of a long imaging sequence) and the maximum deviation from this linear drift are taken.

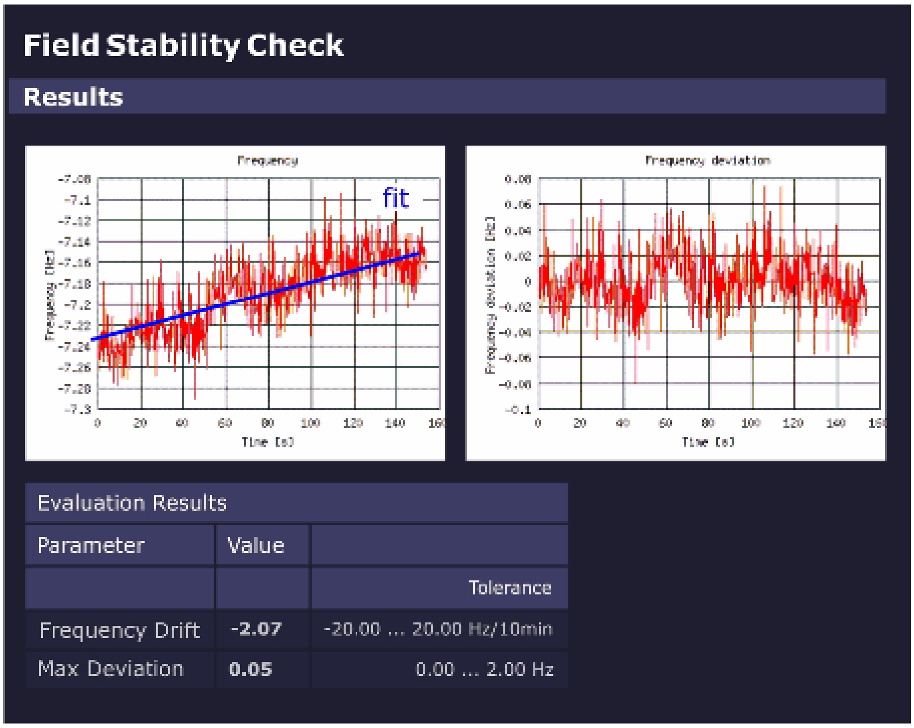
If the selected measurement time is different from 10 min, the drift is calculated for 10 min.

A drift over 10 min should not exceed the pixel bandwidth of a low-bandwidth sequence (e.g., 20 Hz).

The maximum deviation is important for gradient echo sequences to avoid smearing. To allow long echo times, the maximum deviation should be typically less than 2 Hz.

The graphics below shows the frequency variation and the drift-corrected frequency variation across the measurement time.

Fig. 297: Field Stability Check - Results



3.4.3.12 Field Stability Check - Evaluate series

Help ID: 3CD05F02-FE32-4F51-94C3-41BC02E48AEA
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former measurement without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

3.4.3.13
Stability Check - Measure

Help ID: 6CA579E3-92EF-4DEE-8A41-F97C7EF41B1C
Service Level: 7

Expert Mode

Fig. 298: Stability Check Measure - Parameter selection

The Expert Mode function **Stability Check - Measure** evaluates the stability of the MR signal under various sequence conditions.

Different to the Stability Check in the General QA workflow, it allows the following parameter selections:

- Coil type
- Sequence type:
  - Spin-echo
  - Gradient echo (Flash)

  As default, the Spin-echo sequence is selected. This is an exclusive setting so for each **[Go]** in Expert Mode only one sequence selection is possible.
- Repetition time TR and echo time TE

  If the user selects a TR or TE value that is not valid for the selected sequence, an error message is generated when **[Go]** is selected and the valid input ranges are displayed.
- Slice orientation
  - Sagittal
  - Coronal
  - Transverse

  All orientations are selected as default, but a single one, two or all three can be selected.
- Slice shift
  - 0 mm
  - +50 mm

  Both are selected as default. At least one is always selected.

3.4.3.14 Stability Check - Evaluate series

Help ID: 73725B3F-DBC1-40C2-B85C-664143D89A0C
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former measurement without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

3.4.3.15 Long Term Stability Check - Measure

Help ID: DDB58181-821E-4F99-8303-058EFD3295C6
Service Level: 7

Expert Mode

The Expert Mode allows the user:

- to select a coil for measurement (usually the standard Head Coil for the system is used);
- to configure measurement parameters (see below);

Fig. 299: Long Term Stability Check Expert - Parameter selection

Tab. 6 Long Term Stability Check Expert - Parameter description

Measurement Parameter Selections	Description
Coil type	Coil to be used for Long Term Stability measurement (usually standard Head Coil of the system).
Number of slices	▪ Multi slice 3, 11, 15 or 17 slices (default = 17 slices) ▪ Single slice
Slice number	Number of slice to be evaluated (default = central slice).
Repetition time	in ms (default = 1000ms)
Echo time	in ms (default = 30ms)

Measurement Parameter Selections	Description
Flip angle	Flip angle in degrees: ▪ 30 degrees for evaluation of B1-instabilities (i.e. instable transmit components or TX-coil). ▪ for evaluation of other instabilities e.g. B_0, dynamic field perturbations: – 90 degrees (for 1.5 T systems) – 70 degrees (for 3T systems, due to dielectric resonances in the water phantom)
Number of measurements	(default = 512) may be reduced for troubleshooting e.g. 256 to save measurement time. Note: The No. of measurements should not be reduced too much below 256 to maintain a sufficient statistical basis for evaluation.
Number of preparation scans	Preparation scans for the magnetization (default = 5).
Number of warm up repetitions	Warm-up repetitions. Several EPI_FID measurements (default = 60) should be run prior to the “real” measurement to bring the MR system into thermal equilibrium state. No evaluation is performed for the warm-up scans.
Swap orientations	Swap readout and phase-encoding gradient.
Slice orientation	▪ Transversal (= default) ▪ Sagittal ▪ Coronal

Measurement Parameter Selections	Description
Phase correction	▪ None – no correction is applied. ▪ 3-Line – reduction of N/2 ghosts. – the first three echoes (not phase-encoded) are used to correct the further echoes. ▪ 3-Line B0 (= default) – reduction of N/2 - ghost, in addition B_0-field drifts are compensated. – the first three echoes (not phase-encoded) are used to correct the further echoes. ▪ Full – optimal correction for effects accumulating during the echotrain, but time consuming; measurement time is doubled, also for multiple measurements! – there is a prescan with full number of echoes, but not phase-encoded. All echoes in the next scan are corrected by means of that prescan.
Arterial spin labeling evaluation	Switch off/on additional evaluation function for Arterial Spin Labeling (ASL) application.

3.4.3.16 Long Term Stability Check - Evaluate series

Help ID: E13785FD-58F4-47E8-9695-9467FB0975E4
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former Long Term Stability Check without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Enter the Slice number for the evaluation function.
- If applicable, select the Arterial spin labelling evaluation function (select radio button **[enabled]**)

- Press **[Go]** to start the evaluation.

4.1 Maintenance Configuration

Help ID: 60A96B88-4BFA-4FC6-9945-C468E01EC0BB
Service Level: 7

Maintenance Configuration

Within Maintenance Configuration four basic items will be configured:

- Options (optional procedures)
- Condition Based Maintenance
- Measurement Type of Safety Related Tests
- Display of Maintenance Notifications in System Status Overview

Options

The maintenance workflow display the procedures only if the option is installed and configured. Options like MRWP, cooling system, patient table type will be configured in First Installation > MR Configuration > Measurement Configuration. Additional options have to be configured Maintenance Configuration

Following additional option has to be configured:

Configure option	Procedure
Pediatric Coil	Checking warning signs of the pediatric coil (option)
MRWP	Measurement of the MRSC (MRWP) ground wire (annually) Visual inspection of MRWP

Is **Installed** selected, the task will be displayed.

Is **Not Installed** selected, the task procedure will not be displayed.

- Configure the options according to the system configuration.

Condition Based Maintenance

If Condition Based Maintenance is enabled all Maintenance tasks will be controlled by Service Events via GSMS. The maintenance will be initiated by the condition of the system /components and not by a time schedule. The Preventive Maintenance procedures will be faded out.

Prerequisite for Condition Based Maintenance is a daily remote monitoring of the system status. Therefore the data transfer via Siemens Remote Service must be permanently enabled.

If the system has no Siemens Remote Service connection Condition Based Maintenance must be disabled.

Measurement Type of Safety Related Tests

The report of the “initial measurement” of installation serves as a reference document for subsequent repeat measurements. It must therefore be archived by the operator in the System Owner Manual.

First protocol during installation, select > Initial

Repeat protocol during service / maintenance, select > Repeat

Display of Maintenance Notifications in System Status Overview

The System Status Overview displays three status indicators for Maintenance

- Cold Head Maintenance
- Preventive Maintenance
- Safety-related tests

The indicators turn red if the interval for the specific tasks has expired.

This indicator function can be **Enabled** or **Disabled** within Maintenance Configuration.

The default setting of the function is **Disabled**. The three different maintenance indicators will never be displayed.

The maintenance indicators should be **Enabled** only if the customer does not have a service contract and only with the consent of the service manager or technical manager.

4.2
Maintenance Status

Help ID: 63C8D345-4953-4B8D-A403-ABF58F647593
Service Level: 7

Maintenance Status

Maintenance Status displays the status of the Preventive Maintenance and Safety –Related Tests tasks.

Actual Status:

Actual Status displays the current status of the Preventive Maintenance and Safety –Related Tests tasks.

Preview:

Preview provide the possibility to forecast due tasks for a defined period.

The forecast period can be selected with: `Tasks due within next` **nn** `months`

4.3 Safety Related Tests

4.3.1 Measurement Devices

Help ID: A862F1B9-D0C3-44BA-B9FA-4BFCA9D6E6E4
Service Level: 7

Measurement Device

Enter the type and serial numbers of all measurement devices used to install or maintain the system.

Enter the "Next Calibration Due" date, if requested.

If devices have no part numbers enter ‘n.a.’ in the corresponding field.

In case the calibration date is on monthly base, enter the last day of the calibration month in the field ‘Next Calibration Due’.

4.3.2 General Tests

Help ID: 7679F1D0-0F9F-4D5B-90DE-7F866F9F16C0
Service Level: 7

General Information

Safety Information

⚠ WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

General remarks

Based on the risk analysis and the clinical evaluation, the manufacturer is obligated to determine the type and frequency of repair activities as well as the inspection measures required for products to ensure continuously safe and proper operation.

The safety-related tests

- are recurring tests. The scope and schedule of these tests are defined by the manufacturer.
- should determine whether the affected product is in good operating condition, ensuring the safety of patients, users, and other individuals.
- do not replace maintenance activities.

Country-specific laws and regulations have to be taken into consideration, e.g. in Germany the "Medizinprodukte-Betreiberverordung (MPBetreibV)" (Medical Device Operator Regulations), IEC 62353 / DIN EN 62353, etc.

These safety-related tests correspond to the technical safety checks according to "MPBetreibV" (Medical Device Operator Regulations) "Sicherheitstechnische Kontrolle" including IEC 62353 / DIN EN 62353.

The safety-related checks have to be performed **during installation** and at the **intervals** defined in the time schedule.

Documentation

A protocol of the safety-related tests has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter 'n.a.' in the field.

Each test performed successfully has to be confirmed in the test protocol by placing a check mark next to "O.K." Tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with "n.a." If an error occurs which cannot be eliminated, mark the respective test with "not O.K."

The protocol of the installation "First measurement" serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

The MR quality department needs the protocol for legal reasons. Send a copy of the original to:

Tab. 7 Address MR Quality Department

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
MR Quality Department (HC DI MR QT)
P.O. BOX 3260
91050 Erlangen
Germany
mail to: immrqpinstallations.healthcare@siemens.com

The protocols of repeat measurements should be filed in the System Owner Manual as well until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products e.g. in the event damages occur.

Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

Time frame for the safety-related tests

Tab. 8 General tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
System	12	20	
Options and accessories	12	20	
Phantoms	12	10	
Quench tube	12	40	Quench tube outlet sign if missing, sign is part of the sign set 1-7 Adhesive tape for quench tube if missing, p.n. 3866493
Magnet	12	35	Wire or similar to insert into the drain holes, Helium leak detector
Patient fan	12	5	Air filter p.n. 10568825
User documentation	12	5	
Icon and button labels	12	5	
Controlled access area	12	10	Sign if missing, sign is part of the sign set 1-7

Component / activity	Interval (month)	Time required (min)	Required tools and parts
RF room door	12	5	
Warning signs	12	10	
Warning signs for the pediatric coil (optional)	12	5	Sign if missing; refer to SPC
Annual average time including all options: 158 min (2.7 h)			

Tab. 9 Electrical tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the system protective earth resistance	24	30	Protective earth resistance (ground test) meter
Measurement of the fixed patient table protective earth resistance	24	5	Protective earth resistance (ground test) meter
Measurement of the dockable patient table protective earth resistance	12	5	Protective earth resistance (ground test) meter
Measurement of the MRSC protective earth resistance	12	10	Protective earth resistance (ground test) meter
Annual average time including all options: 30 min (0.5 h)			

Tab. 10 Function tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
EPO (if installed)	12	15	
Gradient supervision	12	40	Test spray p.n. 10684091
Patient table	12	35	
Dockable table	12	25	
ERDU	12	15	

Component / activity	Interval (month)	Time required (min)	Required tools and parts
QA	12	70 - 110[1]	QA Phantom set up
Annual average time including all options[2]: 220 min (3.6 h)			

1. Depending on installed options (calculation uses 90 min)
2. Depending on installed options

4.3.2.2 Visual inspection of system

Help ID: 610B250B-E467-4725-9F7F-B7F5560A632C
Service Level: 7

System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
  » The system may not contain damages that impair safety.
  » There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.
- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

4.3.2.3 Inspecting visible cables and cable guides

Help ID: 84378780-0498-4DAB-BB96-BF703F0188DB
Service Level: 7

System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
  » The system may not contain damages that impair safety.
  » There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.
- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

4.3.2.4 Visual inspection of the quench tube

Help ID: DDDD7B14-A3FE-4D41-B7C3-BBBB916DE72F
Service Level: 7

Quench Line OR125

Quench Line - RF Cabin Inspection

Required time: 40 min

Required tools: If missing from the quench line the following spare parts are available from the SPC and must be fitted:-

- Adhesive tape for Quench line (material number **03866493**).
- Quench outlet warning sign.

> Visual inspection of the quench line from the ground only. It is not necessary to remove ceiling tiles etc.

SI Visual inspection of the quench line

- Visually inspect the condition of the quench line.
- Visually check that the warning tape supplied is fitted to the insulation.
- Check that no unauthorized changes to the design of the quench line have been made, both inside or outside the RF room.
- Check the quench line and insulation for damage.
- Check that the connection screws to the magnet elbow assembly are all fitted and tight.
- Check that the isolation bush is fitted between elbow and acoustic decoupler.

Fig. 300: Warning tape

Warning tape is supplied with each magnet.

- Check that the warning tape is attached to the line insulation. (the warning tape should be fitted every 2 to 3 meters along the quench line.

Quench line - Outlet Inspection

⚠ WARNING

Asphyxiation and cold burns

The customer must be notified immediately with corrective actions if the quench line fails to meet 2 critical safety requirements.

» **The quench line outlet is located inside a building with no path for helium gas to escape outside.**

» **The quench outlet has no visible blockage, or restriction of 50 % or greater.**

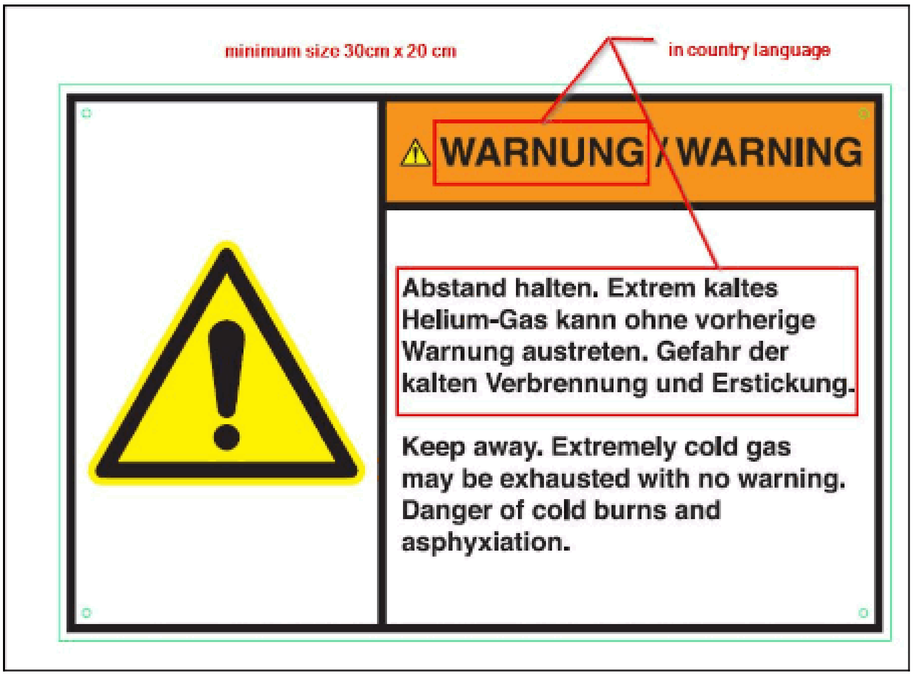
Warning sign requirements

Every outlet must have a warning sign fitted next to the outlet.

If the native country language is not available it is acceptable to source the outlet warning signs locally.

- The warning sign must comply with the ANSI Z535.4 regulations. (See example ANSI regulation below).
- The sign must include a **Warning** symbol and words in the local language and English.
- The minimum size must be 30 cm wide x 20 cm high.

Fig. 301: Outlet warning sign design (example ANS regulation)



- Check that the warning sign is securly fixed and clearly visible next to the outlet of the quench line.

Outlet inspection

CAUTION

Risk of injury

Ensure the outlet can be safely accessed.

» **If the a visual inspection of an outlet located on a rooftop or high on a building cannot be completed safely, access it from ground level.**

- Check the outlet of the quench line for obstructions (e.g. bird's nest).
- Check that a mesh is fitted securely on the quench line outlet.
- Check that water cannot ingress into the outlet (vertical or horizontal).
- Check that the outlet is secuely fixed to the wall.

The outlet cannot be situated where, in case of a quench:

- Helium gas is blown into a public area.
- Helium gas can be drawn into an air inlet.
- Helium gas can enter open windows.

To avoid risk of injury from cold burns and asphyxiation, access to the quench line outlet must be restricted by:

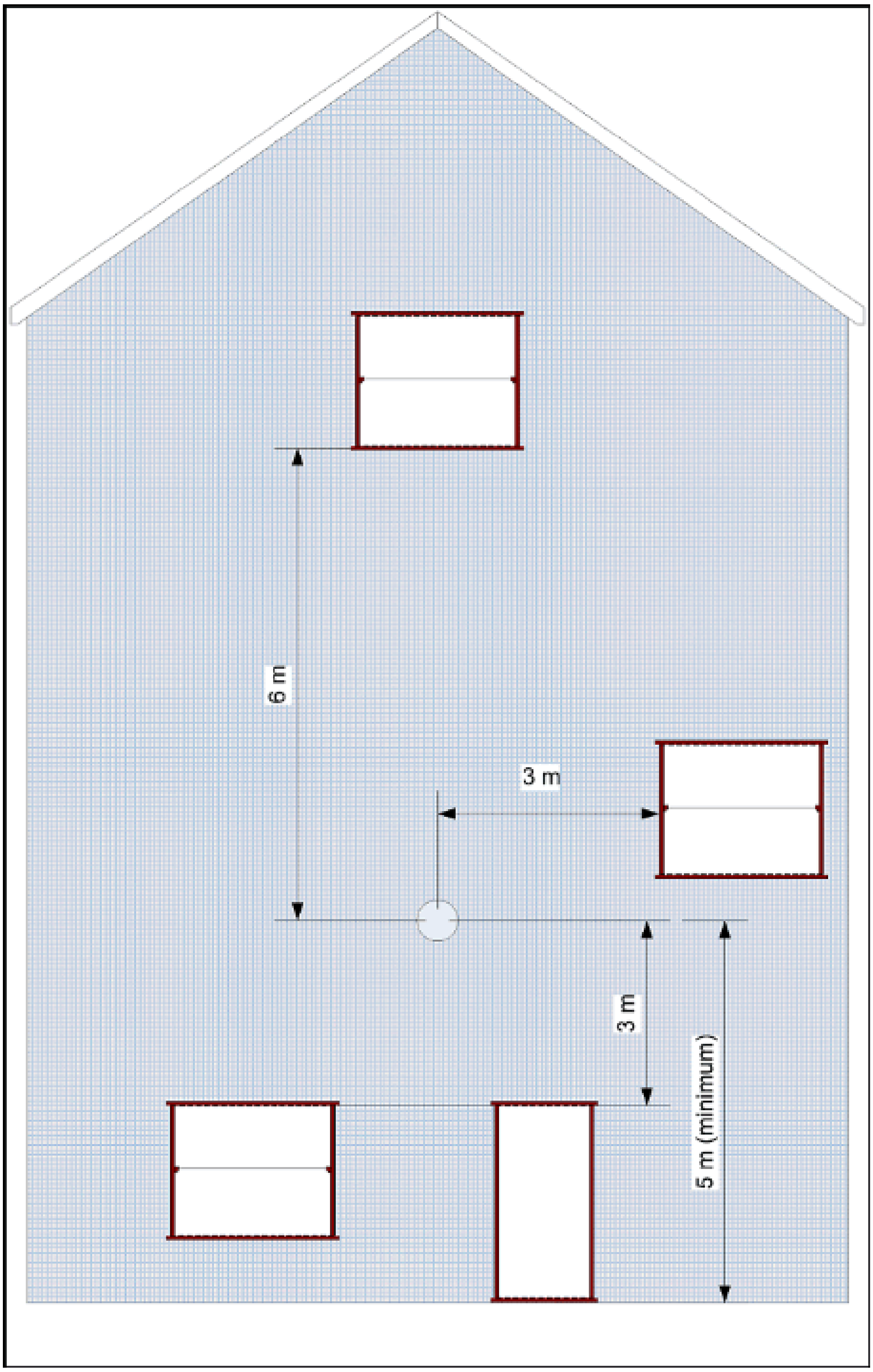
- 3 m on each side and below.

- 6 m vertically above the exit.
- Windows in the restricted access area must be permanently closed. Means of opening the windows must be removed.
- Check that the area next to the outlet has not changed and become a hazard:
  - Air conditioning inlets.
  - Erection (or extensions) of new or temporary buildings, including updates of the building façade.

Outlet Height

- Check that the outlet height is ≥ 5 m for the horizontal outlet designs - Options 1 to 4.
- Check that the outlet height is ≥ 3 m for an angled outlet design.
- A deflector plate can be fitted below outlet options 1, 2 and 3 with the outlet at a height of ≥ 3 m.
- Record the findings in the Protocol.

Fig. 302: Quench outlet - Horizontal safe distance requirements



4.3.2.5 Check for water in the venting

Help ID: A201C5FB-D936-4248-8910-A66300D56E5F
Service Level: 7

Quench Valve

SI Check for water in the venting

Required time: 25 min.

Required tools:

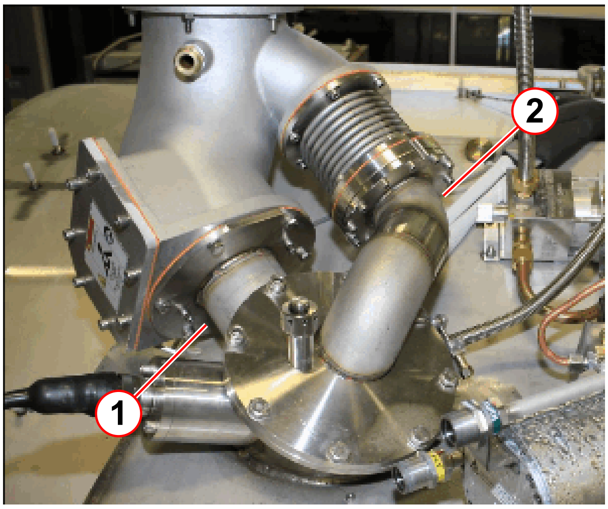
Approximately 10 cm of wire or similar to insert into the drain holes.

Prerequisites:

- Remove the required service access covers. (please refer to instructions > Replacement of Parts > System Covers)
- Set up and secure the service platform to the magnet.

The 2 x drain holes are permanently open. The drain holes allow water condensation to drip from the venting.

Fig. 303: *The position of the two drain holes*



(1) Drain hole at the bottom of the quench valve interface plate
(2) Drain hole at the bottom of the coded vent assembly

Fig. 304: *Quench valve drain hole*

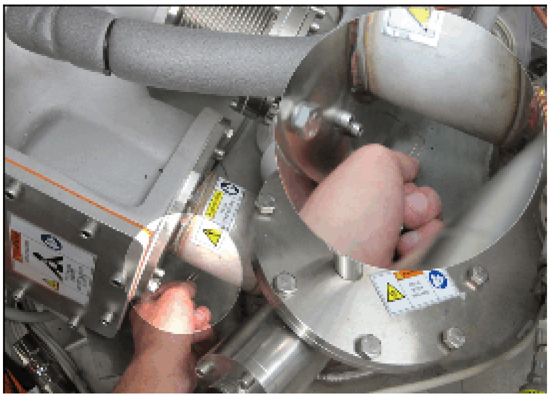
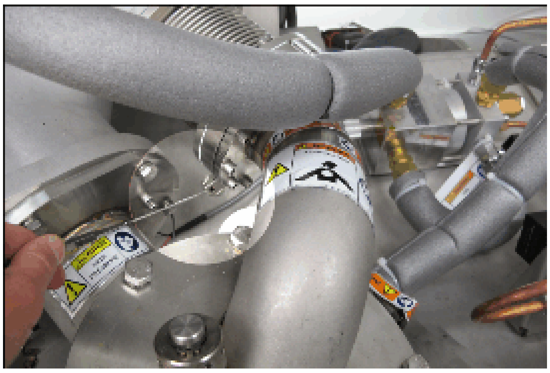


Fig. 305: *Coded vent drain hole*



Quench Valve

- The 2 drain holes are located at the lowest point of the quench manifold and coded vent assembly .

  Pos 1, Location of the Quench valve drain hole.

  Pos 2, Location of the Coded vent drain hole.

  Check to make sure the two holes are **unblocked** by using a piece of wire. e.g. a paper clip

If water is present, determine the source (e.g., damaged or missing protective rain cover, no quench tube insulation or condensation). Advise the customer to initiate corrective action immediately.

4.3.2.6 Check for ice formation at the magnet service turret and venting system

Help ID: 7D899DEE-2BBC-43B9-97B7-9EBE501B5374
Service Level: 7

Ice Formation Check

SI Check for ice formation at the magnet service turret and venting system

Required time: 5 min.

Required tools: N/A

Prerequisites:

- Remove the required service access covers. (please refer to instructions > Replacement of Parts > System Covers)
- Set up and secure the service platform to the magnet.
- The service turret and venting system should be free of ice formation (except after ramping up or filling).

  » Empty out the water from the condensation reservoir located on the floor at the electronics side of the magnet.

4.3.2.7 Magnet system leak check

Help ID: 216EB508-6376-404D-858D-D0187D6FC2B8
Service Level: 7

Turret and Venting Leak Checks

SI Magnet system leak check

Required time: 5 min.

Required tools:

Electronic helium leak detector (material number: 10614850).

Prerequisites: Turret cover is removed (please refer to instructions > Replacement of Parts > System Covers)

Magnet System Leak Check

The magnet is fitted with a recondensing cold head and pressurization heater that maintain the pressure in the helium vessel at a constant **15.5 psi (A)**, providing there are **no leaks** in the venting system.

Large leaks show up as low magnet pressure. Small leaks may not show any change in pressure.

All leaks are measured as a steady loss in the liquid helium level.

Insufficient cooling power of the cold head could also result in helium loss.

If the average power of the heater is = 0 W and the magnet pressure ~ 16 psi (A), check the cold head performance.

- The helium level and magnet pressure must be monitored regularly either by using the **Administration Portal** on the **MRC (MRAWP) Host**.

If the liquid helium level and the magnet pressure have remained constant with no helium loss, no leak check is necessary.

If the magnet has low pressure, or there is evidence of liquid helium loss, then a full turret and venting leak check must be performed.

Leak checking must be performed with an electronic leak detector capable of detecting leaks of a minimum **1 x 10^{-4} mbar l/s.**

Fig. 306: Edwards Leak detector (example)

- Use the Edwards hand-held leak detector (part no **10614850**).

  This is the **recommended equipment** for leaks **greater than 1 x 10^{-4} mbar l/s.**

NOTE: The hand-held device is calibrated in **ml/sec**:-

1 x 10^{-4} mbar l/s **=** 1 x 10^{-4} ml/sec at normal pressure and temperature.

The functionality of the Edwards leak detector can be compromised by magnetic fields. Avoid using the detector near the magnet bore.

All **high-pressure** joints and seals must be checked in case of **low system pressure** and **liquid helium loss.**

Leak check all joints and seals that have been subjected to prolonged low temperatures and subsequent frost build-up. For example, seals which have been cooled to a low temperatures during helium transfers, helium vessel depressurization, or magnet ramping.

Check that the following magnet components are at room temperature before the leak check.

- Cold heads.
- Turret top plate, syphon port and venting.
- 17 psi (A) valve assembly.
- Pressure gauge assembly.

- Connecting pipe couplings.
- Quench valve

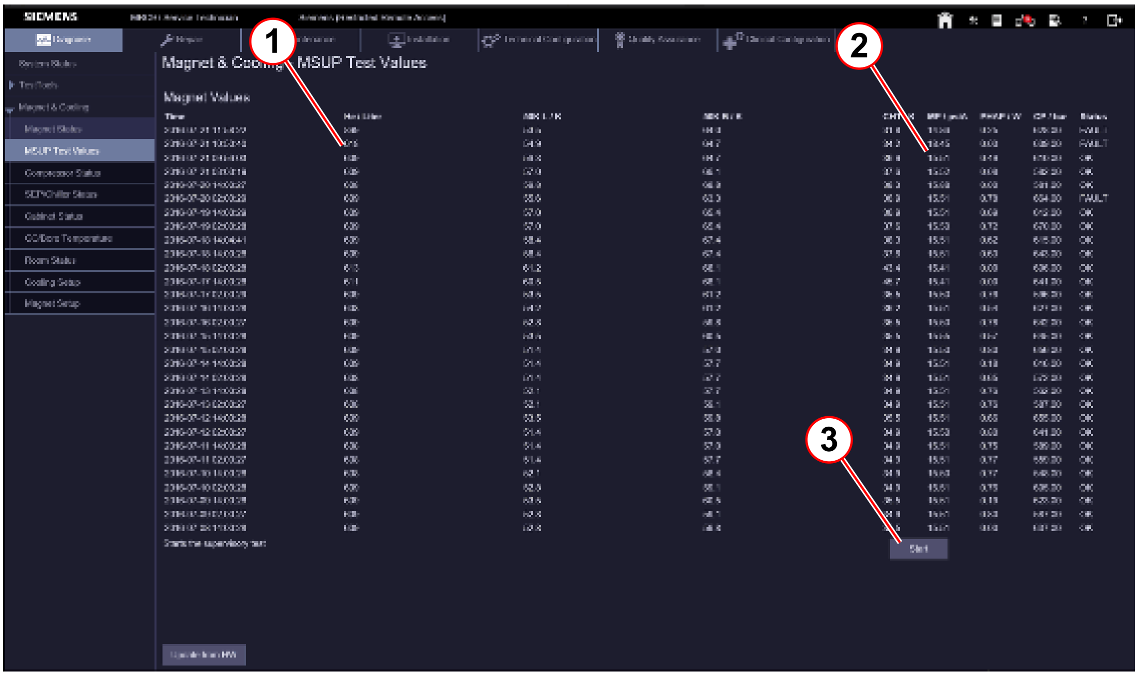
If no leak is found but the system pressure is low, internal components (burst disk ans O-ring) must be checked:

- 17 psi (A) valve assembly.
- Quench valve.
- Coded vent burst disk.

> The magnet must be RUN TO ZERO before checking inside the service hatch or coded burst disk assemblies.

- Check the **liquid helium level** by using the **Administration Portal** on the **MRC (MRAWP) Host**.
  - Access the **Administration Portal** on the MRC.
  - Select the menu item **Diagnose - Magnet & Cooling - Magnet Test Values**.
  - Press the **Start** button to initiate the helium measurement.

Fig. 307: checking liquid helium and magnet pressure levels

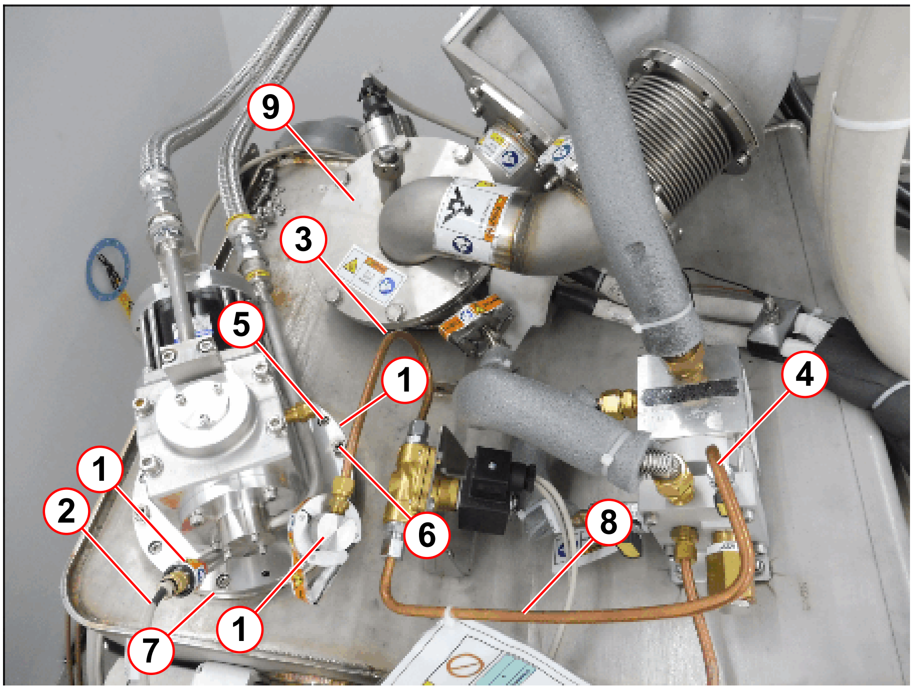


(1) Liquid helium level
(2) Magnet pressure
(3) Start button

- The following checks must be carried out if the magnet shows:
  - a **decreasing** liquid helium level or
  - a magnet pressure **below 15.5 psi (A)**.

Cold Head Checks

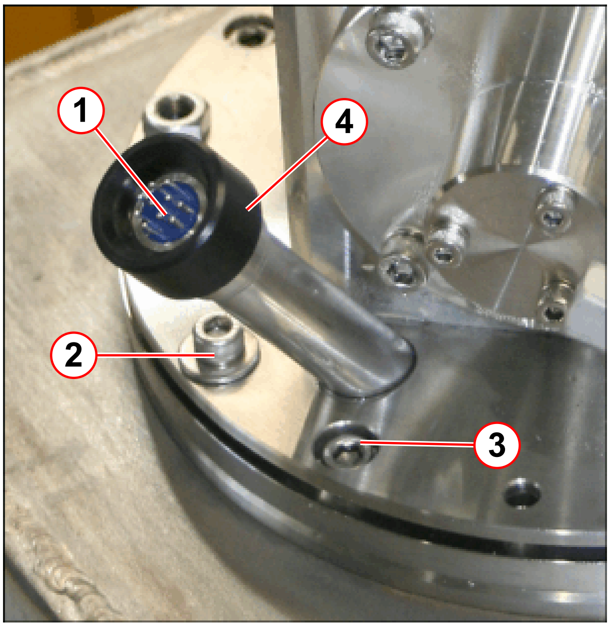
Fig. 308: Leak checking the coldhead seals



(1) Cold head seals, leak check points
(2) Temperature sensor cable
(3) Cold head vent line
(4) Cold head vent line connection at the venting
(5) 8 each, M6 x 16 mm O-ring clamp ring screws, torque to = 8 Nm
(6) 4 each, M6 x 25 mm cold head flange screw,torque to = 4 Nm
(7) O-ring seal
(8) Solinoid to venting pipe
(9) Top plate

- Check the **vent line seals** on the **cold head** and **vent line connection.**

Fig. 309: Cold head 10 pin seal



(1) Leak check point
(2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
(3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
(4) Black sealing cap

- **DO NOT** turn or try to remove the 'black' sealing cap
- Remove the temperature sensor cable (4), with a straight pull away from the black sealing cap.
- Leak check the 10 pin seal connector.
- Immediately reconnect the cable if no leak is found.

The cold head is fitted with **temperature sensors**:

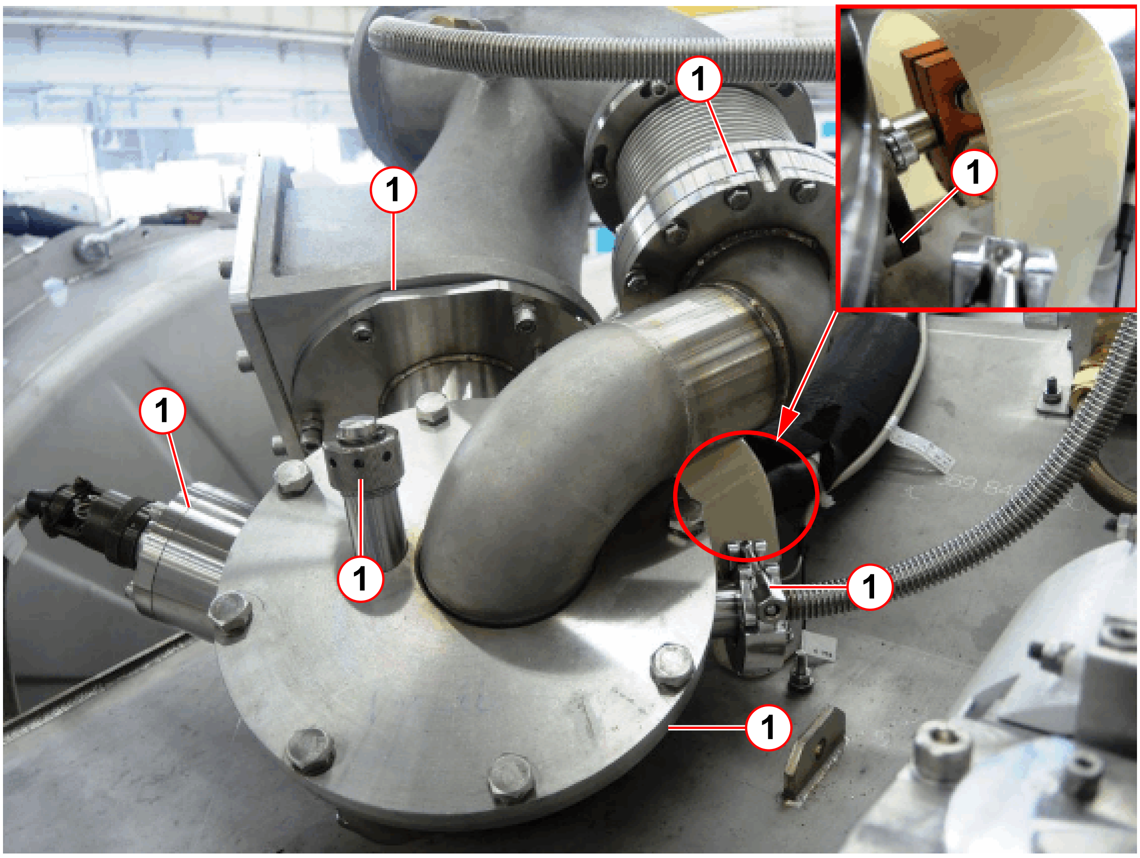
- The feed-through for the wires is sealed with an O-ring.
- Check the feed-through and cold head seals.
- Check the torque of the cold head screws.

If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

Turret and Venting

Fig. 310: Turret and venting - high pressure seals



(1) Turret top plate, syphon port and O-rings

- Check that all clamps on the high-pressure side of the venting are tight. **Do not** try to **overtighten** them - it could cause a leak, or make an existing leak worse.

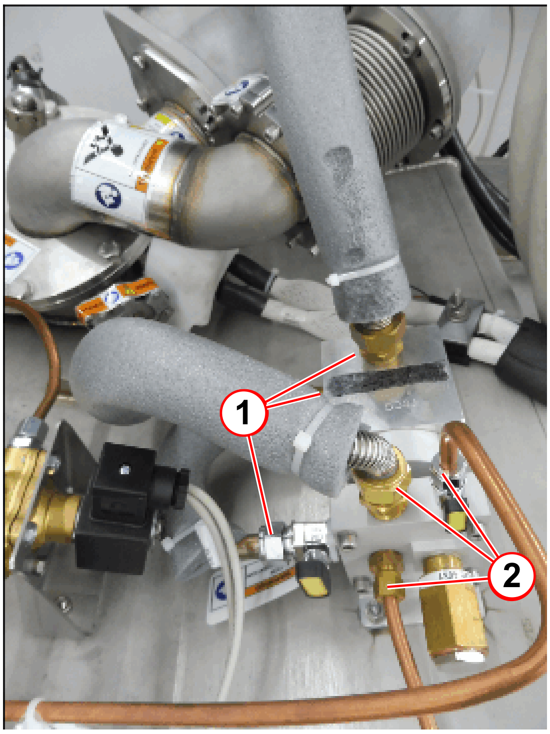
> If no leak is found, but there is a slow, steady liquid helium loss, contact your local Customer Care Center (CCC). Repair requires a specialist.

> Do not use any form of vacuum grease on O-rings and seals.
>
> It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

> Replace a high load seal only with another identical high load seal. High load seals must only be used ONCE for a guaranteed leak tight seal.

Venting Valve Assembly

Fig. 311: Vent valve assembly leak cheakpoints



(1) Low pressure joints
(2) High pressure joints

- Leak check the venting valve **high pressure** assembly.

Pressure Gauge Assembly

Fig. 312: Leak check the pressure line

(1) Leak check points

- Leak check the **high pressure** points.

Contact your local Customer Care Center (CCC) if a leak cannot be resolved.

4.3.2.8 User documentation available and legible

Help ID: F4395154-961A-49BE-888D-C2CBCF8F76E9
Service Level: 7

User documentation

Required time: 5 min

Required tools: n.a.

SI User documentation available and legible

- Ensure that the System Manual is available, legible and valid (e.g. software version).

4.3.2.9 Checking user icons and button labels

Help ID: 5DA2D51C-D4C5-40D0-9F00-703424CE2605
Service Level: 7

User icons, button labels

Required time: 5 min

Required tools: n.a.

SI Checking user icons and button labels

- Inspect the button labeling and user icons for legibility/recognizability (for example "symbol keypad", "**Magnet Stop button**", "Patient Table Stop", etc.).

4.3.2.10 Identification of the controlled access area (0.5 mT area)

Help ID: 92ACE3B3-84A2-415D-8F16-34071EFD7274
Service Level: 7

Identification of the controlled access area

Identification of the controlled access area (0.5 mT area)

Required time: 10 min

Required tools: n.a.

Refer to the Spare Parts Catalogue for the order number for the set of warning signs.

SI Identification of the controlled access area (0.5 mT area)

Fig. 313: *Warning sign "Strong Magnetic Field"*

Fig. 314: *Warning sign " Cardiac Pacemaker"*

Prior to initial Startup, the project manager defines the 0.5 mT area.

- Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone. The signs have to be posted in a way that it is visible even when the door to the controlled area is opened.

SI Checking the warning signs (SIEMENS)

SI Checking the warning signs (customer signs cover all safety aspects as listed on the SIEMENS signs)

The fringe field extends three-dimensional beyond the magnet and can be reduced through shielding. The field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcement in buildings.

Fringe field distribution

> ℹ The field stability over time is < 0.1 ppm/h.

Graphical rendition

Fig. 315: Fringe field Z/Y/X

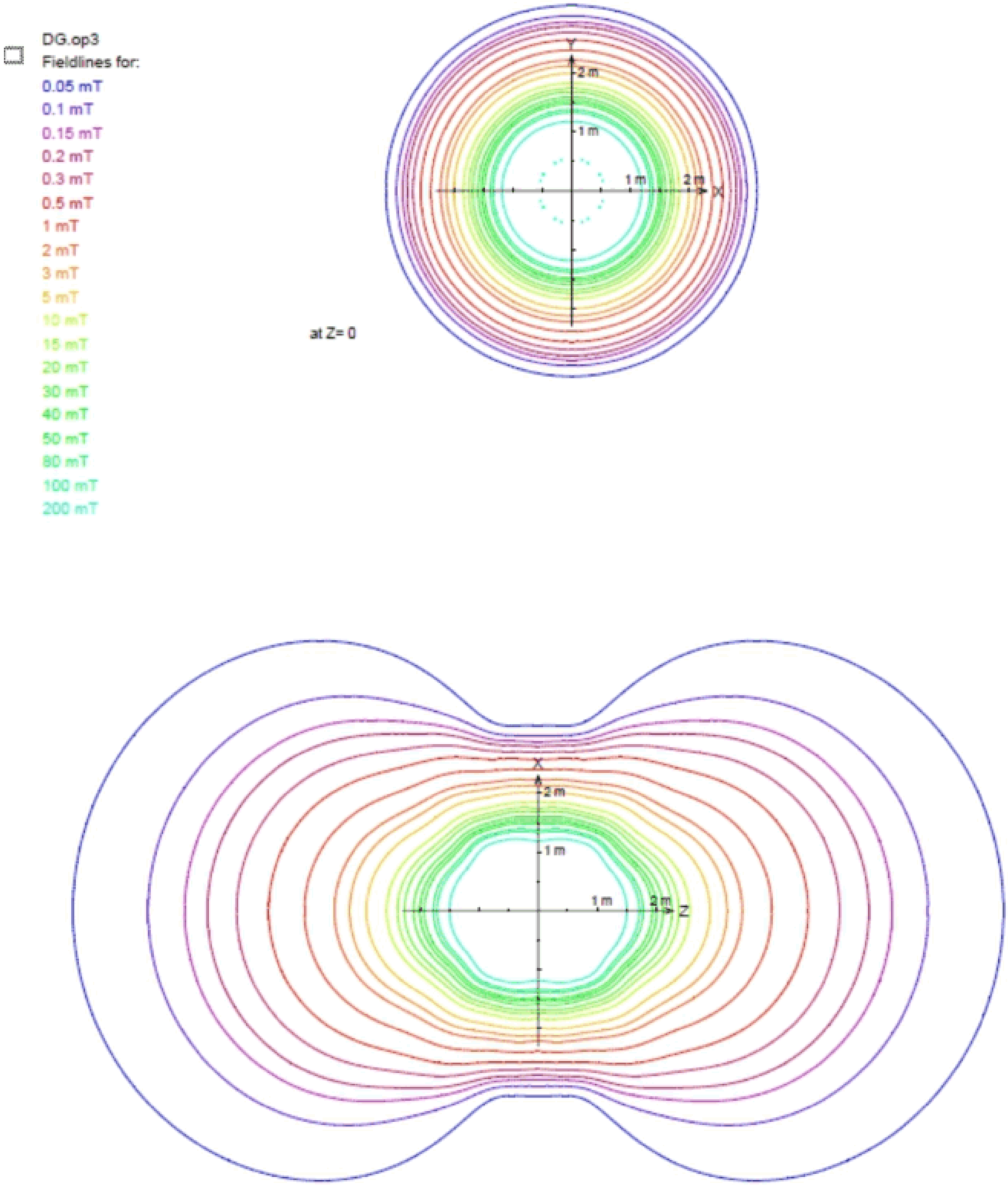


Fig. 316: Fringe field Y/X

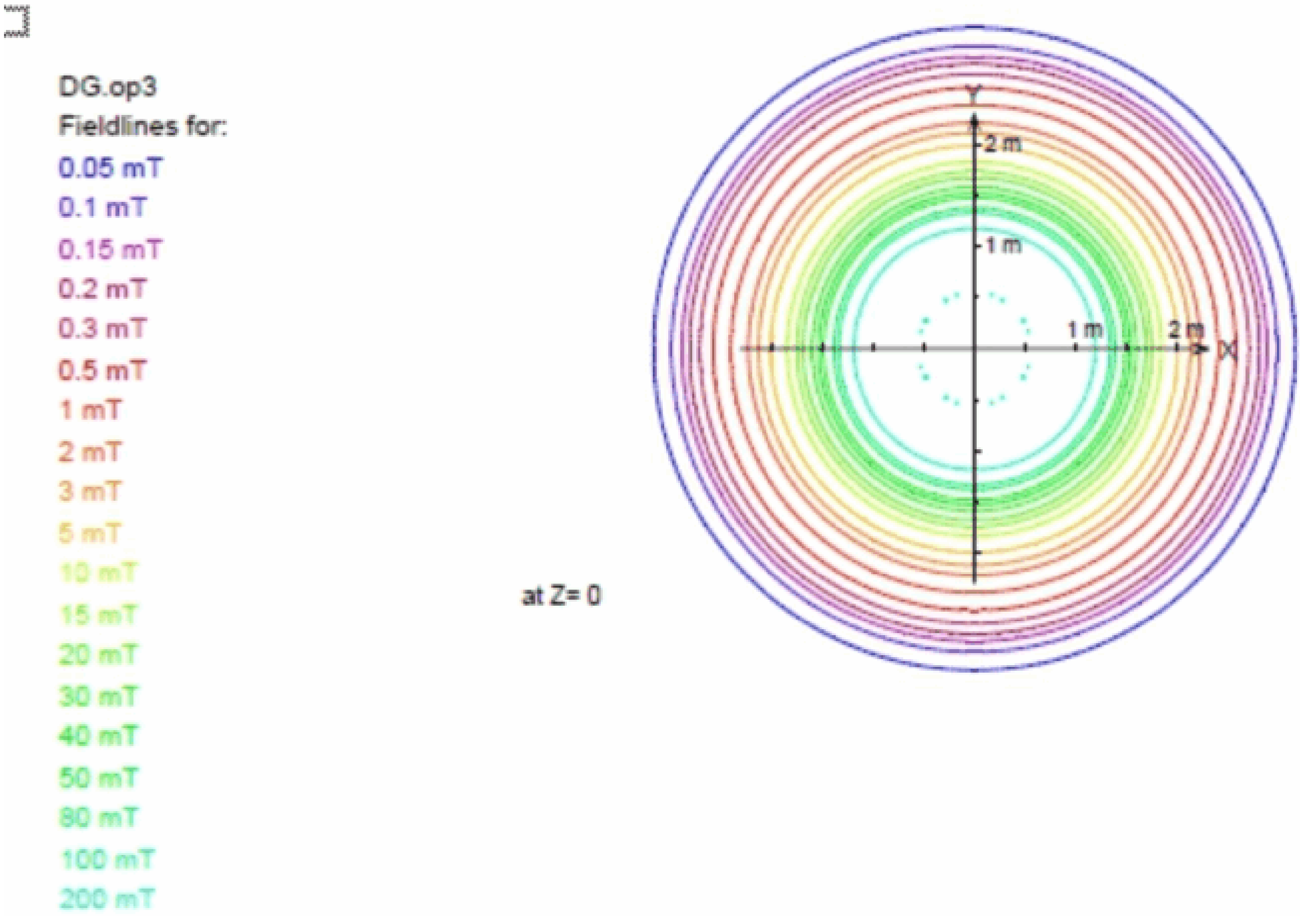
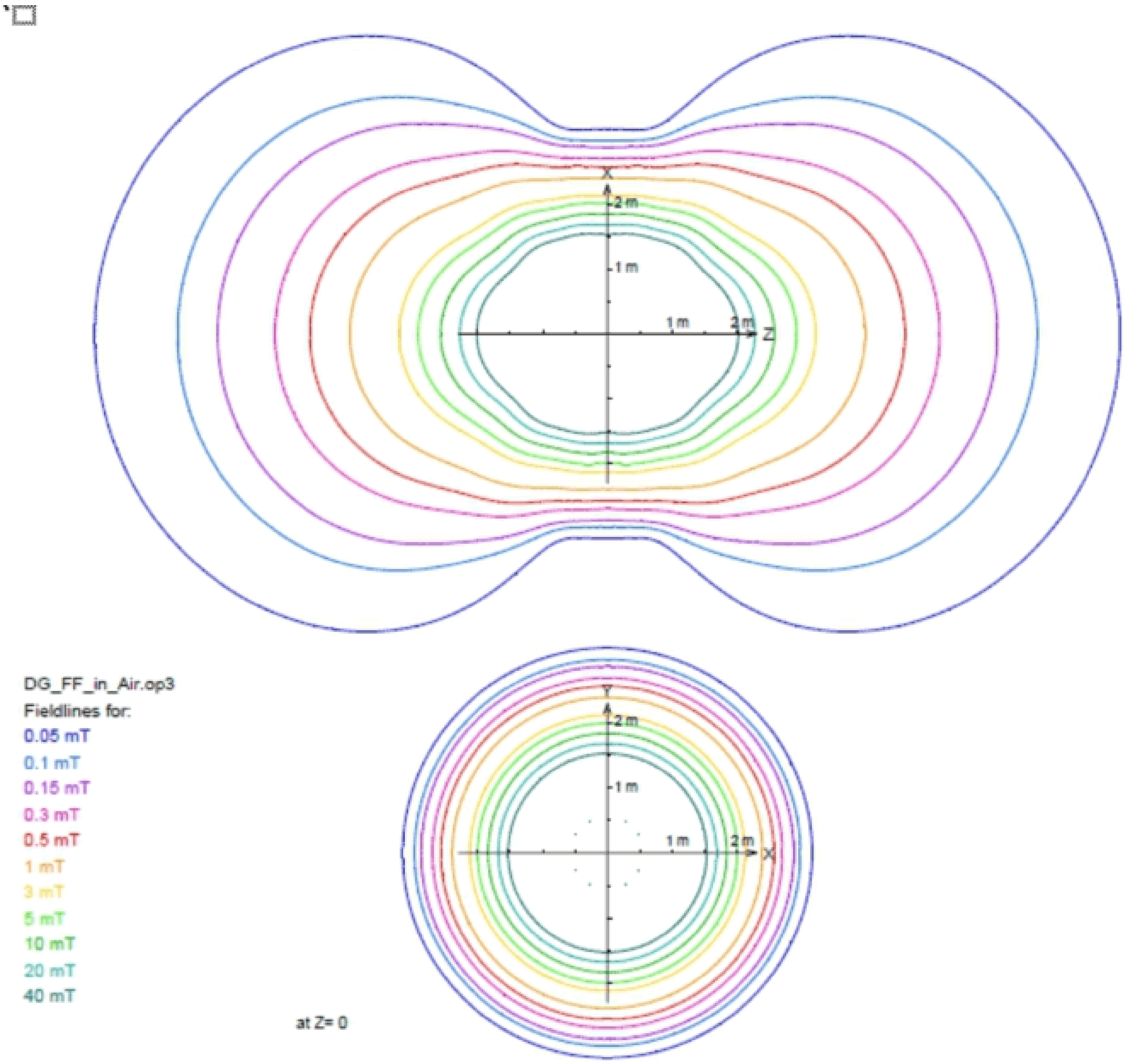


Fig. 317: Fringe field Z/Y



Tablular form

Tab. 11 Fringe field distribution OR125

Fringe field	Distance from the magnet isocenter in the direction of		
	the X-axis in [m]	the Y-axis in [m]	the Z-axis in [m]
200 mT	1.2	1.2	1.5
100 mT	1.3	1.3	1.7
80 mT	1.4	1.4	1.8

50 mT	1.5	1.5	1.9
40 mT	1.5	1.5	2.0
30 mT	1.6	1.6	2.1
20 mT	1.7	1.7	2.3
15 mT	1.7	1.7	2.4
10 mT	1.8	1.8	2.6
5 mT	2.0	2.0	2.9
3 mT	2.16	2.16	3.24
2 mT	2.2	2.2	3.5
1 mT	2.45	2.45	4.05
0.5 mT	**2.65**	**2.65**	**4.65**
0.3 mT	2.8	2.8	5.1
0.2 mT	3.12	3.12	5.72
0.15 mT	3.2	3.2	6.0
0.1 mT	3.8	3.8	6.8
0.05 mT	4.8	4.8	8.1

» The fringe field extends spatially in three dimensions around the magnet isocenter and can be reduced by additional iron shielding. The typical fringe field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcements or other iron masses in buildings.

4.3.2.11 Checking the hearing protection sign

Help ID: 2912BE7A-889E-49FA-8984-92E0D66FA758
Service Level: 7

Hearing protection sign

Required time: 3 min

Required tools or parts: if missing hearing protection sign. The hearing protection sign is part of the warning sign sets, please refer to SPC.

SI Checking the hearing protection sign

- Ensure that the hearing protection sign has been posted at all entrances to the examination room.

Fig. 318: Hearing protection sign

- Install missing signs immediately

4.3.2.12 Checking the warning signs of the pediatric coil (option)

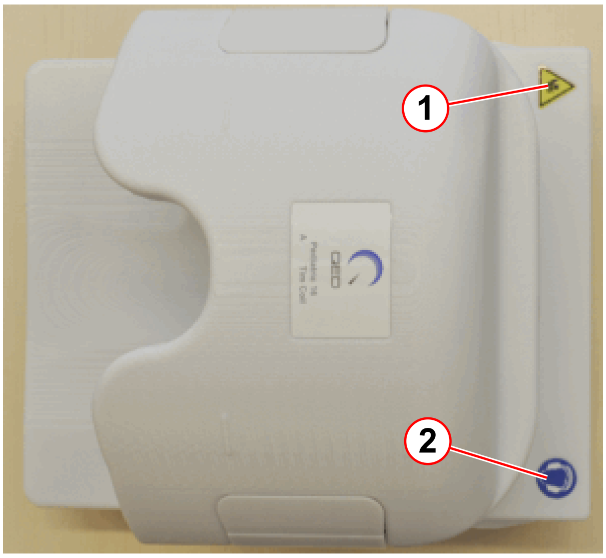
Help ID: B3FE5A47-5F50-474A-B3D0-856E2EEC5F7B
Service Level: 7

Warning Signs of the Pediatric Coil (Option)

Required time: 5 min

Required parts and tools: Sign if missing.

Fig. 319: Pediatric coil



(1) Sign for bruising hazards
(2) Hearing protection sign

SI Checking the warning signs of the pediatric coil

- Ensure that the warning signs (➔ Fig. 320 Page 480), (➔ Fig. 321 Page 480) are affixed to the pediatric coil.
- Replace missing labels immediately.

Fig. 320: Sign for bruising hazards

Fig. 321: Hearing protection sign

4.3.2.13 Checking/Replacing the air filter of the patient fan

Help ID: 715037BE-50B8-4111-8941-E0662B3A42D5
Service Level: 7

Patient fan

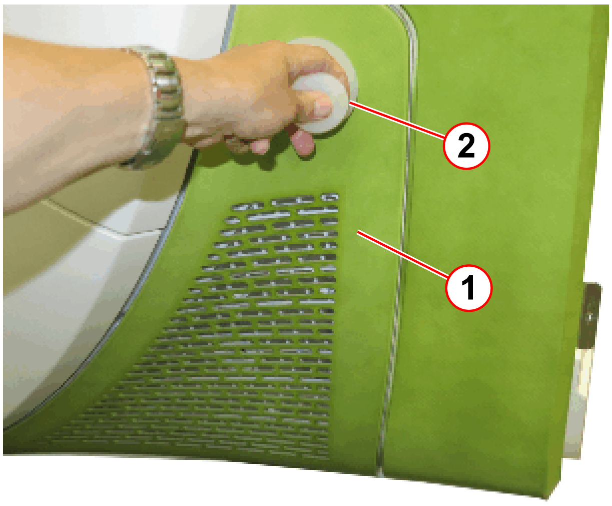
SI Checking the air filter of the patient fan

Required time: 5 min

Required tools and parts: Air filter

- Remove the service cover.

Fig. 322: Air filter of the patient fan



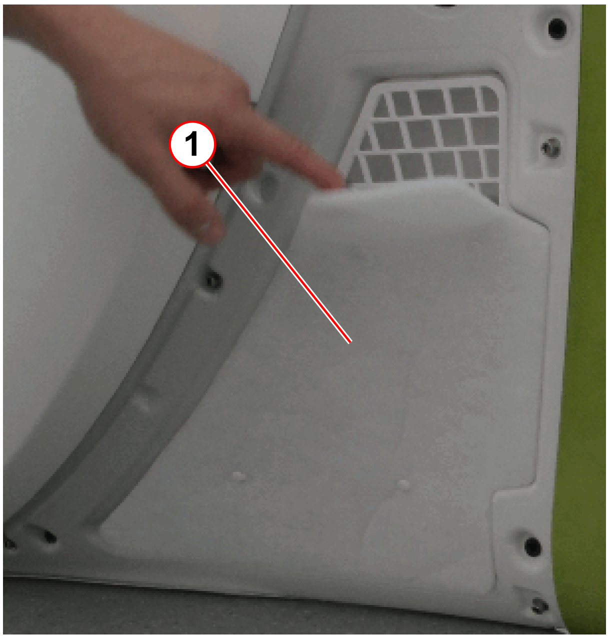
(1) Air filter cover
(2) Suction cup

The air filter cover is fixed with snap locks.

- Attach the suction cup (➔ Fig. 322 Page 481).
- Carefully pull out the cover (➔ 1/Fig. 322 Page 481)(plugged with snap locks).

During first installation of the system it is not necessary to replace the filter.

Fig. 323: Air filter of the patient fan



(1) Filter

- Remove the old filter (➔ 1/Fig. 323 Page 481).
- Install the new air filter.
- Install the air filter cover.

4.3.2.14 Checking the phantoms

Help ID: 97231754-C3EF-4D25-8E86-E53BDA6D0AA9
Service Level: 7

Phantoms

Required time: 10 min

Required tools: n.a.

SI Checking the phantoms

Please read the “MR-Specific Safety Information” prior to inspecting the phantoms.

1. Inspect every phantom used on-site for cracks, leaks, or extensive wear and tear.
   » Replace the phantom if there is any evidence of damage that may affect normal operation.

Defective phantoms must be disposed of. Refer to the “Disposal Instructions”.

2. Inspect every phantom used on-site for excessive air bubbles. Air bubbles can influence your Tune-up results and quality measurements.
3. Ensure that fluid is at an adequate level so that measurement performance is not adversely affected.

Water mixture phantoms with filler cap:

- Refill the phantom, if necessary.
- Phantoms with distilled water mixture - fill with distilled water only.

Water mixture phantoms without filler cap (welded):

- Refilling is not possible. Water mixture phantoms without filler cap must be replaced

Oil phantoms without filler cap (welded):

- Refilling is not possible. Oil phantoms without filler cap must be replaced

Oil phantoms with filler cap:

- Refilling is not possible. Oil phantoms with filler cap must be replaced.

  Do not open the screw cap. It will lead to leakage.

  For example spherical phantom blue D240 p/n 7577666 or p/n 10496685 : The diameter of the air bubble should not exceed 12 cm.

4.3.2.15 Checking the door lock of the RF room

Help ID: F8FC5505-7C34-4697-992F-8812EDC8B0BA
Service Level: 7

RF room door

Required time: 5 min

Required tools: n.a.

Door lock

SI Checking the door lock of the RF room

1. The locking and unlocking mechanism has to function properly.
2. The door handle or knob has to be securely fastened.

> In case of a defect (e.g. defective unlocking mechanism, door handle,), inform the customer to initiate corrective action immediately.

4.3.2.16 Checking the warning signs of the patient table

Help ID: F8D2D88A-F42C-4FED-AC4F-13335A2EB78B
Service Level: 7

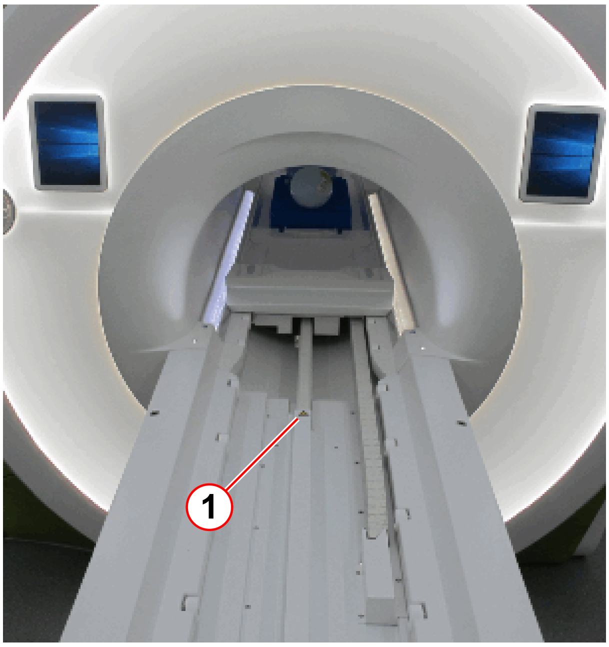
Warning Labels on the Patient Table

Required time: 2 min

Required tools or material: Warning labels, if missing

SI Checking the warning signs on the patient table.

Fig. 324: Patient table



(1) Warning sign bruising hazard

Fig. 325: Warning sign 'bruising hazard'

- Ensure that the warning labels are affixed to the patient table.
  » The labels must be legible.

- Replace missing labels immediately.

4.3.2.17
Visual inspection of Dockable Table

Help ID: 9F3D38A0-145B-41A5-9AA2-47B1D44E421E
Service Level: 7

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

4.3.2.18
Visual inspection of MRWP

Help ID: F7F97A81-696F-4D43-A9F9-56E9061C630E
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

i This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: | Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ____________________

Date: | Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: | Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ____________________

Date: | Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: | Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ______________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ______________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ______________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ______________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

4.3.2.19 Visual inspection of options and accessories

Help ID: 04071914-8D3F-4451-A616-4DA0686A7597
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.

- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

> i This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ______________

Date: Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ______________

Date: Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ______________

Date: Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ______________

Date: Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ________________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ________________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ________________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ________________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ________________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ______________

Date: Name:

4.3.3 Coil Tests

Help ID: 325BD94D-2EA9-4EA8-8E58-A893D618247C
Service Level: 7

Please select the sub-item.

4.3.3.1 Coil Test

Help ID: FFDB078B-9962-4098-8DDF-12398EC04809
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.

- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

> This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ______________

Date: Name:

4.3.4 Electrical Tests

Help ID: 88DDCC38-5176-4E42-919D-65F63A7D3EF4
Service Level: 7

General Information

Safety Information

WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

General remarks

Based on the risk analysis and the clinical evaluation, the manufacturer is obligated to determine the type and frequency of repair activities as well as the inspection measures required for products to ensure continuously safe and proper operation.

The safety-related tests

- are recurring tests. The scope and schedule of these tests are defined by the manufacturer.
- should determine whether the affected product is in good operating condition, ensuring the safety of patients, users, and other individuals.
- do not replace maintenance activities.

Country-specific laws and regulations have to be taken into consideration, e.g. in Germany the "Medizinprodukte-Betreiberverordung (MPBetreibV)" (Medical Device Operator Regulations), IEC 62353 / DIN EN 62353, etc.

These safety-related tests correspond to the technical safety checks according to "MPBetreibV" (Medical Device Operator Regulations) "Sicherheitstechnische Kontrolle" including IEC 62353 / DIN EN 62353.

The safety-related checks have to be performed **during installation** and at the **intervals** defined in the time schedule.

Documentation

A protocol of the safety-related tests has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter 'n.a.' in the field.

Each test performed successfully has to be confirmed in the test protocol by placing a check mark next to "O.K." Tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with "n.a." If an error occurs which cannot be eliminated, mark the respective test with "not O.K."

The protocol of the installation "First measurement" serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

The MR quality department needs the protocol for legal reasons. Send a copy of the original to:

Tab. 12 Address MR Quality Department

Siemens Healthineers Headquarters

Siemens Healthcare GmbH

MR Quality Department (HC DI MR QT)

P.O. BOX 3260

91050 Erlangen

Germany

mail to: immrqpinstallations.healthcare@siemens.com

The protocols of repeat measurements should be filed in the System Owner Manual as well until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products e.g. in the event damages occur.

Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check

Abbr.	Explanation
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

Time frame for the safety-related tests

Tab. 13 General tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
System	12	20	
Options and accessories	12	20	
Phantoms	12	10	
Quench tube	12	40	Quench tube outlet sign if missing, sign is part of the sign set 1-7 Adhesive tape for quench tube if missing, p.n. 3866493
Magnet	12	35	Wire or similar to insert into the drain holes, Helium leak detector
Patient fan	12	5	Air filter p.n. 10568825
User documentation	12	5	
Icon and button labels	12	5	
Controlled access area	12	10	Sign if missing, sign is part of the sign set 1-7
RF room door	12	5	
Warning signs	12	10	
Warning signs for the pediatric coil (optional)	12	5	Sign if missing; refer to SPC
Annual average time including all options: 158 min (2.7 h)			

Tab. 14 Electrical tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the system protective earth resistance	24	30	Protective earth resistance (ground test) meter
Measurement of the fixed patient table protective earth resistance	24	5	Protective earth resistance (ground test) meter
Measurement of the dockable patient table protective earth resistance	12	5	Protective earth resistance (ground test) meter
Measurement of the MRSC protective earth resistance	12	10	Protective earth resistance (ground test) meter
Annual average time including all options: 30 min (0.5 h)			

Tab. 15 Function tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
EPO (if installed)	12	15	
Gradient supervision	12	40	Test spray p.n. 10684091
Patient table	12	35	
Dockable table	12	25	
ERDU	12	15	
QA	12	70 - 110[1]	QA Phantom set up
Annual average time including all options[2]: 220 min (3.6 h)			

1. Depending on installed options (calculation uses 90 min)
2. Depending on installed options

4.3.4.2 Measurement of System Protective Earth Resistance

Help ID: EA79B57C-1247-4A4E-86CA-775A93D56F9B
Service Level: 7

Measurement of the system Protective Earth Resistance (every 2 years)

Required time: 30 min

Required tools: Protective Earth Resistance meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective earth resistance test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

WARNING

The protective earth resistance meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective earth resistance meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

» **Perform protective earth resistance tests using measurement cables of a sufficient length or after the magnet has been ramped down.**

» The protective earth resistance must not exceed **0.3 ohms**.

- Use the **protective earth resistance meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI EPC MR electronic cabinet - ground bolt - line distribution box (limit value ≤ 0.3 Ω)

Value:

SI RF filter plate - ground bolt (limit value ≤ 0.3 Ω)

Value:

SI Magnet cold head compressor - housing (limit value ≤ 0.3 Ω)

Value:

SI Option SEP - metal frame (limit value ≤ 0.3 Ω)

Value:

SI Option IFP - metal frame (limit value ≤ 0.3 Ω)

Value:

SI MRC (MRAWP) - Host - housing screw (limit value ≤ 0.3 Ω)

Value:

SI MRC (MRAWP) - TFT monitor - housing screw at the bottom (limit value ≤ 0.3 Ω)

Value:

SI Multiple socket for the console components (Line Distribution 1 – UPS) - ground screw (limit value ≤ 0.3 Ω)

Value:

SI Multiple socket for the console components (Line Distribution 2 – NO UPS) ground screw (limit value ≤ 0.3 Ω)

Value:

SI RF components on the magnet - RFCEL housing (limit value ≤ 0.3 Ω)

Value:

SI RF components on the magnet - TX-Box housing (limit value ≤ 0.3 Ω)

Value:

SI Magnet - magnet housing (limit value ≤ 0.3 Ω)

Value:

- All other options such as patient monitor, in-room MRC, Digicam, etc.

> Protective earth resistance measurement for PAMO5 CCD-Vario camera is not necessary as it is supplied with low voltage.

SI Option 1 (limit value ≤ 0.3 Ω)

Description:	
Material No.:	
Serial No.:	

Value:

SI Option 2 (limit value ≤ 0.3 Ω)

Description:	
Material No.:	
Serial No.:	

Value:

Enter N.A. in the measured value field if the component/option is not present.

- Enter the values measured for each component in the protocol.

WARNING

Exceeding threshold values.

In the case of an electrical fault, exceeding a threshold value may result in death or serious injury.

» **Test to determine what exceeds the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Work steps of this section were performed by:

		Signature:	
Date:		Name:	

4.3.4.3 Measurement of fixed Patient Table Protective Earth Resistance

Help ID: F1F9635B-52D2-467B-8653-5A705B9EDE15
Service Level: 7

Measurement of the Fixed Patient Table Protective Earth Resistance (every 2 years)

Required time: 5 min

Required tools: Protective Earth Resistance meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective earth resistance test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

WARNING

The protective earth resistance meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective earth resistance meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

» **Perform protective earth resistance tests using measurement cables of sufficient length or after the magnet has been ramped down.**

- Use the **protective earth resistance meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI Installed patient table-metal frame (limit value ≤ 0.3 Ω)

Value:

» The protective earth resistance must not exceed **0.3 ohms**.

- Enter the values measured for each component in the safety-related tests protocol.

WARNING

Exceeding threshold values.

In case of electrical faults, exceeded threshold values may result in death or serious injury.

» **Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Work steps of this section were performed by:

Signature: ____________

Date: Name:

4.3.4.4 Measurement of the Dockable Patient Table Protective Earth Resistance

Help ID: F210BD5D-ED5A-4926-986C-29E3BA11471E
Service Level: 7

Measurement of the Dockable Patient Table Protective Earth Resistance (annually)

Required time: 5 min

Required tools: Protective Earth Resistance meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective earth resistance test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

WARNING

The protective earth resistance meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective earth resistance meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

» **Perform protective earth resistance tests using measurement cables of sufficient length or after the magnet has been ramped down.**

- Use the **protective earth resistance meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI Dockable patient table - metal frame (limit value ≤ 0.3 Ω)

Value:

» The protective earth resistance must not exceed **0.3 ohms**.

- Enter the values measured for each component in the safety-related tests protocol.

WARNING

Exceeding threshold values.

In case of electrical faults, exceeded threshold values may result in death or serious injury.

» **Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Work steps of this section were performed by:

Signature: ____________

Date: Name:

4.3.4.5 Measurement of MRWP (Option) Protective Earth Resistance

```
Help ID: 62AC8880-E3AC-4221-BFF1-B3814F0040A0
Service Level: 7
```

Measurement of the MRSC (MRWP) Protective Earth Resistance (annually)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

Required time: 10 min

Required tools: Protective Earth Resistance meter

Use the protective earth resistance meter to measure the ohmic resistance between the ground pin at the plug of the power distribution for the MRSC components and a non-insulated metallic part of the:

SI MRSC (MRWP) Host - housing screw (limit value ≤ 0.3 Ω)

Value:

SI MRSC (MRWP) TFT monitor - housing screw at the bottom (limit value ≤ 0.3 Ω)

Value:

SI Option 1 (limit value ≤ 0.3 Ω)

Description:	
Material No.:	
Serial No.:	

Value:

WARNING

Exceeding threshold values.

In case of electrical faults, exceeded threshold values may result in death or serious injury.

» **Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

» The protective earth resistance must not exceed **0.3 ohms**.

- Enter the values measured for each component in the protocol.

Enter N.A. in the measured value field if the component/option is not present.

4.3.5 Function Tests

Help ID: 6BBDC5B1-C92A-461D-8C01-E2FDB70596D1
Service Level: 7

General Information

Safety Information

WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

General remarks

Based on the risk analysis and the clinical evaluation, the manufacturer is obligated to determine the type and frequency of repair activities as well as the inspection measures required for products to ensure continuously safe and proper operation.

The safety-related tests

- are recurring tests. The scope and schedule of these tests are defined by the manufacturer.
- should determine whether the affected product is in good operating condition, ensuring the safety of patients, users, and other individuals.
- do not replace maintenance activities.

Country-specific laws and regulations have to be taken into consideration, e.g. in Germany the “Medizinprodukte-Betreiberverordung (MPBetreibV)” (Medical Device Operator Regulations), IEC 62353 / DIN EN 62353, etc.

These safety-related tests correspond to the technical safety checks according to “MPBetreibV” (Medical Device Operator Regulations) “Sicherheitstechnische Kontrolle” including IEC 62353 / DIN EN 62353.

The safety-related checks have to be performed **during installation** and at the **intervals** defined in the time schedule.

Documentation

A protocol of the safety-related tests has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter ’n.a.’ in the field.

Each test performed successfully has to be confirmed in the test protocol by placing a check mark next to “O.K.” Tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with “n.a.” If an error occurs which cannot be eliminated, mark the respective test with “not O.K.”

The protocol of the installation “First measurement” serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

The MR quality department needs the protocol for legal reasons. Send a copy of the original to:

Tab. 16 Address MR Quality Department

Siemens Healthineers Headquarters

Siemens Healthcare GmbH

MR Quality Department (HC DI MR QT)

P.O. BOX 3260

91050 Erlangen

Germany

mail to: immrqpinstallations.healthcare@siemens.com

The protocols of repeat measurements should be filed in the System Owner Manual as well until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products e.g. in the event damages occur.

Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image

Abbr.	Explanation
QSQ	Quality Check System
SW	Software Maintenance

Time frame for the safety-related tests

Tab. 17 General tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
System	12	20	
Options and accessories	12	20	
Phantoms	12	10	
Quench tube	12	40	Quench tube outlet sign if missing, sign is part of the sign set 1-7 Adhesive tape for quench tube if missing, p.n. 3866493
Magnet	12	35	Wire or similar to insert into the drain holes, Helium leak detector
Patient fan	12	5	Air filter p.n. 10568825
User documentation	12	5	
Icon and button labels	12	5	
Controlled access area	12	10	Sign if missing, sign is part of the sign set 1-7
RF room door	12	5	
Warning signs	12	10	
Warning signs for the pediatric coil (optional)	12	5	Sign if missing; refer to SPC
Annual average time including all options: 158 min (2.7 h)			

Tab. 18 Electrical tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the system protective earth resistance	24	30	Protective earth resistance (ground test) meter

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the fixed patient table protective earth resistance	24	5	Protective earth resistance (ground test) meter
Measurement of the dockable patient table protective earth resistance	12	5	Protective earth resistance (ground test) meter
Measurement of the MRSC protective earth resistance	12	10	Protective earth resistance (ground test) meter
Annual average time including all options: 30 min (0.5 h)			

Tab. 19 Function tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
EPO (if installed)	12	15	
Gradient supervision	12	40	Test spray p.n. 10684091
Patient table	12	35	
Dockable table	12	25	
ERDU	12	15	
QA	12	70 - 110[1]	QA Phantom set up
Annual average time including all options[2]: 220 min (3.6 h)			

1. Depending on installed options (calculation uses 90 min)
2. Depending on installed options

4.3.5.2 Checking the emergency power off EPO buttons

Help ID: 0DF6629A-7484-40B5-9484-A1EEAE725C5A
Service Level: 7

Emergency power off switches

Required time: 15 min

Required tools: n.a.

SI Checking the emergency power off switches EPO

Prerequisites:

The system contactor in the EPC and GPA cabinet is switched on.

If a UPS (uninterruptible power supply) is installed, the UPS display shows `O/P: 230 V`. Press the UPS enter button (→ 1/Fig. 326 Page 516) to open display.

The system is not booted (system shutdown or press **F12** during boot).

- Press the emergency power off switch (EPO).
- Check that the **EPC** cabinet is de-energized.
  - » LEDs on the EPC cabinets are off.

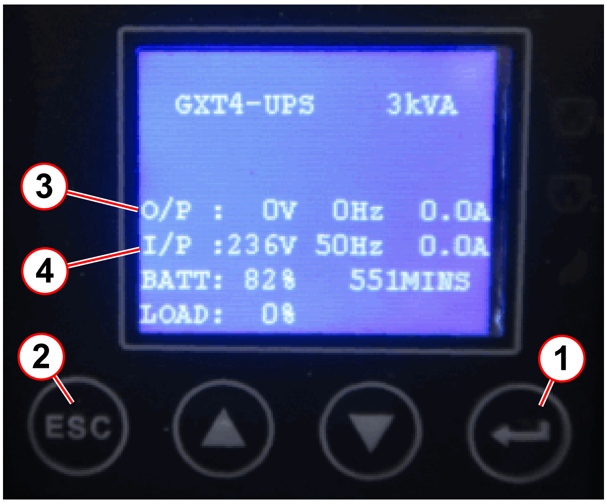
SI EPC cabinet is de-energized.

- Check that the **GPA** cabinet is de-energized.
  - » LEDs on the GPA cabinets are off.

SI GPA cabinet is de-energized.

- If a **UPS** is installed, the UPS output voltage must be switched off.
  - » The UPS display shows `UPS Shutdown Command Received` for few seconds and then `O/P: 0 V` (output voltage is 0V (→ 3/Fig. 326 Page 516)).

Fig. 326: UPS display



(1) Enter button
(2) Escape button
(3) O/P output power
(4) I/P input power

SI Option UPS: display shows output voltage = 0 V

- Check that the **“Line”** LED at the alarm box is off.
  - » If the RF room has an electric or pneumatic door, it must be possible to open it in the event of a power failure.
- Check whether the RF room door can be opened when the system is “powered off.”

Release the EPO switch and reactivate the system by switching on the main power.

After release of the EPO switch, the UPS has to be switched on manually.

- Switch on the UPS: Navigate in the UPS display menu to switch on output power > press **enter** button > **3_CONTROL** > **1_TURN ON & OFF** > **TURN ON UPS** > **Do you want to turn on UPS** > **YES** > **enter**

  » UPS output voltage is on (`O/P: 230 V`)

- **Perform this test on all emergency power-off switches in the equipment room, control room, and the examination room**.

4.3.5.3
Checking the patient table stop function

Help ID: 47B102C7-B251-47D7-9481-0C99802B3920
Service Level: 7

STOP function

Required time: 5 min

Required tools: n.a.

Prerequisites: Table is in the home position.

SI Checking the patient table stop function

- Press the **center move** button.
- While the table is moving toward the isocenter, press the **STOP** button on the right side of the patient table.

  » The patient table stops; the Dot display indicates the table stop.

Fig. 327: Patient table

(1) STOP button

- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by pulling the **STOP** button.
- Then press the up and down buttons simultaneously on the control panel to reset the table.
- Press the center move button again to continue table movement.
- While the table is moving toward the isocenter, press the **STOP** button on the **left** side of the patient table

  » The patient table stops; the display indicates the table stop.

- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by pulling the **STOP** button.
- Then press the up and down buttons simultaneously on the control panel to reset the table.

Fig. 328: Dockable table, control unit

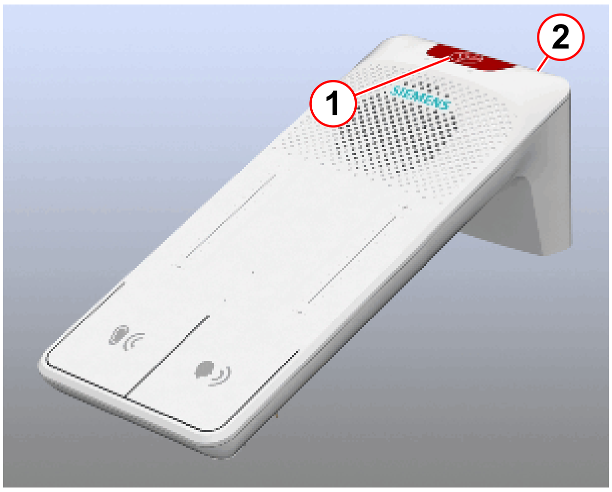
(1) STOP button

- Perform the patient table stop procedure with the **STOP** button for the dockable table **control unit** (➔ 1/Fig. 328 Page 518), if present.
- Perform the patient table stop procedure with the **STOP** button for the **third control panel**, if present.

STOP button on the **intercom**

- Press the **center move** button on the control panel or on the console.
- While the table is moving toward the isocenter, press the **STOP** button on the intercom (➔ 1/Fig. 329 Page 518).

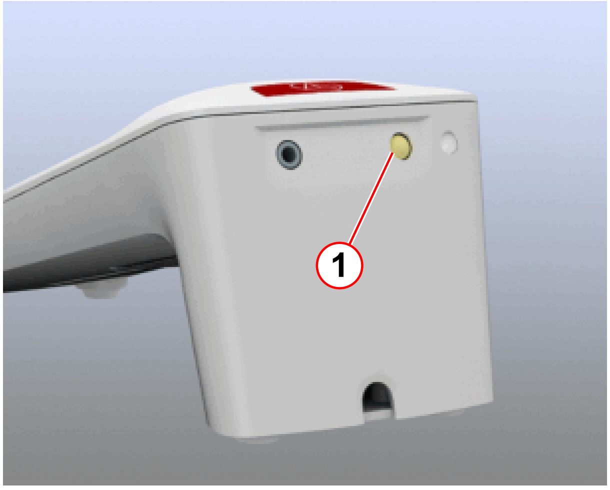
Fig. 329: Intercom



(1) Patient table stop button
(2) Reset patient table stop button

- Make sure that table movement is not possible.
  » On the MRC, the pop-up window opens < `e.g. Sequence was stopped ...`, press **OK**.
- Deactivate the table's **STOP** button by pressing the **reset** button.

Fig. 330: Intercom



(1) Reset patient table stop button

- Then, press the up and down buttons simultaneously on the control panel magnet to reset the table.

4.3.5.4 Checking the emergency release of the patient table

```
Help ID: 113B2538-E842-4718-97F8-01B047E9F761
Service Level: 7
```

Emergency release

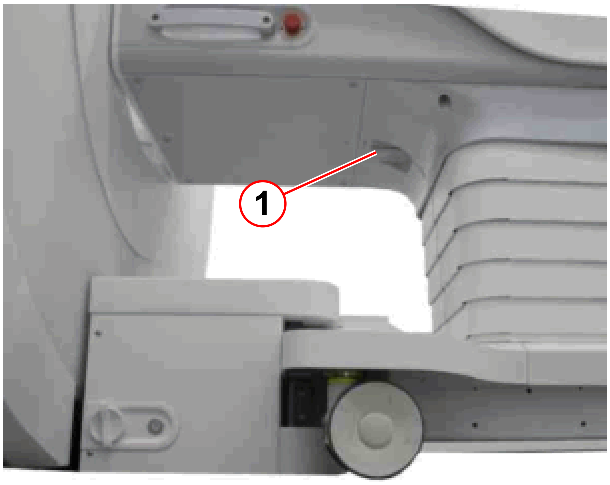
Required time: 5 min

Required tools: n.a.

SI Checking the emergency release of the patient table

- Move the table into center position by pressing the **center move** button.

Fig. 331: Patient table



(1) Emergency release handle

- Turn the emergency release handle (→ 1/Fig. 331 Page 520) on the patient table (a quarter turn clockwise).
- Slide the table all the way out (to the home position).
  » It must be possible to manually pull the table out of the magnet.
  » The emergency release handle will automatically return to the basic position if the table is in the home position.

4.3.5.5 Checking the distance between the patient table and the wall in the RF-room

Help ID: B5A1EAA0-CA61-406E-89D3-4C8DA1CC8ADE
Service Level: 7

Bruising hazard on the service end

SI Checking the distance between the patient table and the wall in the RF room

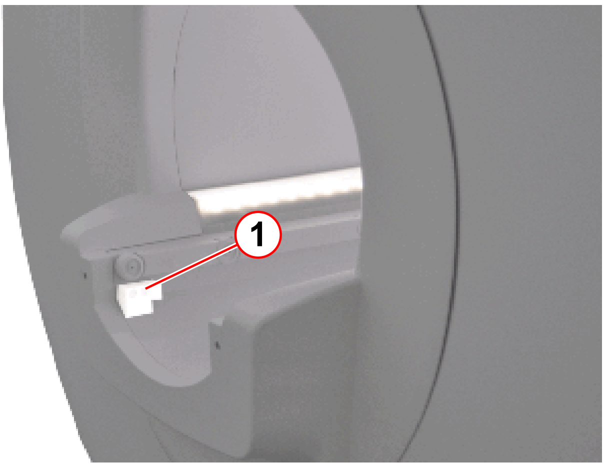
Required time: 5 min

Required tools: Bruising hazard sign, material number 3861718;

tape, material number 3474913; if required

- Check the mechanical end stop on the magnet bore for damage and correct installation.

Fig. 332: Magnet service end

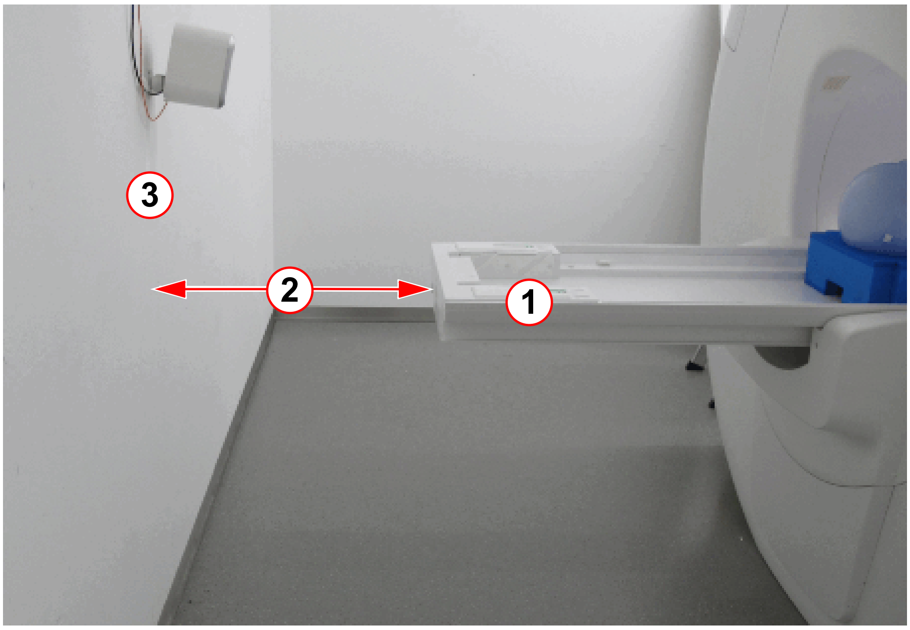


(1) Patient table, an example of the mechanical end stop

- Pull the emergency release handle on the patient table.

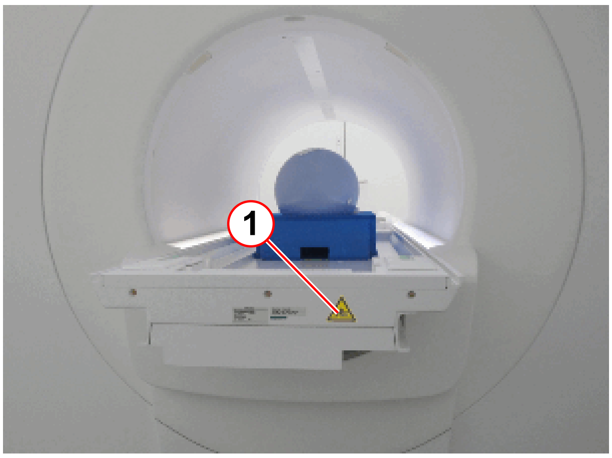
- Manually move the table all the way back (toward the magnet service end) until the table stops in the horizontal end position.
  » If the distance between the patient table and the wall (service end) of the RF room is **greater than** 500 mm, no further actions are necessary. Perform final steps.
  » If the distance between the patient table and the wall (service end) of the RF room is **less than** 500 mm, the bruising hazard sign has been posted at the rear of the patient table and the controlled access area has been secured using yellow and black tape.

Fig. 333: Bruising hazard, controlled access area



(1) Patient table back in horizontal end position
(2) Distance
(3) Wall of the RF room

Fig. 334: Magnet service end / patient table

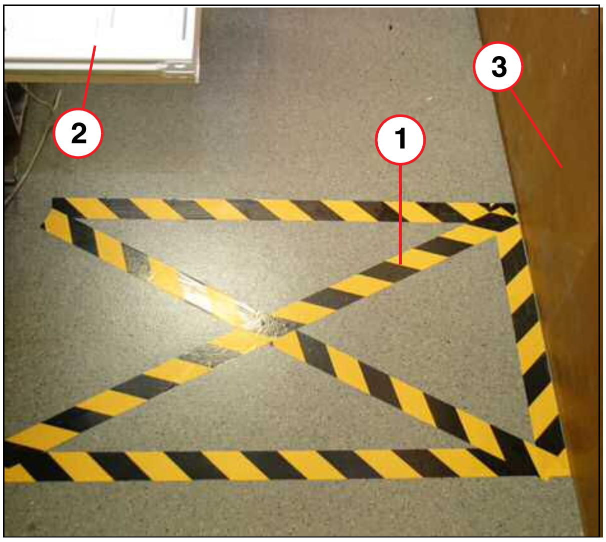


(1) Sign for bruising hazards

Fig. 335: Sign for bruising hazards

- Ensure that the bruising hazard sign has been posted at the rear of the patient table if the distance between the patient table and the wall (service end) is less than 500 mm.
- Install the missing sign immediately.

Fig. 336: Service end area



(1) Marking on the floor
(2) Patient table
(3) Wall of the RF room

- Ensure that the controlled access area has been protected with yellow and black tape if the distance between the patient table and the back wall is less than 500 mm.
- Install the missing tape immediately.

Final steps:

- Slide the table manually back into the home position.
  » The emergency release handle will turn to its original position.
- Confirm the pop-up windows on the MRAWP.
- To reset the table, press the IN and OUT buttons simultaneously on the control panel.

4.3.5.6 Checking the distance between the patient table and the cover

Help ID: 570DED6B-F94B-404E-A53E-5CEE555A981F
Service Level: 7

Distance between the patient table and the cover

SI Checking the distance between the patient table and the cover

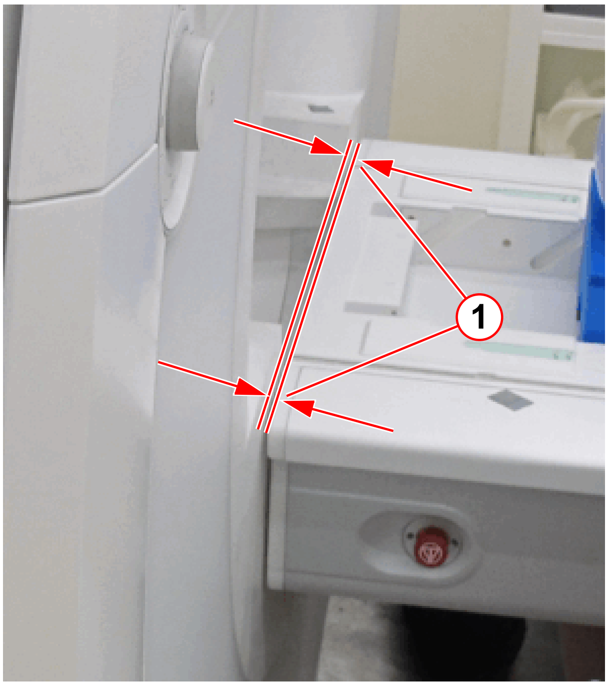
Required time: 5 min

Required tools: 8 mm Allen key or a non-magnetic ruler

Prerequisites: Table is approximately 10 cm below home position.

1. Move the patient table up to the highest position (transfer position vertical/horizontal).

Fig. 337: Patient table clearance



(1) Distance between patient table and cover

2. Measure the distance of the vertical clearance between the tabletop plate and the magnet cover.

   » The distance between the tabletop plate and the magnet cover must be within the range of **< 8 mm**.

   » If the distance is not within range, the patient table must be adjusted. Refer to Installation > Start-up > Check the mechanical adjustment of the patient table.

4.3.5.7 Checking the patient table movement

Help ID: C081FE94-3707-42BD-AB26-FAF61DB7682E
Service Level: 7

Table movement

Required time: 10 min

Required tools: n.a.

SI Checking the patient table movement

General movement

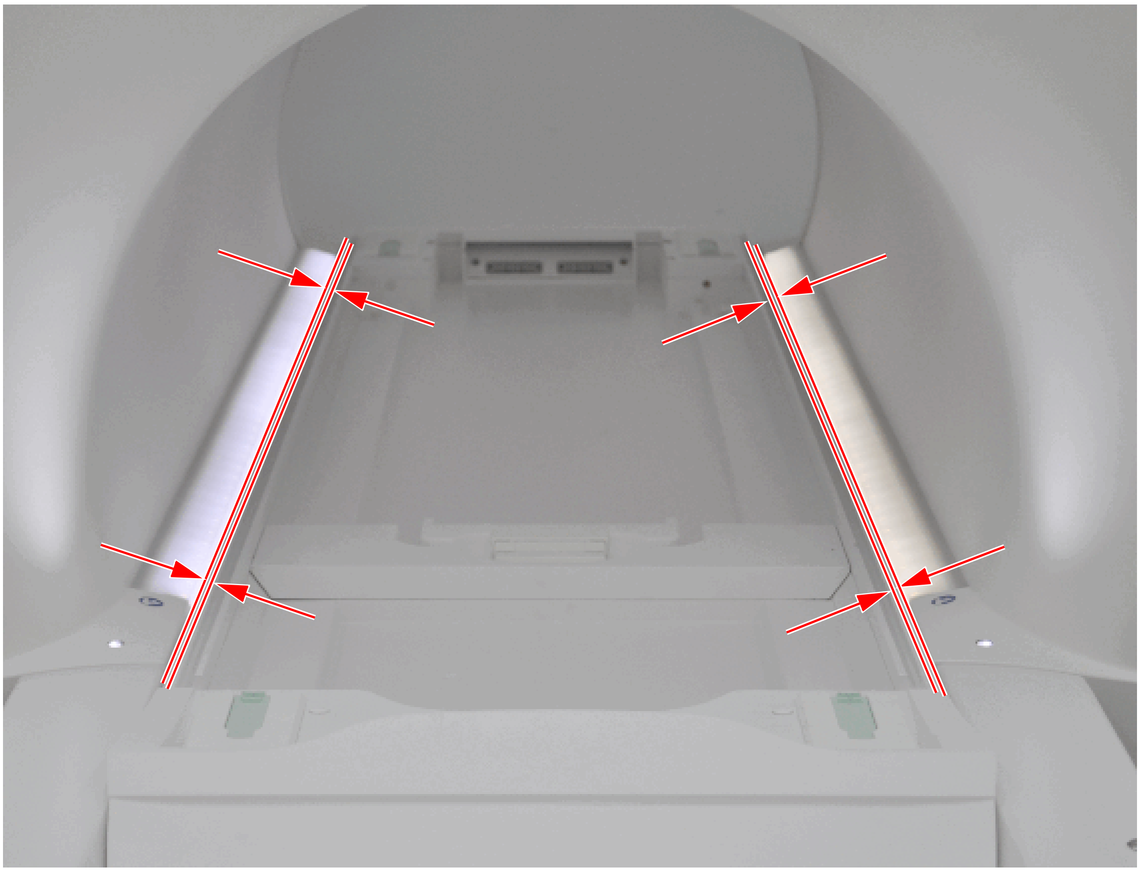
- Move the table all the way out.

  » The table must move without any noticeable noise.

  » The table stops in the horizontal end position.

- Lower the table all the way.
  - » The table must move without any noticeable noise.
  - » The table stops in the vertical end position.
- Move the table all the way up again.
  - » The table must move without any noticeable noise.
  - » The table stops in the vertical end position.
- Move the table into the magnet center.

Fig. 338: Patient table, parallelism



  - » The patient table must move parallel to the body coil.
- Move the table all the way in.
  - » The table must move without any noticeable noise.
  - » The table stops in the horizontal end position.
- Move the table to the home position.

> i If more dockable tables are in use, check the transfer position of these tables too. If the transfer position is not OK, adjust S20; refer to Replacement of Parts - Patient Handling - Patient Table

4.3.5.8 Checking the squeeze bulb

Help ID: AB6BBCC1-32E2-4AB5-B79F-FEE783E5351C
Service Level: 7

Squeeze bulb

Required time: 5 min

Required tools: n.a.

SI Checking the squeeze bulb

- Press the squeeze bulb on the patient table.

The following must occur:

» Continuous tone over the intercom.

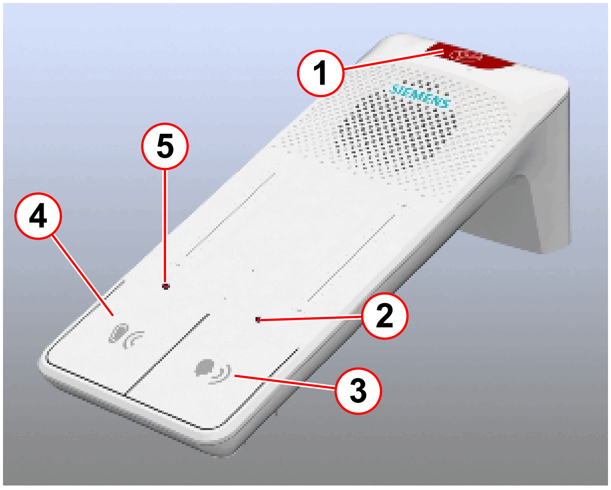
» Red flashing LEDs (→ 2 and 5/Fig. 339 Page 525) on the intercom

» The alarm can be reset by pressing the **Listen** (→ 4/Fig. 339 Page 525) or the **Talk** (→ 3/Fig. 339 Page 525) button on the intercom.

» You can speak with the patient by pressing the **Talk** button on the operating console. If necessary, adjust the volume using the +/- buttons.

» Communication with the patient is established when you press the **Listen** button. If necessary, adjust the volume using the +/- buttons.

Fig. 339: Intercom



(1) Stop button
(2) Talk LED
(3) Talk button
(4) Listen button
(5) Listen LED

4.3.5.9 Checking the side railing switches of the option dockable table

Help ID: 05F8E020-72B6-4784-A23A-272928E3020D
Service Level: 7

Dockable table option: Side railing switch

Required time: 5 min

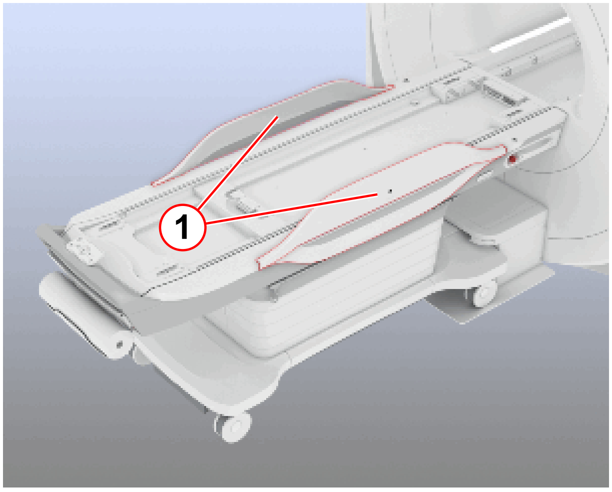
Required tools: n.a.

Prerequisites: Table is docked and in home position.

SI Checking the side railing switches of the option dockable table

- Press the **center move** button.
- While the table is moving toward the isocenter, lift up the side railing on the **left** side of the patient table

Fig. 340: Dockable table



(1) Side rails

» The patient table stops; the Dot display indicates the table stop.

- Make sure that table movement is not possible.
- Move the side railing to the down position.

» Table movement is possible again.

- Press the **center move** button.
- While the table is moving toward the isocenter, lift up the side railing on the **right** side of the patient table.

» The patient table stops; the Dot display indicates the table stop.

- Make sure that table movement is not possible.
- Move the side railing to the down position.

» Table movement is possible again.

- Move the table into the home position.

4.3.5.10 Checking the pressure activated switches of the option dockable table

Help ID: 07AE8919-D1EE-4FCF-8A83-8B9F0EEB3D99
Service Level: 7

Dockable table option: pressure-activated switches

Required time: 5 min

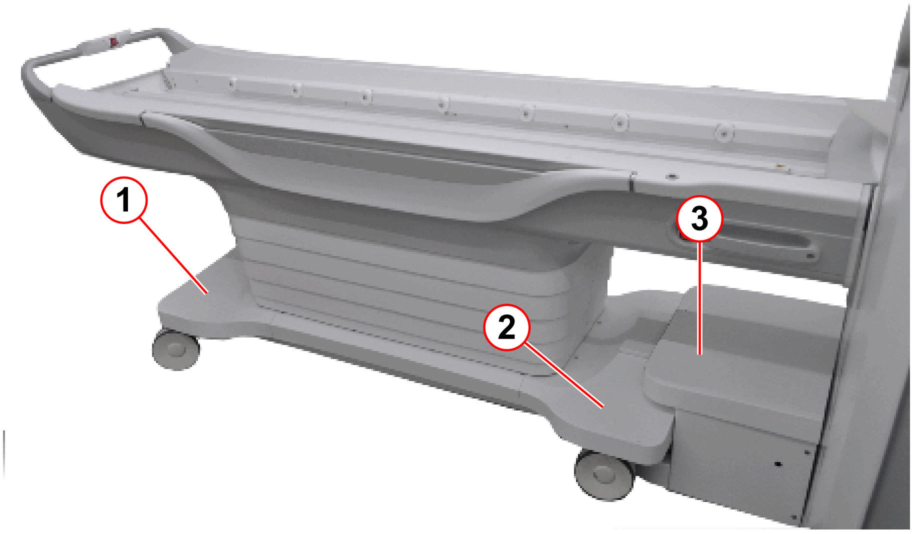
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

There are three pressure-activated switches on the dockable table.

1. Under the chassis foot-end cover
2. Under the chassis head-end cover
3. Under the top cover of the docking station

Fig. 341: Dockable table



(1) Chassis, foot-end cover
(2) Chassis, head- end cover
(3) Top cover of the docking station

SI Checking the pressure activated switches of the optional dockable table

- Move the table downward by pressing the down button.
- During table movement, press the cover of the docking station on the top side to acti-vate switch S60.
  » The table stop movement
- Press the up button
  » The table moves upward
- Press the down button again
- During table movement, press the chassis head-end cover downward to activate switch S60.
  » The table stop movement

- Press the up button
  - » The table moves upward
- Press the down button again
- During table movement, press the chassis foot-end cover downward to activate switch S60.
  - » The table stop movement
- Press the up button
  - » The table moves upward

4.3.5.11 Checking the emergency undock function of the option dockable table

Help ID: 140D3D5F-7A36-4C88-92CE-C2FD58951673
Service Level: 7

Dockable table option: emergency undock function

Required time: 10 min

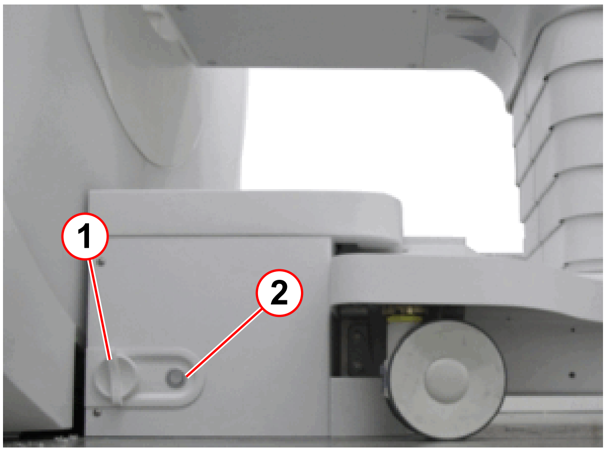
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

SI Checking the emergency undock function of the optional dockable table

- Turn the emergency undocking handle (a quarter turn clockwise).

Fig. 342: Docking station



(1) Emergency undocking handle
(2) Docking reset button

- » The docking station releases the Dockable Table. The table is undocked.

After use of the emergency undocking function docking is not possible. The docking reset button sets the docking station back to normal operation. Docking is possible again.

- Move the Dockable Table away from the docking station.
- Press the docking reset button.

- Move the Dockable Table into the docking station and press the docking button on the patient table control unit.

Fig. 343: Dockable table, control unit

(1) Docking button

» The dockable table is docked.

4.3.5.12 Checking the chassis and the wheel of the option dockable table

Help ID: 1E8DCB2C-C692-456F-AC54-D37D672A5817
Service Level: 7

Dockable table option: chassis and wheels

Required time: 5 min

Required tools: n.a.

SI Checking the chassis and the wheels (dockable table only)

- Undock the Dockable Table from the docking station.
- Move the table by a few meters.
  » The table must move smoothly without any noticeable noise.

Fig. 344: Patient table control unit

(1) Break button

- Press the break button.
  » It must not be possible to move the Dockable Table.

4.3.5.13 Magnet stop button Test (ERDU)

Help ID: E45A510B-E317-4C76-8260-17FED12AEA10
Service Level: 7

ERDU Checks

Required time: 15 min

Required tools: Screwdriver

Prerequisites

WARNING

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorized personnel must be excluded from the imaging suite.

» **Secure the room and prevent unauthorized access.**

- **Restrict access** to the magnet if it is at field when the test is being performed.
- Check that all alarm and error indications show that the system is OK.
- Remove the appropriate covers to gain access to the MSUP and Service Magnet Stop button.

Testing the Magnet Stop button (ERDU)

CAUTION

Magnet quench!

A magnet quench is caused by pressing the service magnet stop button (ERDU).

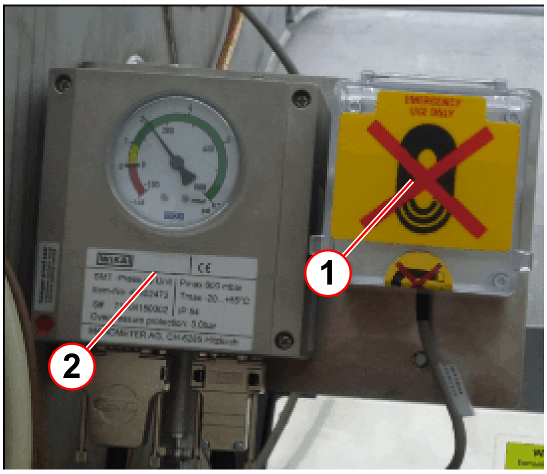
» **Do not operate the Service Magnet Stop button unless an emergency magnet quench is required.**

SI Magnet Stop button test (ERDU)

i The Magnet Stop button locks when depressed.

Turn the button anti-clockwise to unlock.

Fig. 345: Service ERDU location



(1) Service ERDU location
(2) Pressure switch

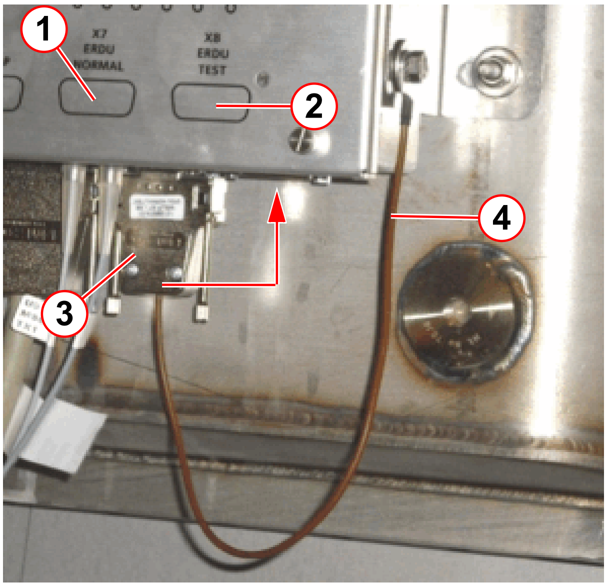
- The Service Magnet Stop button is located next to the pressure switch.
- Check that the button is not activated.
- Check that ERDU test connector is connected to X7.
- Test the Service Magnet Stop button.
  - » Press the button to ensure the MSUP audible alarm sounds
  - » Reset the button - ensure the alarm stops

When the ERDU test connector is disconnected from **BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.**

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When the ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

Fig. 346: ERDU test connector



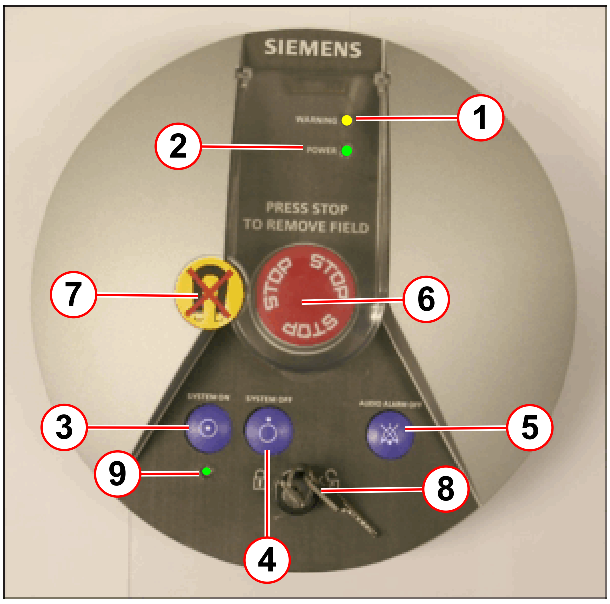
(1) ERDU NORMAL
(2) ERDU TEST
(3) ERDU test connector
(4) ERDU connector strap

The ERDU test connector is fitted to X7 during normal operation.

- Disconnect the test connector from X7 (ERDU NORMAL).
  - » MSUP audible alarm is activated.
- Connect it to X8 (ERDU TEST).
  - » On the alarm box, check the audible alarm sounds.
  - » Press AUDIO ALARM OFF to silence the alarm.
- Check that the ERDU TEST LEDs are NOT illuminated .

If the MSUP audible warning is not sounding, DO NOT CONTINUE.
Investigate the cause of the problem.

Fig. 347: Alarm box - LEDs and buttons

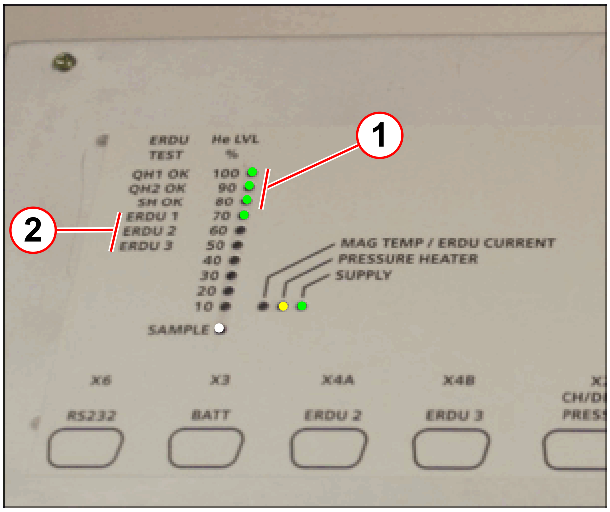


(1) LED WARNING = yellow
(2) LED POWER = green
(3) Button - SYSTEM ON
(4) Button - SYSTEM OFF
(5) Button - AUDIO ALARM OFF
(6) Button - MAGNET STOP (ERDU)
(7) Tamper-proof seal
(8) Power switch locking key
(9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
  - » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
  - » Press AUDIO ALARM OFF to silence the alarm.

Fig. 348: ERDU test - LED indicators (typical)



(1) QH1, QH2, SH = green LED (all ON)
(2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
  - » QH1, QH2, SH and ERDU 1 are NOT illuminated.

Other system ERDU buttons:

- Press one of the stand alone (system) Magnet Stop buttons.
  - Check the audible alarm sounds, and the warning LED flashes on the alarm box.
  - Press AUDIO ALARM OFF to silence the alarm.
- At the MSUP:
  - Check QH1, QH2, SH LEDs are illuminated.
  - Check ERDU 2 LED is illuminated for the magnet room ERDU button, or ERDU 3 LED for the optional 3rd ERDU button.
- Reset the Magnet Stop button.
- Check the LEDs return to normal.
- Repeat for each Magnet Stop button.

Final steps

- Check ALL Magnet Stop buttons are not activated.
- Disconnect the ERDU test connector from X8 (ERDU TEST) and reconnect it to X7 (ERDU NORMAL).
  » On the alarm box, check the audible alarm sounds.
- At the MSUP:
  » Check the MSUP audible alarm is deactivated.
  » Check the MSUP LEDs return to normal.
- At the alarm box:
  » Press AUDIO ALARM OFF to silence the alarm.
- Check the warning LED is not illuminated ‘ON’.

> i If the LED is on, or flashing, the fault must be checked.
> Do not disconnect the ERDU battery.

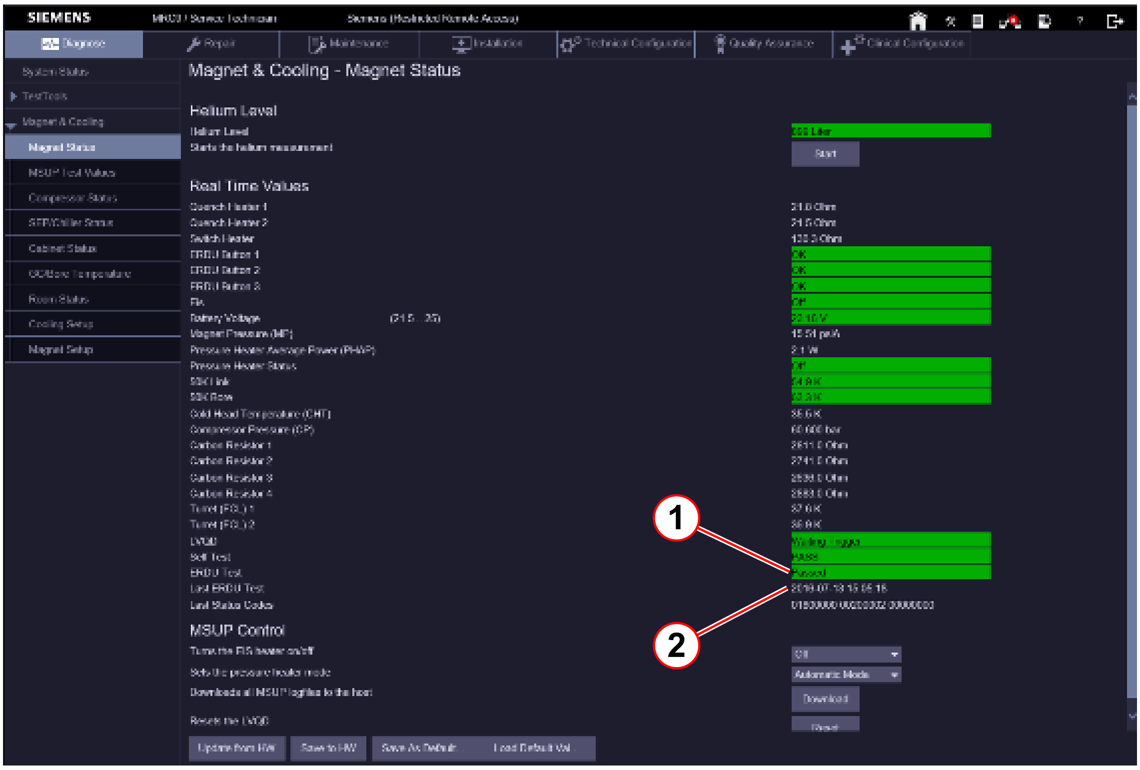
ERDU Test status

- Access the Service Software at the MRC.
- Select the menu item Magnet & Cooling - Magnet Status.

> i Data is only accurate from the last refresh of the page, or from when the page was opened last.

- Check the status of the ERDU Test and the date of the Last ERDU Test.

Fig. 349: ERDU test



(1) ERDU test status

(2) Last test date

4.3.5.14 Checking the gradient supervision

Help ID: 687555C2-3225-4055-A0F3-0D7B406710D5
Service Level: 7

Gradient Supervision

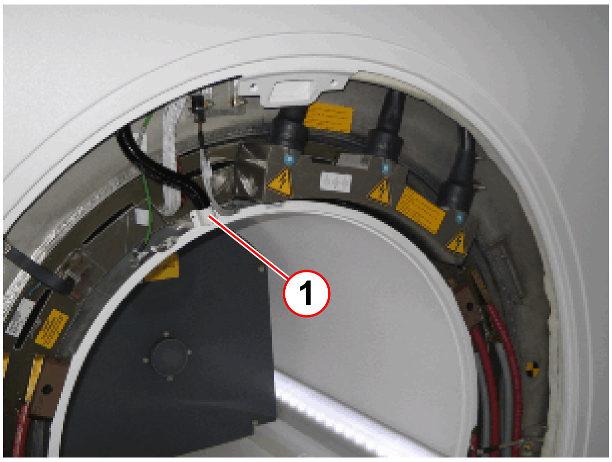
Required time: 30 min

Required tools: Test spray material number 10684091

Prerequisites: The display of the aspiration smoke detector shows OK (→ 1/Fig. 356 Page 538). System is on.

Fig. 350: Test spray

Fig. 351: Magnet service end, gradient connection

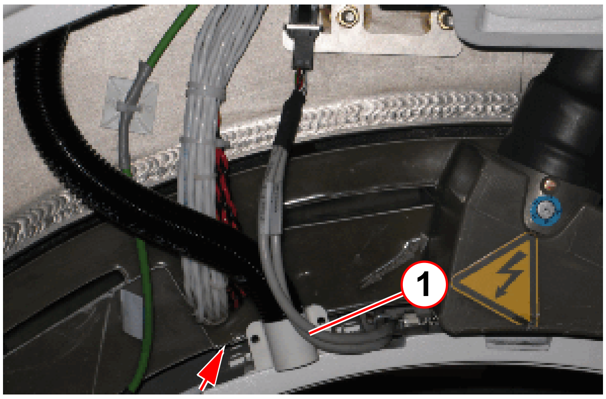


(1) Intake manifold BC

- Remove the rear funnel (service end).

SI Checking the gradient supervision

Fig. 352: Magnet service end, gradient connection



(1) Intake manifold BC

- Spray the test gas dosed in the gap between the gradient coil and body coil toward the intake of the BC manifold for 5 seconds. In this context, “dosed” means not completely pressing the spray button. Distance between spray and intake manifold should be approximately 5 to 10 cm.

NOTICE

On-site smoke detectors

Avoid false alarm of the on-site smoke detector!

» **Check if there is an on-site smoke detector. The test gas must not enter the on-site smoke detector.**

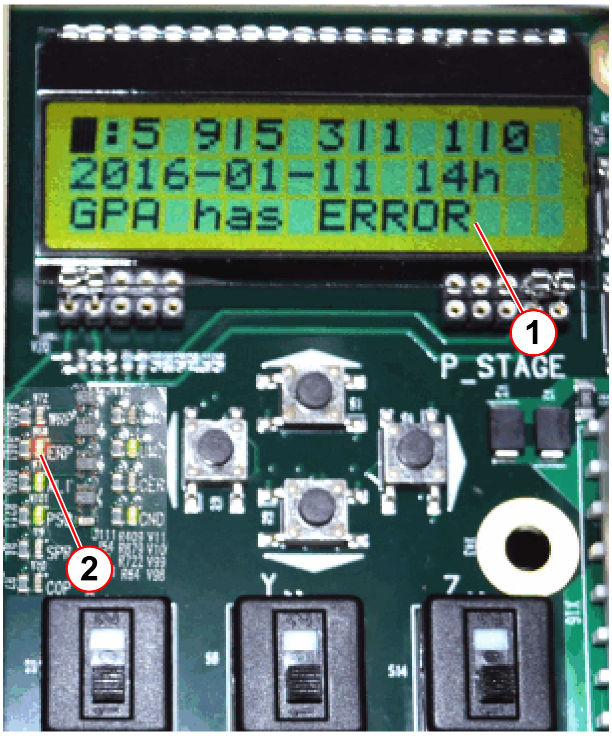
» **Keep the test gas away from on-site smoke detectors.**

» **Keep the spraying as short as possible to minimize test gas concentration in the air.**

Expected test results:

- The test gas is being pulled (driven by two small fans) from the BC manifold and into the aspirating smoke detector. This will take up to 40 seconds (depending on the hose length), then the following results are expected simultaneously:
  - » The red "FIRE" lamp on the VLM device must light up.
  - » The display on the DCC board in the GPA / DGSU shows 'GPA has ERROR' and the ERP LED lights up red.

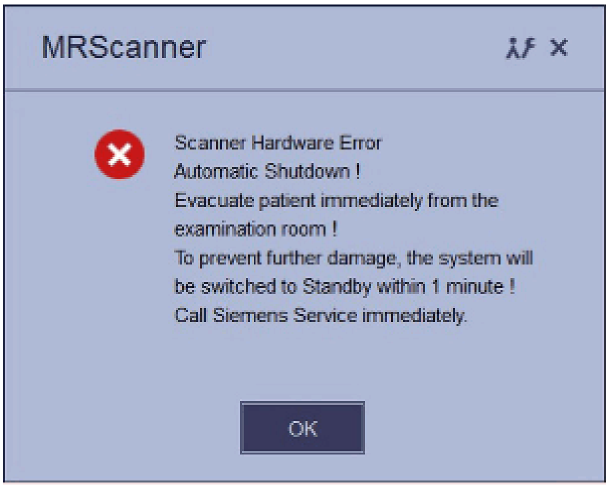
Fig. 353: GPA / DGSU



(1) DCC board display
(2) ERP LED

» An alarm message appears at the main console. (can also be checked in the error log).

Fig. 354: Pop-up for "Automatic Shutdown"



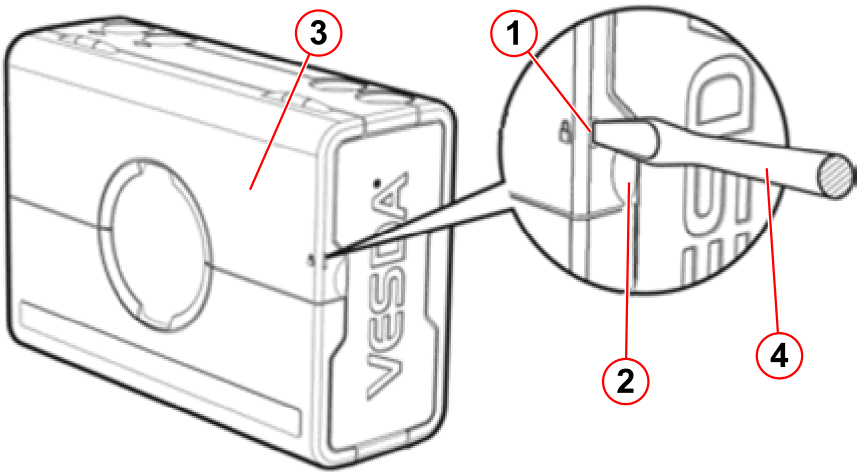
One minute later,

» The system must be switched to stand-by mode automatically.

Reset the system:

- Press a screwdriver into the security tab (→ 1/Fig. 355 Page 537) to open the service access cover with the finger clip (→ 2/Fig. 355 Page 537).

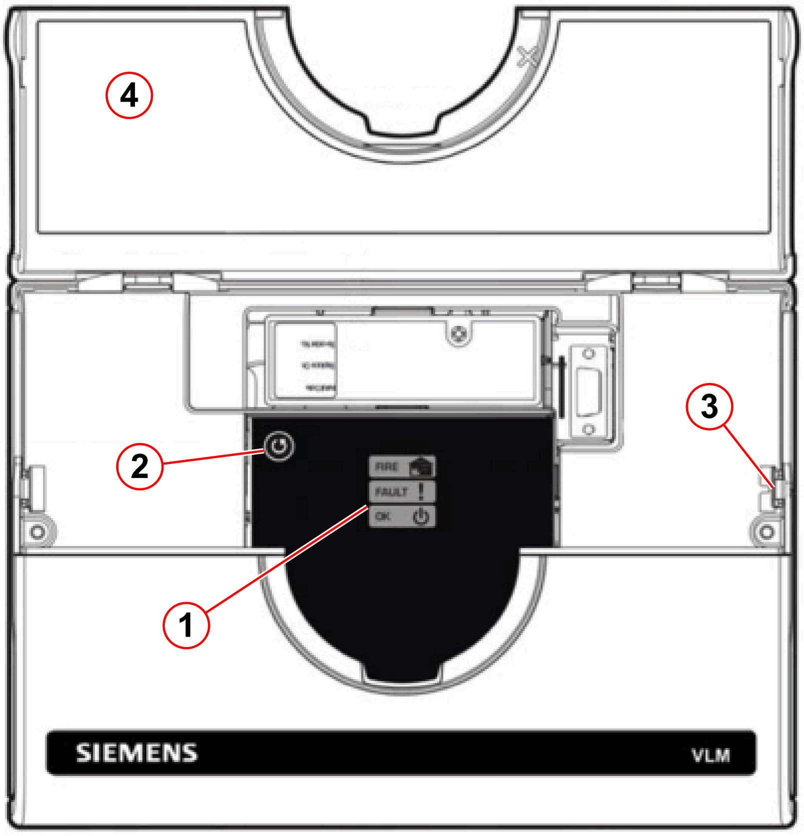
Fig. 355: Aspiration smoke detector ASD



(1) Security tab
(2) Finger clip
(3) Service access cover
(4) Screwdriver

- Reset the alarm by pressing the **reset button** (→ 2/Fig. 356 Page 538)for 2 seconds.

Fig. 356: Aspiration smoke detector



(1) Status display
(2) Reset button
(3) Security tab
(4) Service access cover

> If the reset fails, the concentration of smoke inside the detector chamber is too high. Retry the reset after 1-2 minutes.

» The aspiration smoke detector shows OK (→ 1/Fig. 356 Page 538).

- Start the system by pressing the **ON button** at the alarm box.

4.3.5.15 Quality Assurance

Help ID: 4E85213B-F73D-4CB2-97ED-78F6E63EC002
Service Level: 7

Quality Assurance (QA)

Required time: 70 to 110 min depending on installed options.

Required tools: QA Phantom Setup

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

> Use the Online Help to obtain detailed information.. Press the < **?** > help button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Open the Quality Assurance platform **> Administration Portal > enter password > Quality Assurance > General Quality Assurance**.

SI Quality assurance measurement

- Start the measurement by clicking the “Go” button.
  - » The **Quality Assurance** check is successfully completed when every individual QA step indicates the status **< OK >**.

> If the results are out of tolerance, perform “Tune-Up” or troubleshooting, if necessary. Repeat “Quality Assurance”.

4.3.6 General Quality Assurance

Help ID: 7DFD7C17-DADA-4F81-B6E1-1AF37D0BBC53
Service Level: 3

Introduction

General Quality Assurance (QA) is an automated procedure for:

- Verification of TuneUP adjustments.
- Checking the overall system quality.
- Image quality checks.

> Different from TuneUp, **system parameters are not changed in QA**.

QA Workflow

- General QA is separated into modules, each routine runs as a stand-alone process.
- After start, each procedure reads the specification and configuration parameters automatically.
- QA procedures determine if their measured results are in or out of specifications.

- QA procedures write their individual results to the Service Report.
- The status is returned to the main QA platform and the corresponding status field in the Service UI is updated.

After pressing the **[Go]** button in the QA UI, the **procedures start and perform typical tasks**:

- Each procedure registers the Service Patient (if not already registered).
- If a clinical patient is registered, a pop-up informs the operator to de-register the clinical patient before proceeding.
- The correct phantom setup is displayed by a pop-up and confirmed by the operator (if applicable). If QA is performed remotely, automatic table move is disabled and the operator receives a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Table movement:
  - If **QA is performed locally** at the system, the **table moves automatically** to center the required phantom in the measurement field (change between large/small spherical phantom).
  - If **QA is performed remotely, automatic table move is disabled**. The operator gets a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Configuration and specification values are read.
- Measurements are performed.

QA Function Status

Tab. 20 Possible QA Function Status

Status	Description
To Do	The QA procedure has not yet been performed yet and is pending. OR: The QA procedure needs to be performed again because another procedure (from TuneUp) has requested its execution.
Ok	The QA procedure has been carried out successfully.
Not Ok	The related parameters/results were found to be out of specifications during the last operation.
Error	An error occurred during the last operation.
Cancelled	Procedure canceled by operator.

Evaluation/Results

After end of the QA measurement, evaluations are performed automatically:

- The procedures determine if the measured results are in or out of specifications.
- The results are written to the Service Report.

- The results status is returned to the main QA platform and the corresponding status field is updated.

TuneUp/QA Dependencies

If a TuneUp procedure is successfully completed and has changed its status, the corresponding verification procedure(s) in the Quality Assurance workflow menu is/are changed to status **To Do**.

TuneUp procedures may also affect multiple QA procedures, e.g. after TuneUp Phantom Shim, the QA procedures Phantom Shim Check, Gradient Sensitivity Check, BC Image Brightness and Long Term Stability Check have to be performed.

Standby

In QA, a **Standby** function selection is offered. If selected, the system is switched to standby mode after completion of the QA workflow. All units except for the host computer are switched-off.

> After using the Standby functionality, it is not possible to switch-on the system again remotely.

4.3.6.2 RF Verify MNO

Help ID: 4210C5CB-BBD0-40ED-9200-0E96102D8567
Service Level: 3

General

The procedure RF Verify MNO performs a verification of the Multinuclei RF Characteristics determined during Tune Up.

Individual measurements for all supported nuclei are performed because of the frequency dependency of the broadband RFPA's power output. Forward and reflected DICO power values are measured and checked against their corresponding error tolerances (amplitude- and phase deviation from last saved RF Characteristics).

Measurement

The RF signal is generated at the MNO-TX Box Controller, amplified by the BB RFPA and routed to the dummy load in the MNO TX-Box.

The DICO factor is calculated and checked. Then, the deviation of the amplitude and phase from the linear characteristic is evaluated.

Evaluation/Results

- DICO calibration voltage
- DICO calibration statistics
- plots show the measured characteristics (amplitude and phase)

- maximum power deviation and peak-to-peak phase deviation

4.3.6.3 Feedback Loop Calibration Check

Help ID: 4172C660-AF1B-46AF-A127-DF43D6E9194F
Service Level: 3

General

The **Feedback Loop Calibration Check** procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the "tuncal" service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box.

Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a "Not Ok".

Evaluation/Results

Results in Report:

Tables:

- Delay evaluation
- Dispersion of delays from single BC elements

Plots:

- BC phase plot

4.3.6.4 RX Gain Calibration Check

Help ID: 4A3CD412-40FD-43CD-9DD2-BEF0BA7D4C40
Service Level: 3

General

The **RX Gain Calibration Check** verifies the amplification correction factors for all individual receiver channels which were determined during TuneUP / RX Gain Calibration:

- (Residual) RX channel correction factors
- High- / Low-gain factors

Measurement

The procedure uses loop measurements (without MR measurements). A fixed calibration signal is fed through all RX channels which are switched to high- and low-gain. For the measurement, the stored TuneUp parameters for the given RX channel- and high- / low-gain-correction are applied. The generated raw data is analyzed and all gain correction factors and the quotient of the High- / Low-gain modes are determined.

Evaluation/Results

Results in Report:

- Average gain correction factor over all receive channels.
- Gain correction factors for each individual receive channel.
- High- / Low-gain signals for each individual receive channel. Comparison of amplitude- and phase-difference to the stored TuneUp correction values.
- Noise evaluation.

4.3.6.5 BC Tuning Check

```
Help ID: 12EC839E-96E8-4A6D-8498-C6930BCCCDEF
Service Level: 3
```

General

The **BC Tuning Check** procedure is used to measure and evaluate the central resonance frequencies of the Body Coil resonator systems as well as the coupling between them.

A high efficiency of RF power transmission to the Body Coil requires sufficiently low reflection factors at the TX-Box output.

This condition is usually met, if the reflection factors of the Body Coil resonator systems BC0 and BC1 are high (without load) and equal. However, with load, these reflection factors shall be low.

The other requirement is, coupling between both resonators shall be minimum (i.e. high decoupling values).

Measurement

BC Reflection

First, a **BC Reflection** measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals, covering the full transmit operating frequency range. The result of each measurement is the reflection or r-factor = Ur/Uf which is plotted for both Body Coil systems BC0 and BC1 along the frequency axis.

The evaluation determines:

- each coil's center (tuned) frequency is within a specified range, i.e. is not too far from the center of the operating range.
- that the center frequencies of both coil systems BC0 and BC1 are tuned within specified limits of each other, i.e. the delta frequency is low.

- that the reflection factors are within expected limits. These values provide useful information about the quality (Q-factor) of the BC resonant circuits. They also depend on the coil load and cannot be adjusted.

  The Software detects the resonant frequency always at the point where the reflection of the BC has its minimum.

Transmission

In a second step, a **Transmission** measurement is performed with transmitting a signal into the BC0-system and measuring the coupled signal in the BC1-system and vice versa. The transmission factor describes the amount of RF-coupling between the two Body Coil systems, mainly due to coil construction but also affected by other factors e.g., mechanic position adjustment of the Body Coil within the Gradient Coil's Faraday Shield.

The evaluation checks the transmission value, expressed in decibels (dB), to be within expected limits. **A high negative dB-value indicates minimum coupling** i.e., a good decoupling between BC0 and BC1 system.

Evaluation/Results

Results in Report:

- The reflection related values for both (BC0 and BC1) systems, like the resonance frequency, the reflection factor and the difference between both systems.
- The transmission related value like the frequency and the decoupling between both systems.
- The new and old phase difference.
- Diagrams showing the measured values for the transmission and the reflection behavior as a function of frequency.

4.3.6.6 Cable Length Check

```
Help ID: B19E08C7-B349-43A5-99BF-E46BAB8192A4
Service Level: 3
```

General

The QA step **Cable Length Check** verifies the BC transmit cable phase shifts which have been determined in TuneUp / Cable Length.

The BC transmit cable length may slightly vary due to production tolerances. However, accurate information about the lengths of both BC transmit cables is required for an optimal function of the extended TX B1-field regulation loop in the measurement framework.

Measurement

A measurement with the service sequence rf_loop is performed. An RF pulse is fed to both Body Coil channels with the Body Coil switched to 'detune' mode. The forward and reflected signals are taken from the internal directional couplers (DICO) of the TX-Box and analyzed by the monitoring Software.

From these results, the so-called S-matrix elements which describe the characteristics of the BC transmit path are determined. These values allow, together with the known Body Coil RF-properties, to calculate the cable lengths for both Body Coil channels in terms of the induced signal phase shifts.

Evaluation/Results

In the Service report, the actual, measured phase shifts for both Body Coil channels are displayed, together with the old values from TuneUp. The latter ones are read from the BC coil files. Difference values are calculated and checked against the specified tolerance range.

4.3.6.7 Gradient Regulator Check

Help ID: 99738FE4-5E37-4EFF-A222-CE0B23F5D7EE
Service Level: 3

General

The **Gradient Regulator Check** procedure only measures the gradient waveform with the present PID regulator settings and verifies that these settings are correct. The check analyzes several aspects of the actual current output by comparing it to the expected ideal pulse output. In addition, the procedure will measures the linearity of the gradients by stepping through the full range of the gradient strength from negative to positive amplitudes.

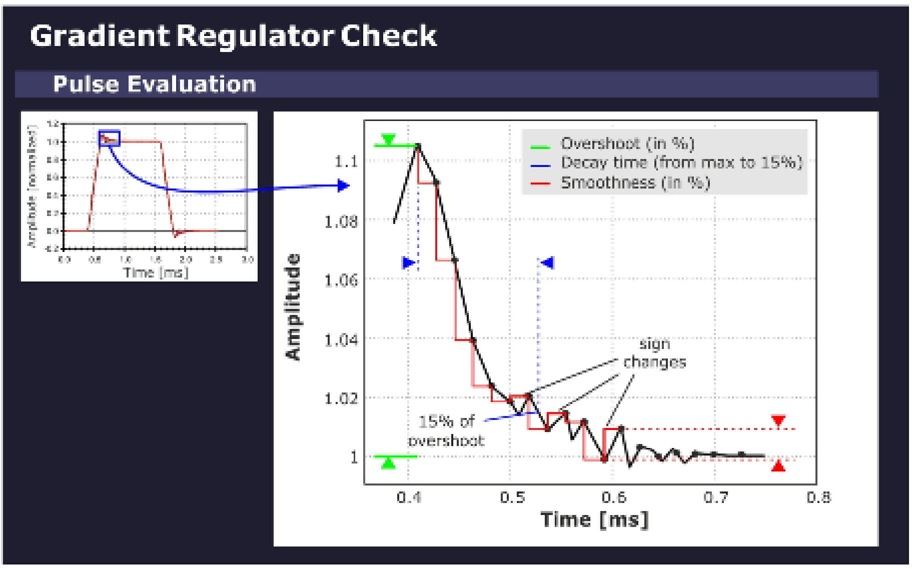
Measurement

The sequence uses a gradient table to generate the gradient pulses. The selected gradient (depending on the selected orientation) steps from the maximum negative to the maximum positive gradient amplitude pulse in linear portions. Each gradient in the table is switched on using a ramp-up and ramp-down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.

Evaluation/Results

Based on the maximum gradient amplitude pulse, the program evaluates the characteristic overshoot that occurs just after the pulse ramp-up. The amplitude of this **overshoot** is determined as a percentage relative to the nominal gradient amplitude. The program also calculates the **decay time** it takes for this exponential-like overshoot to return to a specified value. In addition, a **smoothness calculation** is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.

Fig. 357: Gradient Regulator Overshoot



Results in Report:

- PID gradient regulator parameters.
- Optimization parameters overshoot, decay time and smoothness.
- Gradient current pulse diagram (pulse form).
- Enlarged diagram of overshoot portion and current ripple (top portion).

4.3.6.8 Phantom Shim Check

Help ID: D847AEFF-8EE6-4E1B-83A4-EE3620B311E9
Service Level: 3

General

The **Phantom Shim Check** procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be accurately determined based on the frequency distribution in the volume elements making up the phantom image.

Two measurements are performed with the parameters from the corresponding TuneUp procedure to determine the remaining field terms and the overall homogeneity within the phantom volume. No adjustment of the shim terms is made in this case.

For systems with the advanced shim option, the procedure is performed with shim currents applied.

Measurement

The actual shim status of the system is measured with a 3D-Dess-sequence (Double Echo Steady State with a FISP and PSIF echo). The resulting voxels are cubes which are analyzed

by the Software.

Evaluation/Results

The field homogeneity is evaluated based on the frequency of the voxels. For the check of the B_0-field, it is sufficient to compare the field differences between neighboring voxels. The field difference is obtained by comparing the phases of the magnetization vectors.

The complete data set is evaluated by a differential shim equation. This equation minimizes the difference between the field generated by the3 Gradient Coils (and the optional 5 shim coils) and the measured magnetic field inhomogeneities.

Results in Report:

- Field terms
- MR Frequency
- Gradient Offsets
- Shim Currents (optional)

4.3.6.9 BC Power Losses Check

Help ID: D35D6115-3348-47DF-8FBE-8CBE8F1C7E7C
Service Level: 3

General

A check is performed to verify that the TuneUp of BC Power Losses has been performed successfully. This is done by measuring the Coil Power Loss value again and checking against the Coil Power Loss value in the coil file. Since the default of this value is very low, it would be out of specification if no TuneUp was executed.

In addition, this QA step verifies that the SAR (Specific Absorption Rate) check is activated. The forward and reflected voltages from both TX Box channels are also tested during the measurement.

Measurement

The TuneUp “Coil Power Loss” measurement is performed.

Evaluation/Results

The Coil Power Loss is measured in CP and EP mode.

- Check if the Coil Power Loss value in the coil file is within specifications and comparable to the measured value.
- Read and check the value for the SAR.
- Read and check the forward and reflected voltages from TX-Box and check them for consistency with broadband power meter values.

Results in Report:

- Phantom position and diameter.
- Measured and stored Coil Power Loss values.
- Measured reference amplitude (AdjTra) from the TALES (TX-Box power meter).

- The measured SAR and voltage values.

4.3.6.10 Cross Term Compensation Check

Help ID: 94F6F45A-CD81-4440-9FDE-F6D1AC3824A4
Service Level: 3

General

The **Cross-Term Compensation Check (CTC)** checks the quality of the cross-term compensation previously adjusted in TuneUp.

In QA mode, the parameters of the inverse filter function which compensates the cross-term effect are checked.

Measurement

A CTC measurement is performed.

Evaluation/Results

Results in Report:

- Cross Term Amplitudes and Time Constants.
- Diagrams visualizing the corresponding cross term amplitudes.

4.3.6.11 Eddy Current Compensation Check

Help ID: ACA062B6-B909-438A-A732-F3BF88BA7502
Service Level: 3

General

The **Eddy Current Compensation Check** (ECC) checks the quality of the eddy current compensation adjustment previously performed in TuneUp.

Measurement

The procedure performs a check of the ECC parameters. The existing ECC settings are used for the measurement.

A preparation measurement is performed to check for the correct positioning of the phantom in the magnet center and for sufficient field homogeneity. The used sequence is a double echo. The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity. The second echo uses a readout gradient and calculates the phantom position in the readout direction.

Evaluation/Results

Results in Report:

- B_0 eddy current values for different time ranges.

- 1^{st} order (gradient) eddy current values for different time ranges.

4.3.6.12 Gradient Delay Check

Help ID: 946ABE8C-323C-48C6-B295-5C45C3DD15AC
Service Level: 3

General

The **Gradient Delay Check** procedure determines the delay time between gradient switched on digitally and the time the actual gradient field builds up. Gradient Delay can slightly differ for each gradient axis.

In QA mode, the Gradient Delay adjustment values, previously established in TuneUp, are is verified.

Measurement

Each gradient is measured at 14 different gradient strengths from -19.5 mT/m to +19.5 mT/m. For an ideal gradient system, all delays are the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, that is, high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, there is some kind of nonlinearity present in the gradient system.

Evaluation/Results

Results in Report:

- Current- and calculated- Gradient Delay values in microseconds.
- Maximum Gradient Delay Deviation values.

4.3.6.13 Gradient Sensitivity Check

Help ID: 16C2854F-2DEB-4B6F-BEDA-E6418994A88B
Service Level: 3

General

The **Gradient Sensitivity Check** procedure checks the Gradient Sensitivity values previously adjusted in TuneUp to ensure that the system uses correct the gradient strengths and measures the correct phantom image size in all orientations.

For the measurement, the distortion correction filter is enabled to compensate gradient field non-linearities.

Measurement

- A standard imaging sequence is repeated three times: one measurement for each orientation.

- The three images are then analyzed to determine the exact location of the center of the phantom and the phantom diameters. The standard readout and phase encoding direction for each orientation is swapped to determine the center and diameter of the phantom. The average of these two results determines the final diameter value.
- The measured values are evaluated and compared with the well-known diameters of the phantom for verification of the correctly adjusted Gradient Sensitivities.

Evaluation/Results

Results in Report:

- Measured phantom diameter in all three dimensions.
- Measured phantom centers in all three dimensions.
- Gradient Sensitivities.
- Phantom images in all three slice orientations.

4.3.6.14 BC Image Brightness Check

Help ID: 4B8C4548-E064-495C-B6B2-AC746D517DD4
Service Level: 3

General

This procedure is used to verify the image intensity (brightness) for the Body Coil which was previously adjusted in the TuneUp step BC Image Brightness..

Measurement

The measurement runs the service sequence "Body_Tra_br" and generates images with the Body Coil.

Evaluation/Results

Both Body Coil channels BC0 and BC1 are evaluated independently from each other. Also the sum signal of BC0_BC1 is checked.

From each image the following values are determined:

- Brightness
- FFT Scale Factor
- TX Amplitude
- RX Gain Factor

The brightness value is the signal intensity. The FFT scale factor is a correction factor to adjust the brightness to the middle of the specified value range (min. and max.). The transverse image is used for determining the FFT scale factor.

The transmitter amplitude for the sequence (180 degree flip angle) is also evaluated. This value is coil load specific, so it depends on the phantom.

4.3.6.15 Image Orientation Check

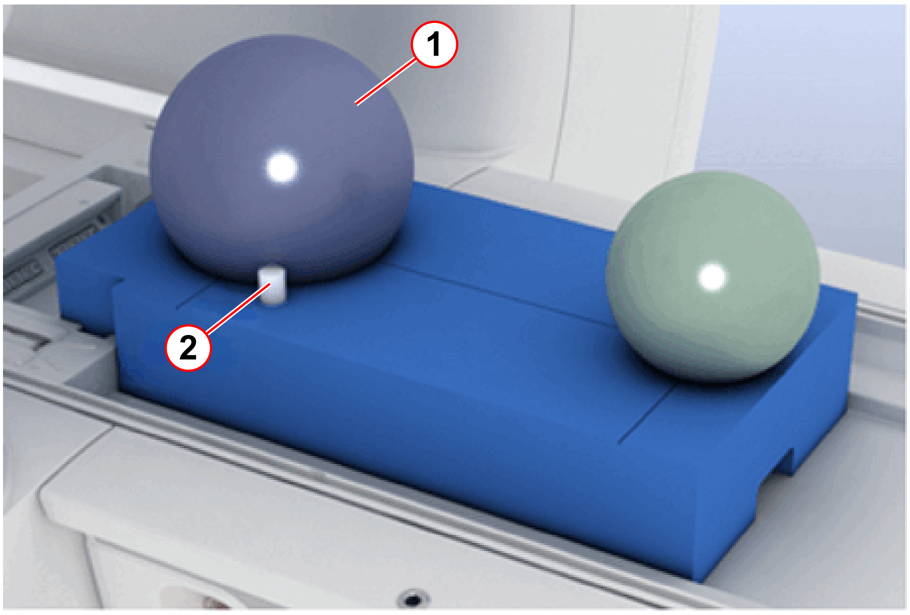
Help ID: F09D8D74-F01E-4A0A-B3FD-A116D3E77C87
Service Level: 3

General

The **Image Orientation Check** verifies the correct position of the phantom setup. It uses an additional, asymmetrically positioned small bottle phantom to determine the correct gradient orientation and polarity.

If any of the gradient power amplifier or -coil connections were swapped or polarities of any gradient axis were wrong, the images would be displayed with incorrect orientation.

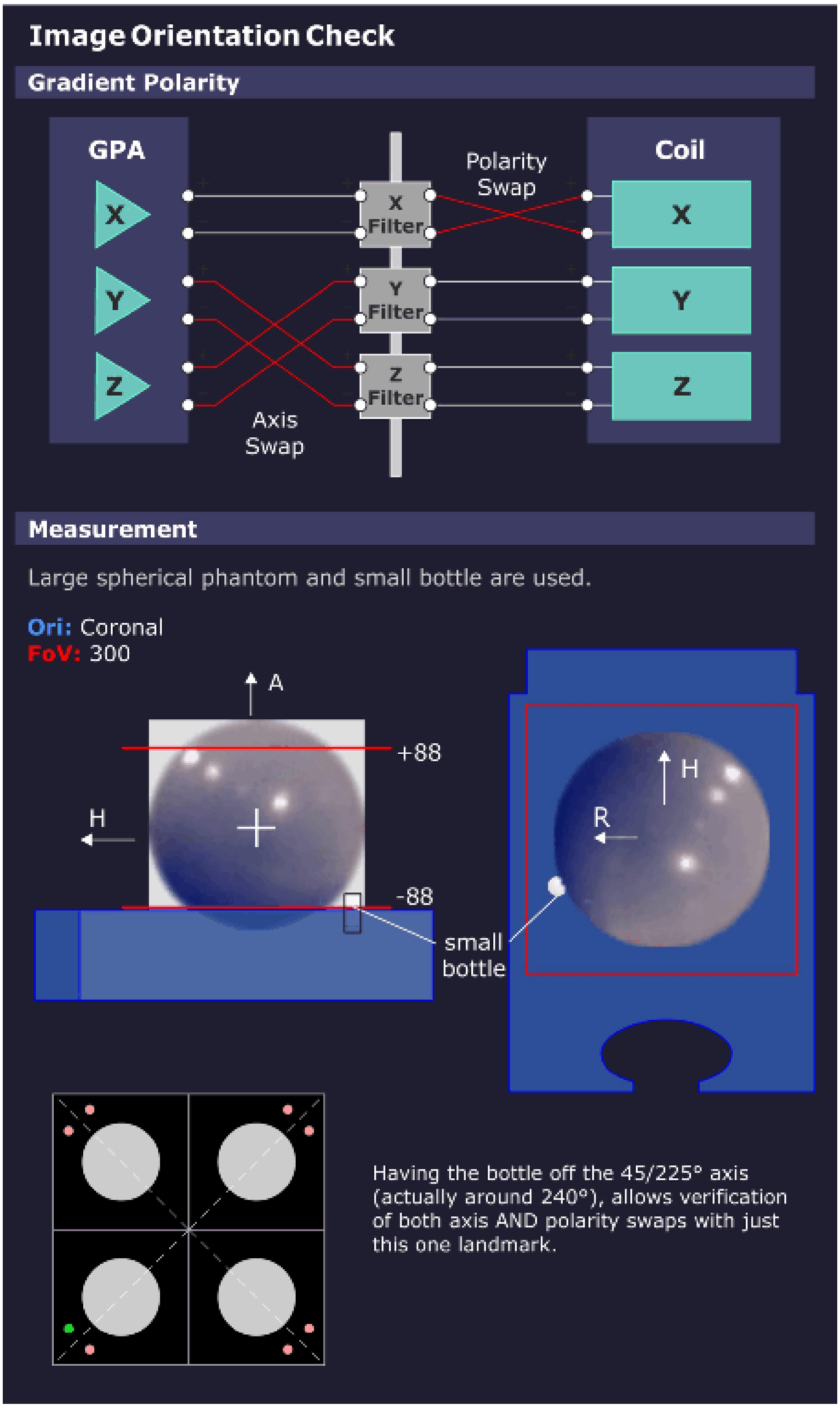
Fig. 358: TuneUP_QA Phantom Setup



(1) Spherical phantom
(2) Small bottle phantom for Image Orientation Check

Measurement

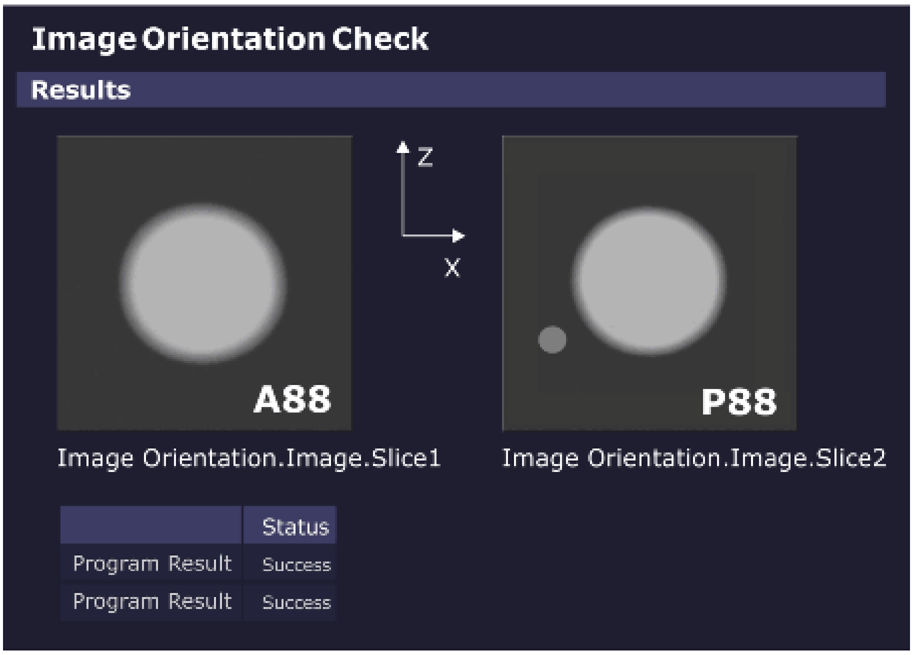
Fig. 359: Image Orientation Check - Measurement



Images are generated from the phantom setup with the additional, asymmetrically positioned small bottle.

Evaluation/Results

Fig. 360: Image Orientation Check - Results



The measured images are automatically evaluated.

If the phantom and bottle are not in the expected position, the procedure returns with the status “Not Ok”.

Results in Report:

- Images showing the phantom geometry (phantom and bottle).

4.3.6.16 Gradient Risetime Check

Help ID: B668AA17-8C5F-4E6C-8908-BB2F522B4F00
Service Level: 3

General

The **Gradient Rise Time Check** measures the minimal gradient Rise Time or the so-called “Slew Rate”.

Measurement

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, five echoes develop because of the bipolar triangle gradient pulses. These echoes should appear, according to the pre-dephasing, exactly at the pulse spikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate could not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Sequence of procedures:

- Verification check of the used phantom.
- Gradient Risetime measurement for all three gradient axes.

Evaluation/Results

Results in Report:

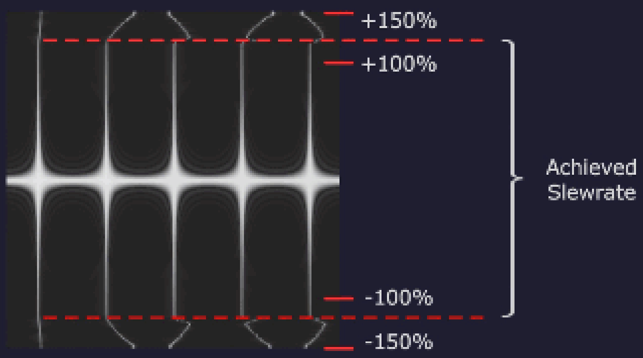
- Graphics of echo maximum versus gradient slew rate, echo 1-5
- Statistics of measured values against specifications
- X/Y/Z-gradient Rise Time results
- X/Y/Z-gradient Rise Time multiple echo output (raw data image, echo 1-5)

Fig. 361: Gradient Risetime Check - Results

Gradient Risetime Check

Results

Gradient Rise Time Results		
Axis	Slew Rate	
		Tolerance
X	112.46 %	≥100 %
Y	111.43 %	≥100 %
Z	112.48 %	≥100 %



X-Gradient Multiple Echo Output

Slewrates and Echopositions						
Echo	Slew Rate Positive Polarity		Slew Rate Neg. Polarity		Diff. of Echo Positions	
		Tolerance		Tolerance		Tolerance
1	**126.68 %**	≥100.00 %	**126.68 %**	≥100.00 %	-0.039	-1.000 ... 1.000
2	**114.67 %**	≥100.00 %	**114.69 %**	≥100.00 %	0.044	-1.000 ... 1.000
3	**113.58 %**	≥100.00 %	**113.58 %**	≥100.00 %	-0.076	-1.000 ... 1.000
4	**112.48 %**	≥100.00 %	**113.58 %**	≥100.00 %	0.041	-1.000 ... 1.000
5	**112.48 %**	≥100.00 %	**112.48 %**	≥100.00 %	-0.039	-1.000 ... 1.000

X-Gradient strength slew rate achieved is 112.483% of specified value
(mean value of the lowest two values of all echoes)

Troubleshooting

In case of problems (result **"Not Ok"**) check the following items:

- Check raw data for spikes. No spikes must be visible.
- Stability/value of mains voltage
- Line resistance
- GPA line transformer and correct line voltage adaptation
- GPA rectifier board
- GPA current sensors (especially if value "Difference of Echo Positions" is out of spec.)
- GPA Power Stages

- Results of QA Regulator Check

4.3.6.17 Spike Check

Help ID: B4E9366D-2E5A-4CE9-969E-3EBDE191949C
Service Level: 3

The **Spike Check** is a quick test to check the MR imaging system for spikes and RF interference.

A “spike stress” measurement is run and its raw data evaluated by a program for spikes. The corresponding image data is evaluated for RF interference.

General

Electrostatic discharges during the measurement may generate some spikes within the raw data. Spikes are individual raw data points or small groups of raw data points with amplitudes above the typical noise level of the MR imaging system. **Spikes** can degrade the MR imaging quality. In this case, the signal-to-noise ratio can be reduced or the image can show a periodic intensity modulation or structure.

One typical source of spikes is the mechanical and electrostatic stress of the MR system caused by the operation of the gradient system.

A simple way to check the MR system for spikes is to run a noise measurement without RF pulses. During the acquisition window massive mechanical and electrostatic stress is generated with rapid switching of gradient pulses with maximum strength.

In case no spikes and no RF interference are present, pure statistical noise shall be received.

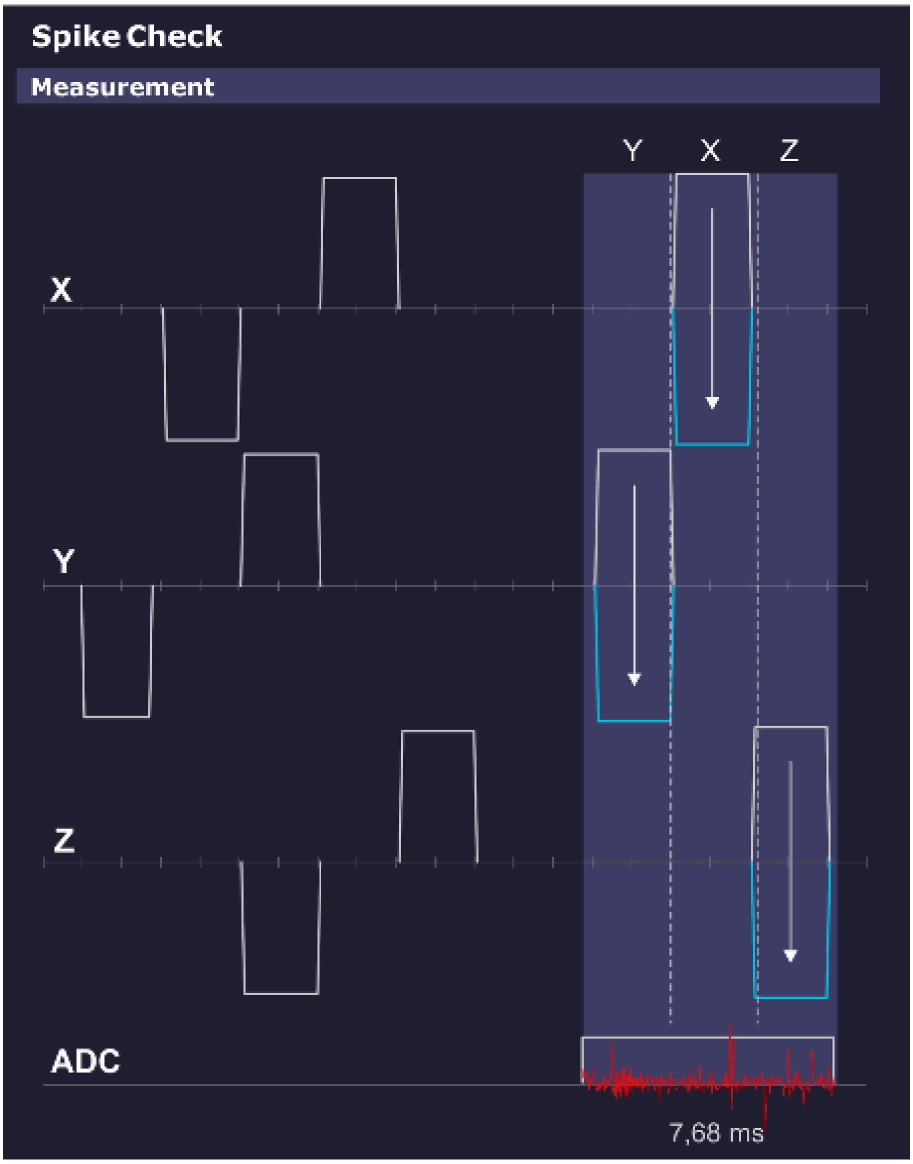
Measurement

This sequence does not apply RF pulses, so no MR signal is generated.

The spike sequence starts with an initial series of strong gradient pulses of both positive and negative polarities on all three gradients intended to cause a buildup of static electrical energy caused by any moving components (cables, covers, etc.).

The ADC is then enabled and again a pattern of gradient pulses with amplitudes ranging from max. positive to max. negative are applied. Any static electrical discharges caused by e.g. moving or loose parts are picked up by the receiving coil (usually the Body Coil), digitized by the ADCs and stored in the raw data.

Fig. 362: Spike Check - Measurement



Evaluation/Results

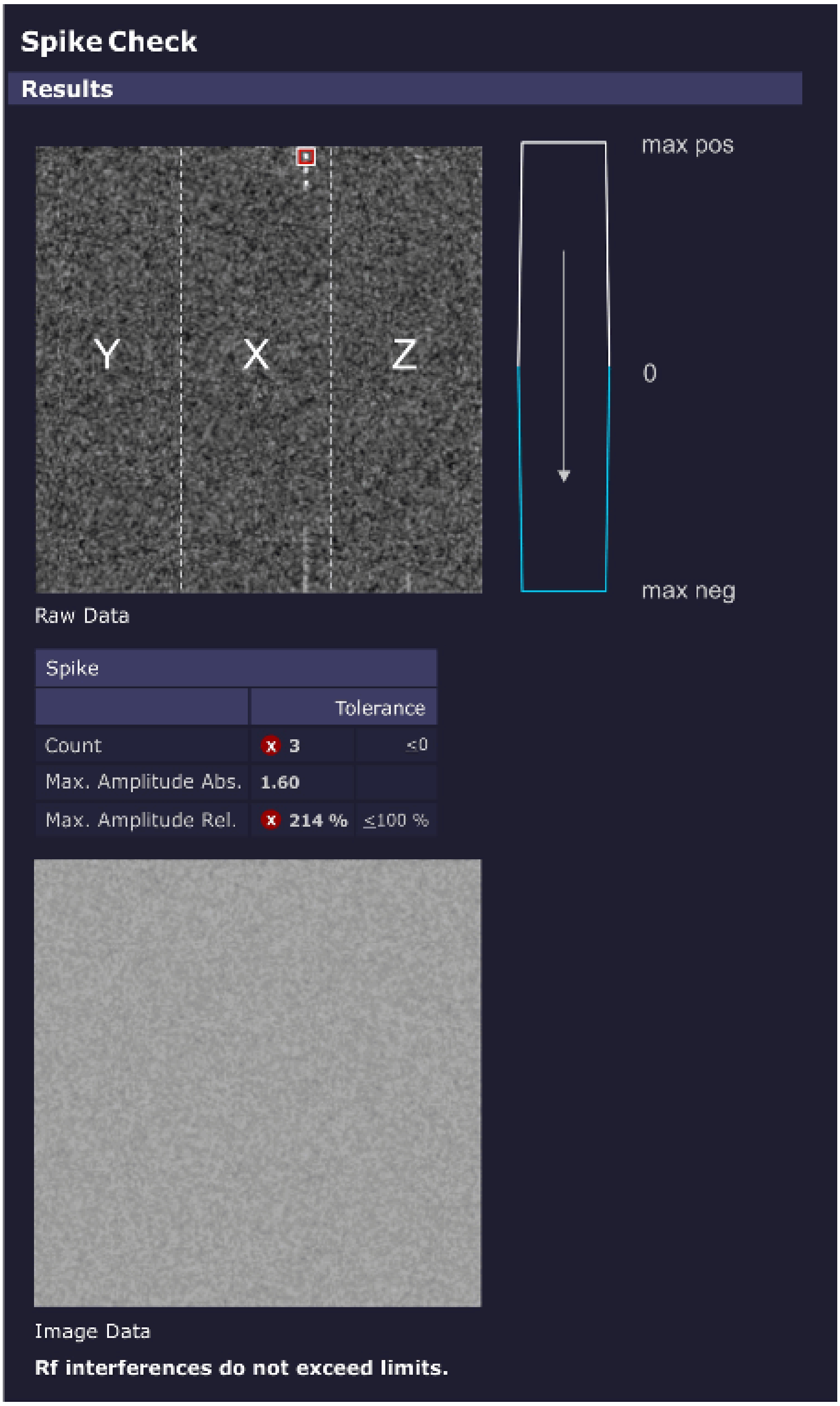
Search for **spikes in raw data** and **RF interference in the image data**.

The evaluation looks for intensities in the raw data set exceeding a threshold (usually noise) and counts both the number and intensities of the spikes and displays them numerically and graphically. The electrical discharges are visible as "spikes" in the raw data.

The position of the spikes in the raw data set relates to the gradient axis that was switched on when the spike occurred. This position may indicate the cause of the spikes.

In case RF interference was present during the acquisition, the disturbances can be seen in the displayed image data.

Fig. 363: Spike Check - Results



Results in Report:

- Raw data image and resulting RF interference image.
- If spikes are found, the following values are displayed:
  - Total number of spikes.

- Magnitude of the highest spike.

4.3.6.18 RF Noise Check

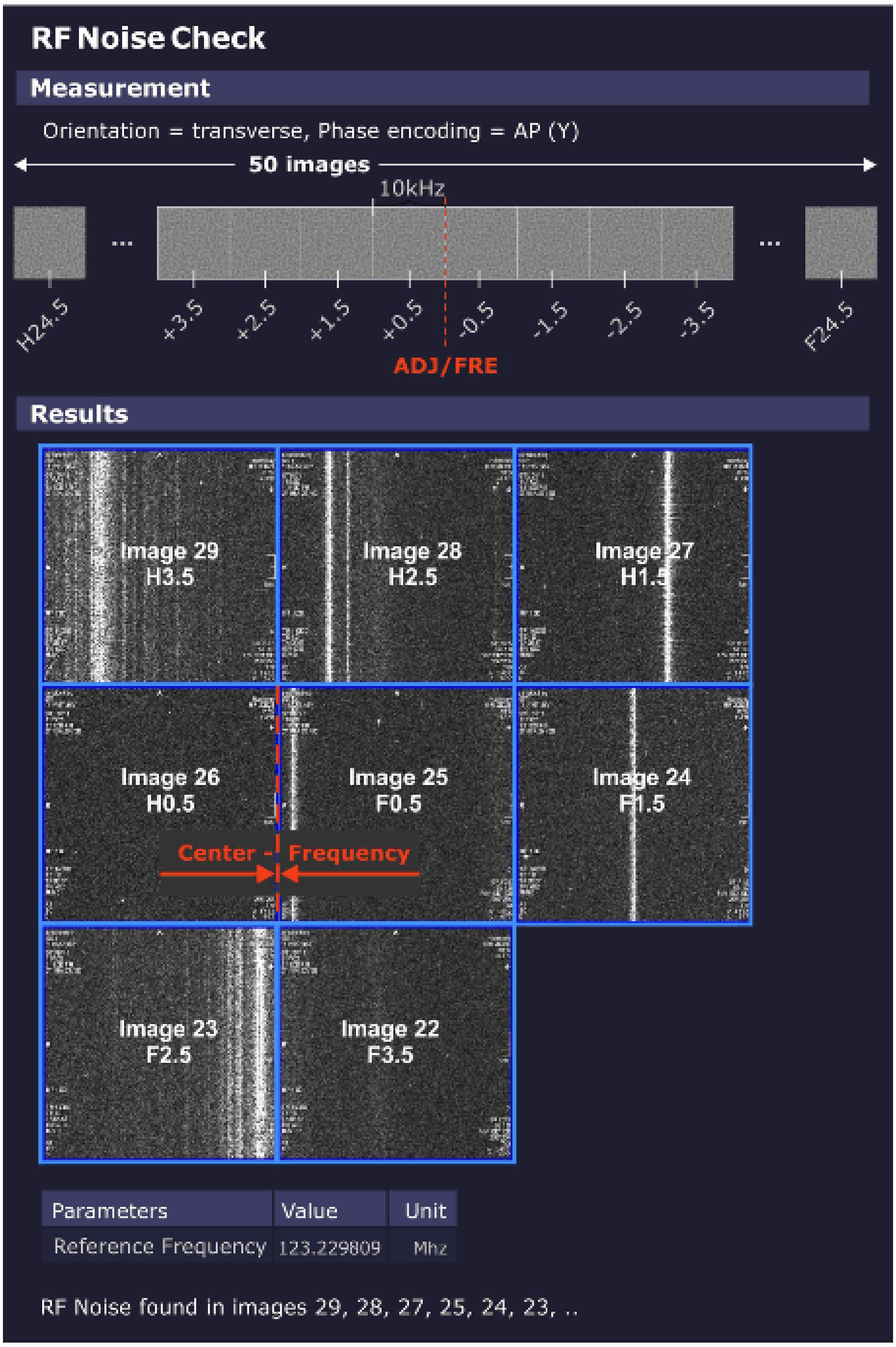
Help ID: AA585552-5E09-43B7-8E85-B9A8394037E2
Service Level: 3

General

The **RF Noise Check** is used to detect RF interferences which can disturb the MR-image.

Measurement

Fig. 364: RF Noise Check



The noise measurement is performed with the service sequence rf_noise. The sequence is without RF or gradient pulses, just an open receiver.

The sequence produces 50 images (i.e., slices). The difference between the images is the variation of the center frequency. Each image covers 10 kHz (256 times 39 Hz) so that the complete receiver bandwidth of +/-250 kHz is covered.

The border between the center slices 25, 26 is the current center MR system frequency.

Evaluation/Results

The evaluation algorithm looks for RF interference in the acquired image data i.e., dots and streak artifacts in phase-encoding direction and noise level.

Results in Report:

- RF interference according to the analyzed frequency bands.

4.3.6.19 PTX Image Check

Help ID: 82D6FBB2-442B-4B71-A95B-FD165371082B
Service Level: 3

General

A parallel transmit (pTX) array consists of 2 or more distinct transmitter channels where amplitude and phases can be modulated independently from each other.

Such a system can generate dedicated local magnetization patterns, e.g. exciting one spatial region of a patient or phantom while leaving other regions untouched.

This mechanism, however, is rather sensible to system instabilities or malfunctions of components in the imaging chain.

In order to detect such deviations, the service procedure runs the dynamic pTX sequence to produce a well-defined local magnetization pattern which is analyzed for image quality.

Measurement

For the measurement, the small spherical 170 mm water-filled phantom is used.

a homogeneous image is produced which allows to verify the correct positioning of the phantom.

The pTX sequence “dyn_pulse” is executed and generates a well-defined target magnetization pattern.

First, a homogeneous image is produced which allows to verify the correct phantom positioning.

Second, a homogeneously excited plane with an embedded dark (non-excited) rectangle is measured.

Subsequently, the Software analyzes the resulting images to detect potential system malfunctions.

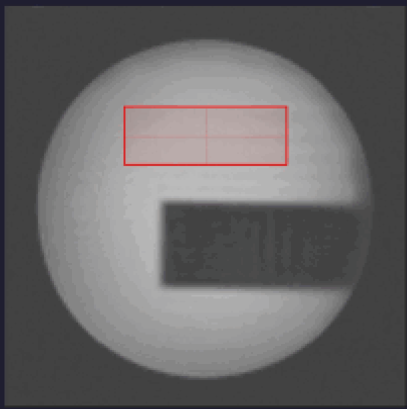
Evaluation/ Results

Fig. 365: PTX Image Check - Results

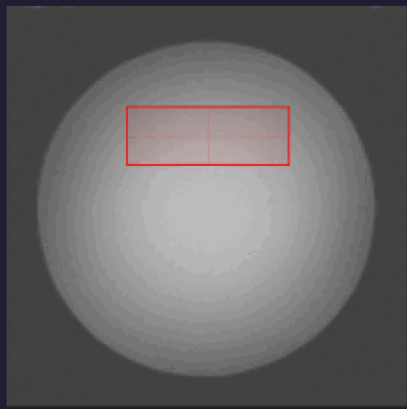
PTX Image Check

Results

Bright area



Protocol pTX, ROI: Bright



Protocol SE, ROI: Bright

Evaluation Results		
Parameter	Value	
		Tolerance
Artefact level	1.4 %	≤10 %
Brightness ratio	124 %	90 ... 130 %

Dark area

Protocol pTX, ROI: Dark

Evaluation Results		
Parameter	Value	
		Tolerance
Dark to bright contrast	87.8 %	≥70 %

First, a ROI is placed entirely in the **bright region** of the excitation pattern. This ROI is analyzed for relative image brightness (which should be similar to the brightness in the homogeneous image) and artifact level.

A second ROI is placed entirely inside the **dark region** (rectangle). The relative brightness of this second ROI, compared with the brightness of the first ROI, defines the image contrast.

4.3.6.20 Stability Check

Help ID: 796762F3-ECF9-458A-83A1-CAFDEC3BA27D
Service Level: 3

General

The **Stability Check** evaluates the stability of the MR signal under various sequence conditions.

A spin echo and a gradient echo are measured 256 times without phase encoding using fixed sequence parameters.

As a result, all raw data lines look alike (if the system is stable and not taking in account stochastic variations).

The stability of the echo signal across the measured lines is evaluated:

- Variations of the amplitudes
- Variations of phases
- Positions of the echoes
- Frequency perturbations

Measurement

In Standard Mode, a **spin echo** and a **gradient echo** are measured 256 times without phase encoding using fixed sequence parameters:

- TR = 301 ms
- TE = 20 ms

All **three orientations** are measured, so each gradient is performed as readout gradient once:

In addition, both **0 and 50 mm slice shift** positions are measured.

Evaluation/Results

The **spin echo sequence** uses a default setting of TE = 20 ms. Due to chosen TR = 301 ms and the inversion effect of the 180° pulse, it is sensitive to instabilities in the range of a few Hz up to 100 Hz which may be caused by e.g.:

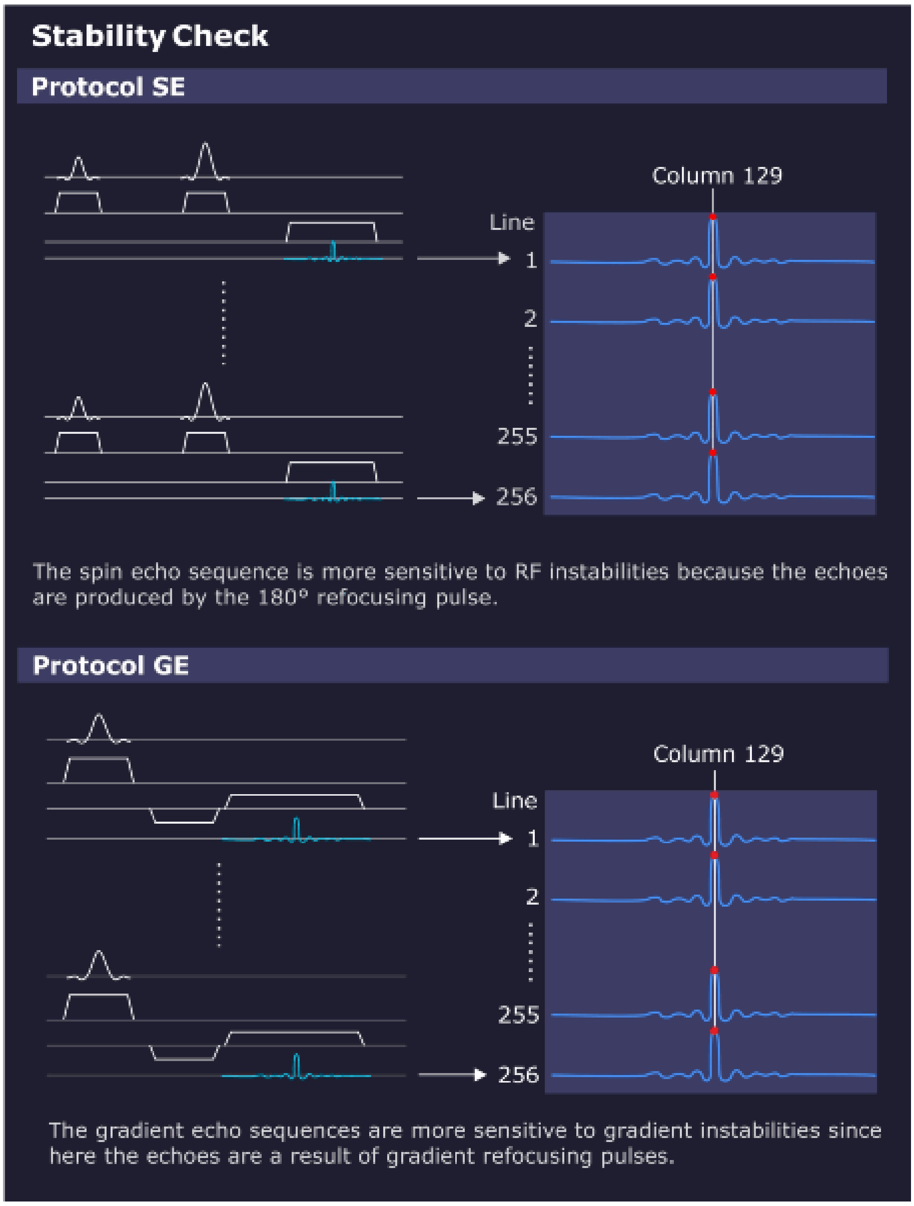
- 50/60 Hz AC power line hum;
- or 16.7 Hz AC trains.

The **gradient echo / flash sequence** also uses a default setting of TE = 20 ms and TR = 301 ms. Due to the chosen parameters and the missing 180° pulse, it is insensitive to 50 Hz line hum. The gradient echo is sensitive from DC up to a few Hz to slow external field interferences caused by e.g.:

- DC-driven local train systems like tram or subway;

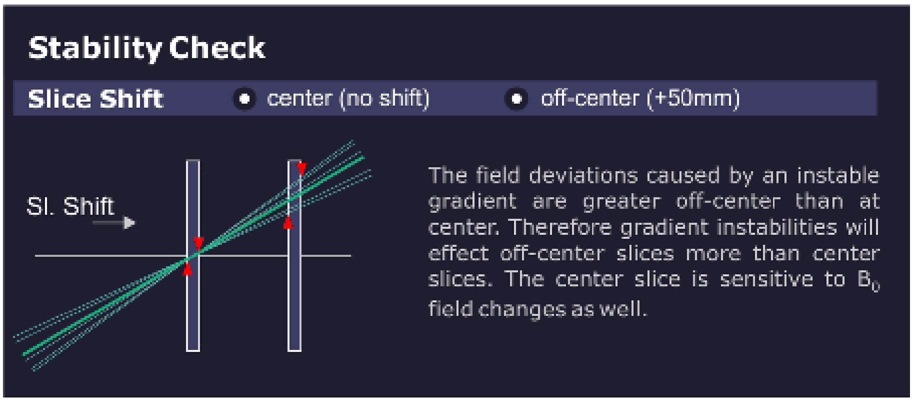
- Magnetic field instabilities;
- or mechanical vibrations.

Fig. 366: Stability Check Evaluation



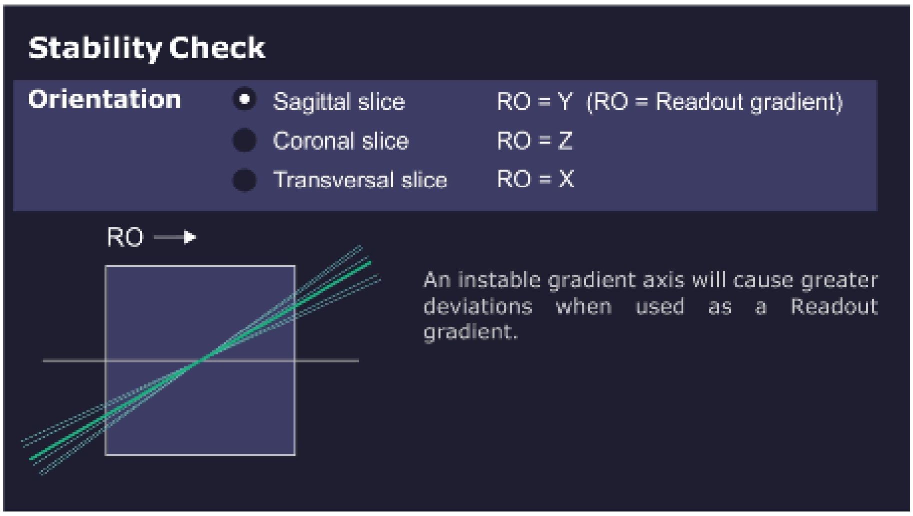
The measurement result of the **center slice** is sensitive to instabilities of the B_0 field. The **50 mm offset slice** is more sensitive to gradient-like instabilities.

Fig. 367: Slice Shift



The **orientation** determines the role of each gradient axis. For example, lets assume there is an instability in the X gradient. The values for the sagittal orientation (X = slice selection) may be in spec. while the spec for the transverse orientation (X = read out) could be out of spec.

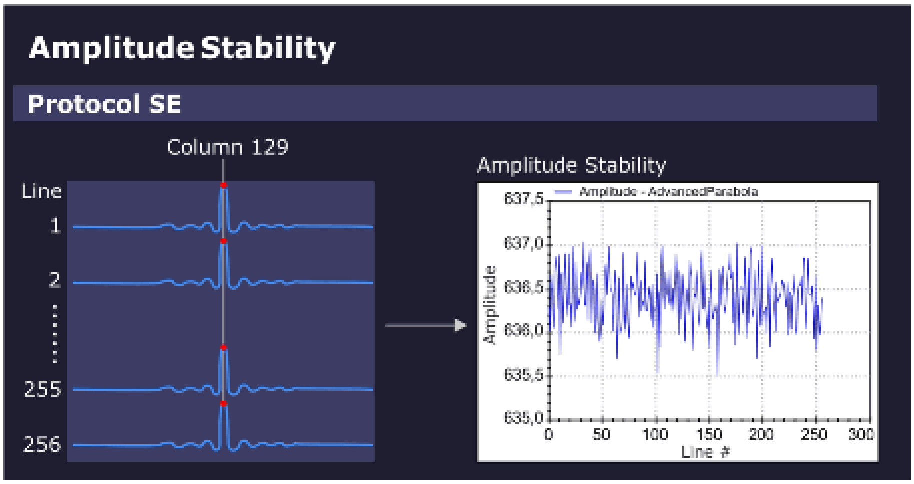
Fig. 368: Orientation



Amplitude stability (spin echo):

The amplitude value in the maximum signal column (e.g. column 129) is displayed in the graphics output.

Fig. 369: Amplitude Stability (Spin Echo)



Phase stability (spin echo):

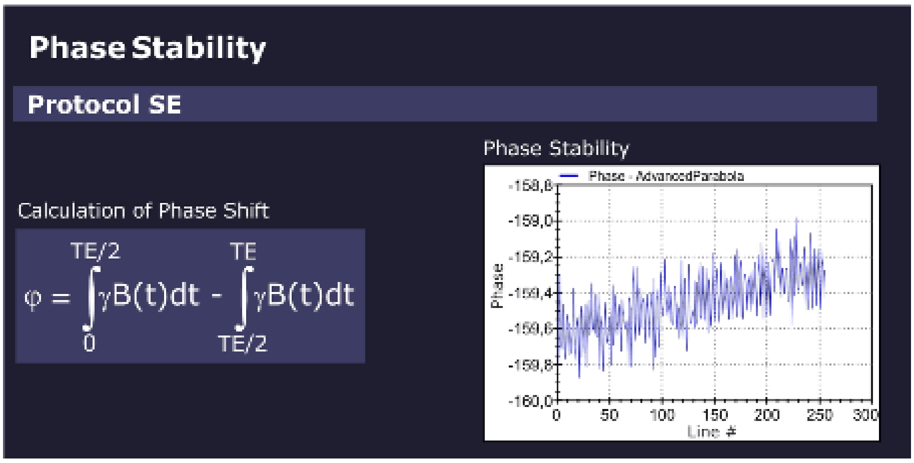
The phase stability of the spin echo is displayed in the graphics output.

The phase shift of a spin echo is determined by the integral across the magnetic field from a 90˚ to a 180˚ pulse minus the integral from a 180˚ pulse to the echo.

For a constant field or a slowly varying field, both integrals cancel out each other.

However, for an oscillating field with a period of TE (20 ms = 1/50 Hz), this expression becomes maximum. In other words, the spin echo measurement with the default setting of TE = 20 ms (and TR = 301 ms) is sensitive to interferences which may be caused by 50/60 Hz AC power line hum or 16.7 Hz trains.

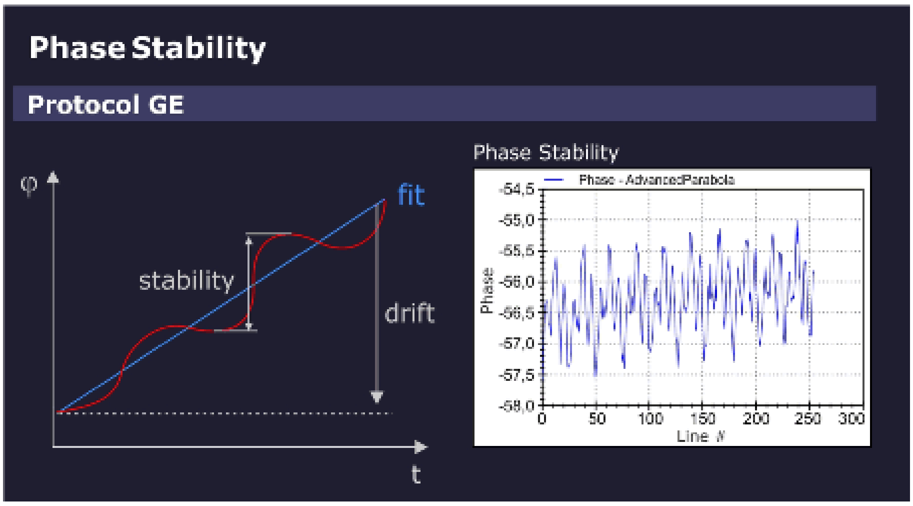
Fig. 370: Phase Stability (Spin Echo)



Phase stability (gradient echo):

For the gradient echo sequence, the phase stability is evaluated by fitting a straight line to the phase data. When you subtract the value of the linear fit from the phase data, the phase stability changes. The slope of the straight line indicates the linear phase drift, caused by slow external field interferences, for example, moving elevators, cars, trucks but also mechanic vibrations.

Fig. 371: Phase Stability (Gradient Echo)



Stability of echo position:

For both measurements, spin echo and gradient echo, the position of the echo peak column is determined for every measurement line and checked against specifications.

The cause of an unstable echo position may be gradient-like interferences from the environment or from the gradient itself.

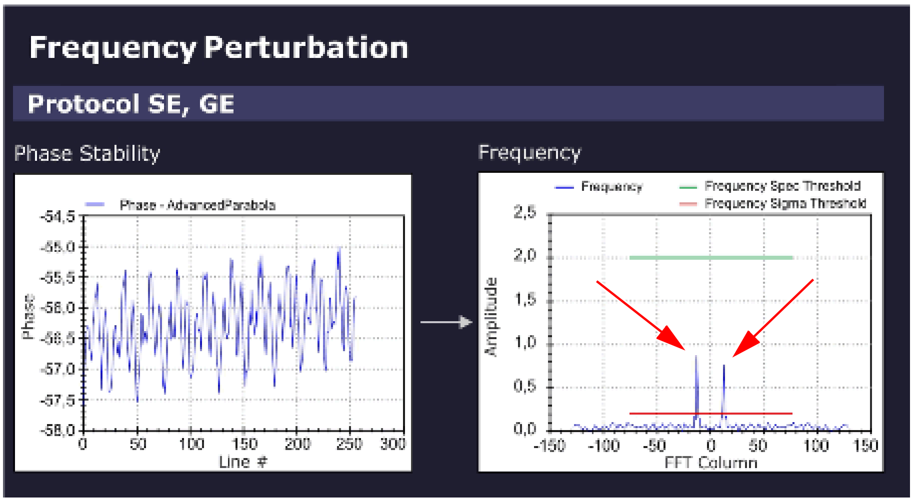
The sensitivity for the direction of the interference is determined by the orientation of the readout gradient.

Frequency perturbation:

The Software differentiates the raw data of the stability measurements and checks for sidebands (ghosts).

In the evaluation result, a frequency perturbation value in percent and the Frequency (peak) position in the FFT Column is displayed.

Fig. 372: Frequency Perturbation (sidebands)



Example: (Frequency Perturbation of 0.8%, Frequency Position -13)

- The sampling rate in the stability measurement is one sampling point every 301 ms (default TR = 301 ms).
- Consequently, the Nyquist-frequency = +- 1/602 ms = +-1.66 Hz. The limiting frequencies at +-128 in the frequency diagram above correspond to +1.66 Hz and – 1.66 Hz, respectively.
- The perturbation frequencies are located at +-13 units, i.e. at 13/128 * 1.66 Hz = +-0.17 Hz.
  » The 300th harmonic of 0.17 Hz yields a 50 Hz interference.

Results in Report:

Measured value	Description
Amplitude Stability	Range of amplitudes (peak to peak)
Phase Stability	Deviation of phase from linear fit
Phase Drift	Drift of phase
Peak Column Stability	Range of peak column
Peak Column Average	Mean peak column
Frequency Perturbation	Relative amplitude of the highest contributing disturber frequency; zero if no frequency is above the relative threshold nor above (noise level * noise factor).
Frequency Position	Position of the highest contributing disturber frequency; zero if no frequency is above the relative (sigma) threshold nor above the (noise level * noise factor).

Troubleshooting

Symptom/Deviation	Possible Cause/Service Action
Amplitude	RF-instability problem, check: ■ Stability of RF System / TX ■ Stability of RF System / RX

Symptom/Deviation	Possible Cause/Service Action
Phase	Magnetic field fluctuation, check: ■ After magnet ramping or after heavy gradient load (e.g. DTI-, EPI- or StabilityLongTermCheck measurement) wait at least 1 hr. before running Stability Check; ■ Run QA-Expert / Field Stability Check.
Depending on orientation	Instability of a gradient axis, check: ■ The Gradient System hardware. ■ Run service sequence “CalcAr” in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.
Independent of orientation	Magnetic field instable or external field interference, check: ■ The Gradient System hardware. ■ See hints “Only in gradient echo”.

Symptom/Deviation	Possible Cause/Service Action
Only slice-shift dependent	Most likely, a gradient problem, check: ▪ The Gradient System hardware ▪ especially, the gradient axis used for slice-selection
Only in gradient echo	1. Static or slowly varying magnetic field problem? In the vicinity of the site, check for: - Trains; interference from power lines and rails. - Elevators. - Car/truck traffic and parking. 2. Very low amplitude due to static magnetic field (B_o) homogeneity problem? - Compare amplitude with spin echo result (both amplitudes should be similar). - Run QA / Phantom Shim Check. 3. Phase drift out of spec., B_o-drift problem? - System hot? Excessive usage prior to Stability Check measurement (e.g. repeated DTI-, EPI- or StabilityLongTermCheck measurements); - System too cold? Wait at least 2 hr before measurement. 4. Peak column stability out of spec. i.e., fluctuation of echo position due to gradient field instability? - Check the Gradient System hardware - Run service sequence “CalcAr” in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.

Additional hints:

- Switch off components one by one to isolate the cause of the instability, e.g.:
  - Cold head
  - Lighting in magnet room
  - Shim PSU
  - Disable gradient axes with DIP switch at DGSU board (first each single axis, then complete gradient power amplifier)
- For periodic instability, vary TR and TE (Stability Check / Expert Mode) to confirm or rule out the frequency of interference:
  - Vary TR to avoid being synchronous to the interference.

- Vary TE to become more sensitive to particular frequencies.
- If an external interference is suspected, repeat measurement at different times e.g. during night.

4.3.6.21 Synthesizer Check

Help ID: 100FDE50-E606-4FB0-B37C-3F6B6E932DAF
Service Level: 3

General

The **Synthesizer Check** procedure is an important test to verify proper operation of the synthesizer.

Validation of the frequency and phase quality of the synthesizer is important for the accuracy of the slice position.

Measurement

- Verification of the frequency and phase operation of the RF reference signal is performed by using the MR signal itself. After demodulation of the received MR signal, the reference signal is compared to the MR signal in terms of frequency and phase.
  - In one test, the reference signal is varied in frequency over the whole range of the MR system.
  - A second test modifies the phase in fixed intervals.

    » With this test, the entire operating range of the oscillator can be tested. The test is performed for one RX-NCO (Numerically Controlled Oscillator) from group 1 and one RX-NCO from group 2. The two RX-NCOs are validated and can be used for testing the other NCOs.
- Loop-measurements between the tested RX-NCOs and the TX-NCOs approve the correct function of the TX-NCOs.
- In additional steps, all other RX-NCOs are tested against the approved TX-NCOs by testing group 1 RX-NCOs and group 2 RX-NCOs separately.

Evaluation/Results

Results in Report:

- Frequency- and phase-deviations for all used NCOs.

4.3.6.22 Long Term Stability Check

Help ID: 8B498548-2A22-4DDB-8B42-C258D604AAF0
Service Level: 3

General

The **Long Term Stability Check** determines the scanner stability which is important for Functional MR Imaging (fMRI).

The fMRI application visualizes activities in the brain. The patient is exposed to a periodic optical, acoustical or other form of stimulation. During the exposure, a series of MR images is acquired. Subsequently, the images are evaluated for signal intensity changes. The stimulated, activated regions of the brain are localized by their typical, very little variations of signal intensity.

However, these little variations are easily superimposed by instabilities of the MR system or the environment. This QA step makes sure, they are below a specified level.

Typical sources of instabilities are:

- Local B_0-field variation caused by temperature variation in the shim unit (shim iron plates) generates frequency instabilities. The result is a virtual movement of the measured object during acquisitions.
- B1-field variations due to instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, also leading to virtual movement of the object.
- Mechanical vibrations, e.g., by the cold head, produce signal intensity variation caused by the movement of the measured object relative to the magnet isocenter.
- Movement of phantom fluid causes phase errors.

Measurement

A series of 512 measurements (EPI_FID sequence) with 11 slices is performed and repeated once again. The first measurement is performed with a low flip angle of 30 degrees for stability evaluation of the B1-field (i.e., caused by instabilities of the TX chain).

The second measurement uses a higher flip angle of 90(70) degrees to minimize the influences of instable RF but focus on B_0-instabilities and gradient-like interferences.

> Wait at least 15 minutes between positioning the phantom and starting the measurement. Even small movements in the phantom liquid have a negative effect to the measurement results.

Evaluation/Results

The Software places an evaluation ROI in the central slice and performs the following graphical and numerical evaluations:

- Mean values of a 15x15 ROI related to the mean value of all measurements as a function of the image number
- Standard deviation of mean values of a 15x15 ROI (linear and logarithmic plot)
- Measurement protocol values

- Statistics of mean values (for ROI size 15x15 and 25x25):
  - Mean Image Intensity
  - Peak to Peak Variation
  - Relative Peak to Peak Variation
  - Standard Deviation
  - Total Relative Linear Drift
  - Standard Deviation from Linear Fit
- Image quality parameters such as SNR and ghosting

4.3.7 Create Protocol/Report

Help ID: 7159C0B3-6DC2-484D-A26F-8195992408B3
Service Level: 7

Create Protocol/Report

Status

Status displays the current status of the procedures.

Overview of critical procedures:

Overview of critical procedures list the tasks which are Not OK or overdue.

Test Result

The tester has to evaluate the test results as defined and listed under **Test Result**.

- Select the applicable evaluation.

Without selection of the Test Result the report cannot be saved.

Remarks:

Remarks provide the possibility to enter notes or to describe deficiencies in detail.

Printed Signature Name

The Tester has to enter his name.

- Enter your name

Without a name the report cannot be saved.

Save and Create Report

After completion of the Safety –Related Tests tasks the entries must be saved. Otherwise the entries will be lost.

- Click **Save and Create Report** to save our entries and generate a report.
  - » The **Report** is now saved and is then part of the Site-specific Data.

Use the **Reports platform** to display or to print the protocols.

4.4 Preventive Maintenance

4.4.1 Measurement Devices

Help ID: 807BE59F-0985-4043-B4B5-C1E49AA03C45
Service Level: 7

Measurement Device

Enter the type and serial numbers of all measurement devices used to install or maintain the system.

Enter the "Next Calibration Due" date, if requested.

If devices have no part numbers enter 'n.a.' in the corresponding field.

In case the calibration date is on monthly base, enter the last day of the calibration month in the field 'Next Calibration Due'.

4.4.2 Preventive Maintenance

Help ID: F2AF3314-035A-4294-8FFB-87DD937AEEA3
Service Level: 7

General information

To ensure the continuous safety, quality, and reliability of these products, periodic maintenance must be performed in accordance with the Maintenance Instructions / Safety-related tests.

Periodic maintenance includes:

- Safety-related tests (not included in these Maintenance Instructions)
- Preventive Maintenance

> i RF room door
>
> The RF room door is not part of the MAGNETOM system maintenance and not considered in these instructions. The RF room door has been maintained according to the manufacturer instructions.

Safety Information

WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

Documentation

A protocol of each maintenance activity must be generated. All fields must be completed.

Each work step or test performed successfully has to be marked “OK” in the test protocol. The designation n.a. (not applicable) indicates that the checkpoint or measured value is not used for this system or that the checkpoint is not due (e.g. interval is more than one year). If an error occurs that cannot be eliminated, mark the respective test with “not OK”.

The Maintenance Protocol has to be archived by the operator in the System Owner Manual. We recommend that country organizations also compile such documentation to ensure optimum product operation, e.g. should damage occur.

Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

Timeframe for the Maintenance Schedule

Tab. 21 Maintenance Schedules Cooling and Consoles

Component / activity	Initial[1] (months)	Interval[2] (months)	Time required (minutes)	Required tools and parts
Computer	18	12	10	Vacuum cleaner
TFT monitor	18	12	10	
Gradient filters	18	12	10	
Cooling	18	12	10	
SEP (optional)	18	12	40	50/ 41 mm open-ended wrench or water pump pliers, needle-nose pliers, cloth to absorb water, bucket, flashlight
Annual average time: 80 minutes				

1. Initial maintenance means the first maintenance schedule after installation of the system. The initial maintenance has to be performed, at the latest, 18 months after installation, unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.
2. Interval means the recurrent maintenance interval after the initial maintenance. The recurrent maintenance has to be performed, at the latest, every 12 months, unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.

Tab. 22 Maintenance Schedules Fixed Patient Table

Component / activity	Initial[1] (months)	Interval[2] (months)	Time required (minutes)	Required tools and parts
Option fixed patient table - spindle and guide rails	18	24	30	Lubricant
Annual average time, fixed patient table: 15 minutes				

1. Initial maintenance means the first maintenance schedule after installation of the system. The initial maintenance has to be performed, at the latest, 18 months after installation, unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.
2. Interval means the recurrent maintenance interval after the initial maintenance. The recurrent maintenance has to be performed, at the latest, every 24 months, unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.

Tab. 23 Maintenance Schedules Option Dockable Patient Table

Component / activity	Initial[1] (months)	Interval[2] (months)	Time required (minutes)	Required tools and parts
Optional dockable table: - spindle and guide rails - docking station	18[3]	24 [4]	60	Lubricant
Dockable table battery	30 [5]	36 [6]	30	Battery
Annual average time dockable table: 40 minutes				

1. Initial maintenance means the first maintenance schedule after installation of the system.
2. Interval means the recurrent maintenance interval after the initial maintenance.
3. The initial maintenance has to be performed, at the latest, 18 months after installation, unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.
4. The recurrent maintenance has to be performed, at the latest, every 24 months unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.
5. The initial maintenance has to be performed, at the latest, 30 months after installation unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.
6. The recurrent maintenance has to be performed, at the latest, every 36 months unless a daily remote monitoring of the system status is performed in which case a condition based maintenance at a later stage is possible.

Cold Head

Tab. 24 Cold Head Replacement

Activity	Interval (months)	Time required	Note
Cold Head Replacement	24 [1]	480	Service personnel training on the Sumitomo cold head is offered by SMT. Instructions for cold head replacement are shown in the video “Sumitomo Refrigerator System”. The video is a supplement to the SMT course.

1. The cold head has to be replaced, at the latest, every 24 months, unless a daily remote monitoring of the magnet refrigerator status is performed in which case a condition based exchange at a later stage is possible.

4.4.2.6 Cleaning the computer

Help ID: 8921EB69-0007-435F-9F8B-9858439EEFAE
Service Level: 7

Computer MRAWP (MRC) - MRWP (MRSC)

Required time: 10 min

Required tools: Vacuum cleaner

PM Cleaning the computer

MRAWP (MRC)

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

MRWP (MRSC) if installed

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

Fig. 373: Example, computer

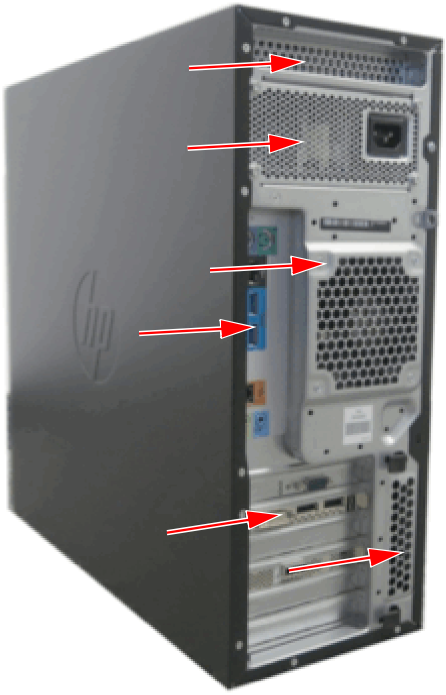


Fig. 374: Example, computer

4.4.2.7
Cooling system - general checks

Help ID: 4A69574F-CD52-4B99-B495-4AD8AAC27C8B
Service Level: 7

Cooling

Required time: 10 min

Required tools: n.a.

PMP Cooling system - general checks

- Navigate to > **Administration Portal** > **Diagnose** > **Magnet & Cooling** > select sub platforms
  - Magnet Status
  - Compressor Status
  - SEP / Chiller Status
  - Cabinet Status
  - Gradient & Bore Temp Status
  - Room Status
- Check the cooling status of the components

4.4.2.8
Checking the gradient filter fans

Help ID: 581AA7E8-A84A-48FB-96D0-6FB8C7647FE6
Service Level: 7

Gradient filters

Required time: 10 min

Required tools: n.a.

The gradient filters are ventilated by 3 fans per axis.

PM Checking the gradient filter fans

Fig. 375: Gradient filter fans

» If one or 2 fans per axis are defective, replace the fan assembly.

» If all 3 fans per axis are defective, the corresponding gradient filter has been damaged due to overheating. Replace the fan assembly **and the corresponding gradient filter.**

4.4.2.9
Cleaning the strainer of the primary water circuit

Help ID: 219A94CE-4DC4-466D-A7B8-BFF08F993A0A
Service Level: 7

SEP (optional)

Required time: 35 min

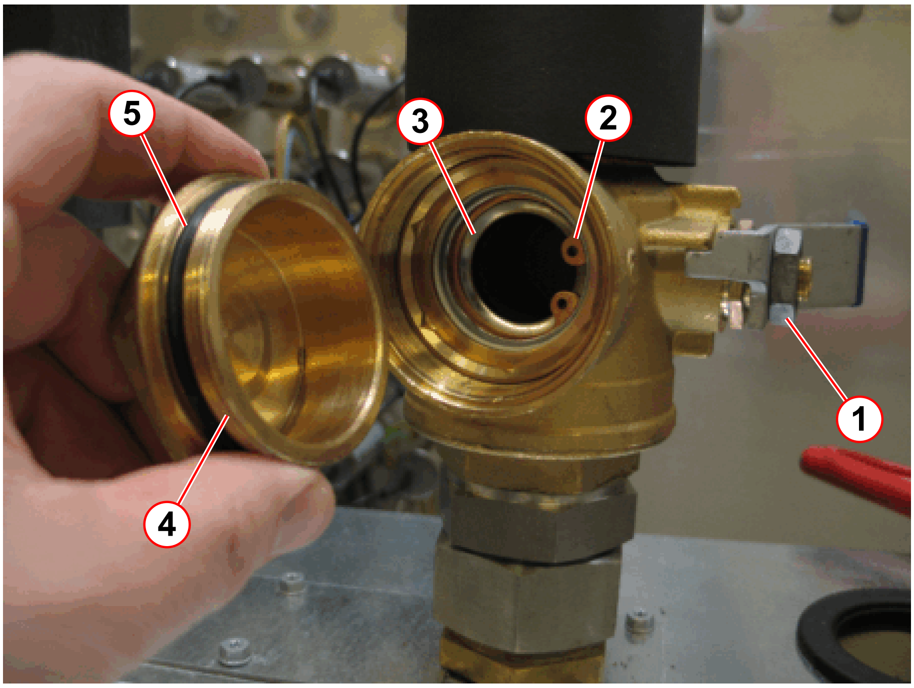
Required tools: 50 mm or 41 open-ended wrench, needle-nose pliers, cloth to absorb water, bucket, flashlight

> Any cooling water containing ethylene glycol must be collected while draining. Return the cooling water to the water circuit during the next filling.
>
> Avoid contact with eyes and skin.

Cleaning the strainer of the primary water circuit

- Switch off the system.
- Switch off the compressor unit using the main power switch (➔ 1/Fig. 378 Page 584).
- Check for leaks.
- Slowly close the valve (➔ 1/Fig. 376 Page 582).

Fig. 376: SEP - valve with strainer



(1) Handle - closed
(2) Circlip
(3) Strainer
(4) Screw cap
(5) O-ring

> The closed valve contains approximately 200 mL of cooling water. Use a bucket to collect the water.

PM Cleaning the strainer of the primary water circuit

- Remove the screw cap (➔ 4/Fig. 376 Page 582) slowly by using an open-ended wrench.
- Use a bucket to collect the water.
- Use a cloth to absorb some water from the strainer hole.
- Remove the circlip (➔ 2/Fig. 376 Page 582) by using needlenose pliers.
- Remove the strainer (➔ 3/Fig. 376 Page 582).

- Inspect the strainer for contamination and clean it by rinsing under tap water.

Fig. 377: Strainer

Reassemble the valve in reverse order.

- Install the strainer.
- Re-attach the strainer with the circlip.
- Check the O-ring (→ 5/Fig. 376 Page 582) of the screw cap.
- Screw the screw cap into the shut-off valve.
- Slowly open the shut-off valve.
- Check for leaks on the screw cap.

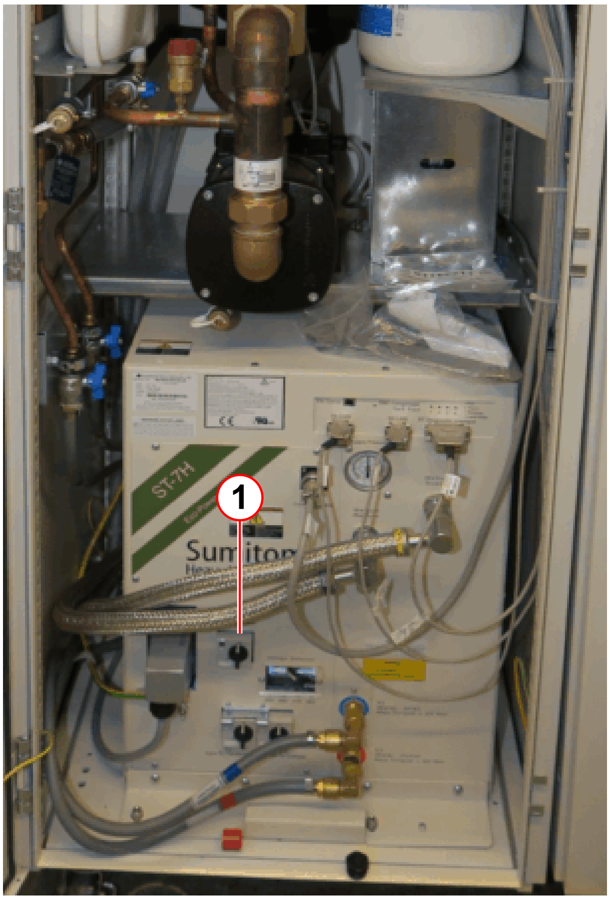
Final Steps

- Switch on the system.
- Open > **Administration Portal** > **Diagnose** > **Magnet & Cooling** > **SEP / Chilller Status**
- Check the **Primary Water - Temperature and Flow**.
- If necessary, advise the customer to initiate corrective action immediately.

Cleaning the strainer of the secondary water circuit

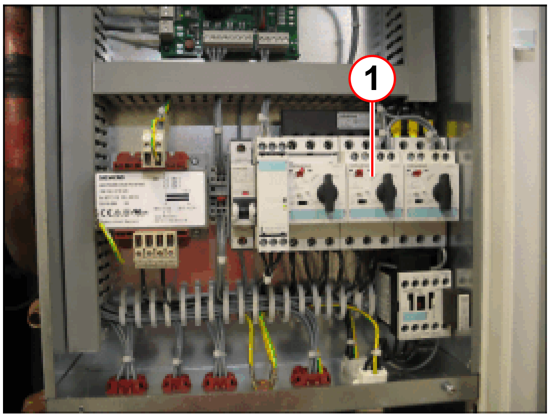
- Switch off the compressor unit using the main power switch (→ 1/Fig. 378 Page 584).
- Switch off the water pump using Q2 .

Fig. 378: SEP



(1) Main power switch

Fig. 379: SEP electronic cabinet



(1) Q2 water pump

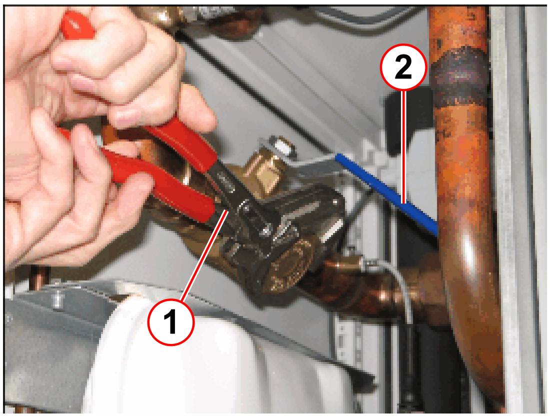
PM Cleaning the strainer of the secondary water circuit

- Slowly close the ball valve (→ Fig. 380 Page 585).

Fig. 380: SEP - ball valve closed

- Remove the valve cap slowly by using an open-ended wrench.

Fig. 381: SEP - ball valve with strainer

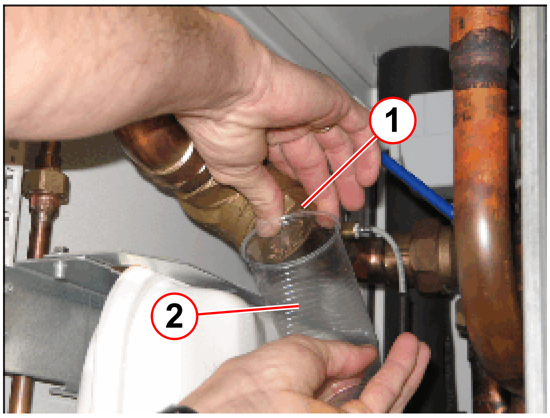


(1) Open-end wrench
(2) Ball valve - closed

> ℹ The closed ball valve contains approximately 200 ml of cooling water. Use a cup to collect the water.

- Use a cup to collect the water (→ 2/Fig. 382 Page 586).

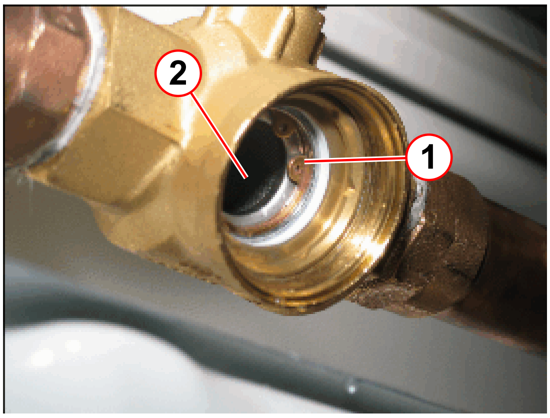
Fig. 382: SEP - ball valve



(1) Valve cap
(2) Cup

- Remove the circlip using the needlenose pliers. (→ 1/Fig. 383 Page 586)

Fig. 383: SEP - strainer



(1) Circlip
(2) Strainer

- Remove the strainer.

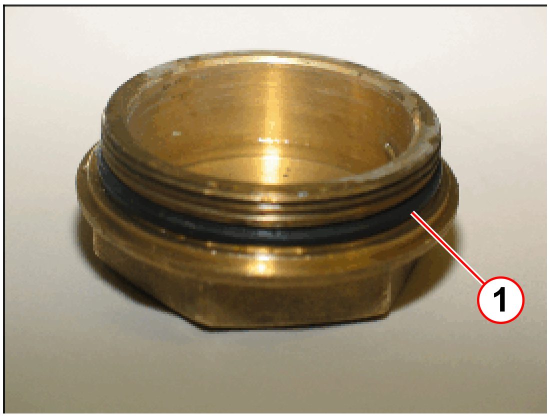
Fig. 384: Strainer

- Inspect the strainer for contamination and clean it by rinsing under tap water.
- Open the drain/fill port valve using the back of the cap.
- Pump water into the circuit.
- Close the valve when the pressure manometer indicates the correct pressure:

Reassemble the strainer.

- Re-install the strainer.
- Re-attach the strainer with the circlip.
- Check the O-ring of the valve cap.

Fig. 385: Ball valve cap



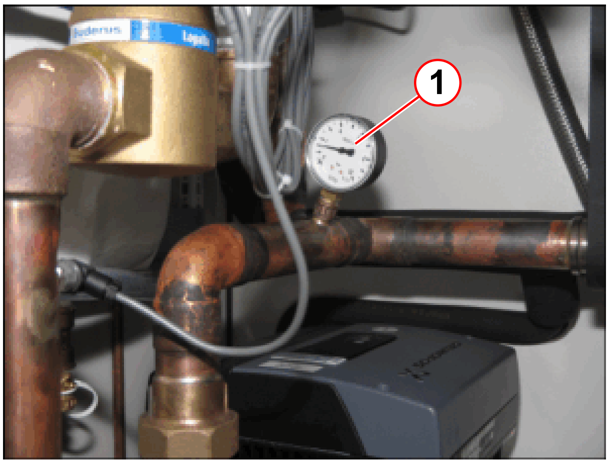
(1) O-ring

- Screw the valve cap into the ball valve.
- Slowly open the ball valve.
- Check for leaks on the valve cap.

Final Steps

- Check the water pressure.
    » Refill water if pressure < 1 bar.

Fig. 386: SEP water pressure



(1) Pressure manometer

	SEP pressure manometer
Static pressure	1.0 $_{-0.5}$ bar

Fig. 387: Pressure manometer

- Switch on the water pump Q3.
- Switch on the compressor unit.
- Check for leaks.

4.4.2.10 Cleaning the strainer of the secondary water circuit

Help ID: 7ED14F43-5673-4263-8184-94A55719501E
Service Level: 7

SEP (optional)

Required time: 35 min

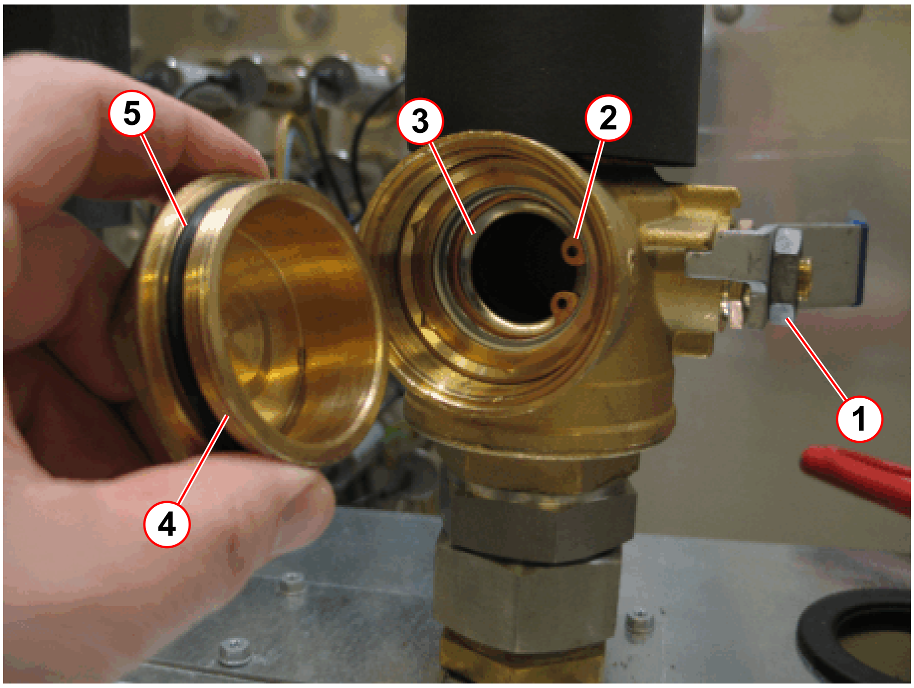
Required tools: 50 mm or 41 open-ended wrench, needle-nose pliers, cloth to absorb water, bucket, flashlight

> Any cooling water containing ethylene glycol must be collected while draining. Return the cooling water to the water circuit during the next filling.
>
> Avoid contact with eyes and skin.

Cleaning the strainer of the primary water circuit

- Switch off the system.
- Switch off the compressor unit using the main power switch (→ 1/Fig. 390 Page 592).
- Check for leaks.
- Slowly close the valve (→ 1/Fig. 388 Page 590).

Fig. 388: SEP - valve with strainer



(1) Handle - closed
(2) Circlip
(3) Strainer
(4) Screw cap
(5) O-ring

> The closed valve contains approximately 200 mL of cooling water. Use a bucket to collect the water.

PM Cleaning the strainer of the primary water circuit

- Remove the screw cap (→ 4/Fig. 388 Page 590) slowly by using an open-ended wrench.
- Use a bucket to collect the water.
- Use a cloth to absorb some water from the strainer hole.
- Remove the circlip (→ 2/Fig. 388 Page 590) by using needlenose pliers.
- Remove the strainer (→ 3/Fig. 388 Page 590).

- Inspect the strainer for contamination and clean it by rinsing under tap water.

Fig. 389: Strainer

Reassemble the valve in reverse order.

- Install the strainer.
- Re-attach the strainer with the circlip.
- Check the O-ring (➔ 5/Fig. 388 Page 590) of the screw cap.
- Screw the screw cap into the shut-off valve.
- Slowly open the shut-off valve.
- Check for leaks on the screw cap.

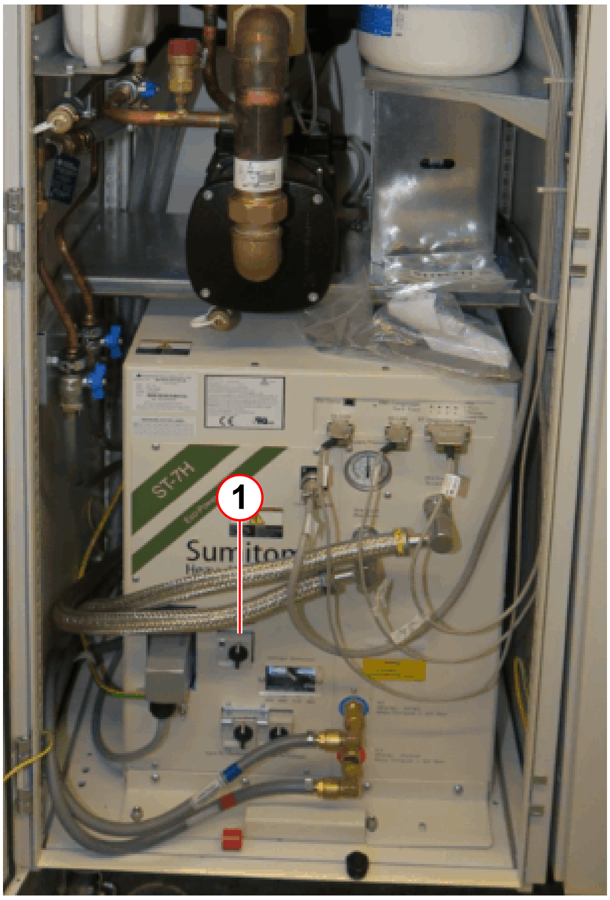
Final Steps

- Switch on the system.
- Open > **Administration Portal** > **Diagnose** > **Magnet & Cooling** > **SEP / Chilller Status**
- Check the **Primary Water - Temperature and Flow**.
- If necessary, advise the customer to initiate corrective action immediately.

Cleaning the strainer of the secondary water circuit

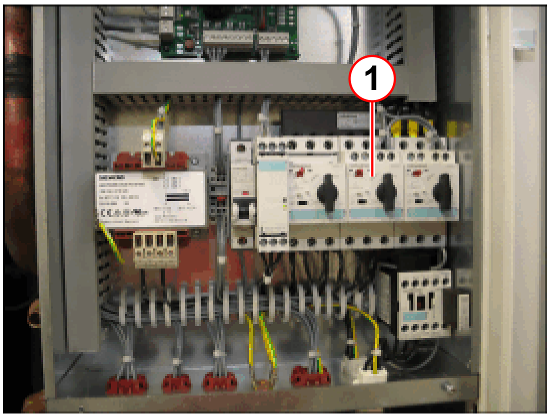
- Switch off the compressor unit using the main power switch (➔ 1/Fig. 390 Page 592).
- Switch off the water pump using Q2 .

Fig. 390: SEP



(1) Main power switch

Fig. 391: SEP electronic cabinet



(1) Q2 water pump

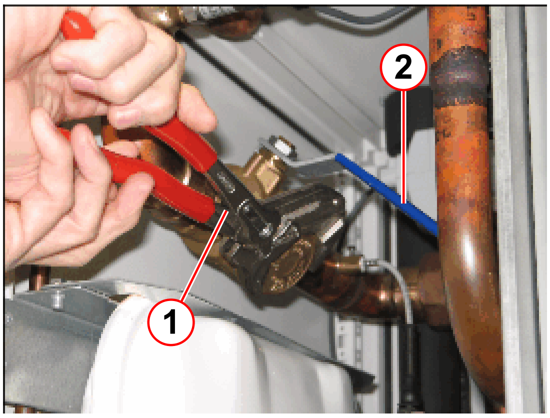
PM Cleaning the strainer of the secondary water circuit

- Slowly close the ball valve (➔ Fig. 392 Page 593).

Fig. 392: SEP - ball valve closed

- Remove the valve cap slowly by using an open-ended wrench.

Fig. 393: SEP - ball valve with strainer

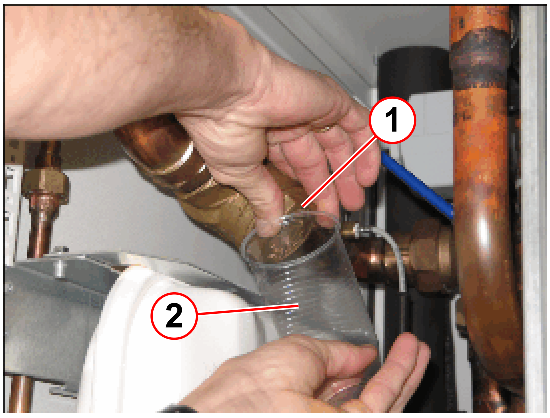


(1) Open-end wrench
(2) Ball valve - closed

> The closed ball valve contains approximately 200 ml of cooling water. Use a cup to collect the water.

- Use a cup to collect the water (➔ 2/Fig. 394 Page 594).

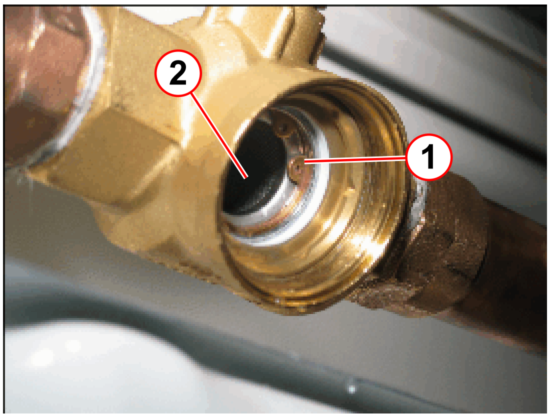
Fig. 394: SEP - ball valve



(1) Valve cap
(2) Cup

- Remove the circlip using the needlenose pliers. (➔ 1/Fig. 395 Page 594)

Fig. 395: SEP - strainer



(1) Circlip
(2) Strainer

- Remove the strainer.

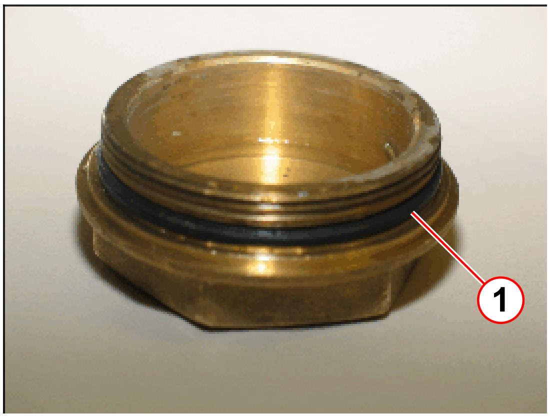
Fig. 396: Strainer

- Inspect the strainer for contamination and clean it by rinsing under tap water.
- Open the drain/fill port valve using the back of the cap.
- Pump water into the circuit.
- Close the valve when the pressure manometer indicates the correct pressure:

Reassemble the strainer.

- Re-install the strainer.
- Re-attach the strainer with the circlip.
- Check the O-ring of the valve cap.

Fig. 397: Ball valve cap



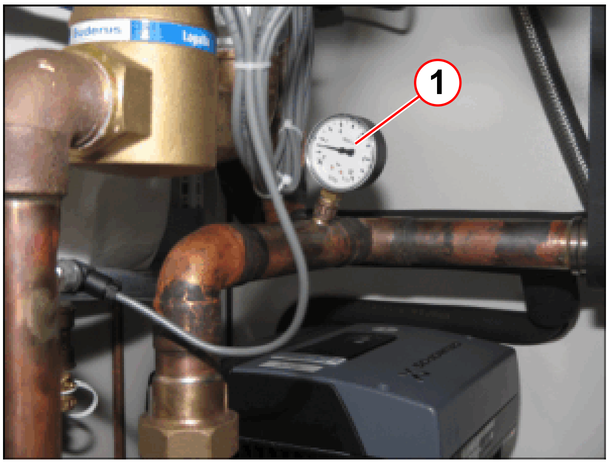
(1) O-ring

- Screw the valve cap into the ball valve.
- Slowly open the ball valve.
- Check for leaks on the valve cap.

Final Steps

- Check the water pressure.
    » Refill water if pressure < 1 bar.

Fig. 398: SEP water pressure



(1) Pressure manometer

	SEP pressure manometer
Static pressure	$1.0_{-0.5}$ bar

Fig. 399: Pressure manometer

- Switch on the water pump Q3.
- Switch on the compressor unit.
- Check for leaks.

4.4.2.11
Replacing the cold head

Help ID: BBFAB5E7-E7E4-42C3-8F27-C01B020DD0AC
Service Level: 7

Cold Head

PM Replace the cold head

Startup Date: ☐ . .

Date of Last Replacement: ☐ DD MMM YYYY

Required time: 8 hours

Required parts and tools: n.a.

The cold head has to be replaced latest every 24 months unless a daily remote monitoring of the magnet refrigerator status is performed in which case a condition based exchange at a later stage is possible.

If there is no daily monitoring of the system:

» Replace the cold head every 24 months

Training is required for cold head replacement. The cold head replacement procedure is not included in this description.

The magnet must be ramped to zero field for the replacement of the cold head.

- Enter the cold head replacement date in the Report.

The electronic low pressure switch should be tested after a cold head exchange.

4.4.2.12
Checking the TFT monitor adjustments

Help ID: F2633193-ECE1-4E54-BB6F-79F947B4CB61
Service Level: 7

TFT monitor

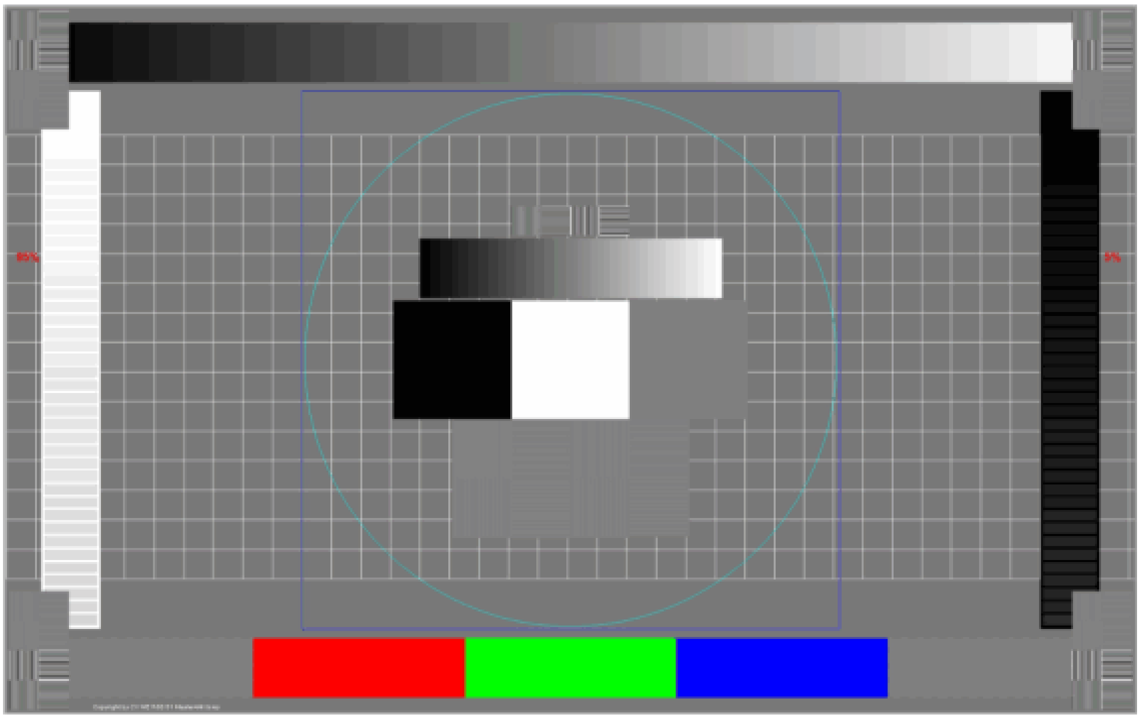
Required time: 10 min

Required tools: n.a.

Calling up the test image

- Simultaneously press **TAB** and **DEL** and ↦ (send to node 1) keys on the console to leave the kiosk mode.
  » Enter user and password
- Open the file explorer (by pressing **Windows** and the **E** key).
- Navigate to **This PC** > **K:** > **SeSo** > **TestImages**
- Double-click the file name: Master4All_1920x1200.bmp
  » Paint opens with the test image.
- Move the **Paint** window over to the left monitor.
- Open the test image in full screen mode (press key **F11** or select tab card **View** > **Full screen)**.

Fig. 400: Test image - Master4All_1920x1200



Check

PMA Checking the TFT monitor

- Check the MRAWP (MRC) TFT monitor using the test image.
  » All colors are displayed.
  » Gray-scale is correctly displayed.
  » Geometry is correct.
- Press the **ESC** key to close the full screen mode.
- Move the **Paint** window over to the right monitor.

- Open the test image in full screen mode (press key **F11** or select tab card **View** > **Full screen)**.
- Check the MRAWP TFT monitor using the test image.
  - » All colors are displayed.
  - » Gray-scale is correctly displayed.
  - » Geometry is correct.
- If the MRWP (MRSC) TFT monitor (option) installed, check the MRWP TFT monitor following the same procedure.

If any adjustment is necessary, refer to 'Replacement of Parts REP'; 'Visually inspecting/ evaluating the TFT monitor' in detail.

Set the normal mode

- Press the **ESC** key to close the full screen mode.
- Close Paint and the file explorer.

4.4.2.13 Lubricating the patient table spindle and guide rails

Help ID: 7A03D002-5AB5-4EBB-8B37-7A022E1DA900
Service Level: 7

Lubrication of the Patient Table Spindle and Guide Rails

Required time: 30 min

Required tools and parts: lubricant

Prerequisites: table is in the home position, dockable table is docked

PM Lubricating the patient table spindle and guide rails

- Lower the table prior to loosening the bolts that hold the upper flanges of the cover for the lifting column to reduce stress on the flange.
- Loosen the cover of the lifting column on the upper fastening and lower the cover.

- Raise table to its maximum height.

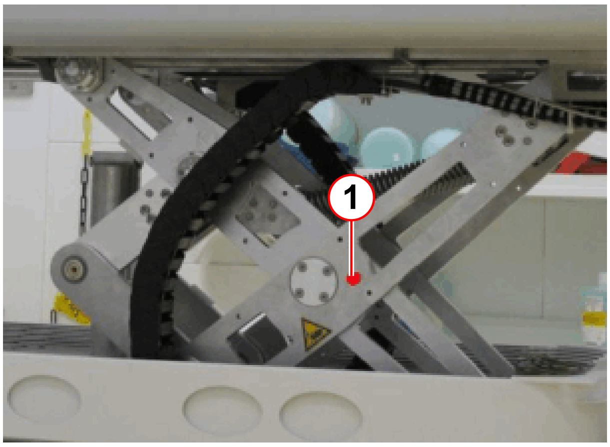
⚠ WARNING

Crushing Hazard

Risk of crushing while moving the table

» **Secure the lifting scissors with the safety pins.**

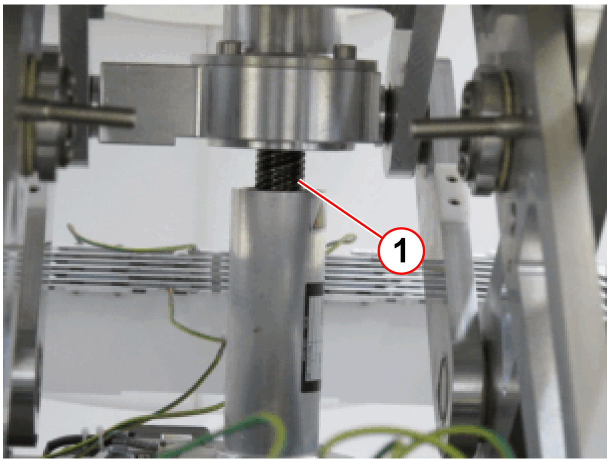
Fig. 401: Patient table scissors



(1) Secure pin

- Switch off fuse F24 (PTAB) on the EPC circuit breaker panel and protect against accidental switch-on.
- Remove the old lubricant from the spindle with a lint-free cloth.
- Lubricate the spindle.

Fig. 402: Patient table

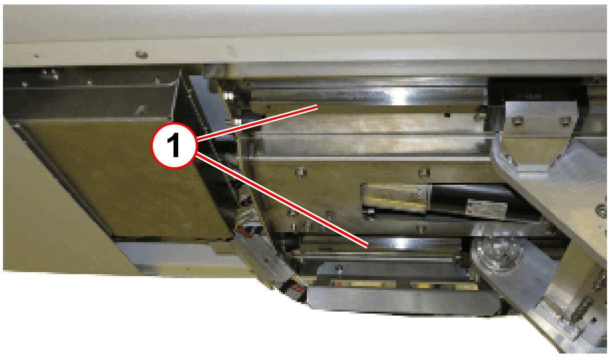


(1) Spindle

- Remove the old lubricant from the guide rails with a lint-free cloth.

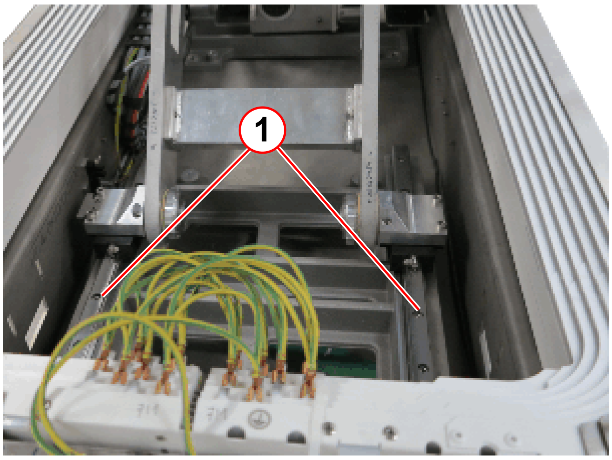
- Lubricate the guide rails.

Fig. 403: Patient table



(1) Guide rails on the upper side

Fig. 404: Patient table



(1) Guide rails

- Switch on fuse F24 (PTAB) on the EPC circuit breaker panel
- Remove the security pin.

WARNING

Crushing Hazard

Risk of crushing while moving the table

» **Keep away from moving parts. Do not bring parts of the body in the crushing zone.**

- Move the table down by using the down button on the control panel.
- Mount the cover of the lifting column.
- Check for smooth upward and downward movement by using buttons on the control panel.

4.4.2.14 Lubricating the docking station spindle (option)

Help ID: C3C93374-8277-49D4-A28F-EE19A98FF470
Service Level: 7

Lubrication of the Docking Station Spindle (Optional)

Required time: 10 min

Required tools and parts: lubricant

Prerequisites: Dockable table is not docked

PM Lubricating the docking station spindle (optional)

- Switch off fuse F24 (PTAB) on the EPC circuit breaker panel and protect against accidental switch-on.

WARNING

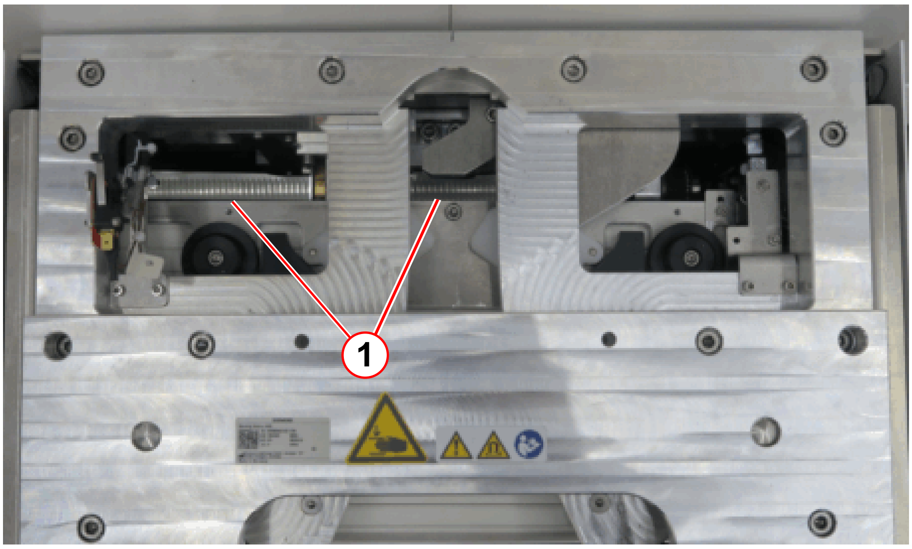
Crushing Hazard

Risk of crushing while moving the spindle

» **Switch off F24.**

- Remove the old lubricant from the spindle with a lint-free cloth.
- Lubricate the spindle.

Fig. 405: Dockable table docking station



(1) Spindle

- Mount the docking station cover.
- Switch on fuse F24 (PTAB) on the EPC circuit breaker panel
- Move the table down by using the down button on the control panel.

4.4.2.15 Replacing the battery of the dockable table

Help ID: 1CB4BDD6-95C2-486F-BFC6-2A955475290C
Service Level: 7

Replacing the Battery of the Dockable Table (Optional)

Required time: 30 min

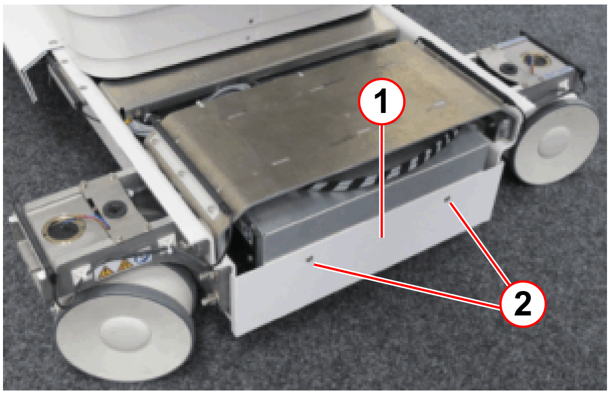
Required tools and parts: Battery

Prerequisites: Table is undocked.

PM Replacing the Battery of the Dockable Table (Optional)

- Move the dockable table outside the magnet room.
- Raise table to its maximum height.
- Switch off the dockable table.
- Remove the base cover rear.

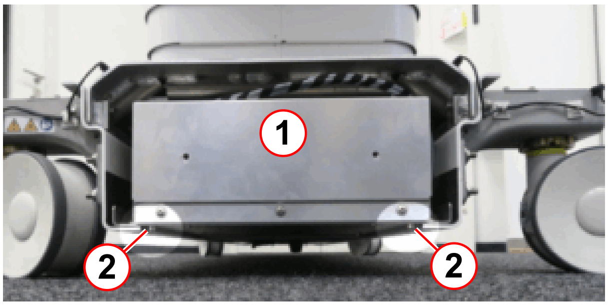
Fig. 406: Dockable table



(1) Battery box cover
(2) Screws

Remove the 3 mm Allen screws.

Fig. 407: Dockable table

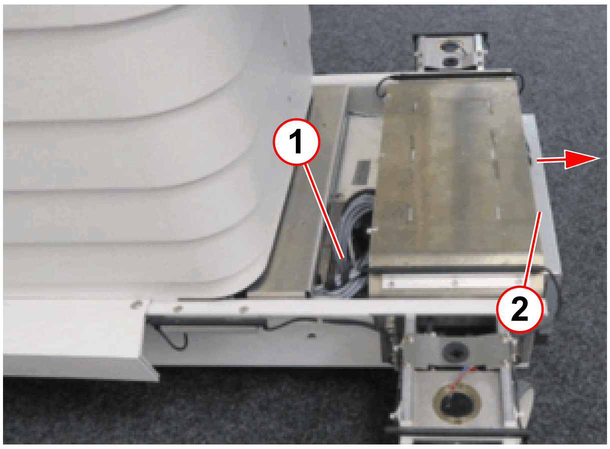


(1) Battery box
(2) Screw

- Remove the 5 mm Allen screws.

Disconnect the cables.

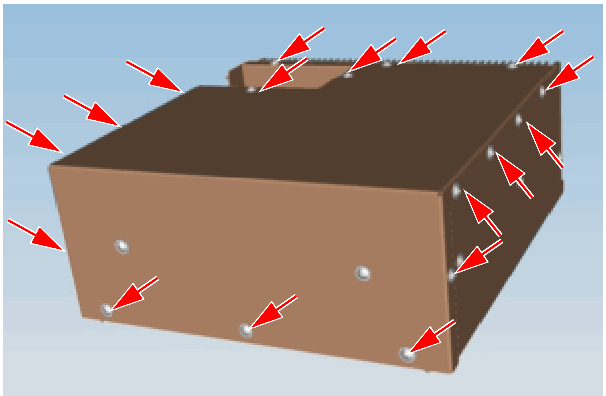
Fig. 408: Dockable table



(1) Cables
(2) Battery box

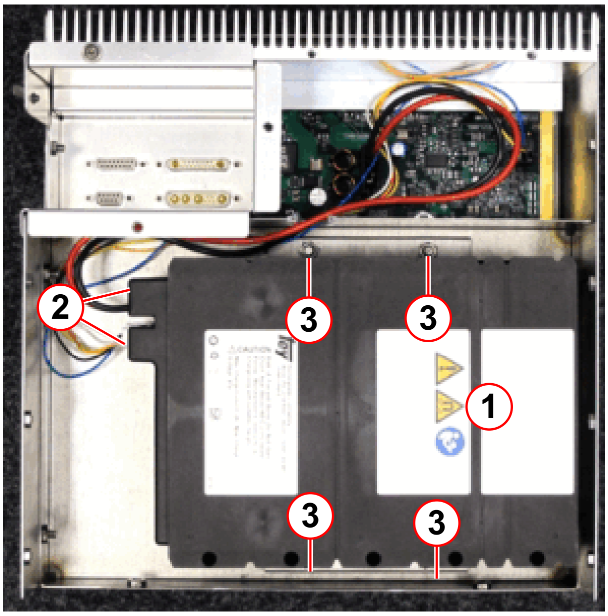
- Slide out the battery box approximately 5 cm.
  » The cable connectors are better accessible.
- Unplug the cable connectors by using a screwdriver.
- Slide out and remove the battery box.

Fig. 409: Dockable table / electronic box cover screws



- Remove the cover screws of the electronic box and remove the cover.

Fig. 410: Battery box



(1) Battery
(2) Cables
(3) Nuts (10 mm)

- Disconnect the cables of the battery.
- Remove the four 10 mm nuts and the two brackets.
- Remove the battery.
- Install the new battery.
- Connect the battery cables.
- Install the brackets and the four 10 mm nuts.

- Attach the battery box cover.
- Slide the battery box in the dockable table.
- Plug the cable connectors by using a screwdriver.
- Install the battery box.
- Install the battery box cover.
- Install the rear base cover.
- Turn the old battery containing lithium over to a disposal company according to local regulations.

CAUTION

Risk of explosion regarding batteries.

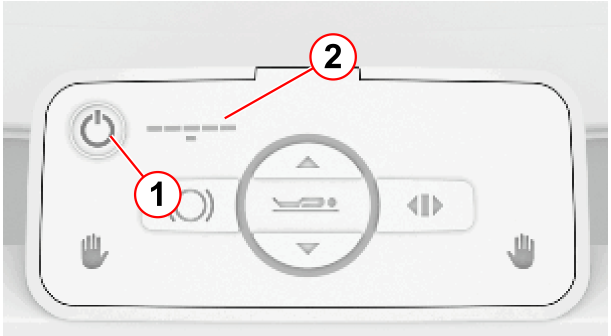
Avoid damage, overheating, and contact with fluids.

» **To avoid short-circuits and heating, cover the terminals with an adequate section of tape or a bag.**

» **Batteries should never be stored or transported with sharp-edged parts or as a bulk shipment to avoid damage.**

» **Observe local laws and regulations.**

Fig. 411: Dockable table control unit



(1) On/off switch
(2) Charge indicator

Final Steps

- Switch on the dockable table.
- Move the table all the way down and then up again.
- Check the charge indicator.
- Dock the dockable table on the docking station to load the battery.

4.4.3 Create Protocol/Report

Help ID: 6FAB638E-A009-4D46-A98A-C89812ADACAF
Service Level: 7

Create Protocol/Report

Status

Status displays the current status of the procedures.

Overview of critical procedures:

Overview of critical procedures list the tasks which are Not OK or overdue.

Test Result

The tester has to evaluate the test results as defined and listed under **Test Result**.

- Select the applicable evaluation.

Without selection of the Test Result the report cannot be saved.

Remarks:

Remarks provide the possibility to enter notes or to describe deficiencies in detail.

Printed Signature Name

The Tester has to enter his name.

- Enter your name

Without a name the report cannot be saved.

Save and Create Report

After completion of the Safety –Related Tests tasks the entries must be saved. Otherwise the entries will be lost.

- Click **Save and Create Report** to save our entries and generate a report.
  » The **Report** is now saved and is then part of the Site-specific Data.

Use the **Reports platform** to display or to print the protocols.

5.1 MR Configuration

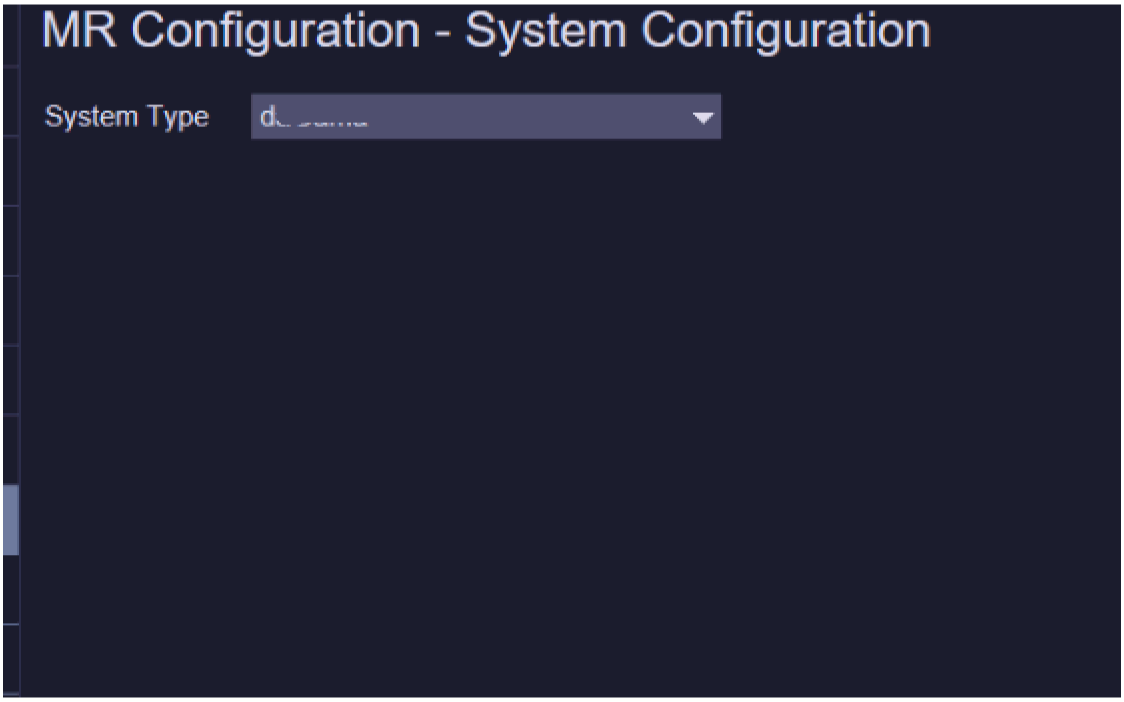
5.1.1 System Configuration

Help ID: AA5789DB-77CD-4C8F-8609-DA2440AD757A
Service Level: 3

With the delivery of the MR system the System Type is pre-configured in the factory.

> This setting is only available in cases like major hardware upgrades from one system type to another system type!

Fig. 412: MR Configuration - System Configuration



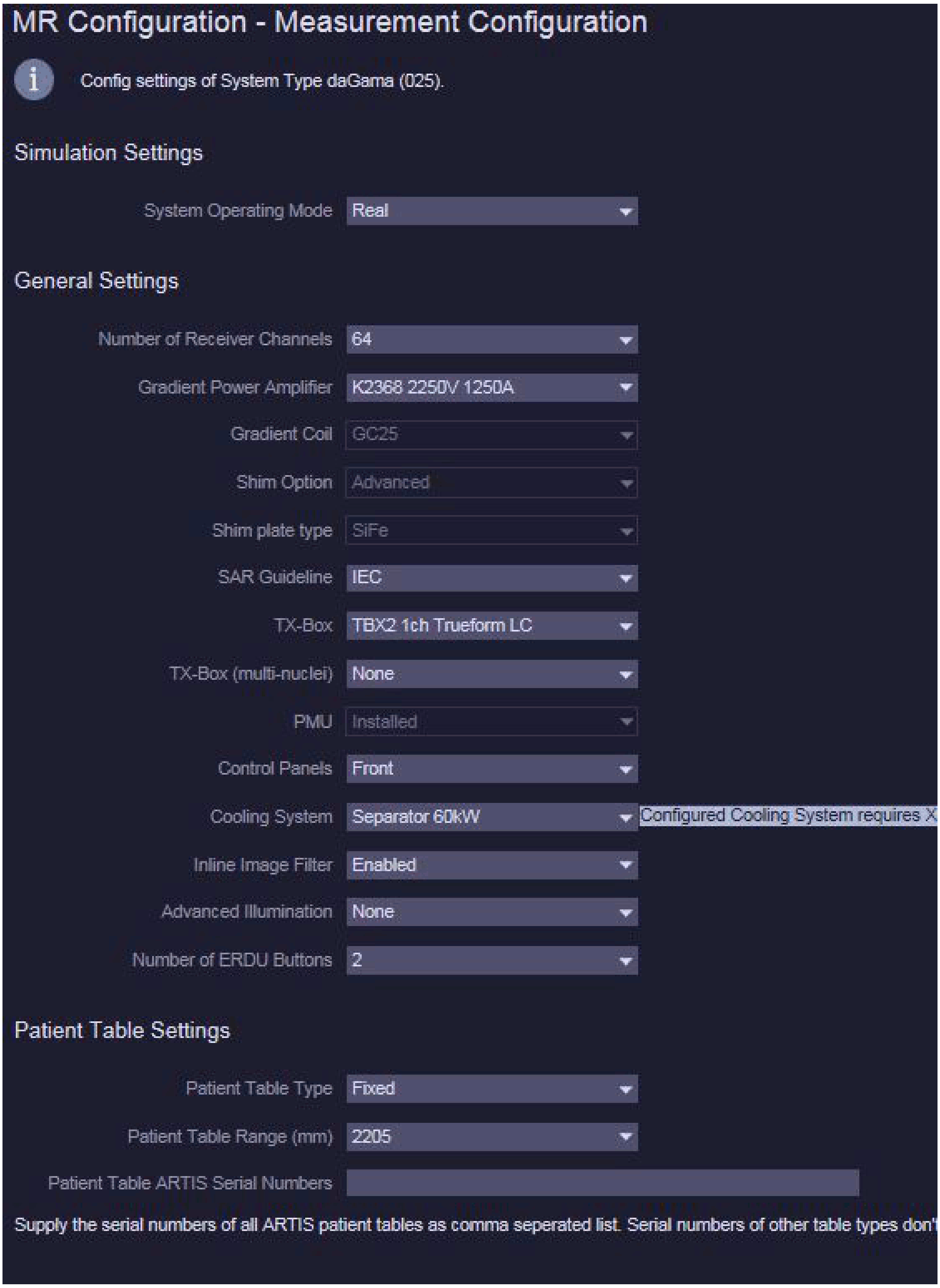
5.1.2 Measurement Configuration

Help ID: 6E389F4F-A325-44E3-8410-77EDC380BB84
Service Level: 3

With the delivery of the MR system the Measurement Configuration is pre-configured in the factory individually for your system type and installed options.

To activate any changes made in the Measurement Configuration a system reboot is required!

Fig. 413: MR Configuration - Measurement Configuration



5.1.3 CAN Box Configuration

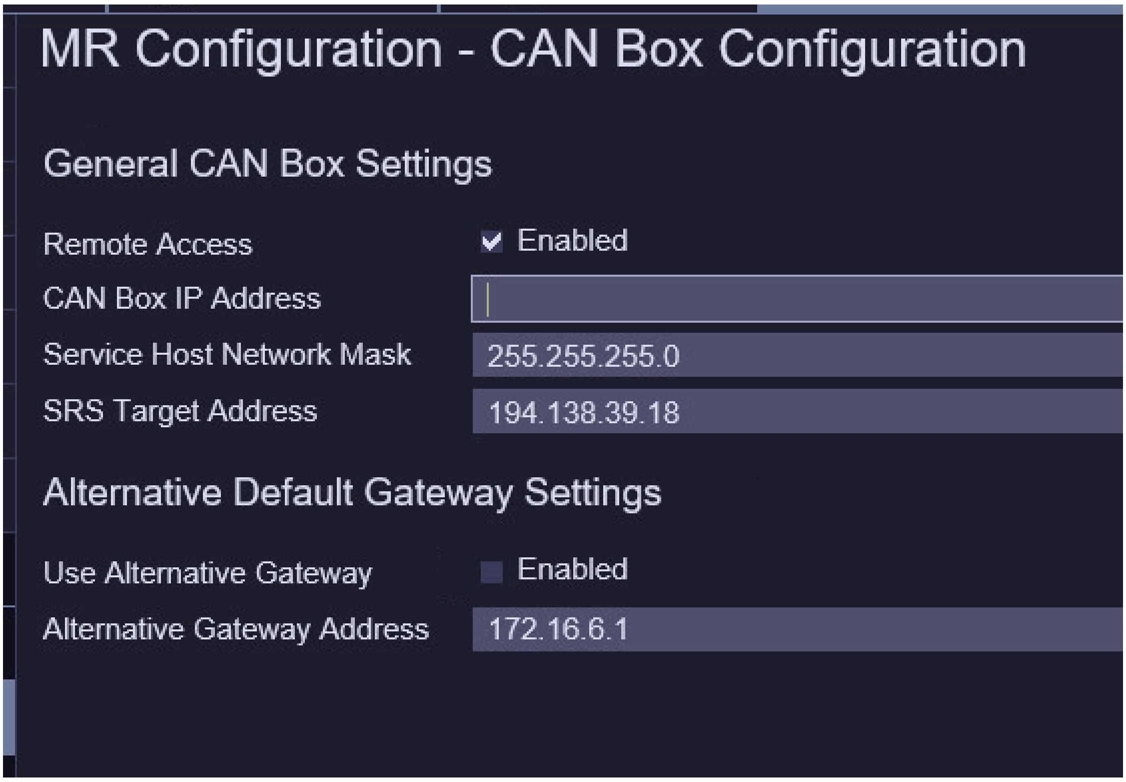
Help ID: D9E7F63B-4429-4F2A-B3F9-215BE8EF38F8
Service Level: 3

CAN Box magnet cooling monitoring and notification unit. The CAN Box is accessible via remote service; the product name is “MR_CAN24” and the system serial number is used.

(ETH 0) X11 at the front of the CAN Box unit is used for the SRS connectivity over the customer LAN.

The check-box **Remote Access** must be set to enable “SRS” connectivity to the CAN Box.

Fig. 414: MR Configuration - CAN Box Configuration



Consult your “SRS Help Desk” regarding configuration and testing of the connectivity.

5.1.4 Coil Configuration

Help ID: AC53C5B9-B1D9-4B27-8555-01B6A2321912
Service Level: 1

Within the “Coil Configuration” you can manage your MR systems coils.

You have the possibility to add, edit, or delete coils in the “Coil Configuration”.

To register new coils, please plug the coils onto the patient table and press **Add Plugged Coils**.

To delete coils from you configuration use **Delete**

Only configured coils will be listed in the **Quality Assurance > Coil** workflow.

Fig. 415: Coil Configuration

MR Configuration - Coil Configuration

Name	Serial Number	Part Number	Revision	Registration Date
FlexLarge 4	1739	10835335	03	2016-09-01 10:06
FlexSmall 4	1731	10835327	03	2016-09-01 10:47
ShoulderLarge 16	1866	10500087	05	2016-09-01 10:56
ShoulderSmall 16	1860	10500088	05	2016-09-01 11:01
Spine 32	2173	10496545	09	2016-09-01 11:07
Body 18	1231	10835325	02	2016-09-01 11:27
Body 18	1617	10835325	02	2016-09-01 11:45
HeadNeck 20	2148	10496500	09	2016-09-01 12:15
HeadNeck 20 TCS	202	10835555	00	2016-09-01 12:33
Body	1006	10676584	00	2016-09-01 14:48
Breast 18 F	1148	10606916	02	2016-09-02 14:59
TxRx CP Head	1034	10606826	02	2016-09-02 16:42
Spine 72 RS	202	10835688	00	2016-09-08 08:10
Spine 72 RS	203	10835688	00	2016-09-08 08:36
Spine 72 RS	201	10835688	00	2016-09-08 09:10

To register new coils, please plug the coils onto the patient table and press [Add Plug

Add Plugged Coils

5.2 Maintenance Configuration

Help ID: 60A96B88-4BFA-4FC6-9945-C468E01EC0BB
Service Level: 7

Maintenance Configuration

Within Maintenance Configuration four basic items will be configured:

- Options (optional procedures)
- Condition Based Maintenance
- Measurement Type of Safety Related Tests
- Display of Maintenance Notifications in System Status Overview

Options

The maintenance workflow display the procedures only if the option is installed and configured. Options like MRWP, cooling system, patient table type will be configured in First Installation > MR Configuration > Measurement Configuration. Additional options have to be configured Maintenance Configuration

Following additional option has to be configured:

Configure option	Procedure
Pediatric Coil	Checking warning signs of the pediatric coil (option)
MRWP	Measurement of the MRSC (MRWP) ground wire (annually) Visual inspection of MRWP

Is **Installed** selected, the task will be displayed.

Is **Not Installed** selected, the task procedure will not be displayed.

- Configure the options according to the system configuration.

Condition Based Maintenance

If Condition Based Maintenance is enabled all Maintenance tasks will be controlled by Service Events via GSMS. The maintenance will be initiated by the condition of the system /components and not by a time schedule. The Preventive Maintenance procedures will be faded out.

Prerequisite for Condition Based Maintenance is a daily remote monitoring of the system status. Therefore the data transfer via Siemens Remote Service must be permanently enabled.

If the system has no Siemens Remote Service connection Condition Based Maintenance must be disabled.

Measurement Type of Safety Related Tests

The report of the “initial measurement” of installation serves as a reference document for subsequent repeat measurements. It must therefore be archived by the operator in the System Owner Manual.

First protocol during installation, select > Initial

Repeat protocol during service / maintenance, select > Repeat

Display of Maintenance Notifications in System Status Overview

The System Status Overview displays three status indicators for Maintenance

- Cold Head Maintenance
- Preventive Maintenance
- Safety-related tests

The indicators turn red if the interval for the specific tasks has expired.

This indicator function can be **Enabled** or **Disabled** within Maintenance Configuration.

The default setting of the function is **Disabled**. The three different maintenance indicators will never be displayed.

The maintenance indicators should be **Enabled** only if the customer does not have a service contract and only with the consent of the service manager or technical manager.

5.3 Measurement Devices

Help ID: 684BBB4D-93EC-4289-AE5E-D51F0D3EEBB4
Service Level: 7

Measurement Device

Enter the type, serial numbers and part numbers of all measurement devices used to install or maintain the system.

Enter the "Next Calibration Due" date, if requested.

If devices have no part numbers enter ‘n.a.’ in the corresponding field.

In case the calibration date is on monthly base, enter the last day of the calibration month in the field ‘Next Calibration Due’.

5.3.2 Siemens Array Shim Device

Help ID: 68CFBB27-0472-4914-9533-47AE08B5802A
Service Level: 7

The data from the Array Shim Device will be automatically taken over from the system.

5.4 Document Version

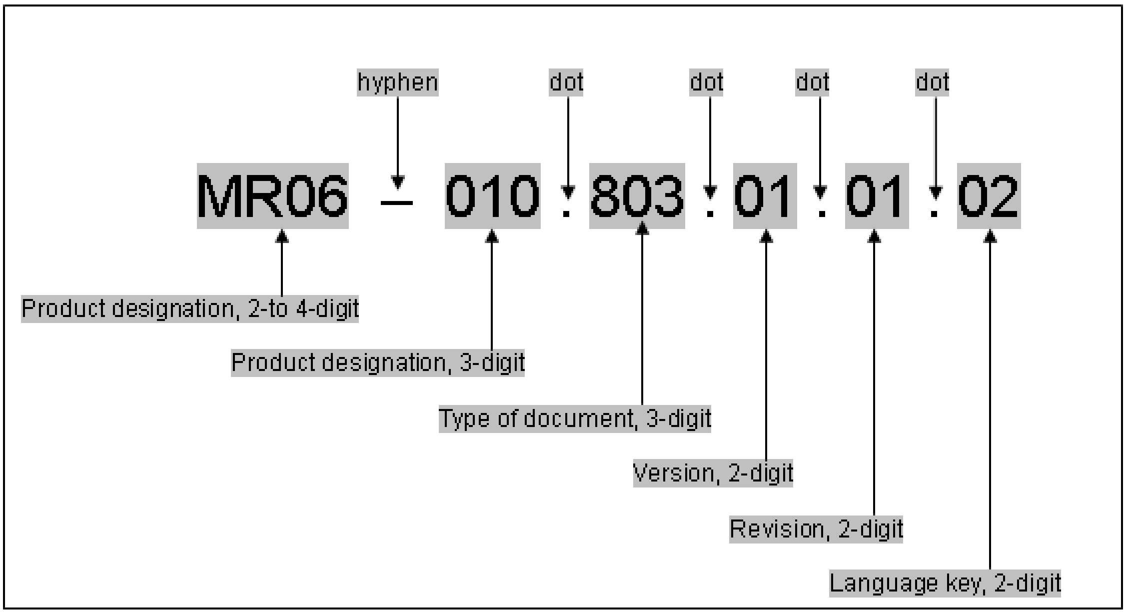
Help ID: E5027B5A-4E54-4826-97FF-E61969B054E8
Service Level: 7

Document version

Please enter the complete print number in the corresponding fields. The print number is located at the bottom of the cover page in the folder.

Please follow the syntax of the print number.

Fig. 416: Syntax print number



5.5 SRT General Tests

Help ID: FDC8FC7E-CA6E-4766-B19F-814F34FE4E50
Service Level: 7

Please select the sub-item.

5.5.1 Visual inspection of system

Help ID: 610B250B-E467-4725-9F7F-B7F5560A632C
Service Level: 7

System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
  - » The system may not contain damages that impair safety.
  - » There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.
- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

5.5.2 Inspecting visible cables and cable guides

Help ID: 84378780-0498-4DAB-BB96-BF703F0188DB
Service Level: 7

System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
  - » The system may not contain damages that impair safety.
  - » There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.

- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

5.5.3 Visual inspection of the quench tube

Help ID: DDDD7B14-A3FE-4D41-B7C3-BBBB916DE72F
Service Level: 7

Quench Line OR125

Quench Line - RF Cabin Inspection

Required time: 40 min

Required tools: If missing from the quench line the following spare parts are available from the SPC and must be fitted:-

- Adhesive tape for Quench line (material number **03866493**).
- Quench outlet warning sign.

> Visual inspection of the quench line from the ground only. It is not necessary to remove ceiling tiles etc.

SI Visual inspection of the quench line

- Visually inspect the condition of the quench line.
- Visually check that the warning tape supplied is fitted to the insulation.
- Check that no unauthorized changes to the design of the quench line have been made, both inside or outside the RF room.
- Check the quench line and insulation for damage.
- Check that the connection screws to the magnet elbow assembly are all fitted and tight.
- Check that the isolation bush is fitted between elbow and acoustic decoupler.

Fig. 417: Warning tape

Warning tape is supplied with each magnet.

- Check that the warning tape is attached to the line insulation. (the warning tape should be fitted every 2 to 3 meters along the quench line.

Quench line - Outlet Inspection

WARNING

Asphyxiation and cold burns

The customer must be notified immediately with corrective actions if the quench line fails to meet 2 critical safety requirements.

» **The quench line outlet is located inside a building with no path for helium gas to escape outside.**

» **The quench outlet has no visible blockage, or restriction of 50 % or greater.**

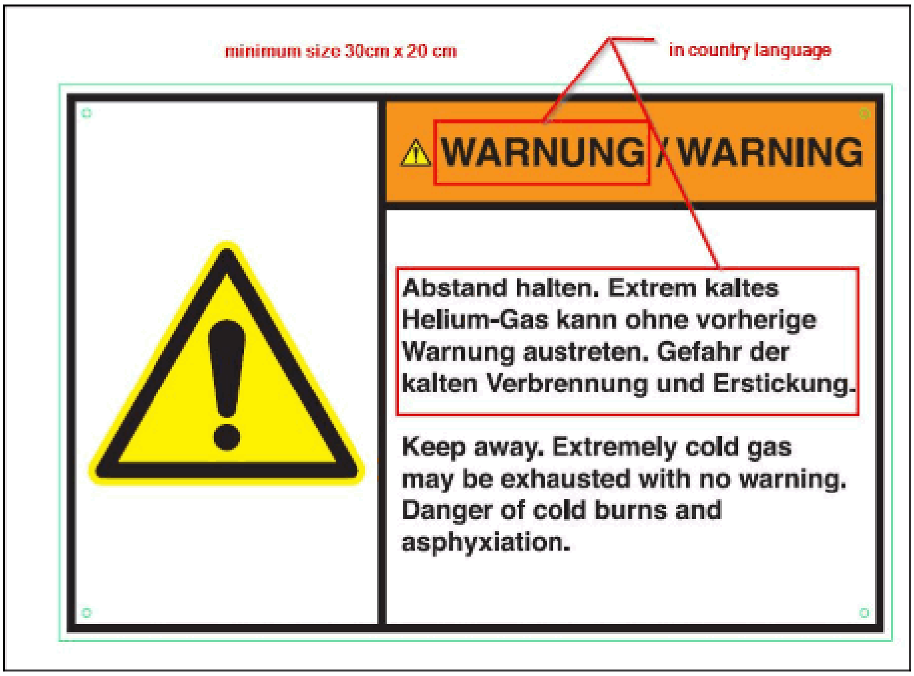
Warning sign requirements

Every outlet must have a warning sign fitted next to the outlet.

If the native country language is not available it is acceptable to source the outlet warning signs locally.

- The warning sign must comply with the ANSI Z535.4 regulations. (See example ANSI regulation below).
- The sign must include a **Warning** symbol and words in the local language and English.
- The minimum size must be 30 cm wide x 20 cm high.

Fig. 418: *Outlet warning sign design (example ANS regulation)*



- Check that the warning sign is securly fixed and clearly visible next to the outlet of the quench line.

Outlet inspection

CAUTION

Risk of injury

Ensure the outlet can be safely accessed.

» **If the a visual inspection of an outlet located on a rooftop or high on a building cannot be completed safely, access it from ground level.**

- Check the outlet of the quench line for obstructions (e.g. bird’s nest).
- Check that a mesh is fitted securely on the quench line outlet.
- Check that water cannot ingress into the outlet (vertical or horizontal).
- Check that the outlet is secuely fixed to the wall.

The outlet cannot be situated where, in case of a quench:

- Helium gas is blown into a public area.
- Helium gas can be drawn into an air inlet.
- Helium gas can enter open windows.

To avoid risk of injury from cold burns and asphyxiation, access to the quench line outlet must be restricted by:

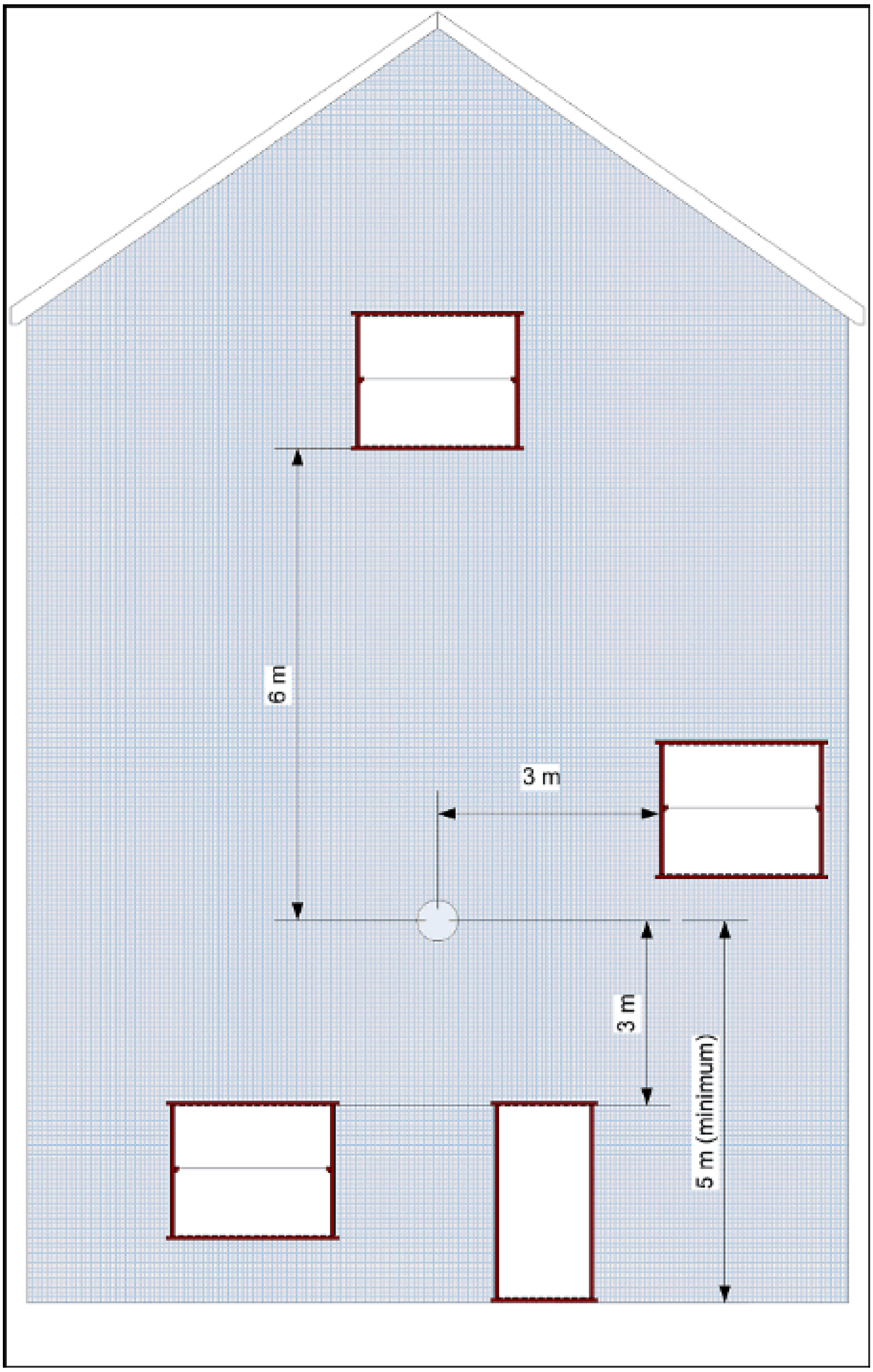
- 3 m on each side and below.

- 6 m vertically above the exit.
- Windows in the restricted access area must be permanently closed. Means of opening the windows must be removed.
- Check that the area next to the outlet has not changed and become a hazard:
  - Air conditioning inlets.
  - Erection (or extensions) of new or temporary buildings, including updates of the building façade.

Outlet Height

- Check that the outlet height is ≥ 5 m for the horizontal outlet designs - Options 1 to 4.
- Check that the outlet height is ≥ 3 m for an angled outlet design.
- A deflector plate can be fitted below outlet options 1, 2 and 3 with the outlet at a height of ≥ 3 m.
- Record the findings in the Protocol.

Fig. 419: Quench outlet - Horizontal safe distance requirements



5.5.4 Check for water in the venting

Help ID: A201C5FB-D936-4248-8910-A66300D56E5F
Service Level: 7

Quench Valve

SI Check for water in the venting

Required time: 25 min.

Required tools:

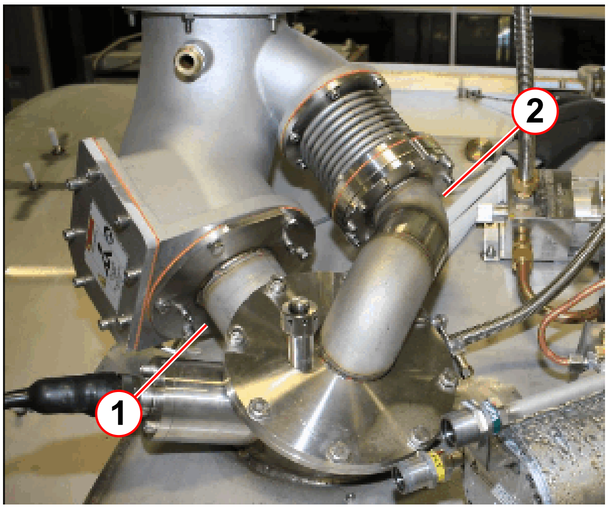
Approximately 10 cm of wire or similar to insert into the drain holes.

Prerequisites:

- Remove the required service access covers. (please refer to instructions > Replacement of Parts > System Covers)
- Set up and secure the service platform to the magnet.

The 2 x drain holes are permanently open. The drain holes allow water condensation to drip from the venting.

Fig. 420: The position of the two drain holes



(1) Drain hole at the bottom of the quench valve interface plate
(2) Drain hole at the bottom of the coded vent assembly

Fig. 421: Quench valve drain hole

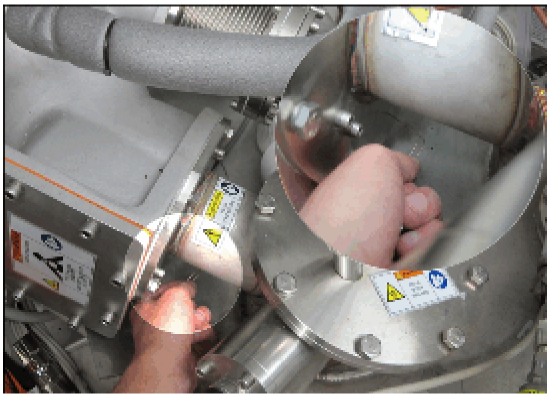
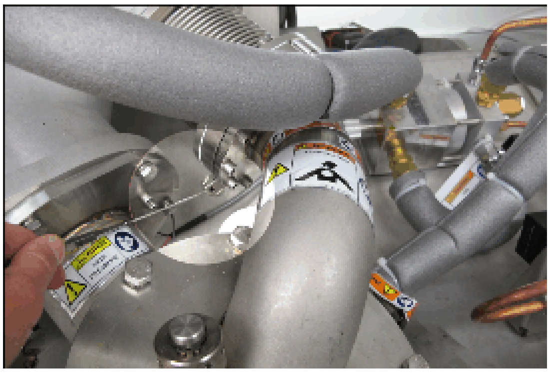


Fig. 422: Coded vent drain hole



Quench Valve

- The 2 drain holes are located at the lowest point of the quench manifold and coded vent assembly .

  Pos 1, Location of the Quench valve drain hole.

  Pos 2, Location of the Coded vent drain hole.

  Check to make sure the two holes are **unblocked** by using a piece of wire. e.g. a paper clip

> If water is present, determine the source (e.g., damaged or missing protective rain cover, no quench tube insulation or condensation). Advise the customer to initiate corrective action immediately.

5.5.5 Check for ice formation at the magnet service turret and venting system

Help ID: 7D899DEE-2BBC-43B9-97B7-9EBE501B5374
Service Level: 7

Ice Formation Check

SI Check for ice formation at the magnet service turret and venting system

Required time: 5 min.

Required tools: N/A

Prerequisites:

- Remove the required service access covers. (please refer to instructions > Replacement of Parts > System Covers)
- Set up and secure the service platform to the magnet.
- The service turret and venting system should be free of ice formation (except after ramping up or filling).
  - » Empty out the water from the condensation reservoir located on the floor at the electronics side of the magnet.

5.5.6 Magnet system leak check

Help ID: 216EB508-6376-404D-858D-D0187D6FC2B8
Service Level: 7

Turret and Venting Leak Checks

SI Magnet system leak check

Required time: 5 min.

Required tools:

Electronic helium leak detector (material number: 10614850).

Prerequisites: Turret cover is removed (please refer to instructions > Replacement of Parts > System Covers)

Magnet System Leak Check

The magnet is fitted with a recondensing cold head and pressurization heater that maintain the pressure in the helium vessel at a constant **15.5 psi (A)**, providing there are **no leaks** in the venting system.

Large leaks show up as low magnet pressure. Small leaks may not show any change in pressure.

All leaks are measured as a steady loss in the liquid helium level.

Insufficient cooling power of the cold head could also result in helium loss.

If the average power of the heater is = 0 W and the magnet pressure ~ 16 psi (A), check the cold head performance.

- The helium level and magnet pressure must be monitored regularly either by using the **Administration Portal** on the **MRC (MRAWP) Host**.

If the liquid helium level and the magnet pressure have remained constant with no helium loss, no leak check is necessary.

If the magnet has low pressure, or there is evidence of liquid helium loss, then a full turret and venting leak check must be performed.

Leak checking must be performed with an electronic leak detector capable of detecting leaks of a minimum **1 x 10^{-4} mbar l/s.**

Fig. 423: Edwards Leak detector (example)

- Use the Edwards hand-held leak detector (part no **10614850**).

  This is the **recommended equipment** for leaks **greater than 1 x 10^{-4} mbar l/s.**

NOTE: The hand-held device is calibrated in **ml/sec**:-

1 x 10^{-4} mbar l/s **=** 1 x 10^{-4} ml/sec at normal pressure and temperature.

The functionality of the Edwards leak detector can be compromised by magnetic fields. Avoid using the detector near the magnet bore.

All **high-pressure** joints and seals must be checked in case of **low system pressure** and **liquid helium loss.**

Leak check all joints and seals that have been subjected to prolonged low temperatures and subsequent frost build-up. For example, seals which have been cooled to a low temperatures during helium transfers, helium vessel depressurization, or magnet ramping.

Check that the following magnet components are at room temperature before the leak check.

- Cold heads.
- Turret top plate, syphon port and venting.

- 17 psi (A) valve assembly.
- Pressure gauge assembly.
- Connecting pipe couplings.
- Quench valve

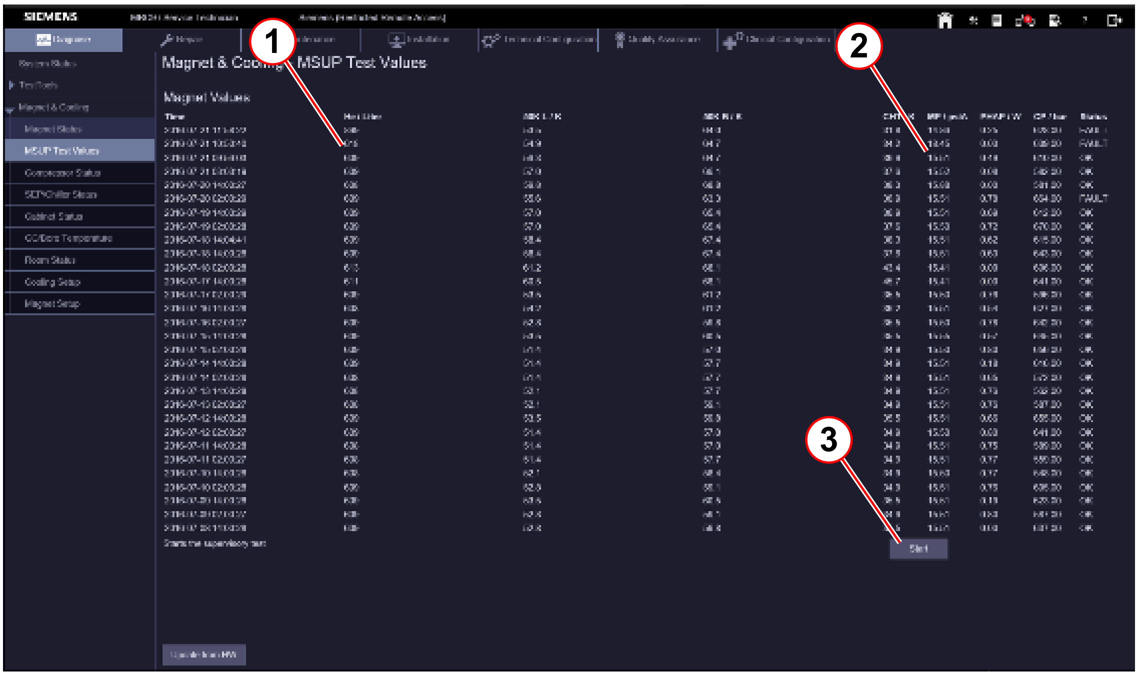
If no leak is found but the system pressure is low, internal components (burst disk ans O-ring) must be checked:

- 17 psi (A) valve assembly.
- Quench valve.
- Coded vent burst disk.

> The magnet must be RUN TO ZERO before checking inside the service hatch or coded burst disk assemblies.

- Check the **liquid helium level** by using the **Administration Portal** on the **MRC (MRAWP) Host**.
  - Access the **Administration Portal** on the MRC.
  - Select the menu item **Diagnose - Magnet & Cooling - Magnet Test Values**.
  - Press the **Start** button to initiate the helium measurement.

Fig. 424: checking liquid helium and magnet pressure levels



(1) Liquid helium level

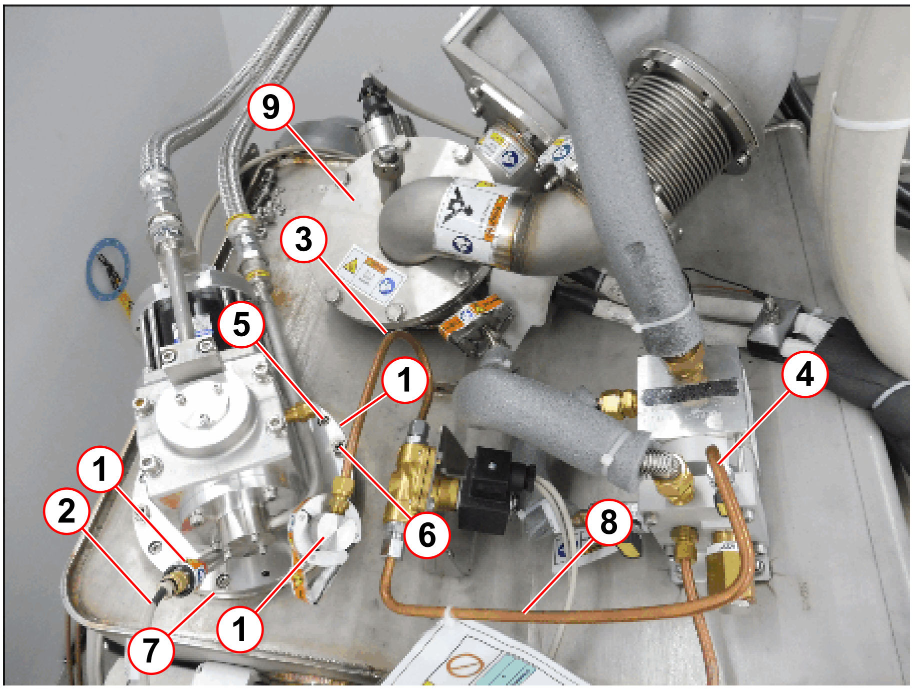
(2) Magnet pressure

(3) Start button

- The following checks must be carried out if the magnet shows:
  - a **decreasing** liquid helium level or
  - a magnet pressure **below 15.5 psi (A)**.

Cold Head Checks

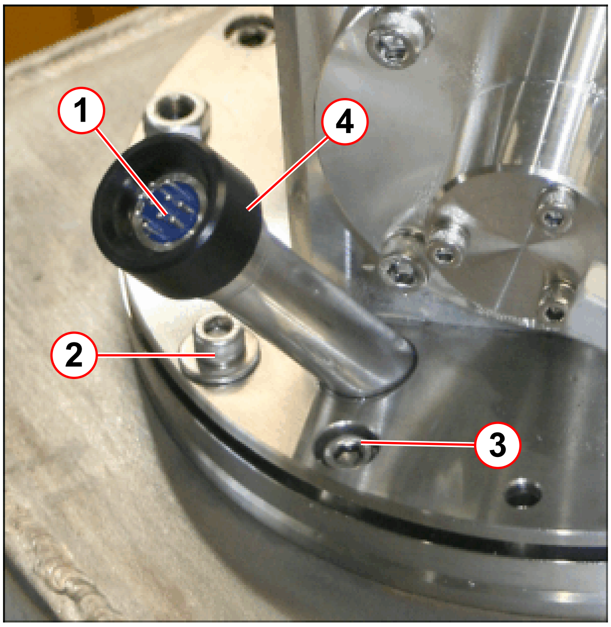
Fig. 425: Leak checking the coldhead seals



(1) Cold head seals, leak check points
(2) Temperature sensor cable
(3) Cold head vent line
(4) Cold head vent line connection at the venting
(5) 8 each, M6 x 16 mm O-ring clamp ring screws, torque to = 8 Nm
(6) 4 each, M6 x 25 mm cold head flange screw,torque to = 4 Nm
(7) O-ring seal
(8) Solinoid to venting pipe
(9) Top plate

- Check the **vent line seals** on the **cold head** and **vent line connection.**

Fig. 426: Cold head 10 pin seal



(1) Leak check point
(2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
(3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
(4) Black sealing cap

- **DO NOT** turn or try to remove the 'black' sealing cap
- Remove the temperature sensor cable (4), with a straight pull away from the black sealing cap.
- Leak check the 10 pin seal connector.
- Immediately reconnect the cable if no leak is found.

The cold head is fitted with **temperature sensors**:

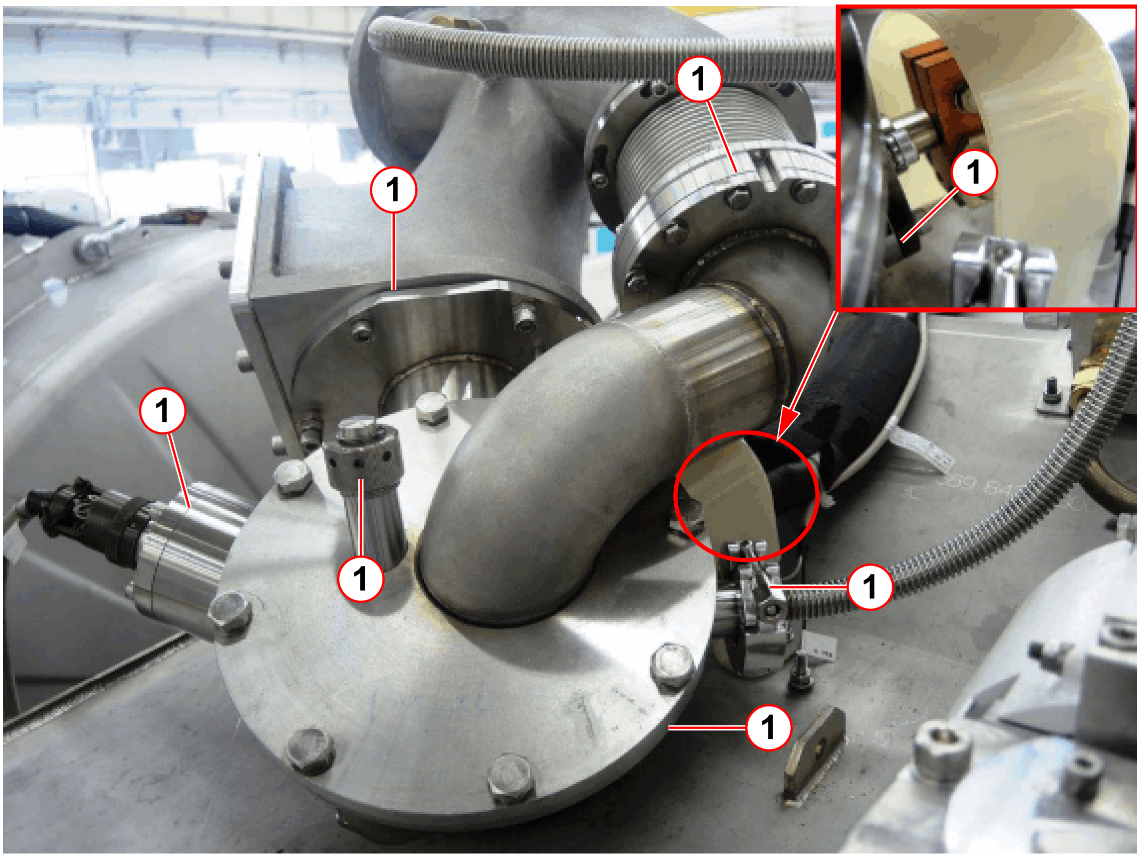
- The feed-through for the wires is sealed with an O-ring.
- Check the feed-through and cold head seals.
- Check the torque of the cold head screws.

If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

Turret and Venting

Fig. 427: Turret and venting - high pressure seals



(1) Turret top plate, syphon port and O-rings

- Check that all clamps on the high-pressure side of the venting are tight. **Do not** try to **overtighten** them - it could cause a leak, or make an existing leak worse.

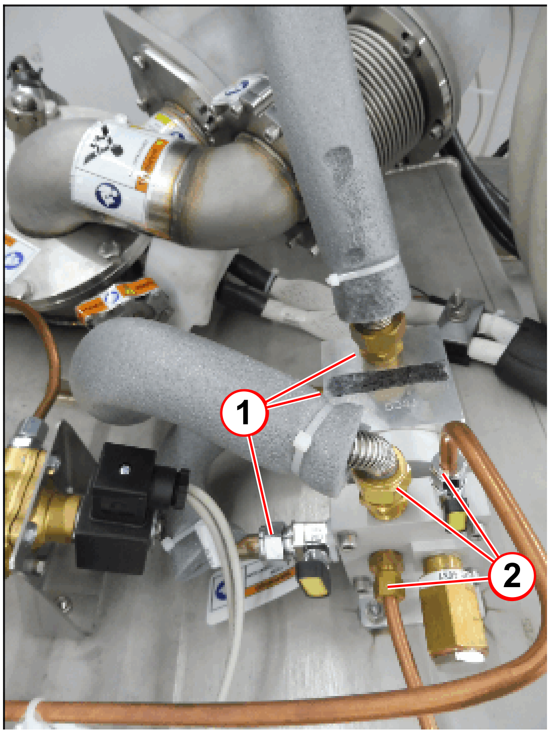
> ℹ If no leak is found, but there is a slow, steady liquid helium loss, contact your local Customer Care Center (CCC). Repair requires a specialist.

> ℹ Do not use any form of vacuum grease on O-rings and seals.
>
> It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

> ℹ Replace a high load seal only with another identical high load seal. High load seals must only be used ONCE for a guaranteed leak tight seal.

Venting Valve Assembly

Fig. 428: Vent valve assembly leak cheakpoints



(1) Low pressure joints
(2) High pressure joints

- Leak check the venting valve **high pressure** assembly.

Pressure Gauge Assembly

Fig. 429: Leak check the pressure line

(1) Leak check points

- Leak check the **high pressure** points.

Contact your local Customer Care Center (CCC) if a leak cannot be resolved.

5.5.7 User documentation available and legible

Help ID: F4395154-961A-49BE-888D-C2CBCF8F76E9
Service Level: 7

User documentation

Required time: 5 min

Required tools: n.a.

SI User documentation available and legible

- Ensure that the System Manual is available, legible and valid (e.g. software version).

5.5.8 Checking user icons and button labels

Help ID: 5DA2D51C-D4C5-40D0-9F00-703424CE2605
Service Level: 7

User icons, button labels

Required time: 5 min

Required tools: n.a.

SI Checking user icons and button labels

- Inspect the button labeling and user icons for legibility/recognizability (for example “symbol keypad”, “**Magnet Stop button**”, “Patient Table Stop”, etc.).

5.5.9 Identification of the controlled access area (0.5 mT area)

Help ID: 92ACE3B3-84A2-415D-8F16-34071EFD7274
Service Level: 7

Identification of the controlled access area

Identification of the controlled access area (0.5 mT area)

Required time: 10 min

Required tools: n.a.

Refer to the Spare Parts Catalogue for the order number for the set of warning signs.

SI Identification of the controlled access area (0.5 mT area)

Fig. 430: Warning sign "Strong Magnetic Field"

Fig. 431: Warning sign " Cardiac Pacemaker"

Prior to initial Startup, the project manager defines the 0.5 mT area.

- Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone. The signs have to be posted in a way that it is visible even when the door to the controlled area is opened.

SI Checking the warning signs (SIEMENS)

SI Checking the warning signs (customer signs cover all safety aspects as listed on the SIEMENS signs)

The fringe field extends three-dimensional beyond the magnet and can be reduced through shielding. The field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcement in buildings.

Fringe field distribution

The field stability over time is < 0.1 ppm/h.

Graphical rendition

Fig. 432: Fringe field Z/Y/X

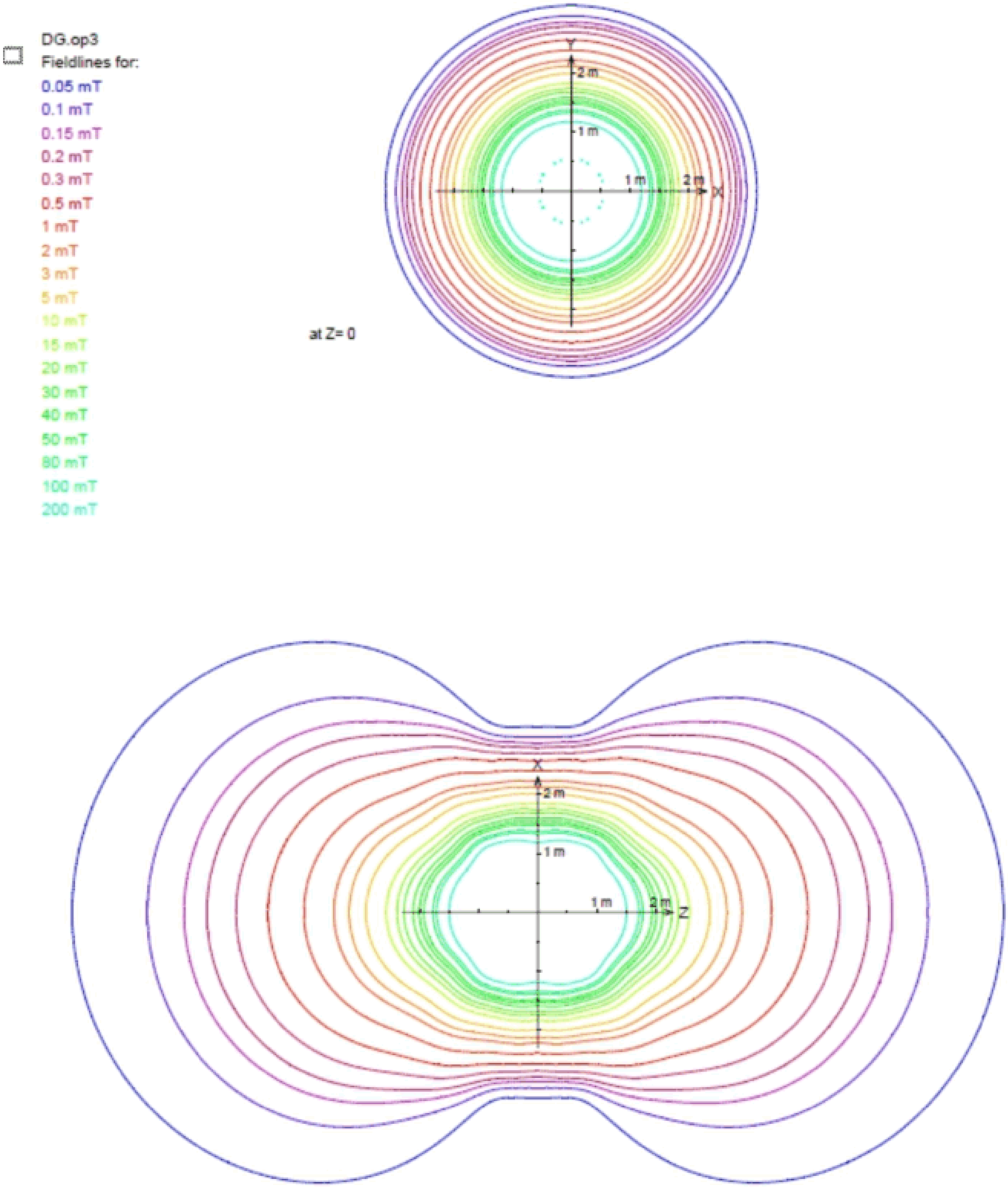


Fig. 433: Fringe field Y/X

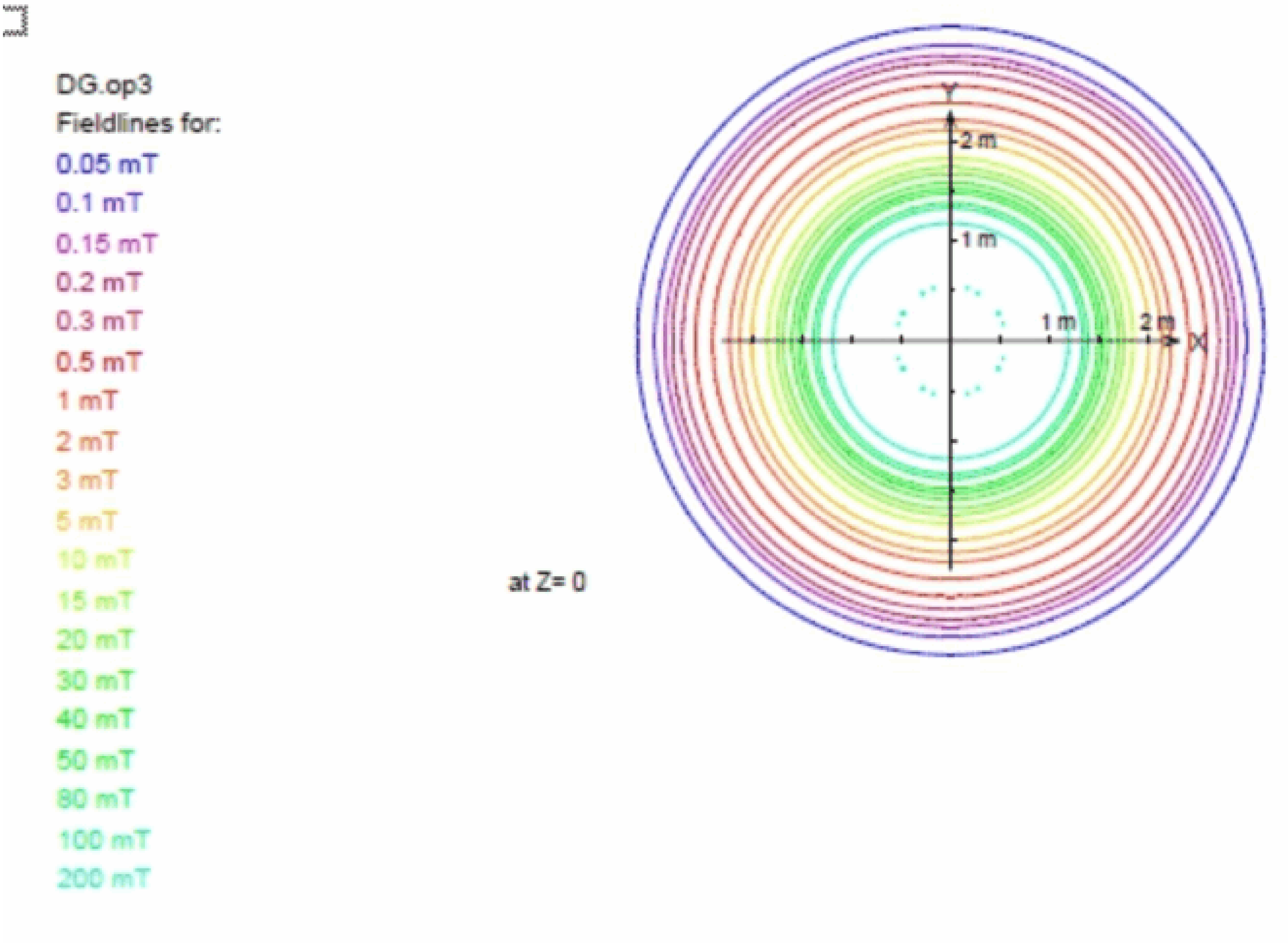
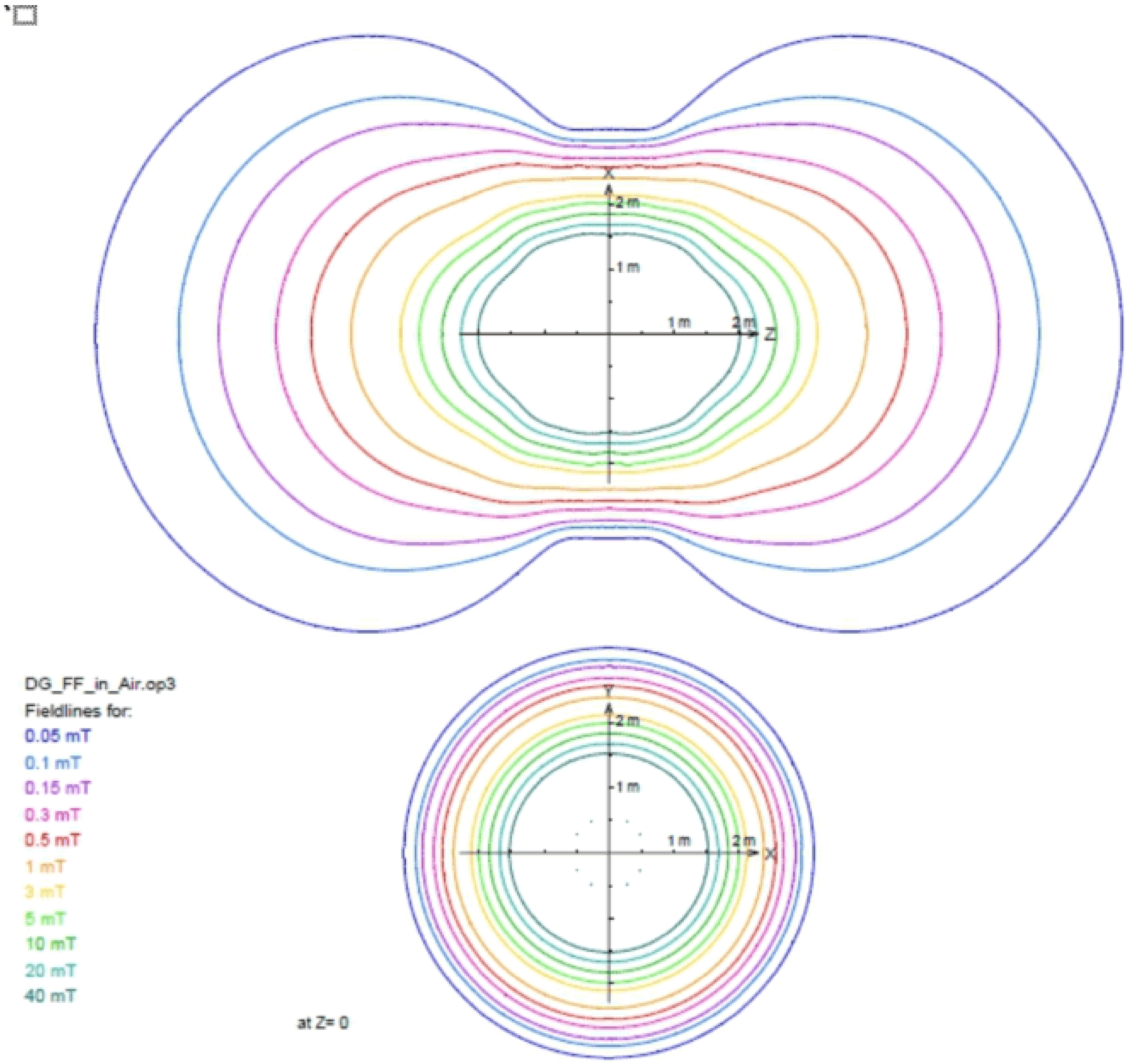


Fig. 434: Fringe field Z/Y



Tablular form

Tab. 25 Fringe field distribution OR125

Fringe field	Distance from the magnet isocenter in the direction of		
	the X-axis in [m]	the Y-axis in [m]	the Z-axis in [m]
200 mT	1.2	1.2	1.5
100 mT	1.3	1.3	1.7
80 mT	1.4	1.4	1.8

50 mT	1.5	1.5	1.9
40 mT	1.5	1.5	2.0
30 mT	1.6	1.6	2.1
20 mT	1.7	1.7	2.3
15 mT	1.7	1.7	2.4
10 mT	1.8	1.8	2.6
5 mT	2.0	2.0	2.9
3 mT	2.16	2.16	3.24
2 mT	2.2	2.2	3.5
1 mT	2.45	2.45	4.05
0.5 mT	**2.65**	**2.65**	**4.65**
0.3 mT	2.8	2.8	5.1
0.2 mT	3.12	3.12	5.72
0.15 mT	3.2	3.2	6.0
0.1 mT	3.8	3.8	6.8
0.05 mT	4.8	4.8	8.1

» The fringe field extends spatially in three dimensions around the magnet isocenter and can be reduced by additional iron shielding. The typical fringe field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcements or other iron masses in buildings.

5.5.10 Checking the hearing protection sign

Help ID: 2912BE7A-889E-49FA-8984-92E0D66FA758
Service Level: 7

Hearing protection sign

Required time: 3 min

Required tools or parts: if missing hearing protection sign. The hearing protection sign is part of the warning sign sets, please refer to SPC.

SI Checking the hearing protection sign

- Ensure that the hearing protection sign has been posted at all entrances to the examination room.

Fig. 435: Hearing protection sign

- Install missing signs immediately

5.5.11 Checking the warning signs of the pediatric coil (option)

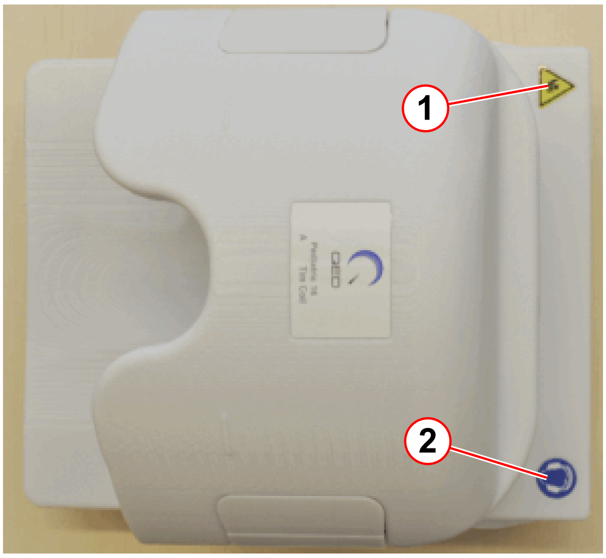
Help ID: B3FE5A47-5F50-474A-B3D0-856E2EEC5F7B
Service Level: 7

Warning Signs of the Pediatric Coil (Option)

Required time: 5 min

Required parts and tools: Sign if missing.

Fig. 436: Pediatric coil



(1) Sign for bruising hazards
(2) Hearing protection sign

SI Checking the warning signs of the pediatric coil

- Ensure that the warning signs (➔ Fig. 437 Page 639), (➔ Fig. 438 Page 639) are affixed to the pediatric coil.
- Replace missing labels immediately.

Fig. 437: Sign for bruising hazards

Fig. 438: Hearing protection sign

5.5.12 Checking/Replacing the air filter of the patient fan

Help ID: 715037BE-50B8-4111-8941-E0662B3A42D5
Service Level: 7

Patient fan

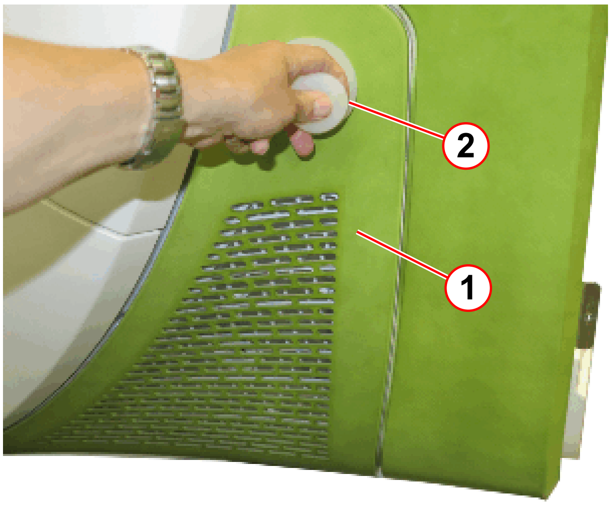
SI Checking the air filter of the patient fan

Required time: 5 min

Required tools and parts: Air filter

- Remove the service cover.

Fig. 439: Air filter of the patient fan



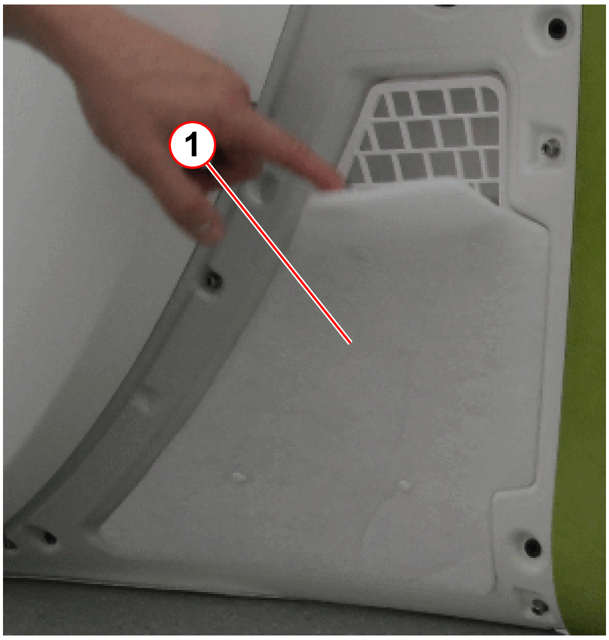
(1) Air filter cover
(2) Suction cup

The air filter cover is fixed with snap locks.

- Attach the suction cup (➔ Fig. 439 Page 640).
- Carefully pull out the cover (➔ 1/Fig. 439 Page 640)(plugged with snap locks).

During first installation of the system it is not necessary to replace the filter.

Fig. 440: Air filter of the patient fan



(1) Filter

- Remove the old filter (→ 1/Fig. 440 Page 641).
- Install the new air filter.
- Install the air filter cover.

5.5.13 Checking the phantoms

Help ID: 97231754-C3EF-4D25-8E86-E53BDA6D0AA9
Service Level: 7

Phantoms

Required time: 10 min

Required tools: n.a.

SI Checking the phantoms

Please read the “MR-Specific Safety Information” prior to inspecting the phantoms.

1. Inspect every phantom used on-site for cracks, leaks, or extensive wear and tear.
   » Replace the phantom if there is any evidence of damage that may affect normal operation.

Defective phantoms must be disposed of. Refer to the “Disposal Instructions”.

2. Inspect every phantom used on-site for excessive air bubbles. Air bubbles can influence your Tune-up results and quality measurements.
3. Ensure that fluid is at an adequate level so that measurement performance is not adversely affected.

Water mixture phantoms with filler cap:

- Refill the phantom, if necessary.
- Phantoms with distilled water mixture - fill with distilled water only.

Water mixture phantoms without filler cap (welded):

- Refilling is not possible. Water mixture phantoms without filler cap must be replaced

Oil phantoms without filler cap (welded):

- Refilling is not possible. Oil phantoms without filler cap must be replaced

Oil phantoms with filler cap:

- Refilling is not possible. Oil phantoms with filler cap must be replaced.

  Do not open the screw cap. It will lead to leakage.

  For example spherical phantom blue D240 p/n 7577666 or p/n 10496685 : The diameter of the air bubble should not exceed 12 cm.

5.5.14 Checking the door lock of the RF room

Help ID: F8FC5505-7C34-4697-992F-8812EDC8B0BA
Service Level: 7

RF room door

Required time: 5 min

Required tools: n.a.

Door lock

SI Checking the door lock of the RF room

1. The locking and unlocking mechanism has to function properly.
2. The door handle or knob has to be securely fastened.

In case of a defect (e.g. defective unlocking mechanism, door handle,), inform the customer to initiate corrective action immediately.

5.5.15 Checking the warning signs of the patient table

Help ID: F8D2D88A-F42C-4FED-AC4F-13335A2EB78B
Service Level: 7

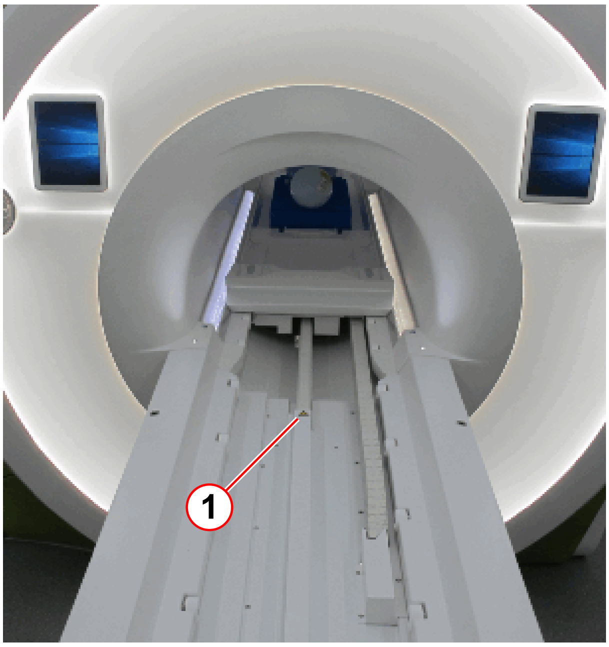
Warning Labels on the Patient Table

Required time: 2 min

Required tools or material: Warning labels, if missing

SI Checking the warning signs on the patient table.

Fig. 441: Patient table



(1) Warning sign bruising hazard

Fig. 442: Warning sign 'bruising hazard'

- Ensure that the warning labels are affixed to the patient table.
  » The labels must be legible.
- Replace missing labels immediately.

5.5.16 Visual inspection of Dockable Table

Help ID: 9F3D38A0-145B-41A5-9AA2-47B1D44E421E
Service Level: 7

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

5.5.17 Visual inspection of MRWP

Help ID: F7F97A81-696F-4D43-A9F9-56E9061C630E
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

> This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: | Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: | Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: | Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: | Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: | Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

5.5.18 Visual inspection of options and accessories

Help ID: 04071914-8D3F-4451-A616-4DA0686A7597
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.

- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

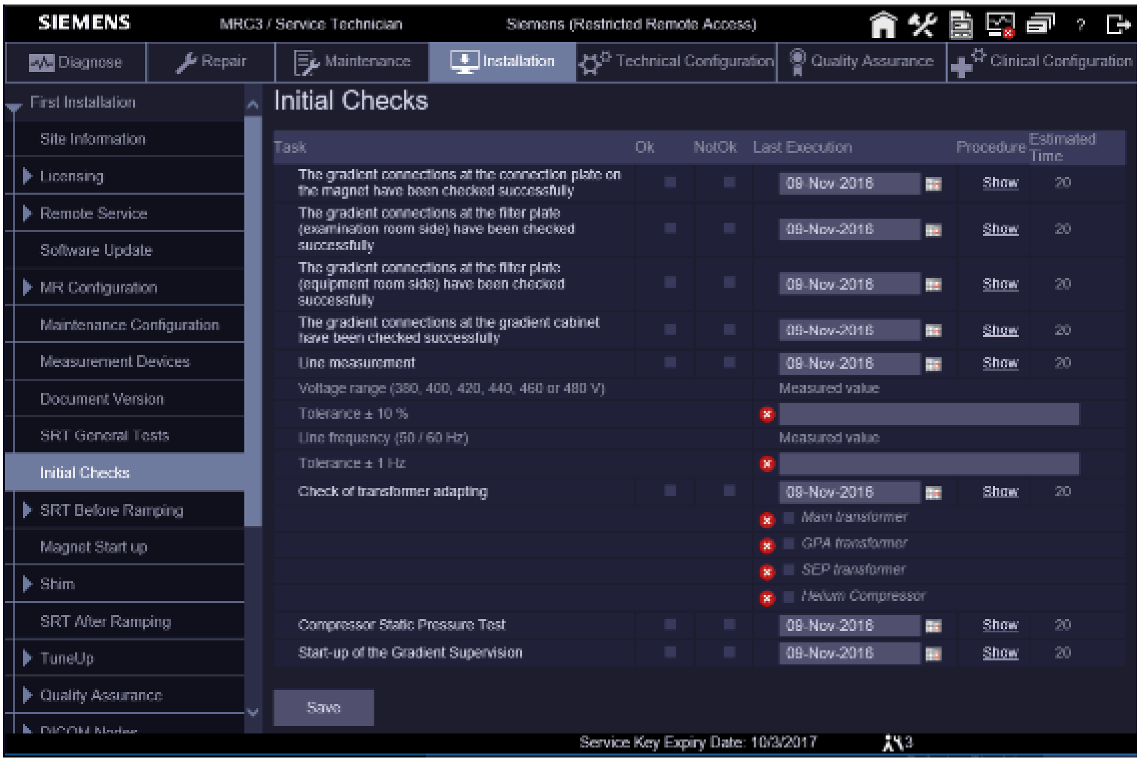
		Signature:	
Date:		Name:	

5.6 Initial Checks

Help ID: 62B2BE53-C0E6-4BC9-8102-412B615FC422
Service Level: 7

Fill out the tasks of the initial checks

Fig. 443: Initial Checks



5.6.1 The gradient connections at the connection plate on the magnet have been checked successfully

Help ID: 6080667E-EF66-4FC0-83FD-EE4AA5FA4162
Service Level: 7

- Mark if the test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.6.2
The gradient connections at the filter plate (examination room side) have been checked successfully

Help ID: 662BE043-1C13-4AD9-B8AE-2A72D85F3695
Service Level: 7

- Mark if the test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.6.3
The gradient connections at the filter plate (equipment room side) have been checked successfully

Help ID: B98DF430-6B52-4377-B148-8D5EE09D66FA
Service Level: 7

- Mark if the test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.6.4
The gradient connections at the gradient cabinet have been checked successfully

Help ID: E0AB05A5-9703-45AD-AB9D-7290BB88EFD6
Service Level: 7

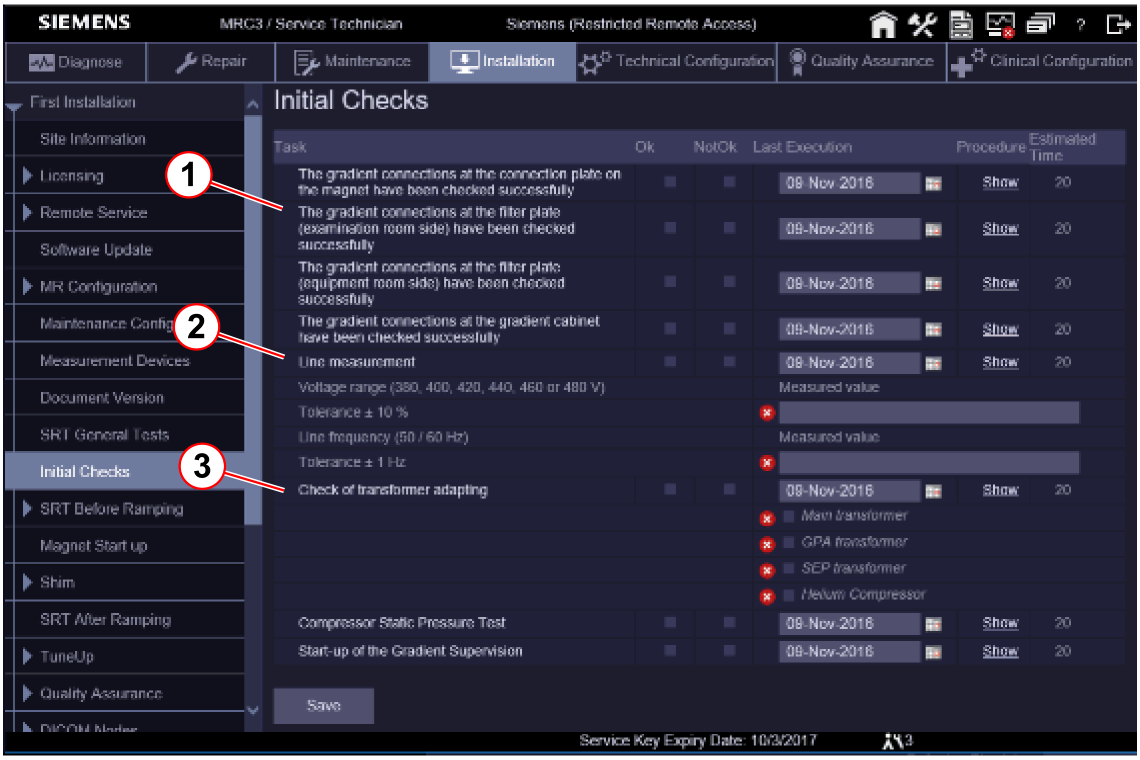
- Mark if the test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.6.5
Line measurement

Help ID: CAB543E1-AC0A-4749-82C5-92CF9054E15E
Service Level: 7

Note the line voltage and line frequency

Fig. 444: Initial Checks



(1) Gradient Connections

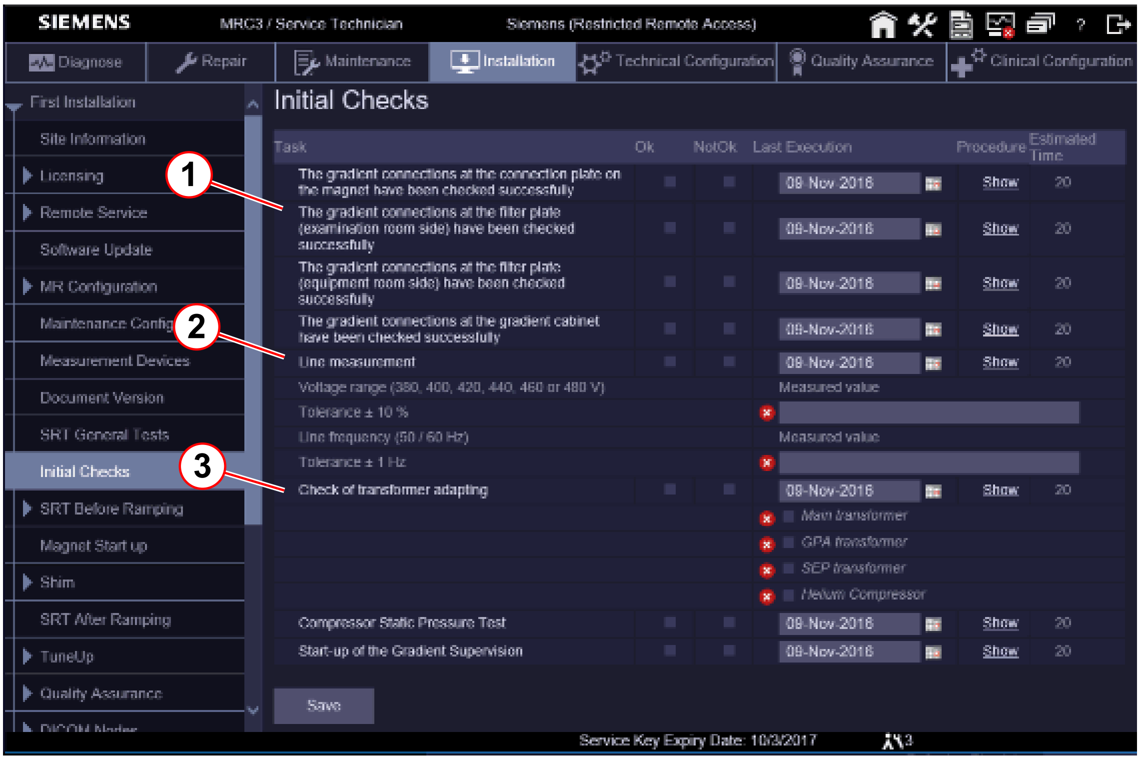
(2) Line Measurement

(3) Transformer Adapting

5.6.6 Check of transformer adapting

Help ID: BA6D0A8F-765D-45E7-84A3-22163135A436
Service Level: 7

Fig. 445: Initial Checks



(1) Gradient Connections
(2) Line Measurement
(3) Transformer Adapting

- Mark the checks:
  - Main Transformer
  - GPA Transformer
  - SEP Transformer
  - Helium Compressor
- Mark if test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.6.7 Compressor Static Pressure Test

Help ID: EA172EC1-B7EE-4200-A30C-F0DE74338564
Service Level: 7

The test is part of Magnet&Cooling / Compressor Status

> The compressor will be switched off for about 60 min.
> The equalization pressure is logged.
> The compressor will be swiched on automatically.

For the

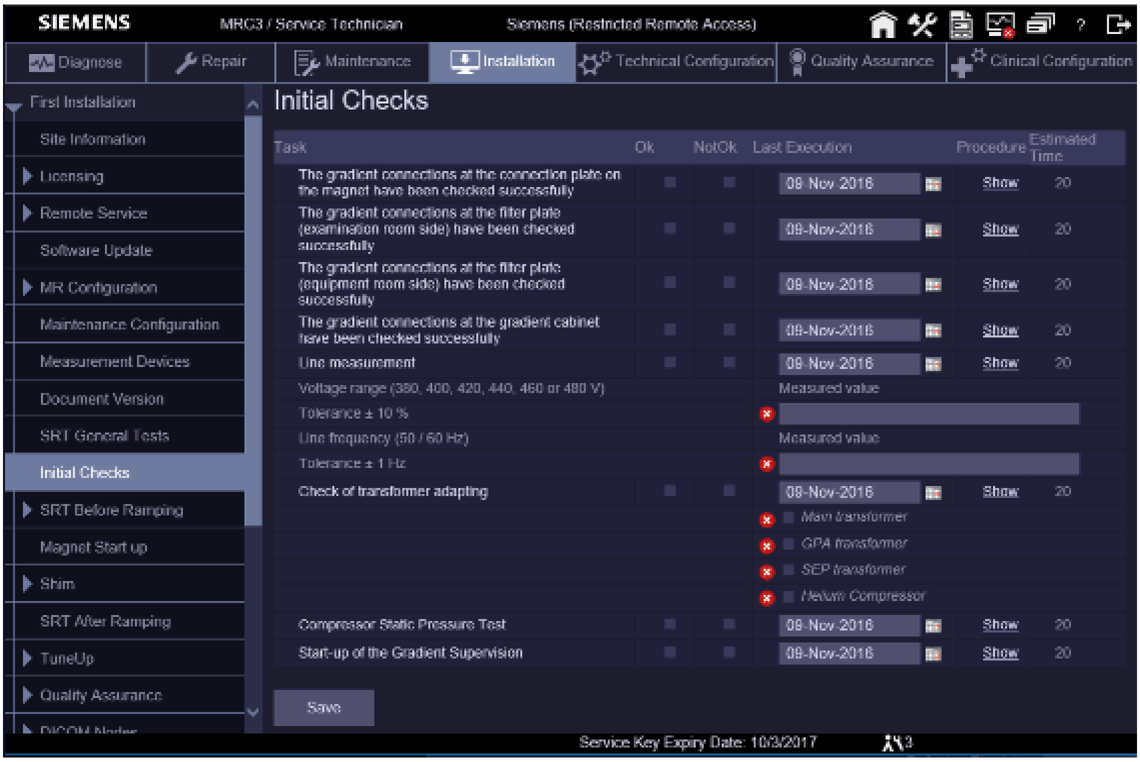


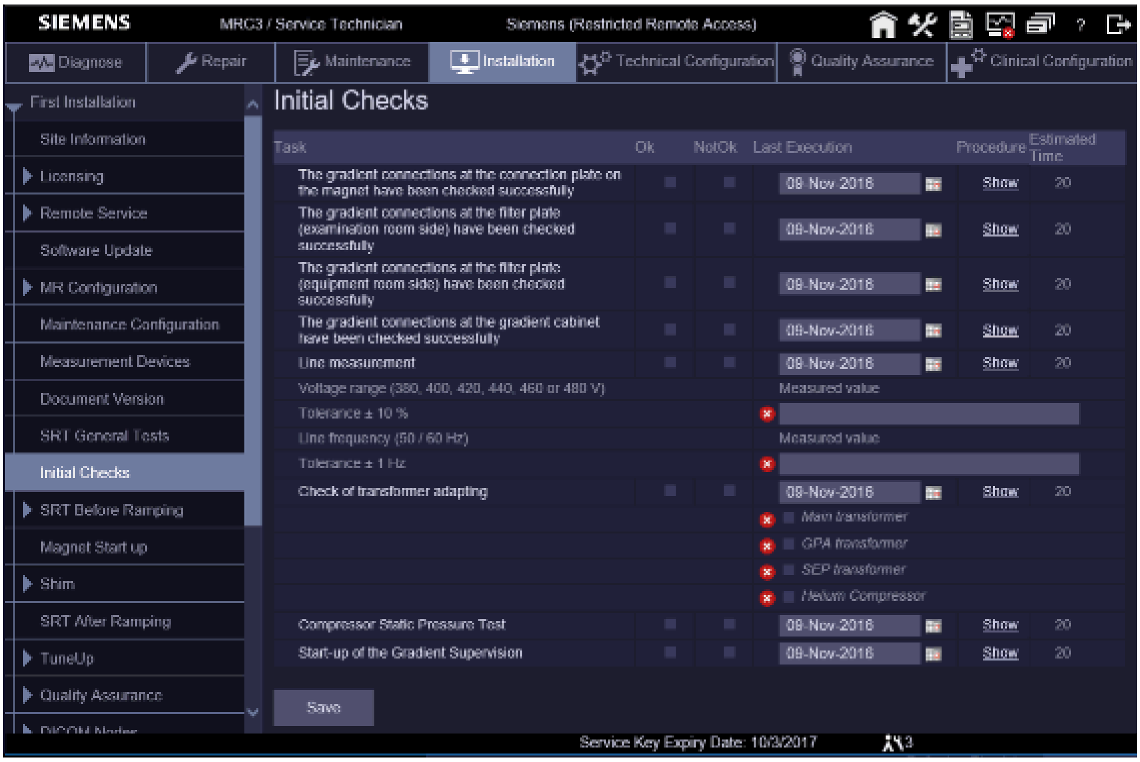
Fig. 446: Initial Checks

- Mark if the test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.6.8 Start-up of the Gradient Supervision

Help ID: C370B435-7AA3-4E0C-AD0D-5F023E0FDC72
Service Level: 7

Fig. 447: Initial Checks



- Mark if the test was **OK** or **NotOk**
- Note the execution date
- Press **Save**

5.7 SRT Before Ramping

5.7.1 Electrical Tests

Help ID: 7DA5DC42-C291-4A82-B4C3-84B74A426643
Service Level: 7

Please select the sub-item.

5.7.1.1 Measurement of System Protective Earth Resistance

Help ID: EA79B57C-1247-4A4E-86CA-775A93D56F9B
Service Level: 7

Measurement of the system Protective Earth Resistance (every 2 years)

Required time: 30 min

Required tools: Protective Earth Resistance meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective earth resistance test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

WARNING

The protective earth resistance meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective earth resistance meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

» **Perform protective earth resistance tests using measurement cables of a sufficient length or after the magnet has been ramped down.**

» The protective earth resistance must not exceed **0.3 ohms**.

- Use the **protective earth resistance meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI EPC MR electronic cabinet - ground bolt - line distribution box (limit value ≤ 0.3 Ω)

Value:

SI RF filter plate - ground bolt (limit value ≤ 0.3 Ω)

Value:

SI Magnet cold head compressor - housing (limit value ≤ 0.3 Ω)

Value:

SI Option SEP - metal frame (limit value ≤ 0.3 Ω)

Value:

SI Option IFP - metal frame (limit value ≤ 0.3 Ω)

Value:

SI MRC (MRAWP) - Host - housing screw (limit value ≤ 0.3 Ω)

Value:

SI MRC (MRAWP) - TFT monitor - housing screw at the bottom (limit value ≤ 0.3 Ω)

Value:

SI Multiple socket for the console components (Line Distribution 1 – UPS) - ground screw (limit value ≤ 0.3 Ω)

Value:

SI Multiple socket for the console components (Line Distribution 2 – NO UPS) ground screw (limit value ≤ 0.3 Ω)

Value:

SI RF components on the magnet - RFCEL housing (limit value ≤ 0.3 Ω)

Value:

SI RF components on the magnet - TX-Box housing (limit value ≤ 0.3 Ω)

Value:

SI Magnet - magnet housing (limit value ≤ 0.3 Ω)

Value:

- All other options such as patient monitor, in-room MRC, Digicam, etc.

Protective earth resistance measurement for PAMO5 CCD-Vario camera is not necessary as it is supplied with low voltage.

SI Option 1 (limit value ≤ 0.3 Ω)

Description:	
Material No.:	
Serial No.:	

Value:

SI Option 2 (limit value ≤ 0.3 Ω)

Description:	
Material No.:	
Serial No.:	

Value:

Enter N.A. in the measured value field if the component/option is not present.

- Enter the values measured for each component in the protocol.

WARNING

Exceeding threshold values.

In the case of an electrical fault, exceeding a threshold value may result in death or serious injury.

» **Test to determine what exceeds the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Work steps of this section were performed by:

Signature:

Date: Name:

5.7.1.2 Measurement of fixed Patient Table Protective Earth Resistance

Help ID: F1F9635B-52D2-467B-8653-5A705B9EDE15
Service Level: 7

Measurement of the Fixed Patient Table Protective Earth Resistance (every 2 years)

Required time: 5 min

Required tools: Protective Earth Resistance meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective earth resistance test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

WARNING

The protective earth resistance meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective earth resistance meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

» **Perform protective earth resistance tests using measurement cables of sufficient length or after the magnet has been ramped down.**

- Use the **protective earth resistance meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI Installed patient table-metal frame (limit value ≤ 0.3 Ω)

Value:

» The protective earth resistance must not exceed **0.3 ohms**.

- Enter the values measured for each component in the safety-related tests protocol.

WARNING

Exceeding threshold values.

In case of electrical faults, exceeded threshold values may result in death or serious injury.

» **Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Work steps of this section were performed by:

Signature: ____________

Date: Name:

5.7.1.3 Measurement of the Dockable Patient Table Protective Earth Resistance

Help ID: F210BD5D-ED5A-4926-986C-29E3BA11471E
Service Level: 7

Measurement of the Dockable Patient Table Protective Earth Resistance (annually)

Required time: 5 min

Required tools: Protective Earth Resistance meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective earth resistance test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

WARNING

The protective earth resistance meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective earth resistance meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

» **Perform protective earth resistance tests using measurement cables of sufficient length or after the magnet has been ramped down.**

- Use the **protective earth resistance meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI Dockable patient table - metal frame (limit value ≤ 0.3 Ω)

Value:

» The protective earth resistance must not exceed **0.3 ohms**.

- Enter the values measured for each component in the safety-related tests protocol.

⚠ WARNING

Exceeding threshold values.

In case of electrical faults, exceeded threshold values may result in death or serious injury.

» **Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Work steps of this section were performed by:

Signature: ____________

Date: Name:

5.7.1.4 Measurement of MRWP (Option) Protective Earth Resistance

Help ID: 62AC8880-E3AC-4221-BFF1-B3814F0040A0
Service Level: 7

Measurement of the MRSC (MRWP) Protective Earth Resistance (annually)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

Required time: 10 min

Required tools: Protective Earth Resistance meter

Use the protective earth resistance meter to measure the ohmic resistance between the ground pin at the plug of the power distribution for the MRSC components and a non-insulated metallic part of the:

SI MRSC (MRWP) Host - housing screw (limit value ≤ 0.3 Ω)

Value:

SI MRSC (MRWP) TFT monitor - housing screw at the bottom (limit value ≤ 0.3 Ω)

Value:

SI Option 1 (limit value ≤ 0.3 Ω)

Description:	
Material No.:	
Serial No.:	

Value:

WARNING

Exceeding threshold values.

In case of electrical faults, exceeded threshold values may result in death or serious injury.

» **Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

» The protective earth resistance must not exceed **0.3 ohms**.

- Enter the values measured for each component in the protocol.

Enter N.A. in the measured value field if the component/option is not present.

5.7.2 Function Tests

Help ID: 948D7432-0ABE-4EEE-8426-3B1D4C086735
Service Level: 7

Please select the sub-item.

5.7.2.1 Checking the emergency power off EPO buttons

Help ID: 0DF6629A-7484-40B5-9484-A1EEAE725C5A
Service Level: 7

Emergency power off switches

Required time: 15 min

Required tools: n.a.

SI Checking the emergency power off switches EPO

Prerequisites:

The system contactor in the EPC and GPA cabinet is switched on.

If a UPS (uninterruptible power supply) is installed, the UPS display shows `O/P: 230 V`. Press the UPS enter button (➔ 1/Fig. 448 Page 670) to open display.

The system is not booted (system shutdown or press **F12** during boot).

- Press the emergency power off switch (EPO).
- Check that the **EPC** cabinet is de-energized.
  » LEDs on the EPC cabinets are off.

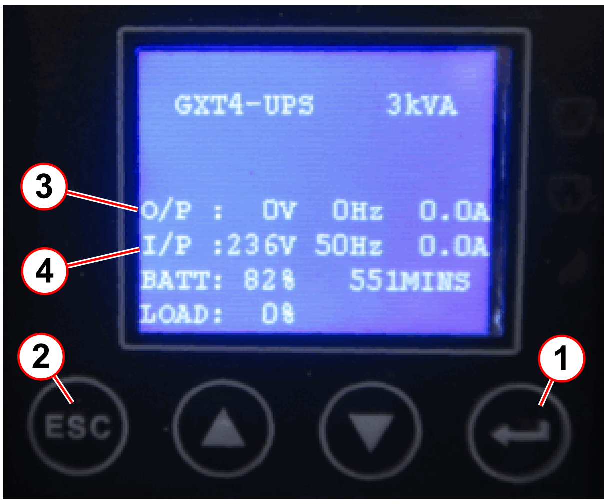
SI EPC cabinet is de-energized.

- Check that the **GPA** cabinet is de-energized.
  » LEDs on the GPA cabinets are off.

SI GPA cabinet is de-energized.

- If a **UPS** is installed, the UPS output voltage must be switched off.
  » The UPS display shows `UPS Shutdown Command Received` for few seconds and then `O/P: 0 V` (output voltage is 0V (➔ 3/Fig. 448 Page 670)).

Fig. 448: UPS display



(1) Enter button
(2) Escape button
(3) O/P output power
(4) I/P input power

SI Option UPS: display shows output voltage = 0 V

- Check that the **"Line"** LED at the alarm box is off.
  » If the RF room has an electric or pneumatic door, it must be possible to open it in the event of a power failure.
- Check whether the RF room door can be opened when the system is "powered off."

Release the EPO switch and reactivate the system by switching on the main power.

After release of the EPO switch, the UPS has to be switched on manually.

- Switch on the UPS: Navigate in the UPS display menu to switch on output power > press **enter** button > **3_CONTROL** > **1_TURN ON & OFF** > **TURN ON UPS** > **Do you want to turn on UPS** > **YES** > **enter**
  » UPS output voltage is on (`O/P: 230 V`)
- **Perform this test on all emergency power-off switches in the equipment room, control room, and the examination room.**

5.7.2.2 Magnet stop button Test (ERDU)

```
Help ID: E45A510B-E317-4C76-8260-17FED12AEA10
Service Level: 7
```

ERDU Checks

Required time: 15 min

Required tools: Screwdriver

Prerequisites

WARNING

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorized personnel must be excluded from the imaging suite.

» **Secure the room and prevent unauthorized access.**

- **Restrict access** to the magnet if it is at field when the test is being performed.
- Check that all alarm and error indications show that the system is OK.
- Remove the appropriate covers to gain access to the MSUP and Service Magnet Stop button.

Testing the Magnet Stop button (ERDU)

CAUTION

Magnet quench!

A magnet quench is caused by pressing the service magnet stop button (ERDU).

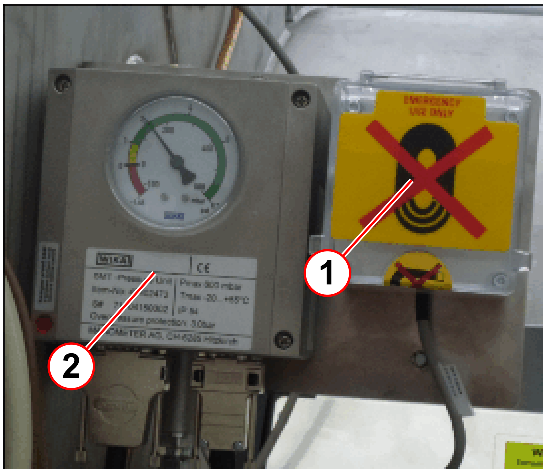
» **Do not operate the Service Magnet Stop button unless an emergency magnet quench is required.**

SI Magnet Stop button test (ERDU)

The Magnet Stop button locks when depressed.

Turn the button anti-clockwise to unlock.

Fig. 449: Service ERDU location



(1) Service ERDU location
(2) Pressure switch

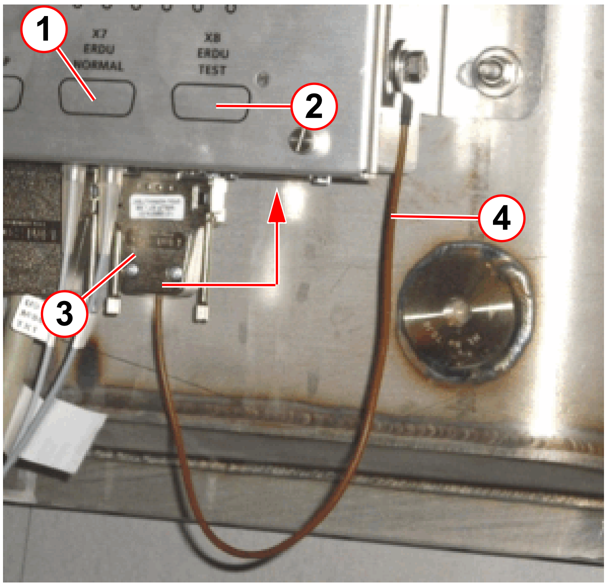
- The Service Magnet Stop button is located next to the pressure switch.
- Check that the button is not activated.
- Check that ERDU test connector is connected to X7.
- Test the Service Magnet Stop button.
  - » Press the button to ensure the MSUP audible alarm sounds
  - » Reset the button - ensure the alarm stops

When the ERDU test connector is disconnected from **BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.**

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When the ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

Fig. 450: ERDU test connector



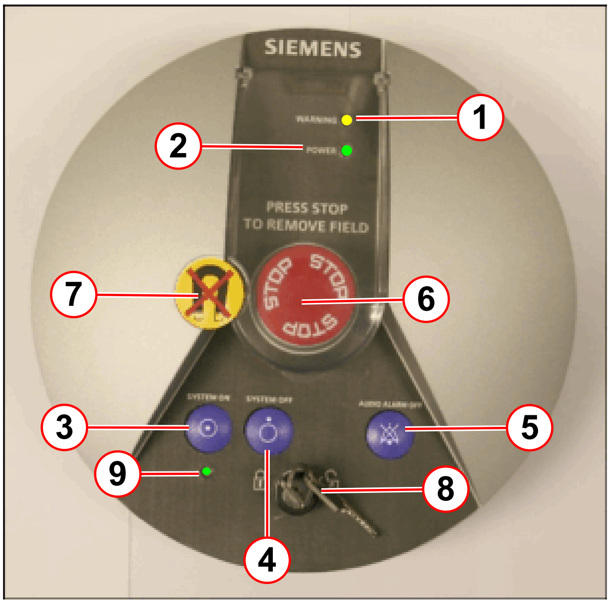
(1) ERDU NORMAL
(2) ERDU TEST
(3) ERDU test connector
(4) ERDU connector strap

The ERDU test connector is fitted to X7 during normal operation.

- Disconnect the test connector from X7 (ERDU NORMAL).
  » MSUP audible alarm is activated.
- Connect it to X8 (ERDU TEST).
  » On the alarm box, check the audible alarm sounds.
  » Press AUDIO ALARM OFF to silence the alarm.
- Check that the ERDU TEST LEDs are NOT illuminated .

> If the MSUP audible warning is not sounding, DO NOT CONTINUE.
>
> Investigate the cause of the problem.

Fig. 451: Alarm box - LEDs and buttons

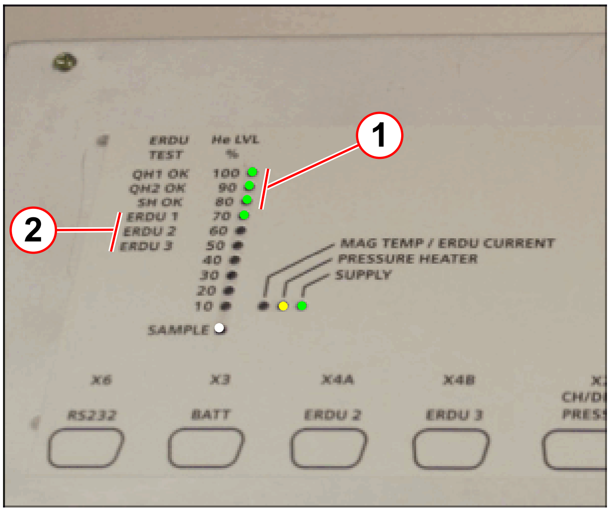


(1) LED WARNING = yellow
(2) LED POWER = green
(3) Button - SYSTEM ON
(4) Button - SYSTEM OFF
(5) Button - AUDIO ALARM OFF
(6) Button - MAGNET STOP (ERDU)
(7) Tamper-proof seal
(8) Power switch locking key
(9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
  - » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
  - » Press AUDIO ALARM OFF to silence the alarm.

Fig. 452: ERDU test - LED indicators (typical)



(1) QH1, QH2, SH = green LED (all ON)
(2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
  - » QH1, QH2, SH and ERDU 1 are NOT illuminated.

Other system ERDU buttons:

- Press one of the stand alone (system) Magnet Stop buttons.
  - Check the audible alarm sounds, and the warning LED flashes on the alarm box.
  - Press AUDIO ALARM OFF to silence the alarm.

- At the MSUP:
  - Check QH1, QH2, SH LEDs are illuminated.
  - Check ERDU 2 LED is illuminated for the magnet room ERDU button, or ERDU 3 LED for the optional 3rd ERDU button.
- Reset the Magnet Stop button.
- Check the LEDs return to normal.
- Repeat for each Magnet Stop button.

Final steps

- Check ALL Magnet Stop buttons are not activated.
- Disconnect the ERDU test connector from X8 (ERDU TEST) and reconnect it to X7 (ERDU NORMAL).
  » On the alarm box, check the audible alarm sounds.
- At the MSUP:
  » Check the MSUP audible alarm is deactivated.
  » Check the MSUP LEDs return to normal.
- At the alarm box:
  » Press AUDIO ALARM OFF to silence the alarm.
- Check the warning LED is not illuminated 'ON'.

> If the LED is on, or flashing, the fault must be checked.
> Do not disconnect the ERDU battery.

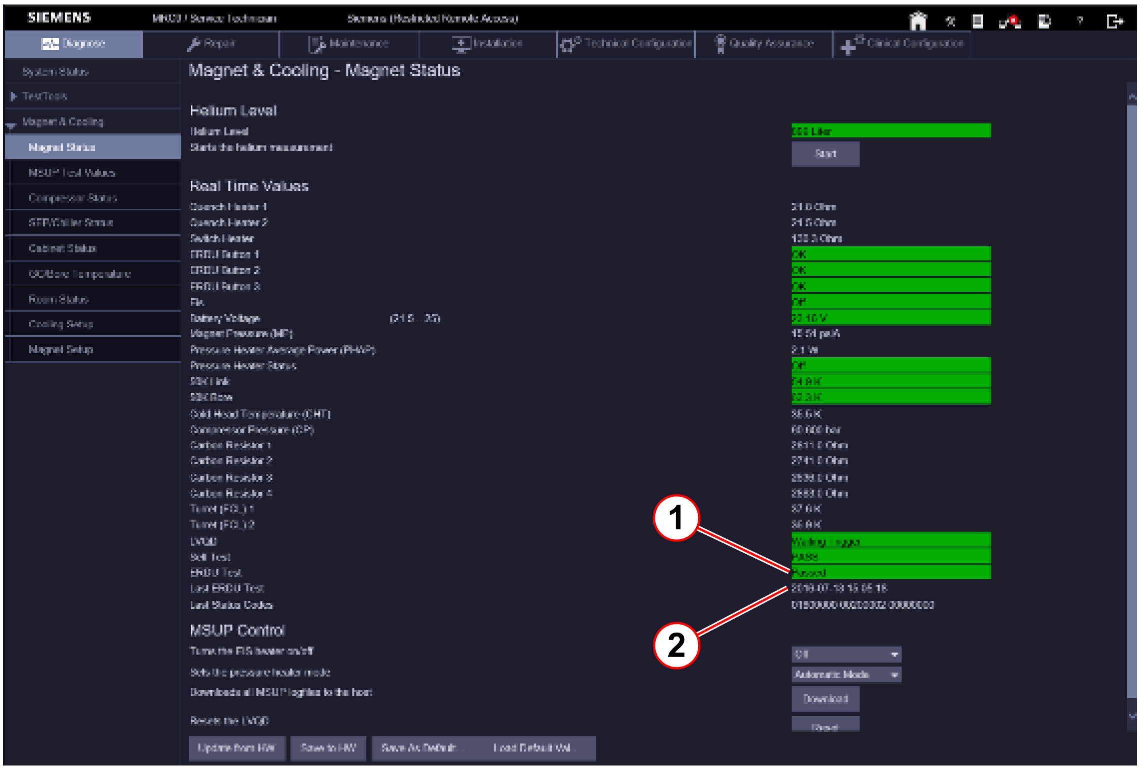
ERDU Test status

- Access the Service Software at the MRC.
- Select the menu item Magnet & Cooling - Magnet Status.

> Data is only accurate from the last refresh of the page, or from when the page was opened last.

- Check the status of the ERDU Test and the date of the Last ERDU Test.

Fig. 453: ERDU test



(1) ERDU test status

(2) Last test date

5.7.2.3 Checking the gradient supervision

Help ID: 687555C2-3225-4055-A0F3-0D7B406710D5
Service Level: 7

Gradient Supervision

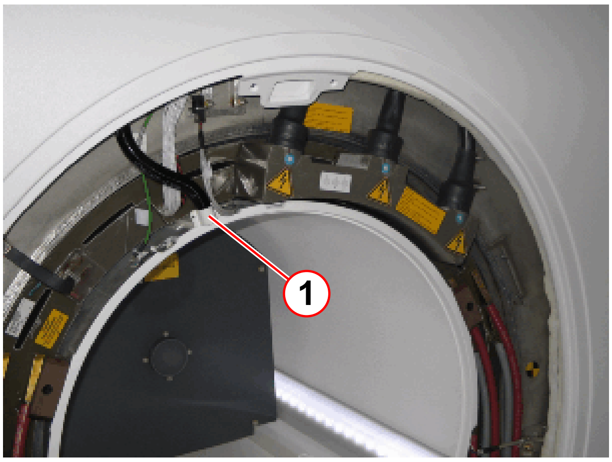
Required time: 30 min

Required tools: Test spray material number 10684091

Prerequisites: The display of the aspiration smoke detector shows OK (➔ 1/Fig. 460 Page 680). System is on.

Fig. 454: Test spray

Fig. 455: Magnet service end, gradient connection

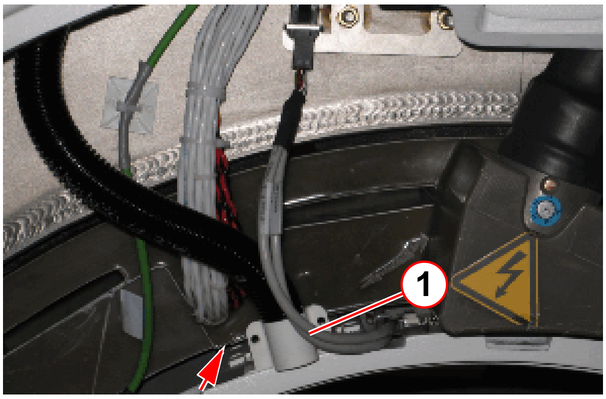


(1) Intake manifold BC

- Remove the rear funnel (service end).

SI Checking the gradient supervision

Fig. 456: Magnet service end, gradient connection



(1) Intake manifold BC

- Spray the test gas dosed in the gap between the gradient coil and body coil toward the intake of the BC manifold for 5 seconds. In this context, “dosed” means not completely pressing the spray button. Distance between spray and intake manifold should be approximately 5 to 10 cm.

NOTICE

On-site smoke detectors

Avoid false alarm of the on-site smoke detector!

» **Check if there is an on-site smoke detector. The test gas must not enter the on-site smoke detector.**

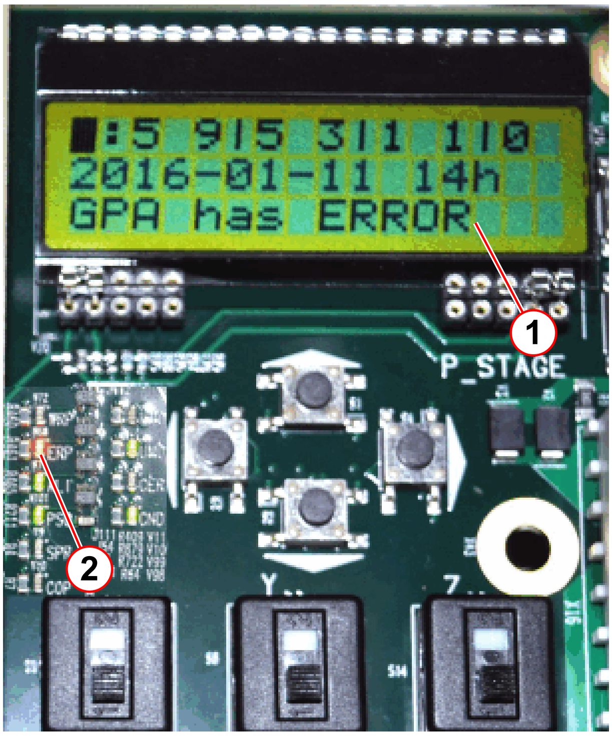
» **Keep the test gas away from on-site smoke detectors.**

» **Keep the spraying as short as possible to minimize test gas concentration in the air.**

Expected test results:

- The test gas is being pulled (driven by two small fans) from the BC manifold and into the aspirating smoke detector. This will take up to 40 seconds (depending on the hose length), then the following results are expected simultaneously:
  - » The red "FIRE" lamp on the VLM device must light up.
  - » The display on the DCC board in the GPA / DGSU shows 'GPA has ERROR' and the ERP LED lights up red.

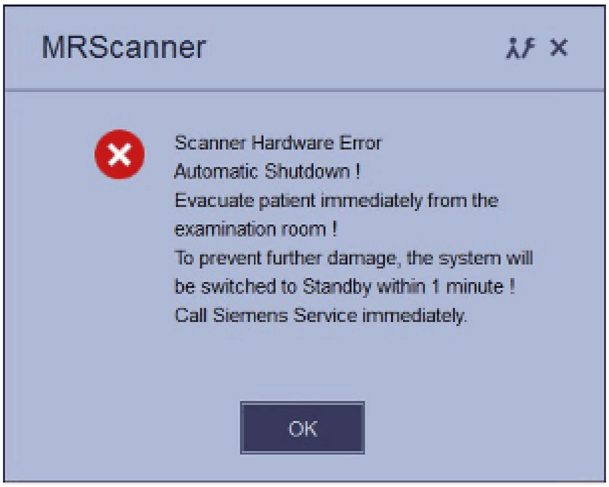
Fig. 457: GPA / DGSU



(1) DCC board display
(2) ERP LED

- » An alarm message appears at the main console. (can also be checked in the error log).

Fig. 458: Pop-up for "Automatic Shutdown"



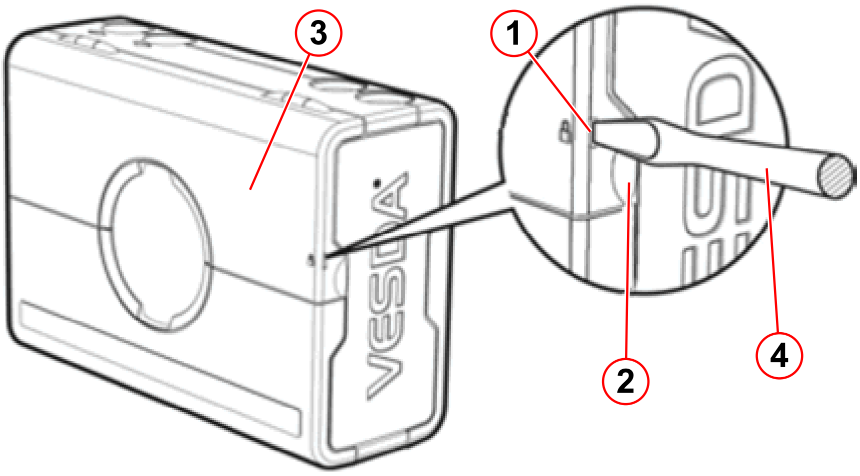
One minute later,

» The system must be switched to stand-by mode automatically.

Reset the system:

- Press a screwdriver into the security tab (➔ 1/Fig. 459 Page 679) to open the service access cover with the finger clip (➔ 2/Fig. 459 Page 679).

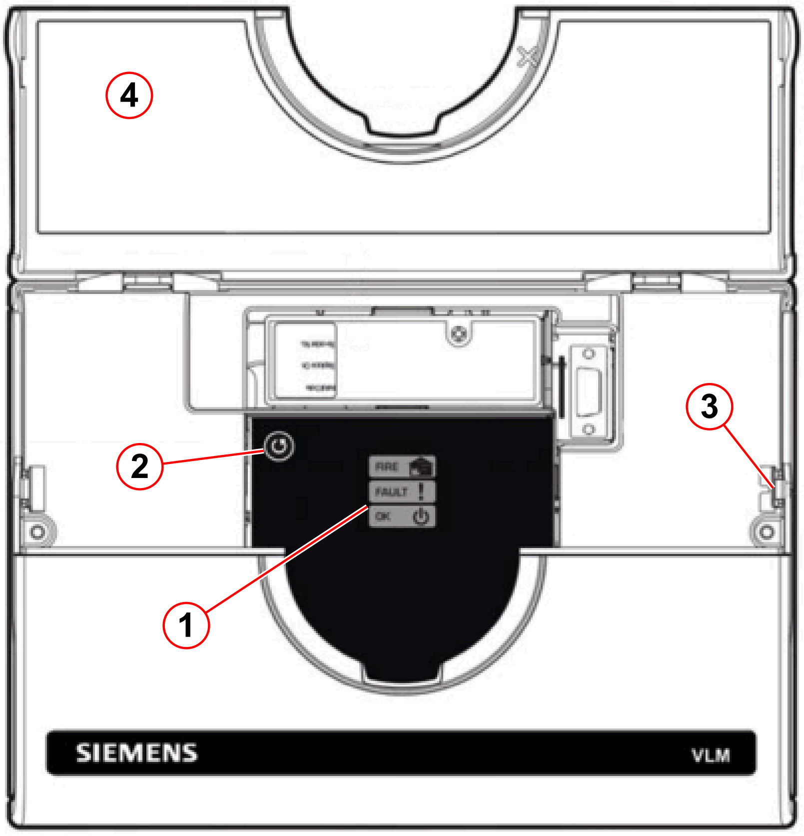
Fig. 459: Aspiration smoke detector ASD



(1) Security tab
(2) Finger clip
(3) Service access cover
(4) Screwdriver

- Reset the alarm by pressing the **reset button** (→ 2/Fig. 460 Page 680)for 2 seconds.

Fig. 460: Aspiration smoke detector



(1) Status display
(2) Reset button
(3) Security tab
(4) Service access cover

If the reset fails, the concentration of smoke inside the detector chamber is too high. Retry the reset after 1-2 minutes.

» The aspiration smoke detector shows OK (→ 1/Fig. 460 Page 680).

- Start the system by pressing the **ON button** at the alarm box.

5.7.2.4 Quality Assurance

Help ID: 4E85213B-F73D-4CB2-97ED-78F6E63EC002
Service Level: 7

Quality Assurance (QA)

Required time: 70 to 110 min depending on installed options.

Required tools: QA Phantom Setup

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

> Use the Online Help to obtain detailed information.. Press the < **?** > help button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Open the Quality Assurance platform **> Administration Portal > enter password > Quality Assurance > General Quality Assurance**.

SI Quality assurance measurement

- Start the measurement by clicking the “Go” button.
  » The **Quality Assurance** check is successfully completed when every individual QA step indicates the status **< OK >**.

> If the results are out of tolerance, perform “Tune-Up” or troubleshooting, if necessary. Repeat “Quality Assurance”.

5.8 Magnet Start up

Help ID: 2F8ACBE3-D7D3-44B1-95CF-9F41995A7417
Service Level: 7

5.8.1 Verifying the electronic low pressure switch

Help ID: DEC96E94-6AE4-4B27-AACA-7D4D5B9B8389
Service Level: 7

This test checks the function of the low pressure switch of the magnet.

The low pressure swich will switch off the helium compressor in case of low pressure in the helium vessel.

The test is described in the Magnet Start-up documentation.

5.8.2 Magnet room empty check

Help ID: E38E07AC-A5EF-403C-87B0-5720D392B964
Service Level: 7

The magnet room has to be inspected for the presence of magnetic parts before ramping up the magnet.

Special care has to be taken by undefined objects.

Refer to the Magnet Start-up / Prerequistites for Ramping the Magnet.

5.8.3 Venting leak check

Help ID: 43CBBBA8-85CB-4A3D-B2EF-F715E0EA288E
Service Level: 7

Perform a turret and venting leak check.

Refer to Magnet Start-up / Prerequisites for Ramping the Magnet.

5.9 Shim

5.9.1 Shim - General

Help ID: 46A7A82A-C6D7-45B8-9C8C-BED652C9CBFF
Service Level: 3

General

Shimming

Superconductive magnets for MR imaging systems must have a homogeneous magnetic field at the required strength. To this end, a shimming technique is used to reduce field inhomogeneity to a minimum.

Underlying Physics

The primary objective of MR systems is to provide diagnostic images for medical purposes. Images of the body can be derived from the information obtained by exciting protons in a magnetic field using a radiofrequency signal. As the protons relax, they release their own radio signal that is detectable and provides information about the chemical contents of the body. The image quality depends on the homogeneity of the magnetic field: the better the homogeneity, the better the image quality. Shimming is an integral part of the magnet installation. It compensates for unit fabrication spread and environmental effects on the system homogeneity.

User selection

The Terminal Program is used for controlling the magnet power supply.

The following shows the correct setup of the Terminal Program.

Terminal Program Setup

Terminal Program must be configured and started **before the 3600 MPSU is switched on.**

5.9.1.2 Magnet Shim

Help ID: 86BF2F55-5481-4532-B078-DFCEBE88F075
Service Level: 3

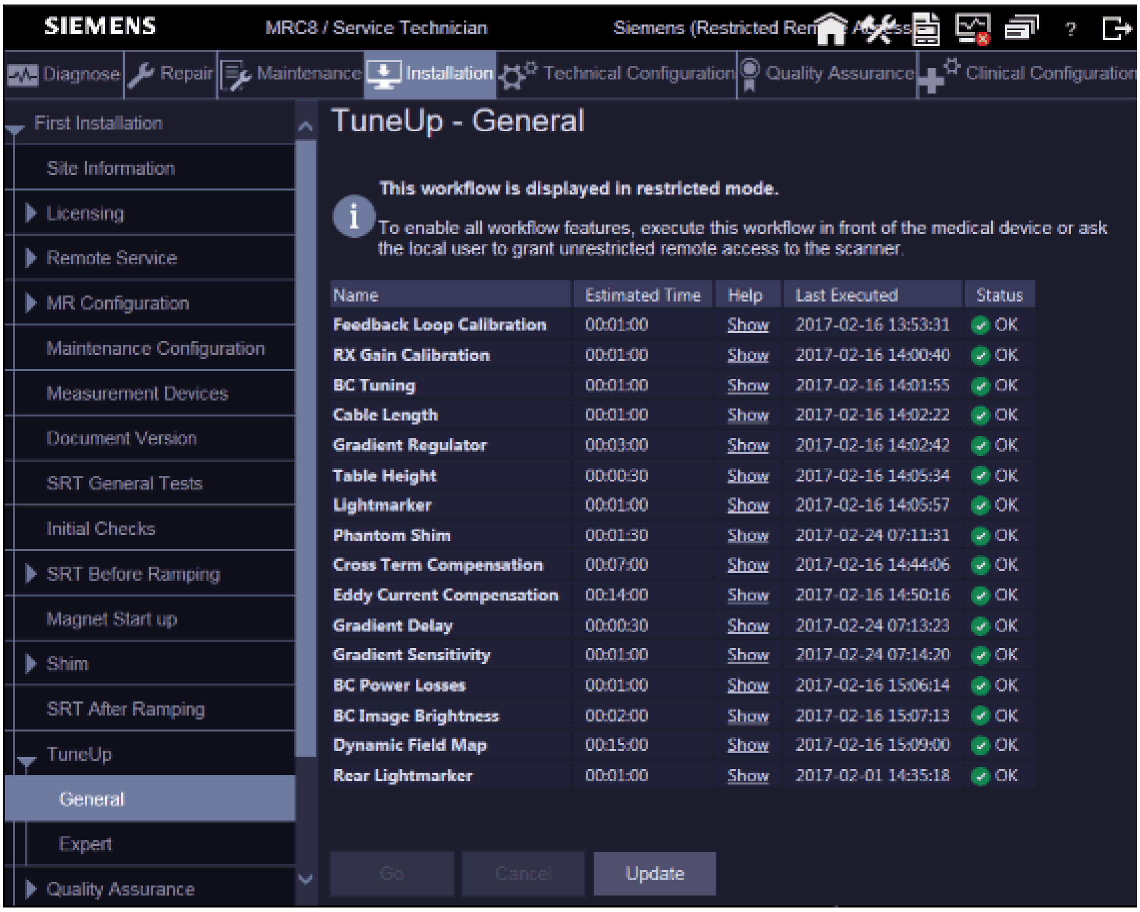
Prerequisites for the Shim Workflow

> i The Feedback Loop Calibration has to be done before starting the shim workflow.

- Select First Installation

- Select TuneUp / General
- Start **Feedback Loop Calibration**

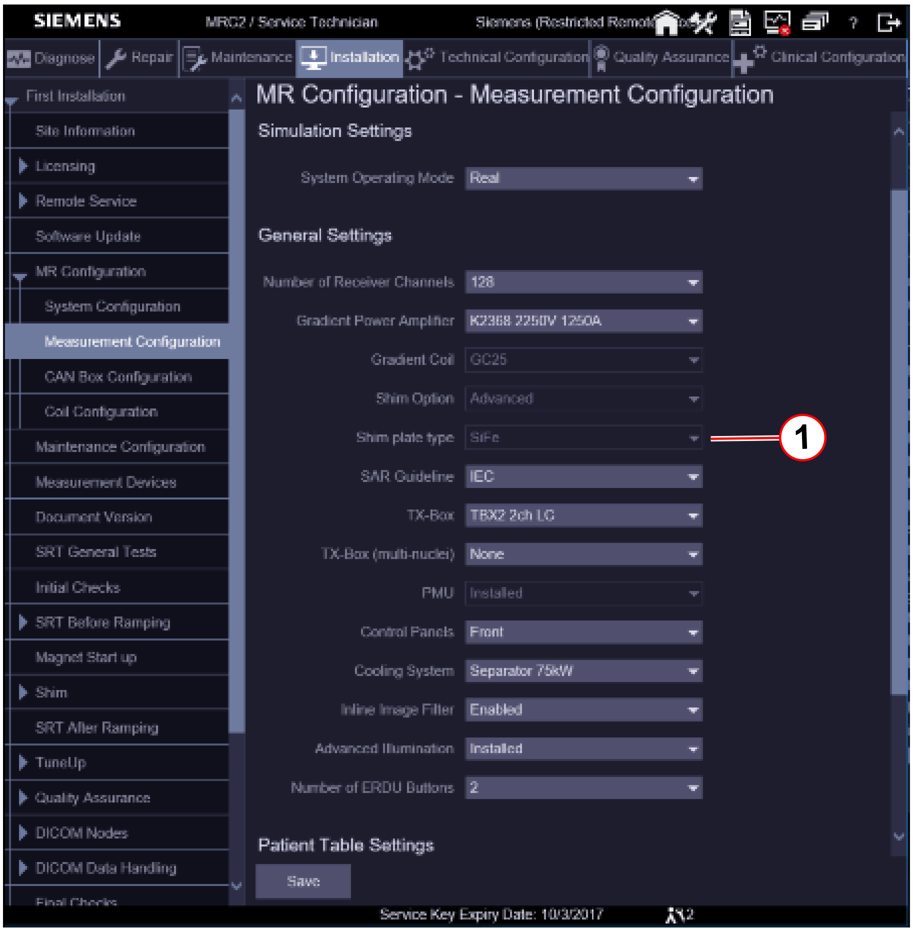
Fig. 461: Feedback Loop Calibration



Passive Shim Workflow

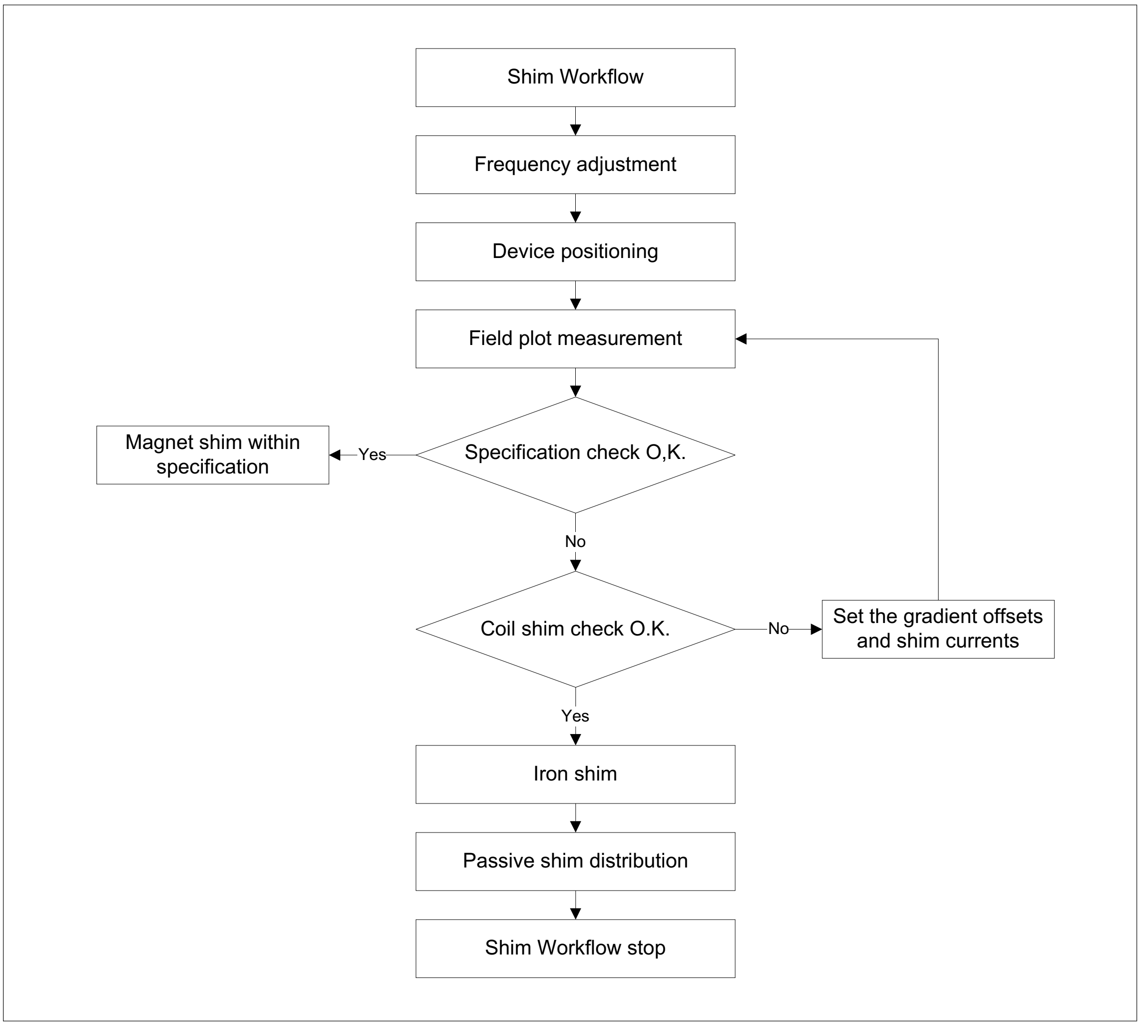
> Magnetom Vida systems are shimmed with Silicon iron (SiFe).
>
> Check the correct shim plate settings. Refer to: MR Configuration / Measurement Configuration / Shim plate type / SiFe.

Fig. 462: Shim plate type



(1) SiFe

Fig. 463: Shim Workflow



The following individual steps are performed in the shim workflow within each workflow mode.

1. Frequency adjustment

- Check to make sure the ASD is connected and no other coils.
- Perform standard frequency adjustment.
- Perform extended frequency search if standard frequency adjustment fails.

  » If the extended frequency search converges, it is shown in the report how to change the magnet current to arrive at the nominal resonance frequency of the system. Once the current is changed, the standard adjustment should converge with the next execution of the program.

2. Device positioning

- Check to make sure the ASD is connected and no other coils.
- Measure frequency of FID signal in all probes for two gradients of opposite strength.
- Check the polarity.
- Calculate position and check whether it is in specification.

- If the position is not in specification, the user is asked to reposition the ASD and then measure again.

3. Field plot measurement

- Check to make sure the ASD is connected and no other coils.
- Ensure that the gradient offsets are applied, and, if applicable, the shim currents as well.
- Check the table position.
- Perform EIS reset.
- Perform adjustment.
- Perform field plot measurement.
- Check data.

4. Specification check

- Create IQShim input file.
- Check to make sure the shim is in specification.
- Set “magnet homogeneity” in the installation protocol according to result.
- Decide whether to continue with coil shim (5.1) or iron shim (5.2) in case shim is not in specification.

5.1 Coil shim

- Create new gradient offsets and shim current.
- Save new offsets/currents if confirmed.

5.2 Iron Shim

- Create an IQShim input file.
- Analyse IQShim output file.
- Report the passive shim distribution result.

Array Shim Device (ASD) kit

Contents of the array shim device box:

- Array Shim Device
  - » Can be used for 1.5T and for 3T
- ASD adapter
  - » Can be used for 1.5T and for 3T
- Array shim cable
  - » Can be used for 1.5T and for 3T
- Shim iron
  - » **Silicon iron for 1.5T and 3T**

Plate type	Part number
Plate 1	10105269

Plate type	Part number
Plate 2	10105268
Plate 3	10105267
Plate 4	10105266
Isolation shim	10117093
Packing shim	8395563

» **Cobalt iron for 3T**

Plate type	Part number
Plate 1	8395522
Plate 2	8395530
Plate 3	8395548
Plate 4	8395555
Isolation shim	10117093
Packing shim	8395563

Do not mix up the cobalt iron and the silicon iron!

Fig. 464: Aera/Skyra Array Shim Device

Fig. 465: Aera/Skyra ASD

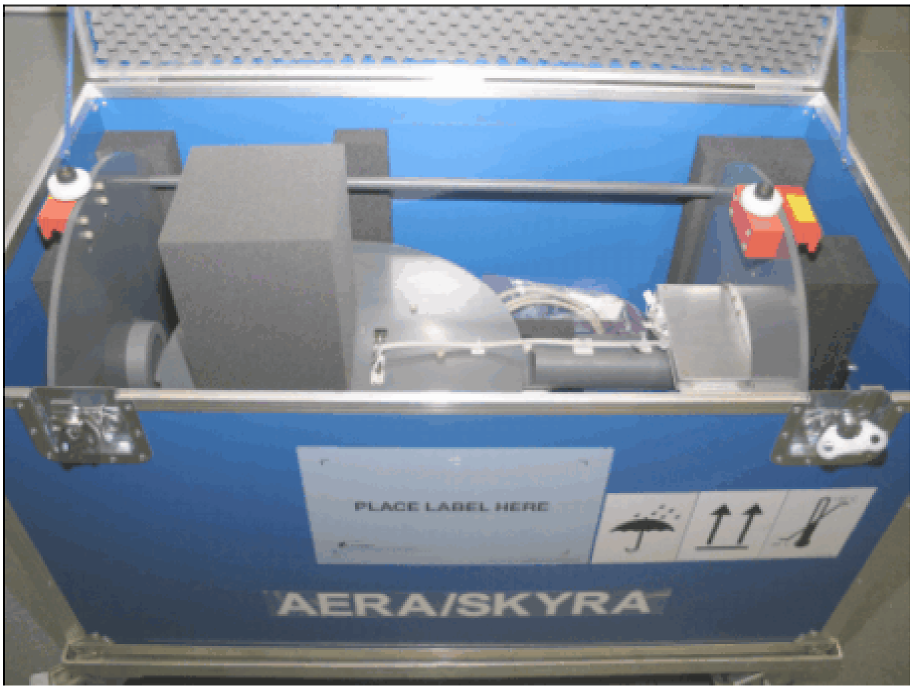


Fig. 466: Aera/Skyra Array Shim Device

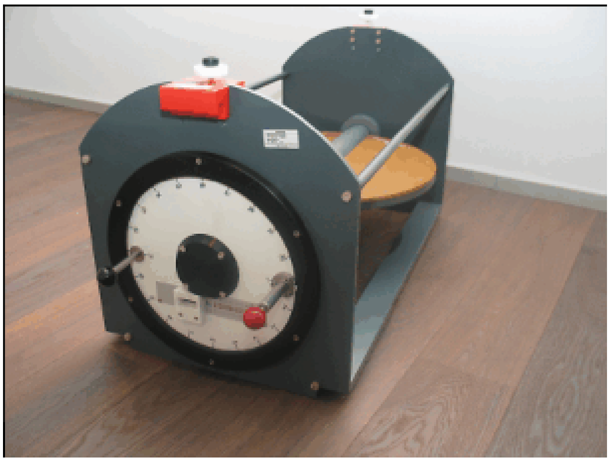
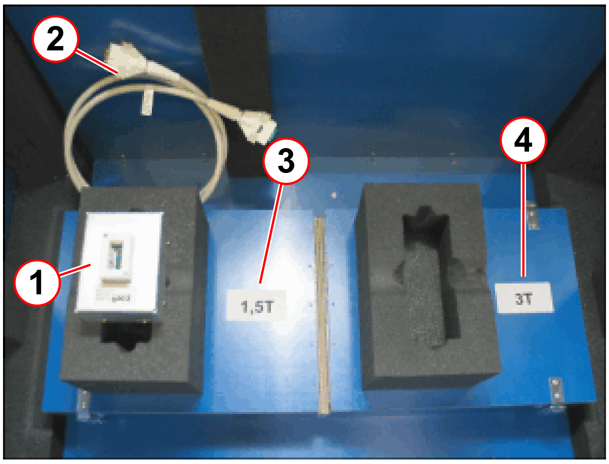


Fig. 467: Array Shim device kit



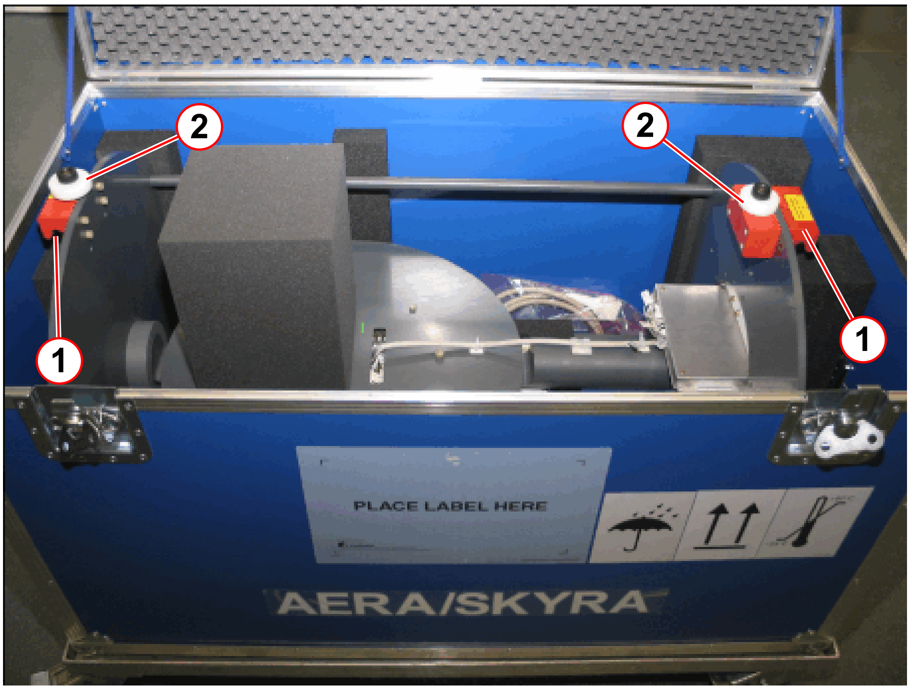
(1) ASD adapter 1.5T, 3.0T
(2) Cable
(3) Shim iron box 1.5T
(4) Shim iron box 3.0T

Installation of Device

- Pick up the array shim device by the two red handles, lift it up carefully, and remove it from the crate.

> To prevent damage to the array shim device, use only the two red handles for transport.
> Two persons are necessary for lifting the array shim device.

Fig. 468: Array shim device



(1) Handle
(2) Bolt

- Move the patient table into the lowest position.
- Slide the array shim device into the magnet.
  Slide it up to the correct position (behind the guide roller 2).

Fig. 469: ASD Position

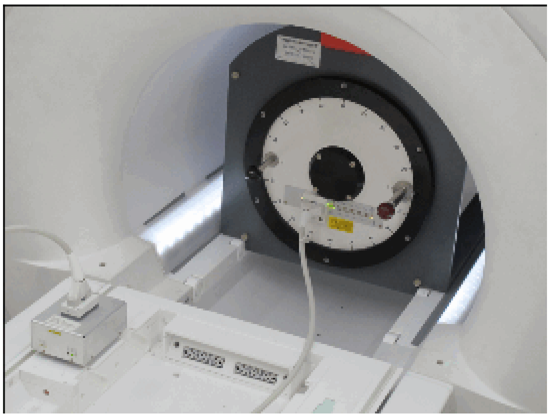
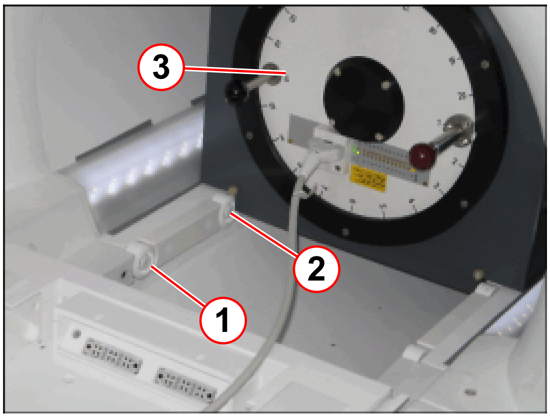
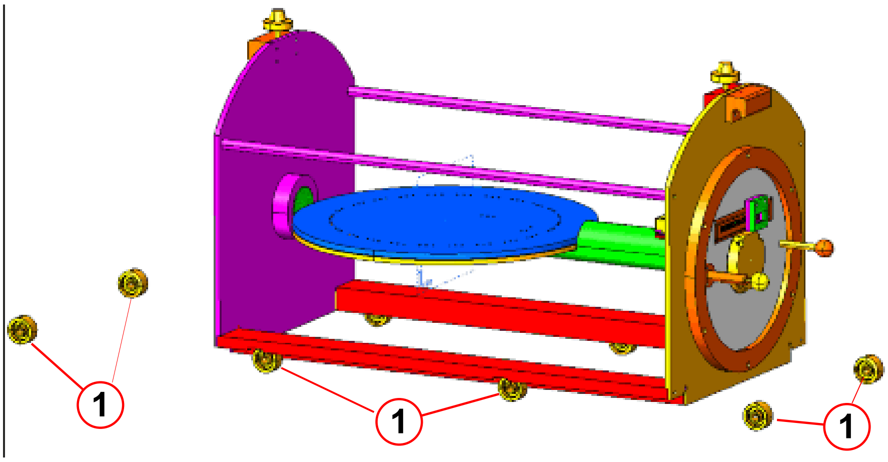


Fig. 470: ASD Position



(1) Guider Roller 1
(2) Guide Roller 2
(3) ASD

Fig. 471: Magnetom Skyra ASD position

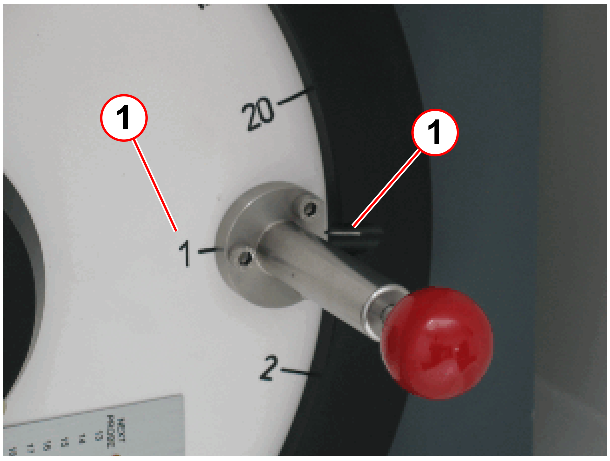


(1) Guide rollers

The array shim device is positioned roughly now..

- Set the array shim device to **position 1**.

Fig. 472: ASD



(1) Position 1

- Pull out the red front knob of the positioning pin, and turn the shim device into position 1.

NOTICE

Do not press the large rotary push button of the control panel if the ASD is installed.

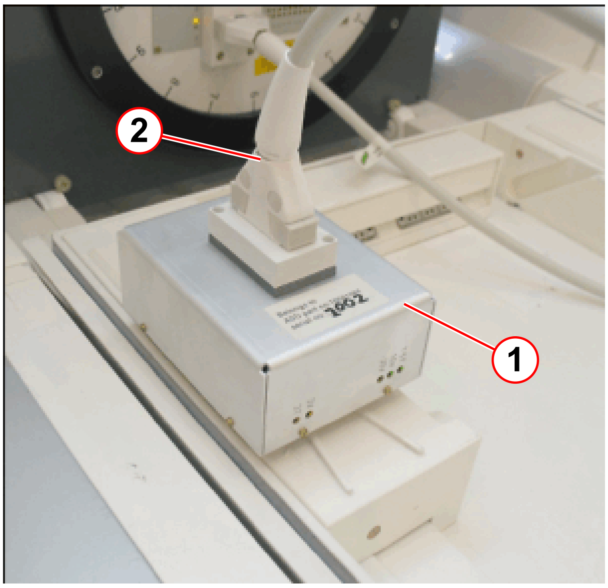
Pressing the large rotary push button will move the patient table to the isocenter automatically.

» **Do not press the large rotary push button if the ASD is installed.**

» **Use the dual stage short range buttons for UP/IN and for OUT/DOWN movement.**

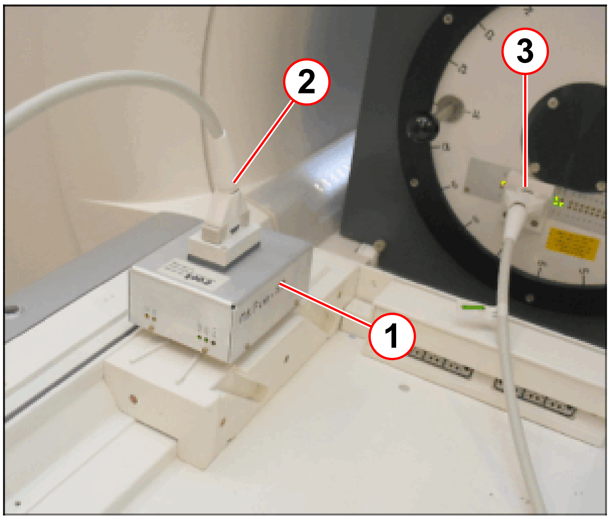
- **Move the patient table into the home position**. The table is slightly magnetic and has to be shimmed as well.
- Connect the ASD adapter to connecter 1 or connector 2 of the patient table.
- Connect the array shim cable to the ASD adapter.
- Connect one end of the array shim cable to the multiplexer of the ASD.

Fig. 473: ASD adapter



(1) ASD adapter
(2) Cable

Fig. 474: ASD adapter

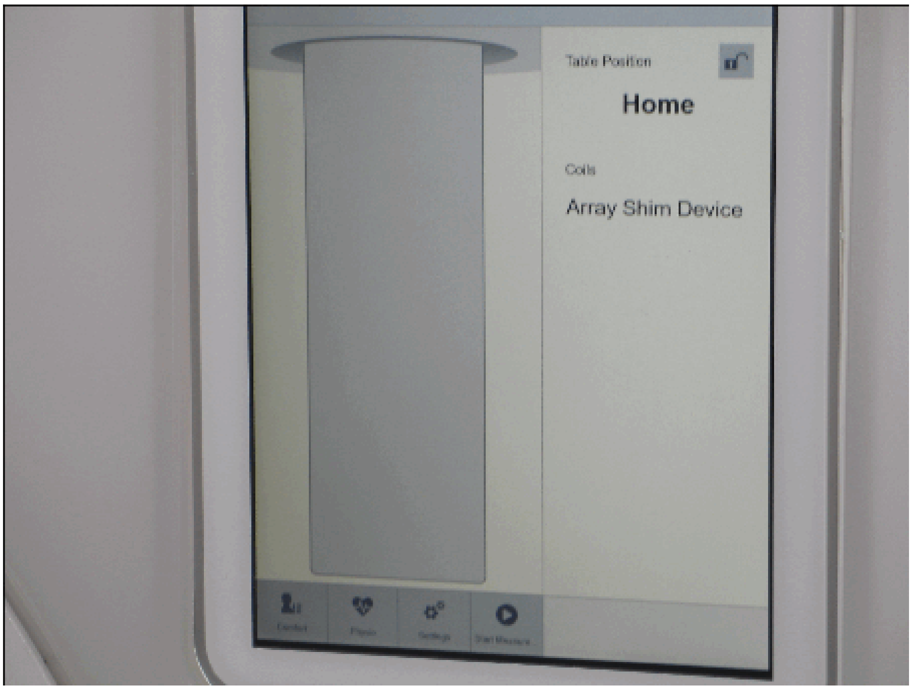


(1) ASD adapter
(2) Cable
(3) Multiplexer

To maintain the correct position of the array shim device, it should not be removed from the magnet during shimming.

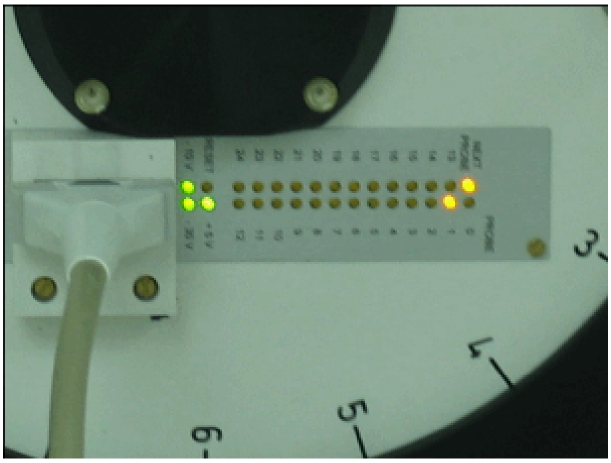
- Indication of the Array Shim Device on the PDD.

Fig. 475: Patient Data Display



- Check the LEDs on the ASD

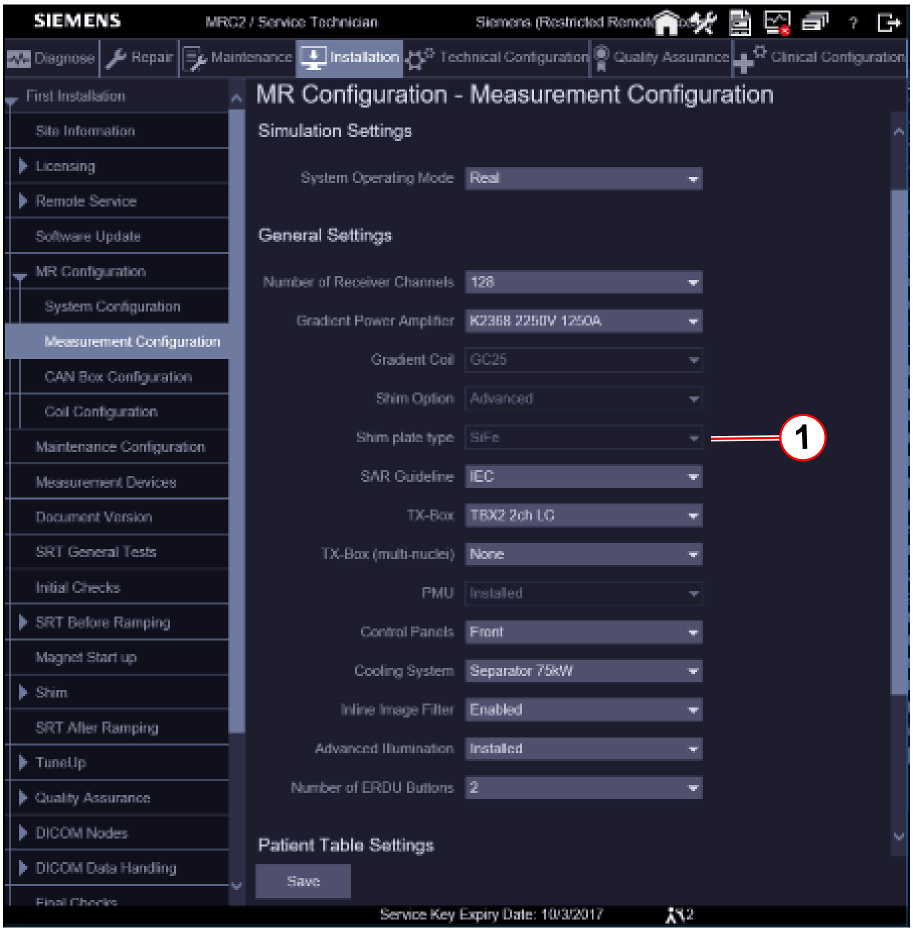
Fig. 476: ASD



Array Shim Measurement

> **i** Magnetom Vida systems are shimmed with Silicon iron (SiFe).
>
> Check the correct shim plate settings. Refer to: MR Configuration / Measurement Configuration / Shim plate type / SiFe.

Fig. 477: Shim plate type



(1) SiFe

The array shim measurement consists of two parts:

- The frequency search that uses the center probe of the array shim device (Frequency range = Center frequency + / - 100kHz).
- The measurement of the 50 cm sphere in 20 steps.

Each measurement consists of 24 measurement points on a 50 cm circle.

The probe signals measured are converted into frequency data and saved to a field data record at the end of the measurement.

This field data record is used for calculating the shim data.

Prerequisites

- MR system is switched on.
- Magnet is at field.
- After the magnet is ramped up, wait for the following amount of time before taking the measurement:
  - At least 30 minutes for the bare and first iteration
  - At least 1 hour for subsequent iterations

- Patient table is in the top position.

Starting the shim workflow

- Start the Service Software UI at the MR console.
- Select **Installation / First Installation / Shim / General**.

Fig. 478: Shim Workflow

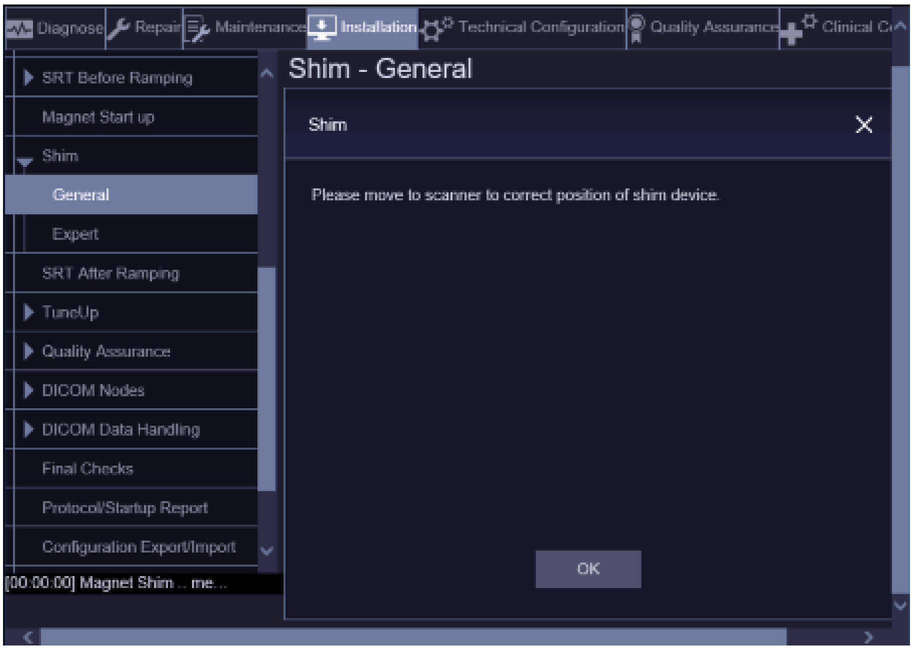
- Start the shim workflow with **Go**.
  » Workflow starts

The shim workflow starts with:

1. Frequency adjustment.
2. Array shim device positioning.
   - The system applies a +1mT/m gradient and measures the frequency at planes 12 and 13.
   - The system applies a -1mT/m gradient and measures the frequency at planes 12 and 13 again.
   - The shim program checks whether the Z gradient is correctly wired.
   - The shim program calculates the offset from the gradient center.

- The shift is displayed on the PDD.

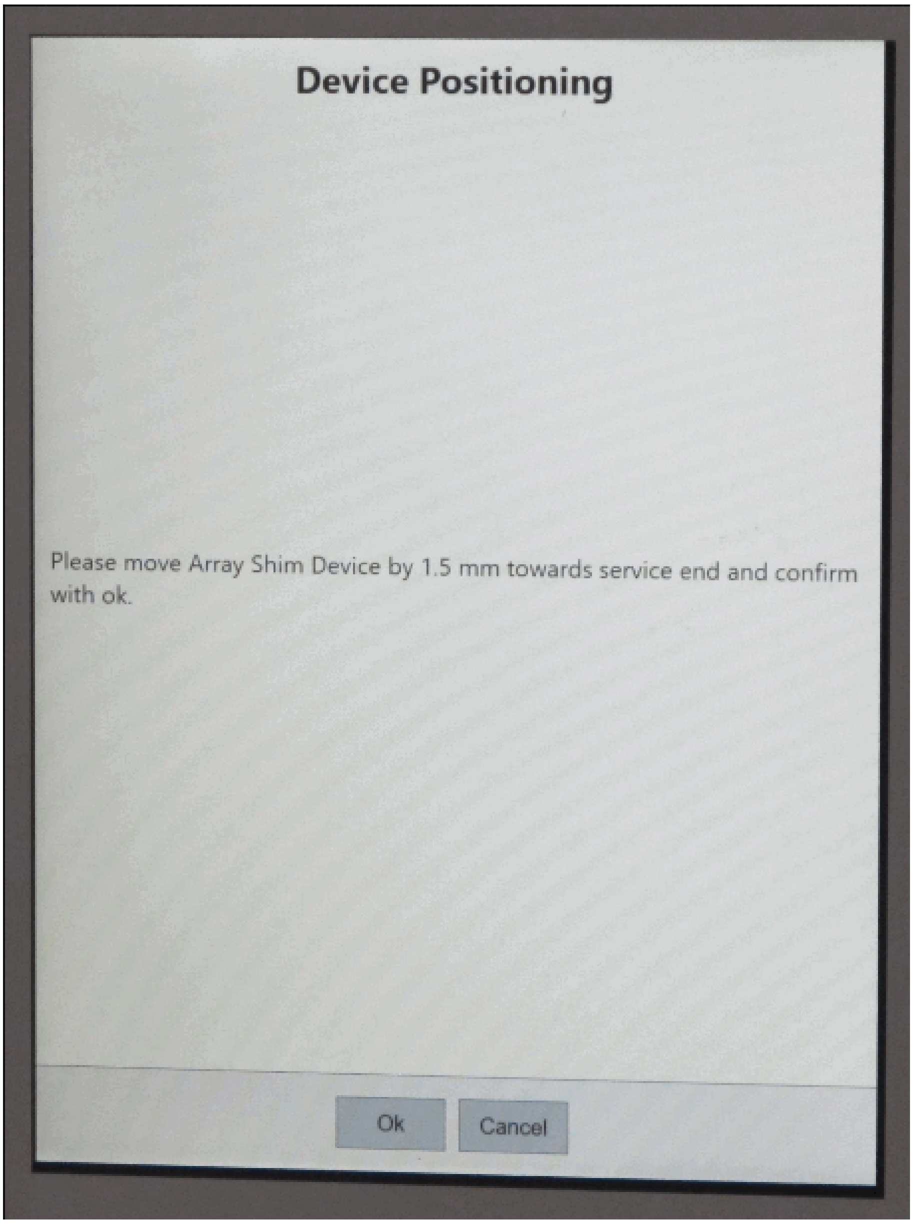
Fig. 479: Shim Workflow



- Press OK and go to the Patient Data Display (PDD).
  » Move the ASD to the correct position (towards service end or towards patient end)

    Press **Ok**

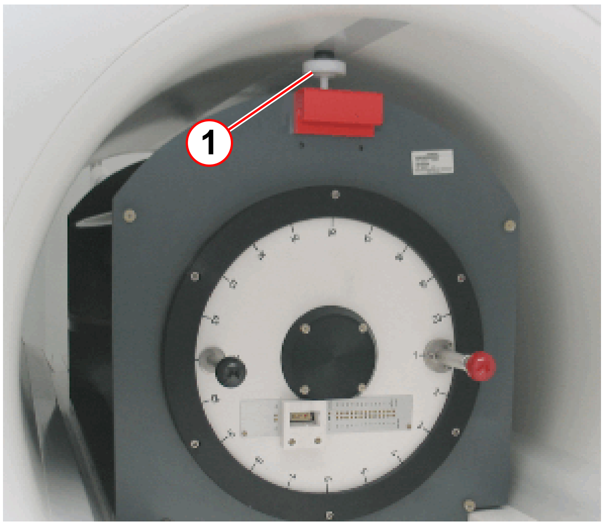
Fig. 480: Device Positioning



» Repeat until the deviation is within +/- 1 mm.

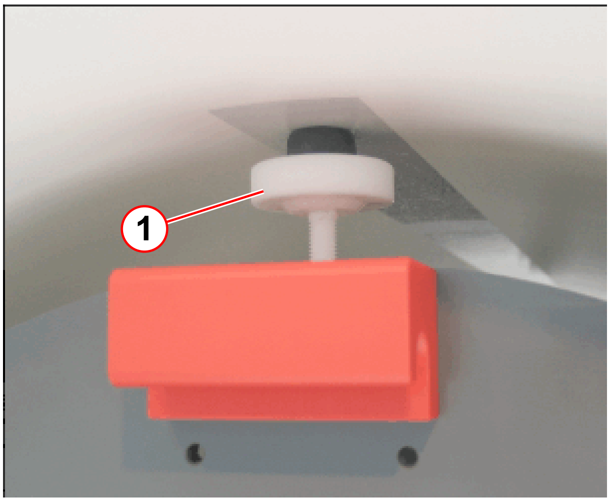
- Secure the ASD by tightening bolt 1 in the resonator at the patient end.
- Secure the ASD by tightening bolt 2 in the resonator at the service end.

Fig. 481: ASD patient end



(1) Tightening bolt 1

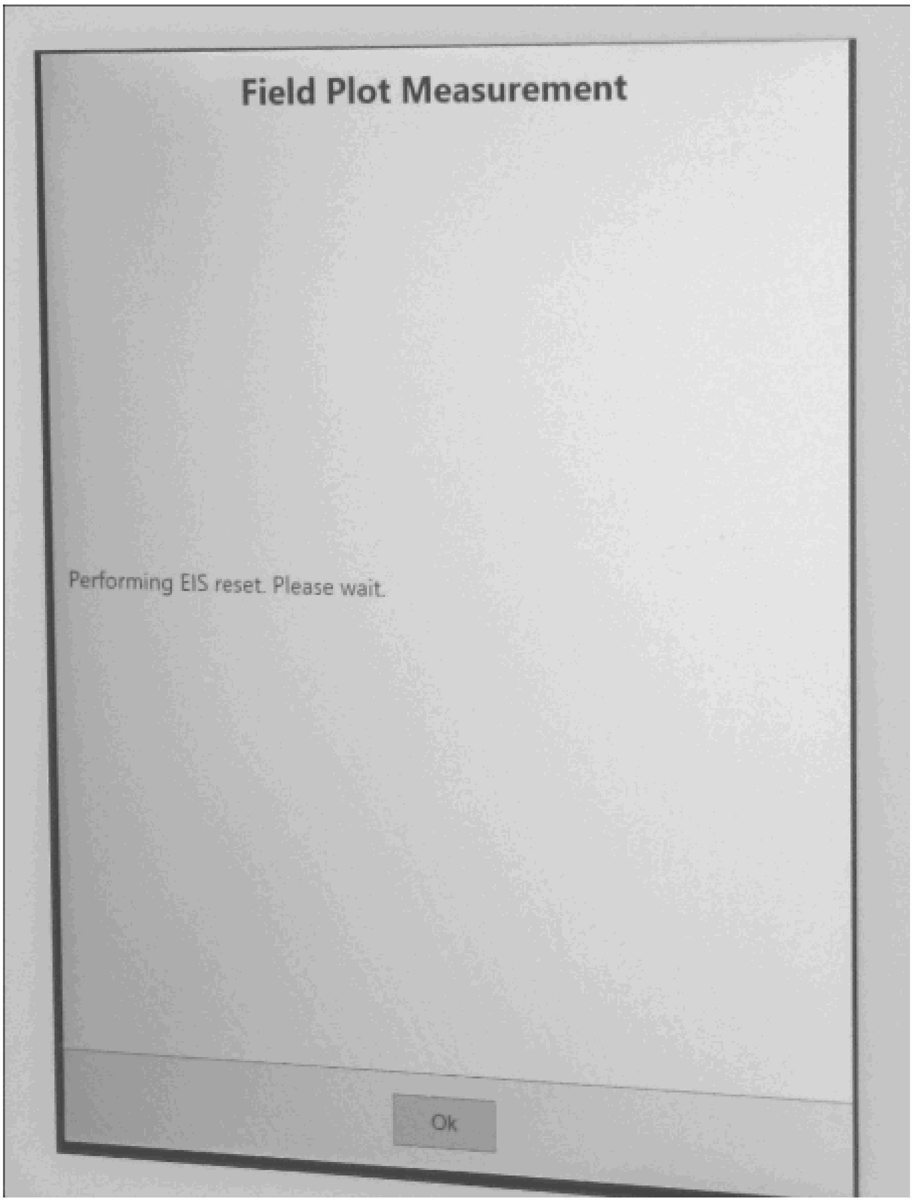
Fig. 482: ASD service end



(1) Tightening bolt 2

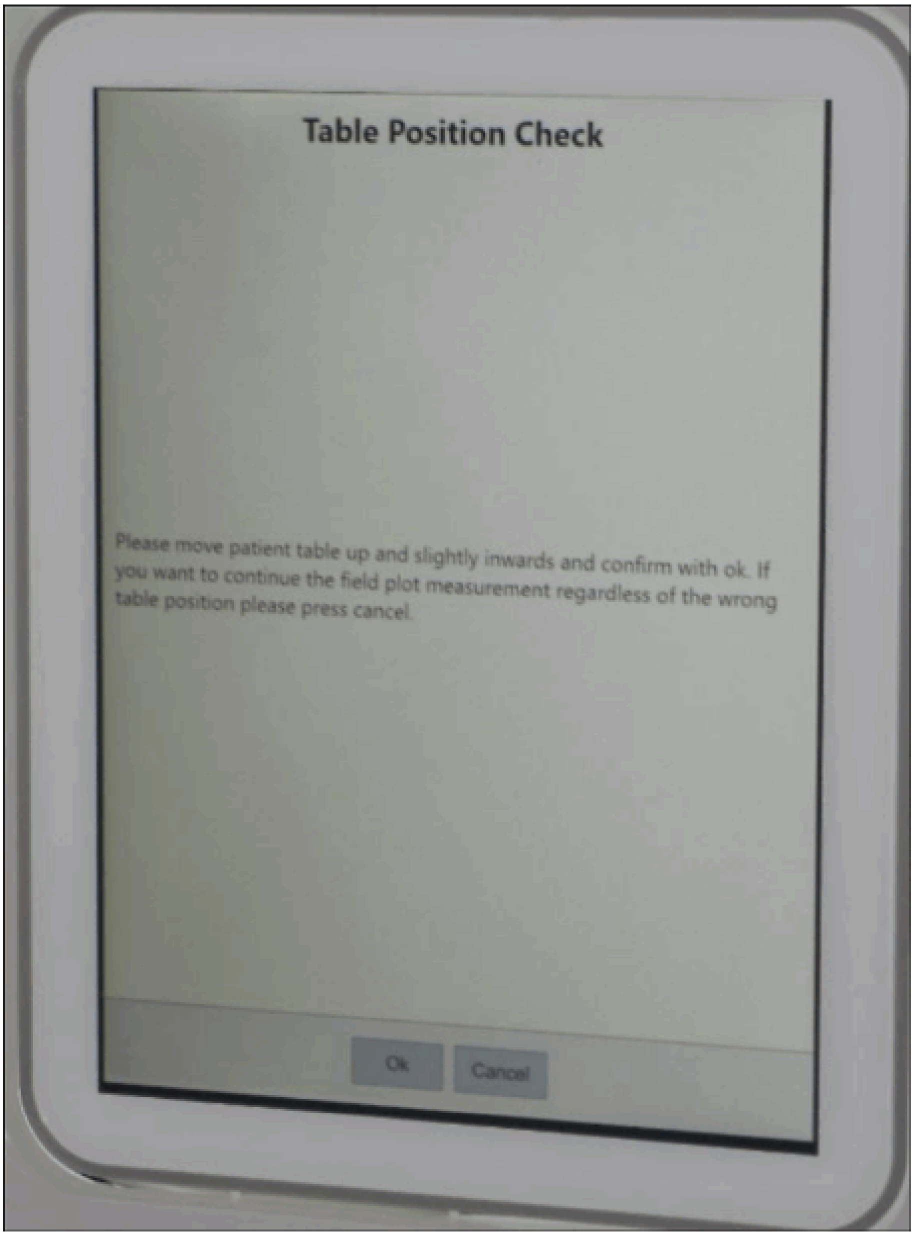
3. EIS reset is performed by the Shim workflow.

Fig. 483: Field Plot Measurement



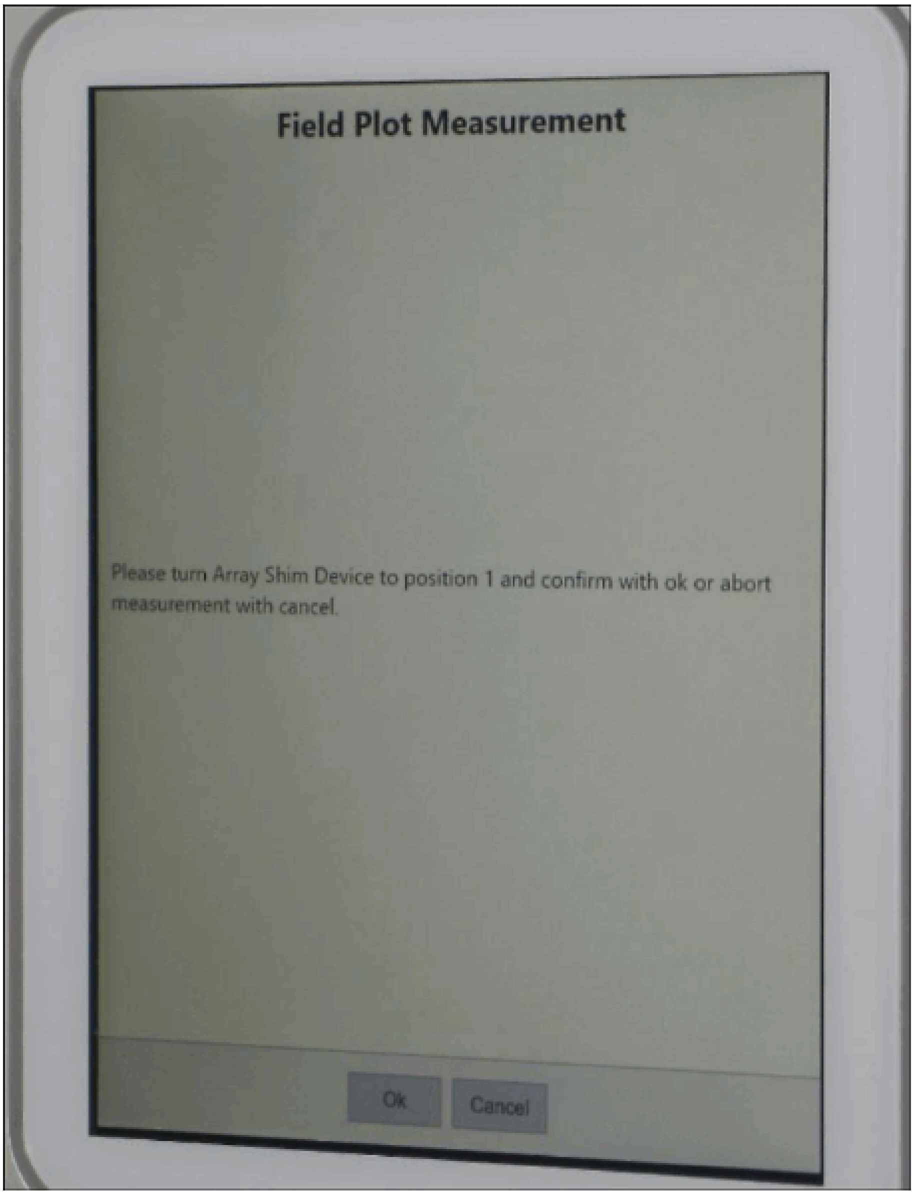
4. Position Check of the patient table
   - If the patient table is not in the correct position following information appears.

Fig. 484: Table Position Check



5. Perform array shim measurement from the patient table operating panel.
   - Start the measurement from position 1.

Fig. 485: Field Plot Measurement

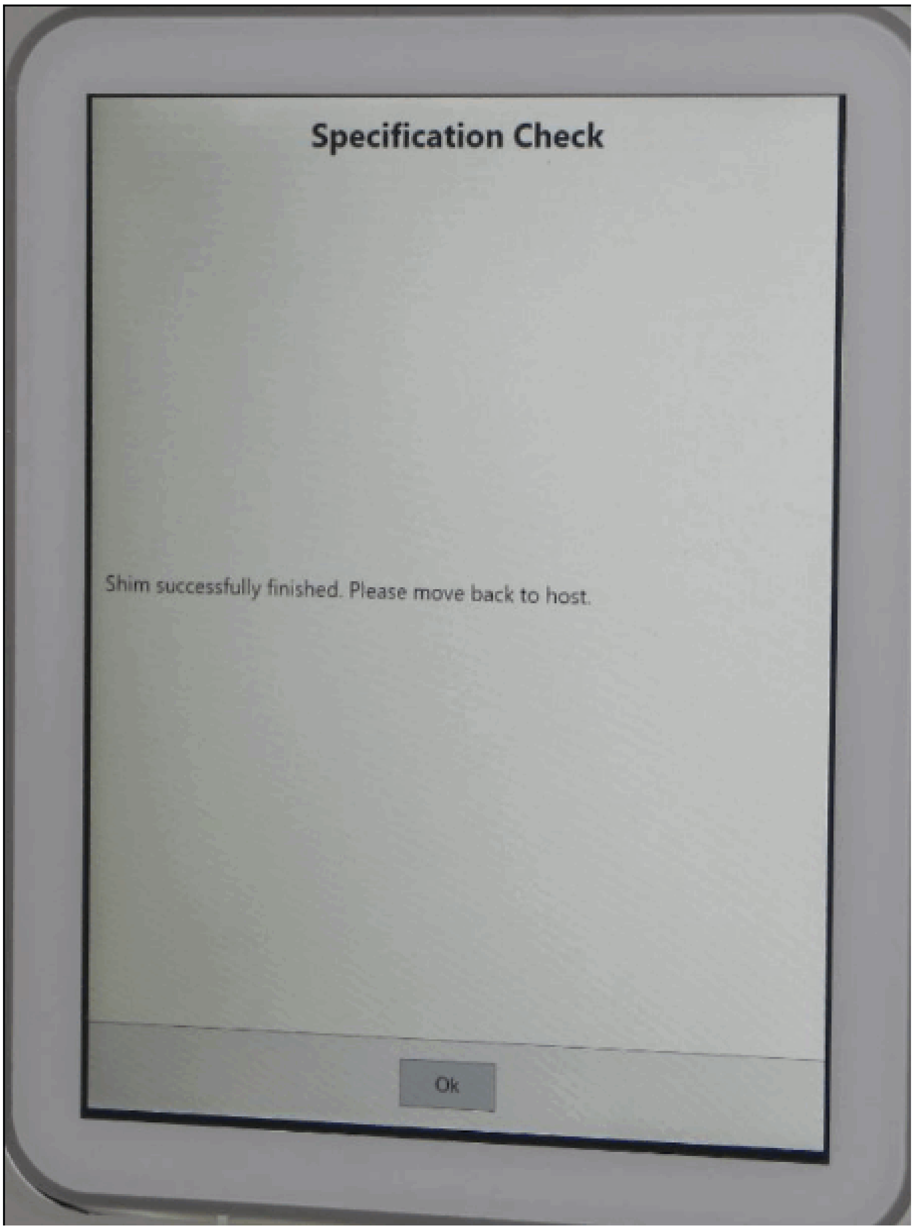


   » Press **Ok**.
   - Turn the ASD device to position 2.
     » Confirm with OK.
   - Repeat this procedure up to position 20.

The measurement has been completed. You can now return to the MR operating console.

The shim program checks the specification values and calculates the passive shim distribution.

Fig. 486: Specification Check



6. Specification check
   - Creates IQShim input file.
   - Checks whether shim is in specification.
   - Decides whether to continue with iron shim in case the shim is not in specification.
7. Coil shim
   - The shim program calculates the gradient offsets and shim currents (E-Shim option)

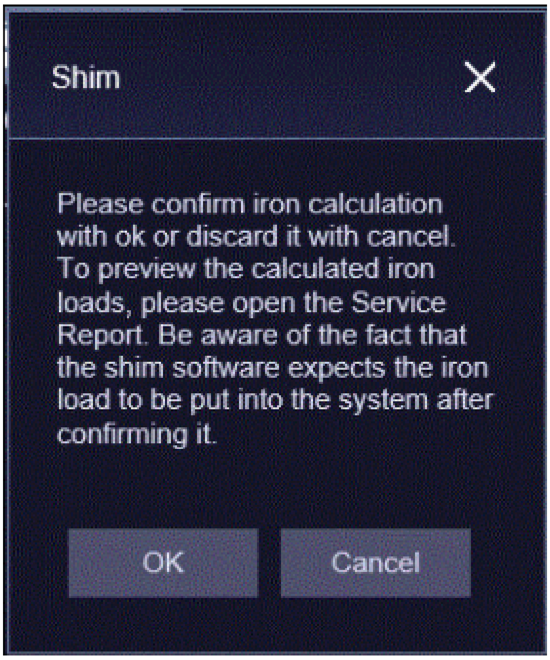
8. Iron shim
   - The shim program calculates the passive shim distribution.
     » Confirm the IQShim output.

When you press OK button, the shim software expects the iron load to be put into the magnet.

OK: The calculated iron will be stored in the database if the OK button is pressed.

Cancel: The calculated iron will not be stored in the database if the Cancel button is pressed.

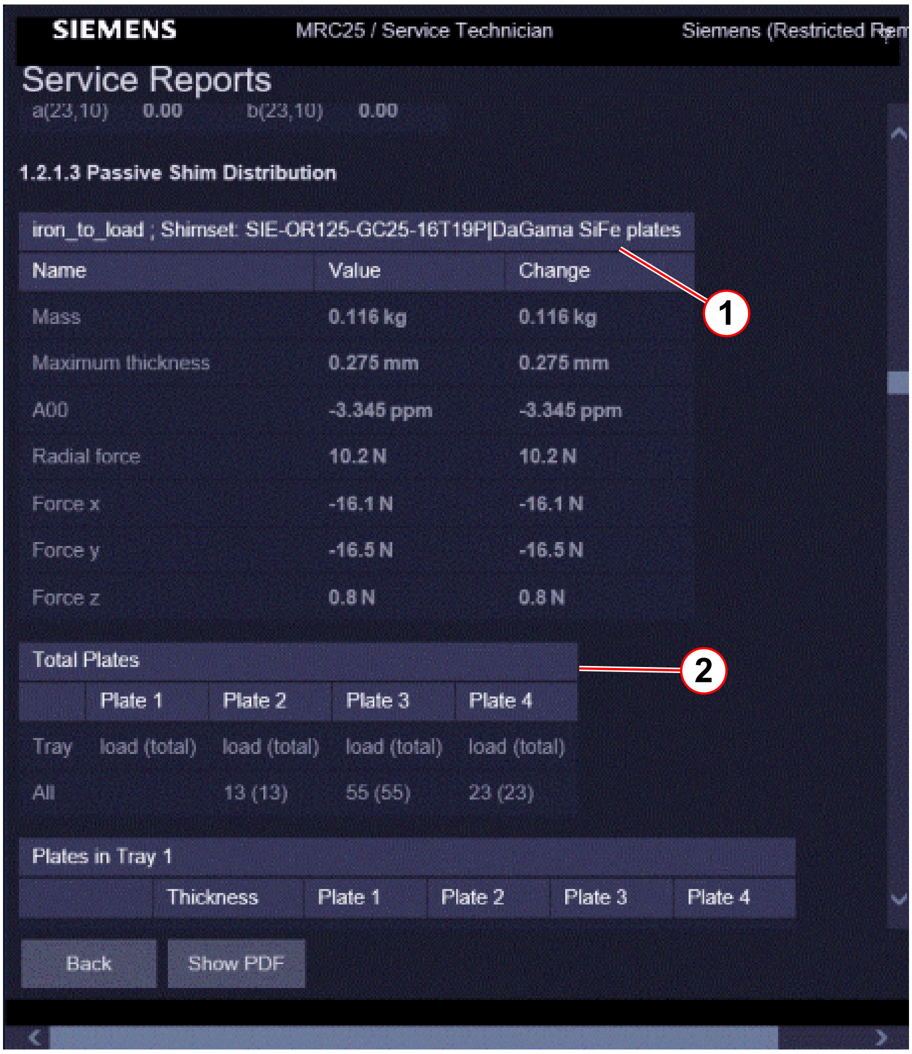
Fig. 487: Confirmation Iron Calculation



The result is saved in the "Service Report" or in the "Shim " directory.

9. Checking the correct shim plate type. (→ 2/Fig. 488 Page 706)
   Checking the calculated passive shim distribution.

Fig. 488: Iron Calculation Report



(1) Shim Plate Type
(2) Total Shim Plates

> **i** Values out of specification are highlighted.

Preparation for Shim Plate Loading

- Print out the distribution list if possible

- Count and sort the number of shim plate 1, shim plate 2, shim plate 3 and shim plate 4.(→ 2/Fig. 488 Page 706).

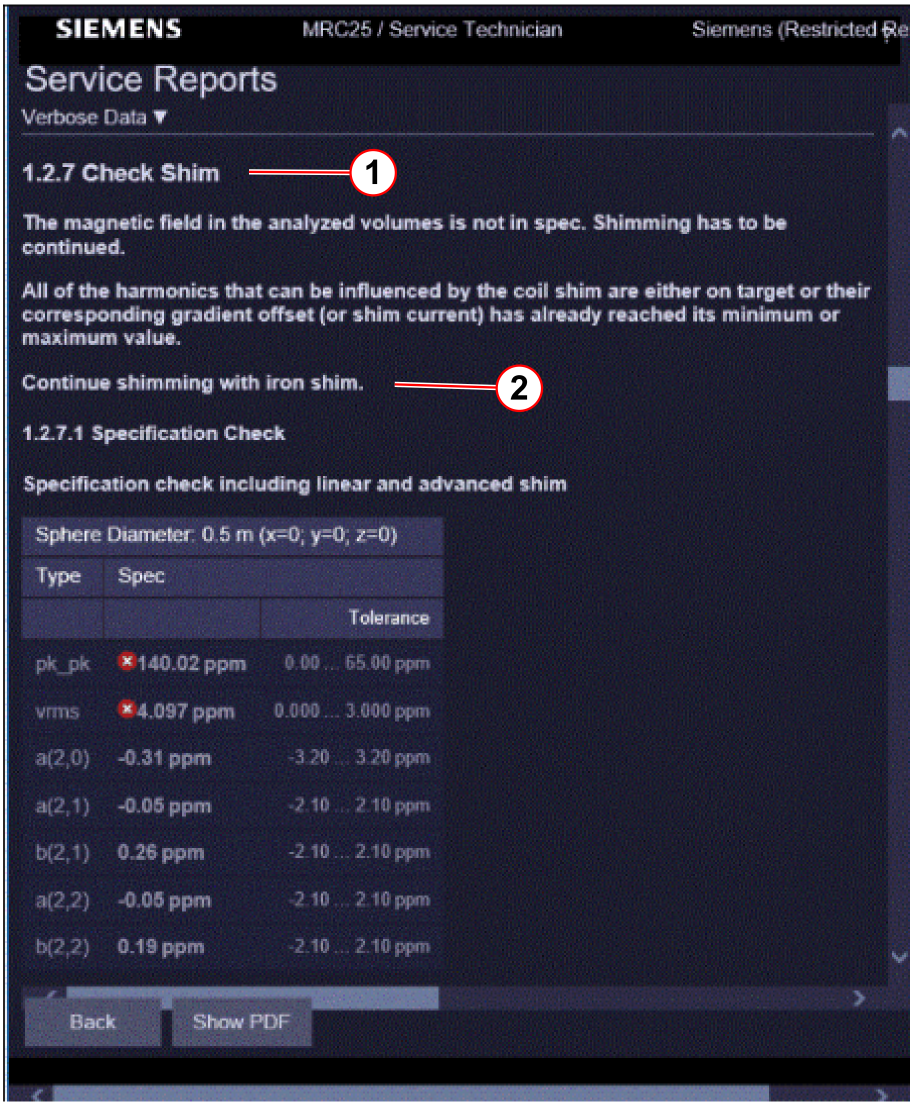
Fig. 489: Iron total

Total Plates				
	Plate 1	Plate 2	Plate 3	Plate 4
Tray	load (total)	load (total)	load (total)	load (total)
All	16 (431)	45 (98)	98 (209)	41 (77)

(1) Plates to load
(2) Plates in total

- Run to zero the magnet for removing and installing the shim trays.
- Repeat the shim workflow until the homogeneity is in specification. (→ 2/Fig. 490 Page 708)

Fig. 490: Shim Check

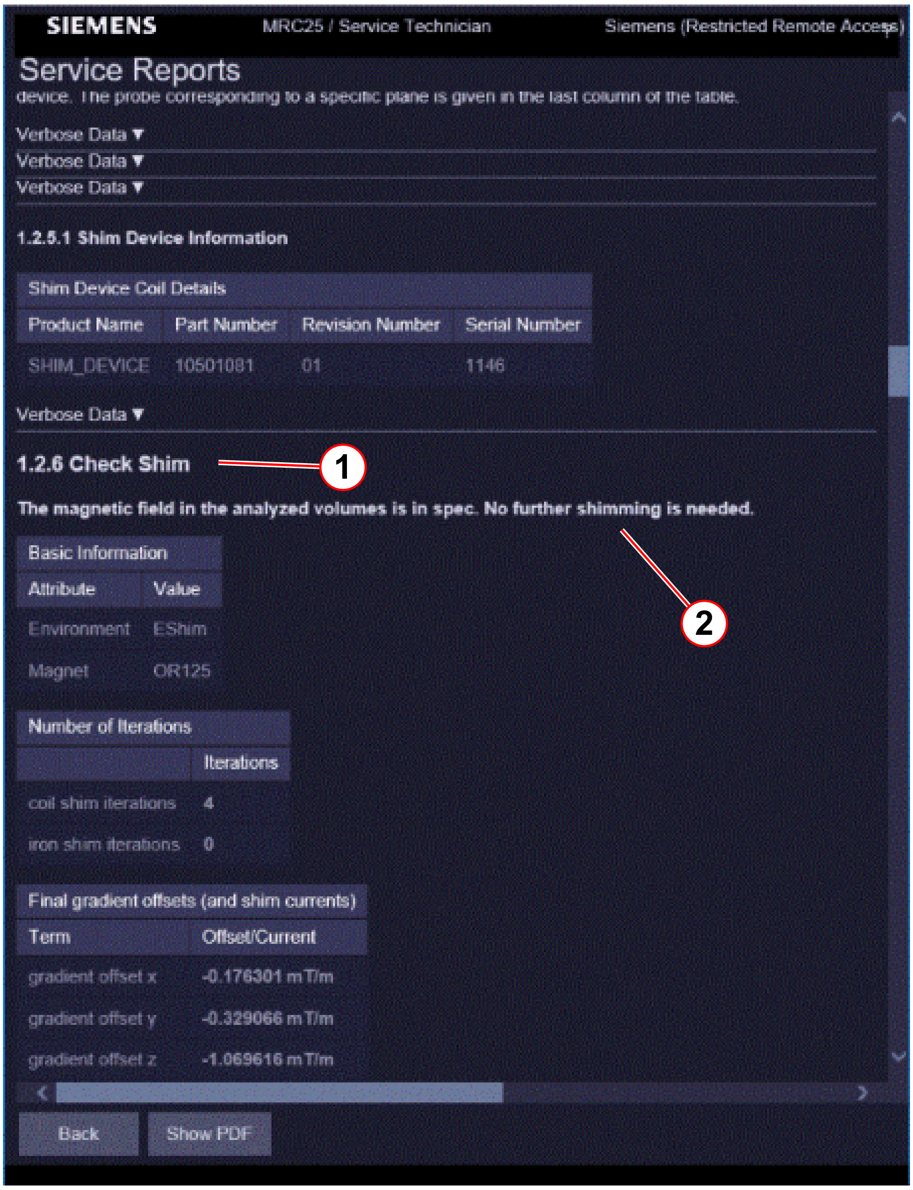


(1) Check Shim
(2) Not in specification

Homogeneity is within specifications.

- When the magnet is shimmed within specifications, the shim is completed.
  - » Output of the IQShim if the magnet shim was successful
  - » No further shimming is needed. (→ 2/Fig. 491 Page 709)

Fig. 491: Shim Check



(1) Check Shim
(2) Shimming is not needed

Removing the shim trays

WARNING

The force created by the fringe field of the magnet will pull out the shim plates and move them to a different location in the shim tray.

Observe the warning. Otherwise death or serious injury can occur.

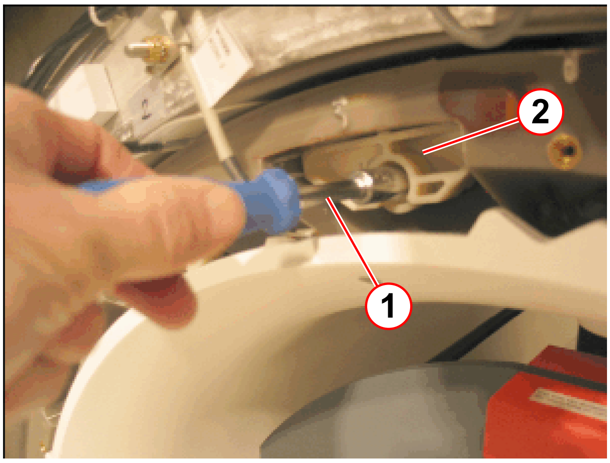
» **Ramp down the magnet before removing and installing the shim trays.**

Ramp down the magnet before removing the shim trays.

The shim trays are accessible from the patient end.

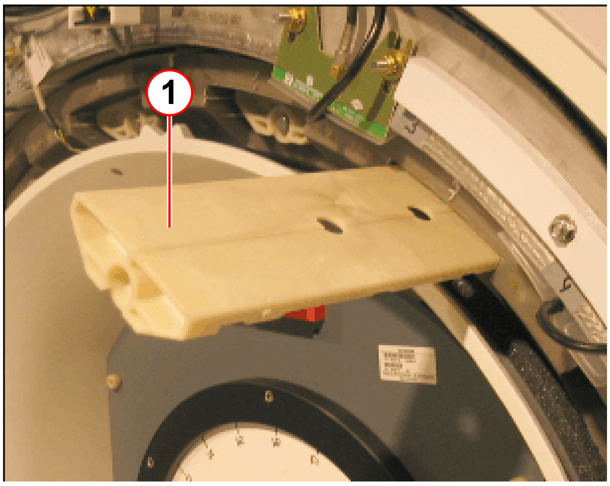
- Remove the funnel.
- Loosen the shim trays with a 10mm socket wrench.
- Remove the mounting screws of the shim trays.

Fig. 492: Shim tray



(1) Socket wrench
(2) Shim tray

Fig. 493: Shim tray



(1) Shim tray

- Carefully pull the shim trays from the gradient coil.

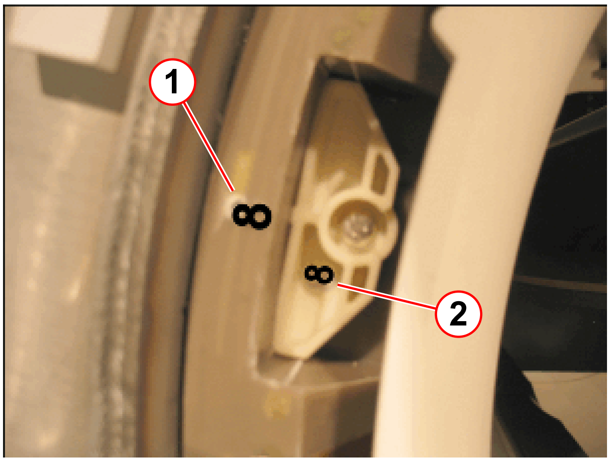
  Each shim tray is identified by a number in front, ranging from **01** to **16**.

When you install the shim trays, ensure that the magnet is ramped down.

When you install the tray, ensure that the number matches the number embossed on the gradient coil.

Tighten the trays with the screws.

Fig. 494: Shim tray



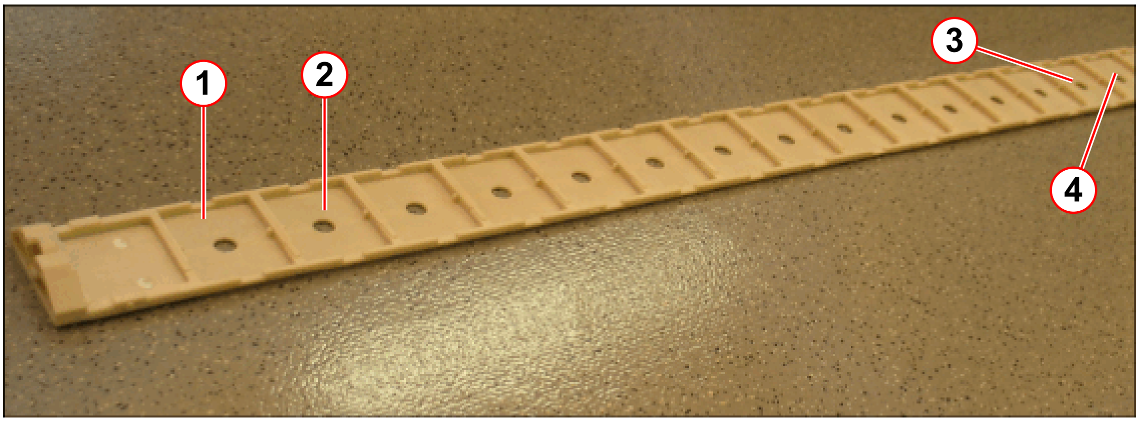
(1) GC label
(2) Shim tray label

Installing the shim iron

Shim kit

The figure shows the shim tray with the position of the trays.

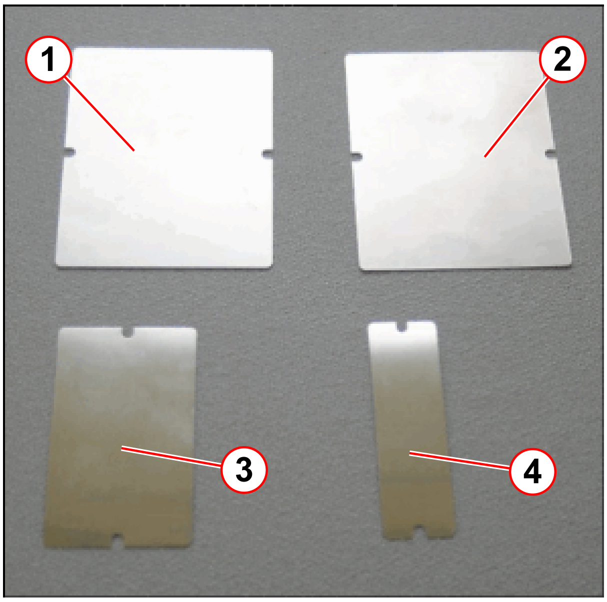
Fig. 495: Shim tray



(1) Position A
(2) Position B
(3) Position P
(4) Position R

The figure shows the shim plates and the packing shims.

Fig. 496: Shim plates



(1) Plate 1 (64.8 x 80 x 0.3 mm)
(2) Plate 2 (64.8 x 80 x 0.1 mm)
(3) Plate 3 (64.8 x 40 x 0.05 mm)
(4) Plate 4 (64.8 x 20 x 0.05 mm)

Fig. 497: Packing shims (75 x 55 x 1 mm, polypropylene molding)

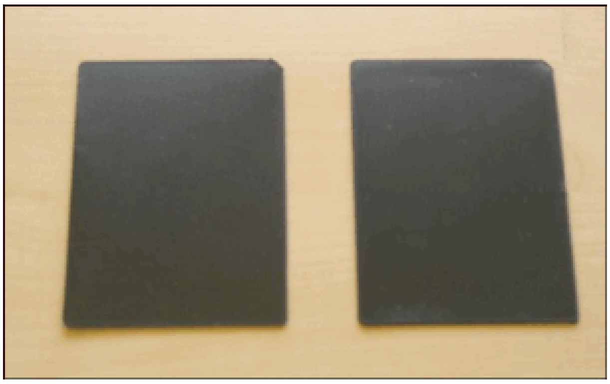
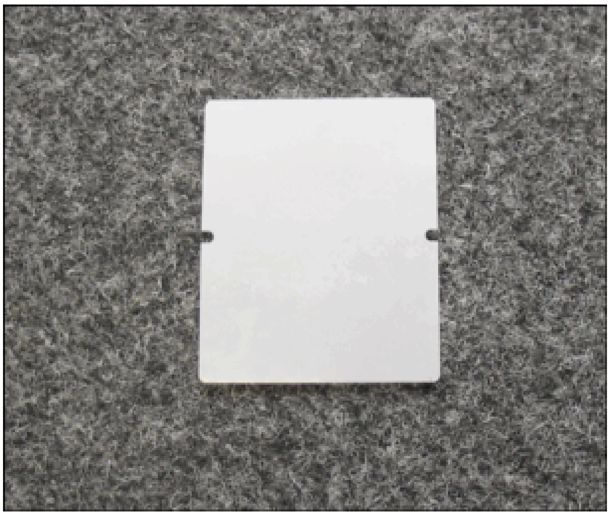


Fig. 498: Insulation shim (80 x 64.8 x 0.19 mm, polyester packer)



Shim plate distribution

Fig. 499: Plates in tray

Plates in Tray 1					
	Thickness	Plate 1	Plate 2	Plate 3	Plate 4
Position	change (total)	load (total)	load (total)	load (total)	load (total)
A	0.000 (0.000)				
B	0.287 (2.987)	0 (9)	2 (2)	3 (3)	1 (1)
C	0.000 (0.000)				
D	0.000 (0.000)				
E	0.000 (0.000)				
F	0.000 (0.000)				
G	0.000 (0.000)				
H	0.000 (0.075)			0 (3)	
J	0.000 (0.237)		0 (2)	0 (1)	0 (1)
K	0.025 (0.025)			1 (1)	
L	0.000 (0.000)				
M	0.000 (0.212)		0 (2)		0 (1)
N	0.000 (0.013)				0 (1)
P	0.000 (0.000)				
R	0.000 (0.000)				
S	0.000 (0.000)				
T	0.000 (0.000)				
U	0.287 (2.987)	0 (9)	2 (2)	3 (3)	1 (1)
V	0.850 (0.850)	2 (2)	2 (2)	2 (2)	
Total Plates		2 (20)	6 (10)	9 (13)	2 (5)

The table shows the number and position of the shim plates:

The number without brackets is the number of plates to be loaded for the actual iteration.

The number in brackets shows the total number of plates.

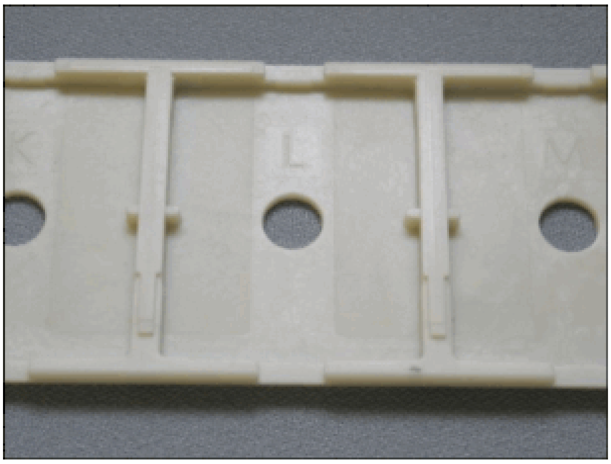
The first position of the shim trays (mounting location at the shim tray) is identified with:

» Magnetom Vida: **A** followed by B, C ,D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V

> **i** Empty pockets should not be covered. The covers of empty pockets may make noise. Store remaining covers on-site for re-shimming.

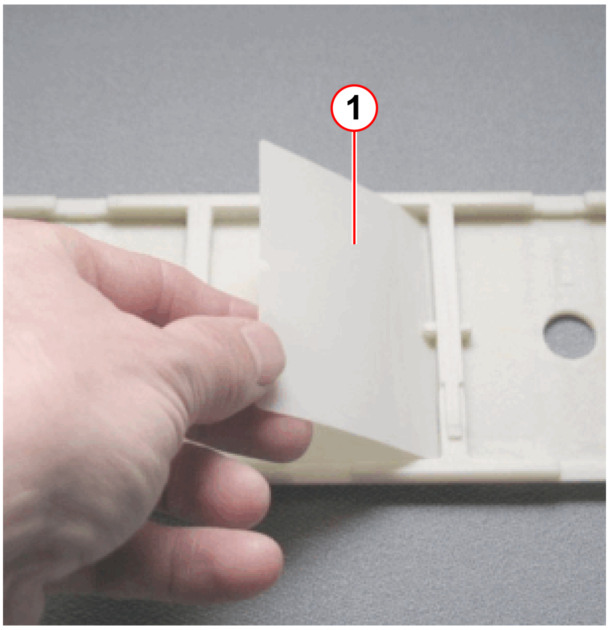
- Install the shim plates as shown in the following figures.

Fig. 500:



- Select the correct shim tray number (1 - 16)and the correct pocket position (A - V).

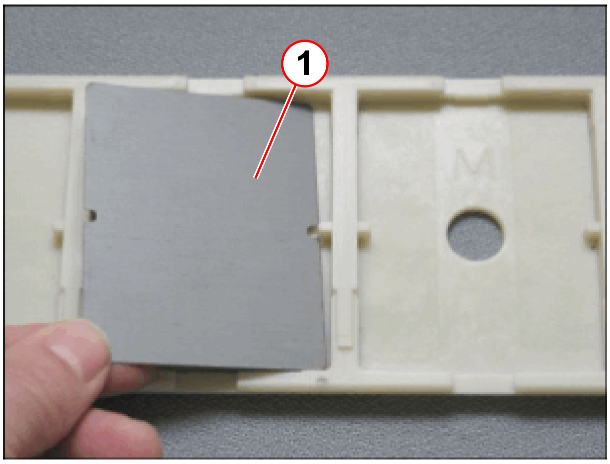
Fig. 501: Insulation shim



(1) Insulation shim

- Install an insulation shim first.

Fig. 502: Shim tray

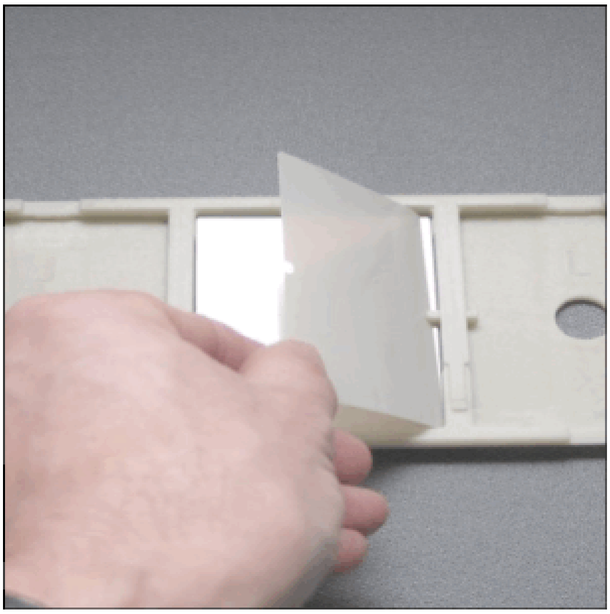


(1) Shim plate

- Install the required plate size
  » Use one layer of insulation shim between each shim plate.

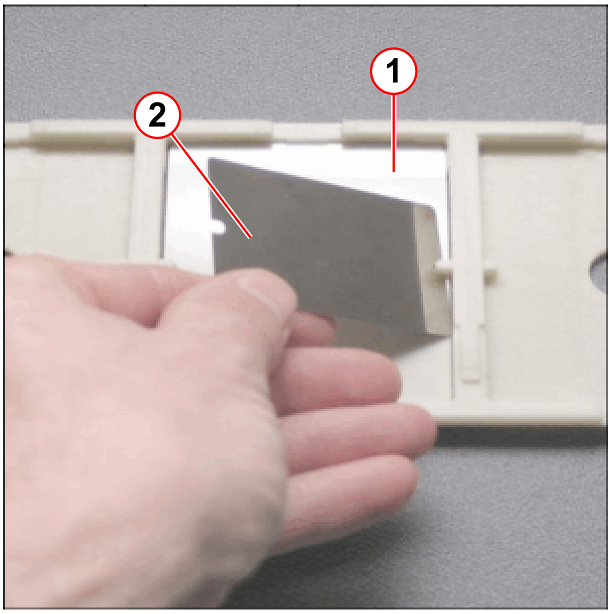
Use one insulation shim between each shim plate.

Fig. 503:



- Install one insulation shim between each shim plate.

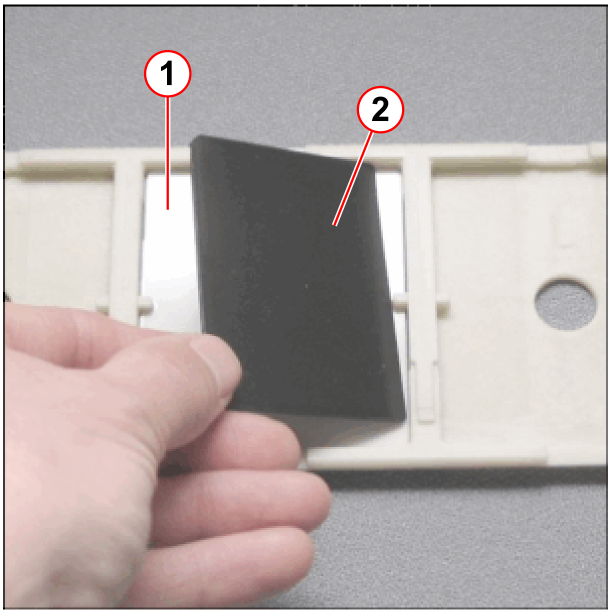
Fig. 504: Shim tray



(1) Insulation shim
(2) Shim plate

- Install the required shim plate.

Fig. 505: Shim tray

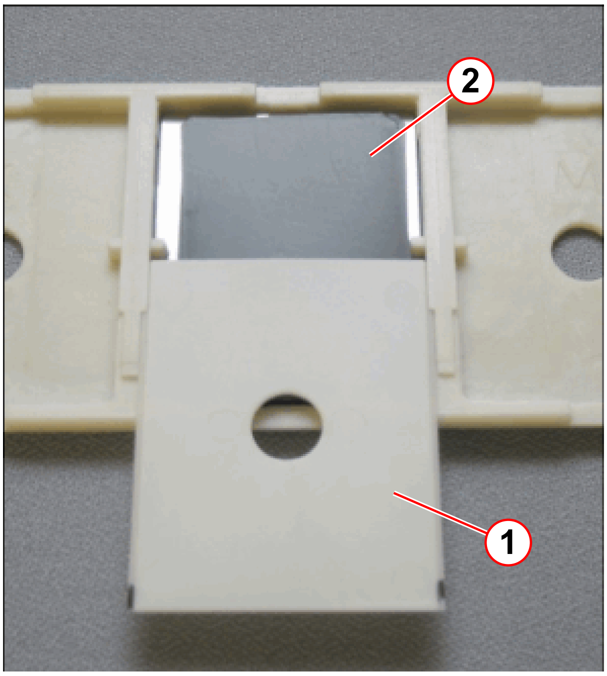


(1) Insulation shim
(2) Packing shim

- Cover the last shim plate with an insulation shim.
- Fill the space between the shim plates and the cover with packing shims.

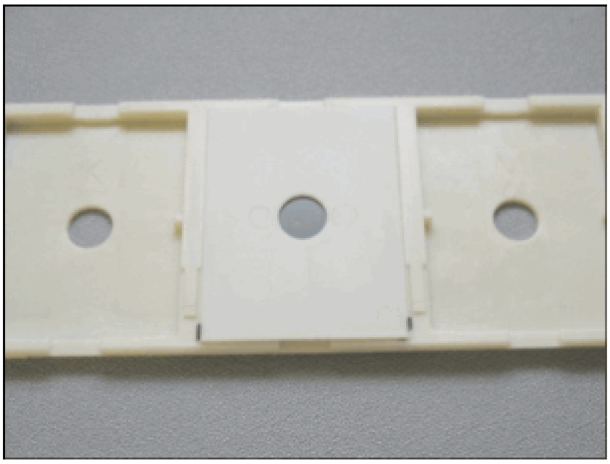
Depending on the space between the shim plates and the cover, use **1 mm packing shims** to keep the shim plates from moving in the shim trays.

Fig. 506: Shim tray



(1) Cover
(2) Packing plate

Fig. 507:

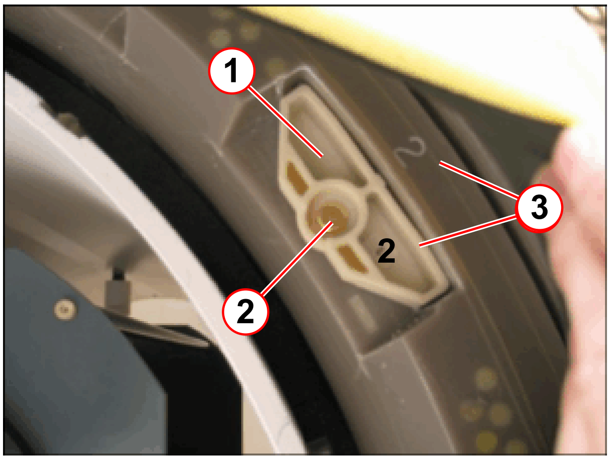


- Close the pocket with the cover.

> ℹ Ensure that the shim plates are securely seated in the shim trays. Loose shim plates cause spikes.

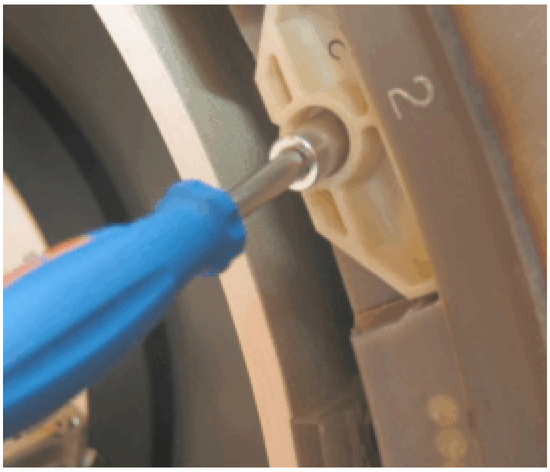
- Install the shim trays in the gradient coil.
- Tighten the shim tray with the screw.

Fig. 508: Shim Tray



(1) Shim tray
(2) Screw
(3) Number

Fig. 509:



When you install the shim trays, ensure that the tray number matches the number embossed on the gradient coil.

Setting the magnet current

To compensate for the additional iron and to ensure that the predefined frequency range of (+/- 100 kHz) is maintained for the array shim, a slightly higher magnet current is set when the magnet is ramped for the first time.

This value is the **Operating current/Measured value** listed under **Operating Parameters** in the **Magnet Acceptance Protocol** by **SMT**.

Magnet current setting

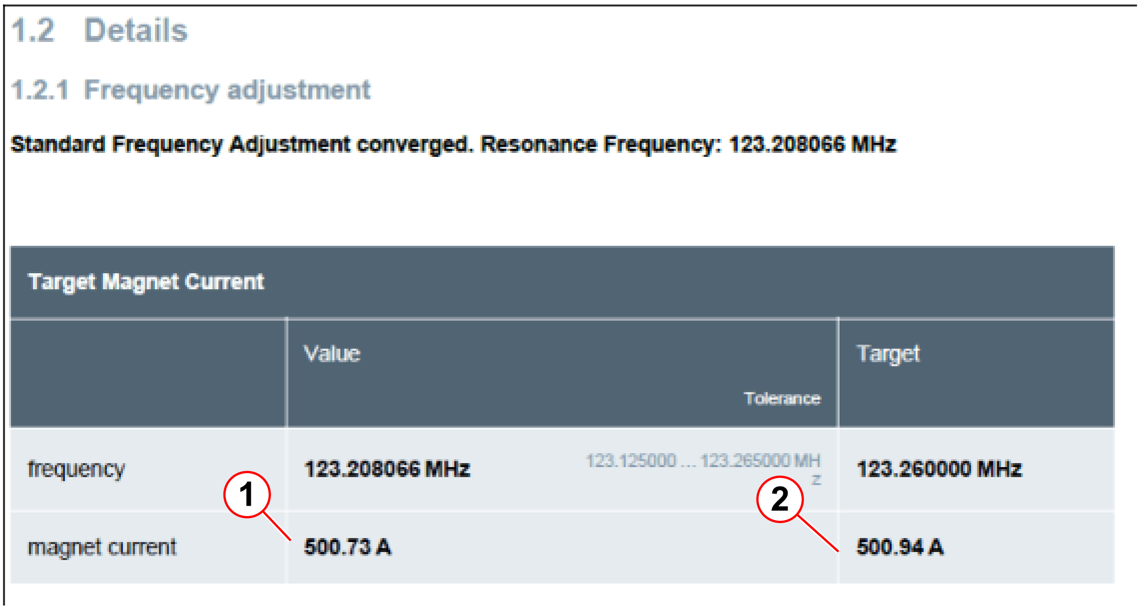
1. Iteration

Use the value listed as Operating current / Measured value under Operating Parameters in the Magnet Acceptance Protocol by SMT.

2. Iteration and following iterations

To ensure that the magnet is ramped at 4 year intervals only, the magnetic field should lie a little under the maximum frequency. Use the calculate current listed in the Magnet Shim Report File.

Fig. 510: Magnet Current



1.2 Details

1.2.1 Frequency adjustment

Standard Frequency Adjustment converged. Resonance Frequency: 123.208066 MHz

Target Magnet Current			
	Value	Tolerance	Target
frequency	**123.208066 MHz**	123.125000 ... 123.265000 MHz	**123.260000 MHz**
magnet current	**500.73 A**		**500.94 A**

(1) Actual Magnet Current
(2) Target Magnet Current

Formula to calculate the current for the maximum field strength:

> Maximum frequency for "Tuning Calibration"
> Magnetom Vida: 123.265 000 MHz.

- Use the following mathematical formula to compute the current for the maximum field strength. The formula takes the tolerances of the different magnet power supplies into consideration:

Magnetom Vida:	I final = (**123.260 000** MHz x **I** actual) / **Fre** actual

> The lower limit of the frequency is 123.100 000 MHz

Magnetom Vida with Multinuclear Spectroskopy Option (MNO)

Magnet OR125 range of frequency:

Target frequency: 123,260 000 MHz

Maximum frequency: 123,265 000 MHz +0 kHz

Minimum frequency: 123,125 000 MHz -0 kHz

Use the mathematical formula below to calculate the current to determine the **target frequency.**

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

OR125	**I final = (123,260 000 MHz x I actual) / Freq actual**

Formula for converting the frequency to Tesla

$$2 * P * f = g * B_0$$

P	=	3.14159
f	=	Frequency in Hz
g	=	Gyromagnetic coefficient of proton, $2.67522128 * 10^8\ s^{-1}\ T^{-1}$
B_0	=	Magnetic field in Tesla

Vertical position of the patient table

The vertical position of the patient table can be adjusted with a reflector bracket.

This is necessary if the front cover funnel is dismounted.

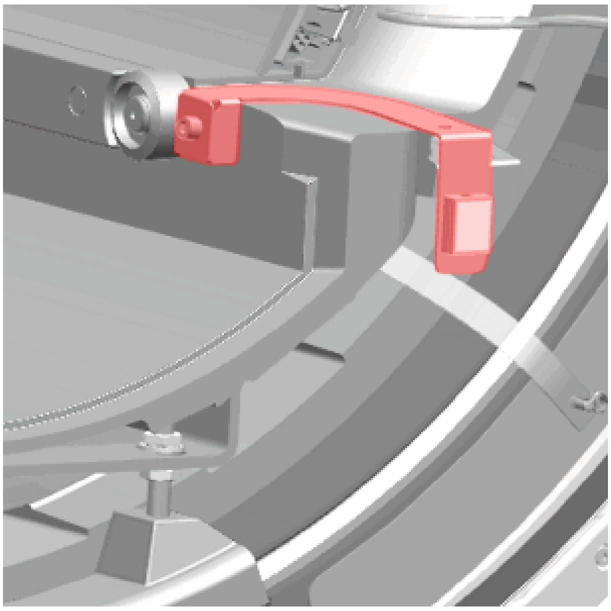
- Take out the reflector bracket from the storgae position behind the front cover.

Fig. 511: Reflector bracket storage position

(1) Reflector bracket

- Fasten the reflector bracket with a screw from the removed funnel.

Fig. 512: Reflector bracket attached to BC



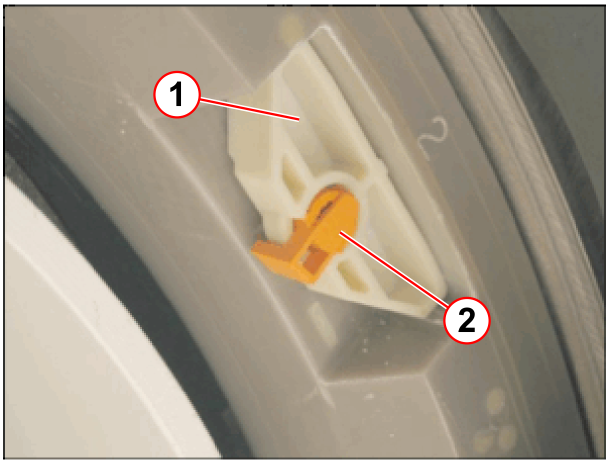
Work after shimming

Locking the shim trays.

> Certain sequences cause extreme transient vibrations of the gradient coil and its surroundings. This could result in components shaking loose and cause severe spiking in the images.

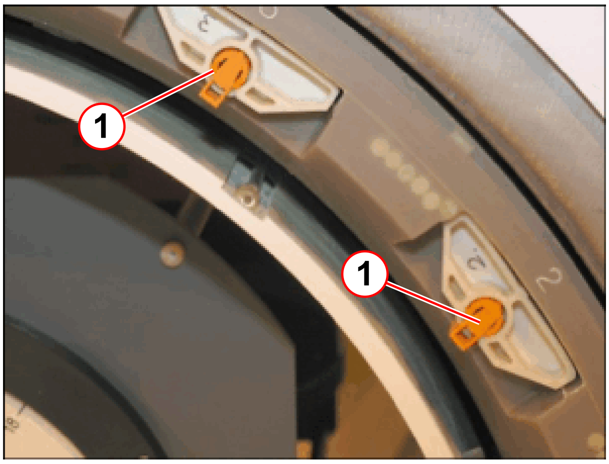
» Therefore, after finishing the passive shim, all fastening screws of the shim trays need to be locked.

Fig. 513: Shim tray



(1) Shim tray
(2) Screw locking device

Fig. 514: Shim tray



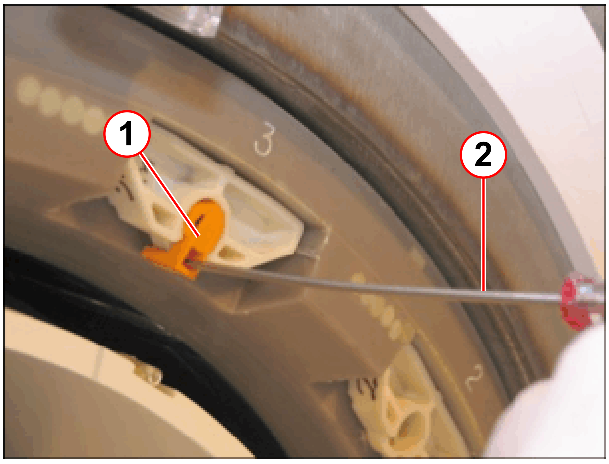
(1) Screw locking device

Unlocking the screw locking device

- Use a flat head screwdriver for unlocking.

Fig. 515: Shim tray

Fig. 516: Shim tray



(1) Screw locking device
(2) Screwdriver

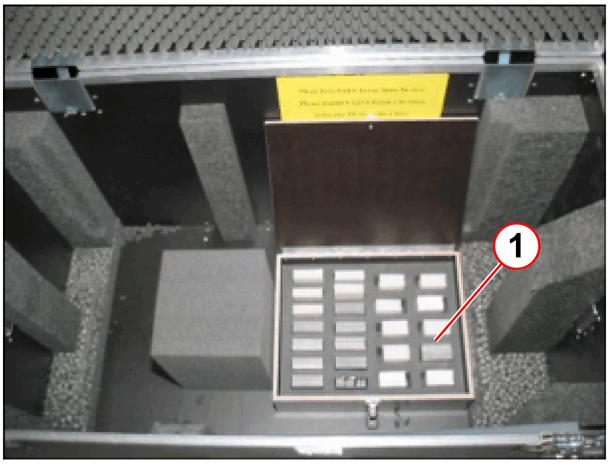
- Install the funnel

Remaining shim plates

The packaged shim plates that are not required should be stored as backup plates in the array shim device kit. These extra shim plates can be used in case the delivered shim plates do not suffice.

Store shim plates only in their original packaging in the array shim device kit.

Fig. 517: Array shim device kit



(1) Shim plates

Fig. 518: Packaged shim plates

Error Handling

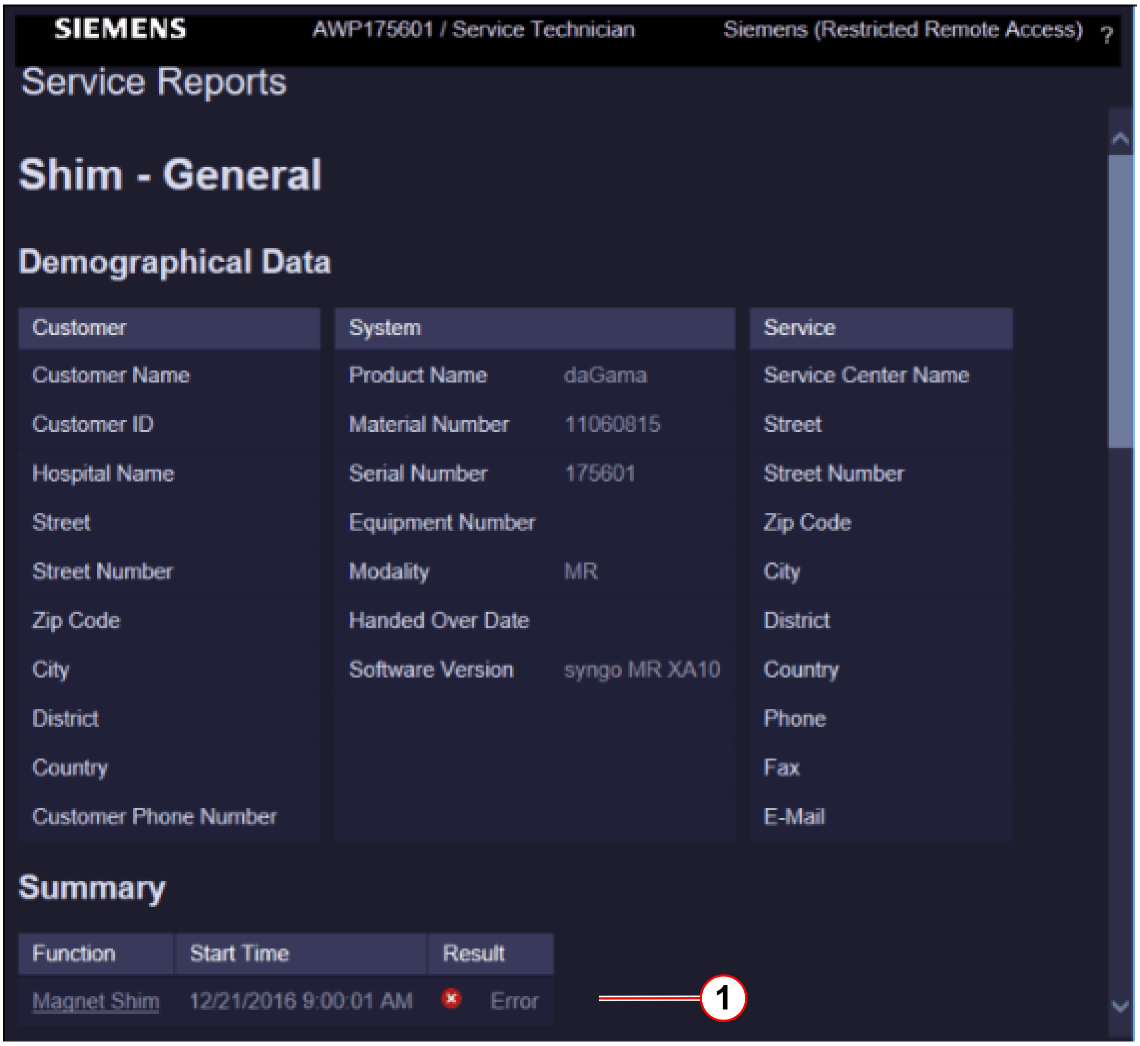
What to do if shim workflow fails

The following test should be performed.

1. Check the shim output for error messages.

   » "Service Reports / Shim - General"

Fig. 519: Shim - General

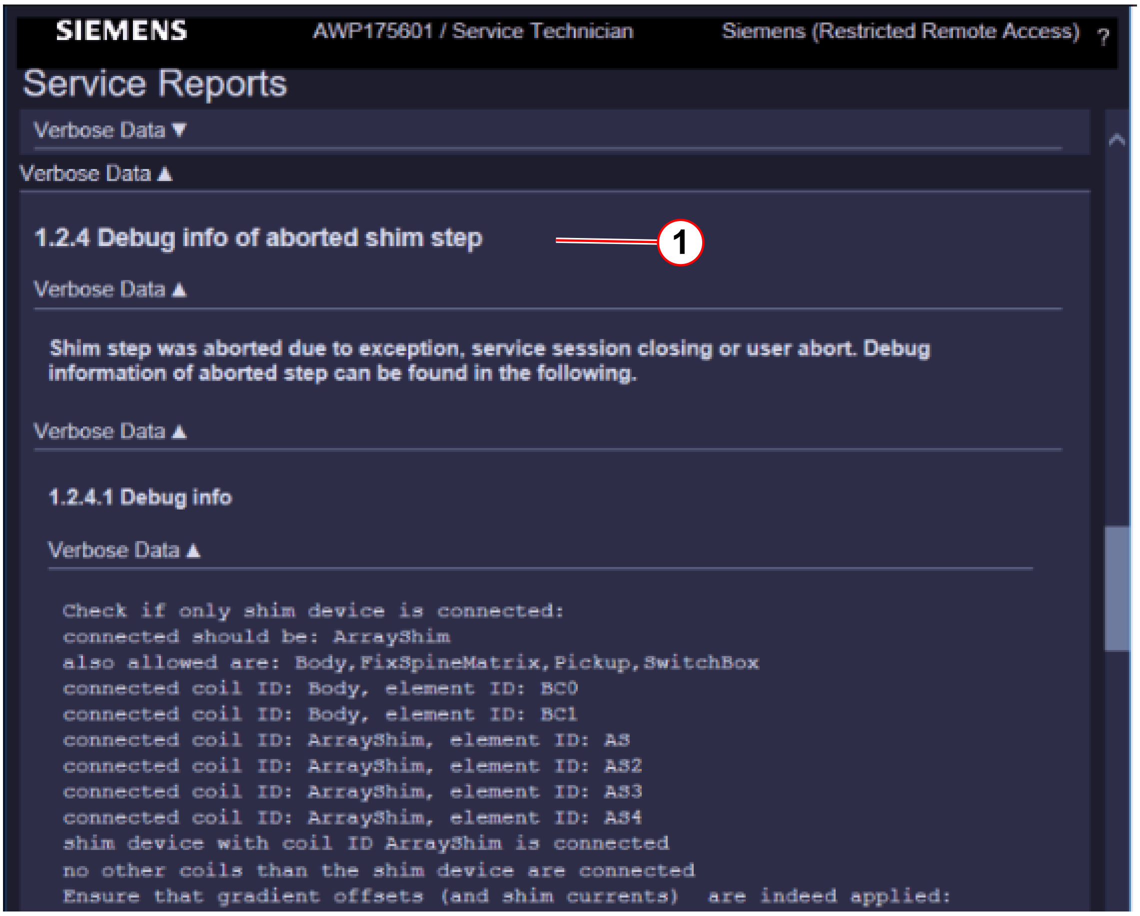


(1) Error

Debug mode

The debug mode shows detailed information.

Fig. 520: Service Report



(1) Debug Information

Shim support

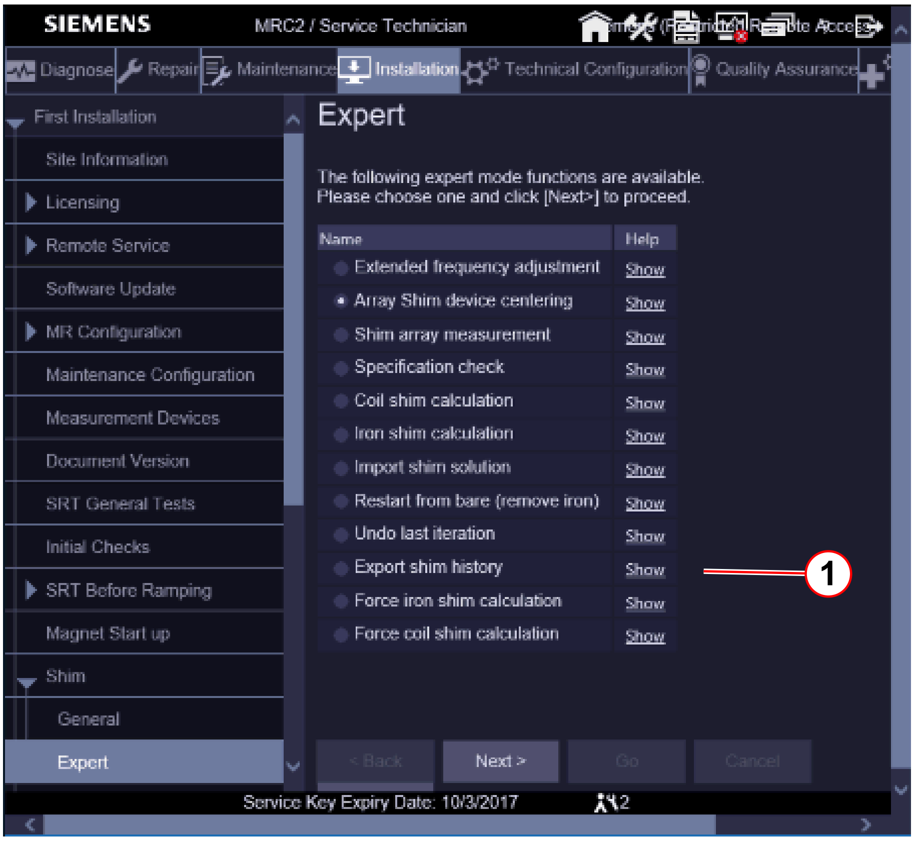
Send the shim folder as a ZIP file to the following address to get shim support:

» magnet.hsc.med@siemens.com

Open the Shim Expert mode:

Select: **Export shim history, Fig.58, Pos1**

Fig. 521: Shim Expert



(1) Export Shim History

This mode saves the current shim history, i.e. shim settings file and all IQShim input and output files, to a “MagnetShim-zip” file.

The zip file is saved in the directory: **K:\Seso\ShimExport\MagnetShim-yyyy-mm-dd-hhmmss.zip**

5.9.2 Shim - Expert

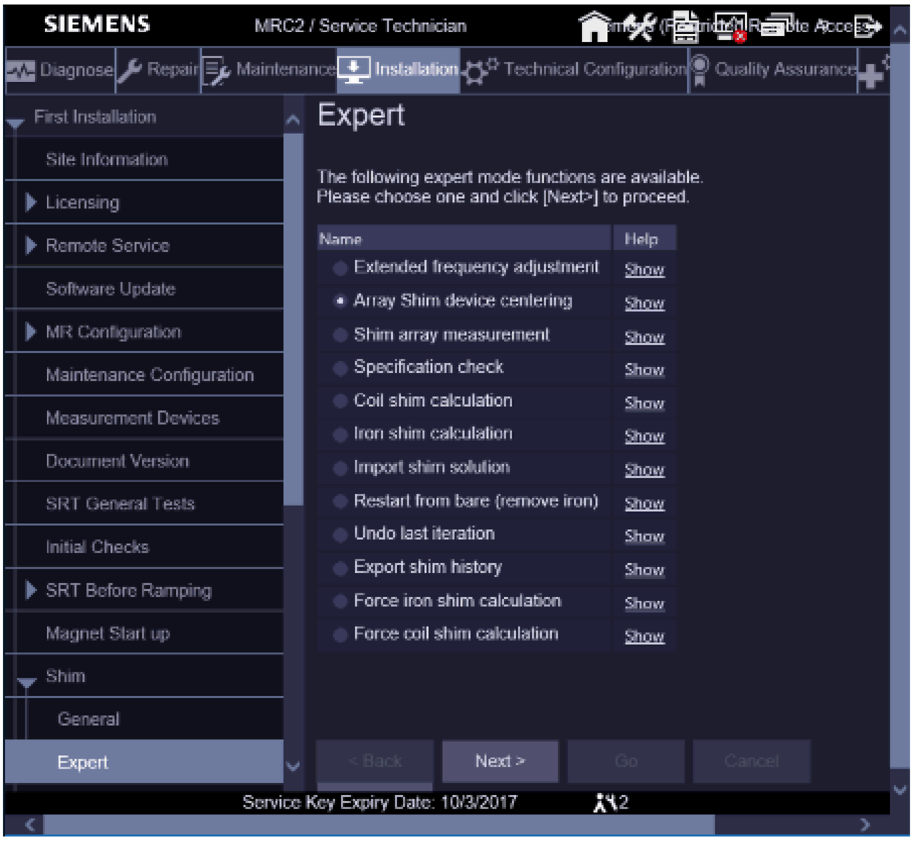
Help ID: EB5078AF-7B40-4D9D-89AF-AA2E282632E1
Service Level: 7

The Expert shim platform allows the individual workflow steps to be started.

Special functions are helpful for troubleshooting.

- Choose the required workflow step.
- Click **Next**

Fig. 522: Shim Expert



5.9.2.1 Extended frequency adjustment

Help ID: 6D7C9E5B-3B13-4ED4-9640-0724C6B5019F
Service Level: 7

An extended frequency adjustment is performed.

5.9.2.2 Array Shim device centering

Help ID: 58A216C8-A96F-4B33-A7D2-079519950845
Service Level: 7

- A gradient centering check is performed.
- Output of required Array Shim Device shift.
- No field measurement afterwards.

5.9.2.3 Shim array measurement

Help ID: CDA931B8-9056-4D95-9FDD-07BDCD61AE1F
Service Level: 7

- Field plot measurement without gradient centering.
- PTAB position check.

- EIS reset.

5.9.2.4 Specification check

Help ID: DFE046D0-340D-4EDE-942A-8BF0F2AFE2CC
Service Level: 7

- Using the existing field plot.
- Run IQShim program.
- Output of harmonics and specification values.

5.9.2.5 Coil shim calculation

Help ID: B3C2E0FD-A78E-4E8D-A64B-604F890EF92A
Service Level: 7

- Using the existing field plot.
- Coil offset calculation is performed.
- Check and eccept or cancel the result.

5.9.2.6 Iron shim calculation

Help ID: F978548E-41D2-412D-930D-83FFACE9C05C
Service Level: 7

- Using the existing field plot.
- Iron shim calculation is performed.
- Check and accept or cancel the result.

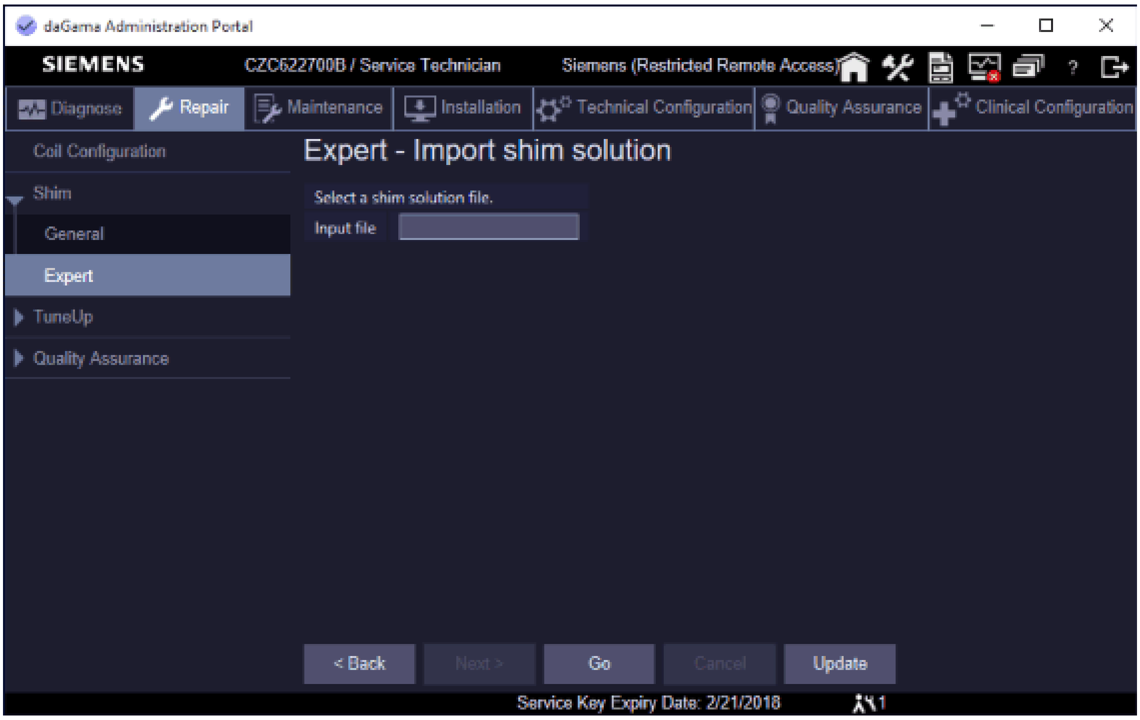
5.9.2.7 Import shim solution

Help ID: EE46BD6D-75B7-4463-B8D4-CFAE47AB3601
Service Level: 7

It may happen that a standard calculation of IQShim is not appropriate for shimming a magnet under certain circumstances. In this case, the current shim history can be sent to a shim expert. The expert performs an IQShim calculation with special parameters and sends the result back, i.e. IQShim output xml file.

The solution can be imported with Expert - Import Shim Solution.

Fig. 523: Import Shim Solution



5.9.2.8 Restart from bare (remove iron)

Help ID: 23BEB07C-6700-418F-AFCE-FDE36AC8785E
Service Level: 7

Remove all iron from the magnet

- Remove all shim iron from the magnet.
- The loaded iron is set to zero in the data base.
- The offset current is set to zero in the data base.

  The old data still remain in the data base for reference.

  Re-set action is logged.
- Iron iteration count is reset to zero
- The shim status is set to “To Do”.

5.9.2.9 Undo last iteration

Help ID: 2C2E208B-0B1A-4159-9B05-521DCFDE12F2
Service Level: 7

- Remove the last iron iteration.
  - » The Iron count is updated
- Old data remained in the database for reference.
- Shim status is set to “To Do”

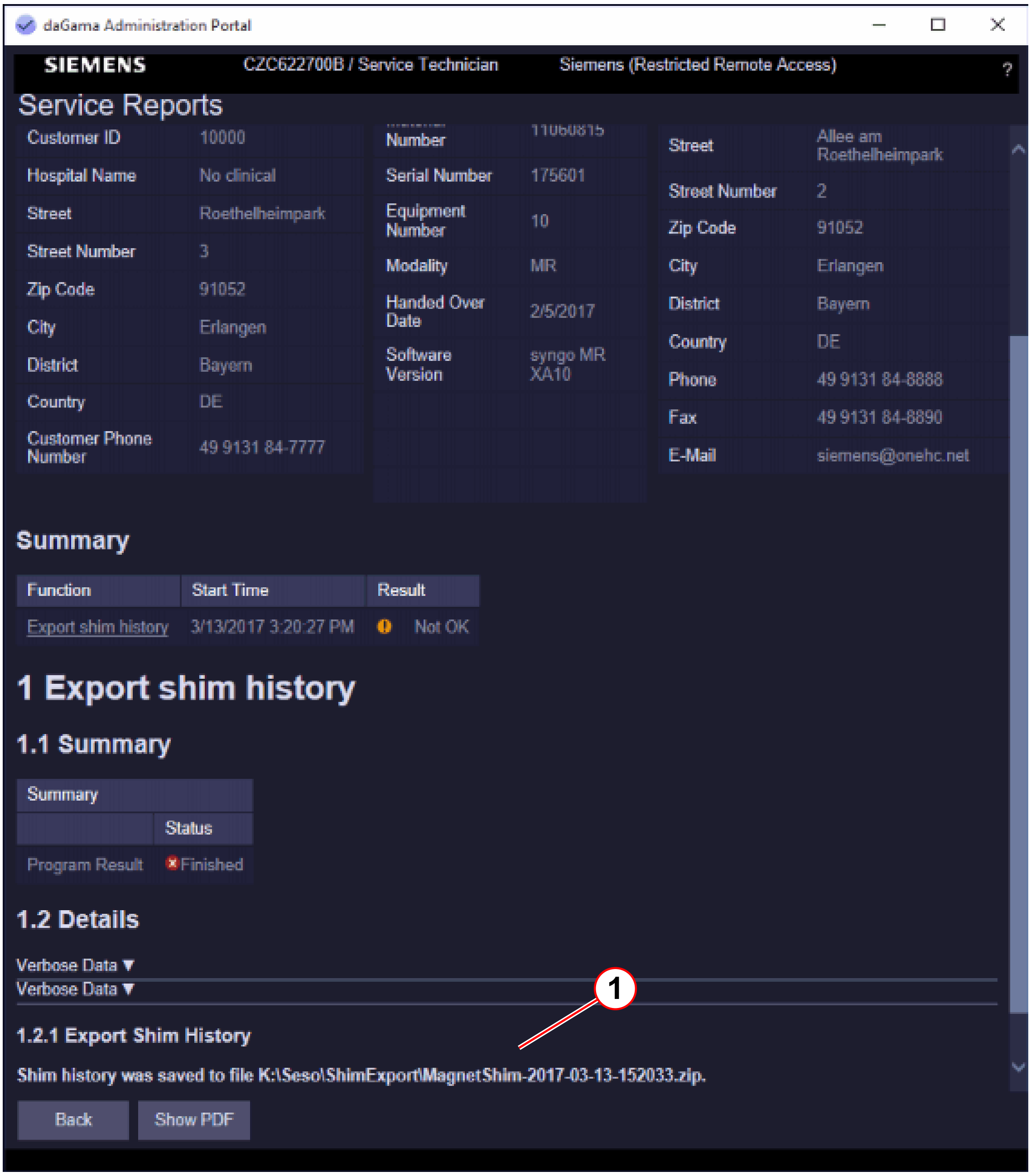
5.9.2.10
Export shim history

Help ID: A584E976-AA36-4210-BFF5-50D6E3190E7A
Service Level: 7

- The content of the Magnet Shim working directory is zipped automatically
- The zip file is saved in the directory: **K:\Seso\ShimExport\MagnetShim-yyyy-mm-dd-hhmmss.zip**

  Refer to Service Reports / Export shim history, Fig 1, Pos1.

Fig. 524: Export Shim History



(1) Log Directory

5.9.2.11 Force iron shim calculation

Help ID: 1F72B798-45D8-424E-9C70-1D35B3BE1634
Service Level: 7

- The force calculation flag is set in the iqshim_input.xml file.
- Plot checks are ignored.
- Iron calculation is performed.
  » Confirm or cancel the result.
- Shim workflow is set to “To Do”.
- Perform a new shim iteration

5.9.2.12 Force coil shim calculation

Help ID: D99B8108-6521-4082-A8BB-96A08505879F
Service Level: 7

- The force offset flag is set in the iqshim_input.xml file.
- Plot checks are ignored.
- Offset calculation is performed.
  » Confirm or cancel the result.
- Shim workflow is set to “To Do”.
- Perform a new shim iteration

5.10 SRT After Ramping

Help ID: B014239D-9A68-4B65-B4D4-803F72520978
Service Level: 7

Safety -related Test after Ramping.

Please select a sub-item.

5.10.1 Checking the patient table stop function

Help ID: 47B102C7-B251-47D7-9481-0C99802B3920
Service Level: 7

STOP function

Required time: 5 min

Required tools: n.a.

Prerequisites: Table is in the home position.

SI Checking the patient table stop function

- Press the **center move** button.
- While the table is moving toward the isocenter, press the **STOP** button on the right side of the patient table.
  » The patient table stops; the Dot display indicates the table stop.

Fig. 525: Patient table

(1) STOP button

- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by pulling the **STOP** button.
- Then press the up and down buttons simultaneously on the control panel to reset the table.
- Press the center move button again to continue table movement.

- While the table is moving toward the isocenter, press the **STOP** button on the **left** side of the patient table
  » The patient table stops; the display indicates the table stop.
- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by pulling the **STOP** button.
- Then press the up and down buttons simultaneously on the control panel to reset the table.

Fig. 526: Dockable table, control unit

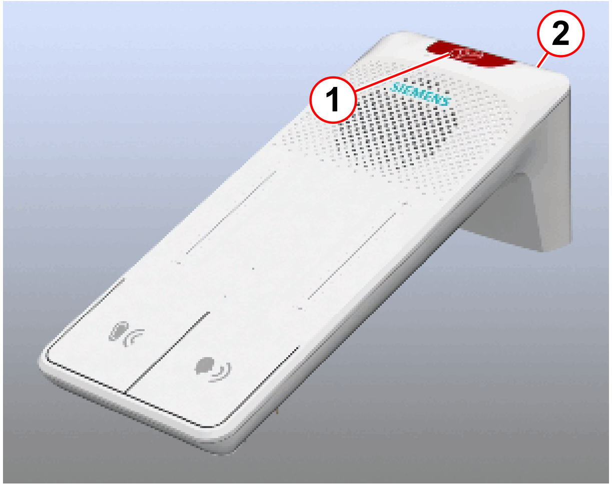
(1) STOP button

- Perform the patient table stop procedure with the **STOP** button for the dockable table **control unit** (➔ 1/Fig. 526 Page 735), if present.
- Perform the patient table stop procedure with the **STOP** button for the **third control panel**, if present.

STOP button on the **intercom**

- Press the **center move** button on the control panel or on the console.
- While the table is moving toward the isocenter, press the **STOP** button on the intercom (➔ 1/Fig. 527 Page 736).

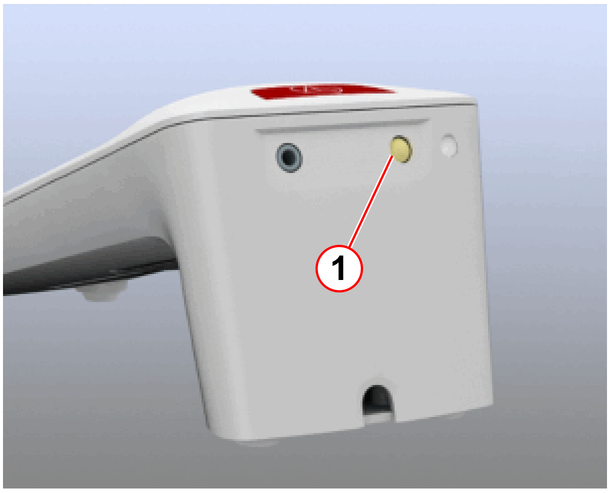
Fig. 527: Intercom



(1) Patient table stop button
(2) Reset patient table stop button

- Make sure that table movement is not possible.
  - » On the MRC, the pop-up window opens `< e.g. Sequence was stopped ...`, press **OK**.
- Deactivate the table's **STOP** button by pressing the **reset** button.

Fig. 528: Intercom



(1) Reset patient table stop button

- Then, press the up and down buttons simultaneously on the control panel magnet to reset the table.

5.10.2 Checking the emergency release of the patient table

Help ID: 113B2538-E842-4718-97F8-01B047E9F761
Service Level: 7

Emergency release

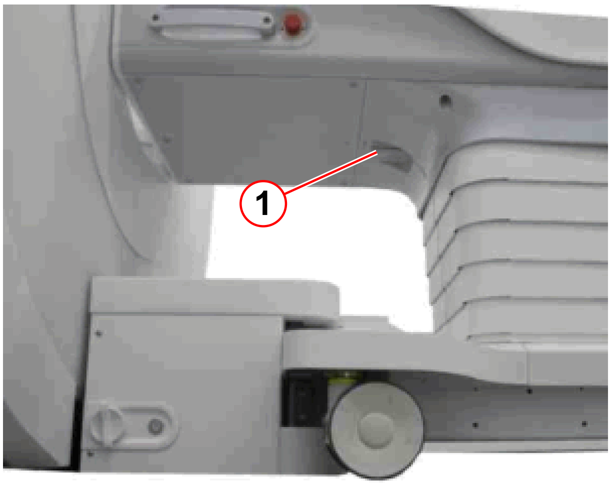
Required time: 5 min

Required tools: n.a.

SI Checking the emergency release of the patient table

- Move the table into center position by pressing the **center move** button.

Fig. 529: Patient table



(1) Emergency release handle

- Turn the emergency release handle (→ 1/Fig. 529 Page 737) on the patient table (a quarter turn clockwise).
- Slide the table all the way out (to the home position).
  » It must be possible to manually pull the table out of the magnet.
  » The emergency release handle will automatically return to the basic position if the table is in the home position.

5.10.3 Checking the distance between the patient table and the wall in the RF-room

Help ID: B5A1EAA0-CA61-406E-89D3-4C8DA1CC8ADE
Service Level: 7

Bruising hazard on the service end

SI Checking the distance between the patient table and the wall in the RF room

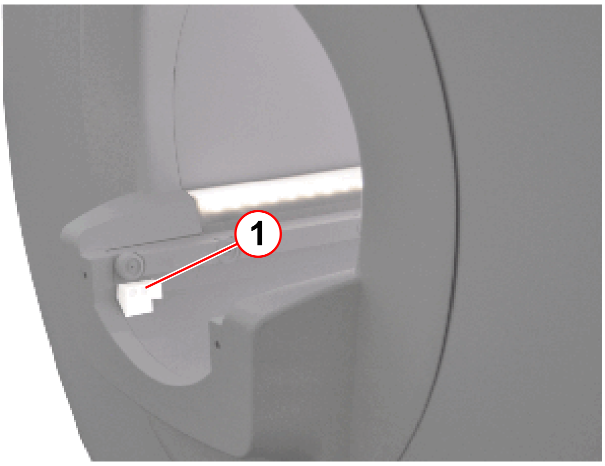
Required time: 5 min

Required tools: Bruising hazard sign, material number 3861718;

tape, material number 3474913; if required

- Check the mechanical end stop on the magnet bore for damage and correct installation.

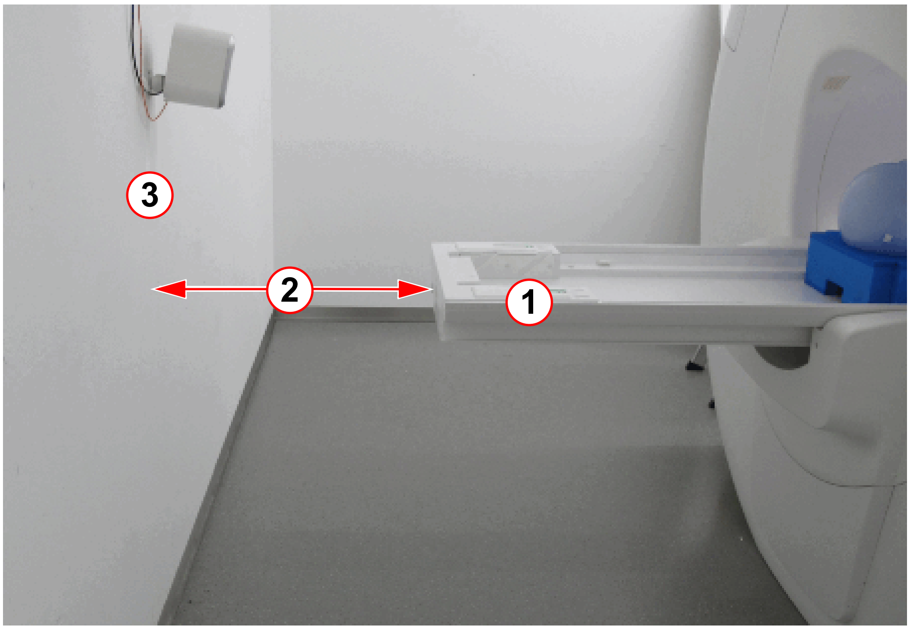
Fig. 530: Magnet service end



(1) Patient table, an example of the mechanical end stop

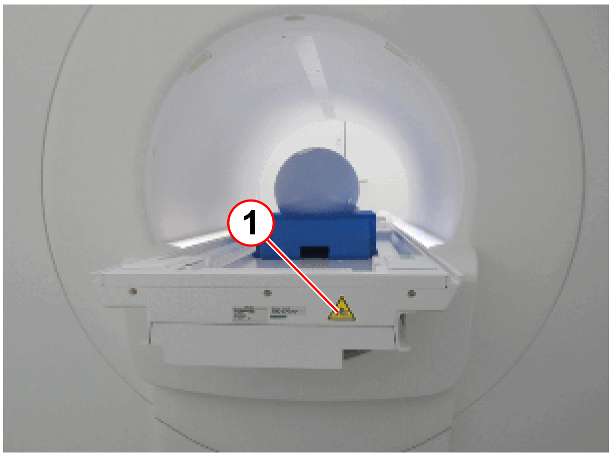
- Pull the emergency release handle on the patient table.
- Manually move the table all the way back (toward the magnet service end) until the table stops in the horizontal end position.
  - » If the distance between the patient table and the wall (service end) of the RF room is **greater than** 500 mm, no further actions are necessary. Perform final steps.
  - » If the distance between the patient table and the wall (service end) of the RF room is **less than** 500 mm, the bruising hazard sign has been posted at the rear of the patient table and the controlled access area has been secured using yellow and black tape.

Fig. 531: Bruising hazard, controlled access area



(1) Patient table back in horizontal end position
(2) Distance
(3) Wall of the RF room

Fig. 532: Magnet service end / patient table

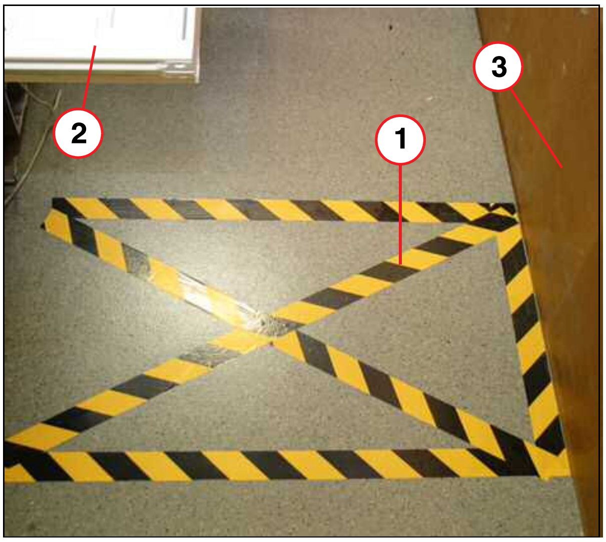


(1) Sign for bruising hazards

Fig. 533: Sign for bruising hazards

- Ensure that the bruising hazard sign has been posted at the rear of the patient table if the distance between the patient table and the wall (service end) is less than 500 mm.
- Install the missing sign immediately.

Fig. 534: Service end area



(1) Marking on the floor
(2) Patient table
(3) Wall of the RF room

- Ensure that the controlled access area has been protected with yellow and black tape if the distance between the patient table and the back wall is less than 500 mm.
- Install the missing tape immediately.

Final steps:

- Slide the table manually back into the home position.
  » The emergency release handle will turn to its original position.
- Confirm the pop-up windows on the MRAWP.
- To reset the table, press the IN and OUT buttons simultaneously on the control panel.

5.10.4 Checking the distance between the patient table and the cover

Help ID: 570DED6B-F94B-404E-A53E-5CEE555A981F
Service Level: 7

Distance between the patient table and the cover

SI Checking the distance between the patient table and the cover

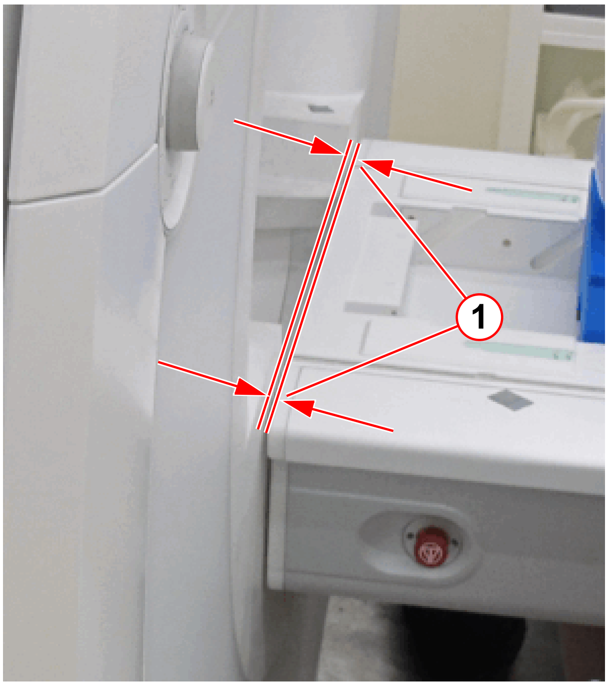
Required time: 5 min

Required tools: 8 mm Allen key or a non-magnetic ruler

Prerequisites: Table is approximately 10 cm below home position.

1. Move the patient table up to the highest position (transfer position vertical/horizontal).

Fig. 535: Patient table clearance



(1) Distance between patient table and cover

2. Measure the distance of the vertical clearance between the tabletop plate and the magnet cover.

   » The distance between the tabletop plate and the magnet cover must be within the range of **< 8 mm**.

   » If the distance is not within range, the patient table must be adjusted. Refer to Installation > Start-up > Check the mechanical adjustment of the patient table.

5.10.5 Checking the patient table movement

Help ID: C081FE94-3707-42BD-AB26-FAF61DB7682E
Service Level: 7

Table movement

Required time: 10 min

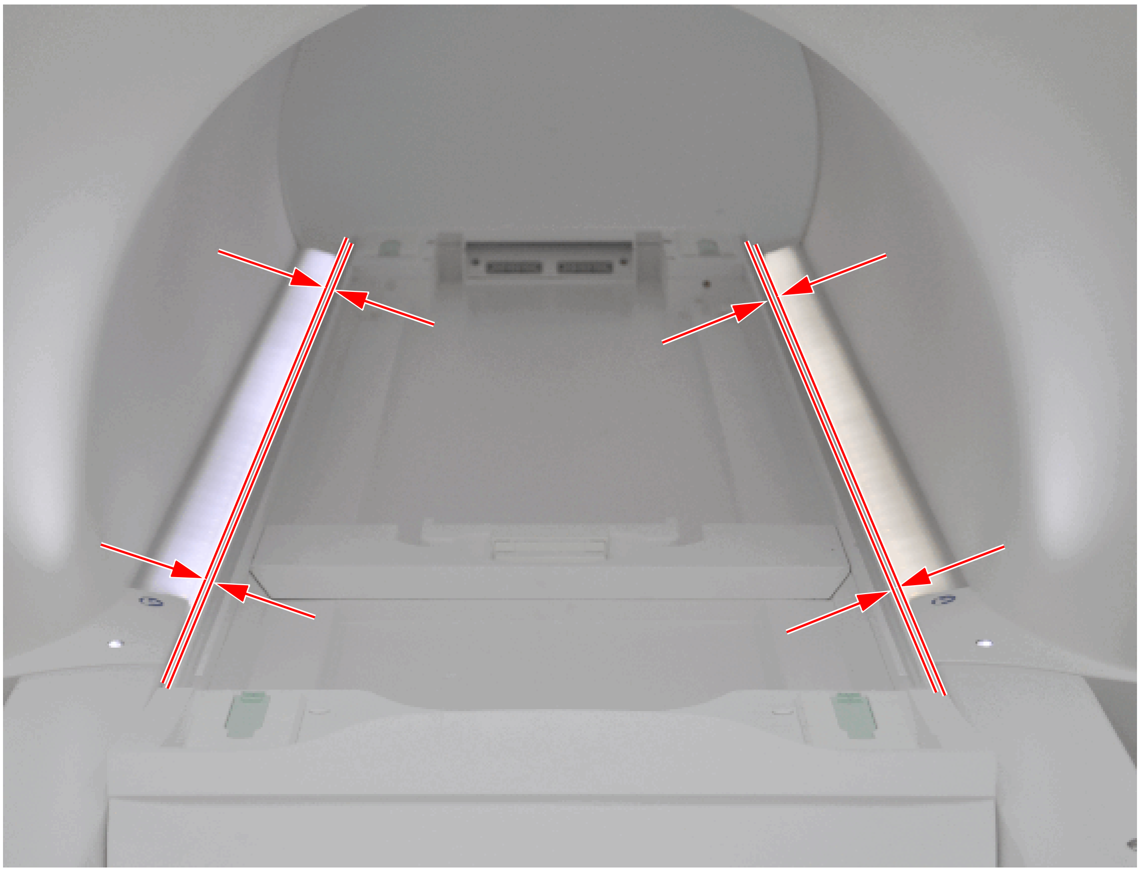
Required tools: n.a.

SI Checking the patient table movement

General movement

- Move the table all the way out.
  - » The table must move without any noticeable noise.
  - » The table stops in the horizontal end position.
- Lower the table all the way.
  - » The table must move without any noticeable noise.
  - » The table stops in the vertical end position.
- Move the table all the way up again.
  - » The table must move without any noticeable noise.
  - » The table stops in the vertical end position.
- Move the table into the magnet center.

Fig. 536: Patient table, parallelism



  - » The patient table must move parallel to the body coil.
- Move the table all the way in.
  - » The table must move without any noticeable noise.
  - » The table stops in the horizontal end position.
- Move the table to the home position.

If more dockable tables are in use, check the transfer position of these tables too. If the transfer position is not OK, adjust S20; refer to Replacement of Parts - Patient Handling - Patient Table

5.10.6 Checking the squeeze bulb

Help ID: AB6BBCC1-32E2-4AB5-B79F-FEE783E5351C
Service Level: 7

Squeeze bulb

Required time: 5 min

Required tools: n.a.

SI Checking the squeeze bulb

- Press the squeeze bulb on the patient table.

The following must occur:

» Continuous tone over the intercom.

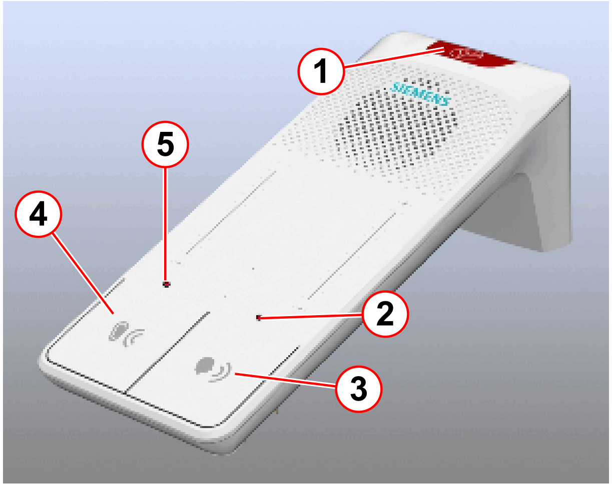
» Red flashing LEDs (→ 2 and 5/Fig. 537 Page 744) on the intercom

» The alarm can be reset by pressing the **Listen** (→ 4/Fig. 537 Page 744) or the **Talk** (→ 3/Fig. 537 Page 744) button on the intercom.

» You can speak with the patient by pressing the **Talk** button on the operating console. If necessary, adjust the volume using the +/- buttons.

» Communication with the patient is established when you press the **Listen** button. If necessary, adjust the volume using the +/- buttons.

Fig. 537: Intercom



(1) Stop button
(2) Talk LED
(3) Talk button
(4) Listen button
(5) Listen LED

5.10.7 Checking the side railing switches of the option dockable table

Help ID: 05F8E020-72B6-4784-A23A-272928E3020D
Service Level: 7

Dockable table option: Side railing switch

Required time: 5 min

Required tools: n.a.

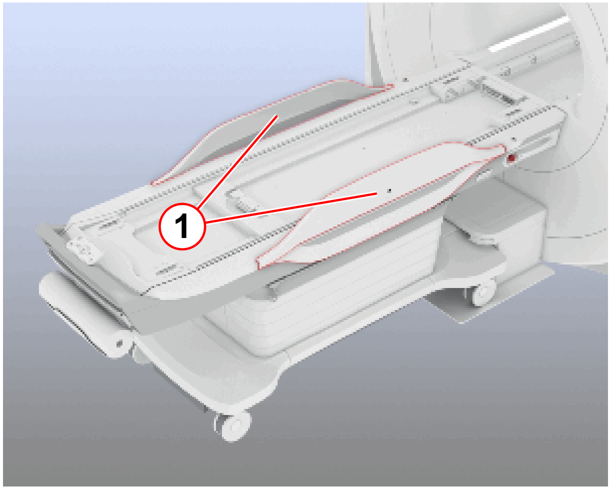
Prerequisites: Table is docked and in home position.

SI Checking the side railing switches of the option dockable table

- Press the **center move** button.

- While the table is moving toward the isocenter, lift up the side railing on the **left** side of the patient table

Fig. 538: Dockable table



(1) Side rails

» The patient table stops; the Dot display indicates the table stop.

- Make sure that table movement is not possible.
- Move the side railing to the down position.

» Table movement is possible again.

- Press the **center move** button.
- While the table is moving toward the isocenter, lift up the side railing on the **right** side of the patient table.

» The patient table stops; the Dot display indicates the table stop.

- Make sure that table movement is not possible.
- Move the side railing to the down position.

» Table movement is possible again.

- Move the table into the home position.

5.10.8 Checking the pressure activated switches of the option dockable table

Help ID: 07AE8919-D1EE-4FCF-8A83-8B9F0EEB3D99
Service Level: 7

Dockable table option: pressure-activated switches

Required time: 5 min

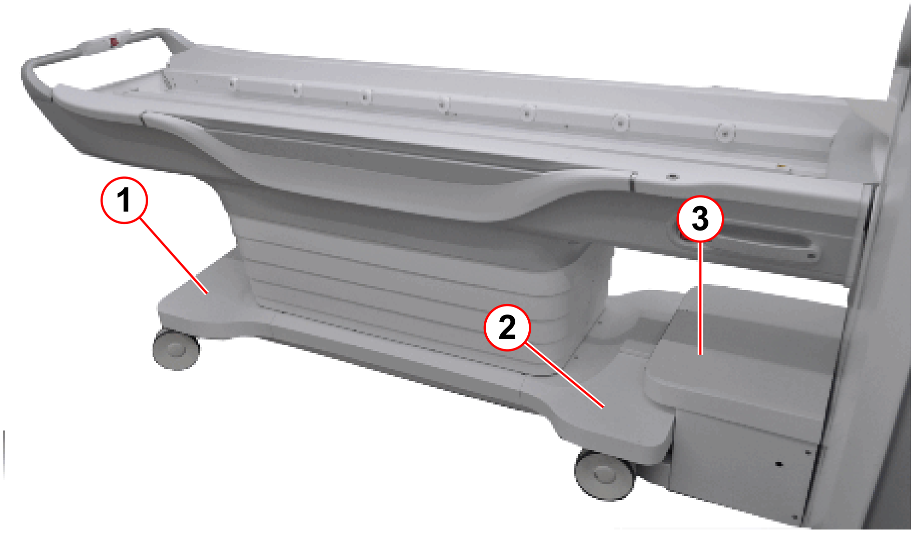
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

There are three pressure-activated switches on the dockable table.

1. Under the chassis foot-end cover
2. Under the chassis head-end cover
3. Under the top cover of the docking station

Fig. 539: Dockable table



(1) Chassis, foot-end cover
(2) Chassis, head- end cover
(3) Top cover of the docking station

SI Checking the pressure activated switches of the optional dockable table

- Move the table downward by pressing the down button.
- During table movement, press the cover of the docking station on the top side to activate switch S60.
  » The table stop movement
- Press the up button
  » The table moves upward
- Press the down button again
- During table movement, press the chassis head-end cover downward to activate switch S60.
  » The table stop movement
- Press the up button
  » The table moves upward
- Press the down button again
- During table movement, press the chassis foot-end cover downward to activate switch S60.
  » The table stop movement

- Press the up button
  » The table moves upward

5.10.9 Checking the emergency undock function of the option dockable table

Help ID: 140D3D5F-7A36-4C88-92CE-C2FD58951673
Service Level: 7

Dockable table option: emergency undock function

Required time: 10 min

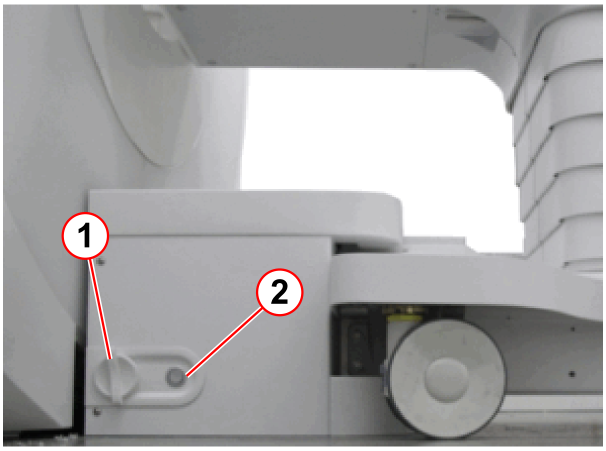
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

SI Checking the emergency undock function of the optional dockable table

- Turn the emergency undocking handle (a quarter turn clockwise).

Fig. 540: Docking station



(1) Emergency undocking handle
(2) Docking reset button

» The docking station releases the Dockable Table. The table is undocked.

> After use of the emergency undocking function docking is not possible. The docking reset button sets the docking station back to normal operation. Docking is possible again.

- Move the Dockable Table away from the docking station.
- Press the docking reset button.
- Move the Dockable Table into the docking station and press the docking button on the patient table control unit.

Fig. 541: Dockable table, control unit

(1) Docking button

» The dockable table is docked.

5.10.10 Checking the chassis and the wheel of the option dockable table

Help ID: 1E8DCB2C-C692-456F-AC54-D37D672A5817
Service Level: 7

Dockable table option: chassis and wheels

Required time: 5 min

Required tools: n.a.

SI Checking the chassis and the wheels (dockable table only)

- Undock the Dockable Table from the docking station.
- Move the table by a few meters.
  » The table must move smoothly without any noticeable noise.

Fig. 542: Patient table control unit

(1) Break button

- Press the break button.
  » It must not be possible to move the Dockable Table.

5.11 TuneUp

5.11.1 TuneUp - General

Help ID: 9A54DF7F-CE66-4190-863F-B65D390E1BA5
Service Level: 3

Introduction

General TuneUp contains a fixed set of procedures (workflow) for calibrating system parameters

Each individual procedure automatically saves data in the corresponding system parameter files, log files and database files.

TuneUp Workflow

- General TuneUP is separated into modules, each routine runs as a stand-alone process.
- After start, each procedure reads the specification and configuration parameters automatically.
- TuneUp procedures determine if their measured results are in or out of specifications.
- TuneUp procedures write their individual results to the Service Report.
- The status is returned to the main TuneUp platform and the corresponding status field in the Service UI is updated.

After pressing the **[Go]** button in the TuneUp UI, the **procedures start and perform typical tasks**:

- Each procedure registers the Service Patient (if not already registered).
- If a clinical patient is registered, a pop-up informs the operator to de-register the clinical patient before proceeding.
- The correct phantom setup is displayed by a pop-up and confirmed by the operator (if applicable). If QA is performed remotely, automatic table move is disabled and the operator receives a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Table movement:
  - If **TuneUp is performed locally** at the system, the **table moves automatically** to center the required phantom in the measurement field (change between large/small spherical phantom).
  - If **TuneUp is performed remotely**, **automatic table move is disabled**. The operator gets a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Configuration and specification values are read.

- Measurements are performed.

TuneUp Function Status

Tab. 26 Possible TuneUp Function Status

Status	Description
To Do	The TuneUp procedure has not yet been performed yet and is pending. OR: The TuneUp procedure has to be performed again because another TuneUp procedure has requested its execution.
Ok	The TuneUp procedure has been carried out successfully.
Not Ok	The related parameters/results were found to be out of specifications during the last operation.
Error	An error occurred during the last operation.
Cancelled	Procedure canceled by operator.

Evaluation/Results

After end of the TuneUp measurement, evaluations are performed automatically:

- The procedures determine if the measured results are in or out of specifications.
- Determined parameters are saved to system files if the procedure results are in specifications.
- The results are written to the Service Report.
- The results status is returned to the main TuneUp platform and the corresponding status field is updated.

TuneUp Verification

All **TuneUp** workflow procedures have their corresponding verification routines in the **Quality Assurance** workflow menu.

TuneUp/TuneUp Dependencies

TuneUp procedures are executed in a listed order because some steps depend on a successful execution of previous steps. For example, the successful execution of “Gradient Regulator” is required for all following TuneUp steps which use an MR-signal.

This means, if the operator runs a Tune Up procedure again, also the status of all depending procedures is reset to **To Do** and they have to be performed again.

In addition, the Software monitors connected MR-system components (for example,TX-Box) for hardware changes. If a change is detected (for example, after parts replacement,) the affected TuneUp adjustment steps are automatically reset to **To Do** status.

TuneUp/QA Dependencies

If a TuneUp procedure is successfully completed and has changed its status, the corresponding verification procedure(s) in the Quality Assurance workflow menu is/are changed to status **To Do**.

TuneUp procedures may also affect multiple QA procedures, for example, after TuneUp Phantom Shim, the QA procedures Phantom Shim Check, Gradient Sensitivity Check, BC Image Brightness and Long Term Stability Check have to be performed.

5.11.1.2 Feedback Loop Calibration

Help ID: C2FE7D73-8E4B-4758-8858-5A5CD39E46B0
Service Level: 3

General

The TX-Box has a built-in feedback loop to compensate for instabilities of the TX B1-field.

The procedure **Feedback Loop Calibration** is used to calibrate internal feedback loop parameters of the TX-Box.

From experience it is known, that the feedback signal phases and the feedback delay depend on the TX-Box loadware version. Thus, an individual phase- and delay-adjustment must be performed in TuneUP although the feedback phase has been initially adjusted during TX-Box fabrication. Improper feedback phase and delay results in suboptimal dynamic behavior and can, in the worst case, even cause oscillation.

For non-pTx systems, a second measurement through the LC path is needed in case the phase difference between LC and BC channels needs calibration.

For systems with multiple transmit channels (pTX), the relative phases of the various channels are free parameters anyhow which are continuously controlled by the measurement system.

After a loadware change at the MR system, it might be necessary to repeat the phase- and delay-adjustment by running Feedback Loop Calibration in TuneUP. The procedure reads out the current delay value of the TX-Box. Then, the phase of second TX-channel is determined at minimum, middle and maximum operation frequency. From the measured phase values, the new delay value and a new file "nb_hybrid_compen.dat" is calculated and loaded into the flash memory of the TX-Box.

Measurement

BC Path

The service sequence “tuncal” is invoked, which runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box. Then, the Software analyzes the frequency dependence of the signal phases.

A linear curve is fitted to the frequency dependency of the signal phases. From the slope of this line, the feedback loop delay is obtained. In order to enhance the precision, the results obtained from both Body Coil elements are combined.

LC Path

A hardware flag determines, if a second measurement through the LC path is needed or not. That measurement calibrates the phase difference between LC and BC channels.

For the measurement, the Service Plug is connected and the service sequence “tuncal” is run through the LC TX-path.

The obtained data set is analyzed in the same way as for the BC TX-path. The difference of the BC and LC phases at the center frequency contains the desired value.

Restart MaRS

After the successful determination of the feedback control loop parameters, a restart of the MaRS Software is required (only needed in TuneUp mode).

The boot process starts the activation of the feedback loop parameters in the flash memory of the TX-Box. Afterward, the feedback control loop is ready for normal operation.

Evaluation/Results

Results in Report:

Tables:

- Feedback loop delay:

  - Delay evaluation

  - Dispersion of delays from single BC elements
- Restart MaRS Software Results

Plots (debug mode):

- BC phase plot

Modified parameters:

- Feedback loop delay

- Feedback loop phase deviation

are saved

5.11.1.3 RX Gain Calibration

Help ID: 66C1477B-627A-4488-A469-B5EC0946F2E6
Service Level: 3

General

The individual receive channels in the RX-electronics have slightly different electrical/frequency characteristics which must be corrected. Otherwise, these variations would cause amplitudes deviations of the individual receive signals leading to image brightness variations in the final image.

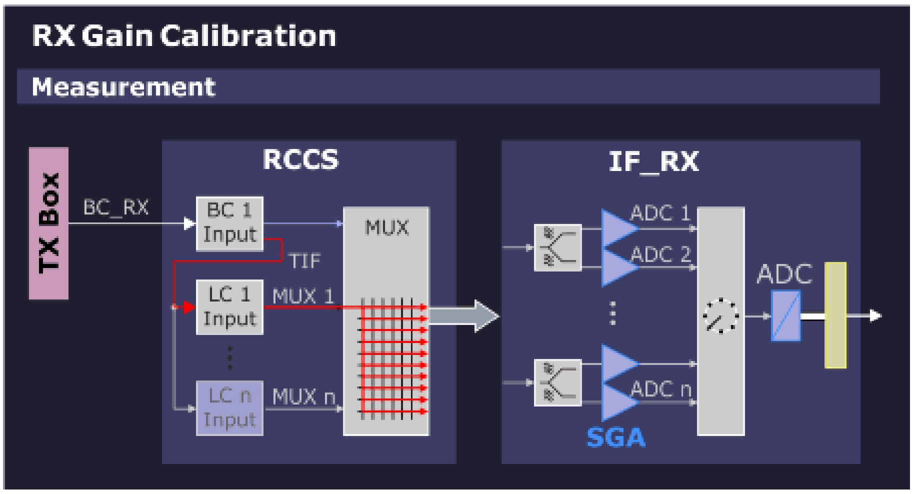
Therefore, it is necessary to normalize the signal intensity of each receive channel so that the image intensity of the final image is the same regardless which receiver channel(s) were used for imaging.

The receiver-related deviations are determined with the RX Gain Calibration with 2 measurements:

- **Channel correction** - determine deviations caused by the switch matrix (in RCCS).
- **High/Low gain correction** - determine amplitude and phase deviations between high- and low-gain settings of the switchable-gain amplifiers (SGA) of the IF_RX board(s).

Measurement

Fig. 543: RX Gain Calibration - Measurement



A fixed calibration signal (BC_RX) is generated in the TX Box and transferred to the RCCS over coax cable. This signal is then fed through the RCCS-internal local coil input multiplexer to the switch matrix where it is sequentially routed through to each of the SGAs and ADCs on the IF_RX boards.

There are 2 measurements made per channel at high- and low-gain settings.

Evaluation/Results

The correction factor of the measured amplitudes for the RX-channels and the quotient of the high/low gain modes are evaluated.

Results in Report:

Tables:

New calibration factors

Plots:

Measured signals

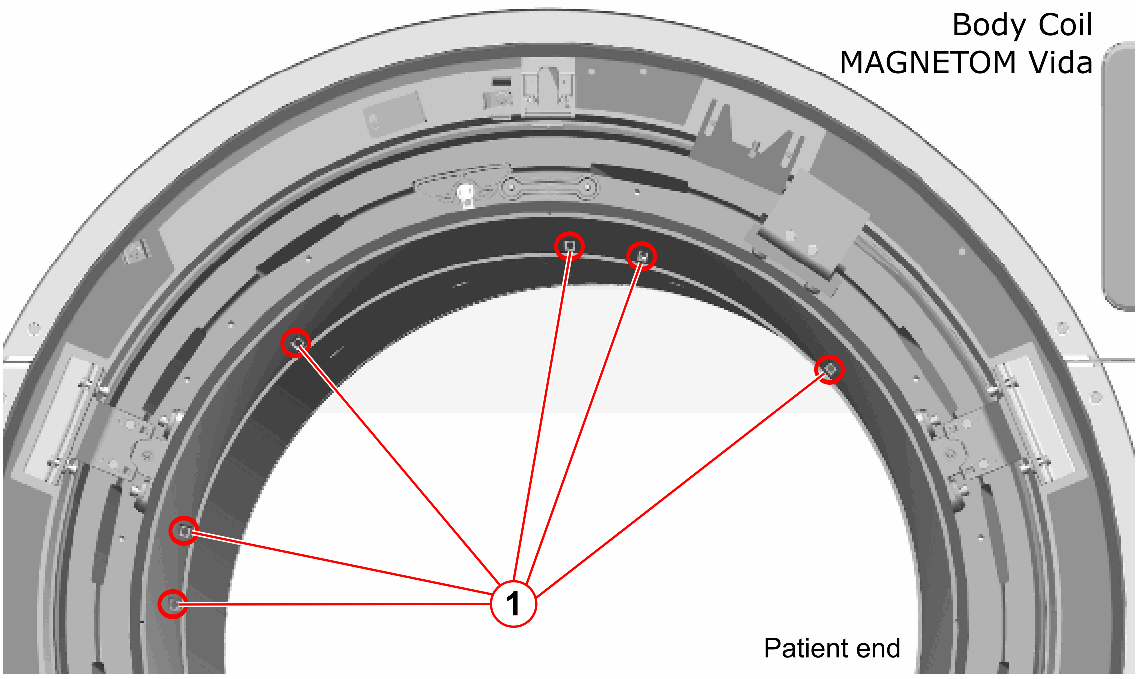
Modified parameters:

The RX calibration factors for all RX channels in high/low gain mode are saved.

5.11.1.4 BC Tuning

Help ID: 03DC2B49-7E45-4948-BAC2-00780F56D2DB
Service Level: 3

Fig. 544: Body Coil - BC Tuning



(1) Location of Trimmer Capacitors

Safety

WARNING

In case a BC Mechanical Position Adjustment at magnet rear-side (service end) is required, you will work close to Gradient Coil connections. These cables carry high voltage!

Risk of electrical shock!

» **Switch-off the GPA with switch S7/ DGSU and circuit breaker F1 in the GPA cabinet.**

» **Subsequently, wait at least 20 minutes to discharge GPA capacitors before starting any BC Mechanical Adjustment.**

» **However, for only adjusting the BC Tuning trimmer capacitors (patient end), the GPA cabinet can be left switched-on.**

NOTICE

The trimmer capacitors have 32 full turns each.

Risk of damage!

» **Do not use force when reaching the capacitor end stops to prevent damage!**

General

The **BC Tuning** procedure is used to measure and, if necessary, adjust the central resonance frequencies of the Body Coil resonator systems and minimize the coupling between them.

A high efficiency of RF power transmission to the Body Coil requires sufficiently low reflection factors at the TX-Box output.

This condition is usually met, if the reflection factors of the Body Coil resonator systems BC0 and BC1 are high (without load) and equal. However, with load, these reflection factors shall be low.

The other requirement is, coupling between both resonators shall be minimum (that is, high decoupling values).

Measurement

BC Reflection

First, a **BC Reflection** measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals, covering the full transmit operating frequency range. The result of each measurement is the reflection or r-factor = Ur/Uf which is plotted for both Body Coil systems BC0 and BC1 along the frequency axis.

The evaluation determines:

- that the tuned frequency of each coil is within a specified range, that is, not too far-off from the operating range center.
- that the center frequencies of both coil systems BC0 and BC1 are tuned closely to each other, that is, the delta frequency is low.
- that the reflection factors are within expected limits. These values provide useful information about the quality (Q-factor) of the BC resonant circuits. They also depend on the coil load and cannot be adjusted.

  The Software detects the resonant frequency always at the point where the reflection of the BC has its minimum.

Transmission

In a second step, a **Transmission** measurement is performed with transmitting a signal into the BC0-system and measuring the coupled signal in the BC1-system and in the opposite direction. The transmission factor describes the amount of RF-coupling between the two Body Coil systems, mainly due to coil construction but also affected by other factors, for example, mechanic position adjustment of the Body Coil within the Gradient Coil Faraday Shield.

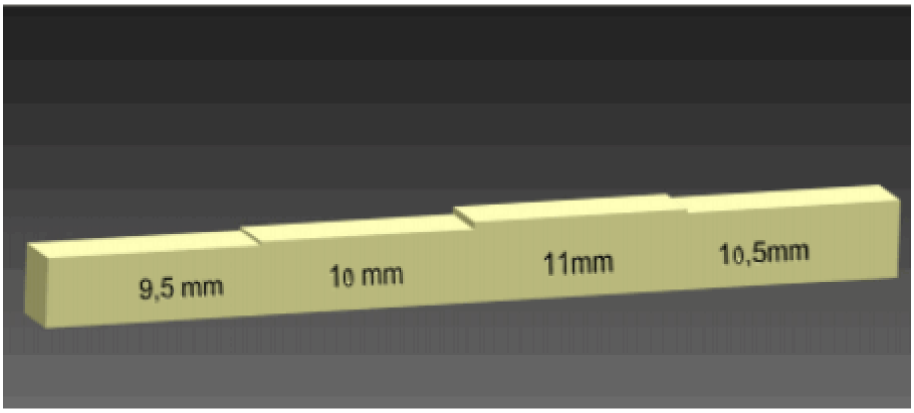
The evaluation checks the transmission value, expressed in decibels (dB), to be within expected limits. **A high negative dB-value indicates minimum coupling**, that is, a good decoupling between BC0 and BC1 system.

Adjustment

Prerequisites

- For BC mechanical position adjustment, the BC feeler gauge (material number 106 76 562) is required.

Fig. 545: BC Feeler gauge (10676562)



- To access the trimmer capacitors at the **Body Coil front-side** remove the service access cover and the front funnel insert.

- Take out the reflector bracket from its storage position behind the magnet front cover.

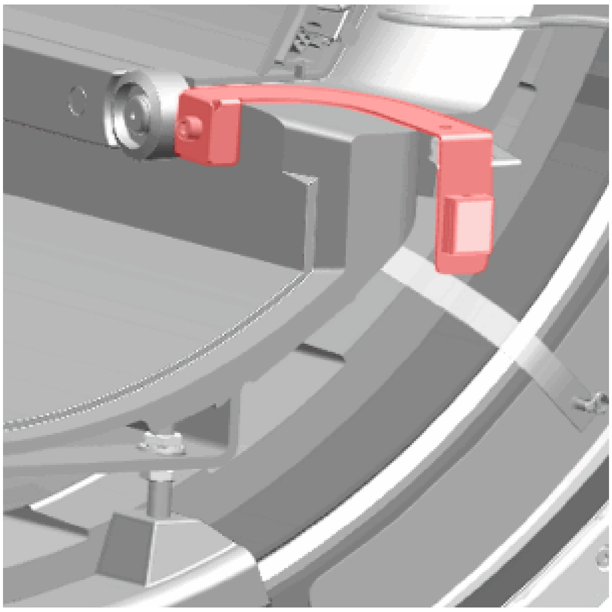
Fig. 546: Reflector bracket storage position

(1) Reflector bracket

> The reflector position defines the drive-in height for the patient table and is adjustable. At first use, check for the correct adjustment to avoid patient table collision.

- Fasten the reflector bracket with a screw from the removed front funnel.

Fig. 547: Reflector bracket attached to BC



Generally, BC Tuning is performed in 3 Steps:

- Step 1: Reset (BC Mechanical Position + Trimmer Capacitors)
- Step 2: Adjust BC Mechanical Position
- Step 3: Adjust Trimmer Capacitors

Exception for new installations or if BC Tuning is already close to specification:

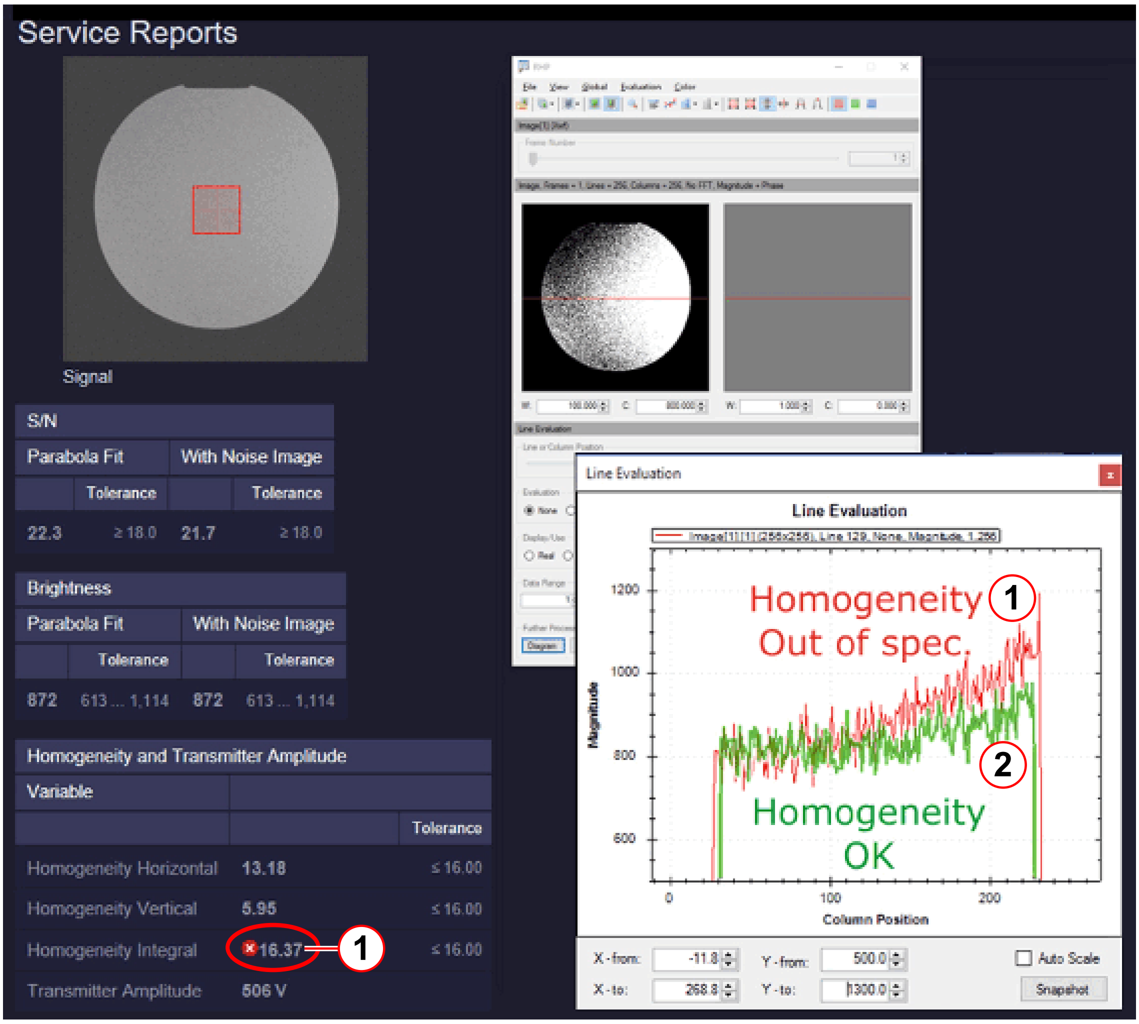
- Skip Steps 1 and 2.
- Perform Step 3 “Adjust Trimmer Capacitors” directly.

Always perform the complete BC Tuning workflow (Step 1-3) after:

- BC was replaced.
- BC Tuning: Transmission value is less than -15dB.
- BC Tuning: Reflection factor difference for BC0 and BC1 is greater than 0.1.
- QA Image Brightness Check fails with homogeneity results out of spec.

- Strong asymmetry in RF-field (B1-field) distribution (for example, strong inhomogeneity in QA Image Brightness Check, transversal orientation).

Fig. 548: QA/ BC Image Brightness Check - Results



(1) Homogeneity NOT OK

(2) Homogeneity OK

Step 1 - Reset

Reset any changes and establish the Body Coil default set up.

Switch off GPA

- Switch off the GPA with switch S7/ DGSU and circuit breaker F1 in the GPA cabinet.
- Wait at least 20 minutes to discharge GPA capacitors before starting BC Mechanical Adjustment!

Set Trimmer Capacitors to Center Position

- Turn all trimmer capacitors carefully in clockwise (“CW”) direction until mechanical stop.

- Now, turn 16 full rotations back in counter-clockwise (“CCW”) direction.

Set BC Mechanical Position to “Center”

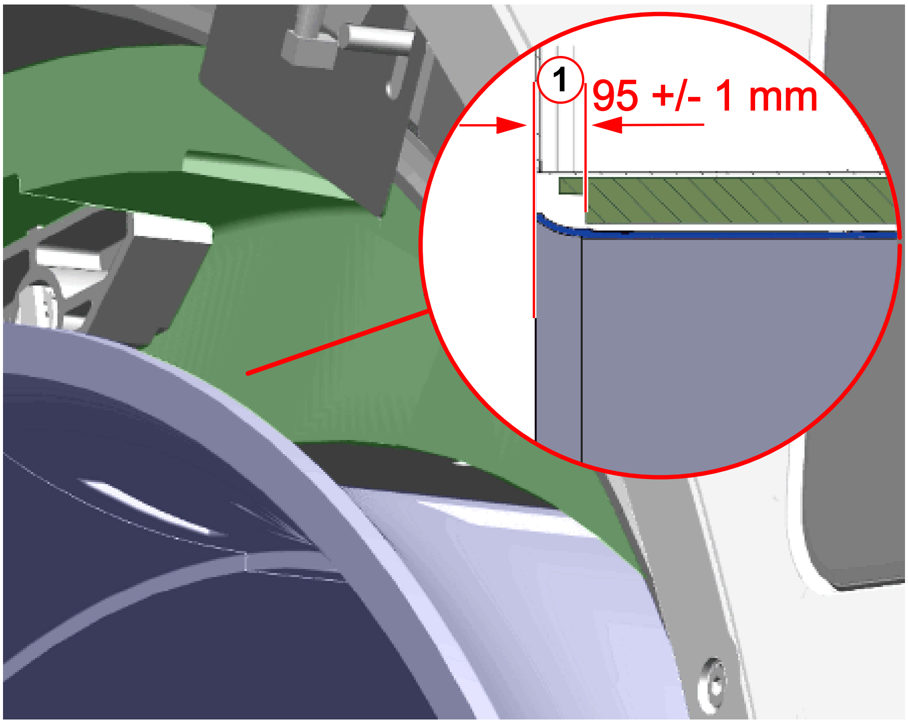
> The Body Coil is aligned with respect to the inner bore of the Gradient Coil (the RF-shield).

> For BC mechanical position adjustment, the BC feeler gauge (material number 106 76 562) is required.

At **Patient end** adjust fixed BC distances to the Gradient Coil:

- In **Z-direction**, adjust **distance of 95 +/-1 mm** between BC tube front surface and recess in GC (see graphics below).

Fig. 549: Body Coil - Front View

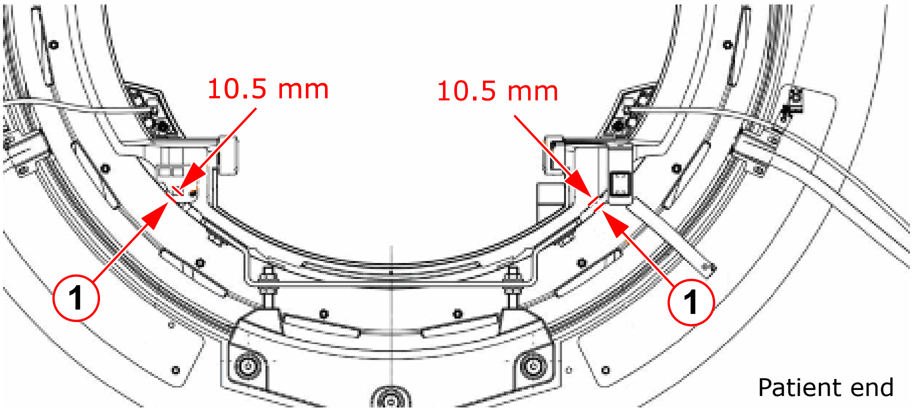


(1) Z-distance 95 +/-1 mm

- In radial direction, **adjust the 2 left/right distances equally to 10.5 mm**.

  Measure at the reference points next to the BC support (from the RF-shield to the 2 flat reference surfaces on the BC bore tube).

Fig. 550: Body Coil - Front View

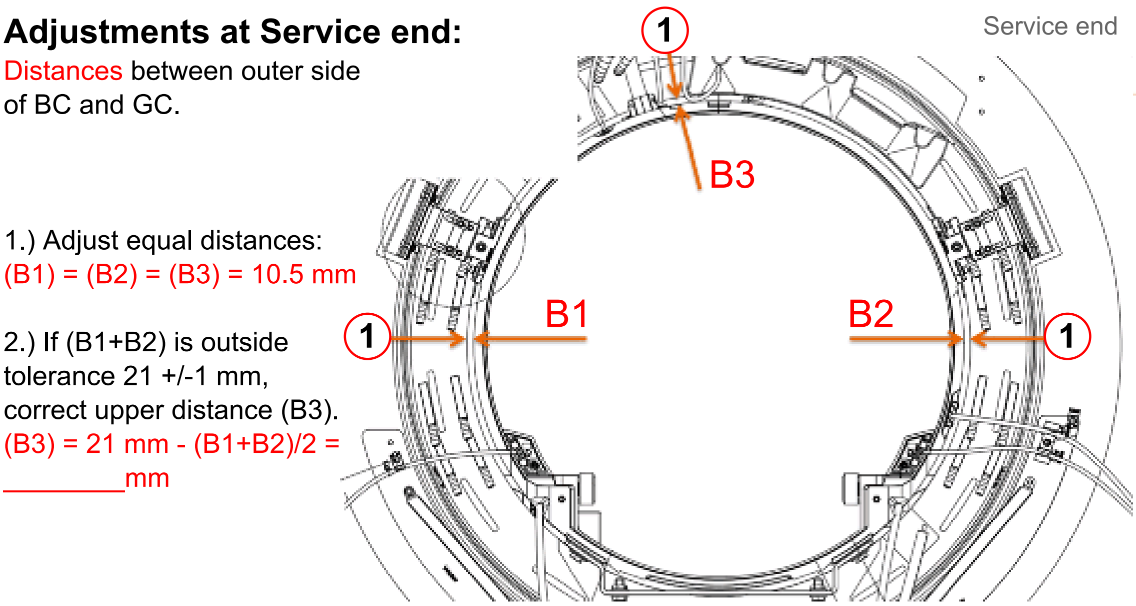


(1) Left/right distance 10.5 mm

At **Service end** center the BC radially inside the GC:

- Adjust **equal left/right distances (B1 = B2) =** 10.5 ± 0.5 mmbetween BC and GC.
- Check the **upper distance (B3)** between BC and GC; **(B3) =** 10.5 ± 0.5 mm.
- Check all distances again **(B1 = B2 = B3) =** 10.5 ± 0.5 mm
- **If (B1 + B2) is outside tolerance** 21 ± 1 mm, correct upper distance: **(B3) = 21 mm – (B1 + B2)/2**

Fig. 551: Body Coil - Service end



(1) Measurement points for distance BC-GC

At service end, the lower BC position (at 6 o'clock) cannot be directly measured because the BC support blocks access.

Due to mechanical tolerances, the actual distance values may slightly differ from the nominal value of 10.5 ± 0.5 mm.

Some calculation is required then, see description above.

Step 2 - Adjust Mechanical Position

The Body Coil Tuning is sensitive to its exact mechanical position inside the Gradient Coil.

A further optimization of the BC mechanical position might be necessary to achieve a successful BC Tuning adjustment with trimmer capacitors.

Switch off GPA

- Switch off the GPA with switch S7/ DGSU and circuit breaker F1 in the GPA cabinet.
- Wait at least 20 minutes to discharge GPA capacitors before starting BC Mechanical Adjustment!

Criteria for a good BC Mechanical Adjustment

1. BC Tuning results (no trimmer adjustments performed yet):
   - **Transmission value better than -25dB**
   - The **reflection factors** for BC0 and BC1 are **maximum**
2. Homogeneous RF-field (B1-field):
   - **QA / BC Image Brightness Check is in spec.**

The BC mechanical adjustment influences the reflection and transmission factors (BC Tuning results) and the B1-homogeneity in the unloaded coil.

Therefore, a balance might be required to satisfy both requirements.

In most cases, a good adjustment of reflection and transmission values also leads to a good B1-field distribution.

Check of BC Tuning Results

Run BC Tuning with the defined QA setup and check:

- Are the "factors (at minimum)" in specification?
  - » Under normal conditions, these **reflection factors without load should be greater than 0.55**.
  - » Otherwise, if the reflection factors without load are much lower then 0.55, the BC mechanical position might be incorrect or additional losses are present (for example, defective capacitors at the Faraday Shield of the GC).
- Is the "factor (at minimum)" **difference for BC0 and BC1 less than 0.1**?

- Is the **transmission value better than -25dB** (that is, larger negative value)?
  - » If all conditions are OK, continue directly with Step 3.
  - » Otherwise, continue with further optimization below.

Further Optimization of Mechanical Position

At Service end, try:

- Move BC up 1 mm

  or
- Move BC down 1 mm

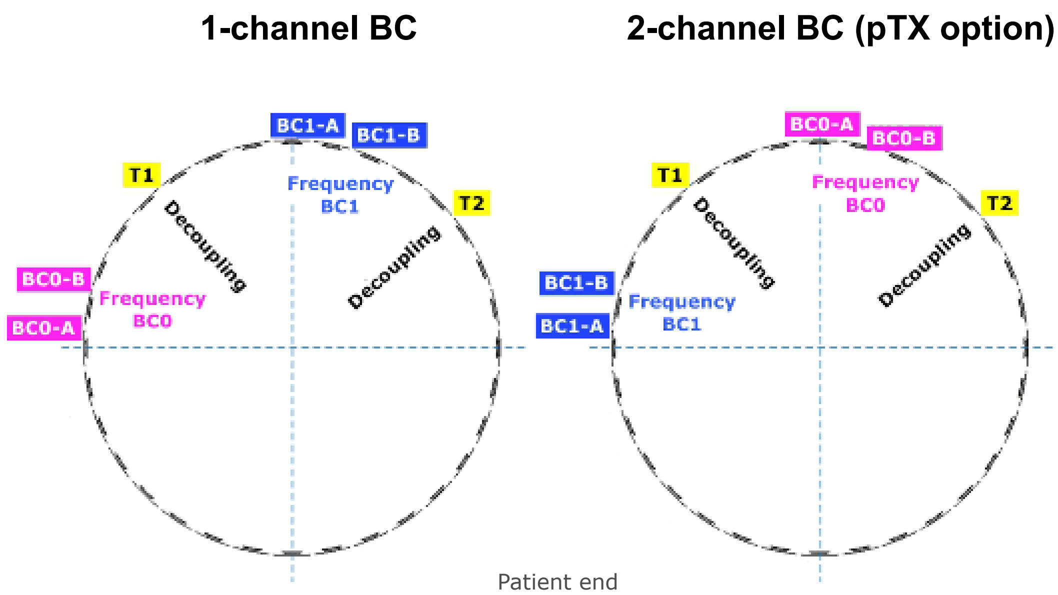
  or
- Move BC left 1 mm

  or
- Move BC right 1 mm
  - » Check after each change if the three BC Tuning Results (see above) have improved.

Step 3 - Adjust Frequency and Decoupling

Adjustment of Trimmer Capacitors

Fig. 552: Trimmer Capacitor assignment



> The Trimmer Capacitor assignment is different for 1-ch BC or 2-ch BC (pTX Option)!

- The BC has **6 trimmer capacitors** which are accessible from patient end.

- The **trimmer capacitors** which belong to the same function (for example, decoupling T1, T2) **are arranged in pairs**.

  Turn a capacitor pair synchronously in the same direction and with same number of turns.
- **Turning a capacitor in clockwise direction** (“CW”) increases its capacitance value which **decreases frequency**.
- More detailed adjustment information is shown at the magnet display during the BC Tuning workflow.

Evaluation/Results

Results in Report:

- The reflection-related values for both (BC0 and BC1) systems (for example, resonance frequency, reflection factor and the difference between both systems).
- The transmission-related values (frequency and decoupling between both systems).
- The new and old phase difference.
- Diagrams showing the values for transmission and reflection as a function of frequency.

5.11.1.5 Cable Length

Help ID: C129565B-356D-4AE3-8F27-3D3ADE42E986
Service Level: 3

General

The **Cable Length** measurement determines the individual length of the BC transmit cables. This information is needed for the extended B1-field regulation loop of the TX-Box.

For an optimal operation of the TX B1-field regulation loop, precise knowledge of the phase situation at the Body Coil is necessary. These signal phases depend mainly on the known RF-properties of the Body Coil itself. However, they also depend on the length of the BC transmit cables which induce an additional phase shift into the TX-path. The actual BC transmit cable lengths may vary slightly (production tolerances) and must be determined at the system by a precise measurement of the induced phase shifts.

Afterward, the exact phase situation at both Body Coil systems is known. This information is used to better analyze and compensate B1-field fluctuations which can easily occur during a measurement (e.g. a BC impedance change caused by thermal drift effects).

Measurement

The Cable Length measurement uses the service sequence rf_loop. An RF pulse is fed to both Body Coil channels with the Body Coil switched to ‘detune’ mode. The forward and reflected signals are taken from the internal directional couplers (DICO) of the TX-Box and analyzed by the monitoring Software.

From these results, the so-called S-matrix elements which describe the characteristics of the BC transmit path are determined. These values allow, together with the known Body Coil RF-properties, to calculate the cable lengths for both Cody Coil channels in terms of the induced signal phase shifts.

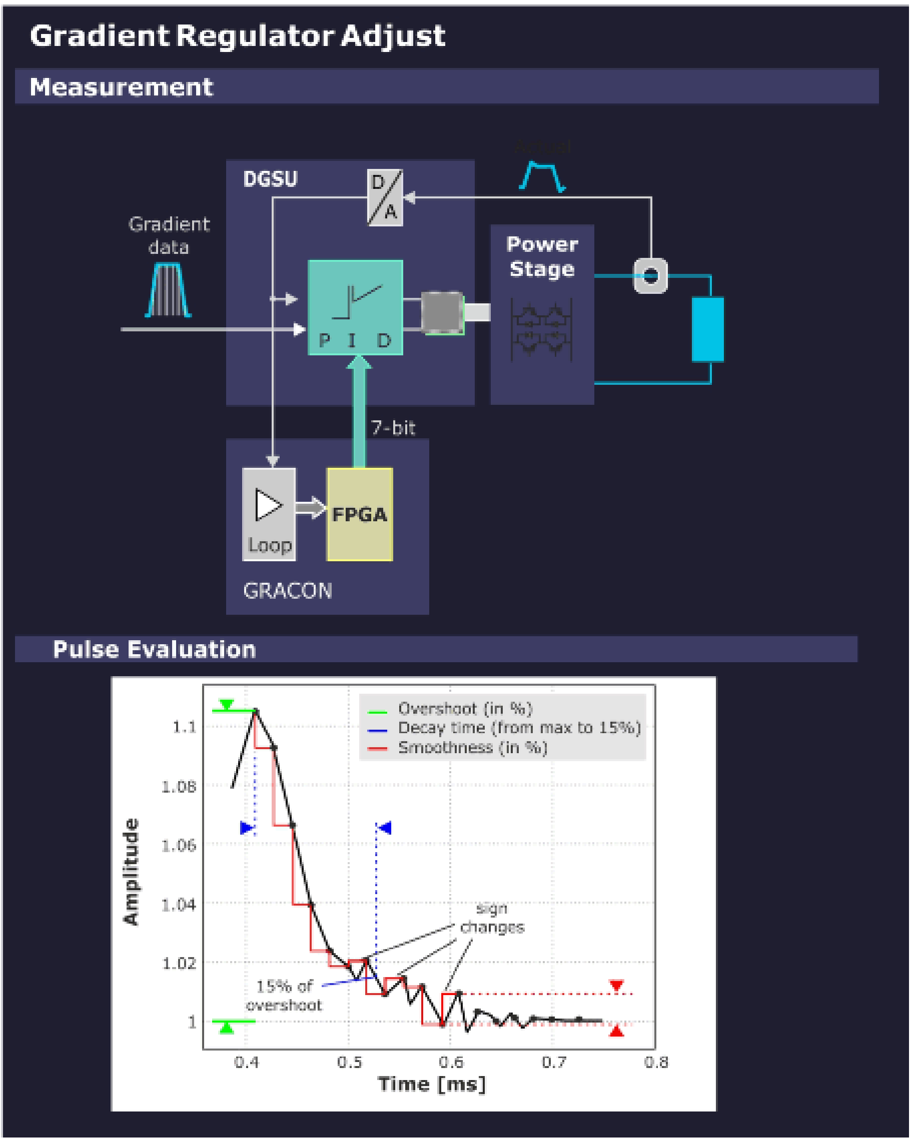
Evaluation/Results

In the report, the measured phase shifts for both Body Coil channels together with their previous values are displayed. The latter ones are read from the BC coil files. In case the phase shifts lie within a configurable specification range, they are saved permanently to the BC coil files.

5.11.1.6 Gradient Regulator

Help ID: 69D468F4-E746-4AD2-A81A-AF7BC2D67D70
Service Level: 3

Fig. 553: Gradient Regulator - Measurement



General

The **Gradient Regulator** adjustment determines the set of PID (P = proportional; I = integral; D = differential) regulator values that reproduce the applied gradient waveforms best. This is achieved iteratively for each axis by repeating the following steps until the results are in specification:

- The system response to an applied waveform is measured.
- New PID values are calculated and applied.

Gradient Regulator is an iterative procedure to improve the PID settings step by step to find the optimum solution. An evaluation is performed after each measurement. The results are plotted before the next iteration starts.

The regulator components are corrected one after another in the order P, I and D. The parts of the PID that have not been subject to the optimization already are deactivated. The initial value for each part (P, I and D) are read from the configuration.

After all regulators have been tuned, it is checked if the measured gradient waveform is in specification. If so, the regulator values are set permanently. If an error occurs or the program is aborted, the old regulator values (saved before the optimization) are restored to the GPA.

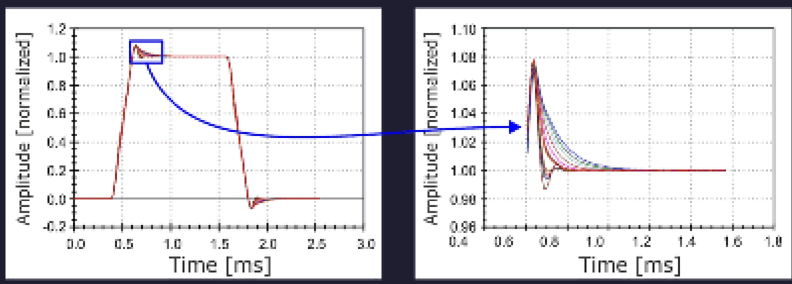
Measurement

- The sequence uses a gradient table to generate the gradient pulses so that the selected gradient (depending on the selected orientation) is switched from the maximum negative to the maximum positive gradient amplitude pulse in linear steps. Each gradient in the table is switched on using a ramp up and ramp down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.
- Based on the maximum gradient amplitude pulse, the program calculates the characteristic overshoot that is detected just after the ramp up time. The amplitude of this overshoot is determined as a percentage relative to the nominal gradient amplitude. Also, the time it takes for this exponential-like overshoot to return to a specified value is calculated. In addition, a smoothness calculation is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.
- Finally, each line in the gradient table is integrated and the resulting mean value for each line is fitted to a linear function. The difference between the data and the linear fit and the maximum deviation are returned as quality parameters.

Evaluation/Results

Fig. 554: Gradient Regulator - Results

Results



Gradient Regulator Results			
Axis	P	I	D
X-Axis	91	90	39
Y-Axis	92	87	38
Z-Axis	92	97	40

P Controller

P Values X-Axis

Iteration	P	I	D	Overshoot	Decay Time	Smoothness
1	70	127	0	x 6.291%	x 0.355ms	x 0.396%
2	78	127	0	x 2.513%	x 0.355ms	x 0.393%
3	91	127	0	0.789%	x 0.355ms	x 0.387%

- PID gradient regulator parameters
- optimization parameters overshoot, decay time and smoothness
- diagrams showing
  - The complete gradient current pulse (pulse form)
  - The enlarged overshoot portion and current ripple (top portion)

Modified/saved parameters:

- PID regulator values of the GPA

5.11.1.7 Table Height

Help ID: 52AB3696-7EB8-453A-8B1D-70BD022B4C0E
Service Level: 3

General

The **Table Height** TuneUp procedure is used to determine the reflector height which is used to find the correct drive-in height for horizontal patient table movement.

The reflector is mounted at the magnet cover front-side and is recognized by an optical sensor at the patient table. The actual reflector position can differ due to mechanical adjustment tolerances but also due to different magnet installation heights.

Knowing the Table Height (i.e. actual reflector position), the Software can optimize the search strategy looking for the reflector (move table up or down?). This saves time for the operator.

Measurement

- The patient table is moved into the Home Position to "wake up" the vertical drive.
- The host computer reads out the actual vertical height value from the patient table controller. A system-specific number of increments is subtracted to determine the "middle of reflector height" position.
- The "middle of reflector height" value is saved at the host computer.
- Prior to each following vertical travel request, the Software can compare the actual vertical table height with the known, determined Table Height (i.e. "middle of reflector height"). Depending on the result, the search strategy to find the reflector is optimized (i.e. table first moves up- or downward).

Evaluation/Results

Results in Report:

- The status of the automatic procedure is displayed.

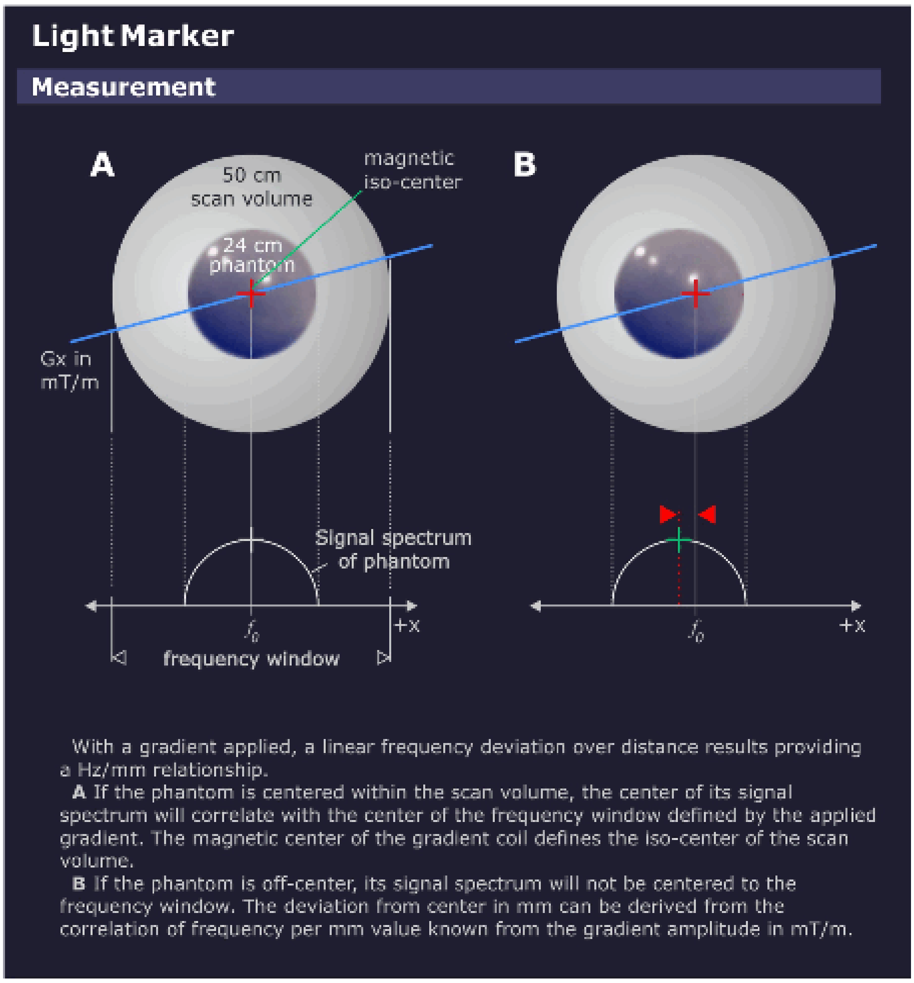
Modified/saved parameters:

The last known "middle of reflector height" is stored in the system.

5.11.1.8 Lightmarker

Help ID: 48907CEE-0E67-4BD6-A412-B297B56D1476
Service Level: 3

Fig. 555: Lightmarker - Measurement



General

The **Light Marker** procedure determines the distance between light marker and gradient center.

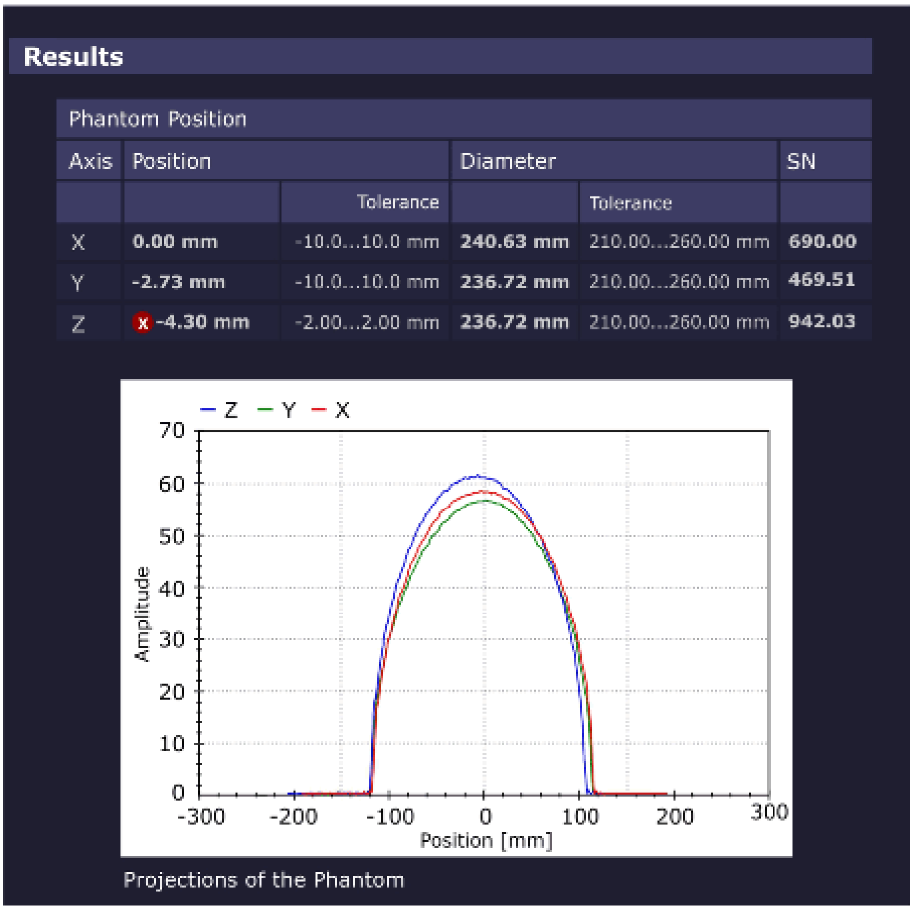
The distance of phantom center to gradient center is evaluated and used to correct the light marker distance.

Measurement

The measurement uses a double echo sequence. The first echo is acquired without applying a readout gradient resulting in an FID. The length of the FID is a rough check of the fundamental imaging volume homogeneity. During the second echo, a readout gradient is applied resulting in a projection of the image onto the readout gradient axis.

Evaluation/Results

Fig. 556: Lightmarker - Results



- The projected image of the phantom is determined, giving the position for the respective axis. This yields a new translation vector to the real gradient isocenter.
- The result of the z-axis measurement is used to determine the table position which is corrected afterward. The table is automatically moved to the correct gradient center. This procedure is repeated iteratively until the phantom position is in specifications.

Results in Report:

- phantom position coordinates.

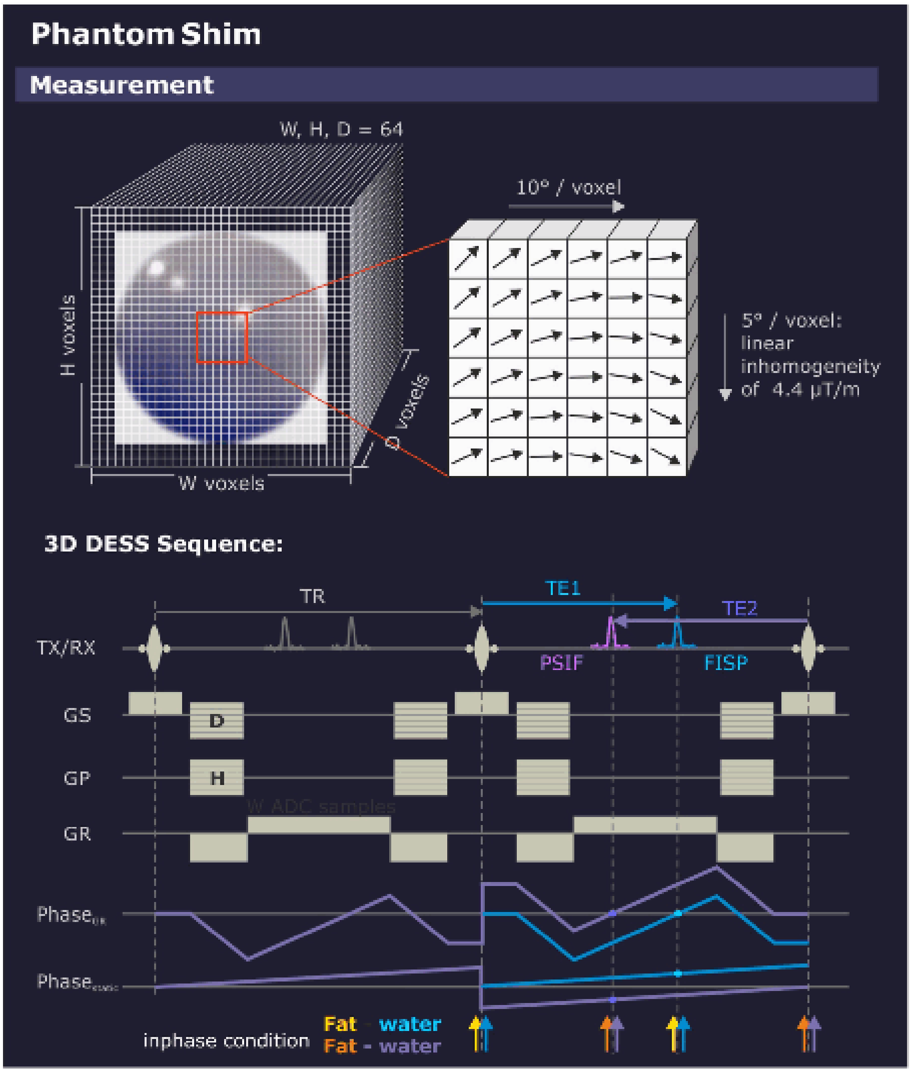
Modified/saved parameters:

- distance from light marker to the gradient isocenter.

5.11.1.9 Phantom Shim

Help ID: 2B324732-3689-4E3A-8FD7-E32FECD8F6CA
Service Level: 3

Fig. 557: Phantom Shim - Measurement



General

The **Phantom Shim** procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be exactly determined based on the frequency distribution in the volume elements making up the phantom image.

Afterward, new gradient offsets and shim coil currents (option) are calculated to improve the field homogeneity. The shim currents are improved iteratively.

The correction currents determined here also serve as default values for the 3D-Shim procedure used for patient imaging.

Measurement

Phantom Shim takes a 3D-measurement, a so-called "Field Map". The resulting voxels are cubes, i.e. the resolution is isotropic. The echo times are chosen for fat and water in phase.

In order to reduce hardware effects (e.g., gradient delays or multiple coil element phase errors), two echoes are measured in one 3D-Dess-sequence (Double Echo Steady State containing an FISP and a PSIF echo) and the phase difference of both echoes is analyzed. Then, the measurement is repeated a second time with opposed gradient polarities to eliminate eddy current effects.

The shim currents are improved iteratively.

Evaluation/Results

Fig. 558: Phantom Shim - Results

Results

Field Terms

Field Term	Value	Tolerance		
Bpp	**1.360**			
Brms	**0.093**			
A10	**-0.020**	-0.70	0.70	ppm
A11	**0.018**	-0.70	0.70	ppm
B11	**0.003**	-0.70	0.70	ppm
A20	**0.046**	-0.70	0.70	ppm
A21	**0.005**	-0.70	0.70	ppm
B21	**0.003**	-0.70	0.70	ppm
A22	**-0.011**	-0.70	0.70	ppm
B22	**0.010**	-0.70	0.70	ppm

Gradient Offset

Axis	Gradient Offset (new)	Gradient Offset (old)	Gradient Offset (change)
GradX	**-0.156** mT/m	-0.155 mT/m	-0.000 mT/m
GradY	**0.311** mT/m	0.311 mT/m	-0.000 mT/m
GradZ	**1.161** mT/m	1.161 mT/m	0.000 mT/m

Shim Currents

Shim Channel	Shim Current (new)	Shim Current (old)	Shim Current (change)
A20	**-2375** mA	-2372 mA	-3 mA
A21	**-161** mA	-160 mA	-1 mA
B21	**72** mA	72 mA	-0 mA
A22	**31** mA	29 mA	2 mA
B22	**41** mA	43 mA	-2 mA

The phase is calculated for each volume element. Based on the phase, the magnetic field can be expanded in terms of spherical harmonics. To reduce calculation times, voxels with little or no signal are discarded.

The individual terms are - up to a sensitivity factor - the parameters to compensate the field terms, including the system frequency, the gradient offsets, and the shim currents.

The impact of the changed currents is checked implicitly with the next iteration to ensure that the homogeneity has improved; a negative impact would suggest wrong cabling that could be checked in more detail with Expert Mode.

Results in Report:

- Field terms
- MR frequency
- Gradient offsets
- Shim currents (optional)

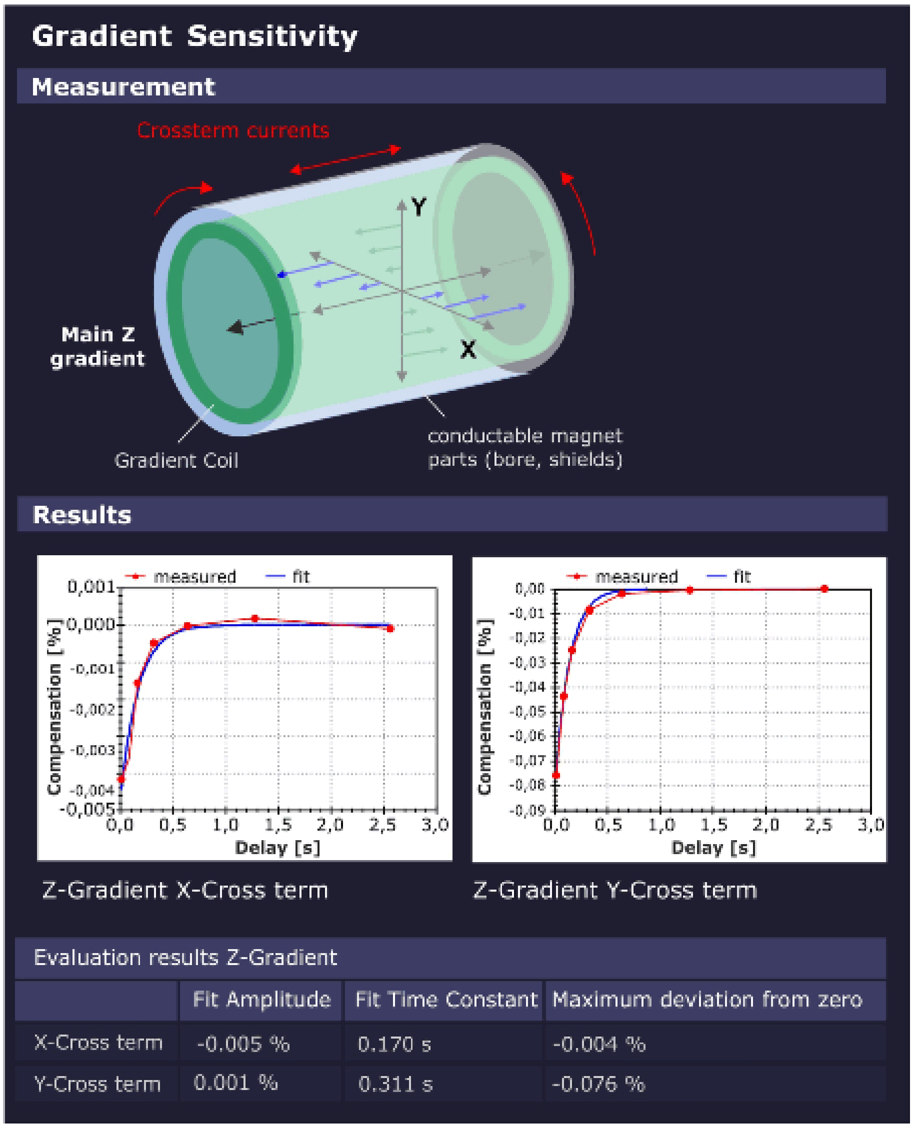
Modified/saved parameters:

- Gradient Offsets and Shim Currents

5.11.1.10
Cross Term Compensation

Help ID: 6E349297-0783-4926-8406-B493D5B788D7
Service Level: 3

Fig. 559: Cross Term Compensation



General

The operation of a single gradient, e.g., Z gradient, produces eddy currents mainly in the Z axis, but also small amounts of eddy currents which are coupled over the conductive magnet components onto the other 2 gradient axes (e.g. X and Y).

These small orthogonal "Cross Term" eddy currents decay exponentially in time, in general with only 1 time constant.

Measurement

The **Cross Term Compensation** procedure measures the decay of these Cross Term eddy-currents and calculates their amplitude and time constant. These data are used later during imaging sequences to apply appropriate, inverse correction gradients in the orthogonal gradient axes.

The result is a much cleaner main gradient with very small residual Cross Terms, improving the final image quality.

Evaluation/Results

Results in Report:

- Cross Term Amplitudes and Time Constants.
- Diagrams visualizing the corresponding Cross Term amplitudes.

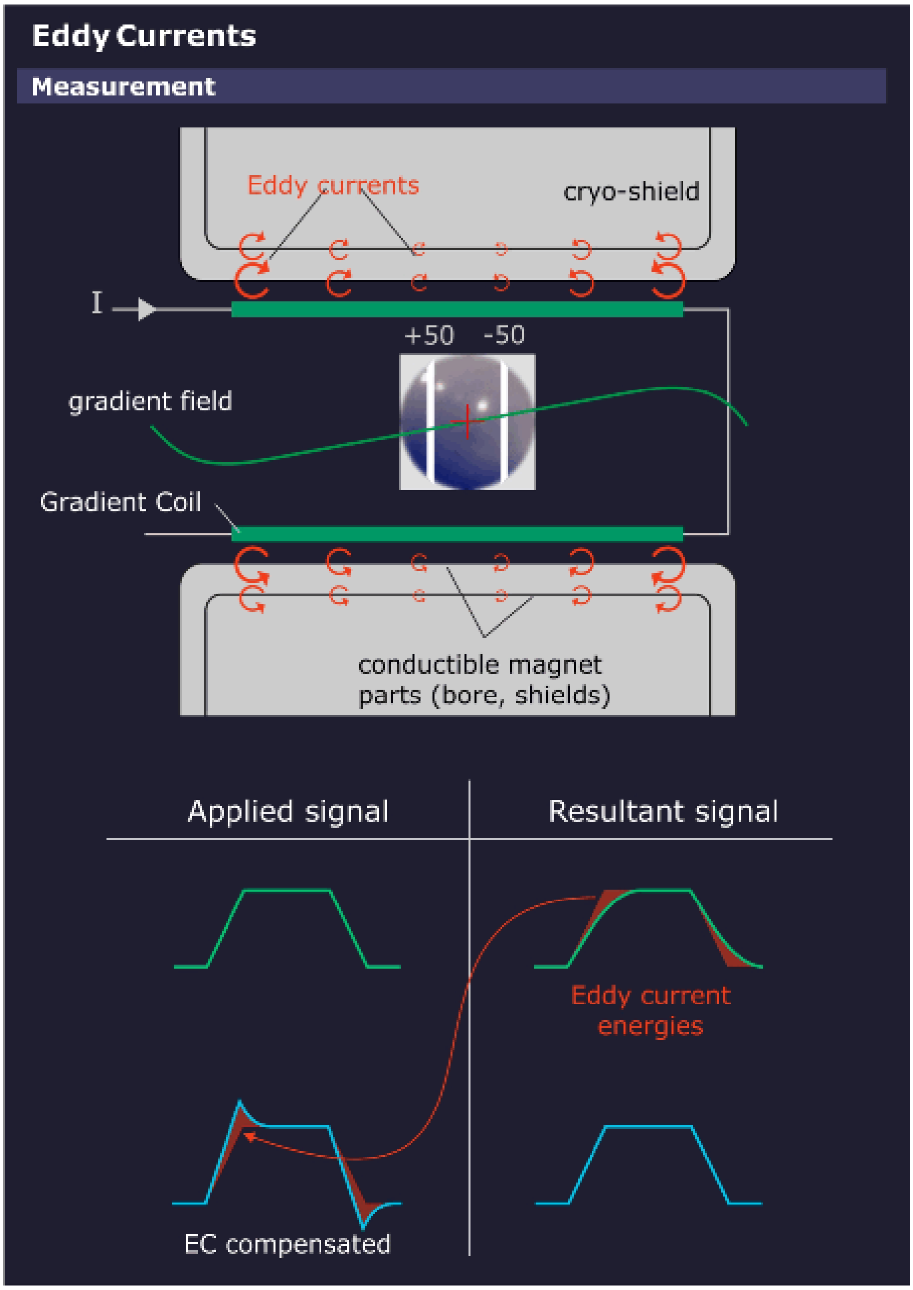
Modified/saved parameters:

- Cross Term compensation parameters

5.11.1.11 Eddy Current Compensation

Help ID: D1587435-76CB-4D71-A21B-091AB06092A9
Service Level: 3

Fig. 560: Eddy Current Compensation - Principle



General

Eddy Current Compensation (ECC) measures the eddy currents induced by all 3 gradients. The measurement requires the **small spherical phantom** rather than the normal body phantom.

The dynamic gradient fields produce eddy currents in all surrounding conductive structures (mainly the Body Coil, magnet bore and the cryogenic shields), resulting in magnetic fields which oppose and distort the applied gradient fields. Eddy currents in the warm components (Body Coil and magnet bore) have relatively short decay times, whereby eddy currents developed in the cold cryo-shields can have relatively long time constants. The eddy currents are analyzed for their time constants and amplitudes to achieve a good compensation. The compensation is made by adding the reciprocal of the measured eddy currents to the gradient pulse, thus neutralizing their effects. For accurate compensation, 5 time constants are used.

Spatial Dependency of Eddy Currents

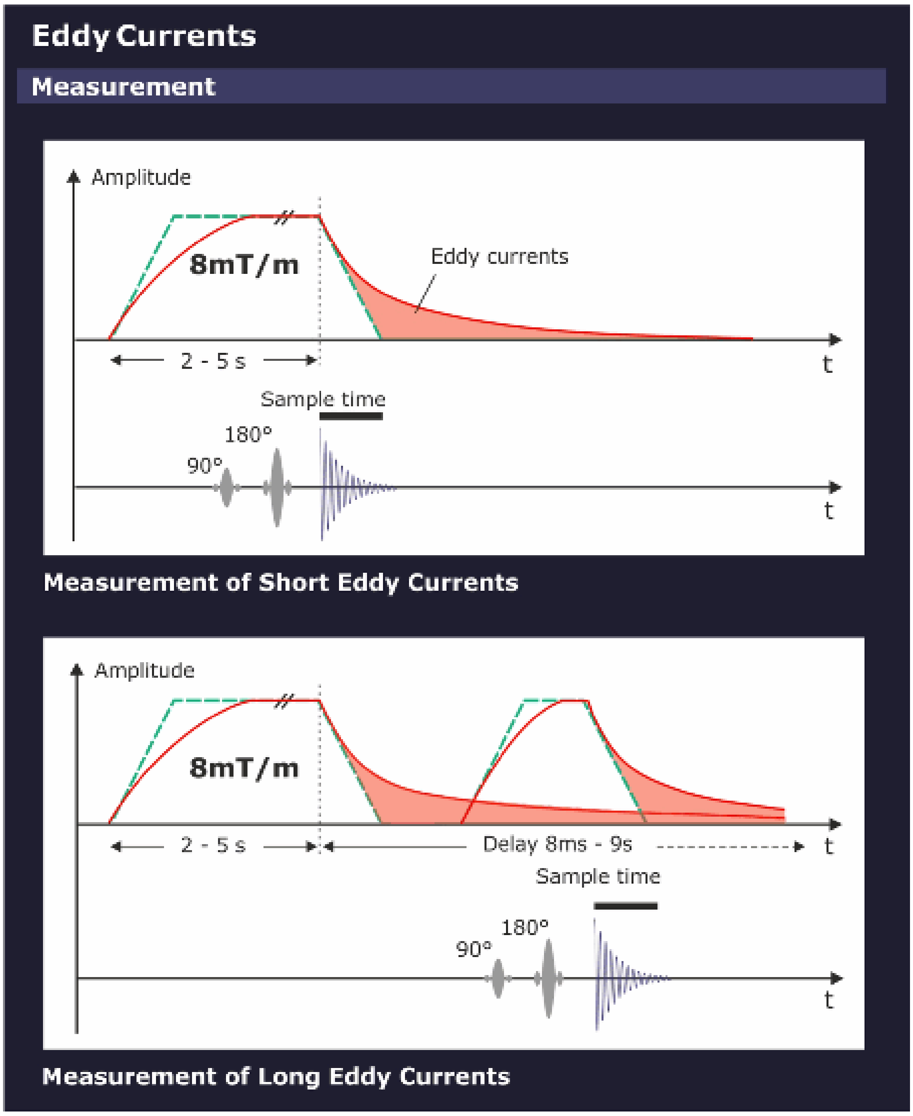
Experience has shown, that for smaller slice shifts it is sufficient to express the eddy fields into a 0^{th} and a 1^{st} order term.

- The **0^{th} order term** can be compensated by changing the synthesizer frequency during measurements.
- The **1^{st} order term** can be compensated by predistorting the gradient waveforms for the GPA.

Term	Description
0^{th}	This term arises from an **asymmetry** of the gradient coil with respect to the magnet bore and cryo shields. This term is space independent and present in the complete imaging volume and adds to the nominal B_0-Field. The amplitude (x) is given in the unit µTm/mT. The time constant of the most significant 0th term component is about 500ms, although there are shorter time constant components as well.
1^{st}	This is the most important term as it has the same symmetry as the gradient field itself. The amplitude is given in % of the applied gradient pulse.
2^{nd} or above	There are also high ordered eddy currents present. But they are usually small and negligible as long as the slice shifts and off-center zooms are not too large. No compensation for the high order eddy currents can be made.

Measurement

Fig. 561: Eddy Current Compensation - Measurement



A **Preparation Measurement** checks for correct positioning of the phantom in the magnet center and for sufficient field homogeneity. This is performed via a double echo sequence.

The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity.

The second echo uses a readout gradient and calculates the phantom position in the readout direction.

The sequence for the measurements of the eddy currents generates several spin echoes with varying delay times.

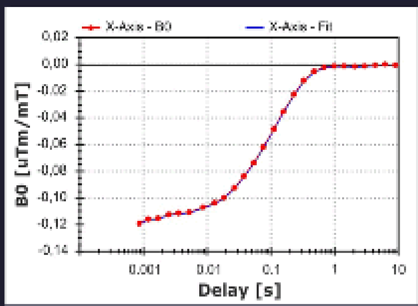
The measurement is made at the moderate slice shifts of 0 ± 50 mmto prevent contributions from higher order terms.

Evaluation/Results

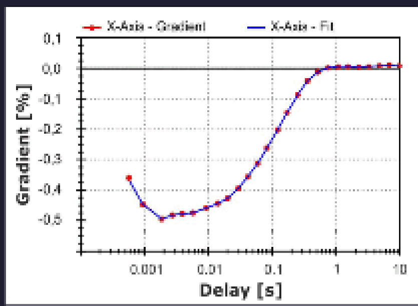
Fig. 562: Eddy Current Compensation - Results

Results

ECC X-Axis Fit Results, Step 1				
Range	B0		Gradient	
		Tolerance		Tolerance
less than 10ms	x **0.132 uTm/mT**	0.00...0.10 uTm/mT	x **0.541%**	0.00...0.10%
more than 10ms	x **0.116 uTm/mT**	0.00...0.10 uTm/mT	x **0.486%**	0.00...0.10%
2-100ms	x **0.059 uTm/mT**	0.00...0.03 uTm/mT	x **0.309%**	0.00...0.040%

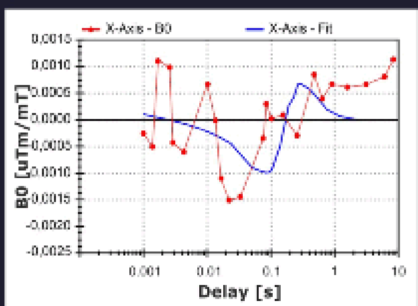


X-Axis B0 Step 1

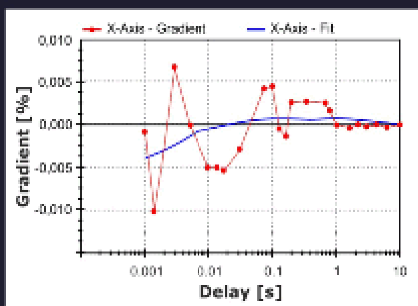


X-Axis Gradient Step 1

ECC X-Axis Fit Results, Step 2				
Range	B0		Gradient	
		Tolerance		Tolerance
less than 10ms	**0.001 uTm/mT**	0.00...0.10 uTm/mT	**0.010%**	0.00...0.10%
more than 10ms	**0.002 uTm/mT**	0.00...0.10 uTm/mT	**0.005%**	0.00...0.10%
2-100ms	**0.003 uTm/mT**	0.00...0.03 uTm/mT	**0.011%**	0.00...0.040%



X-Axis B0 Step 2



X-Axis Gradient Step 2

The 0^{th} and 1^{st} order compensations are calculated.

The decision how many iterations are necessary, is based on the expected improvement for the actual fit.

Results in Report:

- The tables shows for 3 typical time domains (<10ms, >10ms, 2 - 100ms):

  - Relative gradient strength to compensate the first order eddy currents

  - Relative change in B_0 for the 0^{th} order eddy currents.

- Diagrams show the measured data against the fit for both:
  - B_0 eddy currents
  - 1st order eddy currents

Modified/saved parameters:

- Eddy current compensation parameters

5.11.1.12 Gradient Delay

Help ID: 902BC568-F068-432E-9F4E-47CB1EAE3317
Service Level: 3

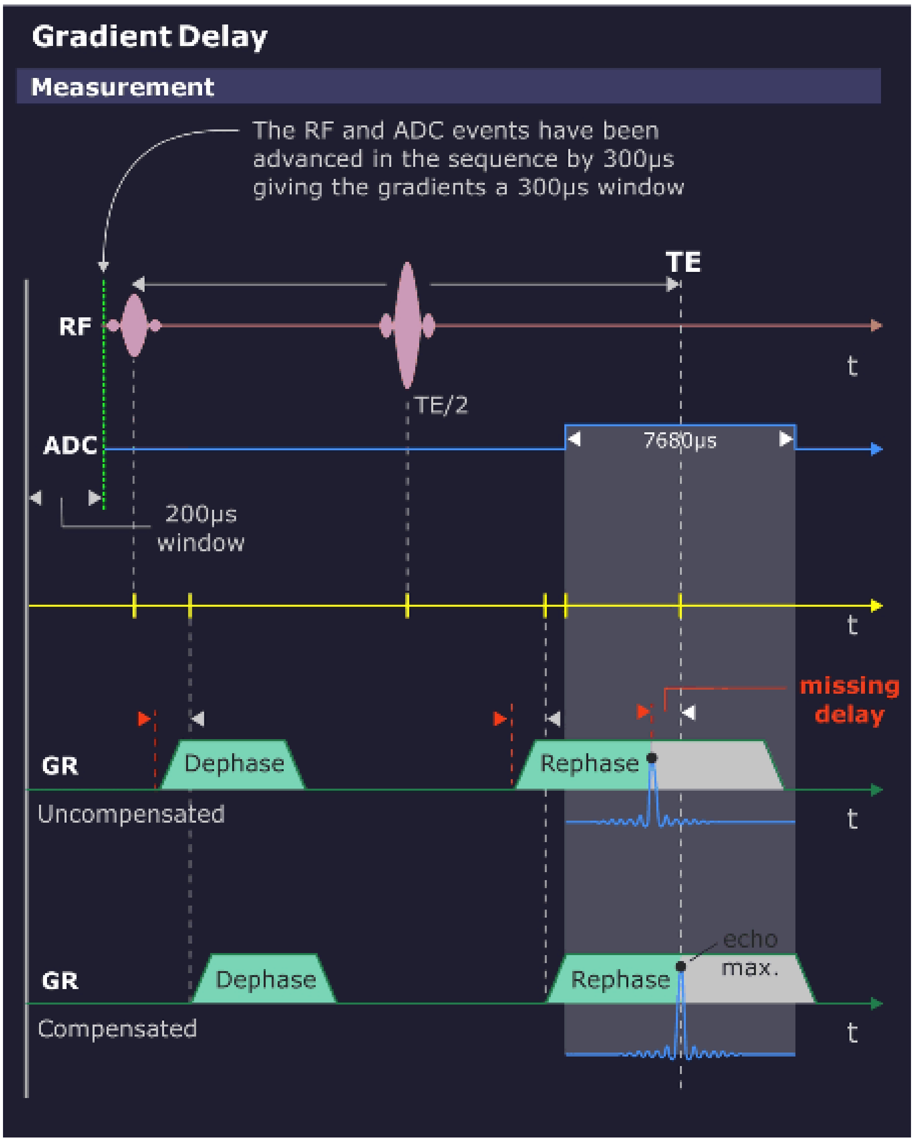
General

The gradient system has a small time delay between the time the gradient is switched on digitally by the measurement control and the time the actual gradient field is present. This delay is in the microseconds range, e.g. 30µs and mainly caused by the coil inductance.

The **Gradient Delay** procedure determines the individual delay time which can slightly differ for each gradient axis. A precise timing of gradient pulses together with the RF pulses is very important for optimum image quality.

Measurement

Fig. 563: Gradient Delay - Measurement



The measurement uses a Spin Echo sequence without phase encoding, hence only 1 line is measured. The measured gradient axis acts as the readout gradient and is switched on once between the 90°- and the 180°-pulse and then again during the readout time of the Spin Echo.

The first gradient pulse causes a dephase of the magnetization vector in the direction of the gradient axis. The amount of dephase is proportional to the time-amplitude integral of the gradient pulse. Subsequently, a 180° refocusing pulse is applied normally resulting in a Spin Echo at TE/2 time later. However, due to the dephasing of the vector by the first gradient pulse, the echo only appears in the center of the ADC window (which is also centered exactly around the TE time point) when an equal rephasing gradient has been applied at the proper time so that the rephasing is finished at the center of the ADC cycle.

For a non-corrected Gradient Delay, the 2 gradient pulses are time-delayed. This fact does not influence the dephasing effect of the first gradient pulse, however, the delayed second gradient pulse causes a rephasing delay and thus a time-delayed echo signal. From the echo signal delay the required correction time can be determined.

Difference Deviation Check

Each gradient is measured at 3 different gradient strengths - 1.5, 4 and 8 mT/m - giving 3 Gradient Delays for each gradient. For an ideal gradient system, all delays are the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, i.e. high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, there is some kind of nonlinearity present in the gradient system.

Evaluation/Results

Fig. 564: Gradient Delay - Results

Results

Gradient Delay Values

	Old Delay		New Delay	
		Tolerance		Tolerance
X Gradient	**20 us**	0.0...170.0 us	**35.6 us**	0.0...170.0 us
Y Gradient	**20 us**	0.0...170.0 us	**33.8 us**	0.0...170.0 us
Z Gradient	**20 us**	0.0...170.0 us	**29.8 us**	0.0...170.0 us

Max Deviation between different Gradient Strengths

	Max Delay Deviation	
		Tolerance
X Gradient	**1.8 us**	≤5.0 us
Y Gradient	**2.0 us**	≤5.0 us
Z Gradient	**1.8 us**	≤5.0 us

Results in Report:

- current and calculated Gradient Delays
- maximum deviation values.

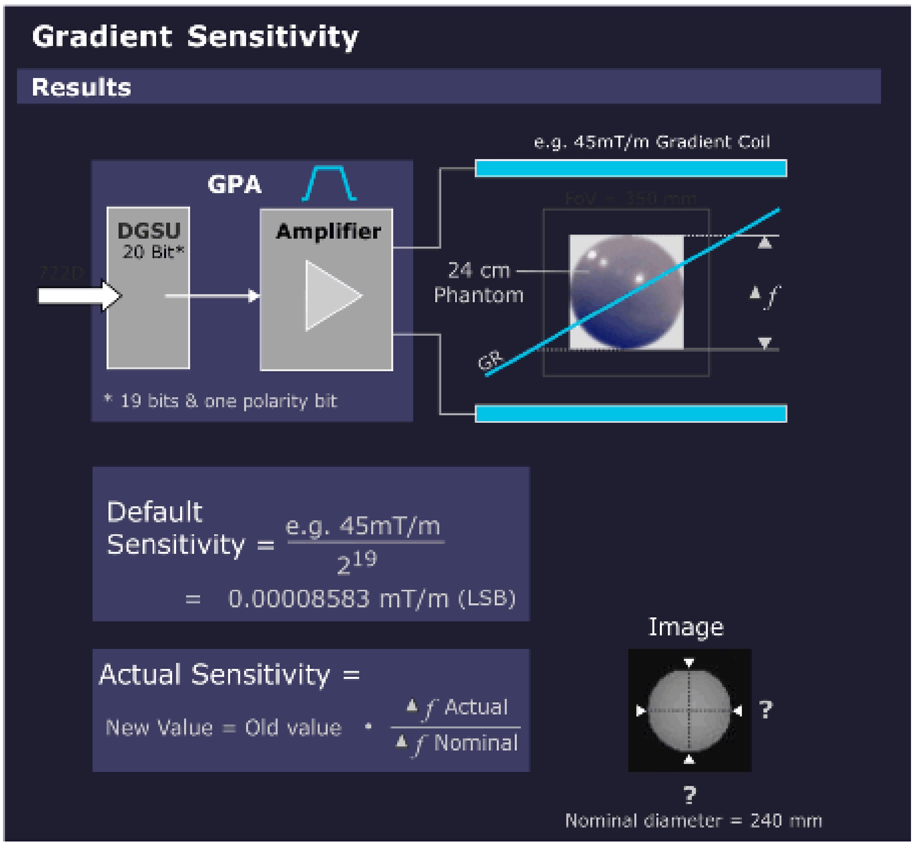
Modified/saved parameters:

- Gradient Delay values

5.11.1.13
Gradient Sensitivity

Help ID: 2E17EBC4-A846-41F7-B775-BA7D410952D4
Service Level: 3

Fig. 565: Gradient Sensitivity - Measurement



General

The **Gradient Sensitivity** procedure calibrates the gradient sensitivities so that the system uses the correct gradient strengths and measure the correct image size in all orientations..

The patient table distance (distance light marker to gradient coil center) is also calculated.

If the phantom is not centered in the magnet within a specified tolerance, the gradient sensitivities cannot be calculated. Repeat the measurement after correcting the table position.

Measurement

- A standard imaging sequence (FLASH with distortion correction) is repeated 3 times, 1 measurement for each orientation.
- The 3 images are analyzed to determine the exact location of the center of the phantom and the phantom diameters. By swapping the standard readout and phase encoding directions in1 orientation, the center and diameter of the phantom for each gradient direction can be determined, once with the gradient as a readout encoder and

again with the gradient as a phase encoder. The average of these 2 results determines the final diameter values.

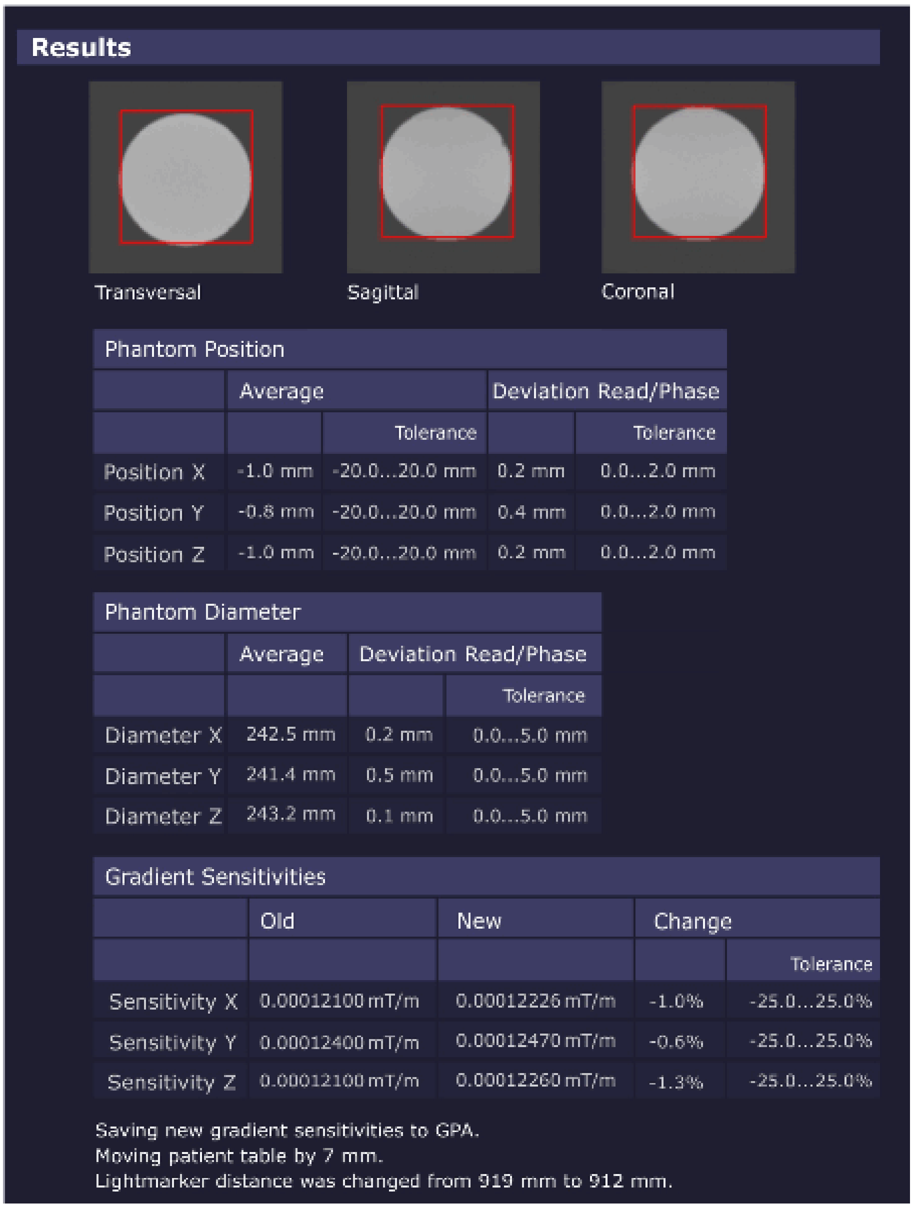
- The measured values are evaluated and compared with the well-known diameters of the phantom for calculating the gradient sensitivities.

The gradient sensitivity is defined as the gradient strength per ADC unit (e.g. 0.00009 mT/m per 1 ADC unit).

> The patient table position is corrected automatically. In case of problems (e.g. if the table moves in the wrong direction), check the Z gradient polarity.

Evaluation/Results

Fig. 566: Gradient Sensitivity - Results



Results

Transversal | Sagittal | Coronal

Phantom Position

	Average		Deviation Read/Phase	
		Tolerance		Tolerance
Position X	-1.0 mm	-20.0...20.0 mm	0.2 mm	0.0...2.0 mm
Position Y	-0.8 mm	-20.0...20.0 mm	0.4 mm	0.0...2.0 mm
Position Z	-1.0 mm	-20.0...20.0 mm	0.2 mm	0.0...2.0 mm

Phantom Diameter

	Average	Deviation Read/Phase	
			Tolerance
Diameter X	242.5 mm	0.2 mm	0.0...5.0 mm
Diameter Y	241.4 mm	0.5 mm	0.0...5.0 mm
Diameter Z	243.2 mm	0.1 mm	0.0...5.0 mm

Gradient Sensitivities

	Old	New	Change	
				Tolerance
Sensitivity X	0.00012100 mT/m	0.00012226 mT/m	-1.0%	-25.0...25.0%
Sensitivity Y	0.00012400 mT/m	0.00012470 mT/m	-0.6%	-25.0...25.0%
Sensitivity Z	0.00012100 mT/m	0.00012260 mT/m	-1.3%	-25.0...25.0%

Saving new gradient sensitivities to GPA.
Moving patient table by 7 mm.
Lightmarker distance was changed from 919 mm to 912 mm.

The images are analyzed to determine the exact location of the center of the phantom and the phantom diameters. The found diameters are checked against the known phantom diameters of 240 mm. The quotient between both gives the correction factor to the actual gradient sensitivity.

Results in Report:

- measured phantom diameter in all 3 dimensions
- measured phantom centers in all 3 dimensions
- gradient sensitivities
- phantom images in all 3 slice orientations

ResulModified/saved parameters:

- Gradient Sensitivity

5.11.1.14
BC Power Losses

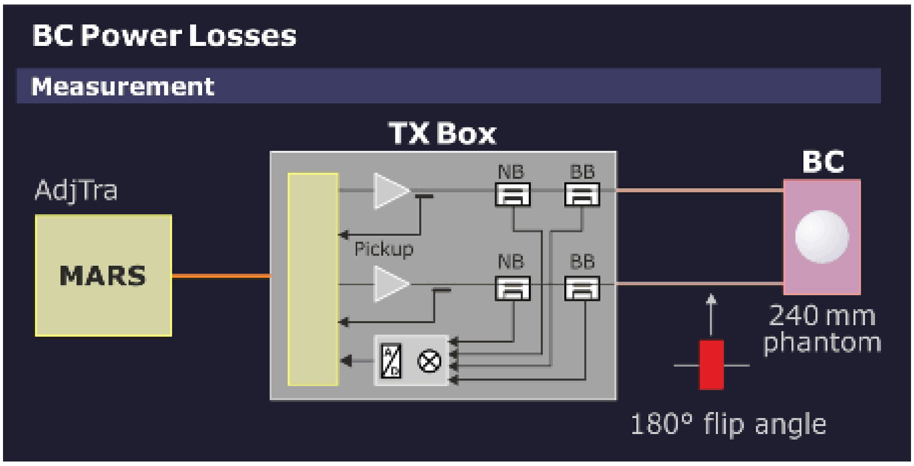
Help ID: 0FBDAC47-1539-4FBF-AD8E-9EE97D158876
Service Level: 3

General

The procedure **BC Power Losses** measures the amount of power loss in the Body Coil and connecting cables. This value is used by the RF Safety Watchdog (RFSWD) monitor to obtain a more accurate SAR measurement.

Measurement

Fig. 567: BC Power Losses - Measurement



- The Coil Power Loss measurement switches the RF Safety Watchdog (RFSWD) into the patient mode.
- At first, a complete inline adjustment with a non-volume selective frequency adjustment is performed to adjust transmitter and DICO. During the transmitter adjustment, the TX-Box internal TALES power reference value is measured and the values of the directional couplers (DICOs) for both transmitting systems.
- The subsequent measurement verifies the position and size of the phantom to ensure that the correct 240 mm spherical phantom is used.

- From the difference of the given theoretic Adj. Tra value for the used phantom and the actual (higher) measured power values, the Software can calculate the Coil Power Losses in the Body Coil and connecting cables.
- For pTX systems (i.e. systems with 2-ch BC; aka TX True Shape option), the Y-Matrix, which describes the characteristics of BC transmit path, is determined by an additional measurement. This provides an alternative means to determine the Coil Power Losses for arbitrary BC excitations.

Evaluation/Results

Fig. 568: BC Power Losses - Results

Results

Coil Power Loss for Polarisation Mode CP

Phantom position and diameter				
	X		Z	
		Tolerance		Tolerance
Center	**-0.82 mm**	-20.00...20.00 mm	**-0.24 mm**	-20.00...20.00 mm
Diameter	**240.79 mm**	220.00...260.00 mm	**240.46 mm**	220.00...260.00 mm

Rf Safety Common	
Name	Value
Sigma Phantom	**1.00**
Reference Amplitude (AdjTra)	**391.73 V**

Rf Safety (Type Sum)		
Name	Value	
		Tolerance
Coil Power Loss	**1,623.30 W**	1200,00...4,000.00 W
Previous Coil Power Loss	**1,000.00 W**	

Rf Safety (Type YMatrix)		
Name	Value	
		Tolerance
Coil Power Loss	**1,625.95 W**	
Coil Power Loss Consistency	**0.16 %**	-10.00...10.00 %

Coil Power Loss for Polarisation Mode EP

Phantom position and diameter				
	X		Z	
		Tolerance		Tolerance
Center	**-0.87 mm**	-20.00...20.00 mm	**-0.18 mm**	-20.00...20.00 mm
Diameter	**240.93 mm**	220.00...260.00 mm	**240.27 mm**	220.00...260.00 mm

Rf Safety Common	
Name	Value
Sigma Phantom	**1.00**
Reference Amplitude (AdjTra)	**4356.54 V**

Rf Safety (Type Sum)		
Name	Value	
		Tolerance
Coil Power Loss	**1,985.00 W**	1200,00...4,000.00 W
Previous Coil Power Loss	**1,000.00 W**	

.
.
.

Results in Report:

- tables showing the phantom position.
- tables showing measured and stored Coil Power Loss values
- tables showing the measured reference power from the TALES

Modified Parameters:

- Body Coil Power Losses in CP- and EP-mode

5.11.1.15
BC Image Brightness

Help ID: 51D260CB-C301-47E2-BBD2-F54667156BC5
Service Level: 3

General

This procedure measures the image brightness of the Body Coil using a standard image measurement. A scaling factor (FFT Scale Factor) is determined and set to ensure that the mean image brightness of the pixels within the phantom is close a specified value.

Measurement

A specified protocol is measured.

Evaluation/Results

The correction factor is calculated for the measured signal levels within the phantom to reach a specified brightness value.

Results in Report:

- Serial and material number of BC (from EEPROM on BC).
- New and old Tx Amplitude.
- New and old FFT Scale Factor.

Modified/saved parameters:

- The FFT scale factor for the coil used is changed in the respective coil file.

5.11.1.16
Dynamic Field Map

Help ID: BCF3C1B6-8197-407E-8991-23DE6FD54E73
Service Level: 3

General

Gradient pulses are always accompanied by unwanted time-dynamic field perturbations caused by e.g. eddy current or hysteresis effects. These field perturbations comprise multiple terms with different spatial dependencies.

The 0^{th} and 1^{st} order terms are analyzed and compensated by the ECC and CTC TuneUp procedures.

The purpose of the **Dynamic Field Map** procedure is to measure and analyze the residual higher order pertubations (>2nd order field terms).

Measurement

The dynamic field map service sequence is executed to measure the dynamic field perturbations using the large spherical phantom.

Evaluation/Results

Results in Report:

- peak-peak values in ppm
- Spherical harmonics up to the 5th order

Modified/saved parameters:

Data for image-based correction of dynamic field perturbations in diffusion imaging are saved.

5.11.1.17 Rear Lightmarker

Help ID: 3EA5E53A-1F90-4308-A6D1-B99EBC2C7D9B
Service Level: 3

General

A second light marker in back of the magnet is available as an option. The procedure" Rear Lightmarker" calibrates the distance between the rear light marker and the Gradient Coil magnetic center.

Measurement

The large phantom must be positioned with the help of the rear light marker and then moved to the gradient center. The measurement uses the TuneUP program "Gradient Sensitivity" (service sequence "gradsens") with three orthogonal slice orientations.

Evaluation/Results

An algorithm determines the Z-position of the phantom center in the image. Using this information, the distance from rear lightmarker to Gradient Coil center is updated and saved.

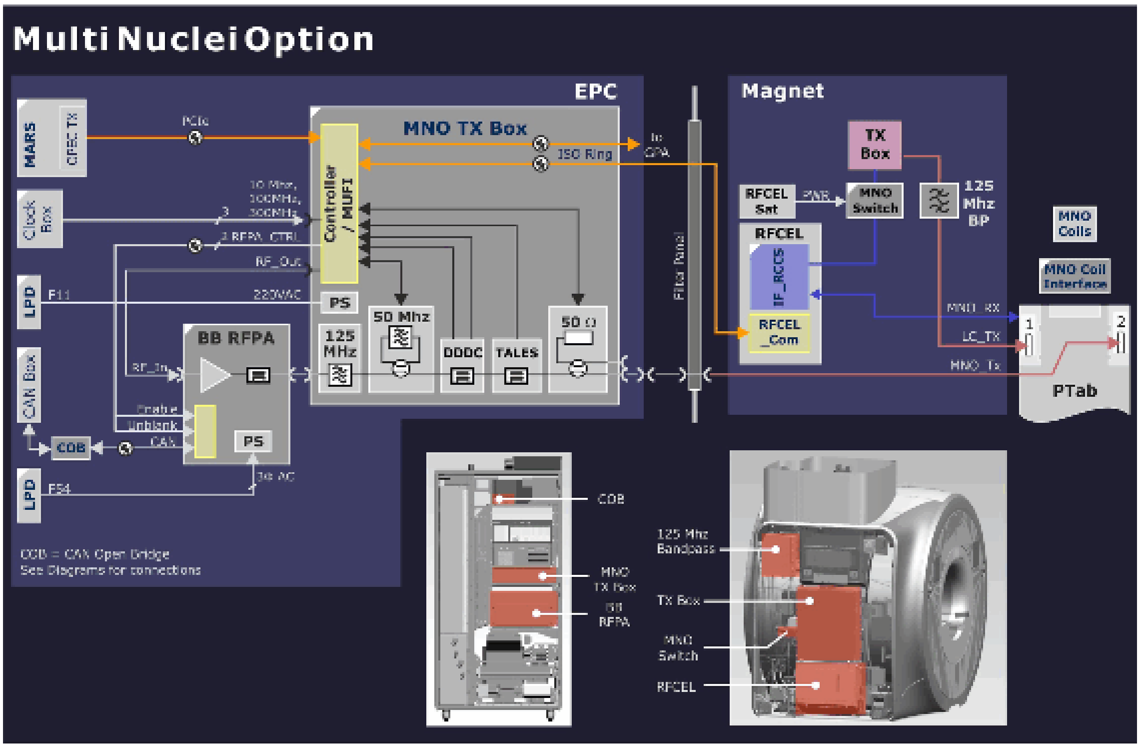
Modified parameters:

- distance from second light marker to gradient center.

5.11.1.18
RF Characteristics MNO

Help ID: 6E1EF408-9472-48A1-87C9-3F22471C6BE0
Service Level: 3

Fig. 569: Multi Nuclei Spectroscopy Option (MNO)



General

The TuneUp step **RF Characteristics MNO** is available only at systems where the Multi Nuclei Option (MNO) for spectroscopy is configured.

These systems have an additional MNO TX-Box and standalone 8 kW broadband RF amplifier (BB RFPA) which can generate hydrogen and several lower x-nucleus frequencies for the MR spectroscopy experiments. Due to the frequency dependency of these MNO components, each individual frequency is calibrated individually with a characteristic curve for each nucleus.

The measurement determines the characteristics (amplitude and phase) at the output of the broadband RFPA. For the measurement, the service plug must be connected to coil plug 2 but no power is transmitted to the plug because the internal 50 Ohm dummy load of the MNO TX-Box is automatically switched to the output of the BB RFPA. Then, the corresponding hydrogen and x-nucleus frequencies are generated at the MNO TX-Box, amplified by the broadband RFPA and fed to the dummy load. Forward and reflected voltages of the BB RFPA are measured at the directional coupler (DICO) and TALES of the MNO TX-Box.

From the measurements, the characteristic curves for each nucleus are calculated and stored in the Software. The data is used to pre-distort the input signal of the BB RFPA with an inverse function, thus a linear output power characteristic can be achieved for all nuclei.

Measurement

The service plug is connected to coil plug 2. The 50 Ohm dummy load of the MNO TX-Box is automatically connected to the BB RFPA output by the Software.

The following measurement steps are performed for every nucleus:

- Calibration voltage search x-nucleus
- RF max. power search x-nucleus
- RF Characteristic measurement x-nucleus

Evaluation/Results

Results in Report:

- Calibration voltage search result
- RF calibration (saved)
- RF maximum power search result
- Measured characteristics (amplitude and phase)
- Calculated characteristic (amplitude and phase)

Modified parameters:

- RF Characteristic data

5.11.2 TuneUp - Expert

Help ID: 244F1380-8367-4997-9BFB-10F63E39A0F3
Service Level: 7

Introduction

Some procedures have an **Expert Mode** platform. These platforms are intended for qualified persons only, because they may provide additional configurations and parameters which require a more detailed knowledge of the underlying physics. In the Expert Mode, an arbitrary coil can be selected for most of the available procedures.

In the Expert Mode, TuneUp is no longer automated. Only one selected procedure at a time is performed.

NOTICE

Expert Mode is used for troubleshooting only!

Any (mis-)adjustments achieved in Expert Mode are also used for standard measurements.

» **For imaging with patients, ensure that all TuneUP and QA steps have been successfully performed in Normal Mode!**

Expert Mode TuneUp procedures may have an impact to the **status** of other TuneUp and QA procedures (dependencies). That means, steps with status “**Done**” may fall back to “**To Do**”.

To reach the “**Done**” status of any TuneUP function, run the procedure successfully in Normal Mode under TuneUp / General.

5.11.2.2 Extended Frequency Adjustment

Help ID: 484CA0A2-E6AC-4DE6-B572-01D58965BD4B
Service Level: 7

Expert Mode

The **Extended Frequency Adjustment** is used to find the system frequency in case the current system frequency is far away from the actual MR resonance frequency.

A larger frequency drift may occur after ramping the magnet (e.g. shim workflow).

Extended Frequency Adjustment can be also used with the Array Shim Device.

Measurement

The procedure performs several non-volume selective frequency adjustments with a moving frequency interval. As a result, the **entire specified frequency range is covered** to find the resonance peak, even if it is far away from the last stored current frequency.

MR system	Lower frequency (Hz)	Upper frequency (Hz)
3 Tesla	123100000	123265000

Evaluation/Results

Results in Report:

- System Frequency

Modified/saved parameters:

- System Frequency

5.11.2.3
Feedback Loop Calibration

Help ID: 6F181F8E-2E47-4CA9-8629-CE1995AA85D7
Service Level: 7

Expert Mode

This Expert Mode step is only available for single TX channel systems.

General

The TX-Box has a built-in feedback loop to compensate for instabilities of the TX B1-field.

The procedure **Feedback Loop Calibration** is used to calibrate internal feedback loop parameters of the TX-Box.

The **Expert Mode** allows for calibrating/checking either feedback loop parameter independently of the actual system status.

5.11.2.4
BC Tuning Expert

Help ID: 1C2F7900-E019-4610-BF22-EE8E0E613846
Service Level: 7

Expert Mode

The **BC Tuning in Expert Mode** offers the possibility to continue the tuning adjustment even if the previously obtained results already fulfill the specification limits.

This method allows further optimization of the BC reflection and transmission values beyond the defined spec. values.

5.11.2.5
Gradient Regulator Expert

Help ID: 6AAB138D-BB8F-459C-84F4-81BC0DA740CD
Service Level: 7

Expert Mode

In the **Expert Mode**, the operator has more control over the adjustment and setting of the regulator values. The selection of a single gradient axis is possible. By default, the X gradient is selected.

Measurement

Three numerical values (P=proportional portion, I=integral portion, D=differential portion; range 1 to 128) display the regulator settings for the selected gradient. When the gradient is selected for the first time, the existing regulator settings are displayed for this gradient. The values are configurable for the operator.

When selecting **[Go]**, these values are loaded into the gradient system as start parameters for the following iterative measurement. At the end, the new, calculated regulator settings are displayed. These settings are loaded when selecting **[Go]** the next time.

5.11.2.6 Phantom Shim Expert

Help ID: 22052F03-68D8-409A-84AD-FDDCA75B3054
Service Level: 7

Expert Mode

In **Expert Mode,** the operator can perform a single phantom shim iteration.

5.11.2.7 Eddy Current Compensation Expert

Help ID: D9B69F8A-96D7-4019-BAC0-3B48D2D0F5AE
Service Level: 7

Expert Mode

In the **Expert Mode** a coil selection is available.

In addition, the Eddy Current Compensation measurement can be performed for one or more gradient axes (by operator selection).

5.11.2.8 RF Characteristics

Help ID: A3FB5B7B-4700-435B-97C2-91525CAA916C
Service Level: 7

Expert Mode

RF Characteristics In Expert Mode operates similar as in the Normal Mode but with a few additional options.

General

The following additional options are available in Expert Mode:

- Refusal to store RF Characteristics after the measurement.
- Modification of maximum RF power allowed.
- Selection of TX-channel, RF path and nucleus (according to available licenses).
- Option to switch-off the frequency adjustment.
- Option to disable the switching to dummy load.

5.12 Quality Assurance

5.12.1 General Quality Assurance

Help ID: 7DFD7C17-DADA-4F81-B6E1-1AF37D0BBC53
Service Level: 3

Introduction

General Quality Assurance (QA) is an automated procedure for:

- Verification of TuneUP adjustments.
- Checking the overall system quality.
- Image quality checks.

Different from TuneUp, **system parameters are not changed in QA**.

QA Workflow

- General QA is separated into modules, each routine runs as a stand-alone process.
- After start, each procedure reads the specification and configuration parameters automatically.
- QA procedures determine if their measured results are in or out of specifications.
- QA procedures write their individual results to the Service Report.
- The status is returned to the main QA platform and the corresponding status field in the Service UI is updated.

After pressing the **[Go]** button in the QA UI, the **procedures start and perform typical tasks**:

- Each procedure registers the Service Patient (if not already registered).
- If a clinical patient is registered, a pop-up informs the operator to de-register the clinical patient before proceeding.
- The correct phantom setup is displayed by a pop-up and confirmed by the operator (if applicable). If QA is performed remotely, automatic table move is disabled and the operator receives a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Table movement:
  - If **QA is performed locally** at the system, the **table moves automatically** to center the required phantom in the measurement field (change between large/small spherical phantom).

- If **QA is performed remotely, automatic table move is disabled**. The operator gets a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.

- Configuration and specification values are read.
- Measurements are performed.

QA Function Status

Tab. 27 Possible QA Function Status

Status	Description
To Do	The QA procedure has not yet been performed yet and is pending. OR: The QA procedure needs to be performed again because another procedure (from TuneUp) has requested its execution.
Ok	The QA procedure has been carried out successfully.
Not Ok	The related parameters/results were found to be out of specifications during the last operation.
Error	An error occurred during the last operation.
Cancelled	Procedure canceled by operator.

Evaluation/Results

After end of the QA measurement, evaluations are performed automatically:

- The procedures determine if the measured results are in or out of specifications.
- The results are written to the Service Report.
- The results status is returned to the main QA platform and the corresponding status field is updated.

TuneUp/QA Dependencies

If a TuneUp procedure is successfully completed and has changed its status, the corresponding verification procedure(s) in the Quality Assurance workflow menu is/are changed to status **To Do**.

TuneUp procedures may also affect multiple QA procedures, e.g. after TuneUp Phantom Shim, the QA procedures Phantom Shim Check, Gradient Sensitivity Check, BC Image Brightness and Long Term Stability Check have to be performed.

Standby

In QA, a **Standby** function selection is offered. If selected, the system is switched to standby mode after completion of the QA workflow. All units except for the host computer are switched-off.

After using the Standby functionality, it is not possible to switch-on the system again remotely.

5.12.1.2 RF Verify MNO

Help ID: 4210C5CB-BBD0-40ED-9200-0E96102D8567
Service Level: 3

General

The procedure RF Verify MNO performs a verification of the Multinuclei RF Characteristics determined during Tune Up.

Individual measurements for all supported nuclei are performed because of the frequency dependency of the broadband RFPA's power output. Forward and reflected DICO power values are measured and checked against their corresponding error tolerances (amplitude- and phase deviation from last saved RF Characteristics).

Measurement

The RF signal is generated at the MNO-TX Box Controller, amplified by the BB RFPA and routed to the dummy load in the MNO TX-Box.

The DICO factor is calculated and checked. Then, the deviation of the amplitude and phase from the linear characteristic is evaluated.

Evaluation/Results

- DICO calibration voltage
- DICO calibration statistics
- plots show the measured characteristics (amplitude and phase)
- maximum power deviation and peak-to-peak phase deviation

5.12.1.3 Feedback Loop Calibration Check

Help ID: 4172C660-AF1B-46AF-A127-DF43D6E9194F
Service Level: 3

General

The **Feedback Loop Calibration Check** procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the "tuncal" service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box.

Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a “Not Ok”.

Evaluation/Results

Results in Report:

Tables:

- Delay evaluation
- Dispersion of delays from single BC elements

Plots:

- BC phase plot

5.12.1.4 RX Gain Calibration Check

Help ID: 4A3CD412-40FD-43CD-9DD2-BEF0BA7D4C40
Service Level: 3

General

The **RX Gain Calibration Check** verifies the amplification correction factors for all individual receiver channels which were determined during TuneUP / RX Gain Calibration:

- (Residual) RX channel correction factors
- High- / Low-gain factors

Measurement

The procedure uses loop measurements (without MR measurements). A fixed calibration signal is fed through all RX channels which are switched to high- and low-gain. For the measurement, the stored TuneUp parameters for the given RX channel- and high- / low-gain-correction are applied. The generated raw data is analyzed and all gain correction factors and the quotient of the High- / Low-gain modes are determined.

Evaluation/Results

Results in Report:

- Average gain correction factor over all receive channels.
- Gain correction factors for each individual receive channel.
- High- / Low-gain signals for each individual receive channel. Comparison of amplitude- and phase-difference to the stored TuneUp correction values.
- Noise evaluation.

5.12.1.5
BC Tuning Check

Help ID: 12EC839E-96E8-4A6D-8498-C6930BCCCDEF
Service Level: 3

General

The **BC Tuning Check** procedure is used to measure and evaluate the central resonance frequencies of the Body Coil resonator systems as well as the coupling between them.

A high efficiency of RF power transmission to the Body Coil requires sufficiently low reflection factors at the TX-Box output.

This condition is usually met, if the reflection factors of the Body Coil resonator systems BC0 and BC1 are high (without load) and equal. However, with load, these reflection factors shall be low.

The other requirement is, coupling between both resonators shall be minimum (i.e. high decoupling values).

Measurement

BC Reflection

First, a **BC Reflection** measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals, covering the full transmit operating frequency range. The result of each measurement is the reflection or r-factor = Ur/Uf which is plotted for both Body Coil systems BC0 and BC1 along the frequency axis.

The evaluation determines:

- each coil's center (tuned) frequency is within a specified range, i.e. is not too far from the center of the operating range.
- that the center frequencies of both coil systems BC0 and BC1 are tuned within specified limits of each other, i.e. the delta frequency is low.
- that the reflection factors are within expected limits. These values provide useful information about the quality (Q-factor) of the BC resonant circuits. They also depend on the coil load and cannot be adjusted.

  The Software detects the resonant frequency always at the point where the reflection of the BC has its minimum.

Transmission

In a second step, a **Transmission** measurement is performed with transmitting a signal into the BC0-system and measuring the coupled signal in the BC1-system and vice versa. The transmission factor describes the amount of RF-coupling between the two Body Coil systems, mainly due to coil construction but also affected by other factors e.g., mechanic position adjustment of the Body Coil within the Gradient Coil's Faraday Shield.

The evaluation checks the transmission value, expressed in decibels (dB), to be within expected limits. **A high negative dB-value indicates minimum coupling** i.e., a good decoupling between BC0 and BC1 system.

Evaluation/Results

Results in Report:

- The reflection related values for both (BC0 and BC1) systems, like the resonance frequency, the reflection factor and the difference between both systems.
- The transmission related value like the frequency and the decoupling between both systems.
- The new and old phase difference.
- Diagrams showing the measured values for the transmission and the reflection behavior as a function of frequency.

5.12.1.6 Cable Length Check

Help ID: B19E08C7-B349-43A5-99BF-E46BAB8192A4
Service Level: 3

General

The QA step **Cable Length Check** verifies the BC transmit cable phase shifts which have been determined in TuneUp / Cable Length.

The BC transmit cable length may slightly vary due to production tolerances. However, accurate information about the lengths of both BC transmit cables is required for an optimal function of the extended TX B1-field regulation loop in the measurement framework.

Measurement

A measurement with the service sequence rf_loop is performed. An RF pulse is fed to both Body Coil channels with the Body Coil switched to 'detune' mode. The forward and reflected signals are taken from the internal directional couplers (DICO) of the TX-Box and analyzed by the monitoring Software.

From these results, the so-called S-matrix elements which describe the characteristics of the BC transmit path are determined. These values allow, together with the known Body Coil RF-properties, to calculate the cable lengths for both Body Coil channels in terms of the induced signal phase shifts.

Evaluation/Results

In the Service report, the actual, measured phase shifts for both Body Coil channels are displayed, together with the old values from TuneUp. The latter ones are read from the BC coil files. Difference values are calculated and checked against the specified tolerance range.

5.12.1.7 Gradient Regulator Check

Help ID: 99738FE4-5E37-4EFF-A222-CE0B23F5D7EE
Service Level: 3

General

The **Gradient Regulator Check** procedure only measures the gradient waveform with the present PID regulator settings and verifies that these settings are correct. The check analyzes several aspects of the actual current output by comparing it to the expected ideal pulse output. In addition, the procedure will measures the linearity of the gradients by stepping through the full range of the gradient strength from negative to positive amplitudes.

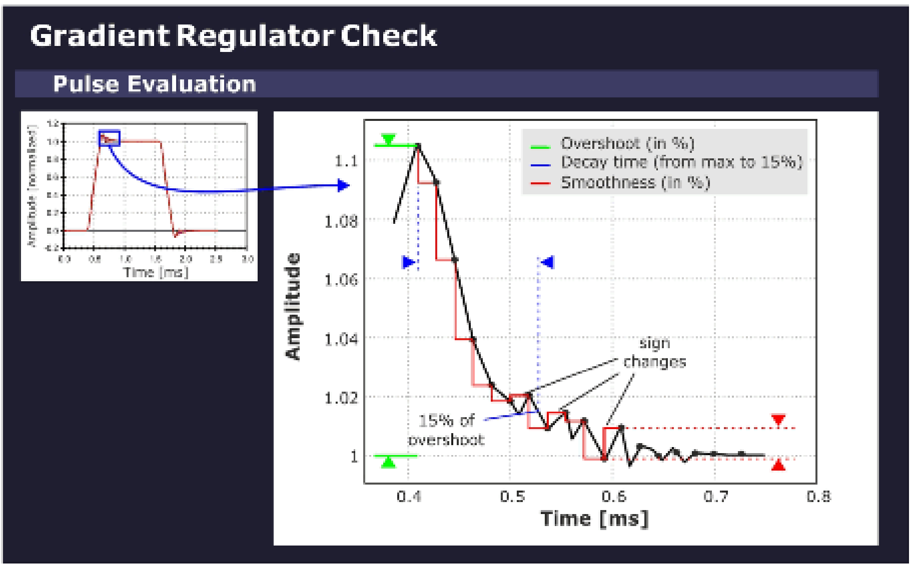
Measurement

The sequence uses a gradient table to generate the gradient pulses. The selected gradient (depending on the selected orientation) steps from the maximum negative to the maximum positive gradient amplitude pulse in linear portions. Each gradient in the table is switched on using a ramp-up and ramp-down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.

Evaluation/Results

Based on the maximum gradient amplitude pulse, the program evaluates the characteristic overshoot that occurs just after the pulse ramp-up. The amplitude of this **overshoot** is determined as a percentage relative to the nominal gradient amplitude. The program also calculates the **decay time** it takes for this exponential-like overshoot to return to a specified value. In addition, a **smoothness calculation** is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.

Fig. 570: Gradient Regulator Overshoot



Results in Report:

- PID gradient regulator parameters.

- Optimization parameters overshoot, decay time and smoothness.
- Gradient current pulse diagram (pulse form).
- Enlarged diagram of overshoot portion and current ripple (top portion).

5.12.1.8 Phantom Shim Check

Help ID: D847AEFF-8EE6-4E1B-83A4-EE3620B311E9
Service Level: 3

General

The **Phantom Shim Check** procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be accurately determined based on the frequency distribution in the volume elements making up the phantom image.

Two measurements are performed with the parameters from the corresponding TuneUp procedure to determine the remaining field terms and the overall homogeneity within the phantom volume. No adjustment of the shim terms is made in this case.

For systems with the advanced shim option, the procedure is performed with shim currents applied.

Measurement

The actual shim status of the system is measured with a 3D-Dess-sequence (Double Echo Steady State with a FISP and PSIF echo). The resulting voxels are cubes which are analyzed

by the Software.

Evaluation/Results

The field homogeneity is evaluated based on the frequency of the voxels. For the check of the B_0-field, it is sufficient to compare the field differences between neighboring voxels. The field difference is obtained by comparing the phases of the magnetization vectors.

The complete data set is evaluated by a differential shim equation. This equation minimizes the difference between the field generated by the3 Gradient Coils (and the optional 5 shim coils) and the measured magnetic field inhomogeneities.

Results in Report:

- Field terms
- MR Frequency
- Gradient Offsets
- Shim Currents (optional)

5.12.1.9
BC Power Losses Check

Help ID: D35D6115-3348-47DF-8FBE-8CBE8F1C7E7C
Service Level: 3

General

A check is performed to verify that the TuneUp of BC Power Losses has been performed successfully. This is done by measuring the Coil Power Loss value again and checking against the Coil Power Loss value in the coil file. Since the default of this value is very low, it would be out of specification if no TuneUp was executed.

In addition, this QA step verifies that the SAR (Specific Absorption Rate) check is activated. The forward and reflected voltages from both TX Box channels are also tested during the measurement.

Measurement

The TuneUp “Coil Power Loss” measurement is performed.

Evaluation/Results

The Coil Power Loss is measured in CP and EP mode.

- Check if the Coil Power Loss value in the coil file is within specifications and comparable to the measured value.
- Read and check the value for the SAR.
- Read and check the forward and reflected voltages from TX-Box and check them for consistency with broadband power meter values.

Results in Report:

- Phantom position and diameter.
- Measured and stored Coil Power Loss values.
- Measured reference amplitude (AdjTra) from the TALES (TX-Box power meter).
- The measured SAR and voltage values.

5.12.1.10
Cross Term Compensation Check

Help ID: 94F6F45A-CD81-4440-9FDE-F6D1AC3824A4
Service Level: 3

General

The **Cross-Term Compensation Check (CTC)** checks the quality of the cross-term compensation previously adjusted in TuneUp.

In QA mode, the parameters of the inverse filter function which compensates the cross-term effect are checked.

Measurement

A CTC measurement is performed.

Evaluation/Results

Results in Report:

- Cross Term Amplitudes and Time Constants.
- Diagrams visualizing the corresponding cross term amplitudes.

5.12.1.11 Eddy Current Compensation Check

Help ID: ACA062B6-B909-438A-A732-F3BF88BA7502
Service Level: 3

General

The **Eddy Current Compensation Check** (ECC) checks the quality of the eddy current compensation adjustment previously performed in TuneUp.

Measurement

The procedure performs a check of the ECC parameters. The existing ECC settings are used for the measurement.

A preparation measurement is performed to check for the correct positioning of the phantom in the magnet center and for sufficient field homogeneity. The used sequence is a double echo. The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity. The second echo uses a readout gradient and calculates the phantom position in the readout direction.

Evaluation/Results

Results in Report:

- B_0 eddy current values for different time ranges.
- 1^{st} order (gradient) eddy current values for different time ranges.

5.12.1.12 Gradient Delay Check

Help ID: 946ABE8C-323C-48C6-B295-5C45C3DD15AC
Service Level: 3

General

The **Gradient Delay Check** procedure determines the delay time between gradient switched on digitally and the time the actual gradient field builds up. Gradient Delay can slightly differ for each gradient axis.

In QA mode, the Gradient Delay adjustment values, previously established in TuneUp, are is verified.

Measurement

Each gradient is measured at 14 different gradient strengths from -19.5 mT/m to +19.5 mT/m. For an ideal gradient system, all delays are the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, that is, high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, there is some kind of nonlinearity present in the gradient system.

Evaluation/Results

Results in Report:

- Current- and calculated- Gradient Delay values in microseconds.
- Maximum Gradient Delay Deviation values.

5.12.1.13 Gradient Sensitivity Check

Help ID: 16C2854F-2DEB-4B6F-BEDA-E6418994A88B
Service Level: 3

General

The **Gradient Sensitivity Check** procedure checks the Gradient Sensitivity values previously adjusted in TuneUp to ensure that the system uses correct the gradient strengths and measures the correct phantom image size in all orientations.

For the measurement, the distortion correction filter is enabled to compensate gradient field non-linearities.

Measurement

- A standard imaging sequence is repeated three times: one measurement for each orientation.
- The three images are then analyzed to determine the exact location of the center of the phantom and the phantom diameters. The standard readout and phase encoding direction for each orientation is swapped to determine the center and diameter of the phantom. The average of these two results determines the final diameter value.
- The measured values are evaluated and compared with the well-known diameters of the phantom for verification of the correctly adjusted Gradient Sensitivities.

Evaluation/Results

Results in Report:

- Measured phantom diameter in all three dimensions.
- Measured phantom centers in all three dimensions.

- Gradient Sensitivities.
- Phantom images in all three slice orientations.

5.12.1.14 BC Image Brightness Check

Help ID: 4B8C4548-E064-495C-B6B2-AC746D517DD4
Service Level: 3

General

This procedure is used to verify the image intensity (brightness) for the Body Coil which was previously adjusted in the TuneUp step BC Image Brightness..

Measurement

The measurement runs the service sequence “Body_Tra_br” and generates images with the Body Coil.

Evaluation/Results

Both Body Coil channels BC0 and BC1 are evaluated independently from each other. Also the sum signal of BC0_BC1 is checked.

From each image the following values are determined:

- Brightness
- FFT Scale Factor
- TX Amplitude
- RX Gain Factor

The brightness value is the signal intensity. The FFT scale factor is a correction factor to adjust the brightness to the middle of the specified value range (min. and max.). The transverse image is used for determining the FFT scale factor.

The transmitter amplitude for the sequence (180 degree flip angle) is also evaluated. This value is coil load specific, so it depends on the phantom.

5.12.1.15 Image Orientation Check

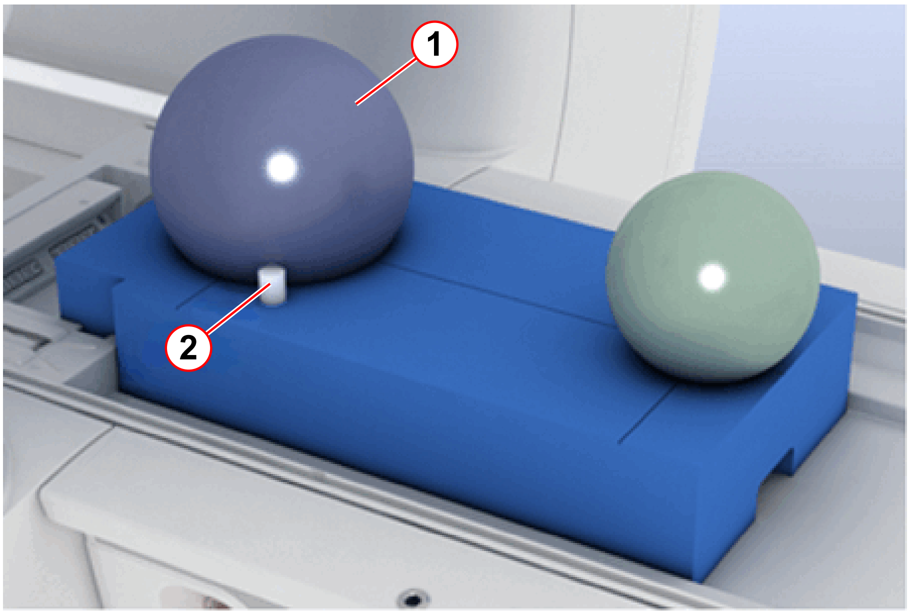
Help ID: F09D8D74-F01E-4A0A-B3FD-A116D3E77C87
Service Level: 3

General

The **Image Orientation Check** verifies the correct position of the phantom setup. It uses an additional, asymmetrically positioned small bottle phantom to determine the correct gradient orientation and polarity.

If any of the gradient power amplifier or -coil connections were swapped or polarities of any gradient axis were wrong, the images would be displayed with incorrect orientation.

Fig. 571: TuneUP_QA Phantom Setup

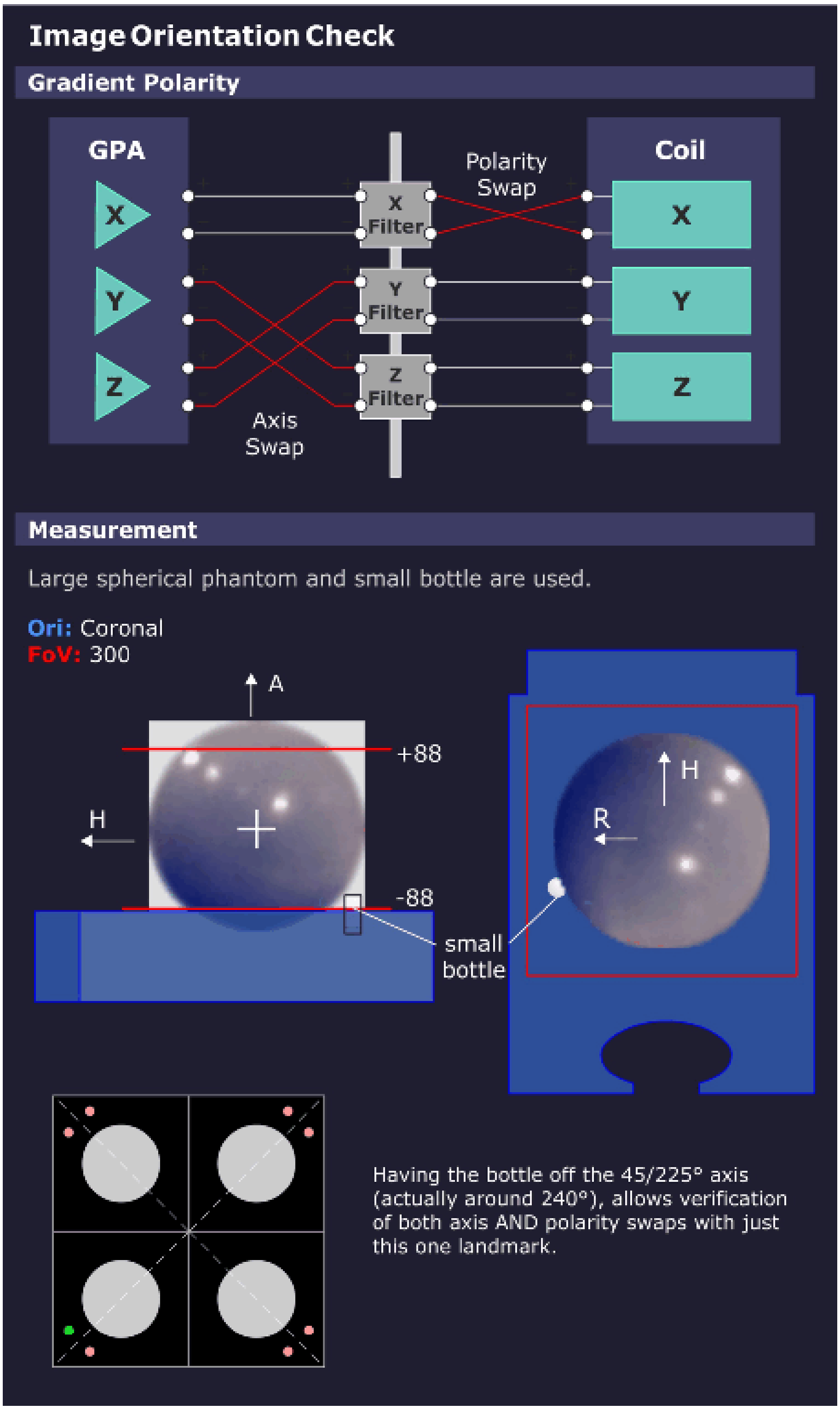


(1) Spherical phantom

(2) Small bottle phantom for Image Orientation Check

Measurement

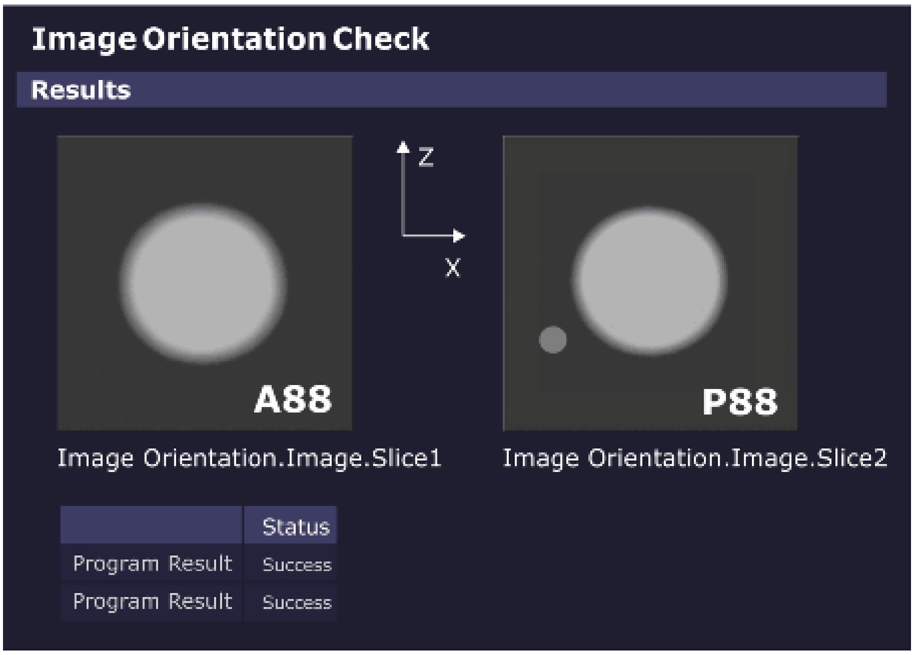
Fig. 572: Image Orientation Check - Measurement



Images are generated from the phantom setup with the additional, asymmetrically positioned small bottle.

Evaluation/Results

Fig. 573: Image Orientation Check - Results



The measured images are automatically evaluated.

If the phantom and bottle are not in the expected position, the procedure returns with the status "Not Ok".

Results in Report:

- Images showing the phantom geometry (phantom and bottle).

5.12.1.16 Gradient Risetime Check

Help ID: B668AA17-8C5F-4E6C-8908-BB2F522B4F00
Service Level: 3

General

The **Gradient Rise Time Check** measures the minimal gradient Rise Time or the so-called "Slew Rate".

Measurement

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, five echoes develop because of the bipolar triangle gradient pulses. These echoes should appear, according to the pre-dephasing, exactly at the pulse spikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate could not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Sequence of procedures:

- Verification check of the used phantom.
- Gradient Risetime measurement for all three gradient axes.

Evaluation/Results

Results in Report:

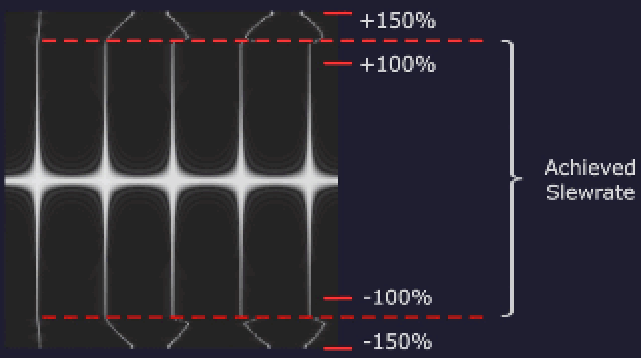
- Graphics of echo maximum versus gradient slew rate, echo 1-5
- Statistics of measured values against specifications
- X/Y/Z-gradient Rise Time results
- X/Y/Z-gradient Rise Time multiple echo output (raw data image, echo 1-5)

Fig. 574: Gradient Risetime Check - Results

Gradient Risetime Check

Results

Gradient Rise Time Results		
Axis	Slew Rate	
		Tolerance
X	112.46 %	≥100 %
Y	111.43 %	≥100 %
Z	112.48 %	≥100 %



X-Gradient Multiple Echo Output

Slewrates and Echopositions						
Echo	Slew Rate Positive Polarity		Slew Rate Neg. Polarity		Diff. of Echo Positions	
		Tolerance		Tolerance		Tolerance
1	**126.68 %**	≥100.00 %	**126.68 %**	≥100.00 %	-0.039	-1.000 ... 1.000
2	**114.67 %**	≥100.00 %	**114.69 %**	≥100.00 %	0.044	-1.000 ... 1.000
3	**113.58 %**	≥100.00 %	**113.58 %**	≥100.00 %	-0.076	-1.000 ... 1.000
4	**112.48 %**	≥100.00 %	**113.58 %**	≥100.00 %	0.041	-1.000 ... 1.000
5	**112.48 %**	≥100.00 %	**112.48 %**	≥100.00 %	-0.039	-1.000 ... 1.000

X-Gradient strength slew rate achieved is 112.483% of specified value (mean value of the lowest two values of all echoes)

Troubleshooting

In case of problems (result **“Not Ok”**) check the following items:

- Check raw data for spikes. No spikes must be visible.
- Stability/value of mains voltage
- Line resistance
- GPA line transformer and correct line voltage adaptation
- GPA rectifier board
- GPA current sensors (especially if value “Difference of Echo Positions” is out of spec.)
- GPA Power Stages

- Results of QA Regulator Check

5.12.1.17 Spike Check

Help ID: B4E9366D-2E5A-4CE9-969E-3EBDE191949C
Service Level: 3

The **Spike Check** is a quick test to check the MR imaging system for spikes and RF interference.

A “spike stress” measurement is run and its raw data evaluated by a program for spikes. The corresponding image data is evaluated for RF interference.

General

Electrostatic discharges during the measurement may generate some spikes within the raw data. Spikes are individual raw data points or small groups of raw data points with amplitudes above the typical noise level of the MR imaging system. **Spikes** can degrade the MR imaging quality. In this case, the signal-to-noise ratio can be reduced or the image can show a periodic intensity modulation or structure.

One typical source of spikes is the mechanical and electrostatic stress of the MR system caused by the operation of the gradient system.

A simple way to check the MR system for spikes is to run a noise measurement without RF pulses. During the acquisition window massive mechanical and electrostatic stress is generated with rapid switching of gradient pulses with maximum strength.

In case no spikes and no RF interference are present, pure statistical noise shall be received.

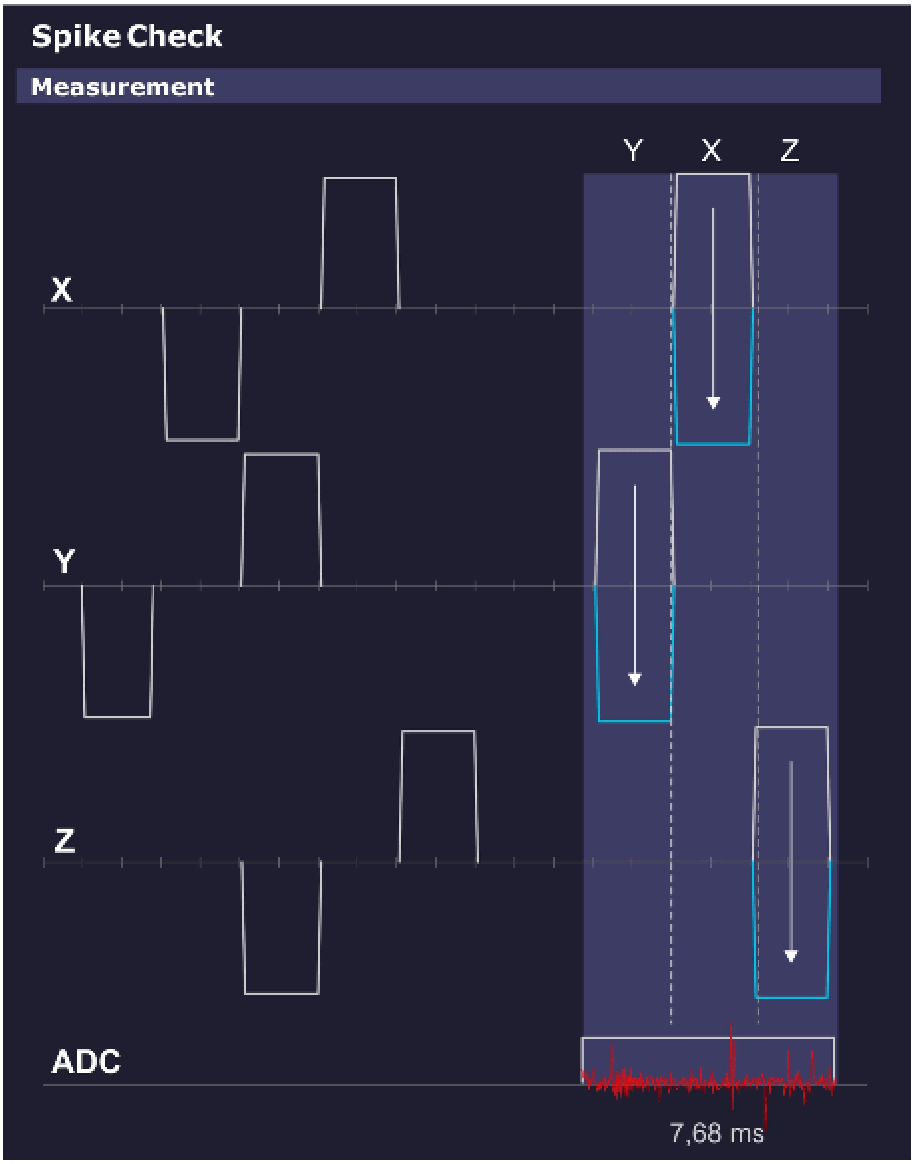
Measurement

This sequence does not apply RF pulses, so no MR signal is generated.

The spike sequence starts with an initial series of strong gradient pulses of both positive and negative polarities on all three gradients intended to cause a buildup of static electrical energy caused by any moving components (cables, covers, etc.).

The ADC is then enabled and again a pattern of gradient pulses with amplitudes ranging from max. positive to max. negative are applied. Any static electrical discharges caused by e.g. moving or loose parts are picked up by the receiving coil (usually the Body Coil), digitized by the ADCs and stored in the raw data.

Fig. 575: Spike Check - Measurement



Evaluation/Results

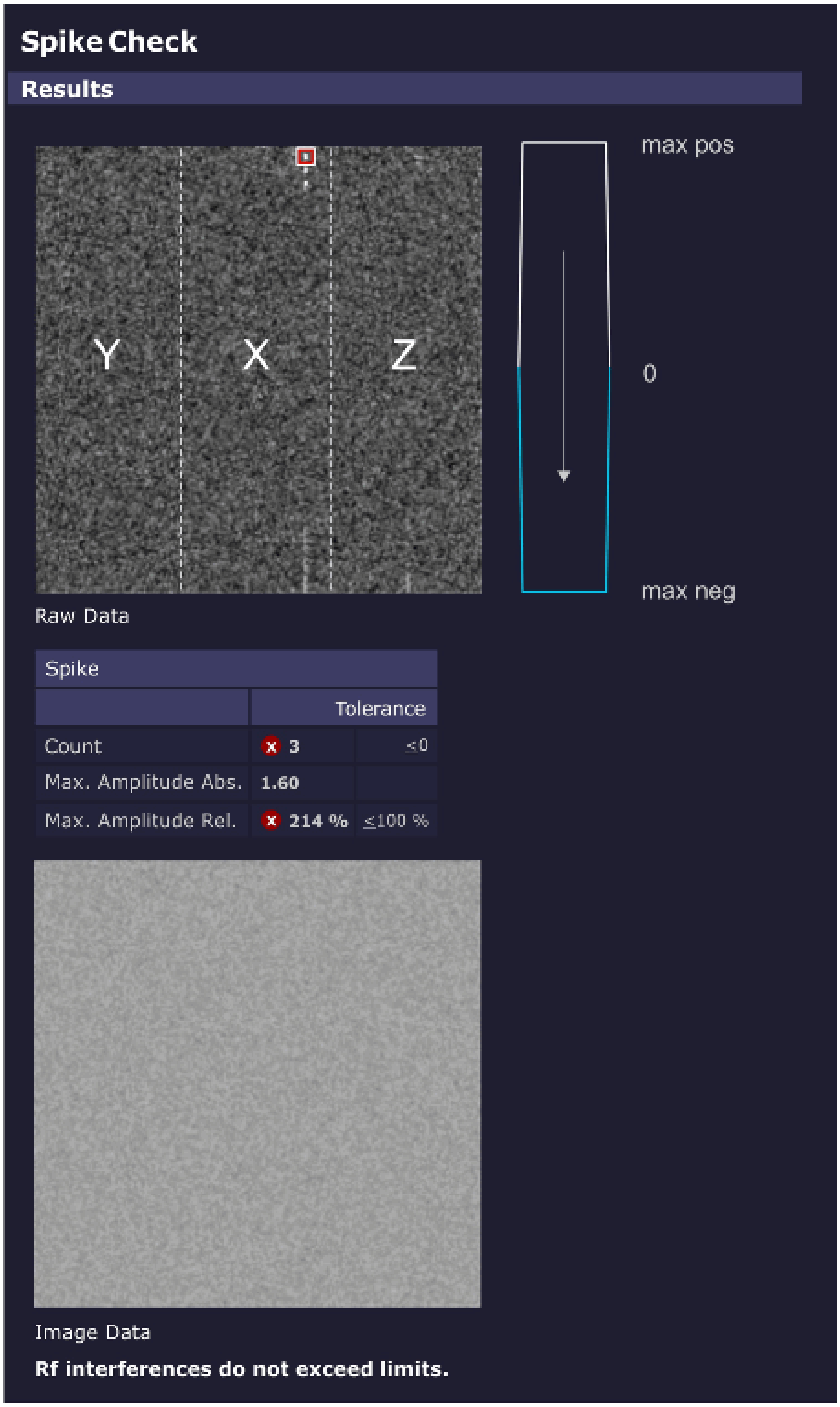
Search for **spikes in raw data** and **RF interference in the image data**.

The evaluation looks for intensities in the raw data set exceeding a threshold (usually noise) and counts both the number and intensities of the spikes and displays them numerically and graphically. The electrical discharges are visible as "spikes" in the raw data.

The position of the spikes in the raw data set relates to the gradient axis that was switched on when the spike occurred. This position may indicate the cause of the spikes.

In case RF interference was present during the acquisition, the disturbances can be seen in the displayed image data.

Fig. 576: Spike Check - Results



Results in Report:

- Raw data image and resulting RF interference image.
- If spikes are found, the following values are displayed:
  - Total number of spikes.

- Magnitude of the highest spike.

5.12.1.18
RF Noise Check

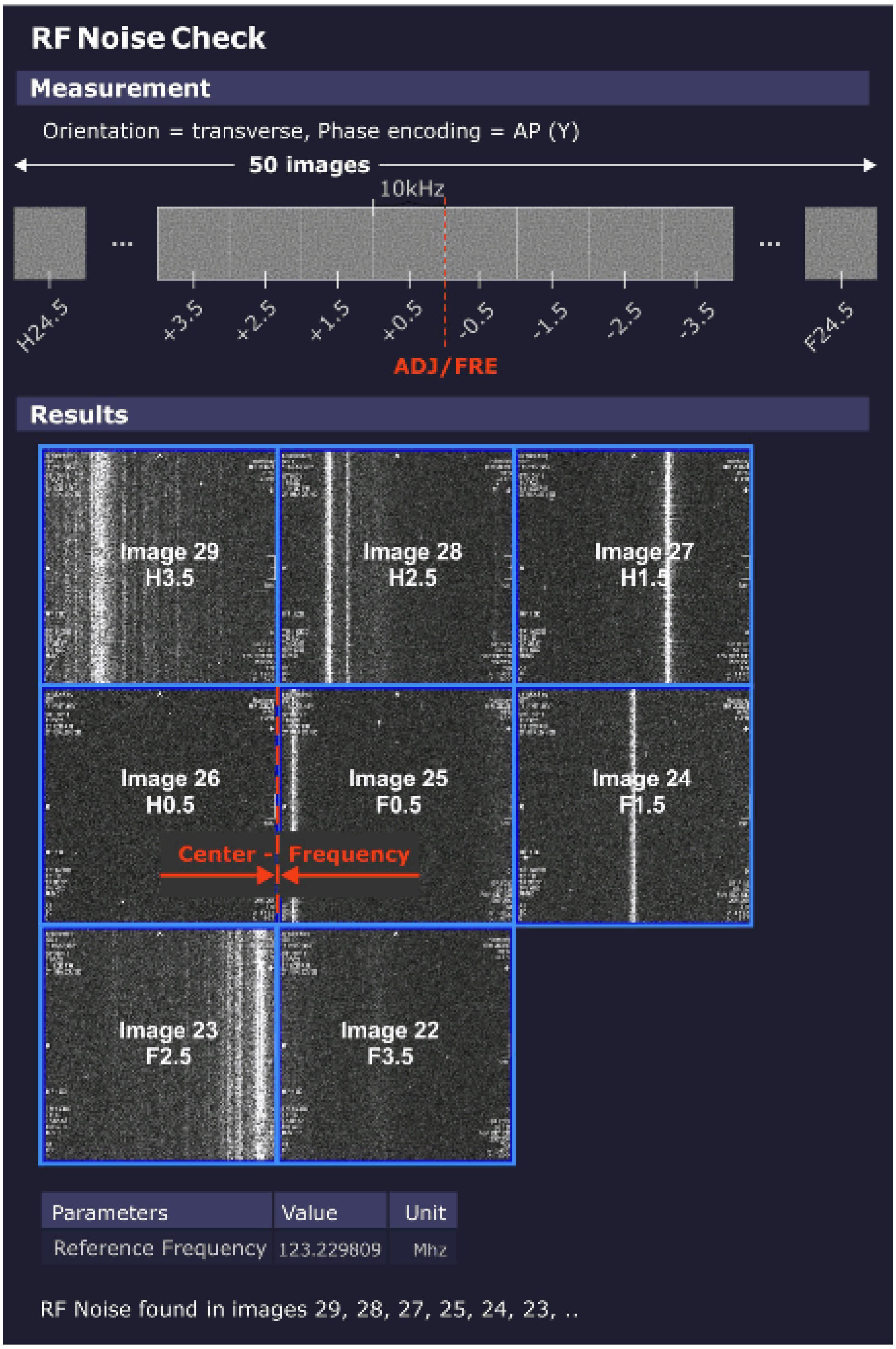
Help ID: AA585552-5E09-43B7-8E85-B9A8394037E2
Service Level: 3

General

The **RF Noise Check** is used to detect RF interferences which can disturb the MR-image.

Measurement

Fig. 577: RF Noise Check



The noise measurement is performed with the service sequence rf_noise. The sequence is without RF or gradient pulses, just an open receiver.

The sequence produces 50 images (i.e., slices). The difference between the images is the variation of the center frequency. Each image covers 10 kHz (256 times 39 Hz) so that the complete receiver bandwidth of +/-250 kHz is covered.

The border between the center slices 25, 26 is the current center MR system frequency.

Evaluation/Results

The evaluation algorithm looks for RF interference in the acquired image data i.e., dots and streak artifacts in phase-encoding direction and noise level.

Results in Report:

- RF interference according to the analyzed frequency bands.

5.12.1.19 PTX Image Check

Help ID: 82D6FBB2-442B-4B71-A95B-FD165371082B
Service Level: 3

General

A parallel transmit (pTX) array consists of 2 or more distinct transmitter channels where amplitude and phases can be modulated independently from each other.

Such a system can generate dedicated local magnetization patterns, e.g. exciting one spatial region of a patient or phantom while leaving other regions untouched.

This mechanism, however, is rather sensible to system instabilities or malfunctions of components in the imaging chain.

In order to detect such deviations, the service procedure runs the dynamic pTX sequence to produce a well-defined local magnetization pattern which is analyzed for image quality.

Measurement

For the measurement, the small spherical 170 mm water-filled phantom is used.

a homogeneous image is produced which allows to verify the correct positioning of the phantom.

The pTX sequence “dyn_pulse” is executed and generates a well-defined target magnetization pattern.

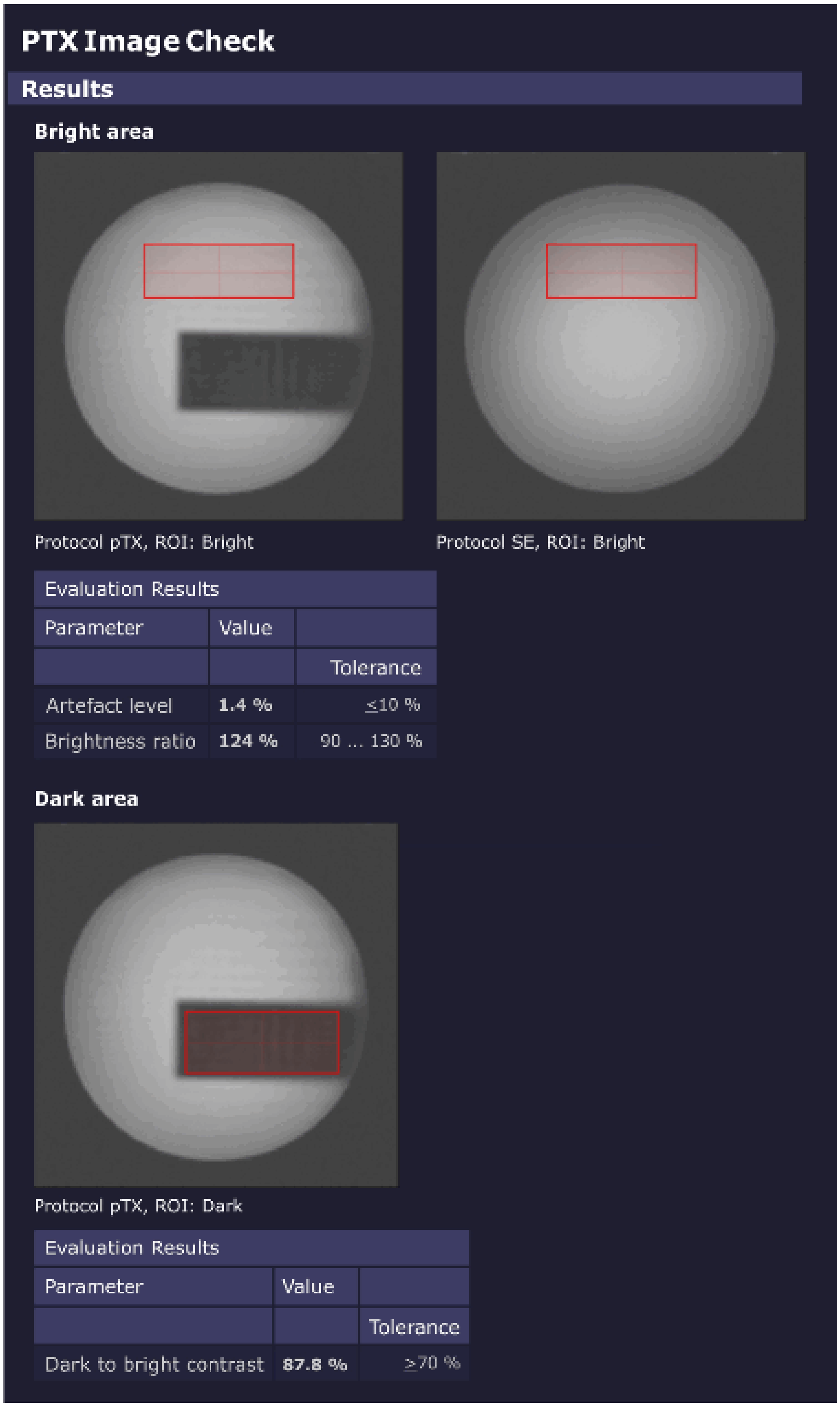
First, a homogeneous image is produced which allows to verify the correct phantom positioning.

Second, a homogeneously excited plane with an embedded dark (non-excited) rectangle is measured.

Subsequently, the Software analyzes the resulting images to detect potential system malfunctions.

Evaluation/ Results

Fig. 578: PTX Image Check - Results



First, a ROI is placed entirely in the **bright region** of the excitation pattern. This ROI is analyzed for relative image brightness (which should be similar to the brightness in the homogeneous image) and artifact level.

A second ROI is placed entirely inside the **dark region** (rectangle). The relative brightness of this second ROI, compared with the brightness of the first ROI, defines the image contrast.

5.12.1.20 Stability Check

Help ID: 796762F3-ECF9-458A-83A1-CAFDEC3BA27D
Service Level: 3

General

The **Stability Check** evaluates the stability of the MR signal under various sequence conditions.

A spin echo and a gradient echo are measured 256 times without phase encoding using fixed sequence parameters.

As a result, all raw data lines look alike (if the system is stable and not taking in account stochastic variations).

The stability of the echo signal across the measured lines is evaluated:

- Variations of the amplitudes
- Variations of phases
- Positions of the echoes
- Frequency perturbations

Measurement

In Standard Mode, a **spin echo** and a **gradient echo** are measured 256 times without phase encoding using fixed sequence parameters:

- TR = 301 ms
- TE = 20 ms

All **three orientations** are measured, so each gradient is performed as readout gradient once:

In addition, both **0 and 50 mm slice shift** positions are measured.

Evaluation/Results

The **spin echo sequence** uses a default setting of TE = 20 ms. Due to chosen TR = 301 ms and the inversion effect of the 180° pulse, it is sensitive to instabilities in the range of a few Hz up to 100 Hz which may be caused by e.g.:

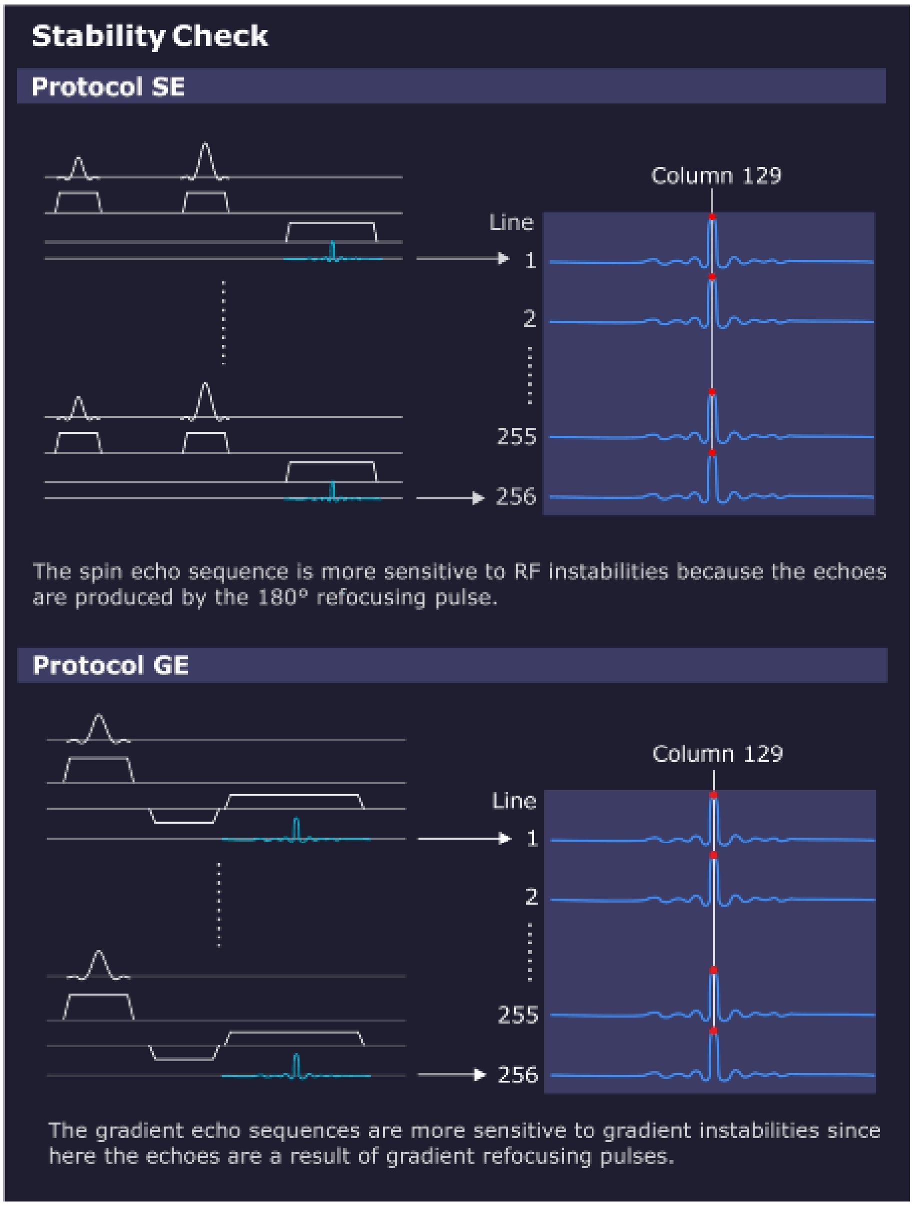
- 50/60 Hz AC power line hum;
- or 16.7 Hz AC trains.

The **gradient echo / flash sequence** also uses a default setting of TE = 20 ms and TR = 301 ms. Due to the chosen parameters and the missing 180° pulse, it is insensitive to 50 Hz line hum. The gradient echo is sensitive from DC up to a few Hz to slow external field interferences caused by e.g.:

- DC-driven local train systems like tram or subway;

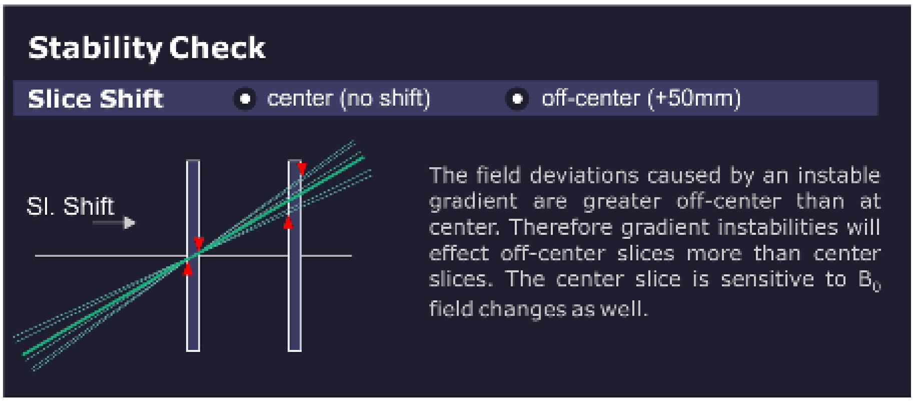
- Magnetic field instabilities;
- or mechanical vibrations.

Fig. 579: Stability Check Evaluation



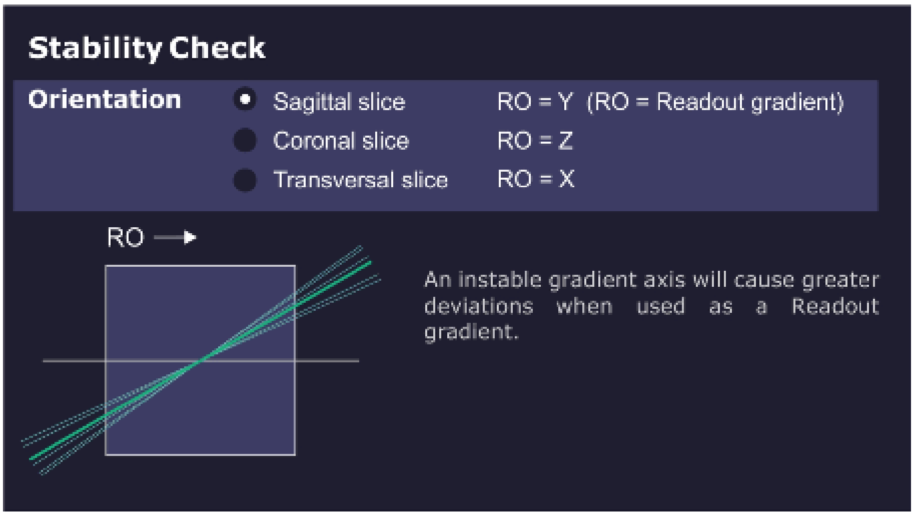
The measurement result of the **center slice** is sensitive to instabilities of the B_0 field. The **50 mm offset slice** is more sensitive to gradient-like instabilities.

Fig. 580: Slice Shift



The **orientation** determines the role of each gradient axis. For example, lets assume there is an instability in the X gradient. The values for the sagittal orientation (X = slice selection) may be in spec. while the spec for the transverse orientation (X = read out) could be out of spec.

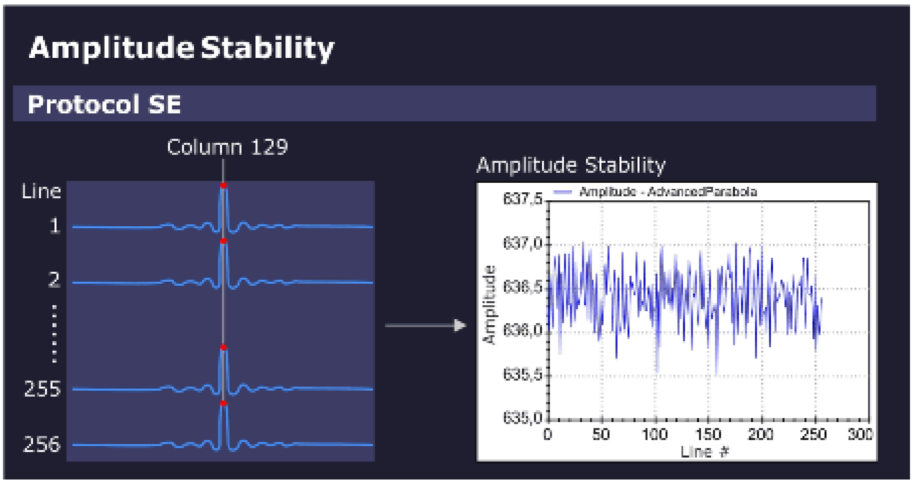
Fig. 581: Orientation



Amplitude stability (spin echo):

The amplitude value in the maximum signal column (e.g. column 129) is displayed in the graphics output.

Fig. 582: Amplitude Stability (Spin Echo)



Phase stability (spin echo):

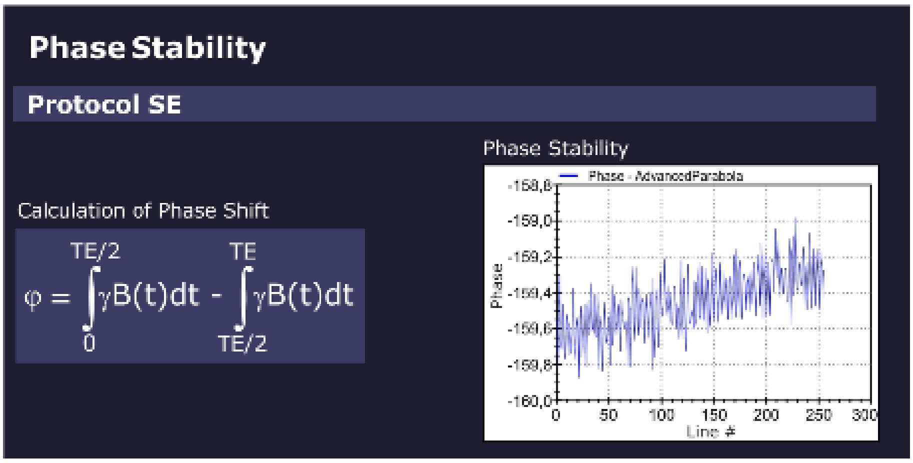
The phase stability of the spin echo is displayed in the graphics output.

The phase shift of a spin echo is determined by the integral across the magnetic field from a 90° to a 180° pulse minus the integral from a 180° pulse to the echo.

For a constant field or a slowly varying field, both integrals cancel out each other.

However, for an oscillating field with a period of TE (20 ms = 1/50 Hz), this expression becomes maximum. In other words, the spin echo measurement with the default setting of TE = 20 ms (and TR = 301 ms) is sensitive to interferences which may be caused by 50/60 Hz AC power line hum or 16.7 Hz trains.

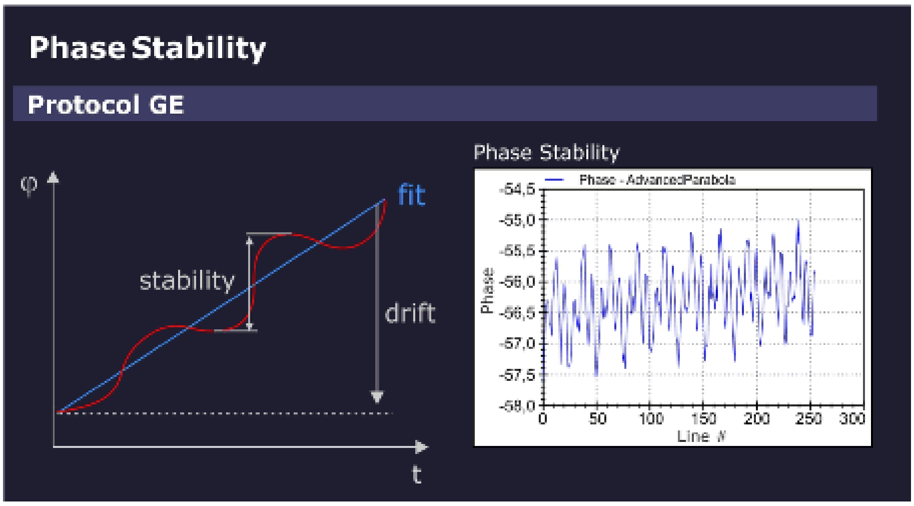
Fig. 583: Phase Stability (Spin Echo)



Phase stability (gradient echo):

For the gradient echo sequence, the phase stability is evaluated by fitting a straight line to the phase data. When you subtract the value of the linear fit from the phase data, the phase stability changes. The slope of the straight line indicates the linear phase drift, caused by slow external field interferences, for example, moving elevators, cars, trucks but also mechanic vibrations.

Fig. 584: Phase Stability (Gradient Echo)



Stability of echo position:

For both measurements, spin echo and gradient echo, the position of the echo peak column is determined for every measurement line and checked against specifications.

The cause of an unstable echo position may be gradient-like interferences from the environment or from the gradient itself.

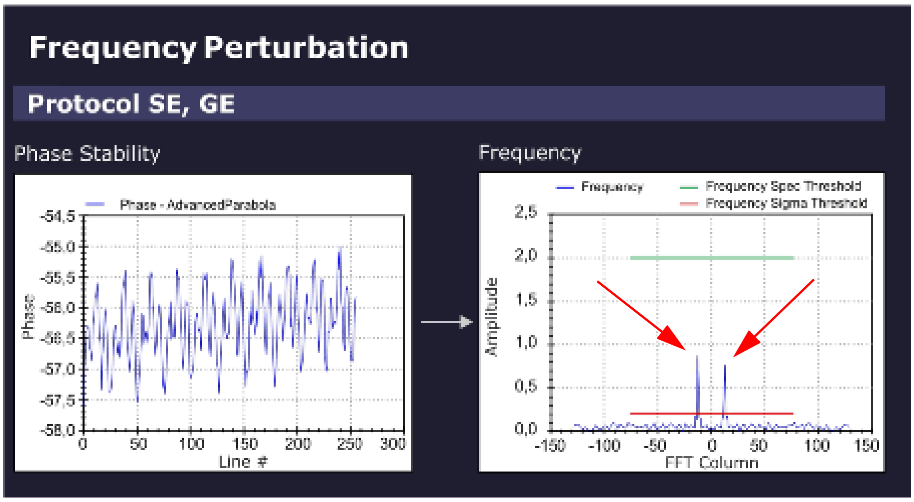
The sensitivity for the direction of the interference is determined by the orientation of the readout gradient.

Frequency perturbation:

The Software differentiates the raw data of the stability measurements and checks for sidebands (ghosts).

In the evaluation result, a frequency perturbation value in percent and the Frequency (peak) position in the FFT Column is displayed.

Fig. 585: Frequency Perturbation (sidebands)



Example: (Frequency Perturbation of 0.8%, Frequency Position -13)

- The sampling rate in the stability measurement is one sampling point every 301 ms (default TR = 301 ms).
- Consequently, the Nyquist-frequency = +- 1/602 ms = +-1.66 Hz. The limiting frequencies at +-128 in the frequency diagram above correspond to +1.66 Hz and – 1.66 Hz, respectively.
- The perturbation frequencies are located at +-13 units, i.e. at 13/128 * 1.66 Hz = +-0.17 Hz.

  » The 300th harmonic of 0.17 Hz yields a 50 Hz interference.

Results in Report:

Measured value	Description
Amplitude Stability	Range of amplitudes (peak to peak)
Phase Stability	Deviation of phase from linear fit
Phase Drift	Drift of phase
Peak Column Stability	Range of peak column
Peak Column Average	Mean peak column
Frequency Perturbation	Relative amplitude of the highest contributing disturber frequency; zero if no frequency is above the relative threshold nor above (noise level * noise factor).
Frequency Position	Position of the highest contributing disturber frequency; zero if no frequency is above the relative (sigma) threshold nor above the (noise level * noise factor).

Troubleshooting

Symptom/Deviation	Possible Cause/Service Action
Amplitude	RF-instability problem, check: ■ Stability of RF System / TX ■ Stability of RF System / RX

Symptom/Deviation	Possible Cause/Service Action
Phase	Magnetic field fluctuation, check: ▪ After magnet ramping or after heavy gradient load (e.g. DTI-, EPI- or StabilityLongTermCheck measurement) wait at least 1 hr. before running Stability Check; ▪ Run QA-Expert / Field Stability Check.
Depending on orientation	Instability of a gradient axis, check: ▪ The Gradient System hardware. ▪ Run service sequence "CalcAr" in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.
Independent of orientation	Magnetic field instable or external field interference, check: ▪ The Gradient System hardware. ▪ See hints "Only in gradient echo".

<table>
<tr><th>Symptom/Deviation</th><th>Possible Cause/Service Action</th></tr>
<tr><td>Only slice-shift dependent</td><td>Most likely, a gradient problem, check:
The Gradient System hardware
especially, the gradient axis used for slice-selection</td></tr>
<tr><td>Only in gradient echo</td><td>Static or slowly varying magnetic field problem?
In the vicinity of the site, check for:
Trains; interference from power lines and rails.
Elevators.
Car/truck traffic and parking.
Very low amplitude due to static magnetic field (B_o) homogeneity problem?
Compare amplitude with spin echo result (both amplitudes should be similar).
Run QA / Phantom Shim Check.
Phase drift out of spec., B_o-drift problem?
System hot? Excessive usage prior to Stability Check measurement (e.g. repeated DTI-, EPI- or StabilityLongTermCheck measurements);
System too cold? Wait at least 2 hr before measurement.
Peak column stability out of spec. i.e., fluctuation of echo position due to gradient field instability?
Check the Gradient System hardware
Run service sequence “CalcAr” in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.</td></tr>
</table>

Additional hints:

- Switch off components one by one to isolate the cause of the instability, e.g.:
  - Cold head
  - Lighting in magnet room
  - Shim PSU
  - Disable gradient axes with DIP switch at DGSU board (first each single axis, then complete gradient power amplifier)
- For periodic instability, vary TR and TE (Stability Check / Expert Mode) to confirm or rule out the frequency of interference:
  - Vary TR to avoid being synchronous to the interference.

  - Vary TE to become more sensitive to particular frequencies.
- If an external interference is suspected, repeat measurement at different times e.g. during night.

5.12.1.21 Synthesizer Check

Help ID: 100FDE50-E606-4FB0-B37C-3F6B6E932DAF
Service Level: 3

General

The **Synthesizer Check** procedure is an important test to verify proper operation of the synthesizer.

Validation of the frequency and phase quality of the synthesizer is important for the accuracy of the slice position.

Measurement

- Verification of the frequency and phase operation of the RF reference signal is performed by using the MR signal itself. After demodulation of the received MR signal, the reference signal is compared to the MR signal in terms of frequency and phase.
  - In one test, the reference signal is varied in frequency over the whole range of the MR system.
  - A second test modifies the phase in fixed intervals.

    » With this test, the entire operating range of the oscillator can be tested. The test is performed for one RX-NCO (Numerically Controlled Oscillator) from group 1 and one RX-NCO from group 2. The two RX-NCOs are validated and can be used for testing the other NCOs.
- Loop-measurements between the tested RX-NCOs and the TX-NCOs approve the correct function of the TX-NCOs.
- In additional steps, all other RX-NCOs are tested against the approved TX-NCOs by testing group 1 RX-NCOs and group 2 RX-NCOs separately.

Evaluation/Results

Results in Report:

- Frequency- and phase-deviations for all used NCOs.

5.12.1.22 Long Term Stability Check

Help ID: 8B498548-2A22-4DDB-8B42-C258D604AAF0
Service Level: 3

General

The **Long Term Stability Check** determines the scanner stability which is important for Functional MR Imaging (fMRI).

The fMRI application visualizes activities in the brain. The patient is exposed to a periodic optical, acoustical or other form of stimulation. During the exposure, a series of MR images is acquired. Subsequently, the images are evaluated for signal intensity changes. The stimulated, activated regions of the brain are localized by their typical, very little variations of signal intensity.

However, these little variations are easily superimposed by instabilities of the MR system or the environment. This QA step makes sure, they are below a specified level.

Typical sources of instabilities are:

- Local B_0-field variation caused by temperature variation in the shim unit (shim iron plates) generates frequency instabilities. The result is a virtual movement of the measured object during acquisitions.
- B1-field variations due to instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, also leading to virtual movement of the object.
- Mechanical vibrations, e.g., by the cold head, produce signal intensity variation caused by the movement of the measured object relative to the magnet isocenter.
- Movement of phantom fluid causes phase errors.

Measurement

A series of 512 measurements (EPI_FID sequence) with 11 slices is performed and repeated once again. The first measurement is performed with a low flip angle of 30 degrees for stability evaluation of the B1-field (i.e., caused by instabilities of the TX chain).

The second measurement uses a higher flip angle of 90(70) degrees to minimize the influences of instable RF but focus on B_0-instabilities and gradient-like interferences.

> i Wait at least 15 minutes between positioning the phantom and starting the measurement. Even small movements in the phantom liquid have a negative effect to the measurement results.

Evaluation/Results

The Software places an evaluation ROI in the central slice and performs the following graphical and numerical evaluations:

- Mean values of a 15x15 ROI related to the mean value of all measurements as a function of the image number
- Standard deviation of mean values of a 15x15 ROI (linear and logarithmic plot)
- Measurement protocol values

- Statistics of mean values (for ROI size 15x15 and 25x25):
  - Mean Image Intensity
  - Peak to Peak Variation
  - Relative Peak to Peak Variation
  - Standard Deviation
  - Total Relative Linear Drift
  - Standard Deviation from Linear Fit
- Image quality parameters such as SNR and ghosting

5.12.2 Coil Tests

Help ID: 33447599-B080-489D-853D-A628FC7C67E8
Service Level: 7

Please select a sub-item.

5.12.2.1 Coil Test

Help ID: FFDB078B-9962-4098-8DDF-12398EC04809
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  - » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

> This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: | Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: | Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: | Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: | Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: | Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ________________

Date: Name:

5.12.3 Coil Quality Assurance

Help ID: 3020AFB6-8C57-418D-AFDD-9141803D9BF4
Service Level: 3

The QA procedure **"Coil Check"** checks the function and quality of each local coil individually.

Before Coil QA, configure new coils in **Admin Portal / Technical Configuration / MR Configuration / Coil Configuration.**

Coils can be also added under **Customer QA**.

5.12.4 Coil Status

Help ID: 59F2EB59-A2D1-4D9C-AEED-F9C79225673D
Service Level: 7

Under **Coil Status**, a status summary of all configured coils at the system is displayed.

The following information is displayed:

- Coil name
- Coil serial number
- Part number (material number)
- Installation date
- A status overview of the individual Coil Quality Assurance measurement
- Date of last execution of the individual Coil Quality Assurance measurements
- Signal to Noise value for each coil

5.12.4.1
Coil Status

Help ID: 9920E976-28FE-45CC-9015-17888738B6A3
Service Level: 7

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (such as a trolley from an old system) are in use.
  - » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

List of options and accessories

> i This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head / Neck coil

Material No.:	
Serial No.:	

SI Spine coil

Material No.:	
Serial No.:	

SI Body 18 coil

Material No.:	
Serial No.:	

SI PA coil

Material No.:	
Serial No.:	

SI Shoulder coil

Material No.:	
Serial No.:	

SI Wrist coil

Material No.:	
Serial No.:	

SI Foot ankle coil

Material No.:	
Serial No.:	

SI Loop coil

Material No.:	
Serial No.:	

SI Flex coil small

Material No.:	
Serial No.:	

SI Flex coil large

Material No.:	
Serial No.:	

SI Knee coil

Material No.:	
Serial No.:	

SI Breast coil

Material No.:	
Serial No.:	

SI Tim adapter

Material No.:	
Serial No.:	

SI Extremity coil

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

Option MRWP

Option present: Yes: ☐ No: ☐

Signature: ____________________

Date: Name:

SI Visual inspection of MRWP (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Dockable table (Option)

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Visual inspection of Dockable Table (Option)

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 1

If an option (for example, a different coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 1

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 2

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 2

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 3

Option present: Yes: ☐ No: ☐

Signature: ____________

Date: Name:

SI Option 3

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________

Date: Name:

Option 4

Option present: Yes: ☐ No: ☐

Signature: ____________________

Date: Name:

SI Option 4

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

Option 5

Option present: Yes: ☐ No: ☐

Signature: ____________________

Date: Name:

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: ____________________

Date: Name:

5.12.5 Quality Assurance - Expert

Help ID: D2F41F74-14FD-41F9-AD9C-601B24C3C3AE
Service Level: 7

Introduction

For some of the QA procedures, an **Expert Mode** is provided.

Depending on the selected QA step, the user interface shows additional parameters on the right side.

> Expert Mode should be only used by qualified operators.
>
> This mode provides additional settings and configurable parameters which require a more detailed knowledge of the underlying physics.

5.12.5.2 RF Verify

Help ID: C7354DE7-20D8-4639-A1DA-911E628FB7AF
Service Level: 7

Expert Mode

In Expert Mode, the measured TuneUp RF Characteristics MNO for hydrogen or the different x-nuclei frequencies can be verified individually.

Available additional options:

- Disable storage of RF Characteristics after the measurement.
- Modification of maximum RF power allowed.
- Selection of TX channel, RF path and nucleus (according to licenses available).
- Option to switch off the frequency adjustment.
- Option to disable the switching to dummy load.

5.12.5.3 Feedback Loop Calibration Check

Help ID: 4433E809-00E7-430B-9826-64F3AD4DFC7B
Service Level: 7

Expert Mode

> This Expert Mode step is only available for single TX channel systems.

Feedback Loop Calibration Check in **Expert Mode** offers the possibility of either checking the feedback loop phase difference, the feedback loop delay, or both values by a single program call up.

General

The Feedback Loop Calibration Check procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the “tuncal” service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box. Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a “Not Ok”.

Evaluation/Results

Results in Report:

- Delay evaluation
- Dispersion of delays from single BC elements
- BC phase plot

5.12.5.4 Phantom Shim Check

Help ID: 897AF1C5-8F5A-4AED-B8F7-65DC5BA57946
Service Level: 7

Expert Mode

A coil selection is available in the Expert Mode.

5.12.5.5 Gradient Risetime Check

Help ID: 16EA805E-B413-4008-B505-8101C4724E20
Service Level: 7

Expert Mode

This procedure determines the minimal rise time or maximum slew rate of the gradient system.

The Expert Mode UI allows the user:

- To select the measurement coil.

- To select the gradient axis.
- To enable additional stress pulses for the measurement.

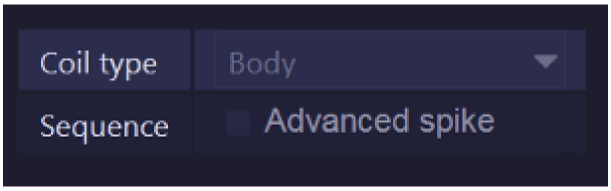
5.12.5.6 Spike Check

Help ID: EDA77FDC-B943-4DBD-838C-2D0545E0D309
Service Level: 7

Expert Mode

For the **Spike Check** in **Expert Mode**, an additional sequence, **Advanced Spike**, is offered.

Fig. 586: Spike Check - Expert



In some cases, spike artifacts may be visible with customer protocols, but not with the regular spike check. If so, it is advisable to reproduce spikes and troubleshoot with the same setup (coils, protocols, position of patient table, Physiological Measurement Unit, etc.) which is used by the customer.

If the artifacts cannot be reproduced at all and also all typical spike sources (see Services Knowledge Base) are ruled out, contact RSC/ HSC.

NOTICE

The "Advanced Spike" procedure contains a powerful gradient pulse scheme which induces high mechanical stress over a wide frequency range. It shall be therefore used very careful and only when all other means are exhausted.

Risk of hardware damage!

» **DO NOT USE the "Advanced Spike" procedure without guidance from RSC/ HSC!**

5.12.5.7 RF Noise Check

Help ID: 3EACA50D-1597-4F09-A708-0AE6878D8DB0
Service Level: 7

Expert Mode

The Expert Mode user interface allows selection of a coil.

In addition, the operator can select either select the full frequency range of +/- 250 kHz or focus on a single frequency band to reduce the measurement time for troubleshooting.

5.12.5.8 Synthesizer Check

```
Help ID: 9D7F9FAC-6FC1-475E-8399-8CC21DFA8A4E
Service Level: 7
```

Expert Mode

A coil selection is available in the Expert Mode.

5.12.5.9 Coil Check - Measure

```
Help ID: C7AD132A-5800-41DD-A778-77EB9BA7C47D
Service Level: 7
```

Expert Mode

Coil Check - Measure provides the option to individually select a coil.

In addition, the uniformity evaluation can be enabled or disabled for the acquired QA image.

5.12.5.10 Coil Check - Evaluate series

```
Help ID: BE987CDD-C2E1-4636-84BD-FED886C9E9D1
Service Level: 7
```

Expert Mode

Coil Check - Evaluate series provides the option to perform a standard Coil Check evaluation with images that are stored in the database. No measurement is performed in this case, only the appropriate Series ID(s) have to be entered.

In addition, uniformity evaluation for the image(s) can be enabled or disabled.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

5.12.5.11 Field Stability Check - Measure

Help ID: B0C73F74-790C-412B-B992-D1B251EF274D
Service Level: 7

Expert Mode

This procedure checks the stability of the magnetic field and corresponding resonance frequency over time. It is useful to evaluate (for example after magnet ramping) if the field is stable enough for imaging or service procedures.

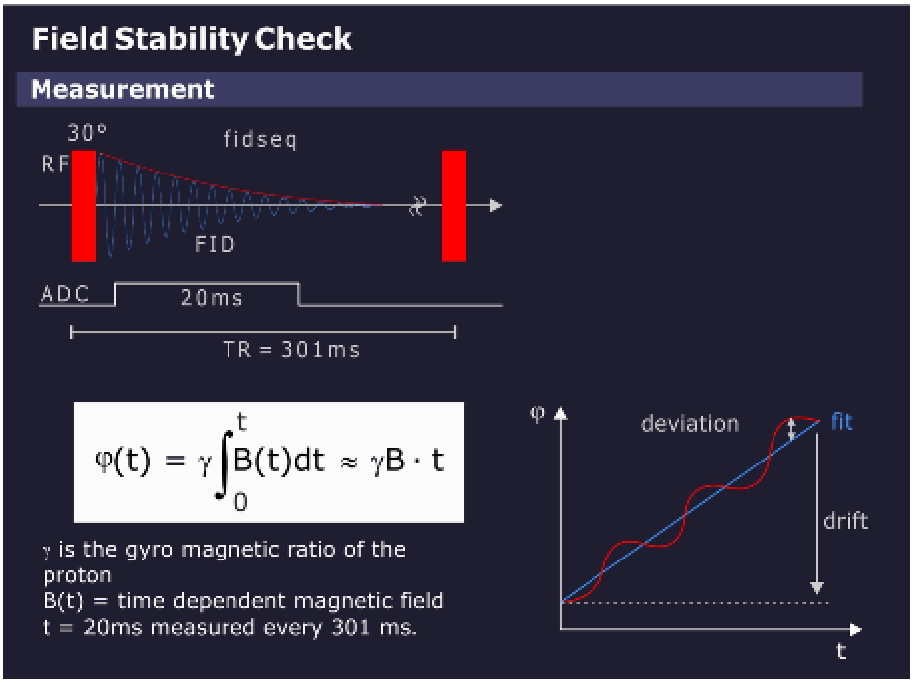
> The Field Stability Check is only available in Expert Mode.

> Allow for sufficient magnet stabilization time before performing Field Stability Check.

Measurement

A series of simple FID (Free Induction Decay) measurements is performed, see graphics below.

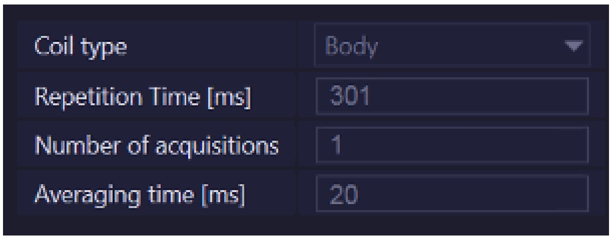
Fig. 587: Field Stability Check - Measurement



A total of 512 FIDs are measured over a 154 second period (repetition time of 301 ms). From each FID, the phase over time is calculated and a linear fit is performed, the slope giving the frequency.

The time range of the fit should be long to get a high accuracy but is usually limited by the length of the FID. A good compromise is 20 ms, which averages over 50 Hz line hum which is usually of no interest in this measurement.

Fig. 588: Field Stability Check - Parameter selection



The following parameter selections are offered:

- Coil type
- Repetition time
- Number of acquisitions
- Averaging time

The measurement time with default parameters is 512 x 301 ms = 154 s.

Evaluation/Results

The result of the field measurement is the field (= frequency) over the measurement time. As a measure of quality, the linear drift over 10 min (= typical measurement time of a long imaging sequence) and the maximum deviation from this linear drift are taken.

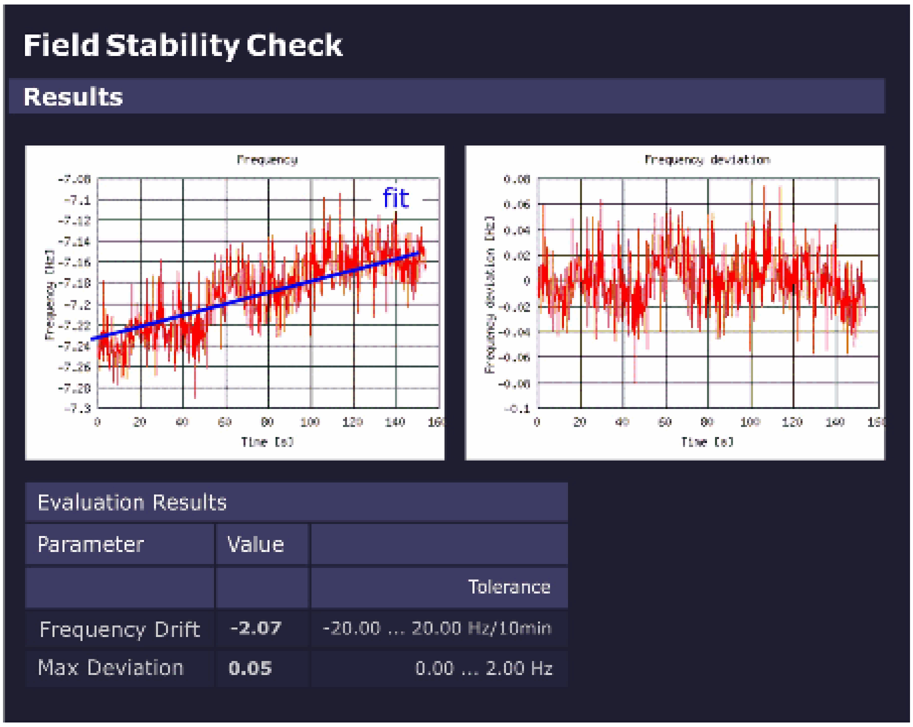
If the selected measurement time is different from 10 min, the drift is calculated for 10 min.

A drift over 10 min should not exceed the pixel bandwidth of a low-bandwidth sequence (e.g., 20 Hz).

The maximum deviation is important for gradient echo sequences to avoid smearing. To allow long echo times, the maximum deviation should be typically less than 2 Hz.

The graphics below shows the frequency variation and the drift-corrected frequency variation across the measurement time.

Fig. 589: Field Stability Check - Results



5.12.5.12
Field Stability Check - Evaluate series

Help ID: 3CD05F02-FE32-4F51-94C3-41BC02E48AEA
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former measurement without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

5.12.5.13
Stability Check - Measure

Help ID: 6CA579E3-92EF-4DEE-8A41-F97C7EF41B1C
Service Level: 7

Expert Mode

Fig. 590: Stability Check Measure - Parameter selection

The Expert Mode function **Stability Check - Measure** evaluates the stability of the MR signal under various sequence conditions.

Different to the Stability Check in the General QA workflow, it allows the following parameter selections:

- Coil type
- Sequence type:
  - Spin-echo
  - Gradient echo (Flash)

  As default, the Spin-echo sequence is selected. This is an exclusive setting so for each **[Go]** in Expert Mode only one sequence selection is possible.
- Repetition time TR and echo time TE

  If the user selects a TR or TE value that is not valid for the selected sequence, an error message is generated when **[Go]** is selected and the valid input ranges are displayed.
- Slice orientation
  - Sagittal
  - Coronal
  - Transverse

  All orientations are selected as default, but a single one, two or all three can be selected.
- Slice shift
  - 0 mm
  - +50 mm

  Both are selected as default. At least one is always selected.

5.12.5.14
Stability Check - Evaluate series

Help ID: 73725B3F-DBC1-40C2-B85C-664143D89A0C
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former measurement without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

5.12.5.15
Long Term Stability Check - Measure

Help ID: DDB58181-821E-4F99-8303-058EFD3295C6
Service Level: 7

Expert Mode

The Expert Mode allows the user:

- to select a coil for measurement (usually the standard Head Coil for the system is used);
- to configure measurement parameters (see below);

Fig. 591: Long Term Stability Check Expert - Parameter selection

Tab. 28 Long Term Stability Check Expert - Parameter description

Measurement Parameter Selections	Description
Coil type	Coil to be used for Long Term Stability measurement (usually standard Head Coil of the system).
Number of slices	▪ Multi slice 3, 11, 15 or 17 slices (default = 17 slices) ▪ Single slice
Slice number	Number of slice to be evaluated (default = central slice).
Repetition time	in ms (default = 1000ms)
Echo time	in ms (default = 30ms)

Measurement Parameter Selections	Description
Flip angle	Flip angle in degrees: ▪ 30 degrees for evaluation of B1-instabilities (i.e. instable transmit components or TX-coil). ▪ for evaluation of other instabilities e.g. B_0, dynamic field perturbations: - 90 degrees (for 1.5 T systems) - 70 degrees (for 3T systems, due to dielectric resonances in the water phantom)
Number of measurements	(default = 512) may be reduced for troubleshooting e.g. 256 to save measurement time. Note: The No. of measurements should not be reduced too much below 256 to maintain a sufficient statistical basis for evaluation.
Number of preparation scans	Preparation scans for the magnetization (default = 5).
Number of warm up repetitions	Warm-up repetitions. Several EPI_FID measurements (default = 60) should be run prior to the “real” measurement to bring the MR system into thermal equilibrium state. No evaluation is performed for the warm-up scans.
Swap orientations	Swap readout and phase-encoding gradient.
Slice orientation	▪ Transversal (= default) ▪ Sagittal ▪ Coronal

Measurement Parameter Selections	Description
Phase correction	▪ None – no correction is applied. ▪ 3-Line – reduction of N/2 ghosts. – the first three echoes (not phase-encoded) are used to correct the further echoes. ▪ 3-Line B0 (= default) – reduction of N/2 - ghost, in addition B_0-field drifts are compensated. – the first three echoes (not phase-encoded) are used to correct the further echoes. ▪ Full – optimal correction for effects accumulating during the echotrain, but time consuming; measurement time is doubled, also for multiple measurements! – there is a prescan with full number of echoes, but not phase-encoded. All echoes in the next scan are corrected by means of that prescan.
Arterial spin labeling evaluation	Switch off/on additional evaluation function for Arterial Spin Labeling (ASL) application.

5.12.5.16 Long Term Stability Check - Evaluate series

Help ID: E13785FD-58F4-47E8-9695-9467FB0975E4
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former Long Term Stability Check without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Enter the Slice number for the evaluation function.
- If applicable, select the Arterial spin labelling evaluation function (select radio button **[enabled]**)

- Press **[Go]** to start the evaluation.

5.13 Final Checks

Help ID: E7FB5804-F5A2-4085-9F4A-ED6A1BF4F13E
Service Level: 7

Please select the sub-item.

5.13.1 Send back the signed Safety Related Test Report to the MRQ Department

Help ID: B929000C-1B4C-49F6-BF85-F1B0917E9732
Service Level: 7

Send back the Safety -Related Tests Report to the MRQ Department

Safety-related Test Report

The protocol of the installation “First measurement” serves as a reference document for subsequent repeat measurements.

The MR quality department needs the protocol for legal reasons.

Tab. 29 Address MR Quality Department

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
MR Quality Department (HC DI MR QT)
P.O. BOX 3260
91050 Erlangen
Germany
mail to: immrqpinstallations.healthcare@siemens.com

5.13.2 Send back the signed Start-up Final Report to the MRQ Department

Help ID: 3B5D44B2-AB46-44FF-AD85-A959C5B603FA
Service Level: 7

Send back the Start-up Report to the MRQ Department

Start-up Report

The ’Start-up Report’ is the final test certificate for the system. The completed ’Start-up Report’ is our verification that the system including all coils and options was properly installed and is operating according to manufacturer’s specifications.

Tab. 30 Address MR Quality Department

Siemens Healthineers Headquarters

Siemens Healthcare GmbH

MR Quality Department (HC DI MR QT)

P.O. BOX 3260

91050 Erlangen

Germany

mail to: immrqpinstallations.healthcare@siemens.com

5.13.3 Send back the Quench Line Acceptance Protocol to the MRQ Department

Help ID: 3FEC563F-6007-422B-AABC-E0E2E18C045F
Service Level: 7

Send back the Quench Line Acceptance Protocol to the MRQ Department

Quench Line Acceptance Protocol

Siemens as the manufacturer of MRI systems has the obligation to verify that risk mitigation measures - such as the correct installation of the quench line – are implemented and to prove the verification by checking (and archiving) written evidence of the installation procedure.

The Quench Line Acceptance Protocol has to filled out completely and signed by the project manager. Please refer to the Planning Guide. Collect the Quench Line Acceptance Protocol from project manager and send it to the MR Quality Department.

Tab. 31 Address MR Quality Department

Siemens Healthineers Headquarters

Siemens Healthcare GmbH

MR Quality Department (HC DI MR QT)

P.O. BOX 3260

91050 Erlangen

Germany

mail to: immrqpinstallations.healthcare@siemens.com

5.13.4
Send back the pictures of magnet room to the MRQ Department

Help ID: 1B816436-9D49-4E7E-ABBF-5C824152B680
Service Level: 7

Send back the Pictures of Magnet Room to the MRQ Department

Pictures of magnet room

A minimum of 4 pictures have to be prepared during start-up prior to the first ramping. The pictures of the magnet room have to be attached to the start-up report. Please refer to the start-up procedure.

Tab. 32 Address MR Quality Department

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
MR Quality Department (HC DI MR QT)
P.O. BOX 3260
91050 Erlangen
Germany
mail to: immrqpinstallations.healthcare@siemens.com

5.14 Protocol/Startup Report

Help ID: 8893957B-3FCC-4E23-91B2-8ABCCCCD31B2
Service Level: 7

Create Protocol/Report

Status

Status displays the current status of the procedures.

Overview of critical procedures:

Overview of critical procedures list the tasks which are Not OK or overdue.

Test Result

The tester has to evaluate the test results as defined and listed under **Test Result**.

- Select the applicable evaluation.

> Without selection of the Test Result the report cannot be saved.

Remarks:

Remarks provide the possibility to enter notes or to describe deficiencies in detail.

Printed Signature Name

The Tester has to enter his name.

- Enter your name

> Without a name the report cannot be saved.

Save and Create Report

After completion of the Safety –Related Tests tasks the entries must be saved. Otherwise the entries will be lost.

- Click **Save and Create Report** to save our entries and generate a report.
  » The **Report** is now saved and is then part of the Site-specific Data.

Use the **Reports platform** to display or to print the protocols.

6.1 MPPS Completion

Help ID: 5AFA6A1A-1E4E-4E4F-ADE6-16E0F3386B96
Service Level: 1

Automatic

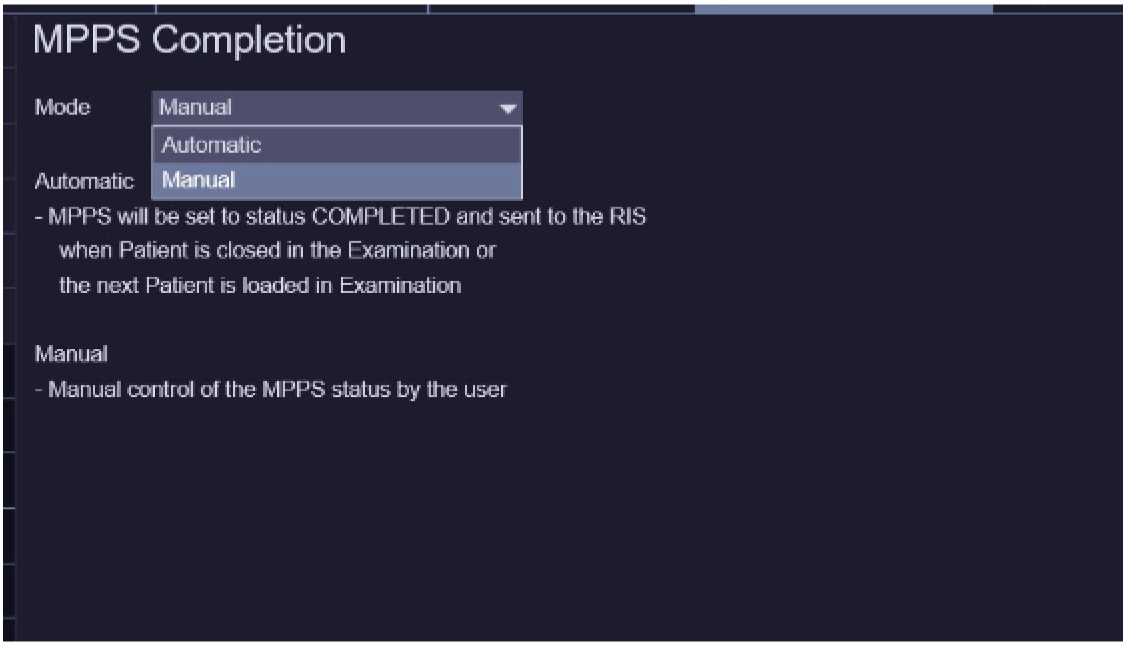
MPPS will be set to status “COMPLETED” and sent to the RIS when closing the patient in the Examination or when the next patient is loaded in Examination.

Manual completion is always possible, regardless if automatic completion is enabled or not!

Manual

When the procedure of a procedure step is completed by the user in the “Examination UI” or in the “Organizer”

Fig. 592: DICOM - MPPS Completion



6.2 DICOM Study Description

Help ID: E896AF24-BD11-48E0-8112-77217DD54237
Service Level: 1

Modality

Dot Cockpit: Region^Examination

Modality including Program level

Dot Cockpit: Region^Examination^Program

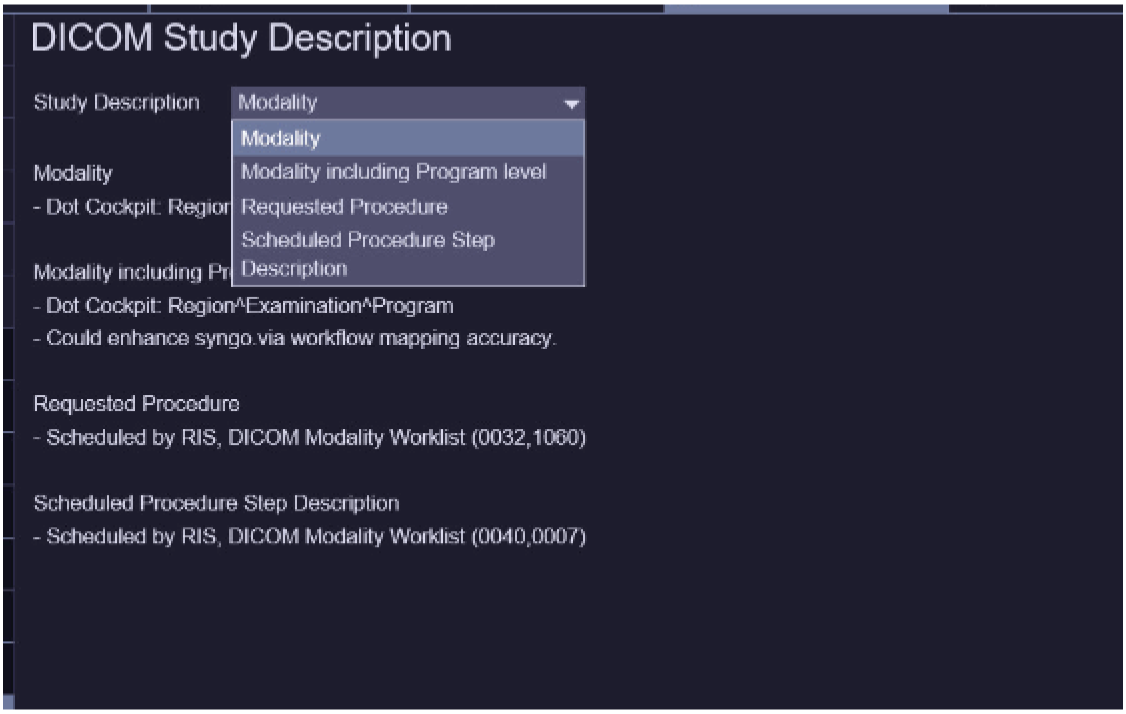
Requested Procedure

Scheduled by RIS, DICOM Modality Worklist (0032,1060)

Scheduled Procedure Step Description

Scheduled by RIS, DICOM Modality Worklist (0040,0007)

Fig. 593: DICOM Study Description



7.1 MR Configuration

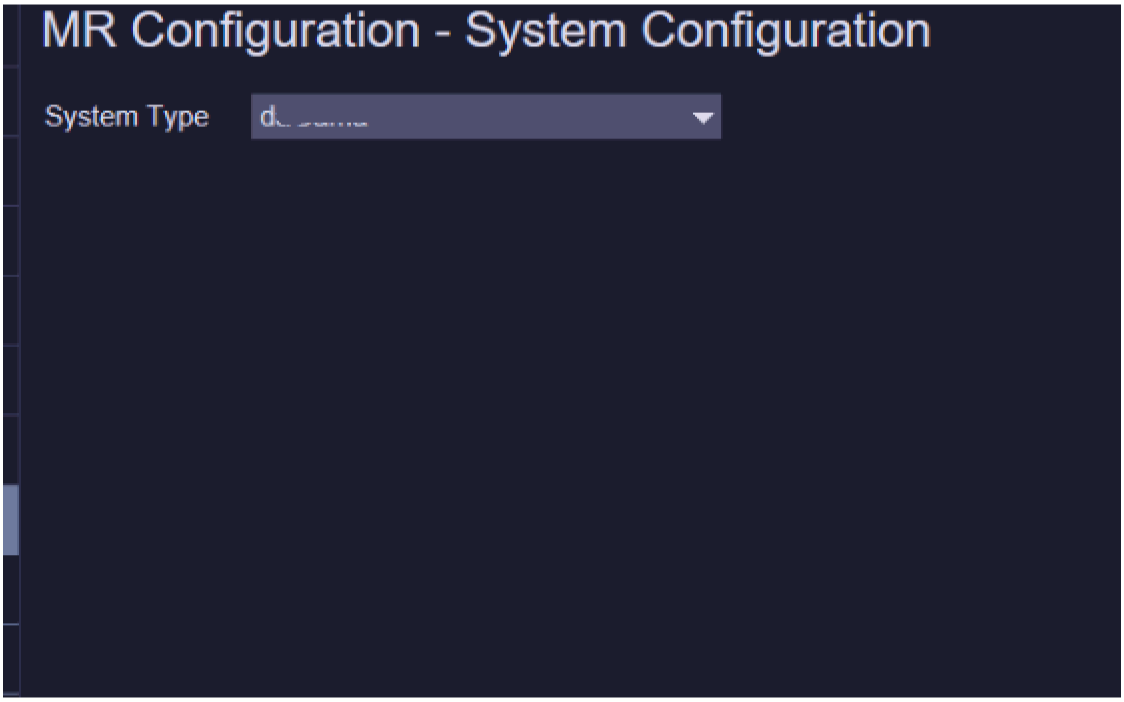
7.1.1 System Configuration

Help ID: AA5789DB-77CD-4C8F-8609-DA2440AD757A
Service Level: 3

With the delivery of the MR system the System Type is pre-configured in the factory.

> This setting is only available in cases like major hardware upgrades from one system type to another system type!

Fig. 594: MR Configuration - System Configuration



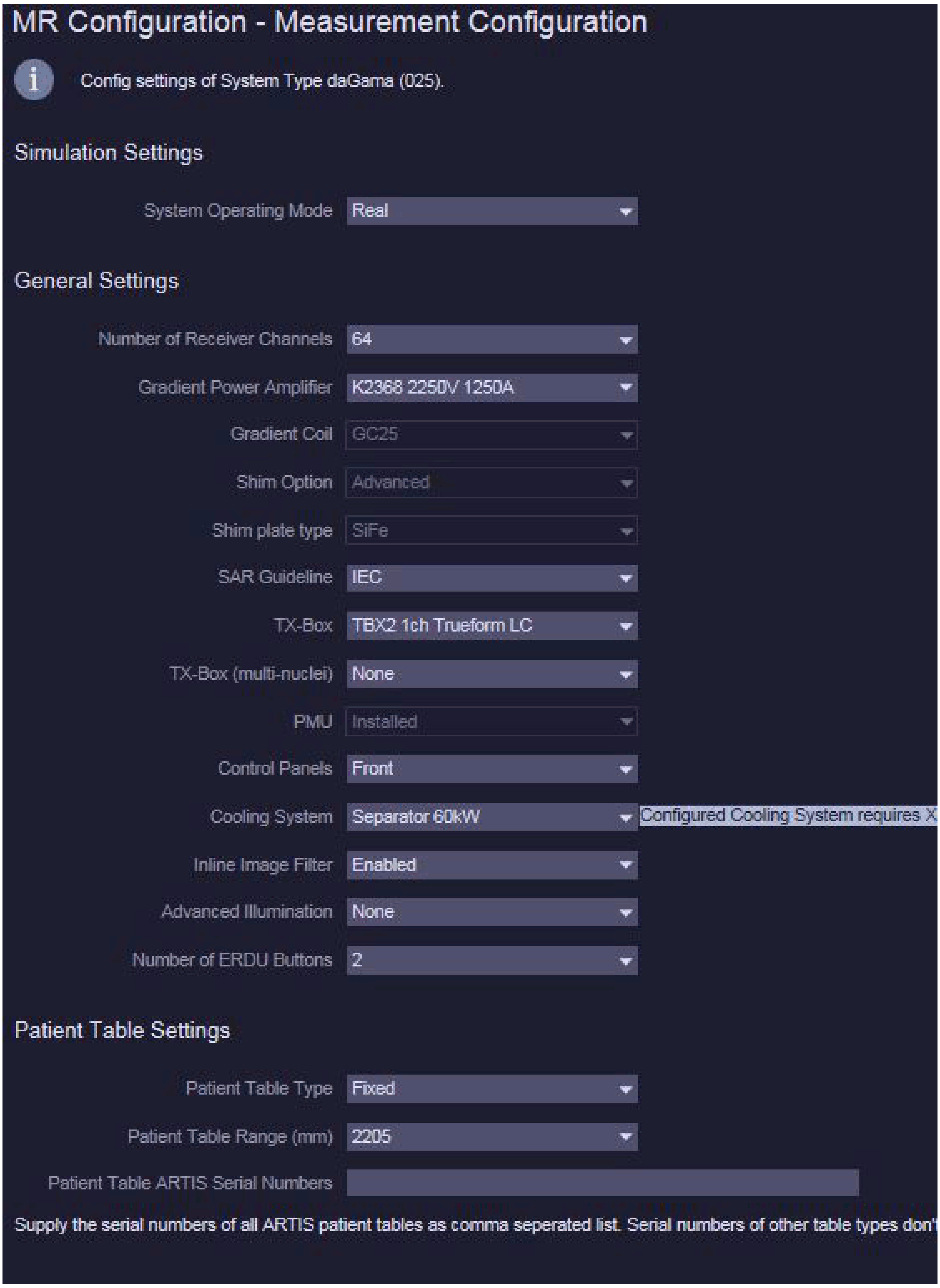
7.1.2 Measurement Configuration

Help ID: 6E389F4F-A325-44E3-8410-77EDC380BB84
Service Level: 3

With the delivery of the MR system the Measurement Configuration is pre-configured in the factory individually for your system type and installed options.

To activate any changes made in the Measurement Configuration a system reboot is required!

Fig. 595: MR Configuration - Measurement Configuration



7.1.3 CAN Box Configuration

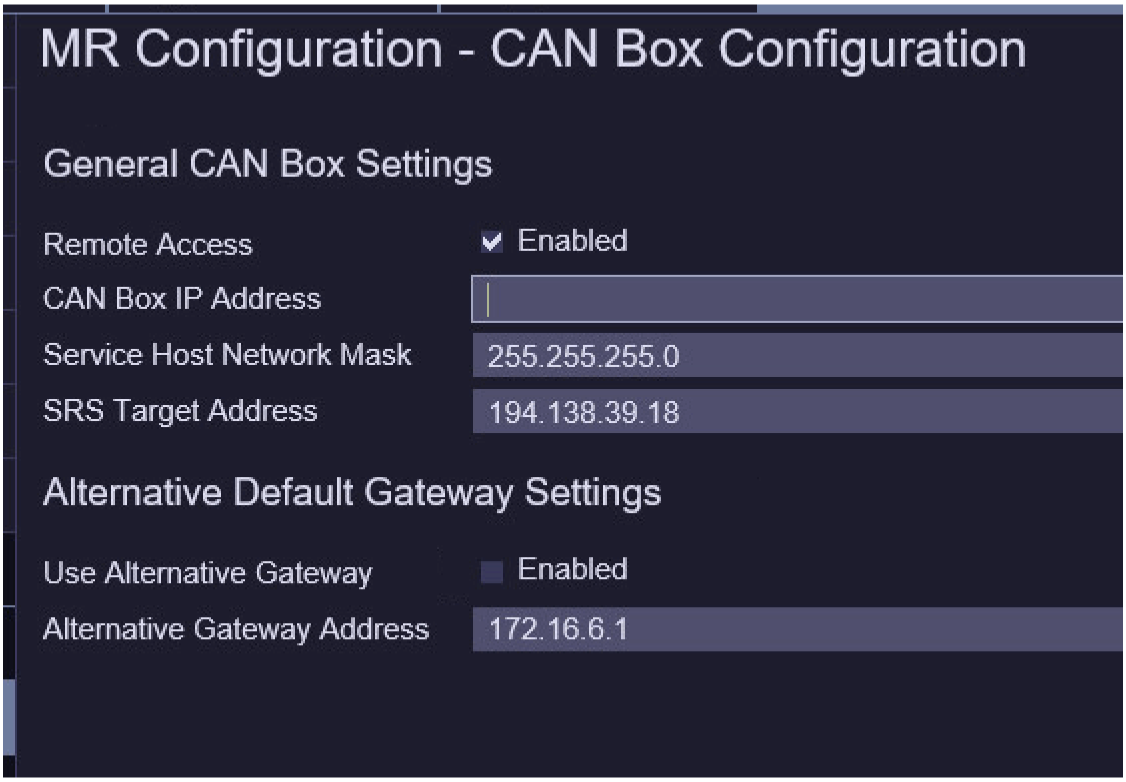
Help ID: D9E7F63B-4429-4F2A-B3F9-215BE8EF38F8
Service Level: 3

CAN Box magnet cooling monitoring and notification unit. The CAN Box is accessible via remote service; the product name is “MR_CAN24” and the system serial number is used.

(ETH 0) X11 at the front of the CAN Box unit is used for the SRS connectivity over the customer LAN.

The check-box **Remote Access** must be set to enable “SRS” connectivity to the CAN Box.

Fig. 596: MR Configuration - CAN Box Configuration



Consult your “SRS Help Desk” regarding configuration and testing of the connectivity.

7.1.4 Coil Configuration

Help ID: AC53C5B9-B1D9-4B27-8555-01B6A2321912
Service Level: 1

Within the “Coil Configuration” you can manage your MR systems coils.

You have the possibility to add, edit, or delete coils in the “Coil Configuration”.

To register new coils, please plug the coils onto the patient table and press **Add Plugged Coils**.

To delete coils from you configuration use **Delete**

> Only configured coils will be listed in the **Quality Assurance > Coil** workflow.

Fig. 597: Coil Configuration

MR Configuration - Coil Configuration

Name	Serial Number	Part Number	Revision	Registration Date
FlexLarge 4	1739	10835335	03	2016-09-01 10:06
FlexSmall 4	1731	10835327	03	2016-09-01 10:47
ShoulderLarge 16	1866	10500087	05	2016-09-01 10:56
ShoulderSmall 16	1860	10500088	05	2016-09-01 11:01
Spine 32	2173	10496545	09	2016-09-01 11:07
Body 18	1231	10835325	02	2016-09-01 11:27
Body 18	1617	10835325	02	2016-09-01 11:45
HeadNeck 20	2148	10496500	09	2016-09-01 12:15
HeadNeck 20 TCS	202	10835555	00	2016-09-01 12:33
Body	1006	10676584	00	2016-09-01 14:48
Breast 18 F	1148	10606916	02	2016-09-02 14:59
TxRx CP Head	1034	10606826	02	2016-09-02 16:42
Spine 72 RS	202	10835688	00	2016-09-08 08:10
Spine 72 RS	203	10835688	00	2016-09-08 08:36
Spine 72 RS	201	10835688	00	2016-09-08 09:10

To register new coils, please plug the coils onto the patient table and press [Add Plug

Add Plugged Coils

7.2 Maintenance Configuration

Help ID: 60A96B88-4BFA-4FC6-9945-C468E01EC0BB
Service Level: 7

Maintenance Configuration

Within Maintenance Configuration four basic items will be configured:

- Options (optional procedures)
- Condition Based Maintenance
- Measurement Type of Safety Related Tests
- Display of Maintenance Notifications in System Status Overview

Options

The maintenance workflow display the procedures only if the option is installed and configured. Options like MRWP, cooling system, patient table type will be configured in First Installation > MR Configuration > Measurement Configuration. Additional options have to be configured Maintenance Configuration

Following additional option has to be configured:

Configure option	Procedure
Pediatric Coil	Checking warning signs of the pediatric coil (option)
MRWP	Measurement of the MRSC (MRWP) ground wire (annually) Visual inspection of MRWP

Is **Installed** selected, the task will be displayed.

Is **Not Installed** selected, the task procedure will not be displayed.

- Configure the options according to the system configuration.

Condition Based Maintenance

If Condition Based Maintenance is enabled all Maintenance tasks will be controlled by Service Events via GSMS. The maintenance will be initiated by the condition of the system /components and not by a time schedule. The Preventive Maintenance procedures will be faded out.

Prerequisite for Condition Based Maintenance is a daily remote monitoring of the system status. Therefore the data transfer via Siemens Remote Service must be permanently enabled.

If the system has no Siemens Remote Service connection Condition Based Maintenance must be disabled.

Measurement Type of Safety Related Tests

The report of the "initial measurement" of installation serves as a reference document for subsequent repeat measurements. It must therefore be archived by the operator in the System Owner Manual.

First protocol during installation, select > Initial

Repeat protocol during service / maintenance, select > Repeat

Display of Maintenance Notifications in System Status Overview

The System Status Overview displays three status indicators for Maintenance

- Cold Head Maintenance
- Preventive Maintenance
- Safety-related tests

The indicators turn red if the interval for the specific tasks has expired.

This indicator function can be **Enabled** or **Disabled** within Maintenance Configuration.

The default setting of the function is **Disabled**. The three different maintenance indicators will never be displayed.

The maintenance indicators should be **Enabled** only if the customer does not have a service contract and only with the consent of the service manager or technical manager.

8.1 Coil Configuration

Help ID: AC53C5B9-B1D9-4B27-8555-01B6A2321912
Service Level: 1

Within the “Coil Configuration” you can manage your MR systems coils.

You have the possibility to add, edit, or delete coils in the “Coil Configuration”.

To register new coils, please plug the coils onto the patient table and press **Add Plugged Coils**.

To delete coils from you configuration use **Delete**

Only configured coils will be listed in the **Quality Assurance > Coil** workflow.

Fig. 598: Coil Configuration

MR Configuration - Coil Configuration

Name	Serial Number	Part Number	Revision	Registration Date
FlexLarge 4	1739	10835335	03	2016-09-01 10:06
FlexSmall 4	1731	10835327	03	2016-09-01 10:47
ShoulderLarge 16	1866	10500087	05	2016-09-01 10:56
ShoulderSmall 16	1860	10500088	05	2016-09-01 11:01
Spine 32	2173	10496545	09	2016-09-01 11:07
Body 18	1231	10835325	02	2016-09-01 11:27
Body 18	1617	10835325	02	2016-09-01 11:45
HeadNeck 20	2148	10496500	09	2016-09-01 12:15
HeadNeck 20 TCS	202	10835555	00	2016-09-01 12:33
Body	1006	10676584	00	2016-09-01 14:48
Breast 18 F	1148	10606916	02	2016-09-02 14:59
TxRx CP Head	1034	10606826	02	2016-09-02 16:42
Spine 72 RS	202	10835688	00	2016-09-08 08:10
Spine 72 RS	203	10835688	00	2016-09-08 08:36
Spine 72 RS	201	10835688	00	2016-09-08 09:10

To register new coils, please plug the coils onto the patient table and press [Add Plug

Add Plugged Coils

8.2 Customer Quality Assurance

Help ID: 48DFD1A1-9248-4E20-8EA8-BACD7573BF22
Service Level: 1

General

A degraded MR image quality may indicate a malfunctioning RF coil. A quality measurement is performed for each RF coil to verify satisfactory operation.

Measurement

Measurement phantoms are used for quality measurements.

> The fluid in the measurement phantoms presents a health hazard. Follow the safety regulations in case of phantom fluid spills.

The quality measurement is performed with the quality program at the syngo Acquisition Workplace. The program checks the signal-to-noise ratio for all RF coils. In addition, it checks the artifact behavior for the Body Coil.

> If the quality measurement results are outside specifications, the phantom fluid may have moved inside the phantom. Measurement phantoms must be positioned for at least 3 minutes at rest on the phantom holder.
>
> If the values of the quality measurement for an RF coil are outside specifications, please repeat the quality measurement.

8.3 Quality Assurance

8.3.1 General Quality Assurance

Help ID: 7DFD7C17-DADA-4F81-B6E1-1AF37D0BBC53
Service Level: 3

Introduction

General Quality Assurance (QA) is an automated procedure for:

- Verification of TuneUP adjustments.
- Checking the overall system quality.
- Image quality checks.

Different from TuneUp, **system parameters are not changed in QA**.

QA Workflow

- General QA is separated into modules, each routine runs as a stand-alone process.
- After start, each procedure reads the specification and configuration parameters automatically.
- QA procedures determine if their measured results are in or out of specifications.
- QA procedures write their individual results to the Service Report.
- The status is returned to the main QA platform and the corresponding status field in the Service UI is updated.

After pressing the **[Go]** button in the QA UI, the **procedures start and perform typical tasks**:

- Each procedure registers the Service Patient (if not already registered).
- If a clinical patient is registered, a pop-up informs the operator to de-register the clinical patient before proceeding.
- The correct phantom setup is displayed by a pop-up and confirmed by the operator (if applicable). If QA is performed remotely, automatic table move is disabled and the operator receives a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.
- Table movement:
  - If **QA is performed locally** at the system, the **table moves automatically** to center the required phantom in the measurement field (change between large/small spherical phantom).

- If **QA is performed remotely**, **automatic table move is disabled**. The operator gets a pop-up to center the required phantom manually or abort the workflow with the **Cancel** button.

■ Configuration and specification values are read.

■ Measurements are performed.

QA Function Status

Tab. 33 Possible QA Function Status

Status	Description
To Do	The QA procedure has not yet been performed yet and is pending. OR: The QA procedure needs to be performed again because another procedure (from TuneUp) has requested its execution.
Ok	The QA procedure has been carried out successfully.
Not Ok	The related parameters/results were found to be out of specifications during the last operation.
Error	An error occurred during the last operation.
Cancelled	Procedure canceled by operator.

Evaluation/Results

After end of the QA measurement, evaluations are performed automatically:

■ The procedures determine if the measured results are in or out of specifications.

■ The results are written to the Service Report.

■ The results status is returned to the main QA platform and the corresponding status field is updated.

TuneUp/QA Dependencies

If a TuneUp procedure is successfully completed and has changed its status, the corresponding verification procedure(s) in the Quality Assurance workflow menu is/are changed to status **To Do**.

TuneUp procedures may also affect multiple QA procedures, e.g. after TuneUp Phantom Shim, the QA procedures Phantom Shim Check, Gradient Sensitivity Check, BC Image Brightness and Long Term Stability Check have to be performed.

Standby

In QA, a **Standby** function selection is offered. If selected, the system is switched to standby mode after completion of the QA workflow. All units except for the host computer are switched-off.

After using the Standby functionality, it is not possible to switch-on the system again remotely.

8.3.1.2 RF Verify MNO

Help ID: 4210C5CB-BBD0-40ED-9200-0E96102D8567
Service Level: 3

General

The procedure RF Verify MNO performs a verification of the Multinuclei RF Characteristics determined during Tune Up.

Individual measurements for all supported nuclei are performed because of the frequency dependency of the broadband RFPA's power output. Forward and reflected DICO power values are measured and checked against their corresponding error tolerances (amplitude- and phase deviation from last saved RF Characteristics).

Measurement

The RF signal is generated at the MNO-TX Box Controller, amplified by the BB RFPA and routed to the dummy load in the MNO TX-Box.

The DICO factor is calculated and checked. Then, the deviation of the amplitude and phase from the linear characteristic is evaluated.

Evaluation/Results

- DICO calibration voltage
- DICO calibration statistics
- plots show the measured characteristics (amplitude and phase)
- maximum power deviation and peak-to-peak phase deviation

8.3.1.3 Feedback Loop Calibration Check

Help ID: 4172C660-AF1B-46AF-A127-DF43D6E9194F
Service Level: 3

General

The **Feedback Loop Calibration Check** procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the “tuncal” service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box.

Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a “Not Ok”.

Evaluation/Results

Results in Report:

Tables:

- Delay evaluation
- Dispersion of delays from single BC elements

Plots:

- BC phase plot

8.3.1.4 RX Gain Calibration Check

Help ID: 4A3CD412-40FD-43CD-9DD2-BEF0BA7D4C40
Service Level: 3

General

The **RX Gain Calibration Check** verifies the amplification correction factors for all individual receiver channels which were determined during TuneUP / RX Gain Calibration:

- (Residual) RX channel correction factors
- High- / Low-gain factors

Measurement

The procedure uses loop measurements (without MR measurements). A fixed calibration signal is fed through all RX channels which are switched to high- and low-gain. For the measurement, the stored TuneUp parameters for the given RX channel- and high- / low-gain-correction are applied. The generated raw data is analyzed and all gain correction factors and the quotient of the High- / Low-gain modes are determined.

Evaluation/Results

Results in Report:

- Average gain correction factor over all receive channels.
- Gain correction factors for each individual receive channel.
- High- / Low-gain signals for each individual receive channel. Comparison of amplitude- and phase-difference to the stored TuneUp correction values.
- Noise evaluation.

8.3.1.5 BC Tuning Check

Help ID: 12EC839E-96E8-4A6D-8498-C6930BCCCDEF
Service Level: 3

General

The **BC Tuning Check** procedure is used to measure and evaluate the central resonance frequencies of the Body Coil resonator systems as well as the coupling between them.

A high efficiency of RF power transmission to the Body Coil requires sufficiently low reflection factors at the TX-Box output.

This condition is usually met, if the reflection factors of the Body Coil resonator systems BC0 and BC1 are high (without load) and equal. However, with load, these reflection factors shall be low.

The other requirement is, coupling between both resonators shall be minimum (i.e. high decoupling values).

Measurement

BC Reflection

First, a **BC Reflection** measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals, covering the full transmit operating frequency range. The result of each measurement is the reflection or r-factor = Ur/Uf which is plotted for both Body Coil systems BC0 and BC1 along the frequency axis.

The evaluation determines:

- each coil's center (tuned) frequency is within a specified range, i.e. is not too far from the center of the operating range.
- that the center frequencies of both coil systems BC0 and BC1 are tuned within specified limits of each other, i.e. the delta frequency is low.
- that the reflection factors are within expected limits. These values provide useful information about the quality (Q-factor) of the BC resonant circuits. They also depend on the coil load and cannot be adjusted.

  The Software detects the resonant frequency always at the point where the reflection of the BC has its minimum.

Transmission

In a second step, a **Transmission** measurement is performed with transmitting a signal into the BC0-system and measuring the coupled signal in the BC1-system and vice versa. The transmission factor describes the amount of RF-coupling between the two Body Coil systems, mainly due to coil construction but also affected by other factors e.g., mechanic position adjustment of the Body Coil within the Gradient Coil's Faraday Shield.

The evaluation checks the transmission value, expressed in decibels (dB), to be within expected limits. **A high negative dB-value indicates minimum coupling** i.e., a good decoupling between BC0 and BC1 system.

Evaluation/Results

Results in Report:

- The reflection related values for both (BC0 and BC1) systems, like the resonance frequency, the reflection factor and the difference between both systems.
- The transmission related value like the frequency and the decoupling between both systems.
- The new and old phase difference.
- Diagrams showing the measured values for the transmission and the reflection behavior as a function of frequency.

8.3.1.6
Cable Length Check

Help ID: B19E08C7-B349-43A5-99BF-E46BAB8192A4
Service Level: 3

General

The QA step **Cable Length Check** verifies the BC transmit cable phase shifts which have been determined in TuneUp / Cable Length.

The BC transmit cable length may slightly vary due to production tolerances. However, accurate information about the lengths of both BC transmit cables is required for an optimal function of the extended TX B1-field regulation loop in the measurement framework.

Measurement

A measurement with the service sequence rf_loop is performed. An RF pulse is fed to both Body Coil channels with the Body Coil switched to 'detune' mode. The forward and reflected signals are taken from the internal directional couplers (DICO) of the TX-Box and analyzed by the monitoring Software.

From these results, the so-called S-matrix elements which describe the characteristics of the BC transmit path are determined. These values allow, together with the known Body Coil RF-properties, to calculate the cable lengths for both Body Coil channels in terms of the induced signal phase shifts.

Evaluation/Results

In the Service report, the actual, measured phase shifts for both Body Coil channels are displayed, together with the old values from TuneUp. The latter ones are read from the BC coil files. Difference values are calculated and checked against the specified tolerance range.

8.3.1.7
Gradient Regulator Check

Help ID: 99738FE4-5E37-4EFF-A222-CE0B23F5D7EE
Service Level: 3

General

The **Gradient Regulator Check** procedure only measures the gradient waveform with the present PID regulator settings and verifies that these settings are correct. The check analyzes several aspects of the actual current output by comparing it to the expected ideal pulse output. In addition, the procedure will measures the linearity of the gradients by stepping through the full range of the gradient strength from negative to positive amplitudes.

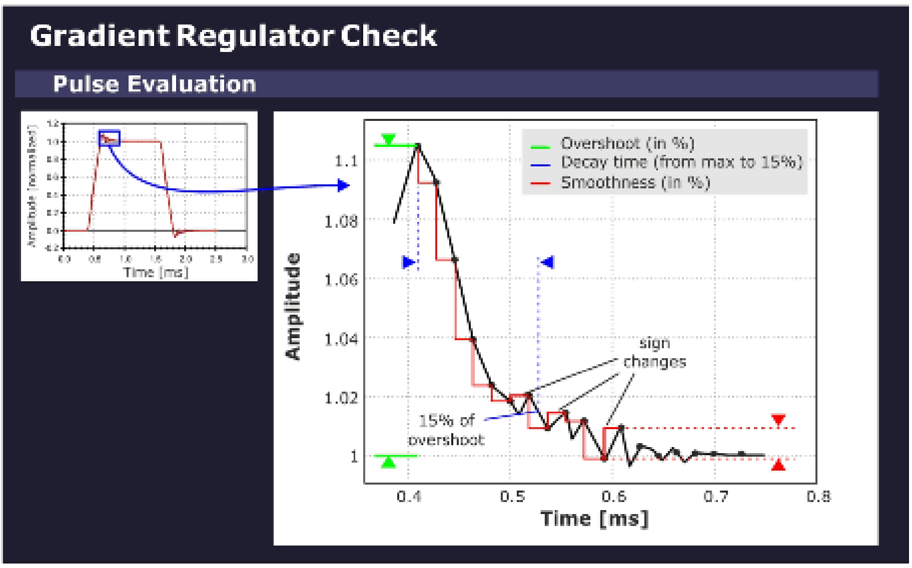
Measurement

The sequence uses a gradient table to generate the gradient pulses. The selected gradient (depending on the selected orientation) steps from the maximum negative to the maximum positive gradient amplitude pulse in linear portions. Each gradient in the table is switched on using a ramp-up and ramp-down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.

Evaluation/Results

Based on the maximum gradient amplitude pulse, the program evaluates the characteristic overshoot that occurs just after the pulse ramp-up. The amplitude of this **overshoot** is determined as a percentage relative to the nominal gradient amplitude. The program also calculates the **decay time** it takes for this exponential-like overshoot to return to a specified value. In addition, a **smoothness calculation** is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.

Fig. 599: Gradient Regulator Overshoot



Results in Report:

- PID gradient regulator parameters.

- Optimization parameters overshoot, decay time and smoothness.
- Gradient current pulse diagram (pulse form).
- Enlarged diagram of overshoot portion and current ripple (top portion).

8.3.1.8 Phantom Shim Check

Help ID: D847AEFF-8EE6-4E1B-83A4-EE3620B311E9
Service Level: 3

General

The **Phantom Shim Check** procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be accurately determined based on the frequency distribution in the volume elements making up the phantom image.

Two measurements are performed with the parameters from the corresponding TuneUp procedure to determine the remaining field terms and the overall homogeneity within the phantom volume. No adjustment of the shim terms is made in this case.

For systems with the advanced shim option, the procedure is performed with shim currents applied.

Measurement

The actual shim status of the system is measured with a 3D-Dess-sequence (Double Echo Steady State with a FISP and PSIF echo). The resulting voxels are cubes which are analyzed

by the Software.

Evaluation/Results

The field homogeneity is evaluated based on the frequency of the voxels. For the check of the B_0-field, it is sufficient to compare the field differences between neighboring voxels. The field difference is obtained by comparing the phases of the magnetization vectors.

The complete data set is evaluated by a differential shim equation. This equation minimizes the difference between the field generated by the3 Gradient Coils (and the optional 5 shim coils) and the measured magnetic field inhomogeneities.

Results in Report:

- Field terms
- MR Frequency
- Gradient Offsets
- Shim Currents (optional)

8.3.1.9 BC Power Losses Check

Help ID: D35D6115-3348-47DF-8FBE-8CBE8F1C7E7C
Service Level: 3

General

A check is performed to verify that the TuneUp of BC Power Losses has been performed successfully. This is done by measuring the Coil Power Loss value again and checking against the Coil Power Loss value in the coil file. Since the default of this value is very low, it would be out of specification if no TuneUp was executed.

In addition, this QA step verifies that the SAR (Specific Absorption Rate) check is activated. The forward and reflected voltages from both TX Box channels are also tested during the measurement.

Measurement

The TuneUp “Coil Power Loss” measurement is performed.

Evaluation/Results

The Coil Power Loss is measured in CP and EP mode.

- Check if the Coil Power Loss value in the coil file is within specifications and comparable to the measured value.
- Read and check the value for the SAR.
- Read and check the forward and reflected voltages from TX-Box and check them for consistency with broadband power meter values.

Results in Report:

- Phantom position and diameter.
- Measured and stored Coil Power Loss values.
- Measured reference amplitude (AdjTra) from the TALES (TX-Box power meter).
- The measured SAR and voltage values.

8.3.1.10 Cross Term Compensation Check

Help ID: 94F6F45A-CD81-4440-9FDE-F6D1AC3824A4
Service Level: 3

General

The **Cross-Term Compensation Check (CTC)** checks the quality of the cross-term compensation previously adjusted in TuneUp.

In QA mode, the parameters of the inverse filter function which compensates the cross-term effect are checked.

Measurement

A CTC measurement is performed.

Evaluation/Results

Results in Report:

- Cross Term Amplitudes and Time Constants.
- Diagrams visualizing the corresponding cross term amplitudes.

8.3.1.11 Eddy Current Compensation Check

Help ID: ACA062B6-B909-438A-A732-F3BF88BA7502
Service Level: 3

General

The **Eddy Current Compensation Check** (ECC) checks the quality of the eddy current compensation adjustment previously performed in TuneUp.

Measurement

The procedure performs a check of the ECC parameters. The existing ECC settings are used for the measurement.

A preparation measurement is performed to check for the correct positioning of the phantom in the magnet center and for sufficient field homogeneity. The used sequence is a double echo. The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity. The second echo uses a readout gradient and calculates the phantom position in the readout direction.

Evaluation/Results

Results in Report:

- B_0 eddy current values for different time ranges.
- 1^{st} order (gradient) eddy current values for different time ranges.

8.3.1.12 Gradient Delay Check

Help ID: 946ABE8C-323C-48C6-B295-5C45C3DD15AC
Service Level: 3

General

The **Gradient Delay Check** procedure determines the delay time between gradient switched on digitally and the time the actual gradient field builds up. Gradient Delay can slightly differ for each gradient axis.

In QA mode, the Gradient Delay adjustment values, previously established in TuneUp, are is verified.

Measurement

Each gradient is measured at 14 different gradient strengths from -19.5 mT/m to +19.5 mT/m. For an ideal gradient system, all delays are the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, that is, high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, there is some kind of nonlinearity present in the gradient system.

Evaluation/Results

Results in Report:

- Current- and calculated- Gradient Delay values in microseconds.
- Maximum Gradient Delay Deviation values.

8.3.1.13 Gradient Sensitivity Check

Help ID: 16C2854F-2DEB-4B6F-BEDA-E6418994A88B
Service Level: 3

General

The **Gradient Sensitivity Check** procedure checks the Gradient Sensitivity values previously adjusted in TuneUp to ensure that the system uses correct the gradient strengths and measures the correct phantom image size in all orientations.

For the measurement, the distortion correction filter is enabled to compensate gradient field non-linearities.

Measurement

- A standard imaging sequence is repeated three times: one measurement for each orientation.
- The three images are then analyzed to determine the exact location of the center of the phantom and the phantom diameters. The standard readout and phase encoding direction for each orientation is swapped to determine the center and diameter of the phantom. The average of these two results determines the final diameter value.
- The measured values are evaluated and compared with the well-known diameters of the phantom for verification of the correctly adjusted Gradient Sensitivities.

Evaluation/Results

Results in Report:

- Measured phantom diameter in all three dimensions.
- Measured phantom centers in all three dimensions.

- Gradient Sensitivities.
- Phantom images in all three slice orientations.

8.3.1.14 BC Image Brightness Check

Help ID: 4B8C4548-E064-495C-B6B2-AC746D517DD4
Service Level: 3

General

This procedure is used to verify the image intensity (brightness) for the Body Coil which was previously adjusted in the TuneUp step BC Image Brightness..

Measurement

The measurement runs the service sequence “Body_Tra_br” and generates images with the Body Coil.

Evaluation/Results

Both Body Coil channels BC0 and BC1 are evaluated independently from each other. Also the sum signal of BC0_BC1 is checked.

From each image the following values are determined:

- Brightness
- FFT Scale Factor
- TX Amplitude
- RX Gain Factor

The brightness value is the signal intensity. The FFT scale factor is a correction factor to adjust the brightness to the middle of the specified value range (min. and max.). The transverse image is used for determining the FFT scale factor.

The transmitter amplitude for the sequence (180 degree flip angle) is also evaluated. This value is coil load specific, so it depends on the phantom.

8.3.1.15 Image Orientation Check

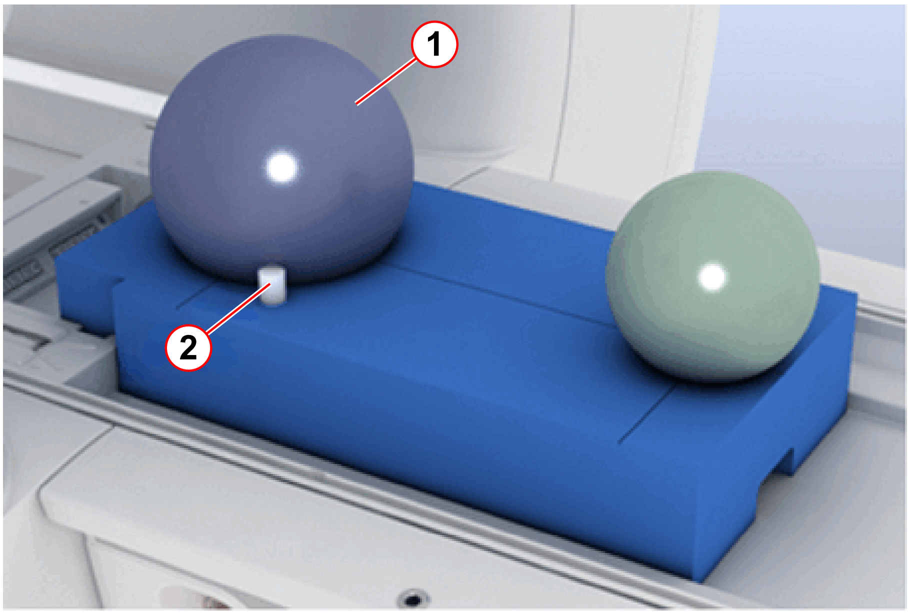
Help ID: F09D8D74-F01E-4A0A-B3FD-A116D3E77C87
Service Level: 3

General

The **Image Orientation Check** verifies the correct position of the phantom setup. It uses an additional, asymmetrically positioned small bottle phantom to determine the correct gradient orientation and polarity.

If any of the gradient power amplifier or -coil connections were swapped or polarities of any gradient axis were wrong, the images would be displayed with incorrect orientation.

Fig. 600: TuneUP_QA Phantom Setup

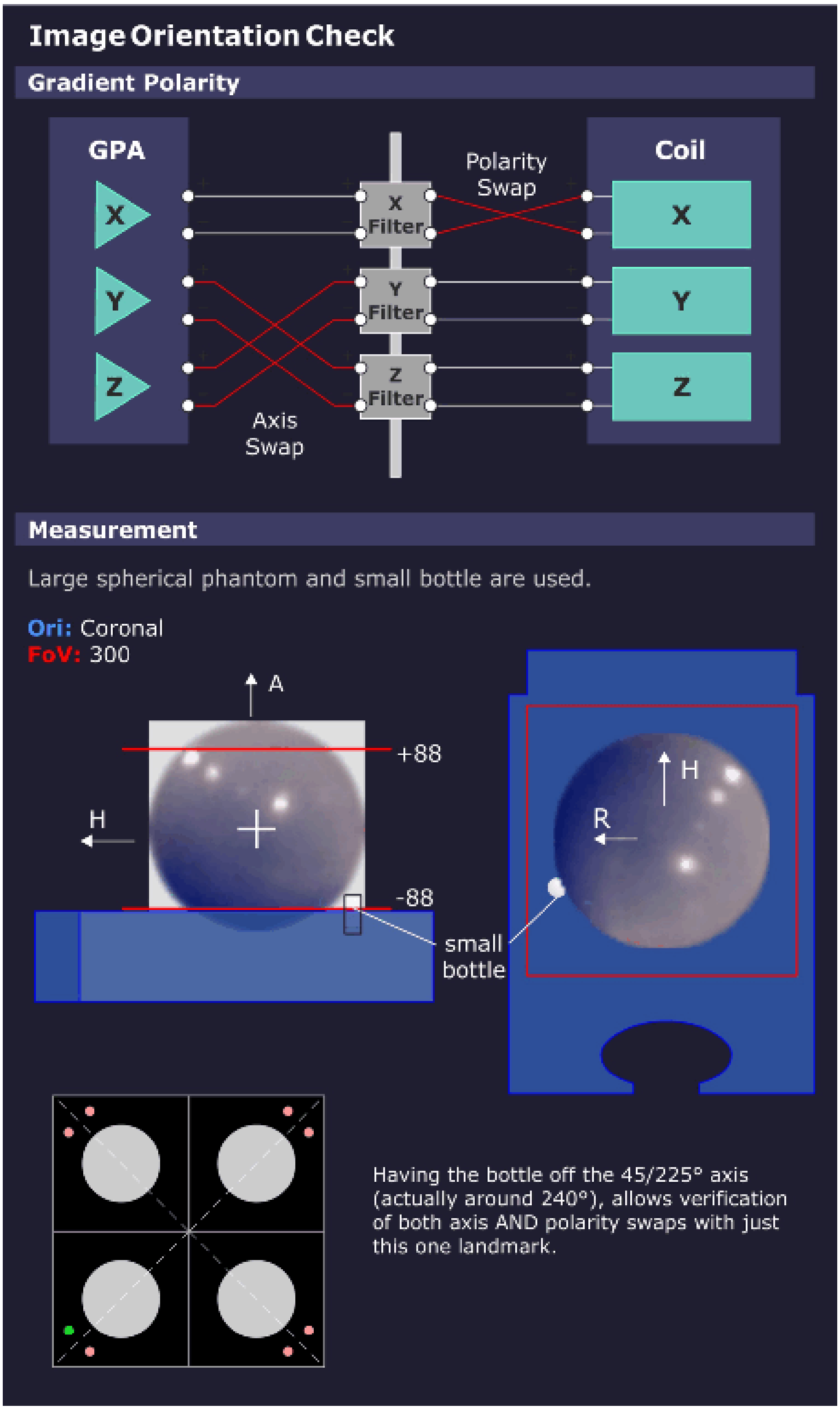


(1) Spherical phantom

(2) Small bottle phantom for Image Orientation Check

Measurement

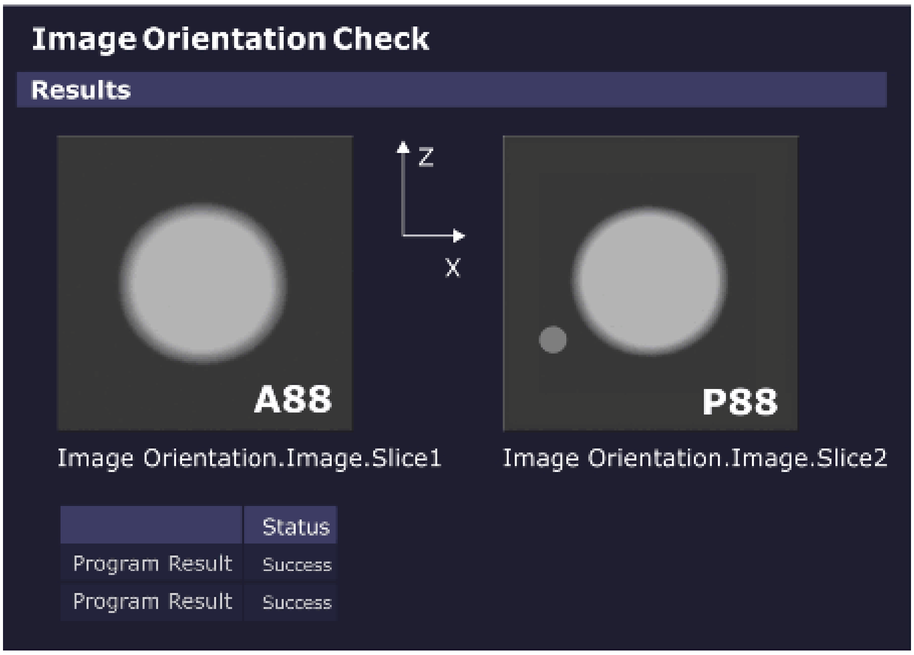
Fig. 601: Image Orientation Check - Measurement



Images are generated from the phantom setup with the additional, asymmetrically positioned small bottle.

Evaluation/Results

Fig. 602: Image Orientation Check - Results



The measured images are automatically evaluated.

If the phantom and bottle are not in the expected position, the procedure returns with the status “Not Ok”.

Results in Report:

- Images showing the phantom geometry (phantom and bottle).

8.3.1.16 Gradient Risetime Check

Help ID: B668AA17-8C5F-4E6C-8908-BB2F522B4F00
Service Level: 3

General

The **Gradient Rise Time Check** measures the minimal gradient Rise Time or the so-called “Slew Rate”.

Measurement

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, five echoes develop because of the bipolar triangle gradient pulses. These echoes should appear, according to the pre-dephasing, exactly at the pulse spikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate could not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Sequence of procedures:

- Verification check of the used phantom.
- Gradient Risetime measurement for all three gradient axes.

Evaluation/Results

Results in Report:

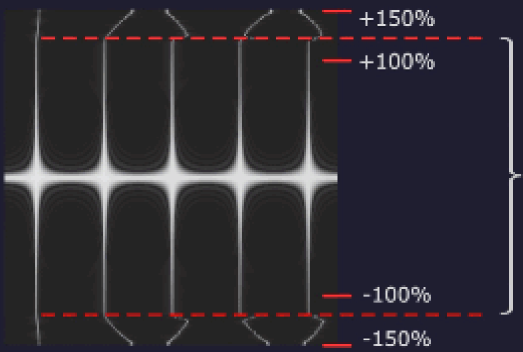
- Graphics of echo maximum versus gradient slew rate, echo 1-5
- Statistics of measured values against specifications
- X/Y/Z-gradient Rise Time results
- X/Y/Z-gradient Rise Time multiple echo output (raw data image, echo 1-5)

Fig. 603: Gradient Risetime Check - Results

Gradient Risetime Check

Results

Gradient Rise Time Results		
Axis	Slew Rate	
		Tolerance
X	112.46 %	≥100 %
Y	111.43 %	≥100 %
Z	112.48 %	≥100 %



X-Gradient Multiple Echo Output

Slewrates and Echopositions						
Echo	Slew Rate Positive Polarity		Slew Rate Neg. Polarity		Diff. of Echo Positions	
		Tolerance		Tolerance		Tolerance
1	**126.68 %**	≥100.00 %	**126.68 %**	≥100.00 %	-0.039	-1.000 ... 1.000
2	**114.67 %**	≥100.00 %	**114.69 %**	≥100.00 %	0.044	-1.000 ... 1.000
3	**113.58 %**	≥100.00 %	**113.58 %**	≥100.00 %	-0.076	-1.000 ... 1.000
4	**112.48 %**	≥100.00 %	**113.58 %**	≥100.00 %	0.041	-1.000 ... 1.000
5	**112.48 %**	≥100.00 %	**112.48 %**	≥100.00 %	-0.039	-1.000 ... 1.000

X-Gradient strength slew rate achieved is 112.483% of specified value
(mean value of the lowest two values of all echoes)

Troubleshooting

In case of problems (result **"Not Ok"**) check the following items:

- Check raw data for spikes. No spikes must be visible.
- Stability/value of mains voltage
- Line resistance
- GPA line transformer and correct line voltage adaptation
- GPA rectifier board
- GPA current sensors (especially if value "Difference of Echo Positions" is out of spec.)
- GPA Power Stages

- Results of QA Regulator Check

8.3.1.17 Spike Check

Help ID: B4E9366D-2E5A-4CE9-969E-3EBDE191949C
Service Level: 3

The **Spike Check** is a quick test to check the MR imaging system for spikes and RF interference.

A "spike stress" measurement is run and its raw data evaluated by a program for spikes. The corresponding image data is evaluated for RF interference.

General

Electrostatic discharges during the measurement may generate some spikes within the raw data. Spikes are individual raw data points or small groups of raw data points with amplitudes above the typical noise level of the MR imaging system. **Spikes** can degrade the MR imaging quality. In this case, the signal-to-noise ratio can be reduced or the image can show a periodic intensity modulation or structure.

One typical source of spikes is the mechanical and electrostatic stress of the MR system caused by the operation of the gradient system.

A simple way to check the MR system for spikes is to run a noise measurement without RF pulses. During the acquisition window massive mechanical and electrostatic stress is generated with rapid switching of gradient pulses with maximum strength.

In case no spikes and no RF interference are present, pure statistical noise shall be received.

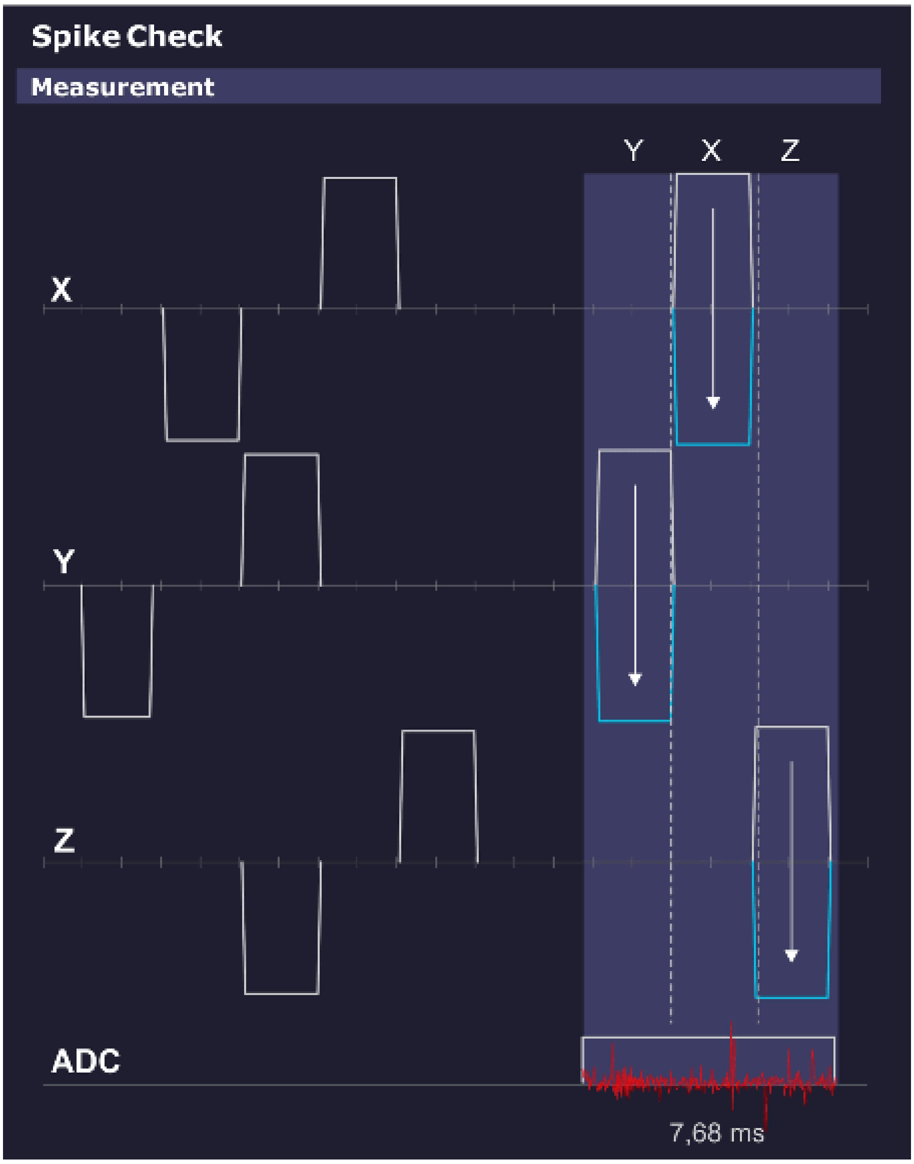
Measurement

This sequence does not apply RF pulses, so no MR signal is generated.

The spike sequence starts with an initial series of strong gradient pulses of both positive and negative polarities on all three gradients intended to cause a buildup of static electrical energy caused by any moving components (cables, covers, etc.).

The ADC is then enabled and again a pattern of gradient pulses with amplitudes ranging from max. positive to max. negative are applied. Any static electrical discharges caused by e.g. moving or loose parts are picked up by the receiving coil (usually the Body Coil), digitized by the ADCs and stored in the raw data.

Fig. 604: Spike Check - Measurement



Evaluation/Results

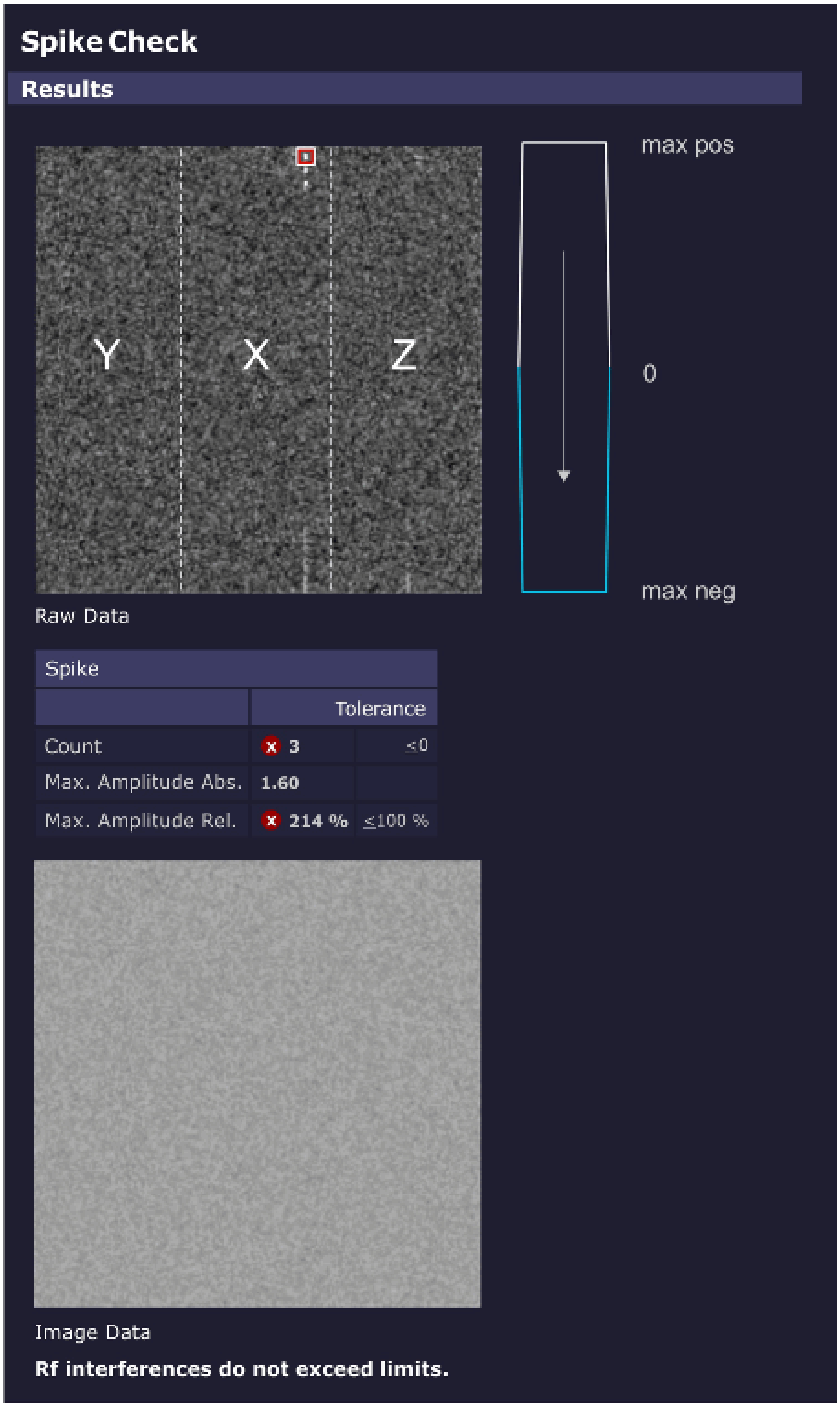
Search for **spikes in raw data** and **RF interference in the image data**.

The evaluation looks for intensities in the raw data set exceeding a threshold (usually noise) and counts both the number and intensities of the spikes and displays them numerically and graphically. The electrical discharges are visible as "spikes" in the raw data.

The position of the spikes in the raw data set relates to the gradient axis that was switched on when the spike occurred. This position may indicate the cause of the spikes.

In case RF interference was present during the acquisition, the disturbances can be seen in the displayed image data.

Fig. 605: Spike Check - Results



Results in Report:

- Raw data image and resulting RF interference image.
- If spikes are found, the following values are displayed:
  - Total number of spikes.

- Magnitude of the highest spike.

8.3.1.18
RF Noise Check

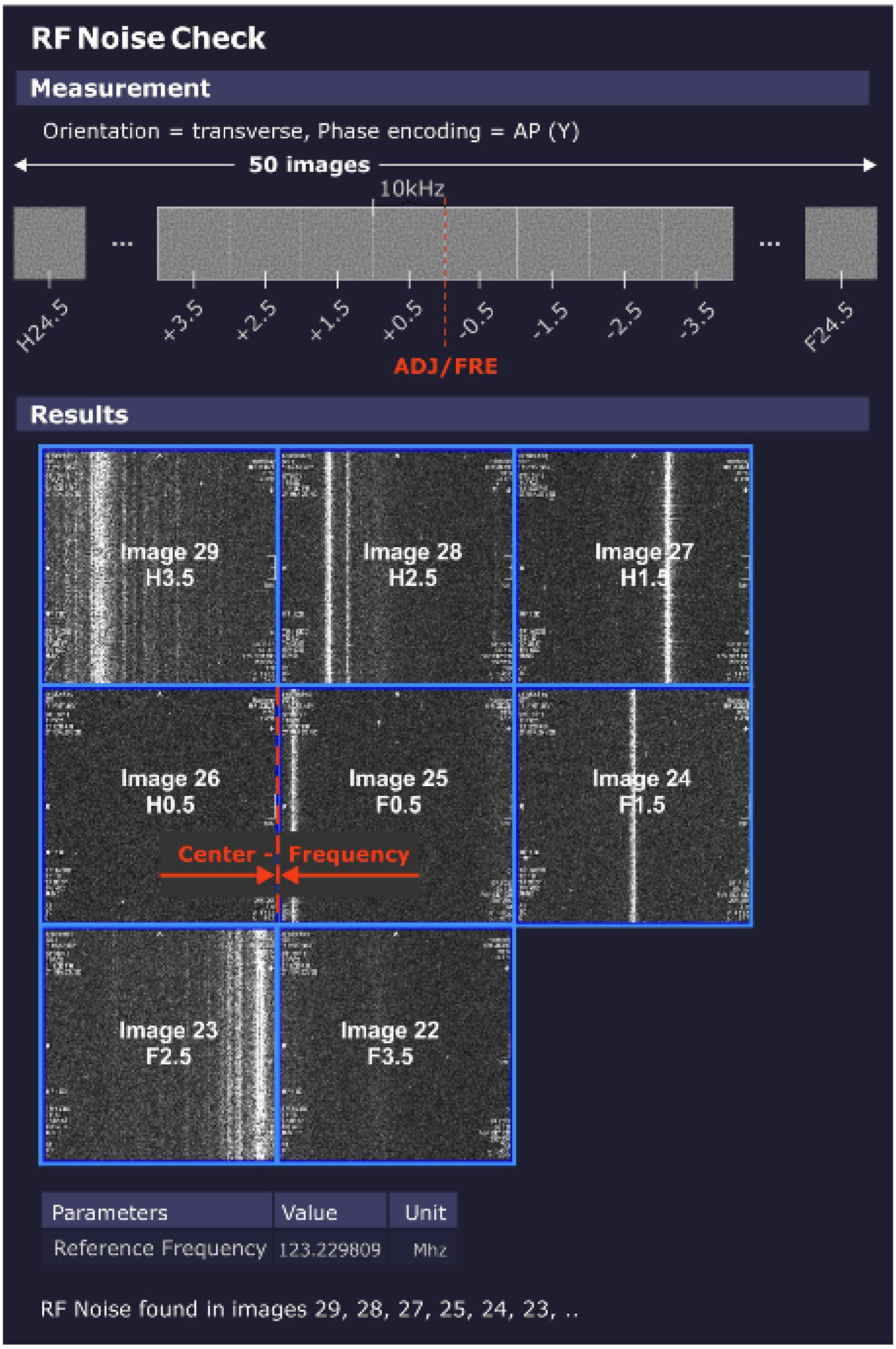
Help ID: AA585552-5E09-43B7-8E85-B9A8394037E2
Service Level: 3

General

The **RF Noise Check** is used to detect RF interferences which can disturb the MR-image.

Measurement

Fig. 606: RF Noise Check



The noise measurement is performed with the service sequence rf_noise. The sequence is without RF or gradient pulses, just an open receiver.

The sequence produces 50 images (i.e., slices). The difference between the images is the variation of the center frequency. Each image covers 10 kHz (256 times 39 Hz) so that the complete receiver bandwidth of +/-250 kHz is covered.

The border between the center slices 25, 26 is the current center MR system frequency.

Evaluation/Results

The evaluation algorithm looks for RF interference in the acquired image data i.e., dots and streak artifacts in phase-encoding direction and noise level.

Results in Report:

- RF interference according to the analyzed frequency bands.

8.3.1.19 PTX Image Check

Help ID: 82D6FBB2-442B-4B71-A95B-FD165371082B
Service Level: 3

General

A parallel transmit (pTX) array consists of 2 or more distinct transmitter channels where amplitude and phases can be modulated independently from each other.

Such a system can generate dedicated local magnetization patterns, e.g. exciting one spatial region of a patient or phantom while leaving other regions untouched.

This mechanism, however, is rather sensible to system instabilities or malfunctions of components in the imaging chain.

In order to detect such deviations, the service procedure runs the dynamic pTX sequence to produce a well-defined local magnetization pattern which is analyzed for image quality.

Measurement

For the measurement, the small spherical 170 mm water-filled phantom is used.

a homogeneous image is produced which allows to verify the correct positioning of the phantom.

The pTX sequence “dyn_pulse” is executed and generates a well-defined target magnetization pattern.

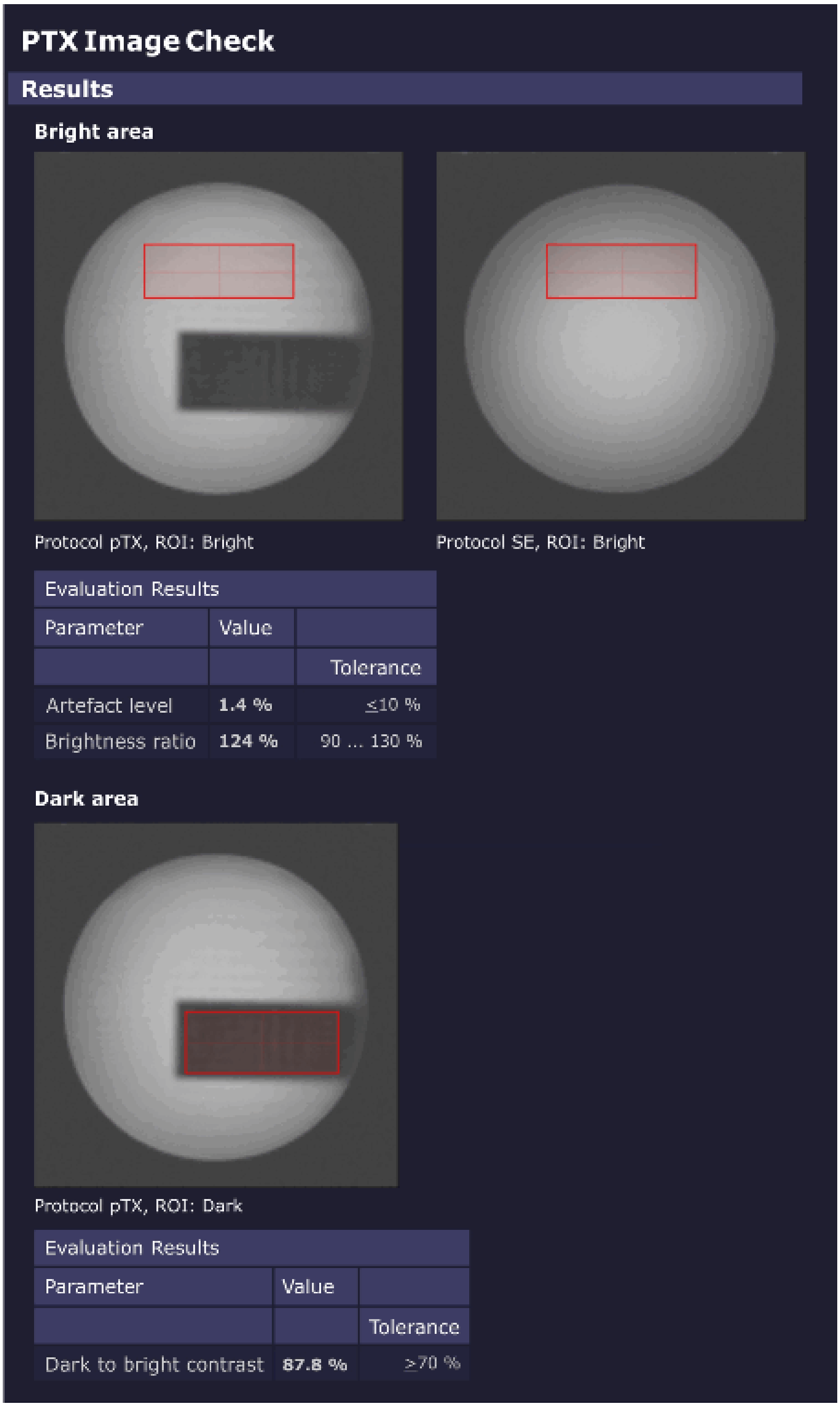
First, a homogeneous image is produced which allows to verify the correct phantom positioning.

Second, a homogeneously excited plane with an embedded dark (non-excited) rectangle is measured.

Subsequently, the Software analyzes the resulting images to detect potential system malfunctions.

Evaluation/ Results

Fig. 607: PTX Image Check - Results



First, a ROI is placed entirely in the **bright region** of the excitation pattern. This ROI is analyzed for relative image brightness (which should be similar to the brightness in the homogeneous image) and artifact level.

A second ROI is placed entirely inside the **dark region** (rectangle). The relative brightness of this second ROI, compared with the brightness of the first ROI, defines the image contrast.

8.3.1.20 Stability Check

Help ID: 796762F3-ECF9-458A-83A1-CAFDEC3BA27D
Service Level: 3

General

The **Stability Check** evaluates the stability of the MR signal under various sequence conditions.

A spin echo and a gradient echo are measured 256 times without phase encoding using fixed sequence parameters.

As a result, all raw data lines look alike (if the system is stable and not taking in account stochastic variations).

The stability of the echo signal across the measured lines is evaluated:

- Variations of the amplitudes
- Variations of phases
- Positions of the echoes
- Frequency perturbations

Measurement

In Standard Mode, a **spin echo** and a **gradient echo** are measured 256 times without phase encoding using fixed sequence parameters:

- TR = 301 ms
- TE = 20 ms

All **three orientations** are measured, so each gradient is performed as readout gradient once:

In addition, both **0 and 50 mm slice shift** positions are measured.

Evaluation/Results

The **spin echo sequence** uses a default setting of TE = 20 ms. Due to chosen TR = 301 ms and the inversion effect of the 180° pulse, it is sensitive to instabilities in the range of a few Hz up to 100 Hz which may be caused by e.g.:

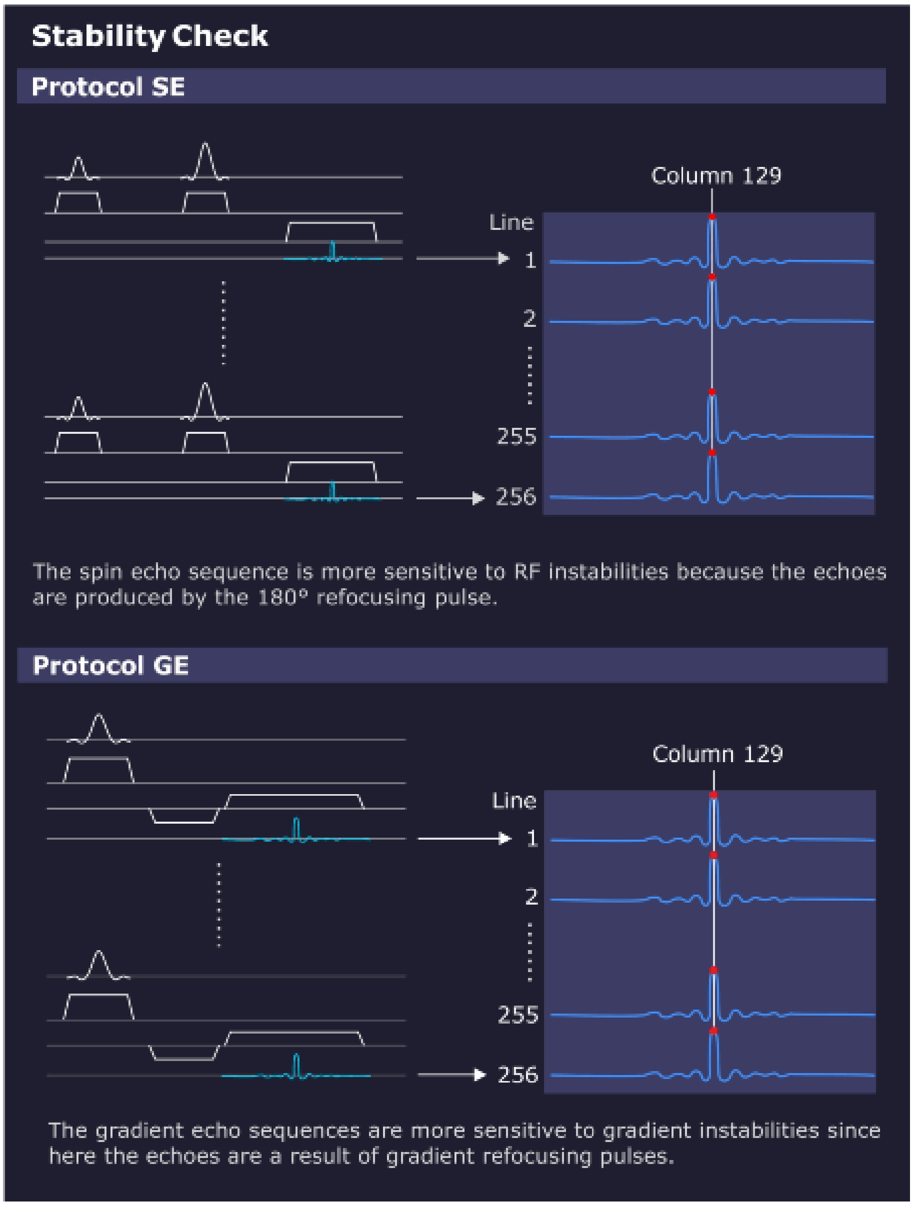
- 50/60 Hz AC power line hum;
- or 16.7 Hz AC trains.

The **gradient echo / flash sequence** also uses a default setting of TE = 20 ms and TR = 301 ms. Due to the chosen parameters and the missing 180° pulse, it is insensitive to 50 Hz line hum. The gradient echo is sensitive from DC up to a few Hz to slow external field interferences caused by e.g.:

- DC-driven local train systems like tram or subway;

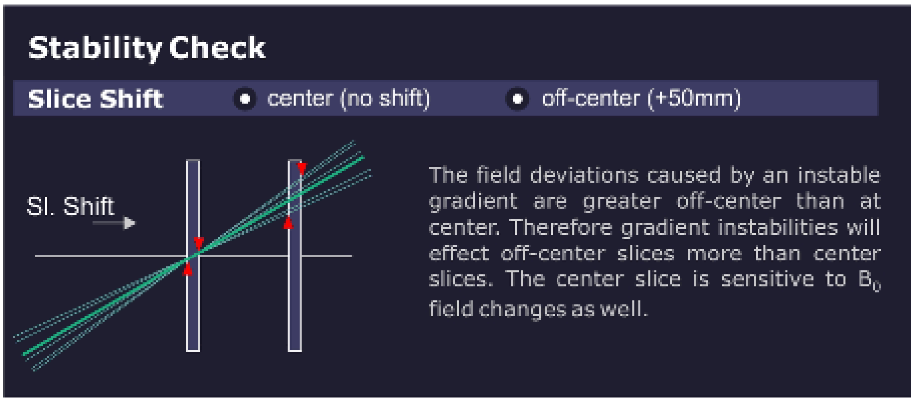
- Magnetic field instabilities;
- or mechanical vibrations.

Fig. 608: Stability Check Evaluation



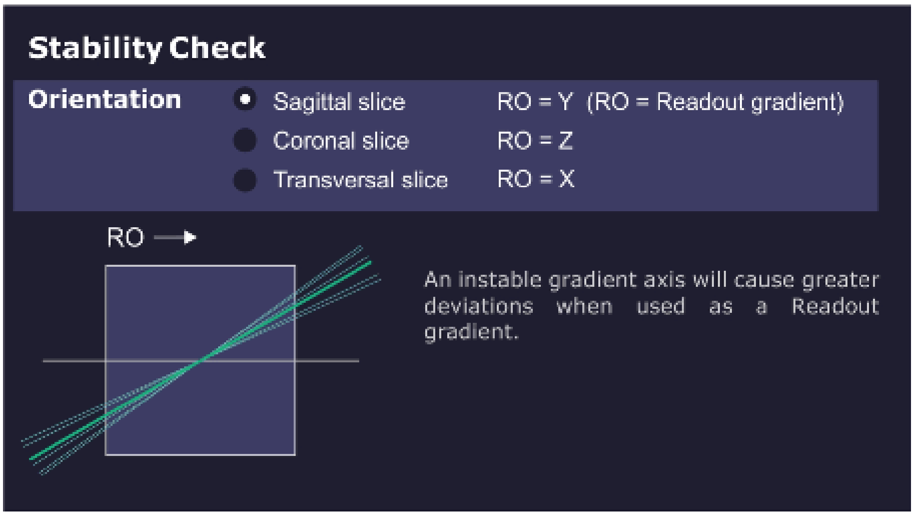
The measurement result of the **center slice** is sensitive to instabilities of the B_0 field. The **50 mm offset slice** is more sensitive to gradient-like instabilities.

Fig. 609: Slice Shift



The **orientation** determines the role of each gradient axis. For example, lets assume there is an instability in the X gradient. The values for the sagittal orientation (X = slice selection) may be in spec. while the spec for the transverse orientation (X = read out) could be out of spec.

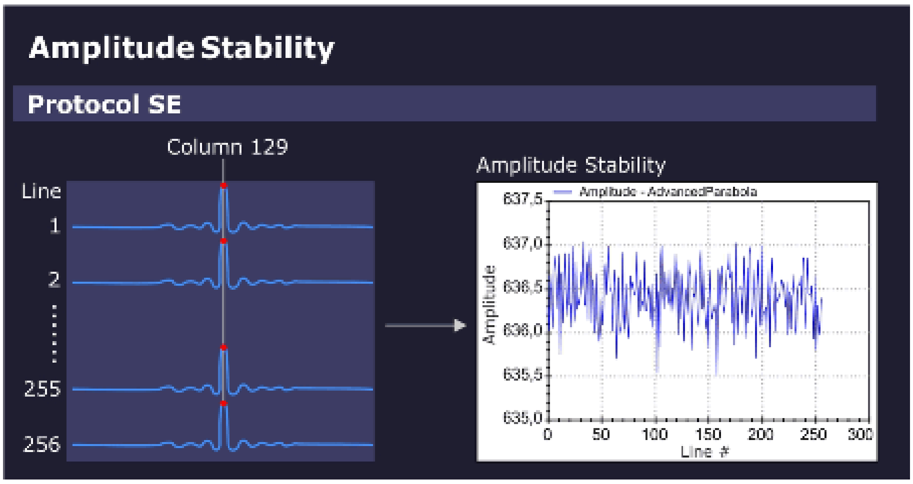
Fig. 610: Orientation



Amplitude stability (spin echo):

The amplitude value in the maximum signal column (e.g. column 129) is displayed in the graphics output.

Fig. 611: Amplitude Stability (Spin Echo)



Phase stability (spin echo):

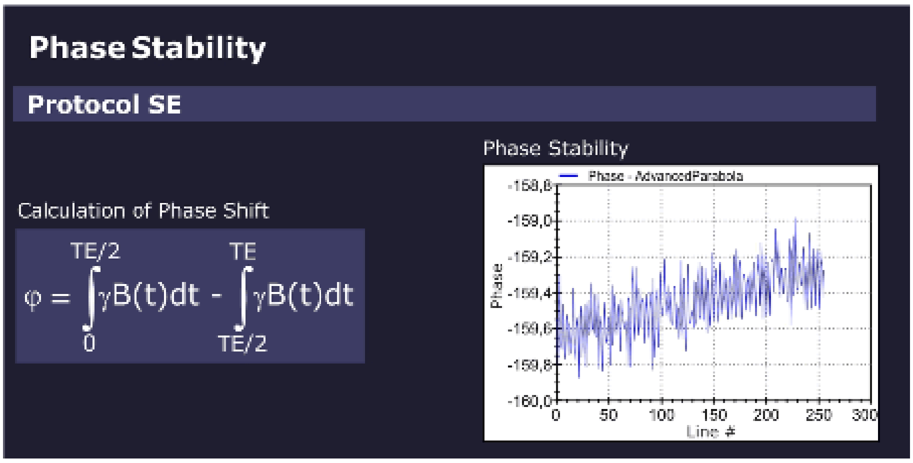
The phase stability of the spin echo is displayed in the graphics output.

The phase shift of a spin echo is determined by the integral across the magnetic field from a 90˚ to a 180˚ pulse minus the integral from a 180˚ pulse to the echo.

For a constant field or a slowly varying field, both integrals cancel out each other.

However, for an oscillating field with a period of TE (20 ms = 1/50 Hz), this expression becomes maximum. In other words, the spin echo measurement with the default setting of TE = 20 ms (and TR = 301 ms) is sensitive to interferences which may be caused by 50/60 Hz AC power line hum or 16.7 Hz trains.

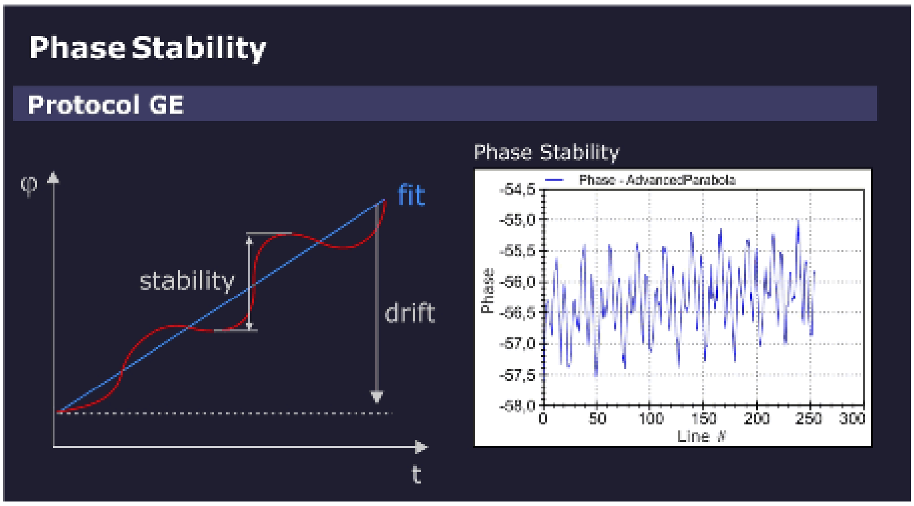
Fig. 612: Phase Stability (Spin Echo)



Phase stability (gradient echo):

For the gradient echo sequence, the phase stability is evaluated by fitting a straight line to the phase data. When you subtract the value of the linear fit from the phase data, the phase stability changes. The slope of the straight line indicates the linear phase drift, caused by slow external field interferences, for example, moving elevators, cars, trucks but also mechanic vibrations.

Fig. 613: Phase Stability (Gradient Echo)



Stability of echo position:

For both measurements, spin echo and gradient echo, the position of the echo peak column is determined for every measurement line and checked against specifications.

The cause of an unstable echo position may be gradient-like interferences from the environment or from the gradient itself.

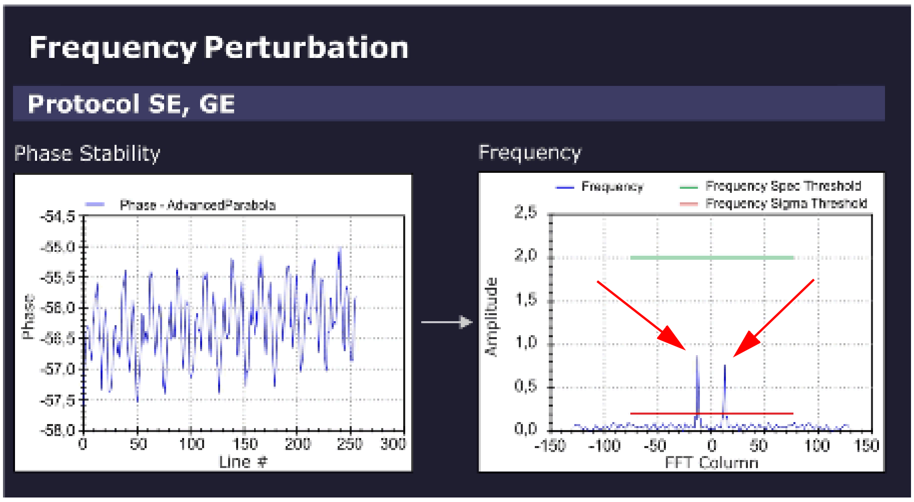
The sensitivity for the direction of the interference is determined by the orientation of the readout gradient.

Frequency perturbation:

The Software differentiates the raw data of the stability measurements and checks for sidebands (ghosts).

In the evaluation result, a frequency perturbation value in percent and the Frequency (peak) position in the FFT Column is displayed.

Fig. 614: Frequency Perturbation (sidebands)



Example: (Frequency Perturbation of 0.8%, Frequency Position -13)

- The sampling rate in the stability measurement is one sampling point every 301 ms (default TR = 301 ms).
- Consequently, the Nyquist-frequency = +- 1/602 ms = +-1.66 Hz. The limiting frequencies at +-128 in the frequency diagram above correspond to +1.66 Hz and – 1.66 Hz, respectively.
- The perturbation frequencies are located at +-13 units, i.e. at 13/128 * 1.66 Hz = +-0.17 Hz.
  » The 300th harmonic of 0.17 Hz yields a 50 Hz interference.

Results in Report:

Measured value	Description
Amplitude Stability	Range of amplitudes (peak to peak)
Phase Stability	Deviation of phase from linear fit
Phase Drift	Drift of phase
Peak Column Stability	Range of peak column
Peak Column Average	Mean peak column
Frequency Perturbation	Relative amplitude of the highest contributing disturber frequency; zero if no frequency is above the relative threshold nor above (noise level * noise factor).
Frequency Position	Position of the highest contributing disturber frequency; zero if no frequency is above the relative (sigma) threshold nor above the (noise level * noise factor).

Troubleshooting

Symptom/Deviation	Possible Cause/Service Action
Amplitude	RF-instability problem, check: ■ Stability of RF System / TX ■ Stability of RF System / RX

Symptom/Deviation	Possible Cause/Service Action
Phase	Magnetic field fluctuation, check: ■ After magnet ramping or after heavy gradient load (e.g. DTI-, EPI- or StabilityLongTermCheck measurement) wait at least 1 hr. before running Stability Check; ■ Run QA-Expert / Field Stability Check.
Depending on orientation	Instability of a gradient axis, check: ■ The Gradient System hardware. ■ Run service sequence “CalcAr” in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.
Independent of orientation	Magnetic field instable or external field interference, check: ■ The Gradient System hardware. ■ See hints “Only in gradient echo”.

Symptom/Deviation	Possible Cause/Service Action
Only slice-shift dependent	Most likely, a gradient problem, check: ■ The Gradient System hardware ■ especially, the gradient axis used for slice-selection
Only in gradient echo	1. Static or slowly varying magnetic field problem? In the vicinity of the site, check for: - Trains; interference from power lines and rails. - Elevators. - Car/truck traffic and parking. 2. Very low amplitude due to static magnetic field (B_o) homogeneity problem? - Compare amplitude with spin echo result (both amplitudes should be similar). - Run QA / Phantom Shim Check. 3. Phase drift out of spec., B_o-drift problem? - System hot? Excessive usage prior to Stability Check measurement (e.g. repeated DTI-, EPI- or StabilityLongTermCheck measurements); - System too cold? Wait at least 2 hr before measurement. 4. Peak column stability out of spec. i.e., fluctuation of echo position due to gradient field instability? - Check the Gradient System hardware - Run service sequence “CalcAr” in all 3 orientations, check images for visible, enhanced ghosting in phase-encoding direction.

Additional hints:

- Switch off components one by one to isolate the cause of the instability, e.g.:
  - Cold head
  - Lighting in magnet room
  - Shim PSU
  - Disable gradient axes with DIP switch at DGSU board (first each single axis, then complete gradient power amplifier)
- For periodic instability, vary TR and TE (Stability Check / Expert Mode) to confirm or rule out the frequency of interference:
  - Vary TR to avoid being synchronous to the interference.

  - Vary TE to become more sensitive to particular frequencies.
- If an external interference is suspected, repeat measurement at different times e.g. during night.

8.3.1.21 Synthesizer Check

Help ID: 100FDE50-E606-4FB0-B37C-3F6B6E932DAF
Service Level: 3

General

The **Synthesizer Check** procedure is an important test to verify proper operation of the synthesizer.

Validation of the frequency and phase quality of the synthesizer is important for the accuracy of the slice position.

Measurement

- Verification of the frequency and phase operation of the RF reference signal is performed by using the MR signal itself. After demodulation of the received MR signal, the reference signal is compared to the MR signal in terms of frequency and phase.
  - In one test, the reference signal is varied in frequency over the whole range of the MR system.
  - A second test modifies the phase in fixed intervals.

    » With this test, the entire operating range of the oscillator can be tested. The test is performed for one RX-NCO (Numerically Controlled Oscillator) from group 1 and one RX-NCO from group 2. The two RX-NCOs are validated and can be used for testing the other NCOs.
- Loop-measurements between the tested RX-NCOs and the TX-NCOs approve the correct function of the TX-NCOs.
- In additional steps, all other RX-NCOs are tested against the approved TX-NCOs by testing group 1 RX-NCOs and group 2 RX-NCOs separately.

Evaluation/Results

Results in Report:

- Frequency- and phase-deviations for all used NCOs.

8.3.1.22 Long Term Stability Check

Help ID: 8B498548-2A22-4DDB-8B42-C258D604AAF0
Service Level: 3

General

The **Long Term Stability Check** determines the scanner stability which is important for Functional MR Imaging (fMRI).

The fMRI application visualizes activities in the brain. The patient is exposed to a periodic optical, acoustical or other form of stimulation. During the exposure, a series of MR images is acquired. Subsequently, the images are evaluated for signal intensity changes. The stimulated, activated regions of the brain are localized by their typical, very little variations of signal intensity.

However, these little variations are easily superimposed by instabilities of the MR system or the environment. This QA step makes sure, they are below a specified level.

Typical sources of instabilities are:

- Local B_0-field variation caused by temperature variation in the shim unit (shim iron plates) generates frequency instabilities. The result is a virtual movement of the measured object during acquisitions.
- B1-field variations due to instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, also leading to virtual movement of the object.
- Mechanical vibrations, e.g., by the cold head, produce signal intensity variation caused by the movement of the measured object relative to the magnet isocenter.
- Movement of phantom fluid causes phase errors.

Measurement

A series of 512 measurements (EPI_FID sequence) with 11 slices is performed and repeated once again. The first measurement is performed with a low flip angle of 30 degrees for stability evaluation of the B1-field (i.e., caused by instabilities of the TX chain).

The second measurement uses a higher flip angle of 90(70) degrees to minimize the influences of instable RF but focus on B_0-instabilities and gradient-like interferences.

> Wait at least 15 minutes between positioning the phantom and starting the measurement. Even small movements in the phantom liquid have a negative effect to the measurement results.

Evaluation/Results

The Software places an evaluation ROI in the central slice and performs the following graphical and numerical evaluations:

- Mean values of a 15x15 ROI related to the mean value of all measurements as a function of the image number
- Standard deviation of mean values of a 15x15 ROI (linear and logarithmic plot)
- Measurement protocol values

- Statistics of mean values (for ROI size 15x15 and 25x25):
  - Mean Image Intensity
  - Peak to Peak Variation
  - Relative Peak to Peak Variation
  - Standard Deviation
  - Total Relative Linear Drift
  - Standard Deviation from Linear Fit
- Image quality parameters such as SNR and ghosting

8.3.2 Coil Quality Assurance

Help ID: 3020AFB6-8C57-418D-AFDD-9141803D9BF4
Service Level: 3

The QA procedure **"Coil Check"** checks the function and quality of each local coil individually.

> Before Coil QA, configure new coils in **Admin Portal / Technical Configuration / MR Configuration / Coil Configuration.**

Coils can be also added under **Customer QA**.

8.3.3 Quality Assurance - Expert

Help ID: D2F41F74-14FD-41F9-AD9C-601B24C3C3AE
Service Level: 7

Introduction

For some of the QA procedures, an **Expert Mode** is provided.

Depending on the selected QA step, the user interface shows additional parameters on the right side.

> Expert Mode should be only used by qualified operators.
>
> This mode provides additional settings and configurable parameters which require a more detailed knowledge of the underlying physics.

8.3.3.2
RF Verify

Help ID: C7354DE7-20D8-4639-A1DA-911E628FB7AF
Service Level: 7

Expert Mode

In Expert Mode, the measured TuneUp RF Characteristics MNO for hydrogen or the different x-nuclei frequencies can be verified individually.

Available additional options:

- Disable storage of RF Characteristics after the measurement.
- Modification of maximum RF power allowed.
- Selection of TX channel, RF path and nucleus (according to licenses available).
- Option to switch off the frequency adjustment.
- Option to disable the switching to dummy load.

8.3.3.3
Feedback Loop Calibration Check

Help ID: 4433E809-00E7-430B-9826-64F3AD4DFC7B
Service Level: 7

Expert Mode

This Expert Mode step is only available for single TX channel systems.

Feedback Loop Calibration Check in **Expert Mode** offers the possibility of either checking the feedback loop phase difference, the feedback loop delay, or both values by a single program call up.

General

The Feedback Loop Calibration Check procedure verifies the internal feedback loop parameters of the TX-Box which are important for an accurate and stable regulation of the B1-transmit field.

Measurement

For the measurement, the “tuncal” service sequence is executed and runs an RF signal with triangular pulse shape over a wide range of frequencies to both BC systems. The signal is measured through the directional couplers (DICOs) of the TX-Box. Then, the Software analyzes the frequency dependence of signal phases.

A linear curve is fitted to the frequency dependence of the signal phases. From the slope of this line, the feedback loop delay is obtained.

This delay is checked against the previous delay value from TuneUP which was stored in the TX-Box. If the deviation is outside the specification range, the step returns a “Not Ok”.

Evaluation/Results

Results in Report:

- Delay evaluation
- Dispersion of delays from single BC elements
- BC phase plot

8.3.3.4 Phantom Shim Check

Help ID: 897AF1C5-8F5A-4AED-B8F7-65DC5BA57946
Service Level: 7

Expert Mode

A coil selection is available in the Expert Mode.

8.3.3.5 Gradient Risetime Check

Help ID: 16EA805E-B413-4008-B505-8101C4724E20
Service Level: 7

Expert Mode

This procedure determines the minimal rise time or maximum slew rate of the gradient system.

The Expert Mode UI allows the user:

- To select the measurement coil.
- To select the gradient axis.
- To enable additional stress pulses for the measurement.

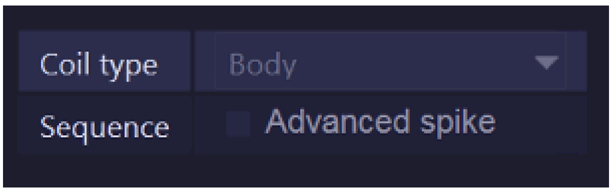
8.3.3.6 Spike Check

Help ID: EDA77FDC-B943-4DBD-838C-2D0545E0D309
Service Level: 7

Expert Mode

For the **Spike Check** in **Expert Mode**, an additional sequence, **Advanced Spike**, is offered.

Fig. 615: Spike Check - Expert



In some cases, spike artifacts may be visible with customer protocols, but not with the regular spike check. If so, it is advisable to reproduce spikes and troubleshoot with the same setup (coils, protocols, position of patient table, Physiological Measurement Unit, etc.) which is used by the customer.

If the artifacts cannot be reproduced at all and also all typical spike sources (see Services Knowledge Base) are ruled out, contact RSC/ HSC.

> **NOTICE**
>
> **The "Advanced Spike" procedure contains a powerful gradient pulse scheme which induces high mechanical stress over a wide frequency range. It shall be therefore used very careful and only when all other means are exhausted.**
>
> **Risk of hardware damage!**
>
> » **DO NOT USE the "Advanced Spike" procedure without guidance from RSC/ HSC!**

8.3.3.7 RF Noise Check

Help ID: 3EACA50D-1597-4F09-A708-0AE6878D8DB0
Service Level: 7

Expert Mode

The Expert Mode user interface allows selection of a coil.

In addition, the operator can select either select the full frequency range of +/- 250 kHz or focus on a single frequency band to reduce the measurement time for troubleshooting.

8.3.3.8 Synthesizer Check

Help ID: 9D7F9FAC-6FC1-475E-8399-8CC21DFA8A4E
Service Level: 7

Expert Mode

A coil selection is available in the Expert Mode.

8.3.3.9 Coil Check - Measure

Help ID: C7AD132A-5800-41DD-A778-77EB9BA7C47D
Service Level: 7

Expert Mode

Coil Check - Measure provides the option to individually select a coil.

In addition, the uniformity evaluation can be enabled or disabled for the acquired QA image.

8.3.3.10
Coil Check - Evaluate series

Help ID: BE987CDD-C2E1-4636-84BD-FED886C9E9D1
Service Level: 7

Expert Mode

Coil Check - Evaluate series provides the option to perform a standard Coil Check evaluation with images that are stored in the database. No measurement is performed in this case, only the appropriate Series ID(s) have to be entered.

In addition, uniformity evaluation for the image(s) can be enabled or disabled.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

8.3.3.11
Field Stability Check - Measure

Help ID: B0C73F74-790C-412B-B992-D1B251EF274D
Service Level: 7

Expert Mode

This procedure checks the stability of the magnetic field and corresponding resonance frequency over time. It is useful to evaluate (for example after magnet ramping) if the field is stable enough for imaging or service procedures.

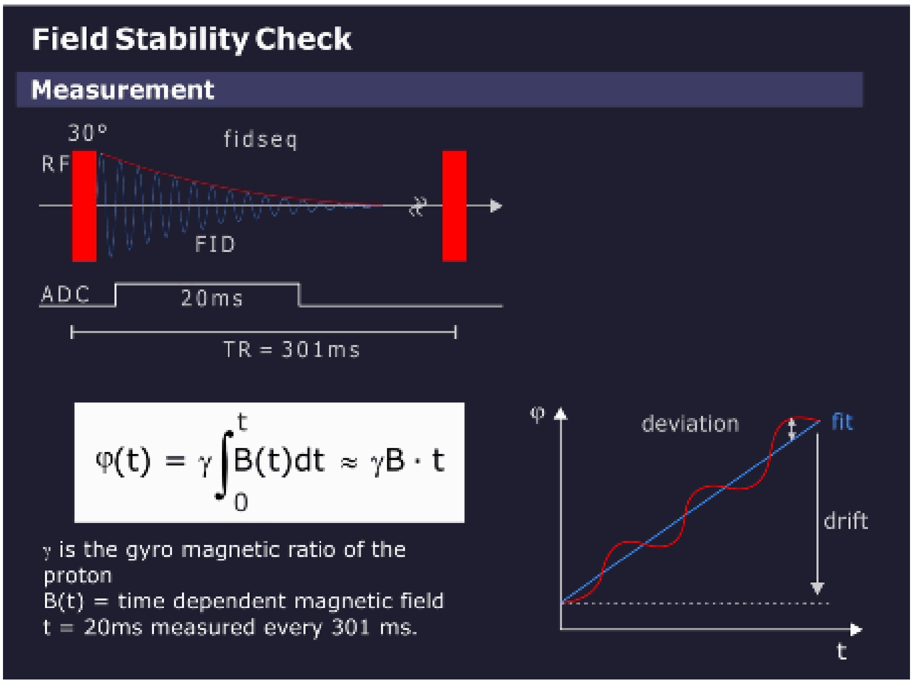
i The Field Stability Check is only available in Expert Mode.

i Allow for sufficient magnet stabilization time before performing Field Stability Check.

Measurement

A series of simple FID (Free Induction Decay) measurements is performed, see graphics below.

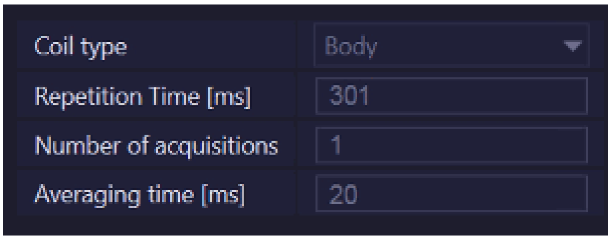
Fig. 616: Field Stability Check - Measurement



A total of 512 FIDs are measured over a 154 second period (repetition time of 301 ms). From each FID, the phase over time is calculated and a linear fit is performed, the slope giving the frequency.

The time range of the fit should be long to get a high accuracy but is usually limited by the length of the FID. A good compromise is 20 ms, which averages over 50 Hz line hum which is usually of no interest in this measurement.

Fig. 617: Field Stability Check - Parameter selection



Coil type	Body
Repetition Time [ms]	301
Number of acquisitions	1
Averaging time [ms]	20

The following parameter selections are offered:

- Coil type
- Repetition time
- Number of acquisitions
- Averaging time

The measurement time with default parameters is 512 x 301 ms = 154 s.

Evaluation/Results

The result of the field measurement is the field (= frequency) over the measurement time. As a measure of quality, the linear drift over 10 min (= typical measurement time of a long imaging sequence) and the maximum deviation from this linear drift are taken.

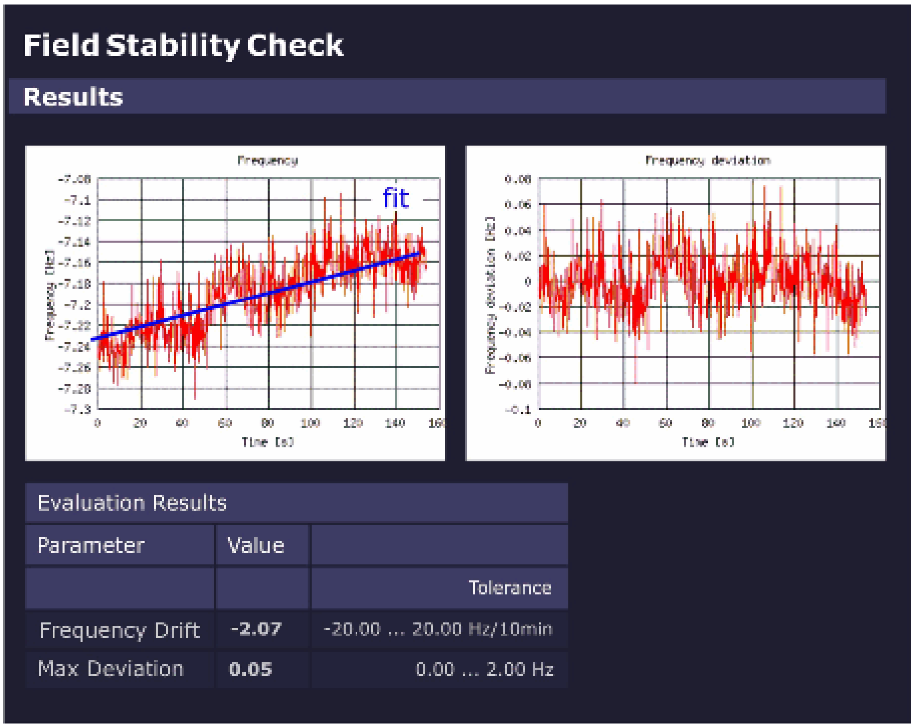
If the selected measurement time is different from 10 min, the drift is calculated for 10 min.

A drift over 10 min should not exceed the pixel bandwidth of a low-bandwidth sequence (e.g., 20 Hz).

The maximum deviation is important for gradient echo sequences to avoid smearing. To allow long echo times, the maximum deviation should be typically less than 2 Hz.

The graphics below shows the frequency variation and the drift-corrected frequency variation across the measurement time.

Fig. 618: Field Stability Check - Results



8.3.3.12 Field Stability Check - Evaluate series

Help ID: 3CD05F02-FE32-4F51-94C3-41BC02E48AEA
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former measurement without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

8.3.3.13
Stability Check - Measure

Help ID: 6CA579E3-92EF-4DEE-8A41-F97C7EF41B1C
Service Level: 7

Expert Mode

Fig. 619: Stability Check Measure - Parameter selection

The Expert Mode function **Stability Check - Measure** evaluates the stability of the MR signal under various sequence conditions.

Different to the Stability Check in the General QA workflow, it allows the following parameter selections:

- Coil type
- Sequence type:
  - Spin-echo
  - Gradient echo (Flash)

  As default, the Spin-echo sequence is selected. This is an exclusive setting so for each **[Go]** in Expert Mode only one sequence selection is possible.
- Repetition time TR and echo time TE

  If the user selects a TR or TE value that is not valid for the selected sequence, an error message is generated when **[Go]** is selected and the valid input ranges are displayed.
- Slice orientation
  - Sagittal
  - Coronal
  - Transverse

  All orientations are selected as default, but a single one, two or all three can be selected.
- Slice shift
  - 0 mm
  - +50 mm

  Both are selected as default. At least one is always selected.

8.3.3.14 Stability Check - Evaluate series

Help ID: 73725B3F-DBC1-40C2-B85C-664143D89A0C
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former measurement without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Press **[Go]** to start the evaluation.

8.3.3.15 Long Term Stability Check - Measure

Help ID: DDB58181-821E-4F99-8303-058EFD3295C6
Service Level: 7

Expert Mode

The Expert Mode allows the user:

- to select a coil for measurement (usually the standard Head Coil for the system is used);
- to configure measurement parameters (see below);

Fig. 620: Long Term Stability Check Expert - Parameter selection

Tab. 34 Long Term Stability Check Expert - Parameter description

Measurement Parameter Selections	Description
Coil type	Coil to be used for Long Term Stability measurement (usually standard Head Coil of the system).
Number of slices	▪ Multi slice 3, 11, 15 or 17 slices (default = 17 slices) ▪ Single slice
Slice number	Number of slice to be evaluated (default = central slice).
Repetition time	in ms (default = 1000ms)
Echo time	in ms (default = 30ms)

<table>
<tr><th>Measurement Parameter Selections</th><th>Description</th></tr>
<tr><td>Flip angle</td><td>Flip angle in degrees:
30 degrees for evaluation of B1-instabilities (i.e. instable transmit components or TX-coil).
for evaluation of other instabilities e.g. B_0, dynamic field perturbations:
90 degrees (for 1.5 T systems)
70 degrees (for 3T systems, due to dielectric resonances in the water phantom)</td></tr>
<tr><td>Number of measurements</td><td>(default = 512) may be reduced for troubleshooting e.g. 256 to save measurement time.

Note:
The No. of measurements should not be reduced too much below 256 to maintain a sufficient statistical basis for evaluation.</td></tr>
<tr><td>Number of preparation scans</td><td>Preparation scans for the magnetization (default = 5).</td></tr>
<tr><td>Number of warm up repetitions</td><td>Warm-up repetitions. Several EPI_FID measurements (default = 60) should be run prior to the “real” measurement to bring the MR system into thermal equilibrium state. No evaluation is performed for the warm-up scans.</td></tr>
<tr><td>Swap orientations</td><td>Swap readout and phase-encoding gradient.</td></tr>
<tr><td>Slice orientation</td><td>Transversal (= default)
Sagittal
Coronal</td></tr>
</table>

Measurement Parameter Selections	Description
Phase correction	▪ None – no correction is applied. ▪ 3-Line – reduction of N/2 ghosts. – the first three echoes (not phase-encoded) are used to correct the further echoes. ▪ 3-Line B0 (= default) – reduction of N/2 - ghost, in addition B_0-field drifts are compensated. – the first three echoes (not phase-encoded) are used to correct the further echoes. ▪ Full – optimal correction for effects accumulating during the echotrain, but time consuming; measurement time is doubled, also for multiple measurements! – there is a prescan with full number of echoes, but not phase-encoded. All echoes in the next scan are corrected by means of that prescan.
Arterial spin labeling evaluation	Switch off/on additional evaluation function for Arterial Spin Labeling (ASL) application.

8.3.3.16 Long Term Stability Check - Evaluate series

Help ID: E13785FD-58F4-47E8-9695-9467FB0975E4
Service Level: 7

Expert Mode

In Expert Mode, the operator has the possibility to evaluate data of a former Long Term Stability Check without performing a measurement.

Workflow:

- Go to the Patient Browser.
- Select the series for evaluation and export it to the local file system.
- Enter the full path of the chosen export folder in the input field **Series ID**.
- Enter the Slice number for the evaluation function.
- If applicable, select the Arterial spin labelling evaluation function (select radio button **[enabled]**)

- Press **[Go]** to start the evaluation.

9.1 MPPS Completion

Help ID: 5AFA6A1A-1E4E-4E4F-ADE6-16E0F3386B96
Service Level: 1

Automatic

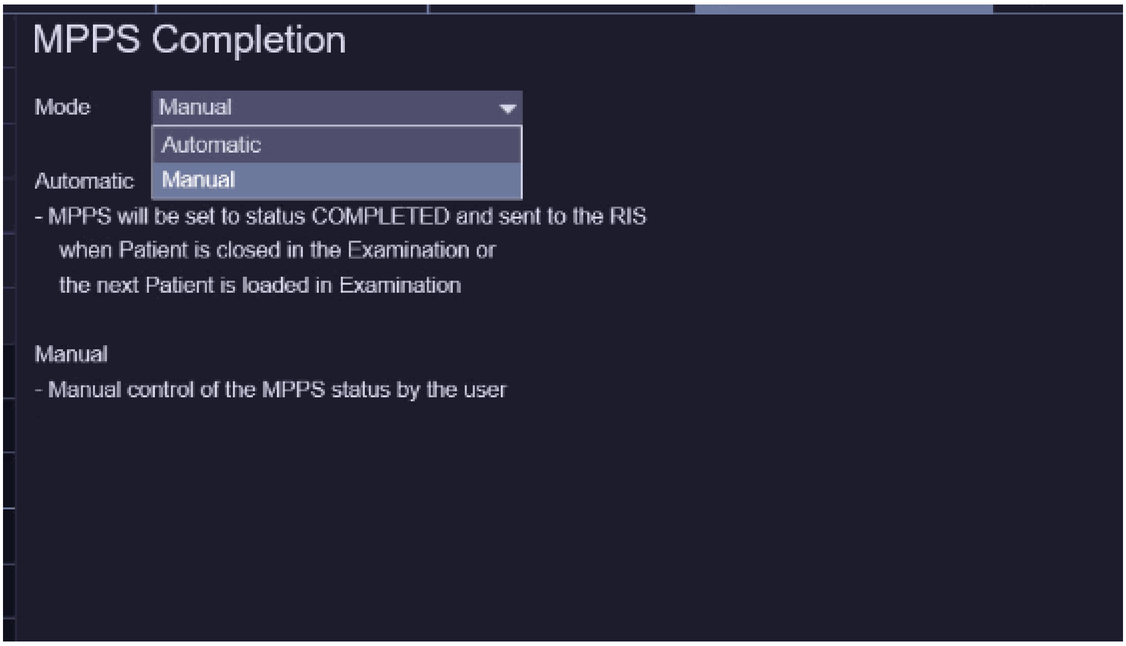
MPPS will be set to status "COMPLETED" and sent to the RIS when closing the patient in the Examination or when the next patient is loaded in Examination.

> Manual completion is always possible, regardless if automatic completion is enabled or not!

Manual

When the procedure of a procedure step is completed by the user in the "Examination UI" or in the "Organizer"

Fig. 621: DICOM - MPPS Completion



9.2 DICOM Study Description

Help ID: E896AF24-BD11-48E0-8112-77217DD54237
Service Level: 1

Modality

Dot Cockpit: Region^Examination

Modality including Program level

Dot Cockpit: Region^Examination^Program

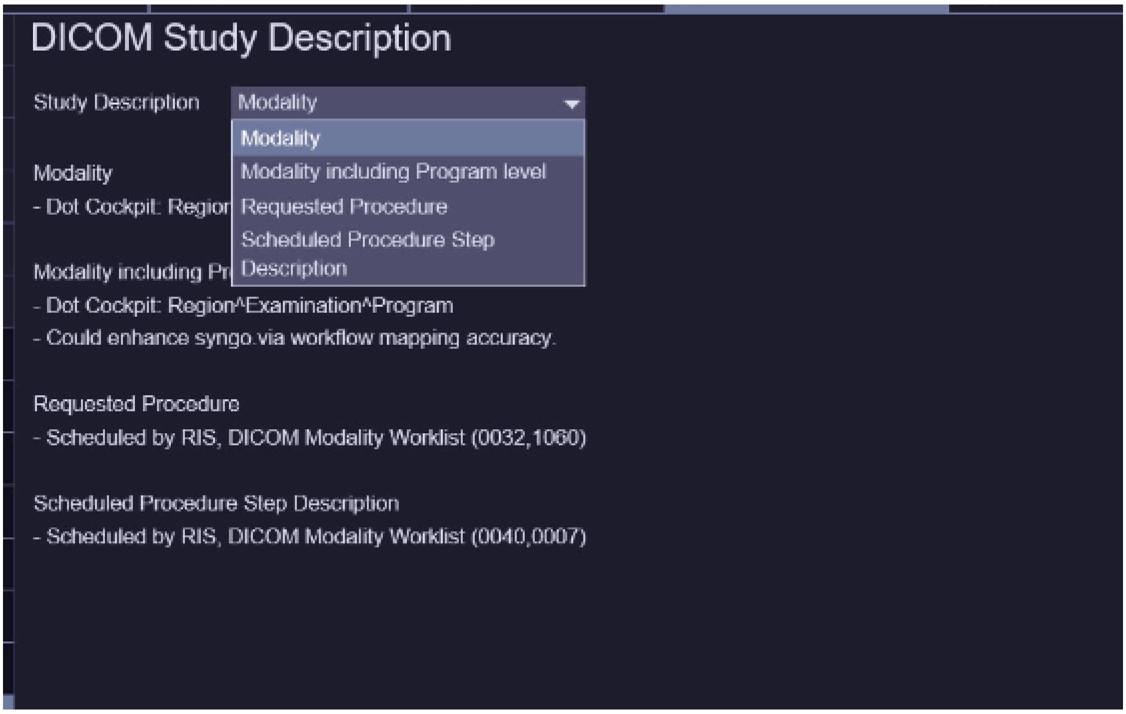
Requested Procedure

Scheduled by RIS, DICOM Modality Worklist (0032,1060)

Scheduled Procedure Step Description

Scheduled by RIS, DICOM Modality Worklist (0040,0007)

Fig. 622: DICOM Study Description



10.1 Supervision

Help ID: 88637B79-E430-42A1-BA0A-0376FD4B2A21
Service Level: 5

General

The button **Tools / Supervision** opens the MR service application **Supervision**.

It allows the user to enable and disable the hardware supervision for various components. During error analysis, this application allows for error messages issued by hardware components to be ignored (Supervision: **OFF**), or to be treated as non existent (Configuration: **Virtual**). The supervision status can be changed whenever individual components should not be monitored during troubleshooting.

This feature for example enables the service engineer to also troubleshoot bus-systems. It is possible to run tests using a minimum hardware configuration (e.g. in RFCEL), meaning removing boards and nodes from the communication links.

> The supervision may only to be used by trained service personnel!

> These manual settings are kept even if the Service Software is closed. The default settings are restored with the next host reboot or restart of the system.
>
> Before patient examinations all components **Supervision** must be set to **On** and the **Configuration** to **Real**!

Fig. 623: Example Supervision

The actual components are different for the different system types.
Also a installed Multi Nucleus option (MNO) will be visible in that list.

The following settings are possible:

- **On**

  The component is supervised and the error messages from the peripheral units are monitored (normal setting).

- **Off**

  The error messages from the peripheral units are not monitored, but the data transfer and acknowledgement from the component are still supervised.

  Use this mode to suppress/ignore incoming error messages from the peripheral units and keep the measurement system operational.
- **Real**

  The component is installed and supervised (normal setting).
- **Virtual**

  The component is treated as existent, but the data transfer to and the acknowledgement from the component is only simulated.

  This mode may be used for troubleshooting when a test measurement must be started without supervision of a certain component. This is not usually necessary, however. Using this mode may lead to other problems, as the simulation is not perfect.

Furthermore the following controls are available:

- The button **X** in the header bar is used to close the window.
- After pressing the **Save** button saves and activate the changed supervision mask.
- With the **Restore factory settings** button everything is reset to normal. After pressing, usually everything is set to "On" and "Real".
- **Update** updates the contents of the window.

Components

Examples of components that can be configured:

Tab. 35 System Components within Supervision (example MAGNETOM Vida)

Group	Component	Description
Host	MRC	MRAWP
MARS	MCIR	MaRS
RCCS	RFCELK2264_RCCS	IF_RCCS
RFIS	RFCELK2264_RFIS	BC_Dyn & LC_Dyn (PIN Control)
RFCEL_Coil	RFCELK2264_Coil	Coil_ID Identification
RFCEL_SAT	RFCELK2264_SAT	RFCEL-Satellite
Intercom	RFCELK2264_INTERCOM	Intercom interface and Intercom board
MOCO	RFCELK2364_MOCO	Motion correction unit
CAN24	CAN24K2304	CAN bus and 24/7 monitoring unit
RFCel-COM	RFCELK2264_COM	Controller of RFCEL_COM

Group	Component	Description
OPEC_TX	OptoExpressCard20_Tx_1	Transmit Opto Express card
DSP	RFCELK2264_DSP	PALI resp. Stimo functionality
VLM	VLMK2309	VLM
HEC	HECK2366	Helium compressor (refrigerator)
PTAB	PTABK2363M	Mobile Patient Table
Controller	PCIe	PCI-bus components
MSUP	MSUPK2306	Magnet Supervision
MPS	MPSK2276	Magnet Power Supply (if connected)
CPAN	CPANK2363_FL	Scanner Control Unit front left-side
CPAN	CPANK2363_FR	Scanner Control Unit front right-side
CPAN	CPANK2363_RR	Scanner Control Unit rear (optional)
Rx	RxK2304_64Ch_1	DIG_RX_1
RX	RxK2304_64Ch_2	DIG_RX_2
OPEC_RX	OptoExpressCard20_Rx_1	Receive Opto Express card
RSR	RFCELK2364_RSR	Respiratory unit
Tx	TxK2304 TxK2304_2Ch	TX-Box 2-CH TX-Box
GPA	GPAK2368	Gradient Power Amplifier
SHIM	SHIMK2259	Shim power supply
PMU	PMUK2301	Physiological Measurement Unit
AirCooling	ACSK2300	Cabinet Cooling Unit
WaterCooling	SEPK2360	System Separator (passive)

10.2 Tools

Help ID: 89679419-B578-45E0-80D4-F3CBB07FC195
Service Level: 7

General

The button **Tools** opens the MR service application **Tools**. It allows for some settings of the measurement system to be changed, which may prove helpful during troubleshooting.

The change of the settings is only valid for the Service Patient. The settings are reset to patient mode with the registration of the next patient and with each system reboot.

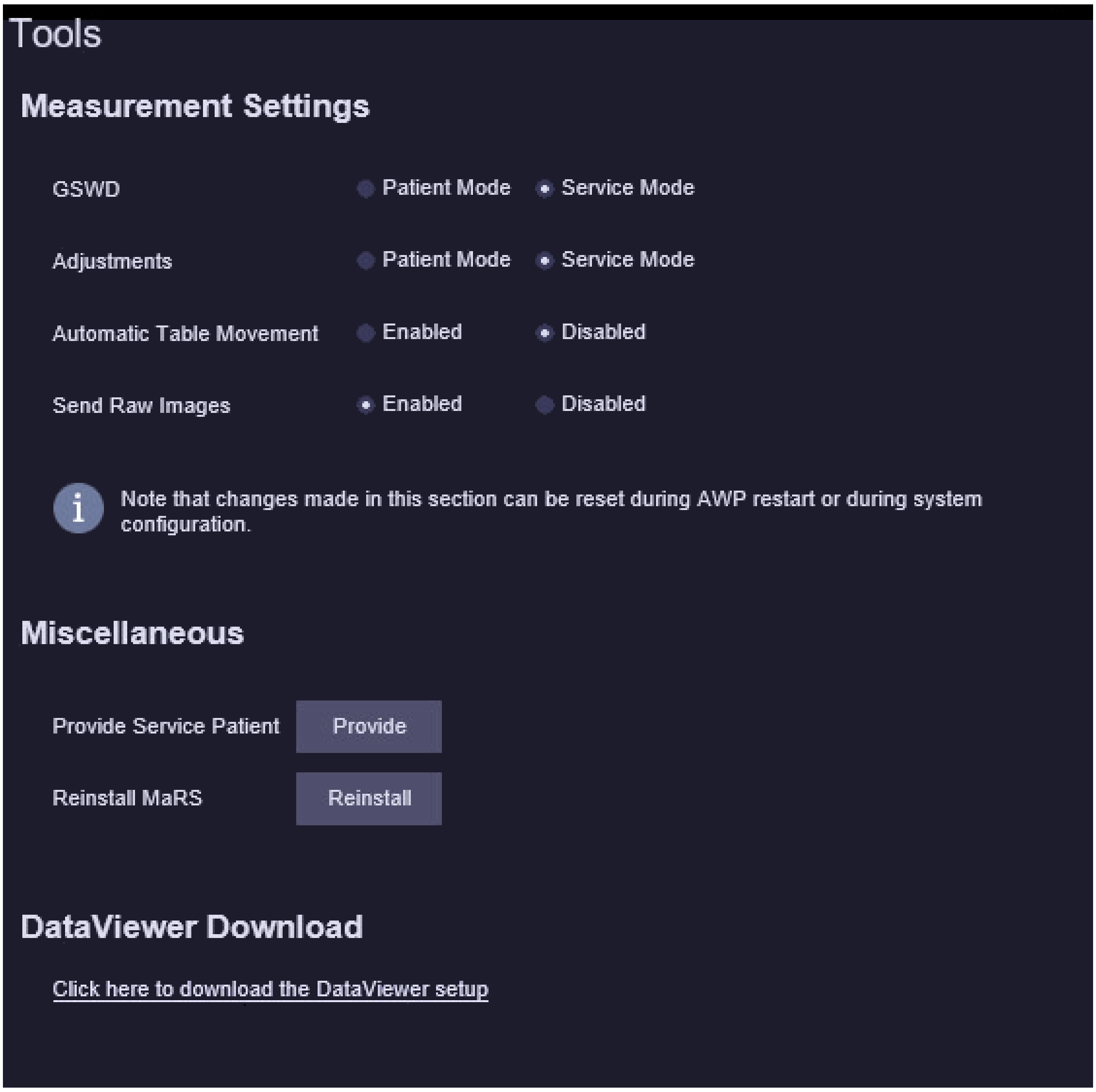
Before you perform any service workflows you need to configure the required Measurement Settings and register the Service Patient below Miscellaneous by selecting Provide Service Patient: **Provide**.

Furthermore the following controls are available:

- The button **Exit** in the header bar is used to close the window.
- The **Save** button in the footer bar is used to save the new settings.
- With the **Default** button in the footer bar everything is reset to normal. After pressing, everything is set to Patientmode.
- Additionally in the footer bar the status of the invoked action (change of setting, registering the Service Patient) is visible.

> The setting of the RFSWD (Patient mode/Service mode) is automatically handeled within TestTools, as the tests check the necessary prerequisites and set the mode accordingly.

Fig. 624: Tools



Measurement Settings

The following settings are possible:

- The **Gradient Safety Watchdog (GSWD)** can be set to
  - **Patient Mode**
    The safety watchdog functions in the same way as with a normal patient.
  - **Service Mode**
    The safety watchdog is disabled as much as necessary for service actions.
- The **Adjustments** can be set to
  - **Patient Mode**
    The adjustments function in the same way as with a normal patient. That means that if the patient table moves, an adjustment of the measurement system is invoked automatically.

- **Service Mode**

  Automatic adjustments of the measurement system after a move of the patient table is disabled.

■ **Automatic Table Movement**

This feature is used in the TuneUp/QA workflow to automatically change between the large and the small spherical phantom.

- **Enabled**

  The table movements required for TuneUp, QA, and TestTools are performed automatically; no pop-up windows with instructions will appear.

- **Disabled**

  The table movements required for TuneUp, QA, and TestTools are only performed automatically within a workflow (e.g. Table Adjust). If a movement is necessary at the beginning, e.g. move to a new phantom setup, the corresponding instructions will appear in a pop-up window.

■ **Send Raw Images**

This allows you to automatically store raw images

- **Enabled**

  The data is stored.

- **Disabled**

  No data is stored.

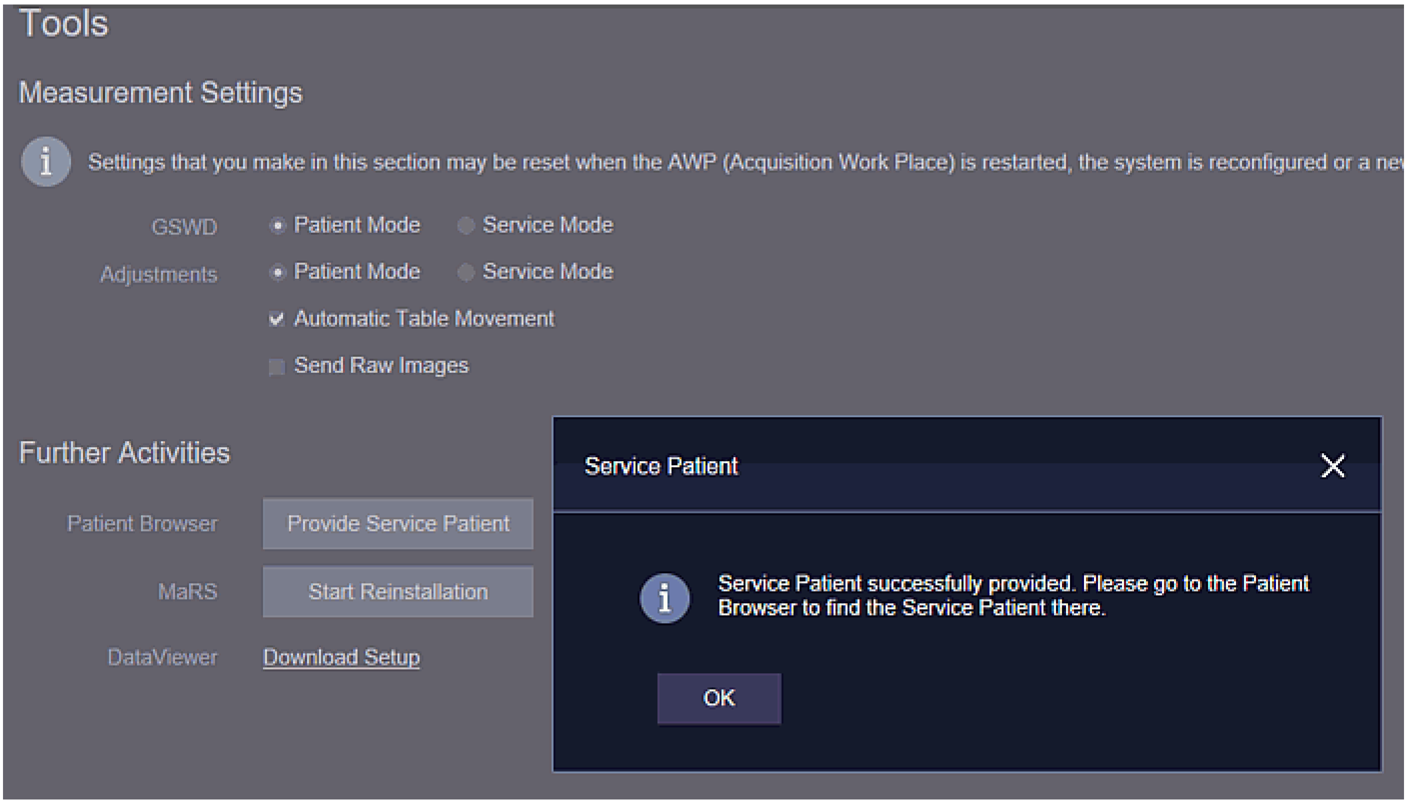
Miscellaneous

This section of the window consists of two controls:

- **Provide Service Patient**

  By clicking **Provide** the Service Patient is registered.

Fig. 625: Activating the Service Patient



For the activation navigate to **Tools button / Tools / Further Activities / Provide Service Patient**.

Afterwards register the Service Patient in the patient browser and fill in or confirm all undefined settings until the **Exam** button is released.

It is possible to **register** the Service Patient so that running a test or QA is not necessary in order to have this patient registered for further troubleshooting or service.

With Tuneup, QA, and TestTools, the Service Patient is automatically registered as soon as raw data is generated. With Tuneup and QA this will almost always be the case, whereas with TestTools mainly the RF tests require the Service Patient.

> Please note that the behaviour of the Service Patient is different to what you are used to from NUMARIS/4.

> Before the PMU functionality test is started, a clinical patient or the service patient has to be registered. Otherwise the physiological display is not accessible.

- **Reinstall MaRS**

  Clicking **Reinstall** will start the software installation of the MaRS.

In case that you suspect any software problems with the MaRS you can trigger a forced SW installation with this selection.

DataViewer Download

To download the "SetupDataViewer.exe" to your Service PC, select the link: **Click here to download the DataViewer setup**

11.1 Startup Reports - Reports Saved

Help ID: F4E87ECD-7276-4E20-8A09-0CC7B44D68C3
Service Level: 1

Start-up Reports Notification in System Status Overview

The System Status Overview displays a status indicator for the Startup Reports.

The status indicator is red if no Start-up Report and Safety-related Tests Report has been saved.

- Save the Start-up Report and Safety-related Tests Report after completing of the system startup.

If at least one Start-up Report and one Safety-related Tests Report are saved the status indicator turns to green.

11.2
Maintenance - Cold Head Maintenance

Help ID: 5BB59823-8168-4EF7-BD0A-8E8FC61F7EAE
Service Level: 1

Display of Maintenance Notifications in System Status Overview

The System Status Overview displays three status indicators for Maintenance

- Cold Head Maintenance
- Preventive Maintenance
- Safety-related tests

The indicators turn red if the interval for the specific tasks has expired.

This indicator function can be **Enabled** or **Disabled** within Maintenance Configuration.

The default setting of the function is **Disabled**. The three different maintenance indicators will never be displayed.

The maintenance indicators should be **Enabled** only if the customer does not have a service contract and only with the consent of the service manager or technical manager.

11.3 Maintenance - Safety Related Tests

Help ID: 65A5B706-9B4D-4270-BA58-B7527438BEEA
Service Level: 1

Display of Maintenance Notifications in System Status Overview

The System Status Overview displays three status indicators for Maintenance

- Cold Head Maintenance
- Preventive Maintenance
- Safety-related tests

The indicators turn red if the interval for the specific tasks has expired.

This indicator function can be **Enabled** or **Disabled** within Maintenance Configuration.

The default setting of the function is **Disabled**. The three different maintenance indicators will never be displayed.

The maintenance indicators should be **Enabled** only if the customer does not have a service contract and only with the consent of the service manager or technical manager.

11.4
Maintenance - Maintenance Tests

Help ID: 5F088AC3-1B51-439F-AA1B-B1FE35B66593
Service Level: 1

Display of Maintenance Notifications in System Status Overview

The System Status Overview displays three status indicators for Maintenance

- Cold Head Maintenance
- Preventive Maintenance
- Safety-related tests

The indicators turn red if the interval for the specific tasks has expired.

This indicator function can be **Enabled** or **Disabled** within Maintenance Configuration.

The default setting of the function is **Disabled**. The three different maintenance indicators will never be displayed.

The maintenance indicators should be **Enabled** only if the customer does not have a service contract and only with the consent of the service manager or technical manager.

11.5 Customer Environment - Helium Level

Help ID: 079DD846-DAF1-4E18-A803-4DD2C565F578
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The Customer Environment status indicator **Helium Level** displays if the current magnet helium level is sufficient to operate the system.

- green icon: the helium level is OK.
- yellow exclamation mark (Warning): contact Siemens Service.
- red box (Error): contact Siemens Service immediately! (risk of magnet quench)

11.6
Customer Environment - Helium Trend

Help ID: ABEB2028-705B-4634-BC51-F5AB690597E2
Service Level: 1

> The System Status Overview page has a refresh delay up to one minute!

The Customer Environment status indicator **Helium Trend** displays if the current magnet helium level is stable.

- green icon: no helium loss.
- red box (Error): contact Siemens Service. (helium loss detected)

11.7
Customer Environment - Cooling

Help ID: 4CB52CB7-8E58-4733-854D-8469019024F8
Service Level: 1

> The System Status Overview page has a refresh delay up to one minute!

The Customer Environment status indicator **Cooling** displays if the current status of the Primary Cooling Circuit is sufficient to operate the system.

- green icon: primary cooling OK.
- yellow exclamation mark (Warning): check the MR-system water cooling that is provided by your building facility.
- red box (Error): check the MR-system water cooling that is provided by your building facility or contact Siemens Service.

11.8 Periphery Groups - Control System

Help ID: 7AED0CF5-21B5-43BB-9C7F-CD1DD5F2F864
Service Level: 1

The SystemStatusOverview page has a refresh delay up to one minute!

The status indicator of the periphery group Control System displays if the related periphery components are all in operational state or not.

- green icon: control system OK.
- yellow exclamation mark (Warning): restart System. if the problem persists, contact Siemens Service.
- red box (Error): restart System. if the problem persists, contact Siemens Service.

11.9 Periphery Groups - Cooling System

Help ID: B9981F58-1513-4D82-9EB8-98BABCBB85D3
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Cooling System** displays if the related periphery components are all in operational state or not.

- green icon: secondary cooling system OK.
- yellow exclamation mark (Warning): contact Siemens Service.
- red box (Error): contact Siemens Service.

11.10 Periphery Groups - Magnet

Help ID: 2D2415E1-0482-46A2-BAB1-4570BA698AD8
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Magnet** displays if the related periphery components are all in operational state or not.

- green icon: magnet periphery OK.
- yellow exclamation mark (Warning): contact Siemens Service.
- red box (Error): contact Siemens Service.

11.11
Periphery Groups - Patient Table

Help ID: C382406A-1645-4E31-BD70-251E8C893D30
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Patient Table** displays if the related periphery components are all in operational state or not.

- green icon: patient table OK.
- yellow exclamation mark (Warning): check the patient table visually, restart System. if the problem persists, contact Siemens Service.
- red box (Error): check the patient table visually, restart System. if the problem persists, contact Siemens Service.

11.12 Periphery Groups - Gradient System

Help ID: E0DF8981-D409-4F7A-B407-7F661A5202CF
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Gradient System** displays if the related periphery components are all in operational state or not.

- green icon: gradient system OK.
- yellow exclamation mark (Warning): wait 3 minutes, restart System. if the problem persists, contact Siemens Service.
- red box (Error): restart System. if the problem persists, contact Siemens Service.

11.13
Periphery Groups - Receive System

Help ID: F6DBD39C-BC30-4586-8B18-5BEBD28E15F6
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Receive System** displays if the related periphery components are all in operational state or not.

- green icon: receive system OK.
- yellow exclamation mark (Warning): wait 3 minutes, restart System. if the problem persists, contact Siemens Service.
- red box (Error): restart System. if the problem persists, contact Siemens Service.

11.14
Periphery Groups - Transmit System

Help ID: 49FE45BA-ED51-4DD1-AFB8-23E9365FEFC3
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Transmit System** displays if the related periphery components are all in operational state or not.

- green icon: transmit system OK.
- yellow exclamation mark (Warning): wait 3 minutes, restart System. if the problem persists, contact Siemens Service.
- red box (Error): restart System. if the problem persists, contact Siemens Service.

11.15 Periphery Groups - Miscellaneous

Help ID: 085FEBEF-760C-4D7E-BEE4-988690FD044D
Service Level: 1

The System Status Overview page has a refresh delay up to one minute!

The status indicator of the periphery group **Miscellaneous** (HWDefault) displays if the related periphery components are all in operational state or not.

- green icon: periphery components OK.
- yellow exclamation mark (Warning): wait 3 minutes, restart System. if the problem persists, contact Siemens Service.
- red box (Error): restart System. if the problem persists, contact Siemens Service.

There are no Hazard IDs in this document.

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